

EGA IV, SECTIONS 1–10

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§1. RELATIVE FINITENESS CONDITIONS. CONSTRUCTIBLE SETS IN PRESCHEMES

In this section, we will resume and complete the exposé of “finiteness conditions” for a morphism of preschemes $f : X \rightarrow Y$ given in (I, 6.3 and 6.6). There are essentially two notions of “finiteness” of a *global* nature on X , that of *quasi-compact* morphism (defined in (I, 6.6.1)) and that of a *quasi-separated* morphism; on the other hand, there are two notions of “finiteness” of a *local* nature on X , that of a morphism *locally of finite type* (defined in (I, 6.6.2)) and that of a morphism *locally of finite presentation*. By combining these local notions with the preceding global notions, we obtain the notion of a morphism *of finite type* (defined in (I, 6.3.1)) and of a morphism *of finite presentation*. For the convenience of the reader, we will give again in this section the principal properties stated in (I, 6.3 and 6.6), referring, of course, to those numbered passages in Chapter I for their proofs.

In nos 1.8 and 1.9, we complete, in the context of preschemes, and making use of the previous notions of finiteness, the results on constructible sets given in (0_{III}, §9).

1.1. Quasi-compact morphisms

Definition (1.1.1). — We say that a morphism of preschemes $f : X \rightarrow Y$ is *quasi-compact* if the continuous map f from the topological space X to the topological space Y is quasi-compact (0, 9.1.1), in other words, if the inverse image $f^{-1}(U)$ of every quasi-compact open subset U of Y is quasi-compact (cf. (I, 6.6.1)).

If $\mathfrak{B}$ is a basis for the topology of Y consisting of affine open sets, then for f to be quasi-compact, it is necessary and sufficient that for all $V \in \mathfrak{B}$, $f^{-1}(V)$ is a *finite union of affine open sets*. For example, if Y is affine and X is quasi-compact, every morphism $f : X \rightarrow Y$ is quasi-compact (I, 6.6.1).

If $f : X \rightarrow Y$ is a quasi-compact morphism, then it is clear that for every open subset V of Y , the restriction of f to $f^{-1}(V)$ is a quasi-compact morphism $f^{-1}(V) \rightarrow V$. Conversely, if (U_α) is an open cover of Y and $f : X \rightarrow Y$ is a morphism such that the restrictions $f^{-1}(U_\alpha) \rightarrow U_\alpha$ are quasi-compact, then f is quasi-compact. As a result, if $f : X \rightarrow Y$ is an S -morphism of S -preschemes, and if there exists an open cover (S_λ) of S such that the restrictions $g^{-1}(S_\lambda) \rightarrow h^{-1}(S_\lambda)$ of f (where g and h are the structure morphisms) are quasi-compact, then f is quasi-compact.

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Proposition (1.1.2). —

- (i) An immersion $X \rightarrow Y$ is quasi-compact if it is closed, or if the underlying space of Y is locally Noetherian or if the underlying space of X is Noetherian.

- (ii) The composition of two quasi-compact morphisms is quasi-compact.
- (iii) If $f : X \rightarrow Y$ is a quasi-compact S -morphism, the same is true of $f_{(S')} : X_{(S')} \rightarrow Y_{(S')}$ for every extension $g : S' \rightarrow S$ of the base prescheme.
- (iv) If $f : X \rightarrow X'$ and $g : Y \rightarrow Y'$ are two quasi-compact S -morphisms,

$$f \times_S g : X \times_S Y \longrightarrow X' \times_S Y'$$

is quasi-compact.

- (v) If the composition $g \circ f$ of two morphisms $f : X \rightarrow Y$, $g : Y \rightarrow Z$ is quasi-compact, and if either g is separated or the underlying space of X is locally Noetherian, then f is quasi-compact.
- (vi) For a morphism f to be quasi-compact, it is necessary and sufficient that f_{red} be so.

For the proof, see (I, 6.6.4). We note that the assertion (vi) is also a consequence of the following more general proposition:

Proposition (1.1.3). — Let $f : X \rightarrow Y$, $g : Y \rightarrow Z$ be two morphisms. If $g \circ f$ is quasi-compact and f is surjective, then g is quasi-compact.

Proof. Indeed, if V is a quasi-compact open subset of Z , $f^{-1}(g^{-1}(V))$ is quasi-compact by hypothesis, and we have $g^{-1}(V) = f(f^{-1}(g^{-1}(V)))$, since f is surjective; so $g^{-1}(V)$ is quasi-compact. $\square$

Corollary (1.1.4). — Let $f : X \rightarrow Y$, $g : Y' \rightarrow Y$ be two morphisms, $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \rightarrow Y'$; suppose g is quasi-compact and surjective (resp. surjective). Then, for f to be quasi-compact (resp. dominant), it suffices that f' be so.

Proof. If $g' : X' \rightarrow X$ is the canonical projection, g' is a surjective morphism (I, 3.5.2), and we have $f \circ g' = g \circ f'$; if f' is quasi-compact (resp. dominant), the same is true of $g \circ f'$ since g is quasi-compact (1.1.2) (resp. surjective); as $f \circ g'$ is quasi-compact (resp. dominant) and g' is surjective, we deduce that f is quasi-compact (1.1.3) (resp. dominant). $\square$

To abbreviate, we will call *maximal point* of a prescheme X every generic point of an irreducible component of X ; if $X = \text{Spec}(A)$ is affine, the maximal points of X are the *minimal prime ideals* of A . IV-1 | 226

Proposition (1.1.5). — Let $f : X \rightarrow Y$ be a quasi-compact morphism. The following conditions are equivalent:

- a) f is dominant;
- b) for every maximal point y of Y , we have $f^{-1}(y) \neq \emptyset$;
- c) for every maximal point y of Y , $f^{-1}(y)$ contains a maximal point of X .

The equivalence of a) and c) was proved in (I, 6.6.5). It is clear that c) implies b); on the other hand, if X' is an irreducible component of X meeting $f^{-1}(y)$, $f(X')$ is irreducible and contains y , so it is contained in the irreducible component $Y' = \overline{\{y\}}$ of Y (0, 2.1.6); if x is the generic point of X' , we therefore necessarily have $f(x) = y$ (0, 2.1.5); so b) implies c).

Proposition (1.1.6). — Let $f' : X' \rightarrow Y$, $f'' : X'' \rightarrow Y$ be two morphisms, X the coproduct prescheme $X' \amalg X''$, $f : X \rightarrow Y$ the morphism equal to f' on X' and to f'' on X'' . For f to be quasi-compact, it is necessary and sufficient that f' and f'' be so.

This follows immediately from the definition.

1.2. Quasi-separated morphisms

Definition (1.2.1). — Let X, Y be two preschemes. We say that a morphism $f : X \rightarrow Y$ is *quasi-separated* (or that X is *quasi-separated over Y*) if the diagonal morphism $\Delta_f = (1_X, 1_X)_Y : X \rightarrow X \times_Y X$ is quasi-compact. We say that a prescheme X is *quasi-separated* if it is quasi-separated over $\text{Spec}(\mathbf{Z})$.

By definition, every separated morphism is quasi-separated, Δ_f being a closed immersion (1.1.2, (i)); a scheme X is a quasi-separated prescheme, being separated over $\text{Spec}(\mathbf{Z})$ (I, 5.4.1).

Proposition (1.2.2). —

- (i) Every monomorphism of preschemes $f : X \rightarrow Y$ (in particular every immersion) is quasi-separated.
- (ii) If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are two quasi-separated morphisms, $g \circ f : X \rightarrow Z$ is quasi-separated.
- (iii) Let X, Y be two S -preschemes, $f : X \rightarrow Y$ a quasi-separated S -morphism. Then, for every base change $g : S' \rightarrow S$, the morphism $f_{(S')} : X_{(S')} \rightarrow Y_{(S')}$ is quasi-separated.
- (iv) If $f : X \rightarrow Y, f' : X' \rightarrow Y'$ are two quasi-separated S -morphisms, then

$$f \times_S f' : X \times_S X' \rightarrow Y \times_S Y'$$

is quasi-separated.

- (v) If $f : X \rightarrow Y, g : Y \rightarrow Z$ are two morphisms such that $g \circ f$ is quasi-separated, then f is quasi-separated.
- (vi) If $f : X \rightarrow Y$ is a quasi-separated morphism, $f_{\text{red}} : X_{\text{red}} \rightarrow Y_{\text{red}}$ is quasi-separated.

Proof. By virtue of (I, 5.5.12), it suffices to prove (i), (ii) and (iii).

The assertion (i) is trivial, every monomorphism being separated (I, 5.5.1). To prove (iii), we reduce immediately to the case where $Y = S$ (I, 3.3.11) and the assertion follows from the fact that $\Delta_{f_{(S')}} = (\Delta_f)_{(S')}$ (I, 5.3.4) and from (1.1.2, (iii)). To prove (ii), consider the projections p and q of $X \times_Y X$ over X ; if $i = (p, q)_Z$, we know that the diagram

$$\begin{array}{ccc} X \times_Y X & \xrightarrow{i} & X \times_Z X \\ \pi \downarrow & & \downarrow f \times_Z f \\ Y & \xrightarrow{\Delta_g} & Y \times_Z Y \end{array}$$

(where π is the structure morphism) is commutative and identifies $X \times_Y X$ with the product, as $(Y \times_Z Y)$ -preschemes, of Y and $X \times_Z X$ (I, 5.3.5). If g is quasi-separated, Δ_g is quasi-compact, and hence so is i (1.1.2, (iii)); if in addition f is quasi-separated, Δ_f is quasi-compact, and hence so is $i \circ \Delta_f$ (1.1.2, (ii)), which is equal to $\Delta_{g \circ f}$. $\square$

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Corollary (1.2.3). —

- (i) Let X be a quasi-separated prescheme. Then every morphism $f : X \rightarrow Y$ is quasi-separated.
- (ii) Let Y be a quasi-separated prescheme. For a morphism $f : X \rightarrow Y$ to be quasi-separated, it is necessary and sufficient that the prescheme X be quasi-separated.

This follows immediately from (1.2.2, (ii) and (v)) applied to X, Y and $\text{Spec}(\mathbf{Z})$.

Proposition (1.2.4). — Let $f : X \rightarrow Y$ be a morphism, $g : Y \rightarrow Z$ a quasi-separated morphism. If $g \circ f$ is quasi-compact, then so is f .

Proof. We know that f is the composite morphism $X \xrightarrow{\Gamma_f} X \times_Z Y \xrightarrow{\text{pr}_2} Y$ (I, 5.3.13); on the other hand, pr_2 is identified with $(g \circ f) \times_Z 1_Y$ (I, 3.3.4) and since $g \circ f$ is quasi-compact, so is pr_2 (1.1.2, (iv)). Finally, we have the commutative diagram

$$(1.2.4.1) \quad \begin{array}{ccc} X & \xrightarrow{\Gamma_f} & X \times_Z Y \\ f \downarrow & & \downarrow f \times 1_Y \\ Y & \xrightarrow{\Delta_g} & Y \times_Z Y \end{array}$$

which identifies X with the product, as $(Y \times_Z Y)$ -preschemes, of Y and $X \times_Z Y$ (I, 5.3.7). As by hypothesis Δ_g is a quasi-compact morphism, so is Γ_f (1.1.2, (iii)); the conclusion then follows from (1.1.2, (ii)). $\square$

Proposition (1.2.5). — *Let $f : X \rightarrow Y$, $g : Y' \rightarrow Y$ be two morphisms, $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \rightarrow Y'$. If g is quasi-compact and surjective and f' quasi-separated, then f is quasi-separated.*

Proof. We have indeed $(X' \times_{Y'} X') = (X \times_Y X)_{(Y')}$ and $\Delta_{f'} = (\Delta_f)_{(Y')}$ (I, 5.3.4); as the projection $X' \times_{Y'} X' \rightarrow X \times_Y X$ is quasi-compact (1.1.2, (iii)) and surjective (I, 3.5.2), we can, by virtue of (I, 3.3.11), apply (1.1.4), which shows that since $\Delta_{f'}$ is a quasi-compact morphism, the same is true of Δ_f . $\square$

Proposition (1.2.6). — *Let $f : X \rightarrow Y$ be a morphism, (U_α) a covering of Y by open subsets such that the induced preschemes U_α are quasi-separated. For f to be quasi-separated, it is necessary and sufficient that each of the induced preschemes on the open subsets $f^{-1}(U_\alpha)$ be quasi-separated.*

Proof. The inverse image in $X \times_Y X$ of U_α is $X_\alpha \times_{U_\alpha} X_\alpha$, writing $X_\alpha = f^{-1}(U_\alpha)$, and the restriction $X_\alpha \rightarrow X_\alpha \times_{U_\alpha} X_\alpha$ of Δ_f is none other than Δ_{f_α} , denoting by f_α the restriction $X_\alpha \rightarrow U_\alpha$ of f ; by virtue of the definition (1.2.1) and the local character of the notion of quasi-compact morphism (1.1.1), for f to be quasi-separated, it is necessary and sufficient that each of the morphisms f_α be so. But since by hypothesis the morphism $U_\alpha \rightarrow \text{Spec}(\mathbf{Z})$ is quasi-separated, saying that f_α is quasi-separated is equivalent to saying that the composite $X_\alpha \xrightarrow{f_\alpha} U_\alpha \rightarrow \text{Spec}(\mathbf{Z})$ is quasi-separated (1.2.2, (ii) and (v)); hence the proposition. $\square$

To verify that a morphism is quasi-separated, we are therefore reduced to giving criteria for a prescheme to be quasi-separated: IV-1 | 228

Proposition (1.2.7). — *Let X be a prescheme, (U_α) a covering of X formed by quasi-compact open subsets. The following properties are equivalent:*

- a) X is quasi-separated.
- b) For every quasi-compact open subset U of X , the canonical immersion $U \rightarrow X$ is quasi-compact (in other words, U is retrocompact in X).
- b') The intersection of two quasi-compact open subsets of X is quasi-compact.
- c) For every pair of indices (α, β) , the intersection $U_\alpha \cap U_\beta$ is quasi-compact.

Proof. The properties b) and b') are trivially equivalent by definition of a quasi-compact morphism. As a quasi-compact open subset is a finite union of affine open subsets, for two quasi-compact open subsets U, V of X , $U \times_Z V$ is a quasi-compact open subset of $X \times_Z X$ (I, 3.2.7), whose inverse image under Δ_X is $U \cap V$, so a) implies b'). It is trivial that b') implies c); finally, if c) is satisfied, the $U_\alpha \times_Z U_\beta$ form a covering of $X \times_Z X$ by quasi-compact open subsets, and we know that for the morphism $\Delta_X : X \rightarrow X \times_Z X$ to be quasi-compact, it suffices that the inverse images $U_\alpha \cap U_\beta$ under Δ_X be quasi-compact (1.1.1); so c) implies a). $\square$

Corollary (1.2.8). — *Every prescheme X whose underlying space is locally Noetherian (for example a locally Noetherian prescheme) is quasi-separated; every morphism $f : X \rightarrow Y$ is then quasi-separated.*

Proposition (1.2.9). — *Let X', X'' be two preschemes, $X = X' \amalg X''$ their coproduct prescheme, $f' : X' \rightarrow Y$, $f'' : X'' \rightarrow Y$ two morphisms, $f : X \rightarrow Y$ the morphism which coincides with f' on X' and with f'' on X'' . For f to be quasi-separated, it is necessary and sufficient that f' and f'' be so.*

Proof. Indeed, $X \times_Y X$ is the coproduct of the four preschemes $X' \times_Y X'$, $X' \times_Y X''$, $X'' \times_Y X'$ and $X'' \times_Y X''$, and Δ_f is the morphism which coincides with $\Delta_{f'}$ on X' and with $\Delta_{f''}$ on X'' ; the proposition then follows immediately from the definitions. $\square$

1.3. Morphisms locally of finite type

(1.3.1). Recall that if A is a ring, a (commutative) A -algebra B is said to be of *finite type* if it is generated by a finite number of elements of B , or, what amounts to the same thing, if it is isomorphic to a quotient of a polynomial algebra $A[T_1, \dots, T_n]$ by an ideal of this algebra. It is clear that for every (commutative) A -algebra A' , $B \otimes_A A'$ is then an A' -algebra of finite type. If B is an A -algebra of finite type and C a B -algebra of finite type, C is an A -algebra of finite type, for if C is a quotient of $B[T_1, \dots, T_n]$ and B a quotient of $A[S_1, \dots, S_m]$, $B[T_1, \dots, T_n]$ is isomorphic to $A[S_1, \dots, S_m, T_1, \dots, T_n]/\mathfrak{a}$, where $\mathfrak{a}$ is the kernel of the surjection $A[S_1, \dots, S_m, T_1, \dots, T_n] \rightarrow B[T_1, \dots, T_n]$; hence C is also a quotient of $A[S_1, \dots, S_m, T_1, \dots, T_n]$, and so it is indeed an A -algebra of finite type.

Definition (1.3.2). — Let $f : X \rightarrow Y$ be a morphism of preschemes, x a point of X , $y = f(x)$. We say that f is of *finite type at x* if there exists an affine open neighbourhood V of y and an affine open neighbourhood U of x such that $f(U) \subset V$ and the ring $A(U)$ is an $A(V)$ -algebra of finite type. We say that f is *locally of finite type* if it is of finite type at every point of X .

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Proposition (1.3.3). — *If Y is a locally Noetherian prescheme and $f : X \rightarrow Y$ a morphism locally of finite type, then X is locally Noetherian.*

For the proof, see (I, 6.3.7).

Proposition (1.3.4). —

- (i) *Every local immersion is locally of finite type.*
- (ii) *If two morphisms $f : X \rightarrow Y$, $g : Y \rightarrow Z$ are locally of finite type, then so is $g \circ f$.*
- (iii) *If $f : X \rightarrow Y$ is an S -morphism locally of finite type, $f_{(S')} : X_{(S')} \rightarrow Y_{(S')}$ is locally of finite type for every extension $S' \rightarrow S$ of the base prescheme.*
- (iv) *If $f : X \rightarrow X'$ and $g : Y \rightarrow Y'$ are two S -morphisms locally of finite type, $f \times_S g$ is locally of finite type.*
- (v) *If the composition $g \circ f$ of two morphisms is locally of finite type, f is locally of finite type.*
- (vi) *If a morphism f is locally of finite type, the same is true of f_{red} .*

For the proof, see (I, 6.6.6).

Corollary (1.3.5). — *Let $f : X \rightarrow Y$ be a morphism locally of finite type. For every morphism $Y' \rightarrow Y$ such that Y' is locally Noetherian, $X \times_Y Y'$ is locally Noetherian.*

Proof. Indeed, $f_{(Y')} : X \times_Y Y' \rightarrow Y'$ is locally of finite type by (1.3.4, (iii)), and it suffices to apply (1.3.3). $\square$

Proposition (1.3.6). — *Let $\varphi : A \rightarrow B$ be a homomorphism of rings. For the corresponding morphism $f : \text{Spec}(B) \rightarrow \text{Spec}(A)$ to be locally of finite type, it is necessary and sufficient that B be an A -algebra of finite type.*

For the proof, see (I, 6.3.3), taking into account that f is quasi-compact and using (I, 6.6.3).

Proposition (1.3.7). — *For a morphism locally of finite type $f : X \rightarrow Y$ to be surjective, it is necessary and sufficient that, for every algebraically closed field Ω , the map $X(\Omega) \rightarrow Y(\Omega)$ corresponding to f (I, 3.4.1) be surjective.*

For the proof, see (I, 6.3.10).

(1.3.8). Let A be a ring. We say that an A -algebra B is *essentially of finite type* if B is A -isomorphic to an A -algebra of the form $S^{-1}C$, where C is an A -algebra of finite type and S a multiplicative subset of C .

Proposition (1.3.9). —

- (i) *If B is an A -algebra essentially of finite type and C a B -algebra essentially of finite type, then C is an A -algebra essentially of finite type.*

- (ii) Let B be an A -algebra essentially of finite type, A' an A -algebra; then $B' = B \otimes_A A'$ is an A' -algebra essentially of finite type.

Proof. (i) Let $B = S'^{-1}B'$ and $C = T'^{-1}C'$, where B' (resp. C') is an A -algebra (resp. B -algebra) of finite type and S' (resp. T') a multiplicative subset of B' (resp. C'); then C' is of the form $S'^{-1}C''$, where C'' is also a B' -algebra of finite type, so C is also a ring of fractions of C'' ; as C'' is an A -algebra of finite type, this proves our assertion.

- (ii) If B is a ring of fractions of an A -algebra of finite type C , B' is a ring of fractions of the A' -algebra of finite type $C \otimes_A A'$, which proves (ii). $\square$

(1.3.10). If B is a local A -algebra essentially of finite type, B is of the form $C_{\mathfrak{q}}$, where C is an A -algebra of finite type and $\mathfrak{q}$ a prime ideal of C (Bourbaki, *Alg. comm.*, chap. II, §5, prop. 11). Let $\mathfrak{p}$ be the inverse image in A of the ideal $\mathfrak{q}$; if we set $S = A - \mathfrak{p}$, $C_{\mathfrak{q}}$ is also the localization of $S^{-1}C$ at a prime ideal; as $S^{-1}C$ is an algebra of finite type over $A_{\mathfrak{p}} = S^{-1}A$, we see that B is also an $A_{\mathfrak{p}}$ -algebra essentially of finite type, and furthermore the homomorphism $A_{\mathfrak{p}} \rightarrow B$ is local.

Proposition (1.3.11). — If B is a local A -algebra essentially of finite type, B is A -isomorphic to a quotient of an A -algebra of the form $C_{\mathfrak{q}}$, where $C = A[T_1, \dots, T_n]$ is a polynomial algebra over A and $\mathfrak{q}$ a prime ideal of C .

Proof. Indeed, by definition, B is isomorphic to $C'_{\mathfrak{q}'}$, where C' is an A -algebra of finite type and $\mathfrak{q}'$ a prime ideal of C' ; but $C' = C/\mathfrak{b}$, where $C = A[T_1, \dots, T_n]$ and $\mathfrak{b}$ is an ideal of C ; so $\mathfrak{q}' = \mathfrak{q}/\mathfrak{b}$, where $\mathfrak{q}$ is a prime ideal of C , and $C'_{\mathfrak{q}'}$ is isomorphic to $C_{\mathfrak{q}}/\mathfrak{b}C_{\mathfrak{q}}$. $\square$

1.4. Morphisms locally of finite presentation

(1.4.1). Given a ring A , we say that an A -algebra B (commutative) is of *finite presentation* if it is isomorphic to a quotient $A[T_1, \dots, T_n]/\mathfrak{a}$ of a polynomial algebra over A by an ideal of finite type $\mathfrak{a}$ of $A[T_1, \dots, T_n]$. For every (commutative) A -algebra A' , $B \otimes_A A'$ is then an A' -algebra of finite presentation, being isomorphic to the quotient of $A'[T_1, \dots, T_n]$ by the image of the ideal $\mathfrak{a} \otimes_A A'$ in this ring, which is evidently an $A'[T_1, \dots, T_n]$ -module of finite type. If B is an A -algebra of finite presentation and C a B -algebra of finite presentation, then C is an A -algebra of finite presentation. Indeed, let $B = A[S_1, \dots, S_m]/\mathfrak{a}$ and $C = B[T_1, \dots, T_n]/\mathfrak{b}$, where $\mathfrak{a}$ (resp. $\mathfrak{b}$) is an ideal of finite type of $A[S_1, \dots, S_m]$ (resp. $B[T_1, \dots, T_n]$); the ring $B[T_1, \dots, T_n]$ is isomorphic to $A[S_1, \dots, S_m, T_1, \dots, T_n]/\mathfrak{a}'$, where $\mathfrak{a}'$ is the canonical image of $\mathfrak{a} \otimes_A A[T_1, \dots, T_n]$, and is therefore an ideal of finite type of $A[S_1, \dots, S_m, T_1, \dots, T_n]$; the ideal $\mathfrak{b}$ of $B[T_1, \dots, T_n]$ is of the form $\mathfrak{b}'/\mathfrak{a}'$, where $\mathfrak{b}'$ is an ideal of $A[S_1, \dots, S_m, T_1, \dots, T_n]$; as $\mathfrak{a}'$ and $\mathfrak{b}'/\mathfrak{a}'$ are modules of finite type over $A[S_1, \dots, S_m, T_1, \dots, T_n]$, so is $\mathfrak{b}'$, and C , being isomorphic to $A[S_1, \dots, S_m, T_1, \dots, T_n]/\mathfrak{b}'$, is therefore an A -algebra of finite presentation. We note finally that if A is *Noetherian*, then so is $A[T_1, \dots, T_n]$; hence every A -algebra of finite type is an algebra of finite presentation.

Definition (1.4.2). — Let $f : X \rightarrow Y$ be a morphism of preschemes, x a point of X , $y = f(x)$. We say that f is of *finite presentation at x* if there exists an affine open neighbourhood V of y and an affine open neighbourhood U of x such that $f(U) \subset V$ and such that the ring $\mathcal{O}(U)$ is an $\mathcal{O}(V)$ -algebra of finite presentation. We say that f is *locally of finite presentation* if it is of finite presentation at every point of X .

If Y is locally Noetherian, it amounts to the same to say that f is locally of finite type or locally of finite presentation.

Proposition (1.4.3). — (i) Every local isomorphism is locally of finite presentation.

- (ii) If two morphisms $f : X \rightarrow Y$, $g : Y \rightarrow Z$ are locally of finite presentation, so is $g \circ f$.

- (iii) If $f : X \rightarrow Y$ is an S -morphism locally of finite presentation, $f_{(S')} : X_{(S')} \rightarrow Y_{(S')}$ is locally of finite presentation for every base extension $S' \rightarrow S$ of the base prescheme.

- (iv) If $f : X \rightarrow X'$ and $g : Y \rightarrow Y'$ are two S -morphisms locally of finite presentation, $f \times_S g$ is locally of finite presentation.

- (v) If the composition $g \circ f$ of two morphisms is locally of finite presentation and if g is locally of finite type, then f is locally of finite presentation.

Proof. Assertion (i) is trivial. To prove (iii), we may limit to the case where $S = Y$ (I, 3.3.11); let x' be a point of $X_{(S')}$, s' and x its projections on S' and X respectively, s the common projection of s' and x on S . By hypothesis, there exists an affine open neighbourhood V (resp. U) of s (resp. x) such that the image of U is contained in V and such that $\mathcal{O}(U)$ is an $\mathcal{O}(V)$ -algebra of finite presentation; let V' be an affine open neighbourhood of s' whose image in S is contained in V ; then $U \times_V V'$ is an affine open neighbourhood of x' whose image in S' is contained in V' and whose ring $\mathcal{O}(U) \otimes_{\mathcal{O}(V)} \mathcal{O}(V')$ is an $\mathcal{O}(V')$ -algebra of finite presentation (1.4.1).

To prove (ii), consider a point $x \in X$, and let $y = f(x)$, $z = g(f(x))$. There is by hypothesis an affine open neighbourhood $W = \text{Spec}(C)$ (resp. $V = \text{Spec}(B)$) of z (resp. y) such that $g(V) \subset W$ and such that B is a C -algebra of finite presentation. On the other hand, there is an affine open neighbourhood $V' = \text{Spec}(B')$ (resp. $U = \text{Spec}(A)$) of y (resp. x) such that $f(U) \subset V'$ and such that A is a B' -algebra of finite presentation. Let $s \in B$ be such that the open set $D(s) = \text{Spec}(B_s)$ in V is a neighbourhood of y contained in V' , and let $U' = f^{-1}(D(s)) \subset U$; we have $U' = \text{Spec}(A \otimes_{B'} B_s)$, and $A \otimes_{B'} B_s$ is a B_s -algebra of finite presentation (1.4.1); on the other hand $B_s = B[T]/(1 - sT)$ is a B -algebra of finite presentation by definition; hence by (1.4.1), $A \otimes_{B'} B_s$ is a C -algebra of finite presentation, which proves (ii).

We know that (iv) follows from (i), (ii), and (iii) (I, 3.5.1). To prove (v), consider the commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{\Gamma_f} & X \times_Z Y \\ & & \downarrow f \times 1 \\ Y & \xrightarrow{\Delta_g} & Y \times_Z Y \end{array}$$

which identifies X with the product of the $(Y \times_Z Y)$ -preschemes Y and $X \times_Z Y$ (I, 5.3.7), Δ_g being the diagonal morphism. If we know that Δ_g is locally of finite presentation, it will follow by (iii) that so is Γ_f . On the other hand, we have the factorisation of f as $X \xrightarrow{\Gamma_f} X \times_Z Y \xrightarrow{p_2} Y$ (I, 5.3.13), where the projection p_2 is equal to $(g \circ f) \times_Z 1_Y$; as $g \circ f$ is assumed locally of finite presentation, so is p_2 by (iv), and it will follow finally from (ii) that f is also locally of finite presentation. We thus see that we are reduced to proving the following. $\square$

Corollary (1.4.3.1). — *Let $g : Y \rightarrow Z$ be a morphism locally of finite type; then the diagonal morphism $\Delta_g : Y \rightarrow Y \times_Z Y$ is locally of finite presentation.*

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Proof. We may reduce to the case where $Z = \text{Spec}(A)$, $Y = \text{Spec}(B)$, B being an A -algebra of finite type; the morphism Δ_g then corresponds to the augmentation homomorphism $\delta : B \otimes_A B \rightarrow B$ such that $\delta(x \otimes y) = xy$. In view of the definition (1.4.2), it suffices to show that the kernel $\mathfrak{J}$ of δ is an ideal of finite type, and this follows from (0, 20.4.4). $\square$

Proposition (1.4.4). — *Let A be a ring, B an A -algebra, B' a polynomial algebra $A[T_1, \dots, T_n]$, $u : B' \rightarrow B$ a surjective homomorphism of A -algebras. For B to be an A -algebra of finite presentation, it is necessary and sufficient that the kernel $\mathfrak{J}$ of u be an ideal of finite type of B' .*

Proof. The condition is sufficient by definition (1.4.1). To see that it is necessary, we remark that the morphism $g : \text{Spec}(B') \rightarrow \text{Spec}(A)$ is locally of finite type; if B is an A -algebra of finite presentation, the morphism $f : \text{Spec}(B) \rightarrow \text{Spec}(B')$ corresponding to u is such that $g \circ f : \text{Spec}(B) \rightarrow \text{Spec}(A)$ is locally of finite presentation, and so it follows from (1.4.3, (v)) that f is also locally of finite presentation. Now, to show that the ideal $\mathfrak{J}$ is of finite type, it suffices to prove that $\mathfrak{J}$ is a B' -module of finite type (II, 6.1.4.1). Returning to the definition (1.4.2), we are finally reduced to proving the following. $\square$

Lemma (1.4.4.1). — *Let $\mathfrak{a}$ be an ideal of a ring C , s an element of C , such that $C_s/\mathfrak{a}_s$ is a C -algebra of finite presentation; then $\mathfrak{a}_s$ is an ideal of finite type in the ring C_s .*

Proof. By hypothesis, there exists a polynomial C -algebra $C' = C[T_1, \dots, T_n]$ and a surjective C -homomorphism $v : C' \rightarrow C_s/\mathfrak{a}_s$ whose kernel $\mathfrak{b}$ is of finite type. Let $p : C_s \rightarrow C_s/\mathfrak{a}_s$ be the canonical homomorphism; for each i , there is a $t_i \in C$ such that $p(t_i/s^k) = v(T_i)$ (we may assume that the exponent k is the same for all indices i). Consider then the C_s -algebra $D = C_s[T_1, \dots, T_n]$, and the C_s -algebra homomorphism $w : D \rightarrow C_s$ such that $w(T_i) = t_i/s^k$; w is evidently surjective, and so is $v' = p \circ w : D \rightarrow C_s/\mathfrak{a}_s$. On the other hand, every polynomial P of D can be written as P/s^m , where $P \in C' = C[T_1, \dots, T_n]$, and we have $v'(P/s^m) = (1/s^m)v(P)$; as $1/s$ is invertible in C_s , the relation $v'(P/s^m) = 0$ in $C_s/\mathfrak{a}_s$ is equivalent to $v(P) = 0$, and hence the kernel $\mathfrak{b}'$ of v' is generated by the image of $\mathfrak{b}$ in the ring D , and *a fortiori* is of finite type in D . But $\mathfrak{b}' = v'^{-1}(0) = w^{-1}(p^{-1}(0)) = w^{-1}(\mathfrak{a}_s)$, and as w is surjective, $\mathfrak{a}_s = w(\mathfrak{b}')$, which proves that $\mathfrak{a}_s$ is an ideal of finite type of C_s . $\square$

Corollary (1.4.5). — *Let X, Y be two preschemes, $j : X \rightarrow Y$ an immersion, U an open subset of Y such that $j(X)$ is closed in U , $\mathcal{J}$ the quasi-coherent ideal of $\mathcal{O}_U$ that defines the closed subprescheme of Y associated to j . For j to be locally of finite presentation, it is necessary and sufficient that $\mathcal{J}$ be an $\mathcal{O}_U$ -module of finite type.*

Proposition (1.4.6). — *Let $\varphi : A \rightarrow B$ be a homomorphism of rings. For the corresponding morphism $f : \text{Spec}(B) \rightarrow \text{Spec}(A)$ to be locally of finite presentation, it is necessary and sufficient that B be an A -algebra of finite presentation.*

Proof. The condition being trivially sufficient, it remains to prove that it is necessary. If f is locally of finite presentation, it already follows from (1.3.6) that B is an A -algebra of finite type, so there exists a surjective homomorphism $u : B' = A[T_1, \dots, T_n] \rightarrow B$ of A -algebras; it follows from (1.4.3, (v)) that the immersion $j : \text{Spec}(B) \rightarrow \text{Spec}(B')$ is locally of finite presentation; hence by (1.4.5), if $\mathfrak{J}$ is the kernel of u , the $\tilde{B}'$ -module $\tilde{\mathfrak{J}}$ is of finite type, and hence by (II, 6.1.4.1) $\mathfrak{J}$ is an ideal of finite type. $\square$

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Proposition (1.4.7). — *Let $\rho : A \rightarrow B$ be a homomorphism of rings making B a finite A -algebra. For B to be an A -algebra of finite presentation, it is necessary and sufficient that B be an A -module of finite presentation.*

We will use the following.

Lemma (1.4.7.1). — *If B is a finite A -algebra, there exists a surjective homomorphism of A -algebras $u : B' \rightarrow B$, where B' is a finite A -algebra of finite presentation and a free A -module.*

Proof. Indeed, there is a finite system $(a_i)_{1 \leq i \leq m}$ of generators of the A -algebra B such that each a_i satisfies a relation $F_i(a_i) = 0$, where $F_i \in A[X]$ is a monic polynomial of degree > 0 ; the A -algebra quotient $B'_i = A[X]/(F_i)$ is therefore free of finite rank over A ; let c_i be the class of X in B'_i . There exists an A -homomorphism of algebras $u_i : B'_i \rightarrow B$ such that $u_i(c_i) = a_i$; it then suffices to take for B' the tensor product $B'_1 \otimes_A B'_2 \otimes_A \dots \otimes_A B'_m$ and to consider the homomorphism $u_1 \otimes u_2 \otimes \dots \otimes u_m : B' \rightarrow B$, which is surjective by construction. Furthermore, if $B'' = A[T_1, \dots, T_m]$ and if b'_i denotes the canonical image of c_i in B' , the A -homomorphism of algebras $v : B'' \rightarrow B'$ such that $v(T_i) = b'_i$ ($1 \leq i \leq m$) is surjective, and its kernel $\mathfrak{b}'$ is the ideal of B'' generated by the $F_i(T_i)$, and therefore of finite type; this proves that B' is an A -algebra of finite presentation. $\square$

Proof of (1.4.7). This lemma being proved, let us keep the same notation, and let $w = u \circ v : B'' \rightarrow B$.¹ Let $\mathfrak{b}$ (resp. $\mathfrak{a}$) be the kernel of w (resp. u); we have $\mathfrak{a} = v(\mathfrak{b})$ since v is surjective. Now by (1.4.4), if B is an A -algebra of finite presentation, $\mathfrak{b}$ is an ideal of finite type of B'' , so $\mathfrak{a}$ is an ideal of finite type of B' , hence an A -module of finite type since B' is a finite A -algebra. As B' is a free A -module, B is an A -module of finite presentation. Conversely, if B is an A -module of finite presentation, $\mathfrak{a}$ is an A -module of finite type (Bourbaki, *Alg. comm.*, chap. I, §2, n° 8, lemme 9) and *a fortiori* an ideal of finite type of B' ; hence, B is by definition a B' -algebra of finite presentation, and as B' is an A -algebra of finite presentation, B is an A -algebra of finite presentation. $\square$

¹The French source prints $v \circ u$; with the displayed domains and codomains, the typed composite is $u \circ v$.

1.5. Morphisms of finite type

Proposition (1.5.1). — *Let $f : X \rightarrow Y$ be a morphism, (U_α) a covering of Y by open affines. The following conditions are equivalent:*

- a) *f is locally of finite type and quasi-compact.*
- b) *For all α , $f^{-1}(U_\alpha)$ is a finite union of affine open sets $V_{\alpha i}$ such that each ring $\mathcal{O}(V_{\alpha i})$ is an $\mathcal{O}(U_\alpha)$ -algebra of finite type.*
- c) *For every affine open subset U of Y , $f^{-1}(U)$ is a finite union of affine open sets V_j such that the rings $\mathcal{O}(V_j)$ are $\mathcal{O}(U)$ -algebras of finite type.*

Proof. For the proof, see (I, 6.3.2) and (I, 6.6.3). □

Definition (1.5.2). — We say that a morphism f is of *finite type* if it satisfies the equivalent conditions of the proposition (1.5.1).

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Proposition (1.5.3). — *Let $f : X \rightarrow Y$ be a morphism of finite type; if Y is Noetherian, so is X .*

Proof. For the proof, see (I, 6.3.7). □

Proposition (1.5.4). —

- (i) *Let $j : X \rightarrow Y$ be an immersion. If j is quasi-compact (which is the case if j is a closed immersion, or if the underlying space of X is Noetherian, or if the space underlying Y is locally Noetherian), then j is of finite type.*
- (ii) *The composition of two morphisms of finite type is of finite type.*
- (iii) *If $f : X \rightarrow Y$ is an S -morphism of finite type, $f_{(S')} : X_{(S')} \rightarrow Y_{(S')}$ is of finite type for every base extension $S' \rightarrow S$ of the base prescheme.*
- (iv) *If $f : X \rightarrow X'$ and $g : Y \rightarrow Y'$ are two S -morphisms of finite type, $f \times_S g$ is of finite type.*
- (v) *If the composition $g \circ f$ of two morphisms is of finite type and if g is quasi-separated or if X is Noetherian, f is of finite type.*
- (vi) *If a morphism f is of finite type, so is f_{red} .*

Proof. This follows immediately (except for the case (v) where X is Noetherian, proved in (I, 6.3.6)) from the definitions and from (1.1.2), (1.2.4), and (1.3.4). □

Corollary (1.5.5). — *Let $f : X \rightarrow Y$ be a morphism of finite type. For every morphism $Y' \rightarrow Y$ such that Y' is Noetherian, $X \times_Y Y'$ is Noetherian.*

Proof. This follows from (1.5.4, (iii)) and from (1.5.3). □

Corollary (1.5.6). — *Let X be a prescheme of finite type over a locally Noetherian prescheme S . Then every S -morphism $f : X \rightarrow Y$ is of finite type.*

Proof. For the proof, see (I, 6.3.9). □

Proposition (1.5.7). — *Let $\varphi : A \rightarrow B$ be a homomorphism of rings. For the corresponding morphism $f : \text{Spec}(B) \rightarrow \text{Spec}(A)$ to be of finite type, it is necessary and sufficient that φ make B an A -algebra of finite type.*

Proof. For the proof, see (I, 6.3.3). □

1.6. Morphisms of finite presentation

Definition (1.6.1). — Let X, Y be two preschemes. We say that a morphism $f : X \rightarrow Y$ is of *finite presentation* if it satisfies the following three conditions:

- (i) f is locally of finite presentation;
- (ii) f is quasi-compact (which, when (i) is satisfied, is equivalent to saying that f is of finite type (1.5.2));
- (iii) f is quasi-separated.

We then also say that X is of *finite presentation over Y* , or a *Y -prescheme of finite presentation*.

Condition (iii) is automatically satisfied if f is separated, or if X is locally Noetherian (1.2.8); when Y is locally Noetherian, it amounts to the same to say that f is of *finite presentation* or that f is of *finite type* (the latter condition implying that X is locally Noetherian (1.3.3)).

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- Proposition (1.6.2).** —
- (i) Every quasi-compact immersion that is locally of finite presentation (in particular every quasi-compact open immersion) is of finite presentation.
 - (ii) The composition of two morphisms of finite presentation is of finite presentation.
 - (iii) Let X, Y be two S -preschemes, $f : X \rightarrow Y$ an S -morphism of finite presentation. Then, for every morphism $S' \rightarrow S$, the morphism $f_{(S')} : X_{(S')} \rightarrow Y_{(S')}$ is of finite presentation.
 - (iv) Let $f : X \rightarrow Y, f' : X' \rightarrow Y'$ be two S -morphisms of finite presentation; then $f \times_S f' : X \times_S X' \rightarrow Y \times_S Y'$ is of finite presentation.
 - (v) If the composition $g \circ f$ of two morphisms is of finite presentation and if g is quasi-separated and locally of finite presentation, then f is of finite presentation.

Proof. This follows immediately from (1.1.2), (1.2.2), (1.2.4), and (1.4.3). $\square$

It follows in particular from (1.6.2, (iii)) that if f is a morphism of finite presentation and U an open subset of Y , the restriction $f^{-1}(U) \rightarrow U$ of f is a morphism of finite presentation. Conversely, let (U_α) be a covering of Y by open affines, and suppose that the restrictions $f^{-1}(U_\alpha) \rightarrow U_\alpha$ of f are morphisms of finite presentation; then f is evidently locally of finite presentation, quasi-compact by virtue of (1.1.1), and quasi-separated by virtue of (1.2.6). We can therefore say that the property for a morphism $f : X \rightarrow Y$ of being of finite presentation is *local on Y* .

If X is a quasi-separated prescheme, every morphism $f : X \rightarrow Y$ is quasi-separated (1.2.3). Hence, if f is quasi-compact and locally of finite presentation, f is of finite presentation.

In particular:

Corollary (1.6.3). — Let $\varphi : A \rightarrow B$ be a homomorphism of rings. For the corresponding morphism $f : \text{Spec}(B) \rightarrow \text{Spec}(A)$ to be of finite presentation, it is necessary and sufficient that φ make B an A -algebra of finite presentation.

Proof. The necessity of the condition follows from (1.4.6). $\square$

Remark (1.6.4). — In the definition (1.6.1), condition (iii) is not a consequence of the two others. Let for example Y be an affine scheme whose underlying space is not Noetherian, and let U be an open subset that is not quasi-compact in Y . Let X be the prescheme obtained by gluing two preschemes Y_1, Y_2 isomorphic to Y along the open subsets U_1, U_2 corresponding to U (0_I, 4.1.7), so that X is the union of two open subsets isomorphic respectively to Y_1 and Y_2 (and that we identify with them), with $Y_1 \cap Y_2 = U$. Furthermore, there is a morphism $f : X \rightarrow Y$ that coincides in Y_i with the isomorphism $Y_i \rightarrow Y$ ($i = 1, 2$). It is clear that f satisfies condition (i) of (1.6.1), being a local isomorphism; it also satisfies (ii), the inverse image of an open quasi-compact subset of Y being the union of its images in Y_1 and Y_2 ; but since $Y_1 \cap Y_2$ is not quasi-compact, f is not quasi-separated (1.2.7).

Proposition (1.6.5). — Let X', X'' be two preschemes, $X = X' \amalg X''$ their sum, $f' : X' \rightarrow Y, f'' : X'' \rightarrow Y$ two morphisms, $f : X \rightarrow Y$ the morphism that coincides with f' in X' and with f'' in X'' . For f to be of finite presentation, it is necessary and sufficient that f' and f'' be so.

Proof. It suffices to show that for f to possess one of the three properties of the definition (1.6.1), it is necessary and sufficient that f' and f'' possess it; this is evident for the property of being locally of finite presentation, which is local on X ; for the property of being quasi-compact, this was seen in (1.1.6), and for the property of being quasi-separated, in (1.2.9). $\square$

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1.7. Improvements of earlier results We will give in this number a list of propositions proved in the preceding chapters, whose statements can be improved with the aid of the new finiteness conditions introduced above.

(1.7.1). In the statements of (I, 6.4.2), (I, 6.4.3), and (I, 6.4.9), we can, instead of assuming that X and Y are of finite type over the field K , assume only that they are *locally of finite type* over K ; for (I, 6.4.2), it suffices to observe that every point of a prescheme X that is closed in every affine open subset of X is closed in X , and an affine scheme locally of finite type over K is *ipso facto* of finite type over K (1.5.1). For (I, 6.4.9), we use (1.3.7) and (1.3.4, (v)).

(1.7.2). In (I, 6.5.1, (ii)), we can, instead of assuming that S is locally Noetherian and Y of finite type over S , assume only that Y is *locally of finite presentation* over S , as one shows immediately by applying the definition (1.4.2). Similarly in (I, 6.5.4, (ii)) and (I, 6.5.5, (ii)) it suffices to assume that f is an S -morphism, Y being *locally of finite presentation* over S .

(1.7.3). In (I, 7.1.11), for the second assertion, instead of assuming S locally Noetherian and Y of finite type over S , we can assume only Y locally of finite presentation over S (the proof depending on (I, 6.5.1, (ii))); similarly in (I, 7.1.12) to (I, 7.1.15).

(1.7.4). In (I, 9.2.2), we can replace the three hypotheses by the single hypothesis (implied by each of the others) that f is *quasi-compact and quasi-separated*, by virtue of (1.2.6) and (1.2.7). We note that in (I, 9.2.1), the hypothesis means exactly that f is quasi-compact and quasi-separated.

(1.7.5). In (I, 9.3.1), (I, 9.3.2), and (I, 9.3.3), we can replace the two hypotheses on X by the (weaker) hypothesis that X is a prescheme *quasi-compact and quasi-separated*, the proof using these two properties (by virtue of (1.2.7)).

(1.7.6). In (I, 9.3.4) and (I, 9.3.5), it suffices to assume X quasi-compact, $\mathcal{F}$ and $\mathcal{G}$ quasi-coherent and of finite type.

(1.7.7). In (I, 9.4.7), we can weaken hypothesis b) by assuming only X *quasi-separated*, since it suffices that the canonical immersion $U \rightarrow X$ be quasi-compact (1.2.7). We can make the same modification in (I, 9.4.8), (I, 9.4.9), and (I, 9.4.10).

(1.7.8). In (I, 9.5.1), the conditions indicated in parentheses in the statement can be replaced by the condition that f be quasi-compact and quasi-separated.

(1.7.9). In (I, 9.6.5), we can replace the hypotheses on X by the weaker hypothesis that X is *quasi-compact and quasi-separated*, as the proof shows, which uses only (I, 9.4.7). In (I, 9.6.6), the hypothesis “ X is a quasi-compact scheme” may be replaced by “ X is a prescheme quasi-compact and quasi-separated”.

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(1.7.10). In (II, 1.7.15), we can suppose only that Y is quasi-compact and quasi-separated, the proof using only (I, 9.6.5).

(1.7.11). In the second assertion of (II, 3.4.5), we can substitute for the two hypotheses on Y the weaker hypothesis that Y is quasi-compact and quasi-separated. Similarly in (II, 3.4.8).

(1.7.12). In (II, 3.8.5), we can similarly suppose only that Y is quasi-compact and quasi-separated, this hypothesis entering through (I, 9.4.7).

(1.7.13). In (II, 4.4.1), (II, 4.4.6), and (II, 4.4.7), it suffices to assume that Y is *quasi-compact and quasi-separated*, in view of (1.2.3) in the proof of (II, 4.4.7). Similarly, in (II, 4.4.10.1), it suffices to assume that Z is quasi-compact and quasi-separated.

(1.7.14). In (II, 4.5.2) and (II, 4.5.5), it suffices to assume X *quasi-compact and quasi-separated*, this hypothesis entering through (I, 9.3.1) and (I, 9.3.2).

(1.7.15). In (II, 4.6.8), it suffices still to assume X *quasi-compact and quasi-separated*, this hypothesis entering through (I, 9.4.7).

(1.7.16). In (II, 5.1.2), it suffices to assume that X is quasi-compact and quasi-separated (the hypothesis entering through (II, 4.5.2) and (II, 4.5.5)). Similarly in (II, 5.1.9), where we use (I, 9.6.5).

(1.7.17). In (II, 5.2.1), it suffices always to assume X quasi-compact and quasi-separated (the hypothesis entering through (II, 4.5.2)).

(1.7.18). In (II, 5.2.2), one must assume f quasi-compact and X quasi-separated; the current proof is in fact insufficient (with the hypotheses made), for from the fact that for every quasi-coherent $\mathcal{O}_X$ -module $\mathcal{F}$ we have $R^1 f_*(\mathcal{F}) = 0$, it does not follow, necessarily, for an affine open subset U of Y , that the same property holds for quasi-coherent $(\mathcal{O}_X|_{f^{-1}(U)})$ -modules, if one does not know that such a module is the restriction of a quasi-coherent $\mathcal{O}_X$ -module (the same remark applying to conditions b) and c')); when f is quasi-compact and X quasi-separated, this extension is possible by virtue of (I, 9.4.7), modified above (1.7.7).

(1.7.19). In (II, 5.3.2), it suffices to assume Y quasi-compact and quasi-separated (this hypothesis entering through (II, 4.4.6) and (II, 4.4.7)). Similarly in (II, 5.5.3, (ii)).

(1.7.20). In (III, 1.2.2), (III, 1.2.3), and (III, 1.2.4), we can replace the two hypotheses on X by the weaker hypothesis that X is a prescheme quasi-compact and quasi-separated, the proof using only (I, 9.3.3).

(1.7.21). In (III, 1.4.10) to (III, 1.4.14), we can replace “separated” by “quasi-separated”, this hypothesis serving only to be able to apply (I, 9.2.2) (see above (1.7.4)). Similarly, in (III, 1.4.15), it suffices to assume that the morphism u is quasi-separated and quasi-compact.

(1.7.22). In (III, 2.1.3), we can again replace the hypotheses by the weaker hypothesis that X is a prescheme quasi-compact and quasi-separated, the reasoning being the same as in (III, 1.2.2).

1.8. Morphisms of finite presentation and constructible sets

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(1.8.1). Let X be a prescheme; we know that every quasi-compact open subset of X is a finite union of affine open subsets, and conversely. For an open subset U of X to be *retrocompact* in X (0III, 9.1.1), it is necessary and sufficient that for every open affine subset V of X , $U \cap V$ be quasi-compact. When X is quasi-separated (1.2.1), every quasi-compact open subset of X is retrocompact in X (1.2.7), so every locally constructible part of X is retrocompact in X (0III, 9.1.13); if in addition X is quasi-compact, constructible subsets and locally constructible subsets of X coincide (0III, 9.1.12), as do constructible open subsets and quasi-compact open subsets (0III, 9.1.5).

Proposition (1.8.2). — *Let X, Y be two preschemes, $f : X \rightarrow Y$ a morphism. For every constructible (resp. locally constructible) part Z of Y , $f^{-1}(Z)$ is a constructible (resp. locally constructible) part of X .*

Proof. Suppose Z locally constructible; for every $x \in X$, there will be an affine open neighbourhood V of $f(x)$ such that $Z \cap V$ is constructible in V ; if the proposition is proved for constructible parts, $f^{-1}(Z) \cap f^{-1}(V)$ will be constructible in $f^{-1}(V)$, so $f^{-1}(Z)$ will be locally constructible. It therefore suffices to consider the case where Z is constructible, and taking into account (0III, 9.1.3), we are reduced to the case where Z is open and *retrocompact* in Y , in other words such that the canonical injection $Z \rightarrow Y$ is a quasi-compact morphism; it then suffices to see that $f^{-1}(Z)$ is *retrocompact* in X , in other words such that the canonical injection $f^{-1}(Z) \rightarrow X$ is a quasi-compact morphism; but this follows from (1.1.2, (iii)), since $f^{-1}(Z) = Z \times_Y X$ (I, 4.4.1). $\square$

Lemma (1.8.3). — *Let X be a prescheme quasi-compact and quasi-separated, Z a constructible part of X . There exists then an affine scheme X' and a morphism of finite presentation $f : X' \rightarrow X$ such that $f(X') = Z$.*

Proof. Since X is quasi-compact, it is a finite union of affine open subsets X_i ($i \in I$), and since X is quasi-separated, it follows from (1.2.7) that each of the canonical immersions $g_i : X_i \rightarrow X$ is of finite presentation; one concludes that if $f_i : X'_i \rightarrow X_i$ is a morphism of finite presentation, the same is true of $g_i \circ f_i : X'_i \rightarrow X$ (1.6.2), and if X' is the prescheme sum of the X'_i , the morphism $h : X' \rightarrow X$ which coincides with $g_i \circ f_i$ in each X'_i is also of finite presentation (1.6.5). Since $Z \cap X_i$ is constructible in X_i (0III, 9.1.8), we see that we are reduced to proving the lemma when X_i is *affine* of ring A . Since Z is then a finite union of sets of the form $U \cap \mathbb{A}^n_V$, where U and V are quasi-compact open subsets (0III, 9.1.3), one sees, by considering yet another suitable sum of preschemes, that one can reduce to the case where $Z = U \cap \mathbb{A}^n_V$; since moreover U is a finite union of affine open subsets, one can even restrict ourselves to the case where U is *affine*. Since V is quasi-compact, it is a finite union of open

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subsets of the form X_{f_i} , where $f_i \in A$; let $\mathfrak{J}$ be the ideal of A generated by the f_i , and let Z' be the closed affine subscheme of X that it defines (and which is by construction of finite presentation over X); one has by definition $V = \mathbb{C}Z'$, so $U \cap \mathbb{C}V = U \cap \mathbb{C}Z'$. Consider the affine scheme $X' = Z' \times_X U$ and let $f : X' \rightarrow X$ be the structural morphism; we just saw that the canonical immersion $Z' \rightarrow X$ is of finite presentation, and it is the same for the open immersion $U \rightarrow X$, which is quasi-compact since U and X are affine; one concludes therefore from (1.6.2, (iv)) that f is of finite presentation, and one has $f(X') = Z' \cap U$ (I, 3.4.8), which completes the proof. $\square$

Theorem (1.8.4). — (Chevalley). — *Let $f : X \rightarrow Y$ be a morphism of finite presentation. For every locally constructible part Z of X , $f(Z)$ is locally constructible in Y .*

Proof. Let $y \in Y$ and V an open affine neighbourhood of y ; since f is quasi-compact and quasi-separated, the same is true of its restriction $f^{-1}(V) \rightarrow V$, so $f^{-1}(V)$ is a quasi-compact and quasi-separated prescheme (1.2.3); since $Z \cap f^{-1}(V)$ is constructible in $f^{-1}(V)$ (1.8.1), we see that one can restrict ourselves to the case where Y is affine and Z constructible; X is then quasi-compact and quasi-separated, so by (1.8.3) there exists a morphism of finite presentation $g : X' \rightarrow X$ such that $Z = g(X')$; since $f \circ g$ is of finite presentation, we see that we are reduced to the case where $Z = X$; in other words, it will suffice to prove the $\square$

Lemma (1.8.4.1). — *Let Y be an affine scheme, $f : X \rightarrow Y$ a morphism quasi-compact and locally of finite presentation; then $f(X)$ is a constructible part of Y .*

Proof. Since X is quasi-compact, it is a finite union of affine open subsets; one can therefore restrict ourselves to the case where $Y = \text{Spec}(A)$, $X = \text{Spec}(B)$, B being an A -algebra of finite presentation (1.4.6). Now we have the $\square$

Lemma (1.8.4.2). — *Let A_0 be a ring, $(A_\lambda)_{\lambda \in L}$ an inductive system of A_0 -algebras, $A = \varinjlim A_\lambda$; if B is an A -algebra of finite presentation, there exist a λ and an A_λ -algebra of finite presentation B_λ such that B is isomorphic to $B_\lambda \otimes_{A_\lambda} A$.*

Proof. By hypothesis, B is isomorphic to an algebra of the form $A[T_1, \dots, T_n]/\mathfrak{J}$, where the T_i are indeterminates and $\mathfrak{J}$ an ideal of finite type; let (F_j) ($1 \leq j \leq m$) be a system of generators of $\mathfrak{J}$. Let $\varphi_\lambda : A_\lambda \rightarrow A$ be the canonical homomorphism. Since L is filtering, there exists λ such that each of the coefficients of each of the polynomials F_j is the image by φ_λ of an element of A_λ . In other words, there exists a system of polynomials $(F_{j\lambda})_{1 \leq j \leq m}$ of $A_\lambda[T_1, \dots, T_n]$ such that $F_j = \varphi_\lambda(F_{j\lambda})$ for $1 \leq j \leq m$. If $\mathfrak{J}_\lambda$ is the ideal of $A_\lambda[T_1, \dots, T_n]$ generated by the $F_{j\lambda}$, $\mathfrak{J}$ is the image of $\mathfrak{J}_\lambda \otimes_{A_\lambda} A$ in $A[T_1, \dots, T_n] = A_\lambda[T_1, \dots, T_n] \otimes_{A_\lambda} A$; the ring $B_\lambda = A_\lambda[T_1, \dots, T_n]/\mathfrak{J}_\lambda$ has the required property. $\square$

We apply this lemma here by considering A as the inductive limit of its sub- $\mathbf{Z}$ -algebras of finite type. One sees that there exist such a sub- $\mathbf{Z}$ -algebra A_0 of A , and an A_0 -algebra of finite presentation B_0 such that B is isomorphic to $B_0 \otimes_{A_0} A$; let us set $X_0 = \text{Spec}(B_0)$, $Y_0 = \text{Spec}(A_0)$, so that $X = X_0 \times_{Y_0} Y$, f being the projection $X \rightarrow Y$; let $f_0 : X_0 \rightarrow Y_0$, $g_0 : Y \rightarrow Y_0$ be the structural morphisms; it follows from (I, 3.4.8) that $f(X) = g_0^{-1}(f_0(X_0))$; taking into account (1.8.2), it therefore suffices to show that $f_0(X_0)$ is constructible. In other words, one is finally reduced to proving the

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Corollary (1.8.5). — *Let Y be a Noetherian prescheme, $f : X \rightarrow Y$ a morphism of finite type. For every constructible part Z of X , $f(Z)$ is a constructible part of Y .*

Proof. One reduces as above to proving that $f(X)$ is constructible. Let us apply the criterion (0III, 9.2.3): one must prove that for every irreducible closed part T of Y , $T \cap f(X) = f(f^{-1}(T))$ is rare in T or contains a non-empty open subset of T ; taking a subscheme of Y whose underlying space is T (I, 5.2.1) and considering its inverse image by f , one sees finally (taking into account (1.5.4, (iii))) that one is reduced to proving that if one supposes Y irreducible and f dominant, $f(X)$ contains a non-empty open subset of Y . One can moreover suppose Y affine; then X is a finite union of affine open subsets V_i , and since $f(X)$ is dense in Y , the same is true of at least one of the $f(V_i)$. One can therefore also suppose X affine; if (X_j) is the (finite) family of irreducible components of X , one sees as above that at least one of the $f(X_j)$ is dense in Y ; one can therefore suppose X

irreducible. Finally, by replacing f by f_{red} (1.5.4, (vi)) one can suppose X and Y integral. Then $X = \text{Spec}(B)$, $Y = \text{Spec}(A)$, where A is an integral Noetherian ring, B an integral A -algebra of finite type containing A (I, 1.2.7). It suffices to show that there exists $g \in A$ such that, for every $y \in D(g)$ (that is to say such that $g \notin \mathfrak{p}_y$) the ideal $\mathfrak{p}_y$ is the intersection of A and a prime ideal of B , for this will show that $D(g) \subset f(X)$. Finally it suffices to prove the $\square$

Lemma (1.8.5.1). — *Let A be an integral ring, B an integral A -algebra of finite type. There exists $g \in A$ such that every homomorphism of A into an algebraically closed field Ω , non-zero on g , extends to a homomorphism of B into Ω .*

Proof. This is a classical result of commutative algebra (Bourbaki, *Alg. comm.*, chap. V, §3, n° 1, cor. 3 of th. 1). $\square$

Corollary (1.8.6). — *Let Y be an irreducible prescheme, η its generic point, $f : X \rightarrow Y$ a morphism locally of finite type. If $f^{-1}(\eta)$ is not empty, there exists a neighbourhood U open of η in Y such that $f^{-1}(y)$ is non-empty for every $y \in U$. If $\mathcal{F}$ is a quasi-coherent $\mathcal{O}_X$ -module of finite type, and if $\mathcal{F}_\eta = \mathcal{F} \otimes_{\mathcal{O}_Y} k(\eta)$ is not zero, then there exists a neighbourhood U' of η in Y such that $\mathcal{F}_y = \mathcal{F} \otimes_{\mathcal{O}_Y} k(y)$ is not zero for every $y \in U'$.*

Proof. If p_y is the canonical projection of the fibre $f^{-1}(y) = X \times_Y \text{Spec}(k(y))$ to X , one has $\mathcal{F}_y = p_y^*(\mathcal{F})$, so $\text{Supp}(\mathcal{F}_y) = p_y^{-1}(\text{Supp}(\mathcal{F})) = \text{Supp}(\mathcal{F}) \cap f^{-1}(y)$ by virtue of (I, 9.1.13.1) and (I, 3.6.1), since $\mathcal{F}$ is of finite type; moreover $\text{Supp}(\mathcal{F})$ is closed in X (I, 5.2.2), and if Z is a closed subscheme of X having as underlying space $\text{Supp}(\mathcal{F})$ (I, 5.2.1), and j the canonical immersion $Z \rightarrow X$, $f \circ j$ is locally of finite type (1.3.4); this proves that the first assertion implies the second. To prove the first assertion, let us first note that one can suppose X and Y reduced (1.3.4, (vi)), in other words Y integral. Let ξ be a point of $f^{-1}(\eta)$; one can replace X and Y by affine open neighbourhoods of ξ and η respectively, and thus suppose that $X = \text{Spec}(B)$, $Y = \text{Spec}(A)$, B being an A -algebra of finite type. Moreover, if Z is the reduced closed subscheme of X having $\overline{\{\xi\}}$ as underlying set (I, 5.2.1), one can, as above, replace X by Z , in other words suppose B integral (I, 5.1.4). Since the morphism f is then dominant, the corresponding homomorphism $\varphi : A \rightarrow B$ is injective (I, 1.2.7); the corollary therefore follows finally from the lemma (1.8.5.1). $\square$

Proposition (1.8.7). — *Let S be a prescheme, $g : X \rightarrow S$, $h : Y \rightarrow S$ two morphisms of finite presentation, $f : X \rightarrow Y$ an S -morphism. For every $s \in S$, let $X_s = g^{-1}(s) = X \times_S \text{Spec}(k(s))$, $Y_s = h^{-1}(s) = Y \times_S \text{Spec}(k(s))$, $f_s = f \times 1 : X_s \rightarrow Y_s$. Then the set of $s \in S$ such that f_s is surjective (resp. radicial) is locally constructible.*

Proof. Let E be the set of $s \in S$ such that f_s is surjective; the set $S - E$ is equal to $h(Y - f(X))$; now f is of finite presentation (1.6.2, (v)) so $f(X)$ is locally constructible in Y (1.8.4); since h is of finite presentation, $h(Y - f(X))$ is locally constructible in S , and therefore also E .

To show that the set E' of $s \in S$ such that f_s is radicial is locally constructible, we will use the $\square$

Lemma (1.8.7.1). — *Let U, V be two preschemes; for a morphism $f : U \rightarrow V$ to be radicial, it is necessary and sufficient that the diagonal morphism $\Delta_f : U \rightarrow U \times_V U$ is surjective. Consequently, every radicial morphism is separated.*

Proof. Indeed, Δ_f being an immersion (I, 5.3.9), is a morphism locally of finite type (1.3.4); to say that Δ_f is surjective therefore means that for every algebraically closed field K , the corresponding map $U(K) \rightarrow (U \times_V U)(K) = U(K) \times_{V(K)} U(K)$ is surjective (1.3.7) and (I, 3.4.2.1); but by definition of a fibre product of sets, this means that the map $U(K) \rightarrow V(K)$ corresponding to f is injective, and this is equivalent to saying that f is radicial (I, 3.5.5).

This being so, Δ_{f_s} is deduced from $\Delta_f : X \rightarrow X \times_Y X$ by the base change $\text{Spec}(k(s)) \rightarrow S$ (I, 5.3.4). Since f is of finite presentation, it is the same for the structural morphism $X \times_Y X \rightarrow Y$ (1.6.2, (iv)), so also for Δ_f (1.6.2, (v)); it therefore suffices to apply the first part of the proposition to Δ_f , using the lemma (1.8.7.1). $\square$

1.9. Pro-constructible sets and ind-constructible sets

Lemma (1.9.1). — *Let $(A_\alpha)_{\alpha \in I}$ be an inductive system of rings with index set I filtering to the right, and let $A = \varinjlim A_\alpha$. For $A = 0$, it is necessary and sufficient that there exists an index γ such that $A_\gamma = 0$ (and one then has $A_\beta = 0$ for every $\beta \geq \gamma$).*

Proof. Indeed, for every $\alpha \in I$, the canonical homomorphism $\varphi_\alpha : A_\alpha \rightarrow A$ sends the unit element to the unit element; to say that $A = 0$ means that $\varphi_\alpha(1) = 0$, or again $\varphi_\alpha(1) = \varphi_\alpha(0)$; one knows that this is equivalent to saying that there exists an index $\gamma \geq \alpha$ such that the homomorphism $\varphi_{\gamma\alpha} : A_\alpha \rightarrow A_\gamma$ is such that $\varphi_{\gamma\alpha}(1) = \varphi_{\gamma\alpha}(0)$, and this last relation is equivalent to $A_\gamma = 0$, whence the lemma. $\square$

Proposition (1.9.2). — *Let B be a ring, $(A_\alpha)_{\alpha \in I}$ an inductive system of B -algebras with index set I filtering to the right, and let $A = \varinjlim A_\alpha$; set $Y = \operatorname{Spec}(B)$, $X = \operatorname{Spec}(A)$, $X_\alpha = \operatorname{Spec}(A_\alpha)$, and let $u : X \rightarrow Y$, $u_\alpha : X_\alpha \rightarrow Y$ be the morphisms corresponding to the structural homomorphisms $B \rightarrow A$, $B \rightarrow A_\alpha$. Then:*

(1.9.2.i). *For $X = \emptyset$, it is necessary and sufficient that there exists $\gamma \in I$ such that $X_\gamma = \emptyset$, and one then has $X_\alpha = \emptyset$ for $\alpha \geq \gamma$.*

(1.9.2.ii). *We have*

$$(1.9.2.1) \quad u(X) = \bigcap_{\alpha \in I} u_\alpha(X_\alpha).$$

The assertion (i) is nothing other than the translation of (1.9.1). To prove (ii), note that, since u factors as $X \rightarrow X_\alpha \xrightarrow{u_\alpha} Y$, the left-hand side is contained in the right-hand side. Conversely, let $y \in Y - u(X)$; one has $u^{-1}(y) = \emptyset$, and $u^{-1}(y)$ is the space underlying the $k(y)$ -prescheme $\operatorname{Spec}(A \otimes_B k(y))$; since in the category of B -modules the functor $\varinjlim$ commutes with the tensor product, $A \otimes_B k(y)$ is the inductive limit of the system of rings $A_\alpha \otimes_B k(y)$; the lemma (1.9.1) shows that there exists α such that $A_\alpha \otimes_B k(y) = 0$, that is to say $u_\alpha^{-1}(y) = \emptyset$, so $y \notin u_\alpha(X_\alpha)$.

Proposition (1.9.3). — *Let X be a quasi-compact and quasi-separated prescheme, let E be a part of X , let $(X_\alpha)_{\alpha \in I}$ be an affine open cover of X , and, for every $\alpha \in I$, let $E_\alpha = E \cap X_\alpha$. The following conditions are equivalent:*

- a) There exists a quasi-compact morphism $f : X' \rightarrow X$ such that $E = f(X')$.*
- a') For every α , there exists a quasi-compact morphism $f_\alpha : X'_\alpha \rightarrow X_\alpha$ such that $E_\alpha = f_\alpha(X'_\alpha)$.*
- a'') For every α , there exist an affine scheme X''_α and a morphism $f_\alpha : X'_\alpha \rightarrow X_\alpha$ such that $f_\alpha(X'_\alpha) = E_\alpha$.*
- b) E is an intersection of constructible parts of X .*
- b') $F = X - E$ is a union of constructible parts of X .*

Proof. It is clear that *b)* and *b')* are equivalent. The condition *a)* implies *a')* by taking for X'_α the prescheme induced on the open subset $f^{-1}(X_\alpha)$ of X' and for f_α the restriction of f . To see that *a')* implies *a'')*, it suffices to remark that X'_α is a union of affine open subsets $X''_{\alpha\lambda}$ ($\lambda \in L$); let X''_α be the prescheme *sum* of the $X''_{\alpha\lambda}$ ($\lambda \in L$), which is an affine scheme; if $g_\alpha : X''_\alpha \rightarrow X'_\alpha$ is the morphism coinciding with the identity in each $X''_{\alpha\lambda}$, it is clear that if one sets $f'_\alpha = f_\alpha \circ g_\alpha$, one has $E_\alpha = f'_\alpha(X''_\alpha)$ since g_α is surjective. To show that *a'')* implies *a)*, note that there is a finite part J of I such that the X_α for $\alpha \in J$ cover X ; let X' be the prescheme sum of the X'_α for $\alpha \in J$, and let $f : X' \rightarrow X$ be the morphism coinciding with $j_\alpha \circ f_\alpha$ for every $\alpha \in J$, where $j_\alpha : X_\alpha \rightarrow X$ is the canonical injection. Since E is the union of the $E \cap X_\alpha$ for $\alpha \in J$, one has $E = f(X')$ and it remains to see that f is a quasi-compact morphism; but the hypothesis that X is quasi-separated implies that j_α is quasi-compact (1.2.7), so $j_\alpha \circ f_\alpha$ is quasi-compact (1.1.1) and (1.1.2), and hence so is f (1.1.6).

It remains therefore to prove the equivalence of *a'')* and *b)*. Let us first prove that *a'')* implies *b)*: let us consider a finite part J of I such that the X_α for $\alpha \in J$ cover X ; it will suffice to show that, for every $\alpha \in J$, E_α is an intersection of constructible parts $F_{\alpha\lambda}$ of X_α ($\lambda \in L_\alpha$); indeed, every $F_{\alpha\lambda}$ is also a constructible part of X by virtue of (0_{III}, 9.1.8, (ii)), since the hypothesis that X is quasi-separated implies that every X_α is retrocompact in X (1.2.7); since E is the union of the E_α for $\alpha \in J$, it is the intersection of the finite unions $\bigcup_{\alpha \in J} F_{\alpha, \lambda(\alpha)}$, over all choices of $\lambda(\alpha) \in L_\alpha$; these finite unions are constructible parts of X , which proves our assertion. We may therefore reduce to the case where $X = \operatorname{Spec}(A)$ and $E = f(X')$, with $X' = \operatorname{Spec}(A')$ affine and $f : X' \rightarrow X$ corresponding to a

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homomorphism of rings $A \rightarrow A'$. Let us note now that A' is the inductive limit of the family (A'_λ) of its sub- A -algebras of finite type, ordered by inclusion; if $X'_\lambda = \text{Spec}(A'_\lambda)$, f factors as $X' \rightarrow X'_\lambda \xrightarrow{f_\lambda} X$ and it follows from (1.9.2, (ii)) that $f(X') = \bigcap_\lambda f_\lambda(X'_\lambda)$; if one shows that $f_\lambda(X'_\lambda)$ is an intersection of constructible parts of X , it will be the same for $f(X')$. We may therefore reduce to the case where A' is an A -algebra of finite type. Now, we have the following lemma: $\square$

Lemma (1.9.3.1). — *Let A be a ring, A' an A -algebra of finite type. Then A' is A -isomorphic to a filtering inductive limit $\varinjlim A''_\mu$ where the A''_μ are A -algebras of finite presentation (1.4.1).*

Proof. Indeed, one can write $A' = B/\mathfrak{J}$, with $B = A[T_1, \dots, T_n]$ and $\mathfrak{J}$ an ideal of B . But $\mathfrak{J}$ is inductive limit of the ideals of finite type $\mathfrak{J}'_\mu$ of B , contained in $\mathfrak{J}$, ordered by inclusion; since in the category of B -modules the functor $\varinjlim$ is exact, one concludes that $A' = \varinjlim B/\mathfrak{J}'_\mu$ up to A -isomorphism. $\square$

The same reasoning as above then permits us to reduce to the case where A' is an A -algebra of finite presentation; but then $f(X')$ is constructible in X by virtue of the theorem of Chevalley (1.8.5), which proves *b*).

Let us prove finally that *b*) implies *a''*). If E is an intersection of a family $(G_\lambda)_{\lambda \in L}$ of constructible parts of X , each E_α is an intersection of the $G_\lambda \cap X_\alpha$, which are constructible in X_α (0III, 9.1.8, (i)), and one is reduced to the case where $X = \text{Spec}(A)$ is affine.

One knows then (1.8.3) that for every $\lambda \in L$ there exists a morphism $f_\lambda : X'_\lambda \rightarrow X$, where $X'_\lambda = \text{Spec}(A'_\lambda)$ is affine, such that $f_\lambda(X'_\lambda) = G_\lambda$. It therefore suffices to prove the following lemma:

Lemma (1.9.3.2). — *Let $(A'_\lambda)_{\lambda \in L}$ be a family of A -algebras, $X = \text{Spec}(A)$, $X'_\lambda = \text{Spec}(A'_\lambda)$, and let $f_\lambda : X'_\lambda \rightarrow X$ be the structural morphism. For every finite part J of L , set $A'_J = \bigotimes_{\lambda \in J} A'_\lambda$ (tensor product of A -algebras), $X'_J = \text{Spec}(A'_J)$, and let $f_J : X'_J \rightarrow X$ be the structural morphism. One then has $f_J(X'_J) = \bigcap_{\lambda \in J} f_\lambda(X'_\lambda)$. If $A' = \varinjlim A'_J$, J running through the set of finite parts of L , $X' = \text{Spec}(A')$ and $f : X' \rightarrow X$ is the structural morphism, one has $f(X') = \bigcap_{\lambda \in L} f_\lambda(X'_\lambda)$.*

Proof. The first assertion follows from (I, 3.4.7); the second follows from (1.9.2, (ii)), which gives the relation $f(X') = \bigcap_J f_J(X'_J)$. $\square$

Definition (1.9.4). — Let X be a topological space. We say that a part E of X is *pro-constructible* (resp. *ind-constructible*) in X if, for every $x \in X$, there exists an open neighbourhood U of x in X such that $E \cap U$ is an intersection (resp. a union) of locally constructible parts of U .

Taking into account (1.2.7), (0III, 9.1.8), and (0III, 9.1.12), the equivalent conditions of the proposition (1.9.3) are therefore expressed by saying that E is *pro-constructible* in X , the locally constructible parts of X being identical to the constructible parts of X under the hypotheses of (1.9.3).

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Proposition (1.9.5). — *Let X be a prescheme.*

- (i) *For a part E of X to be pro-constructible, it is necessary and sufficient that $X - E$ be ind-constructible.*
- (ii) *Every finite union and every intersection of pro-constructible parts of X are pro-constructible. Every finite intersection and every union of ind-constructible parts of X are ind-constructible.*
- (iii) *Every intersection (resp. union) of locally constructible parts of X is pro-constructible (resp. ind-constructible). Conversely, if X is quasi-compact and quasi-separated, every pro-constructible (resp. ind-constructible) part of X is intersection (resp. union) of constructible parts of X .*
- (iv) *Let (U_α) be an open cover of X . For a part E of X to be pro-constructible (resp. ind-constructible) in X , it is necessary and sufficient that for every α , $E \cap U_\alpha$ be pro-constructible (resp. ind-constructible) in U_α .*
- (v) *Every pro-constructible part E of X is retrocompact in X . For a locally closed part E of X to be retrocompact, it is necessary and sufficient that it be pro-constructible in X .*
- (vi) *Let $f : X' \rightarrow X$ be a morphism of preschemes; for every pro-constructible (resp. ind-constructible) part E of X , $f^{-1}(E)$ is pro-constructible (resp. ind-constructible) in X' .*
- (vii) *Let $f : X' \rightarrow X$ be a quasi-compact morphism; for every pro-constructible part E' of X' , $f(E')$ is pro-constructible in X ; in particular $f(X')$ is pro-constructible in X .*
- (viii) *Let $f : X' \rightarrow X$ be a morphism locally of finite presentation; for every ind-constructible part E' of X' , $f(E')$ is ind-constructible in X ; in particular $f(X')$ is ind-constructible in X .*

- (ix) Suppose that X is quasi-compact and quasi-separated; then, for a part E of X to be pro-constructible, it is necessary and sufficient that there exist an affine scheme X' and a (necessarily quasi-compact) morphism $f : X' \rightarrow X$ such that $E = f(X')$.

Proof. Properties (i), (ii), (iv) and the first assertion of (iii) follow from the definition (1.9.4) and are valid for any topological space, using the mutual distributivity of intersection and union and the fact that if T is locally constructible in X , $T \cap U$ is locally constructible in U for every open subset U of X . If X is quasi-compact and quasi-separated and E is pro-constructible in X , there is a finite open cover (U_i) of X by affine open subsets such that $E \cap U_i$ is an intersection of constructible parts $M_{i,\lambda}$ of U_i ($\lambda \in L_i$); the $M_{i,\lambda}$ are also constructible parts of X by virtue of (1.2.7) and (0_{III}, 9.1.8, (ii)), and, since E is the union of the $E \cap U_i$, it is the intersection of the finite unions $\bigcup_i M_{i,\lambda(i)}$ (where $\lambda(i) \in L_i$ for every i), which are constructible parts of X ; this proves the second assertion of (iii). The assertion (ix) follows from (iii) and (1.9.3). To prove the first assertion of (v), we may restrict ourselves to the case where X is affine, and then prove that E is quasi-compact (0_{III}, 9.1.1); but since X is then quasi-separated, there exists by virtue of (ix) a quasi-compact morphism $f : X' \rightarrow X$ such that $E = f(X')$; since X' is quasi-compact, the same is true of E under a continuous map.

To prove (vi), one can restrict ourselves, by virtue of (iv), to the case where X is affine; the conclusion follows then from (iii) and (1.8.2).

To prove (vii) as well, one can restrict ourselves to the case where X is affine; then X' is a finite union of affine open subsets X'_i and $f(E')$ is the union of the $f(E' \cap X'_i)$, so one can also suppose that X' is affine, by virtue of (ii); but then $E' = g(X'')$, where $g : X'' \rightarrow X'$ is a quasi-compact morphism, by virtue of (ix), and one has $f(E') = f(g(X''))$, so $f \circ g$ is quasi-compact, and therefore $f(E')$ is pro-constructible.

One deduces from (vii) the proof of the second assertion of (v): indeed, let E be a locally closed and retrocompact part of X , and consider the sub-prescheme of X having E as underlying space (I, 5.2.1); the canonical injection $j : E \rightarrow X$ is a quasi-compact morphism, and it follows from (vii) that $E = j(E)$ is pro-constructible in X .

Finally, to prove (viii), one can still suppose X affine; moreover, if (X'_α) is an open affine cover of X' such that $E' \cap X'_\alpha$ is union of constructible parts of X'_α ((iii) and (iv)), $f(E')$ is union of the $f(E' \cap X'_\alpha)$, and by virtue of the theorem of Chevalley (1.8.4), each $f(E' \cap X'_\alpha)$ is union of constructible parts, whence the conclusion. $\square$

(1.9.6). Examples (1.9.6). — Every finite part of a prescheme X is pro-constructible: indeed, it suffices to consider the parts $\{x\}$ reduced to a single point (1.9.5, (ii)), and $\{x\}$ is the image of $\text{Spec}(k(x))$ by the canonical morphism $\text{Spec}(k(x)) \rightarrow X$, which is quasi-compact; whence the conclusion by (1.9.5, (vii)). On the other hand, a part $\{x\}$ reduced to a point is not necessarily ind-constructible; for example, let A be a Noetherian integral ring having an infinity of maximal ideals, and let x be the generic point of $X = \text{Spec}(A)$; if $\{x\}$ were ind-constructible in X , it would be constructible by virtue of (1.9.5, (iii)) and therefore would contain a non-empty open subset of X (0_{III}, 9.2.2); but this contradicts the hypothesis that A has an infinity of maximal ideals, by virtue of the Artin–Tate theorem (0, 16.3.3).

Every closed part of a prescheme X is pro-constructible, by (1.9.5, (v)). Every open part of X is therefore ind-constructible, but an open part U of X is pro-constructible only if it is *retrocompact*, by virtue again of (1.9.5, (v)).

Finally, if X is a Noetherian prescheme, therefore quasi-separated (1.2.8), it follows from (1.9.5, (iii)) and (0_{III}, 9.1.7) that the ind-constructible parts of X are the unions of locally closed parts of X .

Theorem (1.9.7). — Let X be a quasi-compact prescheme, E an ind-constructible part of X , $(F_\lambda)_{\lambda \in L}$ a family of pro-constructible parts of X such that $\bigcap_{\lambda \in L} F_\lambda \subset E$; then there exists a finite part J of L such that $\bigcap_{\lambda \in J} F_\lambda \subset E$.

Proof. Let us write $F = X - E$; then the sets F and $F \cap F_\lambda$ are pro-constructible, so one is reduced to the following corollary. $\square$

Corollary (1.9.8). — Let X be a quasi-compact prescheme, $(F_\lambda)_{\lambda \in L}$ a family of pro-constructible parts of X whose intersection is empty; then there exists a finite subfamily $(F_\lambda)_{\lambda \in J}$ whose intersection is empty.

Proof. Covering X by a finite number of affine open subsets, and using (1.9.5, (iv)), one is reduced to the case where X is affine; by virtue of (1.9.5, (ix)), one has $F_\lambda = f_\lambda(X'_\lambda)$, where $X'_\lambda = \text{Spec}(A'_\lambda)$ is affine, and f_λ is a morphism $X'_\lambda \rightarrow X$; then, with the notations of (1.9.3.2), on the one hand, by hypothesis $f(X') = \emptyset$, hence $X' = \emptyset$; but this implies by (1.9.2, (i)) that there exists a finite part J of L such that $X'_J = \emptyset$, hence $f_J(X'_J) = \emptyset$; but $f_J(X'_J) = \bigcap_{\lambda \in J} F_\lambda = \emptyset$. $\square$

Corollary (1.9.9). — *Let X be a quasi-compact prescheme, F a pro-constructible part of X , $(E_\lambda)_{\lambda \in L}$ a family of ind-constructible parts of X such that $F \subset \bigcup_{\lambda \in L} E_\lambda$; then there exists a finite part J of L such that $F \subset \bigcup_{\lambda \in J} E_\lambda$. In particular, from every covering of X by ind-constructible parts, one can extract a finite covering of X .*

Proof. This follows from (1.9.7) by passing to complements. $\square$

Proposition (1.9.10). — *Let X be a prescheme. For a part E of X to be ind-constructible, it is necessary that, for every $x \in X$, the intersection $E \cap \overline{\{x\}}$ be a neighbourhood of x in $\overline{\{x\}}$. If X is locally Noetherian, this condition is sufficient.*

Proof. Suppose E ind-constructible, and let Y be the intersection of $\overline{\{x\}}$ and an open affine subset of X containing x ; then there is therefore a sub-prescheme of X having Y for underlying space (I, 5.2.1), and if $j : Y \rightarrow X$ is the canonical injection, $E \cap Y = j^{-1}(E)$ is ind-constructible in Y (1.9.5, (vi)). As Y is quasi-compact and separated, $E \cap Y$ is therefore a union of constructible parts of Y (1.9.5, (iii)); consequently, there are two open subsets U, V in Y such that $x \in U \cap \overline{V} \subset E \cap Y$ (0_{III}, 9.1.3). But as x is the generic point of the irreducible space Y , V is necessarily empty and we have $U \subset E \cap Y$.

Suppose now X locally Noetherian, and let E be a part of X satisfying the condition of the statement. By virtue of the definition (1.9.4), one can restrict to the case where X is Noetherian. Then, for every $x \in E$, $E \cap \overline{\{x\}}$ contains a non-empty open part of the form $U \cap \overline{\{x\}}$, where U is open in X ; this shows that E is a union of locally closed parts of X , and hence is ind-constructible (1.9.6). $\square$

Proposition (1.9.11). — *Let X be a prescheme, E a part of X . The following properties are equivalent:*

- a) E is locally constructible.
- b) E is ind-constructible and pro-constructible.
- c) E and $X - E$ are pro-constructible.
- d) E and $X - E$ are ind-constructible.

Proof. It clearly suffices to prove that d) implies a), and one can restrict to the case where X is affine. Then, by (1.9.5, (iii)), E (resp. $X - E$) is a union of a family (E_λ) (resp. (E'_μ)) of constructible parts of X ; as the E_λ and the E'_μ form a covering of X , it follows from (1.9.9) that there are finitely many indices λ_i, μ_j such that the E_{λ_i} and the E'_{μ_j} form a covering of X ; this implies that E is a union of the E_{λ_i} , and hence is constructible. $\square$

Corollary (1.9.12). — *Let $f : X \rightarrow Y$ be a surjective morphism, either quasi-compact, or locally of finite presentation. For a part E of Y to be locally constructible (resp. pro-constructible, resp. ind-constructible), it is necessary and sufficient that $f^{-1}(E)$ be so in X .*

Proof. We know that the condition is necessary by (1.8.2) and (1.9.5, (vi)); to prove sufficiency, one is reduced to the case where X is affine by virtue of (1.9.5, (iv)); moreover, by virtue of (1.9.11), one can restrict to the case where $f^{-1}(E)$ is pro-constructible, or the case where $f^{-1}(E)$ is ind-constructible. If f is surjective and quasi-compact, and $f^{-1}(E)$ is pro-constructible, it follows from (1.9.5, (vii)) that $E = f(f^{-1}(E))$ is pro-constructible. If f is surjective and locally of finite presentation, and $f^{-1}(E)$ is ind-constructible, $E = f(f^{-1}(E))$ is ind-constructible by (1.9.5, (viii)), which completes the proof. $\square$

(1.9.13). Let X be a prescheme, $\mathcal{T}$ its topology; it follows from (1.9.5, (i) and (ii)) that the ind-constructible (resp. pro-constructible) parts of X are the open parts (resp. closed parts) for a topology on X , called the *constructible topology*, which we will denote by $\mathcal{T}^{\text{cons}}$; we will denote by X^{cons} the set X equipped with the topology $\mathcal{T}^{\text{cons}}$.

Proposition (1.9.14). — *Let X be a prescheme, $\mathcal{T}$ its topology, $\mathcal{T}^{\text{cons}}$ the constructible topology on X .*

- (i) *The topology $\mathcal{T}^{\text{cons}}$ is finer than $\mathcal{T}$.*
- (ii) *The locally constructible parts of X are identical to the parts that are both open and closed in the space X^{cons} .*
- (iii) *For every morphism $f : X \rightarrow Y$, the underlying map of X^{cons} to Y^{cons} is continuous; we denote it f^{cons} .*
- (iv) *If the morphism $f : X \rightarrow Y$ is quasi-compact, f^{cons} is a closed map; in particular, if f is quasi-compact and bijective, f^{cons} is a homeomorphism.*
- (v) *If the morphism $f : X \rightarrow Y$ is locally of finite presentation, f^{cons} is an open map; in particular, if f is bijective and locally of finite presentation, f^{cons} is a homeomorphism.*
- (vi) *For every open subset U of X , the topology induced by $\mathcal{T}^{\text{cons}}$ on U is identical to the topology of U^{cons} .*

Proof. Indeed, (i) follows from the fact that every open set for $\mathcal{T}$ is open for $\mathcal{T}^{\text{cons}}$ (1.9.6), and (ii) is a restatement of (1.9.11); (iii), (iv), and (v) are restatements respectively of (1.9.5, (vi)), (1.9.5, (vii)), and (1.9.5, (viii)). Finally, to prove (vi), it suffices to remark that the canonical injection $j : U \rightarrow X$ is locally of finite presentation (1.4.3), hence $j^{\text{cons}} : U^{\text{cons}} \rightarrow X^{\text{cons}}$ is a continuous open map. $\square$

Proposition (1.9.15). — *Let X be a prescheme.*

- (i) *For X^{cons} to be quasi-compact, it is necessary and sufficient that X be quasi-compact.*
- (ii) *For X^{cons} to be separated, it is necessary and sufficient that X be quasi-separated, and then X^{cons} is locally compact and totally disconnected.*
- (iii) *For X^{cons} to be compact, it is necessary and sufficient that X be quasi-compact and quasi-separated.*
- (iv) *Every point of X admits, for the topology $\mathcal{T}^{\text{cons}}$, an open and compact neighbourhood.*
- (v) *For a morphism $f : X \rightarrow Y$ to be quasi-compact, it is necessary and sufficient that the map $f^{\text{cons}} : X^{\text{cons}} \rightarrow Y^{\text{cons}}$ be proper.*

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Proof. (i) As $\mathcal{T}^{\text{cons}}$ is finer than $\mathcal{T}$, it is clear that if X^{cons} is quasi-compact, so is X ; the converse follows from (1.9.9).

(ii) Suppose X quasi-separated, and let us show that X^{cons} is totally disconnected: indeed, if x, y are two distinct points of X and if, for example, $x \notin \overline{\{y\}}$, there exists an affine open neighbourhood U of x which does not contain y ; as X is quasi-separated, U is retrocompact in X (1.2.7), hence U and $X - U$ are locally constructible in X , and therefore open in X^{cons} by virtue of (1.9.11), whence our assertion. For every open affine subset U of X , the topology induced by $\mathcal{T}^{\text{cons}}$ is that of U^{cons} by virtue of (1.9.14, (vi)); it follows from the above that X^{cons} is locally compact, since U is open in X^{cons} and U^{cons} is compact. It remains to show that if X^{cons} is separated, then X is quasi-separated; consider indeed an open affine subset U of X ; the topology induced on U by $\mathcal{T}^{\text{cons}}$ is that of U^{cons} (1.9.14, (vi)), hence U is compact for this induced topology, since U^{cons} is compact by the first part of the reasoning. If V is a second open affine subset in X , $U \cap V$ is therefore the intersection of two compact parts of the separated space X^{cons} ; as the topology induced by $\mathcal{T}^{\text{cons}}$ on $U \cap V$ is that of $(U \cap V)^{\text{cons}}$ (1.9.14, (vi)) and the identity $(U \cap V)^{\text{cons}} \rightarrow U \cap V$ is continuous, one concludes that $U \cap V$ is quasi-compact for the topology induced by $\mathcal{T}$; X is therefore quasi-separated by virtue of (1.2.7).

(iii) is an immediate corollary of (i) and (ii).

(iv) For every $x \in X$, an open affine neighbourhood U of x for $\mathcal{T}$ is also a neighbourhood of x for $\mathcal{T}^{\text{cons}}$ and it is compact by virtue of (iii) and (1.9.14, (vi)), the topology induced on U by $\mathcal{T}^{\text{cons}}$ being identical to that of U^{cons} .

(v) Suppose f quasi-compact; then we already know (1.9.14, (iv)) that f^{cons} is a closed map. Let on the other hand y be a point of Y , and let us set

$$Z = f^{-1}(y) = X \times_Y \text{Spec}(k(y));$$

Z is quasi-compact (1.1.2, (iii)) and, as the canonical morphism $p : Z \rightarrow X$ is injective, it follows from (i) and from the fact that the map p^{cons} is continuous that the topology induced by that of X^{cons} on $f^{-1}(y)$ makes $f^{-1}(y)$ a quasi-compact space. This proves that f^{cons} is a proper map (Bourbaki, *Top.*

gén., chap. I^{er}, 3^e éd., § 10, n° 2, th. 1). Conversely, suppose the continuous map f^{cons} proper, and let V be an open quasi-compact subset of Y ; if $h : V \rightarrow Y$ is the canonical injection, $h^{\text{cons}} : V^{\text{cons}} \rightarrow Y^{\text{cons}}$ is continuous and injective and V^{cons} is quasi-compact by (i), hence the topology induced on V by that of Y^{cons} makes V a quasi-compact space. The hypothesis that f^{cons} is proper implies that the topology induced on $f^{-1}(V)$ by that of X^{cons} makes $f^{-1}(V)$ a quasi-compact space (*loc. cit.*, prop. 6), hence $f^{-1}(V)$ is also a quasi-compact subspace of X , which shows that the morphism f is quasi-compact. $\square$

(1.9.16). We will show later how one can, for every prescheme X , equip the space X^{cons} with a sheaf of rings which makes it a prescheme whose local rings are all fields, identical to the residue fields at the points of X . Such preschemes arise naturally in the translation, into the language of schemes, of the constructions of Néron [32] in his theory of the reduction of abelian varieties.

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1.10. Application to open morphisms

Theorem (1.10.1). — *Let X be a prescheme, E an ind-constructible part of X (1.9.4), x a point of X . For x to be interior to E , it is necessary and sufficient that every generisation $(0_{\mathbf{I}}, 2.1.2)$ x' of x belong to E , in other words (I, 2.4.2) that $\text{Spec}(\mathcal{O}_{X,x}) \subset E$.*

Proof. The condition is evidently necessary, every neighbourhood of x containing the generisations of x . To show that it is sufficient, we may evidently restrict to the case where $X = \text{Spec}(A)$ is affine, x being a prime ideal $\mathfrak{p}$ of A . We know then (1.9.5, (ix)) that there exists an affine scheme $X' = \text{Spec}(A')$ and a morphism $f : X' \rightarrow X$ such that $f(X') = X - E$. Let $Y = \text{Spec}(\mathcal{O}_{X,x})$ and $Y' = X' \times_X Y$; the hypothesis means (by virtue of (I, 3.6.5)) that $Y' = \emptyset$, in other words $A' \otimes_A A_{\mathfrak{p}} = 0$. As $A_{\mathfrak{p}} = \varinjlim_t A_t$, where t runs through $A - \mathfrak{p}$ (0_I, 1.4.5), we have $\varinjlim_t A'_t = 0$, since the functor $\varinjlim$ commutes with the tensor product. Consequently, by (1.9.1), there exists $t \in A - \mathfrak{p}$ such that $A'_t = 0$, and $D(t)$ is then an open neighbourhood of x in X contained in E . $\square$

Corollary (1.10.2). — *Let $f : X \rightarrow Y$ be a morphism locally of finite presentation, and let $y \in Y$. For y to be interior to $f(X)$, it is necessary and sufficient that every generisation y' of y belong to $f(X)$.*

Proof. We know indeed (1.9.5, (viii)) that $f(X)$ is ind-constructible in Y and it suffices to apply (1.10.1). $\square$

We will say that a continuous map $f : X \rightarrow Y$ is *open at a point* $x \in X$ if the image under f of every neighbourhood of x in X is a neighbourhood of $f(x)$ in Y .

Proposition (1.10.3). — *Let $f : X \rightarrow Y$ be a morphism, x a point of X , $y = f(x)$. Consider the following conditions:*

- a) *f is open at the point x .*
- b) *For every generisation y' of y , there exists a generisation x' of x such that $f(x') = y'$.*
- b') *The image under f of $\text{Spec}(\mathcal{O}_{X,x})$ is $\text{Spec}(\mathcal{O}_{Y,y})$.*
- c) *For every irreducible closed part Y' of Y containing y , there exists an irreducible component of $X' = f^{-1}(Y')$ containing x and dominating Y' .*

Then we have the implications a) $\Rightarrow$ b) $\Leftrightarrow$ b') $\Leftrightarrow$ c). If in addition f is locally of finite presentation, the four conditions are equivalent.

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Proof. By definition of a generisation, the image under f of $\text{Spec}(\mathcal{O}_{X,x})$ is always contained in $\text{Spec}(\mathcal{O}_{Y,y})$, so b) and b') are equivalent. It is immediate that c) implies b), since if y' is a generisation of y , $Y' = \overline{\{y'\}}$ is an irreducible closed part of Y containing y ; if x' is the generic point of an irreducible component of $f^{-1}(Y')$ containing x and dominating Y' , we have $f(x') = y'$ (0_I, 2.1.5) and x' is a generisation of x . Conversely, b) implies c): indeed, let y' be the generic point of Y' and let x' be a generisation of x such that $f(x') = y'$; let Z' be an irreducible component of $f^{-1}(Y')$ containing x' (and hence x); as $f(x')$ is already dense in Y' , so *a fortiori* is $f(Z')$.

To show that a) implies b'), one can evidently restrict to the case where $X = \text{Spec}(B)$ and $Y = \text{Spec}(A)$ are affine, x being a prime ideal $\mathfrak{q}$ of B , so that $\mathcal{O}_{X,x} = B_{\mathfrak{q}} = \varinjlim_t B_t$,

where t runs through $B - \mathfrak{q}$. One has, by (1.9.2), $f(\mathrm{Spec}(\mathcal{O}_{X,x})) = \bigcap_t f(\mathrm{Spec}(B_t)) = \bigcap_t f(D(t))$, and by hypothesis, for every $t \in B - \mathfrak{q}$, y is interior to $f(D(t))$, hence $f(D(t))$ contains $\mathrm{Spec}(\mathcal{O}_{Y,y})$; whence $\mathrm{Spec}(\mathcal{O}_{Y,y}) \subset f(\mathrm{Spec}(\mathcal{O}_{X,x}))$, which establishes our assertion. Finally, when f is locally of finite presentation, b) implies a) by virtue of (1.10.2) applied to the restriction $U \rightarrow Y$ of f to an arbitrary open neighbourhood U of x . $\square$

Corollary (1.10.4). — *Let $f : X \rightarrow Y$ be a morphism. Consider the following conditions:*

- a) f is open.
- b) For every $x \in X$ and every generisation y' of $y = f(x)$, there exists a generisation x' of x such that $f(x') = y'$.
- b') For every $x \in X$, the image under f of $\mathrm{Spec}(\mathcal{O}_{X,x})$ is $\mathrm{Spec}(\mathcal{O}_{Y,f(x)})$.
- c) For every irreducible closed part Y' of Y , every irreducible component of $f^{-1}(Y')$ dominates Y' .

Then we have the implications a) $\Rightarrow$ b) $\Leftrightarrow$ b') $\Leftrightarrow$ c). If in addition f is locally of finite presentation, the four conditions are equivalent.

Proof. To say that f is open means that f is open at every point $x \in X$, hence the implications a) $\Rightarrow$ b) $\Leftrightarrow$ b') follow from the analogous implications in (1.10.3), as well as the implication b) $\Rightarrow$ a) when f is locally of finite presentation. The implication c) $\Rightarrow$ b) also follows from the analogous implication in (1.10.3); let us finally prove that b) implies c). Indeed, let x' be the generic point of an irreducible component of $f^{-1}(Y')$, and let us show that $y' = f(x')$ is the generic point of Y' . Let y'' be a generisation of y' ; by hypothesis there exists a generisation x'' of x' such that $f(x'') = y''$, and as $x'' \in f^{-1}(Y')$, we have necessarily $x'' = x'$, hence $y'' = y'$, which completes the proof. $\square$

§2. BASE CHANGE AND FLATNESS

This section (unlike § 6) only exceptionally makes use of Noetherian techniques. The nos 1 and 2 are essentially nothing more than translations of elementary properties of flatness in commutative Algebra (cf. Bourbaki, *Alg. comm.*, chap. I^{er}), and are included here for the convenience of references. The following numbers are primarily devoted to properties of “descent” by flat or faithfully flat morphisms: if $g : Y' \rightarrow Y$ is such a morphism, we want to be able to assert that a subset of Y , or an $\mathcal{O}_Y$ -Module, or a morphism $X \rightarrow Y$, has a certain property, when we know that its *inverse image* by g possesses this property. We limit ourselves here to properties that do not make use of the general technique of “descent”, which will be developed in Chapter V.

2.1. Flat Modules on preschemes

(2.1.1). Let $f : X \rightarrow Y$ be a morphism of preschemes, $\mathcal{F}$ an $\mathcal{O}_X$ -Module; recall (0_I, 6.7.1) that $\mathcal{F}$ is said to be *f-flat* (or *Y-flat*) at a point $x \in X$ if $\mathcal{F}_x$ is a flat $\mathcal{O}_{f(x)}$ -module, *f-flat* (or *Y-flat*) if it is *f-flat* at every point $x \in X$, and finally that the morphism f is said to be *flat at the point* $x \in X$ (resp. *flat*) if $\mathcal{O}_X$ is *f-flat* at the point x (resp. *f-flat*). When $f = 1_X$, we will simply say that an $\mathcal{O}_X$ -Module $\mathcal{F}$ is *flat at the point* x (resp. *flat*) if it is *X-flat* at this point (resp. at every point $x \in X$), that is to say if $\mathcal{F}_x$ is a flat $\mathcal{O}_x$ -module (resp. if this is the case for every $x \in X$). Recall that we have shown (III, 1.4.15.1) the following property:

Proposition (2.1.2). — *Let A, B be two rings, $\varphi : A \rightarrow B$ a ring homomorphism, $X = \mathrm{Spec}(B)$, $Y = \mathrm{Spec}(A)$, $f : X \rightarrow Y$ the corresponding morphism, and M a B -module. For $\mathcal{F} = \tilde{M}$ to be *f-flat*, it is necessary and sufficient that M be a flat A -module.*

Proposition (2.1.3). — *Let $f : X \rightarrow Y$ be a morphism of preschemes, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module. The following conditions are equivalent:*

- a) For every base change $g : Y' \rightarrow Y$, if we set $X' = X \times_Y Y'$, the functor $\mathcal{G}' \mapsto \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}'$ from the category of quasi-coherent $\mathcal{O}_{Y'}$ -Modules to that of quasi-coherent $\mathcal{O}_{X'}$ -Modules is exact.
- a') Condition a) is satisfied for all canonical morphisms $g : \mathrm{Spec}(\mathcal{O}_y) \rightarrow Y$ (I, 2.4.1), where y ranges over Y .
- b) $\mathcal{F}$ is *f-flat*.

Proof. The questions being local on X and Y , we can restrict to the case where $Y = \mathrm{Spec}(B)$, $X = \mathrm{Spec}(A)$, $\mathcal{F} = \tilde{M}$, where M is an A -module. It is clear that a) implies a'); condition a') implies

that for every $x \in X$, the functor $N \mapsto M_n \otimes_{B_n} N$ is exact in the category of B_n -modules, n being the ideal $j_{f(x)}$ of B ; this means that M_n is a flat B_n -module, and it follows from (0_I, 6.3.3) and from (2.1.2) that $\mathcal{F}$ is f -flat. Finally, to see that $b)$ implies $a)$, we can also restrict to the case where $Y' = \text{Spec}(A')$ is affine and $\mathcal{G}' = \widetilde{N}'$, where N' is an A' -module; the conclusion then follows from (2.1.2) and from the definition of flatness, since $(M \otimes_A N')^\sim = \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}'$. $\square$

Proposition (2.1.4). — *Let $f : X \rightarrow Y$, $g : Y' \rightarrow Y$ be two morphisms of preschemes, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module; set $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \rightarrow Y'$, $\mathcal{F}' = \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y'}$, and let $g' : X' \rightarrow X$ be the canonical projection. Let x' be a point of X' , and set $x = g'(x')$, $y' = f'(x')$, and $y = g(y') = f(x)$. If $\mathcal{F}$ is f -flat at the point x , $\mathcal{F}'$ is f' -flat at the point x' ; in particular, if $\mathcal{F}$ is f -flat, $\mathcal{F}'$ is f' -flat; if f is flat, f' is flat.*

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Proof. It suffices to prove the first assertion; applying three times (I, 3.6.5), as well as (I, 2.4.4), we can reduce to the case where $Y = \text{Spec}(\mathcal{O}_y)$, $X = \text{Spec}(\mathcal{O}_x)$, $Y' = \text{Spec}(\mathcal{O}_{y'})$, $\mathcal{F} = \widetilde{M}$, where $M = \mathcal{F}_x$. The hypothesis and (2.1.2) then imply that $\mathcal{F}$ is f -flat; in other words, we are reduced to proving a special case of the second assertion, and the latter follows immediately from (2.1.3). $\square$

Proposition (2.1.5). — *Consider a commutative diagram of morphisms of preschemes*

$$\begin{array}{ccc} X & \xleftarrow{g'} & X' \\ f \downarrow & & \downarrow f' \\ Y & \xleftarrow{g} & Y' \\ & & \downarrow h \\ & & Z \end{array}$$

where $X' = X \times_Y Y'$ and $f' = f_{(Y')}$. Let x' be a point of X' , and set $x = g'(x')$, $y' = f'(x')$, $y = f(x) = g(y')$, $z = h(y')$. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$ -Module f -flat at the point x (resp. f -flat), and $\mathcal{G}'$ a quasi-coherent $\mathcal{O}_{Y'}$ -Module h -flat at the point y' (resp. h -flat); then $\mathcal{F} \otimes_{\mathcal{O}_{Y'}} \mathcal{G}'$ is a quasi-coherent $\mathcal{O}_{X'}$ -Module $(h \circ f')$ -flat at the point x' (resp. $(h \circ f')$ -flat).

Proof. As in (2.1.4), we reduce to the case where $X = \text{Spec}(\mathcal{O}_x)$, $Y = \text{Spec}(\mathcal{O}_y)$, $Y' = \text{Spec}(\mathcal{O}_{y'})$ and $Z = \text{Spec}(\mathcal{O}_z)$, and it suffices then to prove that $\mathcal{F} \otimes_{\mathcal{O}_{Y'}} \mathcal{G}'$ is $(h \circ f')$ -flat. Taking into account (2.1.2), the proposition follows from Bourbaki, *Alg. comm.*, chap. I^{er}, § 2, n° 7, prop. 8. $\square$

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Corollary (2.1.6). — *Let $f : X \rightarrow Y$, $g : Y \rightarrow Z$ be two morphisms of preschemes, $\mathcal{F}$ an $\mathcal{O}_X$ -Module. If $\mathcal{F}$ is f -flat at the point $x \in X$ and if g is flat at the point $f(x)$, $\mathcal{F}$ is $(g \circ f)$ -flat at the point x . In particular, if f and g are flat morphisms, so is $g \circ f$.*

Proof. This is the special case of (2.1.5) with $Y' = Y$, $\mathcal{G}' = \mathcal{O}_{Y'}$. $\square$

Corollary (2.1.7). — *If $f : X \rightarrow X'$, $g : Y \rightarrow Y'$ are two flat S -morphisms, $f \times_S g : X \times_S Y \rightarrow X' \times_S Y'$ is a flat morphism.*

Proof. This follows from (2.1.4) and (2.1.6) (cf. (I, 3.5.1)). $\square$

Proposition (2.1.8). — *Let $f : X \rightarrow Y$ be a morphism of preschemes,*

$$0 \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}'' \longrightarrow 0$$

an exact sequence of quasi-coherent $\mathcal{O}_X$ -Modules, such that $\mathcal{F}''$ is Y -flat.

- (i) *For every morphism $g : Y' \rightarrow Y$ and every quasi-coherent $\mathcal{O}_{Y'}$ -Module $\mathcal{G}'$, the sequence $0 \rightarrow \mathcal{F}' \otimes_{\mathcal{O}_Y} \mathcal{G}' \rightarrow \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}' \rightarrow \mathcal{F}'' \otimes_{\mathcal{O}_Y} \mathcal{G}' \rightarrow 0$ of $\mathcal{O}_{X'}$ -Modules (where $X' = X \times_Y Y'$) is exact.*
- (ii) *For $\mathcal{F}$ to be Y -flat, it is necessary and sufficient that $\mathcal{F}'$ be so.*

Proof. We can evidently suppose X , Y , Y' affine; the conclusion then follows from (2.1.2) and from (0_I, 6.1.2). $\square$

Corollary (2.1.9). — Let $\mathcal{L}^\bullet$ be a complex of quasi-coherent $\mathcal{O}_X$ -Modules, i an index such that if $d^i : \mathcal{L}^i \rightarrow \mathcal{L}^{i+1}$ is the derivation operator, $\mathcal{B}^{i+1}(\mathcal{L}^\bullet) = \text{Im}(d^i)$ and $\mathcal{Z}^{i+1}(\mathcal{L}^\bullet) = \text{Coker}(d^i)$ are Y -flat. Then, with the notations of (2.1.8), the canonical homomorphism

$$\mathcal{H}^i(\mathcal{L}^\bullet) \otimes_{\mathcal{O}_Y} \mathcal{G}' \longrightarrow \mathcal{H}^i(\mathcal{L}^\bullet \otimes_{\mathcal{O}_Y} \mathcal{G}')$$

is bijective.

Proof. As the tensor product is right exact, we have

$$\mathcal{Z}^{i+1}(\mathcal{L}^\bullet) \otimes_{\mathcal{O}_Y} \mathcal{G}' = \text{Coker}(d^i \otimes 1) = \mathcal{Z}^{i+1}(\mathcal{L}^\bullet \otimes_{\mathcal{O}_Y} \mathcal{G}')$$

and $\mathcal{Z}^i(\mathcal{L}^\bullet) \otimes_{\mathcal{O}_Y} \mathcal{G}' = \mathcal{Z}^i(\mathcal{L}^\bullet \otimes_{\mathcal{O}_Y} \mathcal{G}')$. Moreover, in the exact sequence

$$0 \longrightarrow \mathcal{B}^{i+1}(\mathcal{L}^\bullet) \longrightarrow \mathcal{L}^{i+1} \longrightarrow \mathcal{Z}^{i+1}(\mathcal{L}^\bullet) \longrightarrow 0$$

$\mathcal{Z}^{i+1}(\mathcal{L}^\bullet)$ is Y -flat, so it follows from (2.1.8, (i)) that the exact sequence

$$0 \longrightarrow \mathcal{B}^{i+1}(\mathcal{L}^\bullet) \otimes_{\mathcal{O}_Y} \mathcal{G}' \longrightarrow \mathcal{L}^{i+1} \otimes_{\mathcal{O}_Y} \mathcal{G}' \longrightarrow \mathcal{Z}^{i+1}(\mathcal{L}^\bullet \otimes_{\mathcal{O}_Y} \mathcal{G}') \longrightarrow 0$$

whence $\mathcal{B}^{i+1}(\mathcal{L}^\bullet) \otimes_{\mathcal{O}_Y} \mathcal{G}' = \text{Im}(d^i \otimes 1) = \mathcal{B}^{i+1}(\mathcal{L}^\bullet \otimes_{\mathcal{O}_Y} \mathcal{G}')$. Then, as in the exact sequence

$$0 \longrightarrow \mathcal{H}^i(\mathcal{L}^\bullet) \longrightarrow \mathcal{Z}^i(\mathcal{L}^\bullet) \longrightarrow \mathcal{B}^{i+1}(\mathcal{L}^\bullet) \longrightarrow 0$$

$\mathcal{B}^{i+1}(\mathcal{L}^\bullet)$ is Y -flat, it follows from (2.1.8, (i)) and from what precedes that the sequence

$$0 \longrightarrow \mathcal{H}^i(\mathcal{L}^\bullet) \otimes_{\mathcal{O}_Y} \mathcal{G}' \longrightarrow \mathcal{Z}^i(\mathcal{L}^\bullet \otimes_{\mathcal{O}_Y} \mathcal{G}') \longrightarrow \mathcal{B}^{i+1}(\mathcal{L}^\bullet \otimes_{\mathcal{O}_Y} \mathcal{G}') \longrightarrow 0$$

is exact, which proves the corollary. $\square$

Corollary (2.1.10). — Let $f : X \rightarrow Y$ be a morphism of preschemes, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module which is Y -flat, $\mathcal{L}_\bullet = (\mathcal{L}_i)$ a left resolution of $\mathcal{F}$ formed of quasi-coherent $\mathcal{O}_X$ -Modules which are Y -flat. Then, for every morphism $g : Y' \rightarrow Y$ and every quasi-coherent $\mathcal{O}_{Y'}$ -Module $\mathcal{G}'$, the complex $\mathcal{L}_\bullet \otimes_{\mathcal{O}_Y} \mathcal{G}' = (\mathcal{L}_i \otimes_{\mathcal{O}_Y} \mathcal{G}')_{i \geq 0}$ is a left resolution of $\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}'$. Moreover, if $\mathcal{Z}_i(\mathcal{L}_\bullet) = \text{Ker}(\mathcal{L}_i \rightarrow \mathcal{L}_{i-1})$, the $\mathcal{Z}_i(\mathcal{L}_\bullet)$ are Y -flat, and we have $\mathcal{Z}_i(\mathcal{L}_\bullet) \otimes_{\mathcal{O}_Y} \mathcal{G}' = \mathcal{Z}_i(\mathcal{L}_\bullet \otimes_{\mathcal{O}_Y} \mathcal{G}') = \text{Ker}(\mathcal{L}_i \otimes_{\mathcal{O}_Y} \mathcal{G}' \rightarrow \mathcal{L}_{i-1} \otimes_{\mathcal{O}_Y} \mathcal{G}')$. Setting $\mathcal{R}_i = \text{Im}(\mathcal{L}_{i+1} \rightarrow \mathcal{L}_i) = \mathcal{Z}_i(\mathcal{L}_\bullet)$; we then have the exact sequences

$$0 \leftarrow \mathcal{F} \leftarrow \mathcal{L}_0 \leftarrow \mathcal{R}_0 \leftarrow 0$$

$$0 \leftarrow \mathcal{R}_i \leftarrow \mathcal{L}_{i+1} \leftarrow \mathcal{R}_{i+1} \leftarrow 0 \quad (i \geq 0)$$

and since $\mathcal{F}$ and $\mathcal{L}_i$ are Y -flat, we deduce from (2.1.8, (ii)) by induction that all the $\mathcal{R}_i$ are also Y -flat; using (2.1.8, (i)), we thus have the exact sequences

$$0 \leftarrow \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}' \leftarrow \mathcal{L}_0 \otimes_{\mathcal{O}_Y} \mathcal{G}' \leftarrow \mathcal{R}_0 \otimes_{\mathcal{O}_Y} \mathcal{G}' \leftarrow 0$$

$$0 \leftarrow \mathcal{R}_i \otimes_{\mathcal{O}_Y} \mathcal{G}' \leftarrow \mathcal{L}_{i+1} \otimes_{\mathcal{O}_Y} \mathcal{G}' \leftarrow \mathcal{R}_{i+1} \otimes_{\mathcal{O}_Y} \mathcal{G}' \leftarrow 0 \quad (i \geq 0)$$

which prove the corollary.

Proposition (2.1.11). — Let $f : X \rightarrow Y$ be a flat morphism, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_Y$ -Module of finite presentation. If $\mathcal{J}$ is the Ideal of $\mathcal{O}_Y$ annihilator of $\mathcal{F}$, $f^*(\mathcal{J})$ is the Ideal of $\mathcal{O}_X$ annihilator of $f^*(\mathcal{F})$.

Proof. We have indeed by definition an exact sequence (0_I, 5.3.7)

$$0 \longrightarrow \mathcal{J} \longrightarrow \mathcal{O}_Y \longrightarrow \text{Hom}_{\mathcal{O}_Y}(\mathcal{F}, \mathcal{F})$$

whence, since f is flat, an exact sequence

$$0 \longrightarrow f^*(\mathcal{J}) \longrightarrow \mathcal{O}_X \longrightarrow f^*(\text{Hom}_{\mathcal{O}_Y}(\mathcal{F}, \mathcal{F}))$$

and since $\mathcal{F}$ is an $\mathcal{O}_Y$ -Module of finite presentation, $f^*(\text{Hom}_{\mathcal{O}_Y}(\mathcal{F}, \mathcal{F}))$ canonically identifies with $\text{Hom}_{\mathcal{O}_X}(f^*(\mathcal{F}), f^*(\mathcal{F}))$ (0_I, 6.7.6), whence the conclusion. $\square$

Proposition (2.1.12). — Let X be a prescheme, $\mathcal{F}$ an $\mathcal{O}_X$ -Module of finite presentation, x a point of X . The following conditions are equivalent:

- a) $\mathcal{F}_x$ is a flat $\mathcal{O}_{X,x}$ -module.
- b) There exists an open neighbourhood U of x such that $\mathcal{F}|_U$ is a locally free $(\mathcal{O}_X|_U)$ -Module.

Proof. Indeed, $\mathcal{F}_x$ is a module of finite presentation over $\mathcal{O}_x$, and $\mathcal{O}_x$ is a local ring; it amounts to the same thing to say that $\mathcal{F}_x$ is a flat $\mathcal{O}_x$ -module or a free $\mathcal{O}_x$ -module (Bourbaki, *Alg. comm.*, chap. II, § 3, n° 2, cor. 2 of prop. 5); whence the conclusion, taking into account (0I, 5.2.7). We note that the proposition is valid for any ringed space with local rings. $\square$

Proposition (2.1.13). — *Let $f : X \rightarrow Y$ be a morphism of preschemes. If f is flat at a point $x \in X$ and if the ring $\mathcal{O}_x$ is reduced (resp. integral and integrally closed), then the ring $\mathcal{O}_{f(x)}$ is reduced (resp. integral and integrally closed). If f is faithfully flat and if X is reduced (resp. normal), then Y is reduced (resp. normal).*

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Proof. Set $\mathcal{O}_{f(x)} = A$, $\mathcal{O}_x = B$. If B is a flat A -module, it is also a faithfully flat A -module (0I, 6.6.2), so A identifies with a subring of B ; if B is reduced, so is A . Suppose now that B is integral and integrally closed, and let L be its field of fractions; then $A \subset B$ is integral; denote by $K \subset L$ its field of fractions. The hypothesis implies that $B \cap K = A$ (Bourbaki, *Alg. comm.*, chap. I, § 3, n° 5, prop. 10). If then $t \in K$ is integral over A , it is also integral over B , so belongs to B by hypothesis, and thus $t \in A$, which proves that A is integrally closed. $\square$

Proposition (2.1.14). — *Let $f : X \rightarrow Y$ be a faithfully flat morphism. If X is locally integral and if the space underlying Y is locally Noetherian, then Y is locally integral.*

Proof. Indeed, every $y \in Y$ is of the form $f(x)$ for some $x \in X$ and by hypothesis $\mathcal{O}_y$ identifies with a subring of $\mathcal{O}_x$ (0I, 6.6.1); as $\mathcal{O}_x$ is integral, so is $\mathcal{O}_y$ and this proves the proposition (I, 5.1.4). $\square$

2.2. Faithfully flat Modules on preschemes

Proposition (2.2.1). — *Let $f : X \rightarrow Y$ be a morphism of preschemes, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module. The following conditions are equivalent:*

- a) *For every base change $g : Y' \rightarrow Y$, if we set $X' = X \times_Y Y'$, the functor $\mathcal{G}' \mapsto \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}'$, from the category of quasi-coherent $\mathcal{O}_{Y'}$ -Modules to that of quasi-coherent $\mathcal{O}_{X'}$ -Modules, is exact and faithful.*
- a') *Condition a) is satisfied for all canonical morphisms $g : \text{Spec}(\mathcal{O}_y) \rightarrow Y$ (I, 2.4.1), where y ranges over Y .*
- a'') *Condition a) is satisfied for all canonical immersions $Y' \rightarrow Y$, where Y' ranges over the set of affine open sub-preschemes of Y .*
- b) *$\mathcal{F}$ is Y -flat and, for every $y \in Y$, if we denote by X_y the fibre $f^{-1}(y)$, the $\mathcal{O}_{X_y}$ -Module $\mathcal{F}_y = \mathcal{F} \otimes_{\mathcal{O}_Y} k(y)$ is nonzero.*

Proof. It is clear that a) implies a') and a''); condition a') implies first that $\mathcal{F}$ is Y -flat (2.1.3); on the other hand it implies that for every $y \in Y$, the functor $N \mapsto \mathcal{F} \otimes_{\mathcal{O}_Y} \tilde{N}$ is faithful in the category of $\mathcal{O}_y$ -modules; taking in particular $N = k(y)$, we obtain the second assertion of b). To show that b) implies a), we can restrict to the case where Y is affine, since the question is local over Y . Likewise, to prove that a'') implies a), we are reduced to proving that, when Y is affine, the fact that $\mathcal{G} \mapsto \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}$ is an exact and faithful functor implies condition a). In other words, we are reduced to proving the following more precise proposition: $\square$

Proposition (2.2.2). — *Let $Y = \text{Spec}(A)$ be an affine scheme, $f : X \rightarrow Y$ a morphism of preschemes, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module. Condition a) of (2.2.1) is equivalent to each of the following:*

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- b') *$\mathcal{F}$ is Y -flat and, for every closed point y of Y , we have $\mathcal{F}_y \neq 0$.*
- c) *The functor $\mathcal{G} \mapsto \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}$ from the category of quasi-coherent $\mathcal{O}_Y$ -Modules to that of quasi-coherent $\mathcal{O}_X$ -Modules is exact and faithful.*

Proof. If b') holds and y is a closed point of Y , there is at least one $x \in f^{-1}(y)$ such that $(\mathcal{F}_y)_x \neq 0$; let $U = \text{Spec}(B)$ be an affine open neighbourhood of x , and write $\mathcal{F}|_U = \tilde{M}$, where M is a B -module. Then b') implies that $M/\mathfrak{i}_y M$ is nonzero and hence, since M is a flat A -module (2.1.2), that $M \otimes_A A_y$ is a faithfully flat A_y -module (0I, 6.4.5). The relation $(\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}) \otimes_{\mathcal{O}_Y} \mathcal{O}_y = 0$ for a closed point y of Y implies

$$(\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{O}_y) \otimes_{\mathcal{O}_Y} \mathcal{G}_y = 0,$$

and therefore $\mathcal{G}_y = 0$. But if $\mathcal{G}_y = 0$ for every closed point y of Y , we conclude that $\mathcal{G} = 0$: indeed, if $\mathcal{G} = \tilde{N}$, the annihilator of every element of N is contained in no maximal ideal of A , and hence is all

of A . The relation $\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G} = 0$ implies therefore $\mathcal{G} = 0$, in other words the functor $\mathcal{G} \mapsto \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}$ is faithful; we know moreover that this functor is exact (2.1.3), which shows that $b')$ implies $c)$.

Finally, to see that $c)$ implies $a)$, we can restrict to the case where Y' is also affine. The morphism $g : Y' \rightarrow Y$ is then affine, and so is the projection $g' : X' \rightarrow X$ (II, 1.5.5). Moreover, the functor $\mathcal{H}' \mapsto g'_*(\mathcal{H}')$ is exact in the category of quasi-coherent $\mathcal{O}_{X'}$ -Modules (I, 1.6.4); if $g'_*(\mathcal{H}') = 0$, then $\mathcal{H}' = 0$, so this functor is also faithful. Consequently, to show that $\mathcal{G}' \mapsto \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}'$ is exact and faithful, it suffices to show that the functor

$$\mathcal{G}' \longmapsto g'_*(g'^*(\mathcal{F}) \otimes_{\mathcal{O}_{X'}} f'^*(\mathcal{G}'))$$

is exact and faithful. The fact that g is affine implies that we have a canonical isomorphism

$$(2.2.2.1) \quad \mathcal{F} \otimes_{\mathcal{O}_X} f^*(g_*(\mathcal{G}')) \xrightarrow{\sim} g'_*(g'^*(\mathcal{F}) \otimes_{\mathcal{O}_{X'}} f'^*(\mathcal{G}')).$$

Indeed, we know (II, 1.5.2) that we have a canonical isomorphism

$$f^*(g_*(\mathcal{G}')) \xrightarrow{\sim} g'_*(f'^*(\mathcal{G}')),$$

and on the other hand a canonical homomorphism $\mathcal{F} \rightarrow g'_*(g'^*(\mathcal{F}))$ (0I, 4.4.3.2); composing the homomorphism $\mathcal{F} \otimes_{\mathcal{O}_X} f^*(g_*(\mathcal{G}')) \rightarrow g'_*(g'^*(\mathcal{F})) \otimes_{\mathcal{O}_X} g'_*(f'^*(\mathcal{G}'))$ with the canonical homomorphism (0I, 4.2.2.1), we deduce the homomorphism (2.2.2.1); the verification that it is an isomorphism is immediate by reducing to the case where X is affine. This being so, the functor $\mathcal{G}' \mapsto g_*(\mathcal{G}')$ is exact and faithful, and by hypothesis the same is true of the functor $\mathcal{G} \mapsto \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G} = \mathcal{F} \otimes_{\mathcal{O}_X} f^*(\mathcal{G})$ from quasi-coherent $\mathcal{O}_Y$ -Modules to quasi-coherent $\mathcal{O}_X$ -Modules. Their composite is therefore exact and faithful, which completes the proof of (2.2.1) and (2.2.2). $\square$

Corollary (2.2.3). — *Let $X = \text{Spec}(B)$, $Y = \text{Spec}(A)$ be two affine schemes, $f : X \rightarrow Y$ a morphism, $\mathcal{F} = \bar{M}$ a quasi-coherent $\mathcal{O}_X$ -Module. For $\mathcal{F}$ to satisfy the equivalent conditions of (2.2.1) (or (2.2.2)), it is necessary and sufficient that the A -module M be faithfully flat.*

Proof. Indeed, condition $c)$ of (2.2.2) means that the functor $N \mapsto M \otimes_A N$ from the category of A -modules to that of B -modules is exact and faithful, and the conclusion follows from (0I, 6.4.1). $\square$

Definition (2.2.4). — When the equivalent conditions of (2.2.1) are satisfied, we say that the quasi-coherent $\mathcal{O}_X$ -Module $\mathcal{F}$ is *faithfully flat relative to f* (or to Y).

We note that this notion is local over Y , but *not* over X ; in particular we can have $\mathcal{F}_x = 0$ for certain $x \in X$, in other words $\text{Supp}(\mathcal{F})$ is not necessarily equal to X . However, it follows immediately from criterion (2.2.1)(b) that for every $y \in Y$, there exists at least one $x \in f^{-1}(y)$ such that $(\mathcal{F}_y)_x = \mathcal{F}_x \otimes_{\mathcal{O}_y} k(y) \neq 0$, and a fortiori $\mathcal{F}_x \neq 0$; in other terms:

Corollary (2.2.5). — *If $\mathcal{F}$ is a quasi-coherent $\mathcal{O}_X$ -Module, faithfully flat relative to f , then*

$$f(\text{Supp}(\mathcal{F})) = Y.$$

In particular, f is a surjective morphism.

This result admits a partial converse:

Corollary (2.2.6). — *Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$ -Module of finite type. For $\mathcal{F}$ to be faithfully flat relative to f , it is necessary and sufficient that $\mathcal{F}$ be f -flat and that $f(\text{Supp}(\mathcal{F})) = Y$.*

Proof. Indeed, by (I, 9.1.13) and (I, 3.6.1), we have $\text{Supp}(\mathcal{F}_y) = f^{-1}(y) \cap \text{Supp}(\mathcal{F})$, so in this case criterion (2.2.1)(b) is precisely the condition of the corollary. In particular, the $\mathcal{O}_X$ -Module $\mathcal{O}_X$ is faithfully flat relative to f if and only if f is flat and surjective, that is to say if and only if the morphism f is faithfully flat (0I, 6.7.8). $\square$

Let us spell out the properties intervening in definition (2.2.4):

Proposition (2.2.7). — *Let $f : X \rightarrow Y$ be a morphism of preschemes, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module faithfully flat relative to f . Then, for a sequence $\mathcal{G}' \rightarrow \mathcal{G} \rightarrow \mathcal{G}''$ of quasi-coherent $\mathcal{O}_Y$ -Modules to be exact, it is necessary and sufficient that the corresponding sequence $\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}' \rightarrow \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G} \rightarrow \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}''$ be so. In particular, for a homomorphism $u : \mathcal{G} \rightarrow \mathcal{G}'$ of quasi-coherent $\mathcal{O}_Y$ -Modules to be injective (resp. surjective, bijective, zero), it is necessary and sufficient that $1_{\mathcal{F}} \otimes f^*(u) : \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G} \rightarrow \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}'$ be*

so. For a quasi-coherent $\mathcal{O}_Y$ -Module $\mathcal{G}$ to be zero, it is necessary and sufficient that $\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G} = 0$. For a quasi-coherent $\mathcal{O}_Y$ -Module $\mathcal{G}$, the map $\mathcal{G}' \mapsto \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}'$ from the set of quasi-coherent $\mathcal{O}_Y$ -submodules of $\mathcal{G}$ to the set of quasi-coherent $\mathcal{O}_X$ -submodules of $\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}$ is injective.

Proof. To prove the last assertion, that is to say that for two quasi-coherent sub- $\mathcal{O}_Y$ -Modules $\mathcal{G}'$, $\mathcal{G}''$ of $\mathcal{G}$, the relation $\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}' = \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}''$ implies $\mathcal{G}' = \mathcal{G}''$, we can reduce to the case where $\mathcal{G}' \subset \mathcal{G}''$ (by replacing $\mathcal{G}''$ by $\mathcal{G}' + \mathcal{G}''$), and it then suffices to apply the second assertion of the statement to the injection $u : \mathcal{G}' \rightarrow \mathcal{G}''$. $\square$

Corollary (2.2.8). — Let $f : X \rightarrow Y$ be a faithfully flat morphism, $\mathcal{G}$ a quasi-coherent $\mathcal{O}_Y$ -Module. The canonical map

$$(2.2.8.1) \quad \Gamma(Y, \mathcal{G}) \longrightarrow \Gamma(X, f^*(\mathcal{G}))$$

is injective.

Proof. Indeed, there are canonical identifications (0_I, 5.1.1)

$$\begin{aligned} \Gamma(Y, \mathcal{G}) &\simeq \operatorname{Hom}_{\mathcal{O}_Y}(\mathcal{O}_Y, \mathcal{G}), \\ \Gamma(X, f^*(\mathcal{G})) &\simeq \operatorname{Hom}_{\mathcal{O}_X}(\mathcal{O}_X, f^*(\mathcal{G})). \end{aligned}$$

By virtue of (2.2.1) and (2.2.4), the hypothesis implies that the functor $\mathcal{G} \mapsto f^*(\mathcal{G})$ is exact and faithful in the category of quasi-coherent $\mathcal{O}_Y$ -Modules, and thus a homomorphism $u : \mathcal{O}_Y \rightarrow \mathcal{G}$ is zero if and only if $f^*(u) : f^*(\mathcal{O}_Y) = \mathcal{O}_X \rightarrow f^*(\mathcal{G})$ is zero. $\square$

Remark (2.2.9). — The assertions of (2.2.7) and (2.2.8) are still true when the $\mathcal{O}_Y$ -Modules $\mathcal{G}'$, $\mathcal{G}$, $\mathcal{G}''$ appearing there are arbitrary (not necessarily quasi-coherent). Indeed, for every $y \in Y$, there exists $x \in f^{-1}(y)$ such that $\mathcal{F}_x$ is a flat $\mathcal{O}_y$ -module faithfully flat, and thus the functor $\mathcal{G}_y \mapsto \mathcal{G}_y \otimes_{\mathcal{O}_y} \mathcal{F}_x$ is faithful; since moreover for every $x \in f^{-1}(y)$ the functor $\mathcal{G}_y \mapsto \mathcal{G}_y \otimes_{\mathcal{O}_y} \mathcal{F}_x$ is exact, we immediately deduce the conclusion.

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Proposition (2.2.10). — Let $f : X \rightarrow Y$, $g : Y' \rightarrow Y$, $h : Y' \rightarrow Z$ be three morphisms of preschemes, $X' = X \times_Y Y'$, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module faithfully flat relative to Y , $\mathcal{G}'$ a quasi-coherent $\mathcal{O}_{Y'}$ -Module. For $\mathcal{G}'$ to be flat (resp. faithfully flat) relative to Z , it is necessary and sufficient that $\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}'$ be a flat (resp. faithfully flat) $\mathcal{O}_{X'}$ -Module relative to Z .

Proof. We know already that if $\mathcal{G}'$ is Z -flat, so is $\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}'$ (2.1.5). Consider a base change $Z'' \rightarrow Z$ and set $X'' = X' \times_Z Z''$; if $\mathcal{G}'$ is faithfully flat relative to Z , the functor

$$(2.2.10.1) \quad \mathcal{H}'' \mapsto \mathcal{H}'' \otimes_Z \mathcal{G}' \longrightarrow (\mathcal{H}'' \otimes_Z \mathcal{G}') \otimes_{\mathcal{O}_Y} \mathcal{F} = \mathcal{H}'' \otimes_Z (\mathcal{G}' \otimes_{\mathcal{O}_Y} \mathcal{F})$$

from the category of $\mathcal{O}_{Z''}$ -Modules to that of $\mathcal{O}_{X''}$ -Modules is composed of two exact and faithful functors, so this functor is also exact and faithful. Conversely, if this composite functor is exact (resp. exact and faithful), the same is true of $\mathcal{H}'' \mapsto \mathcal{H}'' \otimes_Z \mathcal{G}'$, since the functor $\mathcal{M}' \mapsto \mathcal{M}' \otimes_{\mathcal{O}_Y} \mathcal{F}$ from quasi-coherent $\mathcal{O}_{Y'}$ -Modules to quasi-coherent $\mathcal{O}_{X'}$ -Modules is exact and faithful by hypothesis. $\square$

Corollary (2.2.11). —

- (i) Let $f : X \rightarrow Y$, $g : Y' \rightarrow Y$ be two morphisms, $X' = X \times_Y Y'$, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module. If $\mathcal{F}$ is faithfully flat relative to Y , $\mathcal{F}' = \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y'}$ is faithfully flat relative to Y' .
- (ii) Let $f : X \rightarrow Y$, $g : Y \rightarrow Z$ be two morphisms, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module faithfully flat relative to Y . For g to be a faithfully flat morphism, it is necessary and sufficient that $\mathcal{F}$ be faithfully flat relative to Z .
- (iii) Let $f : X \rightarrow Y$, $g : Y \rightarrow Z$ be two morphisms, $\mathcal{G}$ a quasi-coherent $\mathcal{O}_Y$ -Module. Suppose the morphism f is faithfully flat. For $\mathcal{G}$ to be flat (resp. faithfully flat) relative to Z , it is necessary and sufficient that $f^*(\mathcal{G})$ be flat (resp. faithfully flat) relative to Z .
- (iv) Let $f : X \rightarrow Y$, $g : Y \rightarrow Z$ be two morphisms, x a point of X . Suppose f is flat at the point x . For $g \circ f$ to be flat at the point x , it is necessary and sufficient that g be flat at the point $f(x)$.

Proof. To prove (i), we apply (2.2.10), replacing Z by Y' and $\mathcal{G}'$ by $\mathcal{O}_{Y'}$. To prove (ii), we apply (2.2.10) taking $Y' \rightarrow Y$ the identity and replacing $\mathcal{G}'$ by $\mathcal{O}_Y$. To prove (iii), we apply (2.2.10), again taking $Y' \rightarrow Y$ to be the identity, and replacing $\mathcal{F}$ by $\mathcal{O}_X$ and $\mathcal{G}'$ by $\mathcal{G}$. Finally, (iv) is deduced from (ii)

by replacing X by $\operatorname{Spec}(\mathcal{O}_x)$, $\mathcal{F}$ by $\mathcal{O}_x$, Y by $\operatorname{Spec}(\mathcal{O}_{f(x)})$ and Z by $\operatorname{Spec}(\mathcal{O}_{g(f(x))})$, and taking into account (0_I, 6.6.2). $\square$

Corollary (2.2.12). — *Let Y be an affine scheme, $f : X \rightarrow Y$ a quasi-compact morphism, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module. If $\mathcal{F}$ is faithfully flat relative to f , there exists an affine scheme X' and a surjective local isomorphism $g : X' \rightarrow X$ such that $g^*(\mathcal{F})$ is faithfully flat relative to $f \circ g$.*

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Proof. Indeed, X is quasi-compact, hence a finite union of affine open subsets X_i ; it suffices to take for X' the affine scheme given by the disjoint sum of the preschemes induced on the open subsets X_i , and for $g : X' \rightarrow X$ the canonical morphism. It is clear that g is a surjective local isomorphism, hence faithfully flat, and the hypothesis implies that $g^*(\mathcal{F})$ is faithfully flat relative to $f \circ g$, by virtue of (2.2.11)(iii). $\square$

Corollary (2.2.13). —

- (i) *Let $f : X \rightarrow Y$, $g : Y' \rightarrow Y$ be two morphisms, $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \rightarrow Y'$. If f is a faithfully flat morphism, so is f' .*
- (ii) *If $f : X \rightarrow X'$, $g : Y \rightarrow Y'$ are two faithfully flat S -morphisms,*

$$f \times_S g : X \times_S Y \longrightarrow X' \times_S Y'$$

is faithfully flat.

- (iii) *Let $f : X \rightarrow Y$, $g : Y \rightarrow Z$ be two morphisms such that f is faithfully flat. For g to be a flat (resp. faithfully flat) morphism, it is necessary and sufficient that $g \circ f$ be a flat (resp. faithfully flat) morphism.*

Proposition (2.2.14). — *Let $f : X \rightarrow Y$ be a faithfully flat morphism and quasi-compact. If X is locally Noetherian, so is Y .*

Proof. The question being local on Y , we can suppose $Y = \operatorname{Spec}(A)$ affine; as f is quasi-compact, it follows from (2.2.12) that we can also restrict to the case where $X = \operatorname{Spec}(B)$ is affine. Then B is a Noetherian ring by hypothesis, and a faithfully flat A -module (2.2.3); so A is Noetherian (0_I, 6.5.2). $\square$

Proposition (2.2.15). — *Let $f : X \rightarrow Y$ be a faithfully flat morphism. If $\mathfrak{S}(X)$ and $\mathfrak{S}(Y)$ denote respectively the sets of sub-preschemes of X and of Y , the map $Z \mapsto f^{-1}(Z)$ of $\mathfrak{S}(Y)$ to $\mathfrak{S}(X)$ is injective.*

Proof. Since f is surjective, on the underlying sets of a sub-prescheme Z of Y we have $f(f^{-1}(Z)) = Z$. On the other hand, if U is an open subset of Y containing Z and in which Z is closed, then $f^{-1}(U)$ is open in X , $f^{-1}(Z)$ is closed in $f^{-1}(U)$, and the restriction $f^{-1}(U) \rightarrow U$ of f is a faithfully flat morphism. We can therefore restrict to considering only the closed sub-preschemes of Y . Now, if Z is a closed sub-prescheme of Y corresponding to a quasi-coherent Ideal $\mathcal{J}$ of $\mathcal{O}_Y$, we know that $f^{-1}(Z)$ corresponds to the quasi-coherent Ideal $f^*(\mathcal{J})\mathcal{O}_X$ of $\mathcal{O}_X$ (I, 4.4.5), and since f is flat, $f^*(\mathcal{J})\mathcal{O}_X$ identifies with $f^*(\mathcal{J})$. But the map $\mathcal{J} \mapsto f^*(\mathcal{J})$ from the set of quasi-coherent Ideals of $\mathcal{O}_Y$ to the set of quasi-coherent Ideals of $\mathcal{O}_X$ is injective (2.2.7), whence the conclusion. $\square$

Corollary (2.2.16). — *Let X, Y be two S -preschemes; if $S' \rightarrow S$ is a faithfully flat morphism, the map $f \mapsto f_{(S')}$ from $\operatorname{Hom}_S(X, Y)$ to $\operatorname{Hom}_{S'}(X_{(S')}, Y_{(S')})$ is injective.*

Proof. We have $X_{(S')} \times_{S'} Y_{(S')} = (X \times_S Y)_{(S')}$ (I, 3.3.10), so the morphism projection $X_{(S')} \times_{S'} Y_{(S')} \rightarrow X \times_S Y$ is faithfully flat (2.2.13). The elements of $\operatorname{Hom}_S(X, Y)$ correspond bijectively to the sub-preschemes of $X \times_S Y$ that are their graphs (I, 5.3.11); if Z_f is the graph of $f \in \operatorname{Hom}_S(X, Y)$, then $Z_{f_{(S')}} = g^{-1}(Z_f)$ (I, 5.3.12). It suffices therefore to apply (2.2.15). $\square$

Proposition (2.2.17). — *Let A be a ring, B an A -algebra such that B is a faithfully flat A -module and of finite presentation. Then the structural homomorphism $\varphi : A \rightarrow B$ is an isomorphism of the A -module A onto a direct summand of the A -module B . If A is a local ring, B is a free A -module and there exists a basis of this module containing the identity element of B .*

Proof. By virtue of Bourbaki, *Alg. comm.*, chap. II, § 3, n° 3, prop. 12, it suffices to prove the proposition when A is a local ring; we know then (*loc. cit.*, n° 2, cor. 2 of prop. 5) that B is an A -module free of finite type, and the conclusion follows from *loc. cit.*, prop. 5. $\square$

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2.3. Topological properties of flat morphisms

Lemma (2.3.1). — Let $f : X \rightarrow Y$ be a quasi-compact and quasi-separated morphism, $g : Y' \rightarrow Y$ a flat morphism; set $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \rightarrow Y'$. Then, for every quasi-coherent $\mathcal{O}_X$ -Module $\mathcal{F}$, the canonical homomorphism

$$(2.3.1.1) \quad g^*(f_*(\mathcal{F})) \longrightarrow f'_*(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_{X'})$$

is bijective.

Proof. This is a special case of (III, 1.4.15) (improved in (1.7.21)). $\square$

Proposition (2.3.2). — Let S be a prescheme, $f : X \rightarrow Y$ a quasi-compact and quasi-separated S -morphism, and let Z be the sub-prescheme of Y which is the closed image of X under f ((I, 9.5.3) and (1.7.8)). Let $j : Z \rightarrow Y$ be the canonical injection, so that $f = j \circ g$, where $g : X \rightarrow Z$ is a morphism (loc. cit.). Let $h : S' \rightarrow S$ be a flat morphism, and set $f' = f_{(S')} : X_{(S')} \rightarrow Y_{(S')}$. Then $j' = j_{(S')} : Z_{(S')} \rightarrow Y_{(S')}$ identifies with the canonical injection of the sub-prescheme of $Y_{(S')}$ which is the closed image of $X_{(S')}$ under f' .

Proof. Since the morphism $Y_{(S')} \rightarrow Y$ is flat (2.1.4), we may reduce to the case $S = Y$ (I, 3.3.11). If $f = (\psi, \theta)$, then Z is the closed sub-prescheme of S defined by the quasi-coherent Ideal $\mathcal{J} \subset \mathcal{O}_S$ which is the kernel of

$$\theta : \mathcal{O}_S \longrightarrow f_*(\mathcal{O}_X)$$

(I, 9.5.2). Since h is flat, the quasi-coherent Ideal $\mathcal{J} \mathcal{O}_{S'}$ identifies with the kernel of

$$h^*(\theta) : \mathcal{O}_{S'} \longrightarrow h^*(f_*(\mathcal{O}_X)).$$

If $f' = (\psi', \theta')$, then $\theta' : \mathcal{O}_{S'} \rightarrow f'_*(\mathcal{O}_{X'})$ is the composite of $h^*(\theta)$ with the canonical homomorphism

$$h^*(f_*(\mathcal{O}_X)) \longrightarrow f'_*(\mathcal{O}_{X'})$$

of (2.3.1.1). The conclusion follows from (2.3.1) and (I, 9.5.2), since f is quasi-compact and quasi-separated (1.1.2 and 1.2.2). $\square$

(2.3.3). We will say that a morphism $f : X \rightarrow Y$ is *quasi-flat* if there exists a quasi-coherent $\mathcal{O}_X$ -Module $\mathcal{F}$ of finite type which is f -flat and whose support is equal to X . We will say that f is *quasi-faithfully flat* if it is quasi-flat and surjective. Every flat (resp. faithfully flat) morphism is quasi-flat (resp. quasi-faithfully flat), for then $\mathcal{F} = \mathcal{O}_X$ satisfies the preceding conditions.

It follows immediately from (2.1.4) and from (I, 9.1.13), that if f is quasi-flat, then, for every morphism $Y' \rightarrow Y$, the morphism $f_{(Y')} : X \times_Y Y' \rightarrow Y'$ is quasi-flat. Similarly (taking into account (I, 3.5.2)), if f is quasi-faithfully flat, so is $f_{(Y')}$.

Proposition (2.3.4). — Let $f : X \rightarrow Y$ be a quasi-flat morphism (2.3.3). Then f possesses the following properties (which are equivalent by virtue of (1.10.4)):

- (i) For every $x \in X$ and every generization y' of $y = f(x)$, there exists a generization x' of x such that $f(x') = y'$.
- (ii) For every $x \in X$, the image under f of $\text{Spec}(\mathcal{O}_{X,x})$ is $\text{Spec}(\mathcal{O}_{Y,y})$.
- (iii) For every closed irreducible part Y' of Y , every irreducible component of $f^{-1}(Y')$ dominates Y' .

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Proof. It suffices, for example, to prove (ii). By hypothesis, there is an $\mathcal{O}_X$ -Module quasi-coherent of finite type $\mathcal{F}$, which is f -flat and such that $\text{Supp}(\mathcal{F}) = X$. For every $x \in X$, $\mathcal{F}_x$ is then an $\mathcal{O}_x$ -module of finite type, nonzero, and moreover $\mathcal{F}_x$ is an $\mathcal{O}_y$ -module flat, for the homomorphism $\rho : \mathcal{O}_y \rightarrow \mathcal{O}_x$. As the latter is local and $\mathcal{F}_x \neq 0$, the Nakayama lemma shows that $\mathcal{F}_x \otimes_{\mathcal{O}_y} k(y) \neq 0$, so $\mathcal{F}_x$ is a faithfully flat $\mathcal{O}_y$ -module (0, 6.4.1). It follows that for every prime ideal $\mathfrak{q}$ of $\mathcal{O}_y$, there exists a prime ideal $\mathfrak{p}$ of $\mathcal{O}_x$ such that $\mathfrak{q} = \rho^{-1}(\mathfrak{p})$ (0, 6.5.1), which proves (ii). $\square$

Corollary (2.3.5). — Let $f : X \rightarrow Y$ be a morphism satisfying the equivalent conditions (i), (ii), (iii) of (2.3.4) (in particular a quasi-flat or open morphism (1.10.4)).

- (i) Let Z, Z' be two closed irreducible subsets of Y such that $Z \subset Z'$, and let T be an irreducible component of $f^{-1}(Z)$; then there exists an irreducible component T' of $f^{-1}(Z')$ containing T (and dominating Z').
- (ii) For every irreducible component T of X , $\overline{f(T)}$ is an irreducible component of Y .

(iii) Suppose Y irreducible, and, if y is its generic point, suppose that $f^{-1}(y)$ is irreducible. Then X is irreducible.

Proof. (i) It suffices to apply (2.3.4), (i) by taking for x the generic point of T ($y = f(x)$ is then the generic point of Z) and for y' the generic point of Z' .

(ii) It is clear that $\overline{f(T)} = Z$ is irreducible, and by virtue of (i), Z cannot be strictly contained in a closed irreducible part of Y .

(iii) By virtue of (ii), every irreducible component of X dominates Y , and hence meets $f^{-1}(y)$, where y is the generic point of Y . The conclusion then follows from (0, 2.1.8). $\square$

Proposition (2.3.6). — *Let Y be a prescheme whose set of irreducible components is locally finite (which is the case if the underlying space is locally Noetherian (cf. (I, 6.1.9))).*

- (i) *Every closed subset W of Y stable by generization (0, 2.1.2) is open. In particular, every connected component of Y is open.*
- (ii) *Let $f : X \rightarrow Y$ be a closed morphism satisfying in addition the equivalent conditions (i), (ii), (iii) of (2.3.4) (which will be the case if f is quasi-flat or open). Then the image under f of every connected component C of X is a connected component of Y .*
- (iii) *Let $f : X \rightarrow Y$ be a morphism satisfying the equivalent conditions (i), (ii), (iii) of (2.3.4), and such that moreover for every $y \in Y$, the set $f^{-1}(y)$ is finite (which will be the case if f is quasi-finite or radicial). Then the set of irreducible components of X is locally finite.*

Proof. (i) If $y \in W$, the generic points η_i ($1 \leq i \leq m$) of the irreducible components Y_i of Y containing y belong to W by hypothesis. Since W is closed, these components themselves are contained in W ; and since there is by hypothesis a neighbourhood U of y which is the union of the $U \cap Y_i$ (0, 2.1.6), we have $U \subset W$, so W is open. Since for every $y \in Y$, $\overline{\{y\}}$ is connected, a connected component of Y is stable by generization, so the second assertion follows immediately from the first.

(ii) Since C is closed in X , $f(C)$ is closed in Y by hypothesis. Moreover, since C is stable by generization, the hypothesis on f implies that $f(C)$ is stable by generization; so, by virtue of (i), $f(C)$ is open and closed, and since it is connected, it is a connected component of Y .

(iii) Let $x \in X$. By hypothesis, there is an open neighbourhood U of $y = f(x)$ which meets only finitely many irreducible components Y_i of Y ($1 \leq i \leq n$); let y_i be the generic point of Y_i . For every irreducible component Z of X meeting $f^{-1}(U)$, the generic point z of Z is necessarily in one of the sets $f^{-1}(y_i)$ (2.3.4). Since each of these sets is finite by hypothesis, this proves our assertion. $\square$

Proposition (2.3.7). — *Let $f : X \rightarrow Y$ be a morphism, $g : Y' \rightarrow Y$ a quasi-flat morphism, $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \rightarrow Y'$.*

- (i) *If f is quasi-compact and dominant, so is f' .*
- (ii) *If every irreducible component of X dominates an irreducible component of Y , then every irreducible component of X' dominates an irreducible component of Y' .*

Proof. Denote by $g' : X' \rightarrow X$ the canonical projection, which is a quasi-flat morphism (2.3.3).

(i) We know already (1.1.2) that f' is quasi-compact; moreover, if y' is a maximal point of Y' , $y = g(y')$ is a maximal point of Y , as follows from (2.3.4), (iii). By hypothesis (1.1.5), there exists $x \in X$ such that $f(x) = y$; so (I, 3.4.7), there exists $x' \in X'$ such that $f'(x') = y'$.

(ii) Let x' be a maximal point of X' , and let $x = g'(x')$; it follows from (2.3.4), (ii) that x is a maximal point of X , and by hypothesis $y = f(x)$ is a maximal point of Y . Set $y' = f'(x')$, so that $g(y') = y$; it remains to show that y' is a maximal point of Y' . By virtue of (I, 5.1.7) and (2.3.3), we may reduce to the case where X and Y are reduced, so that $\mathcal{O}_x$ and $\mathcal{O}_y$ are fields. By virtue of (I, 3.6.5), applied twice, and (I, 2.4.4), we may further reduce to the case where $Y = \text{Spec}(\mathcal{O}_y)$ and $Y' = \text{Spec}(\mathcal{O}_{y'})$. Then f is flat because $\mathcal{O}_y$ is a field (2.1.2), and the same is true of f' (2.1.4); hence (2.3.4), (ii) shows that $y' = f'(x')$ is a maximal point of Y' . $\square$

Corollary (2.3.8). — (Zariski). *Let A, B be two Noetherian local rings, $\mathfrak{m}, \mathfrak{n}$ their maximal ideals respectively, $\varphi : A \rightarrow B$ a local homomorphism. Suppose the following hypotheses satisfied:*

- 1° *B is an A -algebra essentially of finite type (1.3.8).*
- 2° *The $\mathfrak{m}$ -adic completion $\widehat{A}$ of A for the $\mathfrak{m}$ -adic topology is integral.*
- 3° *φ is injective.*

Then the $\mathfrak{m}$ -adic topology of A is induced by the $\mathfrak{n}$ -adic topology of B .

Proof. Set $B' = B \otimes_A \hat{A}$; by hypothesis 1°, B' is of the form $S^{-1}(C \otimes_A \hat{A})$, where C is an A -algebra of finite type and S is a multiplicative subset of C ; hence B' is Noetherian. As A identifies with a subring of $\hat{A}$ (0, 7.3.5), A is integral by hypothesis 2°. Hypothesis 3° implies that there exists a prime ideal $\mathfrak{q}$ of B inducing the zero ideal of A (0, 1.5.8); consequently, the local homomorphism $A \rightarrow B/\mathfrak{q}$ is injective. We can therefore restrict to proving the conclusion of (2.3.8) under the added hypothesis that B is a local ring which is *integral*. Apply (2.3.7), (ii) to $Y = \text{Spec}(A)$, $X = \text{Spec}(B)$, $Y' = \text{Spec}(\hat{A})$, and $X' = \text{Spec}(B')$. Since $Y' \rightarrow Y$ is flat and the integral scheme X dominates the integral scheme Y , every irreducible component of X' dominates Y' . If y , x , and y' are the closed points of Y , X , and Y' respectively, there is a point $x' \in X'$ (in fact unique) lying above x and y' (I, 3.4.9), and $\text{Spec}(\mathcal{O}_{x'})$ therefore dominates $\text{Spec}(\mathcal{O}_{y'})$. Consequently, there is a commutative diagram of local homomorphisms of Noetherian local rings

$$\begin{array}{ccc} B = \mathcal{O}_x & \longrightarrow & \mathcal{O}_{x'} \\ \uparrow \varphi & & \uparrow v \\ A = \mathcal{O}_y & \xrightarrow{u} & \mathcal{O}_{y'} = \hat{A} \end{array}$$

in which u and v are injective (I, 1.2.7). Identify A and $\hat{A}$ with subrings of $\mathcal{O}_{x'}$, and let $\mathfrak{r}$ be the maximal ideal of $\mathcal{O}_{x'}$. Then $\bigcap_k (\mathfrak{r}^k \cap \hat{A}) = 0$ (0, 7.3.5); since $\hat{A}$ is complete and these ideals are open in $\hat{A}$, it follows (Bourbaki, *Alg. comm.*, chap. III, § 2, n° 7, prop. 8) that the topology of $\hat{A}$ is induced by the $\mathfrak{r}$ -preadic topology of $\mathcal{O}_{x'}$. *A fortiori*, the same is true of the topology of A (0, 7.3.5). Moreover, $\mathfrak{n}^k \cap A \subset \mathfrak{r}^k \cap A$, so the $\mathfrak{n}$ -preadic topology of B induces on A a topology finer than the $\mathfrak{m}$ -preadic topology; but $\mathfrak{m}^k \subset \mathfrak{n}^k \cap A$, so these two topologies are identical. $\square$

Remark (2.3.9). — We will see further on (7.8.3, (vii)) that for the Noetherian local rings A most commonly used in Algebraic Geometry, the hypothesis that A is integral and integrally closed implies that the same is true of $\hat{A}$. This is why, in Algebraic Geometry over a base field, one generally states (2.3.8) under the hypothesis that A is *integral and integrally closed*.

Theorem (2.3.10). — Let $f : X \rightarrow Y$ be a quasi-flat morphism (2.3.3). Then, for every pro-constructible part (1.9.4) Z of Y , we have $\overline{f^{-1}(Z)} = f^{-1}(\overline{Z})$.

Proof. Since f is continuous, we have $\overline{f^{-1}(Z)} \subset f^{-1}(\overline{Z})$ and it amounts to proving that for every $x \in X$ such that $f(x) \in \overline{Z}$, x is adherent to $f^{-1}(Z)$; it is clear that the question is local on Y , so we can suppose Y affine. By virtue of the hypothesis, there exists an affine scheme Y' and a morphism $g : Y' \rightarrow Y$ such that $g(Y') = Z$ (1.9.5, (ix)). Let Y_1 be the closed image of Y' under g (I, 9.5.3), and let X_1 be the closed sub-prescheme $f^{-1}(Y_1)$ of X ; if $f_1 : X_1 \rightarrow Y_1$ is the restriction of f to X_1 , we know that f_1 is quasi-flat (2.3.3); we can therefore replace X, Y by X_1, Y_1 respectively, in other words suppose that g is *dominant*. Set then $X' = X \times_Y Y'$, and let f' and g' be the projections of X' into Y' and X respectively, so that we have the commutative diagram

$$\begin{array}{ccc} X & \xleftarrow{g'} & X' \\ f \downarrow & & \downarrow f' \\ Y & \xleftarrow{g} & Y' \end{array}$$

Since f is quasi-flat and g is quasi-compact and dominant, it follows from (2.3.7), (i) (with the roles of f and g interchanged) that g' is dominant, which proves the theorem. $\square$

Corollary (2.3.11). — Let f be a quasi-flat and quasi-compact morphism, F a closed part of X such that $F = f^{-1}(f(F))$; then $F = f^{-1}(\overline{f(F)})$.

Proof. Let Y' be the reduced sub-prescheme of X having F as underlying space (I, 5.2.1), and let $j : Y' \rightarrow X$ be the canonical injection. Then $f \circ j$ is quasi-compact (1.1.2), so $Z = f(F)$ is pro-constructible in Y (1.9.5, (vii)), and the corollary follows from the fact that F is closed. $\square$

We can further write the result of (2.3.11) in the form $F = f^{-1}(f(X) \cap \overline{f(F)})$. In other words, the closed subsets of the subspace $f(X)$ of Y are the parts $Z \subset f(X)$ such that $f^{-1}(Z)$ is closed in X ; this also means that the topology induced by Y on $f(X)$ is the quotient of that of X by the equivalence relation defined by f . In particular: $\square$

Corollary (2.3.12). — Let $f : X \rightarrow Y$ be a quasi-faithfully flat morphism (2.3.3) and quasi-compact. Then the topology of Y is the quotient of that of X by the equivalence relation defined by f , in other words, for $Z \subset Y$ to be open (resp. closed) in Y , it is necessary and sufficient that $f^{-1}(Z)$ be open (resp. closed) in X .

Proof. Indeed, we then have $f(X) = Y$. $\square$

Corollary (2.3.13). — Let X, Y be two S -preschemes, $f : X \rightarrow Y$ an S -morphism faithfully flat and quasi-compact. For Y to be separated over S , it is necessary and sufficient that the canonical immersion $j : X \times_Y X \rightarrow X \times_S X$ (I, 5.3.10) be closed.

Proof. Let us note for this that we have the commutative diagram (I, 5.3.5)

$$\begin{array}{ccc} X \times_Y X & \xrightarrow{j} & X \times_S X \\ \pi \downarrow & & \downarrow f \times_S f \\ Y & \xrightarrow{\Delta_Y} & Y \times_S Y \end{array}$$

identifying $X \times_Y X$ with the product of the $(Y \times_S Y)$ -preschemes Y and $X \times_S X$. Since f is surjective, so are π and $f \times_S f$; hence the inverse image of the diagonal $\Delta_Y(Y)$ under $f \times_S f$ is the image $j(X \times_Y X)$ (I, 3.4.8). Since $f \times_S f$ is faithfully flat and quasi-compact (1.1.2 and 2.2.13), it suffices to apply (2.3.12) to this morphism. $\square$

Corollary (2.3.14). — Let $f : X \rightarrow Y$ be a quasi-faithfully flat and quasi-compact morphism, and let Z be a part of Y . For Z to be a locally closed part pro-constructible of Y , it is necessary and sufficient that $f^{-1}(Z)$ be a locally closed part pro-constructible of X .

Proof. We already know (1.9.12) that, for Z to be a pro-constructible part of Y , it is necessary and sufficient that $f^{-1}(Z)$ be a pro-constructible part of X . The condition is evidently necessary. To prove that it is sufficient, consider the closed reduced sub-prescheme Y_1 of Y having $\overline{Z}$ as underlying space and let X_1 be its inverse image, so that $f^{-1}(\overline{Z}) = \overline{f^{-1}(Z)}$ by virtue of (2.3.10). As $f^{-1}(Z)$ is locally closed in X , it is open in $\overline{f^{-1}(Z)}$, so in X_1 ; now, the morphism $f_1 : X_1 \rightarrow Y_1$ deduced from f by restriction is quasi-faithfully flat and quasi-compact (1.1.2 and 2.3.3); we deduce from (2.3.12) that Z is open in $\overline{Z}$, which shows that Z is locally closed in Y . $\square$

Remark (2.3.15). — It does not suffice, in (2.3.12), to suppose only that f is faithfully flat. For example, take for Y a reduced irreducible algebraic curve (II, 7.4.2), and let X be the prescheme sum $\coprod_{\eta \in Y} \text{Spec}(\mathcal{O}_\eta)$ (I, 3.1). Let $f : X \rightarrow Y$ be the canonical morphism, which is faithfully flat (I, 2.4.2). If η denotes the generic point of Y , then $Z = \{\eta\}$ is not open in Y (II, 7.4.3), whereas $f^{-1}(Z) = \text{Spec}(\mathcal{O}_\eta)$ is open in X , since $\mathcal{O}_\eta$ is a field.

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2.4. Universally open morphisms and flat morphisms

(2.4.1). We have already defined (II, 5.4.9) the notion of *universally closed* morphism; in the same manner, we give the following definitions:

Definition (2.4.2). — We say that a morphism of preschemes $f : X \rightarrow Y$ is *universally open* (resp. *universally bicontinuous*, resp. a *universal homeomorphism*) if, for every morphism $g : Y' \rightarrow Y$, the morphism $f_{(Y')} : X \times_Y Y' \rightarrow Y'$ is open (resp. a homeomorphism onto its image, resp. a homeomorphism onto Y').

We will see further on (14.3.2) that when Y is locally Noetherian, the definition of universally open morphism given here is equivalent to the definition (III, 4.3.9) for morphisms of finite type; the reader can verify that we do not use the latter definition before § 14.

Proposition (2.4.3). —

- (i) An immersion (resp. an open immersion, resp. a closed immersion) is universally bicontinuous (resp. universally open, resp. universally closed).
- (ii) The composite of two universally open morphisms (resp. universally closed morphisms, resp. universally bicontinuous morphisms, resp. universal homeomorphisms) has the same property.
- (iii) If $f : X \rightarrow Y$ is a universally open (resp. universally closed, resp. universally bicontinuous, resp. a universal homeomorphism) S -morphism, then so is $f_{(S')} : X_{(S')} \rightarrow Y_{(S')}$ after every base change $S' \rightarrow S$.
- (iv) If the S -morphisms $f : X \rightarrow X'$ and $g : Y \rightarrow Y'$ are both universally open (resp. universally closed, resp. universally bicontinuous, resp. universal homeomorphisms), then so is

$$f \times_S g : X \times_S Y \longrightarrow X' \times_S Y'.$$

- (v) Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be morphisms, with f surjective. If $g \circ f$ is universally open (resp. universally closed, resp. universally bicontinuous, resp. a universal homeomorphism), then so is g .
- (vi) A morphism f is universally open (resp. universally closed, resp. universally bicontinuous, resp. a universal homeomorphism) if and only if f_{red} has the same property.
- (vii) Let (U_α) be an open covering of Y . A morphism $f : X \rightarrow Y$ is universally open (resp. universally closed, resp. universally bicontinuous, resp. a universal homeomorphism) if and only if, for every α , its restriction $f^{-1}(U_\alpha) \rightarrow U_\alpha$ has the same property.

Proof. Assertion (i) follows from (I, 4.3.2). Assertion (ii) follows immediately from the definitions. The same is true of (iii), after reducing to the case $Y = S$ and $Y' = S'$, as permitted by (I, 3.3.11); assertion (iv) follows from (ii), (iii), and (I, 3.5.1).

To prove (v), note that after every morphism $Z' \rightarrow Z$, the base change

$$f_{(Z')} : X_{(Z')} \longrightarrow Y_{(Z')}$$

is surjective (I, 3.5.2). It is therefore enough to use the following purely topological fact: if f is surjective and $g \circ f$ is open (resp. closed, resp. a homeomorphism onto its image, resp. a bijective homeomorphism), then g has the corresponding property. In the open and closed cases this is Bourbaki, *Top. gén.*, chap. I^{er}, 3^e éd., § 5, n° 1, prop. 1. In the other two cases we may replace Z by $g(Y)$ and thus suppose that $g(f(X)) = g(Y) = Z$. Then $g \circ f$ is a homeomorphism of X onto Z ; since f is surjective, g is bijective, and since the preceding case shows that g is open, g is a homeomorphism of Y onto Z .

To prove (vi), note that a morphism g is open (resp. closed, resp. a homeomorphism onto its image, resp. a bijective homeomorphism) if and only if g_{red} has the same property. On the other hand, for every morphism $Y'' \rightarrow Y$, we have

$$(X_{\text{red}} \times_{Y_{\text{red}}} Y''_{\text{red}})_{\text{red}} = (X \times_Y Y'')_{\text{red}}$$

by (I, 5.1.8); hence the preceding remark shows that if f_{red} is universally open (resp. universally closed, resp. universally bicontinuous, resp. a universal homeomorphism), then so is f . The converse is proved in the same manner, noting here that for every morphism $Y'' \rightarrow Y_{\text{red}}$, we have $(X_{\text{red}} \times_{Y_{\text{red}}} Y'')_{\text{red}} = (X \times_Y Y'')_{\text{red}}$ (I, 5.1.8).

Finally, the necessity of condition (vii) follows immediately from (iii). Conversely, suppose the condition satisfied, and let $g : Y' \rightarrow Y$ be a morphism; then the $g^{-1}(U_\alpha) = U'_\alpha$ form an open covering of Y' and if one notes f_α the restriction $f^{-1}(U_\alpha) \rightarrow U_\alpha$ of f , and f' the morphism $f_{(Y')}$, the restriction $f'^{-1}(U'_\alpha) \rightarrow U'_\alpha$ is nothing other than $(f_\alpha)_{(U'_\alpha)}$. We can therefore restrict to proving that f' is open (resp. closed, resp. a homeomorphism onto its image, resp. a homeomorphism onto Y'), which is immediate. $\square$

Proposition (2.4.4). — *A universally bicontinuous morphism $f : X \rightarrow Y$ is radicial (and hence separated (1.7.7.1)).*

Proof. Indeed, f is universally injective by hypothesis (I, 3.5.11). $\square$

Proposition (2.4.5). —

- (i) An integral, surjective, and radicial morphism $f : X \rightarrow Y$ is a universal homeomorphism.

- (ii) Conversely, suppose that Y is locally Noetherian. If a morphism of finite type $f : X \rightarrow Y$ is a universal homeomorphism, then it is finite, surjective, and radicial.

Proof. (i) The three properties of f are preserved by base change (I, 3.5.2), (I, 3.5.7), and (II, 6.1.5). Since an integral morphism is closed (II, 6.1.10), f is a homeomorphism of X onto Y .

(ii) The morphism f is of finite type and universally closed by hypothesis, and it is separated by (2.4.4); hence it is proper (II, 5.4.1). For every $y \in Y$, the fiber $f^{-1}(y)$ consists of a single point, so (III, 4.4.2) implies that f is finite. It is plainly surjective, and it is radicial because it is universally injective (I, 3.5.11). $\square$

Theorem (2.4.6). — *Let $f : X \rightarrow Y$ be a quasi-flat morphism (2.3.3) and locally of finite presentation. Then f is universally open. In particular, a flat morphism locally of finite presentation is universally open.*

Proof. We know that for every base change $Y' \rightarrow Y$, $f_{(Y')}$ is quasi-flat (2.3.3) and locally of finite presentation (1.4.3, iii). It suffices therefore to prove that f is an open morphism. But this follows from the criterion (1.10.4) for morphisms locally of finite presentation, conditions $b)$, $b')$, and $c)$ of (1.10.4) being nothing other than conditions (i), (ii), (iii) of (2.3.4). $\square$

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Corollary (2.4.7). — *For every prescheme Y , the structural morphism*

$$\mathbf{V}_Y^n \longrightarrow Y, \quad \mathbf{V}_Y^n = Y \times_{\mathrm{Spec}(\mathbf{Z})} \mathrm{Spec}(\mathbf{Z}[T_1, \dots, T_n]),$$

where $\mathbf{V}_Y^n$ is also denoted by $Y[T_1, \dots, T_n]$, is universally open.

Proof. Indeed, for $Y = \mathrm{Spec}(A)$, $Y[T_1, \dots, T_n] = \mathrm{Spec}(A[T_1, \dots, T_n])$, and $A[T_1, \dots, T_n]$ is a free A -algebra of finite presentation. $\square$

Remark (2.4.8). —

- (i) A faithfully flat and quasi-compact morphism $f : X \rightarrow Y$ is *not necessarily open*, even when X and Y are Noetherian. For example, let Y be a reduced irreducible algebraic curve (II, 7.4.2) with generic point y , let $X = Y \amalg \mathrm{Spec}(k(y))$, and let f be the canonical morphism. It is clear that f is flat and surjective, hence faithfully flat, and it is quasi-compact. But the image under f of the open subset $\mathrm{Spec}(k(y))$ of X is $\{y\}$, which is not open in Y (II, 7.4.3).
- (ii) For every prescheme X , the canonical morphism $f : X_{\mathrm{red}} \rightarrow X$ is a closed immersion and a universal homeomorphism by (2.4.3), (vi). But when X is locally Noetherian, f is flat only if X is reduced, in which case $f = 1_X$ (2.2.17).

Proposition (2.4.9). — *Let Y be a discrete prescheme. Then every morphism $f : X \rightarrow Y$ is universally open.*

Proof. The question is local on Y (2.4.3), (vii), so we may assume that the underlying space of Y consists of a single point. Replacing f by f_{red} (2.4.3), (vi), we may further assume that Y is the spectrum of a field k . For every base change $Y' \rightarrow Y$, the inverse images in $X' = X \times_Y Y'$ of the affine open subsets of X cover X' , so we may also suppose that $X = \mathrm{Spec}(B)$ is affine, with B a k -algebra. It suffices then to prove that for every k -algebra A' , if we set $B' = B \otimes_k A'$, the image of every open subset of the form $U = D(t)$ ($t \in B'$) is open in $\mathrm{Spec}(A')$. Now B is the filtered direct limit of the increasing family (B_α) of its sub- k -algebras of finite type. Since filtered direct limits commute with tensor products, B' is the direct limit of the A' -algebras $B_\alpha \otimes_k A' = B'_\alpha$. There exists α such that t is the image in B' of an element $t_\alpha \in B'_\alpha$, so $D(t)$ is the inverse image under the canonical morphism $u_\alpha : \mathrm{Spec}(B') \rightarrow \mathrm{Spec}(B'_\alpha)$ of the open subset $U_\alpha = D(t_\alpha)$ (I, 1.2.2.2). But since k is a field, B_α is a k -algebra of finite presentation and a k -module flat, hence B'_α is an A' -algebra of finite presentation and a flat A' -module; we thus conclude from (2.4.6) that the image of U_α by $f'_\alpha : \mathrm{Spec}(B'_\alpha) \rightarrow \mathrm{Spec}(A')$ is open in $\mathrm{Spec}(A')$. It remains to prove that $f'(U) = f'_\alpha(U_\alpha)$. The inclusion $f'(U) \subset f'_\alpha(U_\alpha)$ is clear, so it remains to show that, for every point $y' \in f'_\alpha(U_\alpha)$, the intersection $V = U \cap f'^{-1}(y')$ is nonempty. Now $V = u_\alpha^{-1}(V_\alpha)$, where $V_\alpha = U_\alpha \cap f'^{-1}_\alpha(y')$; in other words, V_α is a non-empty open subset of the prescheme $\mathrm{Spec}(B'_\alpha \otimes_{A'} k(y')) = \mathrm{Spec}(B_\alpha \otimes_k k(y'))$ and V is its inverse image by the morphism $v : \mathrm{Spec}(B \otimes_k k(y')) \rightarrow \mathrm{Spec}(B_\alpha \otimes_k k(y'))$. Since B_α is a subalgebra of B and $k(y')$ is flat over the field k , the homomorphism $B_\alpha \otimes_k k(y') \rightarrow B \otimes_k k(y')$ is injective. Thus v is dominant (I, 1.2.7), which shows that V is nonempty. $\square$

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Corollary (2.4.10). — Let k be a field, X, Y two k -preschemes; then the morphism projection $X \times_k Y \rightarrow X$ is universally open. In particular, for every extension K of k , the morphism projection $X_{(K)} \rightarrow X$ is universally open.

Proof. It suffices to apply (2.4.9) to the structural morphism $Y \rightarrow \text{Spec}(k)$. $\square$

Remark (2.4.11). — If $f : X \rightarrow Y$ is an open morphism, then for every subset E of Y we have $f^{-1}(\overline{E}) = \overline{f^{-1}(E)}$ (Bourbaki, *Top. gén.*, chap. I, 3^e éd., § 5, n^o 4, prop. 7). This remark applies for example when f is a morphism flat locally of finite presentation (2.4.6), or a morphism projection $X \times_k Y \rightarrow X$ where X, Y are preschemes over a field k (2.4.10), and generalises then (2.3.10).

2.5. Permanence of properties of Modules by faithfully flat descent

Proposition (2.5.1). — Let $f : X \rightarrow Y$ be a morphism, let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$ -Module, and let $g : Y' \rightarrow Y$ be faithfully flat. Set

$$X' = X \times_Y Y', \quad f' = f_{(Y')} : X' \rightarrow Y',$$

let $g' : X' \rightarrow X$ be the canonical projection, and set $\mathcal{F}' = g'^*(\mathcal{F})$. Then $\mathcal{F}$ is flat (resp. faithfully flat) relative to f if and only if $\mathcal{F}'$ is flat (resp. faithfully flat) relative to f' .

Proof. Apply (2.2.10), replacing its $X, Y', Z, \mathcal{F}$, and g' respectively by $Y', X, Y, \mathcal{O}_{Y'}$, and $\mathcal{F}$. This identifies the required flatness (resp. faithful flatness) condition after base change with the corresponding condition relative to $g \circ f'$. On the other hand, g' is faithfully flat (2.2.13) and

$$g \circ f' = f \circ g'.$$

The assertion therefore follows from (2.2.11, iii). $\square$

Proposition (2.5.2). — Let $f : X' \rightarrow X$ be a morphism faithfully flat and quasi-compact, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module, and $\mathcal{F}' = f^*(\mathcal{F})$. Consider, for a quasi-coherent Module, the property of being:

- (i) of finite type;
- (ii) of finite presentation;
- (iii) locally free of finite type;
- (iv) locally free of rank n .

If **P** denotes any one of these properties, then $\mathcal{F}$ has property **P** if and only if $\mathcal{F}'$ does.

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Proof. For a quasi-coherent $\mathcal{O}_X$ -Module to be locally free of finite type, it is necessary and sufficient that it be flat over X and of finite presentation (Bourbaki, *Alg. comm.*, chap. II, § 5, no. 2, cor. 2 of th. 1, taking (2.1.2) into account). By (2.5.1), applied with the identity morphism for f , $\mathcal{F}$ is flat over X if and only if $\mathcal{F}'$ is flat over X' . Thus case (iii) follows from cases (i) and (ii). The same reduction proves (iv), since $f^*(\mathcal{O}_X^n) = \mathcal{O}_{X'}^n$: if $\mathcal{F}$ and $\mathcal{F}'$ are locally free of finite type and $x = f(x')$, their ranks at x and x' are equal, and f is surjective.

To treat cases (i) and (ii), note that the question is local on X , so we may suppose that X is affine. By (2.2.12), there is an affine scheme X'' and a faithfully flat local isomorphism $g : X'' \rightarrow X'$. Consequently, it is equivalent for $f^*(\mathcal{F})$ or $g^*f^*(\mathcal{F})$ to have property **P**. We are therefore reduced to

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$$X = \text{Spec}(A), \quad X' = \text{Spec}(A').$$

In view of (2.2.3) and (II, 6.1.4.1), it remains only to prove the following lemma. $\square$

Lemma (2.5.3). — Let A be a ring, let A' be a faithfully flat A -algebra, and let M be an A -module. Set $M' = M \otimes_A A'$. Then M is of finite type (resp. of finite presentation) if and only if M' is.

Proof. For the proof, see Bourbaki, *Alg. comm.*, chap. I^{er}, § 3, no. 6, prop. 11. $\square$

Remark (2.5.4). — The assertions of (2.5.2) for properties (i) and (ii) remain valid if f is assumed only quasi-faithfully flat (2.3.3) and quasi-compact. Indeed, one is reduced (Bourbaki, *loc. cit.*) to the following lemma.

Lemma (2.5.4.1). — *Let $\rho : A \rightarrow A'$ be a homomorphism of rings such that the corresponding morphism $f : \operatorname{Spec}(A') \rightarrow \operatorname{Spec}(A)$ is surjective. Suppose that there is an A' -module N' of finite type which is flat over A and has support $\operatorname{Spec}(A')$. If $u : P \rightarrow Q$ is a homomorphism of A -modules such that*

$$u \otimes 1 : P \otimes_A A' \longrightarrow Q \otimes_A A'$$

is surjective, then u is surjective.

Proof. Tensoring the assumed surjection with N' first shows that

$$u \otimes 1_{N'} : (P \otimes_A A') \otimes_{A'} N' \longrightarrow (Q \otimes_A A') \otimes_{A'} N'$$

is surjective. Let $\mathfrak{q}$ be a prime ideal of A' and put $\mathfrak{p} = \rho^{-1}(\mathfrak{q})$. After localization, the corresponding map is

$$u_{\mathfrak{p}} \otimes 1 : P_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} N'_{\mathfrak{q}} \longrightarrow Q_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} N'_{\mathfrak{q}}$$

(0, 1.5.4). By the support hypothesis $N'_{\mathfrak{q}} \neq 0$, and $N'_{\mathfrak{q}}$ is flat over $A_{\mathfrak{p}}$ (0, 6.3.1). Nakayama's lemma gives $\mathfrak{q}N'_{\mathfrak{q}} \neq N'_{\mathfrak{q}}$, hence a fortiori $\mathfrak{p}N'_{\mathfrak{q}} \neq N'_{\mathfrak{q}}$. Thus $N'_{\mathfrak{q}}$ is faithfully flat over $A_{\mathfrak{p}}$ (0, 6.4.1), and $u_{\mathfrak{p}}$ is surjective. Since f is surjective, every $\mathfrak{p} \in \operatorname{Spec}(A)$ occurs in this way; therefore u is surjective (Bourbaki, *Alg. comm.*, chap. II, § 3, no. 3, th. 1). $\square$

Proposition (2.5.5). — *Let $f : X \rightarrow Y$ be a morphism, let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$ -Module of finite type which is flat relative to f , and let $\mathcal{G}$ be a quasi-coherent $\mathcal{O}_Y$ -Module of finite type. For $y \in Y$, set*

$$\mathcal{F}_y = \mathcal{F} \otimes_{\mathcal{O}_Y} k(y).$$

A point $x \in X$ is a maximal point of $\operatorname{Supp}(\mathcal{F} \otimes_Y \mathcal{G})$ if and only if $y = f(x)$ is a maximal point of $\operatorname{Supp}(\mathcal{G})$ and x is a maximal point of $\operatorname{Supp}(\mathcal{F}_y)$ in $f^{-1}(y)$. When this holds, we have

(2.5.5.1).

$$\operatorname{length}((\mathcal{F} \otimes_Y \mathcal{G})_x) = \operatorname{length}(\mathcal{G}_y) \operatorname{length}((\mathcal{F}_y)_x).$$

Proof. It is clear that

$$f(\operatorname{Supp}(\mathcal{F} \otimes_Y \mathcal{G})) \subset \operatorname{Supp}(\mathcal{G})$$

(0, 5.2.2); hence the image under f of every irreducible component of $\operatorname{Supp}(\mathcal{F} \otimes_Y \mathcal{G})$ is contained in an irreducible component of $\operatorname{Supp}(\mathcal{G})$.

We may reduce to the case $\operatorname{Supp}(\mathcal{G}) = Y$. Indeed (Err_{III}, 30), since $\mathcal{G}$ is of finite type, there is a closed subscheme $Y' \subset Y$ with underlying space $\operatorname{Supp}(\mathcal{G})$ and a quasi-coherent $\mathcal{O}_{Y'}$ -Module $\mathcal{G}_1$ of finite type such that, for the canonical injection $j : Y' \rightarrow Y$,

$$\mathcal{G} = j_*(\mathcal{G}_1).$$

Set $X' = X \times_Y Y'$, let $f' : X' \rightarrow Y'$ be the projection, and write $\mathcal{F}' = \mathcal{F} \otimes_Y \mathcal{O}_{Y'}$. Then $\mathcal{F}'$ is flat relative to f' , and, under the canonical closed immersion $X' \rightarrow X$,

$$\operatorname{Supp}(\mathcal{F}' \otimes_{Y'} \mathcal{G}_1) = \operatorname{Supp}(\mathcal{F} \otimes_Y \mathcal{G}).$$

Suppose therefore that $\operatorname{Supp}(\mathcal{G}) = Y$. If Z is an irreducible component of $\operatorname{Supp}(\mathcal{G})$, then

$$f^{-1}(Z) \subset \operatorname{Supp}(\mathcal{F} \otimes_Y \mathcal{G})$$

(I, 9.1.13); and (2.3.4) shows that every irreducible component of $f^{-1}(Z)$ dominates Z . Thus, if x is the generic point of an irreducible component of $\operatorname{Supp}(\mathcal{F} \otimes_Y \mathcal{G})$ contained in $f^{-1}(Z)$, then $y = f(x)$ is the generic point of Z . Moreover, by (0, 2.1.8), x is the generic point of an irreducible component of

$$f^{-1}(y) = \operatorname{Supp}(\mathcal{F}_y)$$

(I, 9.1.13); conversely, every generic point of one of these components is the generic point of an irreducible component of $f^{-1}(Z)$.

It remains to prove (2.5.5.1). We have

$$(\mathcal{F} \otimes_Y \mathcal{G})_x = \mathcal{F}_x \otimes_{\mathcal{O}_{Y,y}} \mathcal{G}_y$$

(I, 9.1.12) and $(\mathcal{F}_y)_x = \mathcal{F}_x / \mathfrak{m}_y \mathcal{F}_x$; hence we are reduced to the following lemma. $\square$

Lemma (2.5.5.2). — *Let A and B be local rings, $\rho : A \rightarrow B$ a local homomorphism, and $\mathfrak{m}$ the maximal ideal of A . Let M be an A -module and N a B -module which is faithfully flat over A and such that $N/\mathfrak{m}N$ has finite length as a B -module. Then*

(2.5.5.3).

$$\text{length}_B(M \otimes_A N) = \text{length}_A(M) \cdot \text{length}_B(N/\mathfrak{m}N).$$

Proof. If M has infinite length, then so does $M \otimes_A N$. Indeed, every strictly increasing chain of n submodules M_i of M ($1 \leq i \leq n$) yields pairwise distinct sub- B -modules $M_i \otimes_A N$ of $M \otimes_A N$. Since $N \neq \mathfrak{m}N$, formula (2.5.5.3) holds in this case.

Suppose that M has finite length. If $M = 0$, both sides of (2.5.5.3) vanish, so assume $M \neq 0$. Each $\mathfrak{m}^k M$ has finite length, and if $\mathfrak{m}^k M \neq 0$, Nakayama's lemma gives $\mathfrak{m}^{k+1} M \neq \mathfrak{m}^k M$. Hence some r satisfies $\mathfrak{m}^r M = 0$. The modules $\mathfrak{m}^k M \otimes_A N$ identify with sub- B -modules of $M \otimes_A N$, and

$$\frac{\mathfrak{m}^k M \otimes_A N}{\mathfrak{m}^{k+1} M \otimes_A N} \simeq \frac{\mathfrak{m}^k M}{\mathfrak{m}^{k+1} M} \otimes_{A/\mathfrak{m}} \frac{N}{\mathfrak{m}N}$$

for $0 \leq k \leq r-1$. Its length as a B -module is the product of $\text{length}_B(N/\mathfrak{m}N)$ and the dimension of the $(A/\mathfrak{m})$ -vector space $\mathfrak{m}^k M / \mathfrak{m}^{k+1} M$; the latter dimension is the length of that quotient as an A -module. Summing over $0 \leq k \leq r-1$ gives (2.5.5.3). $\square$

Remark (2.5.5.4). — When N is an A -module of finite type, it is faithfully flat over A if and only if it is flat and $N \neq 0$; indeed, Nakayama's lemma then shows that $\mathfrak{m}N \neq N$.

Lemma (2.5.6). — Let B be a not necessarily commutative ring, let V and W be isomorphic left B -modules, set

$$C = \text{End}_B(V), \quad M = \text{Hom}_B(V, W),$$

and equip M with its canonical right C -module structure. Then M is isomorphic to the right regular module C_d . Moreover, for $u \in M$, the following conditions are equivalent:

- a) $\{u\}$ is a basis of the C -module M ;
- b) u is an isomorphism from V onto W .

Proof. If u is an isomorphism from V onto W , then $v \mapsto u \circ v$ is a bijection from C to M , so b) implies a). Conversely, suppose that $\{u\}$ is a basis of M . By hypothesis there is an isomorphism $u' : V \rightarrow W$, and $\{u'\}$ is also a basis of M . Hence $u = u' \circ w$ for some invertible $w \in C$, that is, for some automorphism w of V . Thus u is an isomorphism. $\square$

Corollary (2.5.7). — Under the hypotheses of (2.5.6), suppose in addition that one of the following conditions holds:

- (i) V and W are Noetherian B -modules;
- (ii) V and W are modules of finite presentation over a commutative subring of B .

Then conditions a) and b) of (2.5.6) are also equivalent to:

- a') u is a generator of the C -module M ;
- b') u is an epimorphism of V onto W .

Proof. An epimorphism from an A -module E onto itself is bijective in either of the following cases: 1° E is Noetherian (Bourbaki, Alg., chap. VIII, § 2, no. 2, lemma 3); 2° A is commutative and E is of finite presentation (8.9.3)². Therefore b) and $b')$ are equivalent. If u generates M and $\{u'\}$ is a basis, there is $v \in C$ such that $u' = u \circ v$, so u is surjective. Hence $a')$ implies $b')$; and a) plainly implies $a')$. This proves the corollary. $\square$

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Proposition (2.5.8). — Let A be a commutative semilocal ring, let B be an A -algebra (not necessarily commutative), and let V and W be B -modules. Let A' be a commutative A -algebra which is faithfully flat as an A -module, and set

$$B' = B \otimes_A A', \quad V' = V \otimes_A A', \quad W' = W \otimes_A A'.$$

Thus B' is an A' -algebra and V', W' are B' -modules. Suppose in addition that one of the following conditions holds:

- (i) A and A' are Noetherian, and V and W are A -modules of finite type;

²The reader may verify that (2.5.7) and (2.5.8) are not used before § 9.

(ii) B is an A -module of finite type, and V is both a projective B -module of finite type and an A -module of finite presentation.

If V' and W' are isomorphic as B' -modules, then V and W are isomorphic as B -modules.

Proof. In case (ii), W' , being isomorphic to V' , is an A' -module of finite type. It follows that W is an A -module of finite type (Bourbaki, *Alg. comm.*, chap. I^{er}, § 3, no. 6, prop. 11). Thus in both cases V and W are A -modules of finite type.

(2.5.8.1). Under either hypothesis (i) or (ii), $\text{Hom}_B(V, W)$ is an A -module of finite type.

This is immediate in case (i), because $\text{Hom}_A(V, W)$ is then an A -module of finite type and $\text{Hom}_B(V, W)$ is an A -submodule. In case (ii), V is a direct factor of a free B -module B^n . Consequently, $\text{Hom}_B(V, W)$ is a direct factor of

$$\text{Hom}_B(B^n, W) = W^n.$$

It is therefore of finite type over A .

Set

$$C = \text{End}_B(V), \quad M = \text{Hom}_B(V, W).$$

These are A -modules of finite type in both cases. Under either hypothesis, the canonical homomorphism

(2.5.8.2).

$$\text{Hom}_A(V, W) \otimes_A A' \longrightarrow \text{Hom}_{A'}(V', W')$$

is bijective (Bourbaki, *Alg. comm.*, chap. II, § 2, no. 10, prop. 11). Since A' is flat over A ,

$$\text{Hom}_B(V, W) \otimes_A A'$$

identifies canonically with a sub- A' -module of $\text{Hom}_A(V, W) \otimes_A A'$. Its image under (2.5.8.2) lies in $\text{Hom}_{B'}(V', W')$: if $u \in \text{Hom}_B(V, W)$ and $a' \in A'$, the image of $u \otimes a'$ is the homomorphism $u' : V' \rightarrow W'$ satisfying $u'(x \otimes 1) = u(x) \otimes a'$, and

$$u'((b \otimes 1)(x \otimes 1)) = u(bx) \otimes a' = bu(x) \otimes a' = (b \otimes 1)(u(x) \otimes a').$$

(2.5.8.3). Under either hypothesis (i) or (ii), the homomorphism

(2.5.8.4).

$$\text{Hom}_B(V, W) \otimes_A A' \longrightarrow \text{Hom}_{B'}(V', W')$$

is bijective.

For $b \in B$, let $h(b)$ (resp. $h'(b)$) denote the A -endomorphism $x \mapsto bx$ of V (resp. W). Let $(b_\alpha)_{\alpha \in I}$ be a system of generators of the A -algebra B . The map

$$u \longmapsto (h'(b_\alpha) \circ u - u \circ h(b_\alpha))_{\alpha \in I}$$

from $\text{Hom}_A(V, W)$ to $(\text{Hom}_A(V, W))^I$ is A -linear, and its kernel is precisely $\text{Hom}_B(V, W)$. Thus

$$0 \longrightarrow \text{Hom}_B(V, W) \longrightarrow \text{Hom}_A(V, W) \longrightarrow (\text{Hom}_A(V, W))^I.$$

The same reasoning applies after replacing A, B, V, W by A', B', V', W' . Moreover, there is a commutative diagram

(2.5.8.5).

$$\begin{array}{ccccccc} 0 & \longrightarrow & \text{Hom}_B(V, W) \otimes_A A' & \longrightarrow & \text{Hom}_A(V, W) \otimes_A A' & \longrightarrow & (\text{Hom}_A(V, W))^I \otimes_A A' \\ & & \downarrow r & & \downarrow s & & \downarrow t \\ 0 & \longrightarrow & \text{Hom}_{B'}(V', W') & \longrightarrow & \text{Hom}_{A'}(V', W') & \longrightarrow & (\text{Hom}_{A'}(V', W'))^I \end{array}$$

Here r is (2.5.8.4), s is (2.5.8.2), and t is the composite

$$(\text{Hom}_A(V, W))^I \otimes_A A' \xrightarrow{w} (\text{Hom}_A(V, W) \otimes_A A')^I \xrightarrow{s^I} (\text{Hom}_{A'}(V', W'))^I$$

where w is the canonical homomorphism (Bourbaki, *Alg.*, chap. II, 3^e ed., § 3, no. 7). Since A' is flat over A , the rows of (2.5.8.5) are exact.

The map s is an isomorphism, hence so is s^I . In case (ii) we may take I finite, and then w is bijective (Bourbaki, *loc. cit.*, prop. 7). In case (i), let B_1 (resp. B_2) be the image of B under h (resp. h') in $\text{End}_A(V)$ (resp. $\text{End}_A(W)$). Both images are A -modules of finite type, so here too I may be chosen finite. Thus t , and consequently r , is bijective in both cases.

(2.5.8.6). It follows from (2.5.8.4) that, if

$$C' = C \otimes_A A', \quad M' = M \otimes_A A'$$

then there are canonical bijections

$$C' \xrightarrow{\sim} \text{End}_{B'}(V'), \quad M' \xrightarrow{\sim} \text{Hom}_{B'}(V', W')$$

the first an isomorphism of A' -algebras and the second, through the first, an isomorphism of right C' -modules.

(2.5.8.7). *Reduction to the case $V = B_s$.* The hypothesis that V' and W' are isomorphic B' -modules implies that C'_d and M' are isomorphic right C' -modules (2.5.6). To prove (2.5.8), it is enough to find $u \in M$ which generates M as a right C -module. Indeed, $u' = u \otimes 1$ then generates M' as a right C' -module. In case (i), V' and W' are Noetherian A' -modules; in case (ii), they are isomorphic A' -modules of finite presentation. Corollary (2.5.7) therefore shows that u' is a B' -isomorphism $V' \rightarrow W'$. Since A' is faithfully flat over A , u is bijective (0, 6.4.1).

In case (i), C and M are A -modules of finite type. In case (ii), C is a direct factor of V^n by (2.5.8.1), hence a projective A -module of finite type. After changing notation, we are therefore reduced to proving (2.5.8) when $V = B_s$; it is enough to prove that the B -module W has a generator. In this case B is an A -module of finite type.

(2.5.8.8). *Reduction to the case where A and A' are fields and B is a simple A -algebra with centre A .*

Let τ be the radical of the semilocal ring A . It is enough to prove that $W/\tau W$ is a cyclic $(B/\tau B)$ -module. Indeed, a surjection

$$B_s/\tau B_s \longrightarrow W/\tau W$$

composes with $B_s \rightarrow B_s/\tau B_s$ and, since B_s is free as a B -module, lifts to a map $u : B_s \rightarrow W$. The displayed surjection is $u \otimes 1$ after tensoring over A with A/τ . Since W is of finite type over A , Nakayama's lemma shows that u is surjective (Bourbaki, *Alg. comm.*, chap. II, § 3, no. 2, cor. 1 of prop. 4).

Put

$$\begin{aligned} A_1 &= A/\tau, & A'_1 &= A' \otimes_A A_1 = A'/\tau A', \\ B_1 &= B/\tau B = B \otimes_A A_1, & W_1 &= W/\tau W = W \otimes_A A_1. \end{aligned}$$

The hypotheses remain valid after replacing A , A' , B , $V = B_s$, and W by A_1 , A'_1 , B_1 , $V_1 = (B_1)_s$, and W_1 . Moreover, A'_1 is faithfully flat over A_1 , and the corresponding base changes of V_1 and W_1 remain isomorphic. We may therefore suppose that A is a finite product of fields.

The ring B is then Artinian. Let $\mathfrak{N}$ be its radical. It is enough, by the same Nakayama argument, to prove that $W/\mathfrak{N}W$ is cyclic over $B/\mathfrak{N}$. Since

$$\begin{aligned} B'/\mathfrak{N}B' &= (B/\mathfrak{N}) \otimes_A A', \\ W'/\mathfrak{N}W' &= (W/\mathfrak{N}W) \otimes_A A', \end{aligned}$$

and the latter is isomorphic to the right regular module over the former, we may further suppose that $\mathfrak{N} = 0$, that is, that B is a semisimple A -algebra.

Write $A = \prod_{i=1}^n k_i$. Then $A' = \prod_{i=1}^n A'_i$, where each A'_i is a k_i -algebra and is annihilated by the other factors k_j . Faithful flatness implies $A'_i \neq 0$. Choose a field quotient k'_i of each A'_i and set $A'' = \prod_i k'_i$. Then A'' is faithfully flat over A and is a quotient of A' . After the further base change,

$$B'' = B \otimes_A A'', \quad W'' = W \otimes_A A'',$$

the module W'' remains isomorphic to B''_s . Hence we may also suppose that A' is a finite product of fields.

Let Z be the centre of B . It is a finite product of fields and an A -module of finite type; B and W are finite, hence projective, Z -modules. Put $Z' = Z \otimes_A A'$. Then

$$B' = B \otimes_Z Z', \quad W' = W \otimes_Z Z',$$

and Z' is faithfully flat over Z . We may replace A by Z and thus suppose that A is the centre of B .

If k_i are the field factors of A , then $B = \prod_i B_i$, with B_i a simple ring of centre k_i , and $W = \bigoplus_i W_i$, where the other factors B_j annihilate W_i . We may suppose likewise that $A' = \prod_i k'_i$, where k'_i/k_i is a field extension. For every i ,

$$(B_i \otimes_{k_i} k'_i)_s \quad \text{and} \quad W_i \otimes_{k_i} k'_i$$

are isomorphic modules over $B_i \otimes_{k_i} k'_i$. It therefore suffices to treat $n = 1$: A and A' are fields, and B is a simple algebra with centre A .

(2.5.8.9). End of the proof. Every B -module is a direct sum of modules isomorphic to a minimal ideal of B (Bourbaki, *Alg.*, chap. VIII, § 5, props. 6 and 8). Thus two B -modules of finite rank over A are isomorphic if and only if they have the same rank over A . By hypothesis,

$$[W' : A'] = [B'_s : A'].$$

But

$$[W' : A'] = [W : A], \quad [B'_s : A'] = [B_s : A],$$

so $[W : A] = [B_s : A]$. This completes the proof. □

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2.6. Permanence of set-theoretic and topological properties of morphisms by faithfully flat descent

Proposition (2.6.1). — *Let $f : X \rightarrow Y$ be an S -morphism of S -preschemes, and let $g : S' \rightarrow S$ be a surjective morphism. Set*

$$X' = X_{(S')}, \quad Y' = Y_{(S')}, \quad f' = f_{(S')} : X' \rightarrow Y'.$$

Consider, for a morphism, the property of being:

- (i) *surjective;*
- (ii) *injective;*
- (iii) *with finite fibres (as sets);*
- (iv) *bijective;*
- (v) *radicial.*

If $\mathbf{P}$ denotes any one of these properties and f' has property $\mathbf{P}$, then so does f .

Proof. The projection $Y' \rightarrow Y$ is itself surjective (I, 3.5.2); hence, by (I, 3.3.11), we may reduce to the case $Y = S$ and $Y' = S'$. For $y \in Y$ (resp. $y' \in Y'$), denote by X_y (resp. $X'_{y'}$) the fibre prescheme $f^{-1}(y)$ (resp. $f'^{-1}(y')$) (I, 3.6.2). If $y' \in Y'$ and $y = g(y')$, there is a canonical isomorphism

$$X'_{y'} \xrightarrow{\sim} X_y \otimes_{k(y)} k(y')$$

(I, 3.6.4). Since $\text{Spec}(k(y')) \rightarrow \text{Spec}(k(y))$ is surjective, so is the projection $X'_{y'} \rightarrow X_y$ (I, 3.5.2). Therefore, if $X'_{y'}$ is nonempty (resp. has at most one point, resp. is a finite set), the same is true of X_y . Since $g : Y' \rightarrow Y$ is surjective, this proves the assertion in cases (i), (ii), and (iii); case (iv) follows from (i) and (ii).

Finally, for (v), it is enough to show that if f' is universally injective (I, 3.5.11), then so is f . Let $Y_1 \rightarrow Y$ be any morphism and set

$$X_1 = X \times_Y Y_1, \quad f_1 = f_{(Y_1)}.$$

Also set

$$\begin{aligned} Y'_1 &= Y_1 \times_Y Y', \\ X'_1 &= X' \times_{Y'} Y'_1 = X_1 \times_{Y_1} Y'_1, \\ f'_1 &= f'_{(Y'_1)} = (f_1)_{(Y'_1)} : X'_1 \rightarrow Y'_1. \end{aligned}$$

The projection $g_1 = g_{(Y_1)} : Y'_1 \rightarrow Y_1$ is surjective (I, 3.5.2); and, since f' is universally injective, f'_1 is injective. Case (ii) then shows that f_1 is injective, which proves the assertion. □

Proposition (2.6.2). — *With the notation of (2.6.1), suppose that $g : S' \rightarrow S$ is faithfully flat and quasi-compact. Consider, for a morphism, the property of being:*

- (i) open;
- (ii) closed;
- (iii) quasi-compact and a homeomorphism onto its image subspace;
- (iv) a homeomorphism.

If $\mathbf{P}$ denotes any one of these properties and f' has property $\mathbf{P}$, then so does f .

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Proof. The morphism $Y' \rightarrow Y$ is faithfully flat and quasi-compact (2.2.13) and (1.1.2); hence, by (I, 3.3.11), we may reduce to the case $Y = S$, $Y' = S'$. Let $g' : X' \rightarrow X$ be the projection. For every subset M of X we have

$$g^{-1}(f(M)) = f'(g'^{-1}(M))$$

(I, 3.4.8). If f' is open (resp. closed), then, for every open (resp. closed) subset M of X , the subset $f'(g'^{-1}(M))$ is open (resp. closed) in Y' . Since g is faithfully flat and quasi-compact, (2.3.12) implies that $f(M)$ is open (resp. closed) in Y . This proves cases (i) and (ii).

Let us prove (iii), which also yields (iv) in view of (2.6.1, (iv)). By (2.6.1, (ii)), f is injective; it remains to prove that f , considered as a map from X onto $f(X)$, is quasi-compact and open. Since f' is quasi-compact, so is f (1.1.4). It is therefore enough to prove, for every closed subset Z of X , that

$$Z = f^{-1}(\overline{f(Z)}).$$

Since g' is surjective, this is equivalent to

$$\begin{aligned} g'^{-1}(Z) &= g'^{-1}(f^{-1}(\overline{f(Z)})) \\ &= f'^{-1}(g^{-1}(\overline{f(Z)})). \end{aligned}$$

The composite of the canonical injection $Z \rightarrow X$ with f is quasi-compact, where Z denotes here the reduced closed subscheme of X with underlying space Z . Applying (2.3.10) to the subset $f(Z)$ of Y , the image of the morphism $f|_Z$, gives

$$g^{-1}(\overline{f(Z)}) = \overline{g^{-1}(f(Z))}.$$

If $Z' = g'^{-1}(Z)$, the required formula is consequently equivalent to

$$Z' = f'^{-1}(\overline{f'(Z')}),$$

which follows from the hypothesis that f' is a homeomorphism from X' onto $f'(X')$. $\square$

Remark (2.6.3). — In cases (i) and (ii), the conclusions of (2.6.2) remain valid if g is assumed only quasi-faithfully flat (2.3.3) and quasi-compact. Indeed, by (2.1.4), (I, 3.5.2), and (I, 9.1.13.1), we may again reduce to the case $Y = S$, $Y' = S'$, and the conclusion follows from (2.3.12). In cases (iii) and (iv), the conclusions remain valid if g is assumed only quasi-faithfully flat, provided that f is already assumed quasi-compact; the proof then uses only (2.3.10) and the surjectivity of g .

Finally, if g is faithfully flat and locally of finite presentation, or if g is surjective and S is discrete, the conclusion of (2.6.2) remains valid when $\mathbf{P}$ is the property

(iii bis) of being a homeomorphism onto its image subspace;
as follows from the proof of (2.6.2) and Remark (2.4.11).

Corollary (2.6.4). — With the notation of (2.6.1), suppose that $g : S' \rightarrow S$ is faithfully flat and quasi-compact. Consider, for a morphism, the property of being:

- (i) universally open;
- (ii) universally closed;
- (iii) quasi-compact and universally bicontinuous;
- (iv) a universal homeomorphism;
- (v) quasi-compact;
- (vi) quasi-compact and dominant.

If $\mathbf{P}$ denotes any one of these properties, then f has property $\mathbf{P}$ if and only if f' does.

Proof. Properties (v) and (vi) are only given for the record, being consequences of (1.1.4), (1.1.6), and (2.3.7). For the other properties, necessity follows from (2.4.3). Conversely, suppose, for example, that f' is universally open, and let $Y_1 \rightarrow Y$ be any morphism. Set

$$X_1 = X \times_Y Y_1, \quad f_1 = f_{(Y_1)}.$$

Also set

$$\begin{aligned} Y'_1 &= Y_1 \times_Y Y', \\ X'_1 &= X' \times_{Y'} Y'_1 = X_1 \times_{Y_1} Y'_1, \\ f'_1 &= f'_{(Y'_1)} = (f_1)_{(Y'_1)} : X'_1 \longrightarrow Y'_1. \end{aligned}$$

The projection $g_1 = g_{(Y_1)} : Y'_1 \rightarrow Y_1$ is faithfully flat and quasi-compact (2.2.13) and (1.1.2). Since f' is universally open, f'_1 is open, and (2.6.2) then shows that f_1 is open. Hence f is universally open. The same argument applies in the other cases.

One may again replace “faithfully flat” by “quasi-faithfully flat”. Moreover, when g is locally of finite presentation, or when g is surjective and S is discrete, property (iii) may be replaced by (iii bis) universally bicontinuous.

□

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2.7. Permanence of various properties of morphisms under faithfully flat descent

Proposition (2.7.1). — *Let $f : X \rightarrow Y$ be an S -morphism of S -preschemes, and let $g : S' \rightarrow S$ be faithfully flat and quasi-compact. Set*

$$X' = X_{(S')}, \quad Y' = Y_{(S')}, \quad f' = f_{(S')} : X' \longrightarrow Y'.$$

Consider, for a morphism, the property of being:

- (i) separated;
- (ii) quasi-separated;
- (iii) locally of finite type;
- (iv) locally of finite presentation;
- (v) of finite type;
- (vi) of finite presentation;
- (vii) proper;
- (viii) an isomorphism;
- (ix) a monomorphism;
- (x) an open immersion;
- (xi) a quasi-compact immersion;
- (xii) a closed immersion;
- (xiii) affine;
- (xiv) quasi-affine;
- (xv) finite;
- (xvi) quasi-finite;
- (xvii) integral.

If $\mathbf{P}$ denotes any one of these properties, then f has property $\mathbf{P}$ if and only if f' does.

Proof. It was proved in Chapters I and II and in Chapter IV, § 1, that if f has one of the properties $\mathbf{P}$ above, then so does f' (without any hypothesis on $g : S' \rightarrow S$). It remains to prove the converse. Since the projection $Y' \rightarrow Y$ is

a faithfully flat and quasi-compact morphism (2.2.13) and (1.1.2), we may, by (I, 3.3.11), reduce to **IV-2 | 30**
the case $S = Y, S' = Y'$.

(i) To say that f is separated means that its diagonal $\Delta_f : X \rightarrow X \times_Y X$ is closed. Since $\Delta_{f'} = (\Delta_f)_{(Y')}$ (I, 5.3.4), if $\Delta_{f'}$ is closed, then so is Δ_f by (2.6.2); hence f is separated.

(ii) This was already proved under weaker hypotheses (1.2.5).

(iii) and (iv). The question is plainly local on X and Y and, in view of (2.2.12), it is therefore enough to prove the following lemma.

Lemma (2.7.1.1). — *Let A be a ring, B an A -algebra, and A' an A -algebra that is faithfully flat as an A -module; put $B' = B \otimes_A A'$. Then B is an A -algebra of finite type (resp. of finite presentation) if and only if B' is an A' -algebra of finite type (resp. of finite presentation).*

Necessity is already known without any hypothesis on A' (1.3.4), (1.3.6), (1.4.3), and (1.4.6). Suppose that B' is an A' -algebra of finite type. Let $(B_\alpha)_{\alpha \in I}$ be the filtered increasing family of finite-type A -subalgebras of B . Then

$$B = \varinjlim_{\alpha} B_{\alpha}, \quad B' = \varinjlim_{\alpha} (B_{\alpha} \otimes_A A'),$$

because tensor product commutes with inductive limits. If (x'_i) is a finite set of generators of the A' -algebra B' , there is an index α such that every x'_i belongs to the subalgebra $B_{\alpha} \otimes_A A'$ of B' . Hence $B' = B_{\alpha} \otimes_A A'$, and faithful flatness gives $B = B_{\alpha}$ (0, 6.4.1).

Now suppose that B' is an A' -algebra of finite presentation. What precedes shows that B is an A -algebra of finite type. Thus there are a polynomial A -algebra $C = A[T_1, \dots, T_m]$ and a surjective A -homomorphism of algebras $C \rightarrow B$. Let $\mathfrak{J}$ be its kernel. We have an exact sequence

$$0 \longrightarrow \mathfrak{J} \longrightarrow C \longrightarrow B \longrightarrow 0,$$

and, since A' is flat over A , an exact sequence

$$0 \longrightarrow \mathfrak{J}' \longrightarrow C' \longrightarrow B' \longrightarrow 0,$$

where

$$C' = C \otimes_A A' = A'[T_1, \dots, T_m], \quad \mathfrak{J}' = \mathfrak{J} \otimes_A A'$$

and $\mathfrak{J}'$ is identified with an ideal of C' . Since B' is of finite presentation over A' , the C' -module $\mathfrak{J}'$ is of finite type (1.4.4). But $\mathfrak{J}' = \mathfrak{J} \otimes_C C'$, and C' is faithfully flat over C (2.2.13) and (2.2.3). It follows that $\mathfrak{J}$ is a C -module of finite type (Bourbaki, *Algèbre commutative*, Chap. I, § 3, no. 6, Prop. 11). Hence B is an A -algebra of finite presentation.

(v) This follows from (iii) and (2.6.4, (v)), by (1.5.2).

(vi) This follows similarly from (iv), (v), and (ii), by (1.6.1).

(vii) This follows from (i), (v), and (2.6.4, (ii)), by (II, 5.4.1).

(viii) Since f' is an isomorphism, it is a universal homeomorphism; hence so is f (2.6.4). In particular, f is quasi-compact and separated (2.4.4). Write $f = (\psi, \theta)$, where ψ is therefore a homeomorphism. It remains to prove that

$$\theta : \mathcal{O}_Y \longrightarrow f_*(\mathcal{O}_X)$$

is an isomorphism of $\mathcal{O}_Y$ -modules. If $f' = (\psi', \theta')$, then

$$\theta' : \mathcal{O}_{Y'} \longrightarrow f'_*(\mathcal{O}_{X'})$$

is the composite of $g^*(\theta)$ with the canonical homomorphism

$$g^*(f_*(\mathcal{O}_X)) \longrightarrow f'_*(\mathcal{O}_{X'})$$

(2.3.2). The latter is bijective by (2.3.1). Thus, if θ' is bijective, then so is $g^*(\theta)$; since g is faithfully flat, θ is bijective (2.2.7). This proves (viii).

(ix) This follows from (viii), (I, 5.3.4), and (I, 5.3.8), which reduces monomorphisms to isomorphisms.

(x) If f' is an open immersion, then $f'(X')$ is open in Y' , and

$$f'(X') = g^{-1}(f(X))$$

(I, 3.4.8). It follows from (2.3.12) that $f(X)$ is open. Replacing Y (resp. Y') by the subprescheme induced on $f(X)$ (resp. $f'(X')$), and using (1.1.2) and (2.2.13), we reduce to the case where f' is an isomorphism. Then f is an isomorphism by (viii), proving (x).

(xi) If f' is a quasi-compact immersion, then it is a quasi-compact and quasi-separated morphism (1.2.2); hence so is f , by (ii) and (2.6.4, (v)). Let Z be the closed subprescheme of Y that is the closed image of X under f (1.7.8), and write $f = j \circ q$, where $j : Z \rightarrow Y$ is the canonical injection. Then $f' = j' \circ q'$, with $j' = j_{(Y')}$ and $q' = q_{(Y')}$. Moreover, j' identifies with the canonical injection $Z' \rightarrow Y'$ of the closed image Z' of X' under f' (2.3.2). The hypothesis on f' now says that q' is an open immersion (I, 9.5.10); hence so is q , by (x), and therefore f is an immersion.

(xii) To say that f (resp. f') is a closed immersion is to say that it is a quasi-compact immersion and a closed morphism. Thus (xii) follows from (xi) and (2.6.2, (ii)).

(xiii) and (xiv). Suppose that f' is affine (resp. quasi-affine). Then f' is quasi-compact and quasi-separated (II, 5.1.1), and so is f by (ii) and (2.6.4, (v)). Put

$$\mathcal{A} = f_*(\mathcal{O}_X), \quad \mathcal{A}' = f'_*(\mathcal{O}_{X'}).$$

By (2.3.1), the canonical homomorphism of $\mathcal{O}_{Y'}$ -algebras $g^*(\mathcal{A}) \rightarrow \mathcal{A}'$ is bijective. Consequently, if $h : Z = \operatorname{Spec}(\mathcal{A}) \rightarrow Y$ is the structure morphism, then the structure morphism $h' : Z' = \operatorname{Spec}(\mathcal{A}') \rightarrow Y'$ identifies with $h_{(Y')}$ (II, 1.5.2). Let $u : X \rightarrow Z$ (resp. $u' : X' \rightarrow Z'$) be the canonical Y -morphism (resp. Y' -morphism) corresponding to the identity homomorphism of $\mathcal{A}$ (resp. $\mathcal{A}'$) (II, 1.2.7). We have a commutative diagram

$$\begin{array}{ccc} X & \longleftarrow & X' \\ u \downarrow & & \downarrow u' \\ Z & \xleftarrow{g'} & Z' \\ h \downarrow & & \downarrow h' \\ Y & \xleftarrow{g} & Y' \end{array}$$

and $h' \circ u' = f'$. It follows from (II, 1.2.7) that $u' = u_{(Z')}$. The projection $g' : Z' \rightarrow Z$ is faithfully flat and quasi-compact (1.1.2) and (2.2.13). The hypothesis on f' says that u' is an isomorphism (resp. an open immersion) (II, 5.1.6); by (viii) (resp. (x)), u is an isomorphism (resp. an open immersion). This proves (xiii) (resp. (xiv)).

(xv) If f' is finite, it is affine; hence f is affine by (xiii). With the notation used there, $\mathcal{A}'$ is an $\mathcal{O}_{Y'}$ -module of finite type and is isomorphic to $g^*(\mathcal{A})$. It follows from (2.5.2) that $\mathcal{A}$ is an $\mathcal{O}_Y$ -module of finite type, and therefore f is finite. IV-2 | 32

(xvi) To say that f is quasi-finite is to say that f is of finite type and that $f^{-1}(y)$ is finite for every $y \in Y$ (II, 6.2.2) and (I, 6.4.4). The conclusion therefore follows from (v) and (xv), applied to the fibres.

(xvii) As in (xv), one first sees that f is affine. We may reduce to the case

$$Y = \operatorname{Spec}(A), \quad Y' = \operatorname{Spec}(A'), \quad X = \operatorname{Spec}(B), \quad X' = \operatorname{Spec}(B'),$$

where $B' = B \otimes_A A'$. The ring B is the inductive limit of its finite-type A -subalgebras B_α ; hence

$$B' = \varinjlim_{\alpha} B'_{\alpha}, \quad B'_{\alpha} = B_{\alpha} \otimes_A A',$$

and each B'_{α} is a finite-type A' -algebra. By hypothesis, B' is integral over A' , so each B'_{α} is an A' -module of finite type. Hence B_{α} is an A -module of finite type (2.5.2), and is therefore integral over A . Since every element of B lies in some B_{α} , the algebra B is integral over A . $\square$

Corollary (2.7.2). — *Under the hypotheses and notation of (2.7.1), suppose that f is quasi-compact. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$ -module and, if $g' : X' \rightarrow X$ is the projection, let $\mathcal{L}' = g'^*(\mathcal{L})$ be its inverse image. Then $\mathcal{L}$ is ample (resp. very ample) for f if and only if $\mathcal{L}'$ is ample (resp. very ample) for f' .*

Proof. Necessity holds without any hypothesis on $g : S' \rightarrow S$ (II, 4.4.10) and (II, 4.6.13). To prove sufficiency, we may, as in (2.7.1), reduce to the case $S = Y, S' = Y'$. The hypothesis on $\mathcal{L}'$ implies that f' is quasi-compact and separated (II, 4.6.1); hence so is f , by (2.6.4, (v)) and (2.7.1, (i)). Put

$$\mathcal{E} = f_*(\mathcal{L}), \quad \mathcal{E}' = f'_*(\mathcal{L}').$$

By (2.3.1), the canonical homomorphism

$$u : g^*(\mathcal{E}) \longrightarrow \mathcal{E}'$$

is bijective. Suppose first that $\mathcal{L}'$ is very ample for f' . The canonical homomorphism $\sigma' : f'^*(\mathcal{E}') \rightarrow \mathcal{L}'$ is then surjective, and

$$r' = r_{\mathcal{L}', \sigma'} : X' \longrightarrow \mathbf{P}(\mathcal{E}')$$

is a necessarily quasi-compact immersion (II, 4.4.4, (b)) and (1.1.2, (v)). If

$$h : \mathbf{P}(\mathcal{E}) \longrightarrow Y, \quad h' : \mathbf{P}(\mathcal{E}') \longrightarrow Y'$$

are the structure morphisms, the bijectivity of u implies that h' identifies with $h_{(Y')}$ (II, 4.1.3). On the other hand, the projection $g' : X' \rightarrow X$ is faithfully flat (2.2.13), and $f \circ g' = g \circ f'$. Let $\sigma : f^*(\mathcal{E}) \rightarrow \mathcal{L}$ be the canonical homomorphism. One readily verifies that $g'^*(\sigma)$ is the composite

$$f'^*(g^*(f_*(\mathcal{L}))) \xrightarrow{f'^*(u)} f'^*(f'_*(\mathcal{L}')) \xrightarrow{\sigma'} \mathcal{L}'.$$

This may be checked, for example, by reducing to the case where Y and Y' are affine. Since σ' is surjective and $f'^*(u)$ is bijective, $g'^*(\sigma)$ is surjective; hence σ is surjective by (2.2.7). Thus the morphism

$$r = r_{\mathcal{L}, \sigma} : X \rightarrow \mathbf{P}(\mathcal{E})$$

is everywhere defined (II, 3.7.4). Put $P = \mathbf{P}(\mathcal{E})$ and $P' = \mathbf{P}(\mathcal{E}')$, and let $g'' : P' \rightarrow P$ be the projection. Then r' identifies with $r_{(P')}$ (II, 4.2.10), while g'' is faithfully flat and quasi-compact (1.1.2) and (2.2.13). By (2.7.1, (xi)), r is an immersion, and therefore $\mathcal{L}$ is very ample (II, 4.4.4, (b)).

Now suppose that $\mathcal{L}'$ is ample for f' . To prove that $\mathcal{L}$ is ample for f , we may reduce to the case where Y is affine (II, 4.6.4); by (2.2.12) and (II, 4.6.13), we may also suppose that Y' is affine. Then X and X' are quasi-compact schemes, and we may apply the criterion of (II, 4.6.8, (c)). Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$ -module of finite type, and let

$$\sigma : f^*f_*(\mathcal{F} \otimes \mathcal{L}^{\otimes n}) \rightarrow \mathcal{F} \otimes \mathcal{L}^{\otimes n}$$

be the canonical homomorphism. As above, $g'^*(\sigma)$ is the composite

$$f'^*(g^*f_*(\mathcal{F} \otimes \mathcal{L}^{\otimes n})) \xrightarrow{f'^*(u)} f'^*f'_*(\mathcal{F}' \otimes \mathcal{L}'^{\otimes n}) \xrightarrow{\sigma'} \mathcal{F}' \otimes \mathcal{L}'^{\otimes n},$$

where $\mathcal{F}' = g'^*(\mathcal{F})$, using (0, 4.3.3.1), and where u is the canonical homomorphism

$$g^*f_*(\mathcal{F} \otimes \mathcal{L}^{\otimes n}) \rightarrow f'_*(\mathcal{F}' \otimes \mathcal{L}'^{\otimes n}).$$

The homomorphism u is bijective for every n by (2.3.1). Moreover, $\mathcal{F}'$ is quasi-coherent and of finite type, so the ampleness of $\mathcal{L}'$ for f' gives an n_0 such that σ' is surjective for $n \geq n_0$. Thus $g'^*(\sigma)$ is surjective for $n \geq n_0$; since g' is faithfully flat, σ is surjective for these values of n (2.2.7). This completes the proof. $\square$

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- Remarks (2.7.3).** — (i) It follows from (2.6.1), (2.6.4), and (2.5.4.1) that the conclusions of (2.7.1) remain valid in cases (i), (iii), (v), (vii), and (xvi) when g is assumed only quasi-compact and *quasi-faithfully flat* (2.3.3). We already noted that case (ii) of (2.7.1) is valid under the sole assumptions that g is surjective and quasi-compact.
- (ii) Under the hypotheses and notation of (2.7.1), it can happen that f is proper and f' is projective while f is not quasi-projective. Indeed, Hironaka [34] gave an example of a proper nonprojective morphism $f : X \rightarrow Y$, where X and Y are regular algebraic schemes (0, 4.1.4) over the same field k , with Y projective over k . Moreover, Y is the union of two affine open subsets Y_i ($i = 1, 2$) such that

$$f_i : X \times_Y Y_i \rightarrow Y_i$$

is projective for $i = 1, 2$. Let $Y' = Y_1 \amalg Y_2$ be their coproduct as preschemes. The canonical morphism $g : Y' \rightarrow Y$, whose restrictions are the canonical injections of Y_1 and Y_2 into Y , is faithfully flat and is quasi-compact by (I, 5.5.10). Yet $f' : X \times_Y Y' \rightarrow Y'$, whose restriction to each Y_i is f_i , is projective (II, 5.5.6), whereas f is not. Consequently there exists an invertible $\mathcal{O}_{X'}$ -module $\mathcal{L}'$ that is f' -ample but is not of the form $g'^*(\mathcal{L})$ for an invertible $\mathcal{O}_X$ -module $\mathcal{L}$, where $g' : X' \rightarrow X$ is the projection, by (2.7.2).

- (iii) Under the hypotheses of (2.7.1), it can happen that f' is a local isomorphism while f is not a local immersion. Indeed, let k be a field, let $\bar{k}$ be an algebraic closure of k , and let $K \neq k$ be a finite separable extension of k . The structure morphism $f : X \rightarrow Y$, where $X = \text{Spec}(K)$ and $Y = \text{Spec}(k)$, is not a local immersion. But if $Y' = \text{Spec}(\bar{k})$, then $Y' \rightarrow Y$ is faithfully flat and quasi-compact, and f' is a local isomorphism, since $X' = X \times_Y Y'$ is a coproduct of finitely many schemes isomorphic to Y' .

2.8. Preschemes over a regular base of dimension 1; closure of a closed subscheme of the generic fibre

Proposition (2.8.1). — Let Y be a locally Noetherian, regular, irreducible prescheme of dimension 1, with generic point η . Let $f : X \rightarrow Y$ be a morphism, let $X_\eta = f^{-1}(\eta)$ be the fibre at the generic point, and let $i : X_\eta \rightarrow X$ be the canonical morphism. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$ -module, put $\mathcal{F}_\eta = i^*(\mathcal{F})$, let $\mathcal{G}'$ be an $\mathcal{O}_{X_\eta}$ -module quotient of $\mathcal{F}_\eta$, and let $\overline{\mathcal{G}'}$ be the image of $\mathcal{F}$ under the composite homomorphism (0, 4.4.3.2)

$$\mathcal{F} \xrightarrow{\rho_{\mathcal{F}}} i_*(i^*(\mathcal{F})) \longrightarrow i_*(\mathcal{G}').$$

Then $\overline{\mathcal{G}'}$ is a quasi-coherent $\mathcal{O}_X$ -module quotient of $\mathcal{F}$, flat over Y , such that

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$$i^*(\overline{\mathcal{G}'}) = \mathcal{G}',$$

and it is the unique $\mathcal{O}_X$ -module quotient of $\mathcal{F}$ having these properties.

Proof. Since i is quasi-compact and quasi-separated (1.1.2) and (1.2.2), it follows from (1.7.4) that, for every quasi-coherent $\mathcal{O}_{X_\eta}$ -module $\mathcal{H}'$, the $\mathcal{O}_X$ -module $i_*(\mathcal{H}')$ is quasi-coherent. Moreover, for every open subset U of X , by definition (0, 3.4.1),

$$i_*(\mathcal{H}')|U = (i|U \cap X_\eta)_*(\mathcal{H}'|U \cap X_\eta).$$

If the proposition is proved when X and Y are affine, the general case follows by gluing, in view of the uniqueness assertion in the affine case. In other words, we are reduced to proving the following lemma.

Lemma (2.8.1.1). — Let A be a Noetherian regular integral ring of dimension 1 (0, 17.3.6), let K be its field of fractions, let M be an A -module, and let N' be a K -module quotient of

$$M_{(K)} = M \otimes_A K$$

by a sub- K -module P' . Let $\overline{N'}$ be the image of M under the composite homomorphism

$$M \longrightarrow M_{(K)} \longrightarrow N'.$$

Then $\overline{N'}$ is a flat A -module, and it is the unique quotient module N of M that is flat over A and for which the kernel of the surjective homomorphism

$$M_{(K)} \longrightarrow N_{(K)}$$

is equal to P' .

For every maximal ideal $\mathfrak{m}$ of A , the ring $A_{\mathfrak{m}}$ is a regular local ring of dimension 1, hence a discrete valuation ring. Therefore an A -module N is flat if and only if it is torsion-free (0, 6.3.4). Since N' is a K -vector space, it is torsion-free as an A -module, and the same is therefore true of its submodule $\overline{N'}$. Moreover, one immediately verifies that $\overline{N'}_{(K)}$ identifies with N' .

Conversely, let N be an A -module quotient of M having the properties in the statement. Since N is flat over A , the canonical homomorphism

$$N \longrightarrow N_{(K)} = N \otimes_A K$$

is injective. Since $N_{(K)}$ identifies with N' , the conclusion follows from the commutativity of the diagram

$$\begin{array}{ccc} M & \longrightarrow & N \\ \downarrow & & \downarrow \\ M_{(K)} & \longrightarrow & N_{(K)}. \end{array}$$

□

Corollary (2.8.2). — Under the conditions of (2.8.1), the module $\mathcal{F}$ is flat over Y if and only if the canonical homomorphism

$$\mathcal{F} \longrightarrow i_*(\mathcal{F}_\eta) = i_*(i^*(\mathcal{F}))$$

is injective.

(2.8.3). The formation of the $\mathcal{O}_X$ -module $\overline{\mathcal{G}}'$ is *functorial* in $\mathcal{F}$ and $\mathcal{G}'$. More precisely, let $\mathcal{F}_1$ and $\mathcal{F}_2$ be two quasi-coherent $\mathcal{O}_X$ -modules, let $u : \mathcal{F}_1 \rightarrow \mathcal{F}_2$ be an $\mathcal{O}_X$ -homomorphism, let $\mathcal{G}'_i$ be an $\mathcal{O}_{X_\eta}$ -module quotient of $(\mathcal{F}_i)_\eta$ for $i = 1, 2$, and let $v : \mathcal{G}'_1 \rightarrow \mathcal{G}'_2$ be a homomorphism making the diagram

$$\begin{array}{ccc} (\mathcal{F}_1)_\eta & \xrightarrow{i^*(u)} & (\mathcal{F}_2)_\eta \\ \downarrow & & \downarrow \\ \mathcal{G}'_1 & \xrightarrow{v} & \mathcal{G}'_2 \end{array}$$

commutative. Such a homomorphism v , when it exists, is uniquely determined by this property. Then the diagram

$$\begin{array}{ccc} \mathcal{F}_1 & \xrightarrow{u} & \mathcal{F}_2 \\ \downarrow & & \downarrow \\ i_*(\mathcal{G}'_1) & \xrightarrow{i_*(v)} & i_*(\mathcal{G}'_2) \end{array}$$

is commutative, and consequently there is a unique homomorphism

$$w : \overline{\mathcal{G}}'_1 \rightarrow \overline{\mathcal{G}}'_2$$

making the diagram

$$\begin{array}{ccc} \mathcal{F}_1 & \xrightarrow{u} & \mathcal{F}_2 \\ \downarrow & & \downarrow \\ \overline{\mathcal{G}}'_1 & \xrightarrow{w} & \overline{\mathcal{G}}'_2 \end{array}$$

commutative.

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Proposition (2.8.4). — Suppose that Y satisfies the hypotheses of (2.8.1). Let X_1 and X_2 be two Y -preschemes, let $\mathcal{F}_i$ be a quasi-coherent $\mathcal{O}_{X_i}$ -module, and let $\mathcal{G}'_i$ be an $\mathcal{O}_{(X_i)_\eta}$ -module quotient of $(\mathcal{F}_i)_\eta$, for $i = 1, 2$. Then

(2.8.4.1).

$$\overline{\mathcal{G}}'_1 \otimes_{\mathcal{O}_Y} \overline{\mathcal{G}}'_2 = \overline{\mathcal{G}'_1 \otimes_{k(\eta)} \mathcal{G}'_2}.$$

Proof. Put $X = X_1 \times_Y X_2$. The left-hand side of (2.8.4.1) is a quasi-coherent $\mathcal{O}_X$ -module flat over Y (0, 6.2.1); its inverse image on X_η is

$$\mathcal{G}'_1 \otimes_{k(\eta)} \mathcal{G}'_2$$

(I, 9.1.5), and it is a quotient of $\mathcal{F}_1 \otimes_{\mathcal{O}_Y} \mathcal{F}_2$. The conclusion therefore follows from the uniqueness assertion of (2.8.1). $\square$

Proposition (2.8.5). — Suppose that X and Y satisfy the hypotheses of (2.8.1), and let Z' be a closed subprescheme of X_η . Then there exists a unique closed subprescheme $\overline{Z}'$ of X that is flat over Y and satisfies

$$i^{-1}(\overline{Z}') = Z'.$$

Proof. Let $\mathcal{J}'$ be the quasi-coherent ideal of $\mathcal{O}_{X_\eta}$ defining Z' . Apply (2.8.1) with

$$\mathcal{F} = \mathcal{O}_X, \quad \mathcal{G}' = \mathcal{O}_{X_\eta} / \mathcal{J}'.$$

If $\overline{\mathcal{G}}' = \mathcal{O}_X / \mathcal{J}$, then

$$\mathcal{J}' = i^*(\mathcal{J})\mathcal{O}_{X_\eta},$$

and hence $i^{-1}(\overline{Z}') = Z'$ (I, 4.4.5). $\square$

The prescheme $\overline{Z'}$ is the *closed image* of Z' under the composite morphism

$$Z' \longrightarrow X_\eta \xrightarrow{i} X,$$

where the first arrow is the canonical injection (I, 9.5.3). Its underlying space is the closure of Z' in X (I, 9.5.4), which justifies the notation. We also say that $\overline{Z'}$ is the closure subscheme of X of Z' .

Corollary (2.8.6). — *Let X_1 and X_2 be two Y -preschemes, and let Z'_i be a closed subscheme of $(X_i)_\eta$, for $i = 1, 2$. Then*

(2.8.6.1).

$$\overline{Z'_1} \times_Y \overline{Z'_2} = \overline{Z'_1 \times_{k(\eta)} Z'_2}.$$

Proof. This follows from (2.8.4) and (2.8.5). $\square$

§3. ASSOCIATED PRIME CYCLES AND PRIMARY DECOMPOSITIONS

In this section, we mainly give the translation of results on modules presented in Bourbaki, *Alg. comm.*, chap. IV, which we follow very closely. The notions that follow seem to be of interest only in the case of *locally Noetherian* preschemes. IV-2 | 36

3.1. Associated prime cycles of a module

Definition (3.1.1). — Let X be a prescheme and $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -module. We say that a point $x \in X$ is *associated* to $\mathcal{F}$ if the maximal ideal $\mathfrak{m}_x$ of $\mathcal{O}_x$ is associated to the $\mathcal{O}_x$ -module $\mathcal{F}_x$ (in other words, if $\mathfrak{m}_x$ is the annihilator of an element of $\mathcal{F}_x$). We denote by $\text{Ass}(\mathcal{F})$ the set of $x \in X$ associated to $\mathcal{F}$. We say that a closed irreducible subset Z of X is an *associated prime cycle* of $\mathcal{F}$ if its generic point is associated to $\mathcal{F}$. When $\mathcal{F} = \mathcal{O}_X$, we also say that the points (resp. *prime cycles*) associated to $\mathcal{F}$ are associated to the *prescheme* X .

We say that an associated prime cycle of $\mathcal{F}$ (resp. X) is *embedded* if it is contained in another associated prime cycle of $\mathcal{F}$ (resp. X) (in other words, if it is not *maximal* in the set of these cycles).

If X is locally Noetherian, the *irreducible components* of X are none other than the *maximal* (or *non-embedded*) prime cycles associated to X .

It is clear that if $x \in \text{Ass}(\mathcal{F})$, then $\mathcal{F}_x \neq 0$, in other words

$$(3.1.1.1) \quad \text{Ass}(\mathcal{F}) \subset \text{Supp}(\mathcal{F}).$$

If $x \in X$ is associated to $\mathcal{F}$, it is evidently associated to $\mathcal{F}|U$ for every open neighbourhood U of x , and conversely, if it is associated to $\mathcal{F}|U$ for one of these neighbourhoods, it is associated to $\mathcal{F}$.

We note finally that for a quasi-coherent $\mathcal{O}_X$ -module $\mathcal{F}$, the existence of embedded associated prime cycles is a *local* question, for if y and z are two points of $\text{Ass}(\mathcal{F})$ such that $y \in \overline{\{z\}}$, every neighbourhood of y contains z .

Proposition (3.1.2). — *Let A be a Noetherian ring, M an A -module, $X = \text{Spec}(A)$, $\mathcal{F} = \widetilde{M}$; for a point $x \in X$ to be associated to $\mathcal{F}$, it is necessary and sufficient that the prime ideal $\mathfrak{j}_x$ of A be associated to the module M (in other words, be the annihilator of an element $f \in M$).*

Proof. This follows from the definition (3.1.1) and from Bourbaki, *loc. cit.*, § 1, no 2, cor. of prop. 5, applied to $S = A - \mathfrak{j}_x$. $\square$

Proposition (3.1.3). — *Let X be a locally Noetherian prescheme, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -module, x a point of X , and Z the reduced closed sub-prescheme of X having $\overline{\{x\}}$ as underlying space (I, 5.2.1). The following conditions are equivalent:*

- a) $x \in \text{Ass}(\mathcal{F})$.
- b) There exist an open neighbourhood U of x and a section $f \in \Gamma(U, \mathcal{F})$ such that $U \cap Z$ is equal to $\text{Supp}(\mathcal{O}_U \cdot f)$.
- b') There exist an open neighbourhood U of x and a section $f \in \Gamma(U, \mathcal{F})$ such that $U \cap Z$ is an irreducible component of $\text{Supp}(\mathcal{O}_U \cdot f)$.
- c) There exist an open neighbourhood U of x and a submodule of $\mathcal{F}|U$ isomorphic to $\mathcal{O}_Z|U$ ($\mathcal{O}_Z$ being identified with a quotient of $\mathcal{O}_X$).

c') There exist an open neighbourhood U of x and a coherent submodule $\mathcal{G}$ of $\mathcal{F}|_U$ such that $U \cap Z$ is an irreducible component of $\text{Supp}(\mathcal{G})$.

Proof. It is clear that c) implies b), for it suffices to take for f the element of $\Gamma(U, \mathcal{F})$ which corresponds to the unit section of $\mathcal{O}_Z|_U$. As $U \cap Z$ is irreducible (0, 2.1.6), b) implies b'), and b') implies c') since $\mathcal{O}_X$ is coherent (0, 5.3.4). To see that c') implies a), we can reduce to the case where $U = X = \text{Spec}(A)$ is affine, A being Noetherian, and where $\mathcal{F} = \tilde{M}$, where M is an A -module, and $\mathcal{G} = \tilde{N}$, where N is a sub-module of M , of finite type. We then know that the minimal elements of $\text{Supp}(\mathcal{G})$ are the maximal points of $V(\text{Ann}(N))$ (0, 1.7.4), and these are also the minimal elements of $\text{Ass}(N)$ (Bourbaki, *Alg. comm.*, chap. IV, § 1, no 3, cor. 1 of prop. 7); as $\text{Ass}(N) \subset \text{Ass}(M) = \text{Ass}(\mathcal{F})$, we see that c') implies a). Finally, a) implies c) by (3.1.2), taking X affine again, $\mathcal{F} = \tilde{M}$, and Z defined by the ideal $j_x A$ (with the notation of (3.1.2)). $\square$

Corollary (3.1.4). — *Let X be a locally Noetherian prescheme and $\mathcal{F}$ a coherent $\mathcal{O}_X$ -module. Then the maximal prime cycles associated to $\mathcal{F}$ are the irreducible components of $\text{Supp}(\mathcal{F})$, and the generic points of these components are the points $x \in X$ such that $\mathcal{F}_x$ is an $\mathcal{O}_x$ -module of finite nonzero length.*

Proof. Indeed, if x is the generic point of an irreducible component Z of $\text{Supp}(\mathcal{F})$, the equivalence of a) and c') in (3.1.3) shows that x belongs to $\text{Ass}(\mathcal{F})$, and Z is an associated prime cycle of $\mathcal{F}$, necessarily maximal by (3.1.1.1); the converse follows trivially from (3.1.1.1). Finally, the last assertion is local and follows from Bourbaki, *Alg. comm.*, chap. IV, § 2, no 5, cor. 2 of prop. 7. $\square$

Corollary (3.1.5). — *Let X be a locally Noetherian prescheme and $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -module. For $\mathcal{F} = 0$, it is necessary and sufficient that $\text{Ass}(\mathcal{F}) = \emptyset$.*

Proof. The question being local, we reduce to the case where X is affine, and the conclusion follows immediately from (3.1.2) and from Bourbaki, *Alg. comm.*, chap. IV, § 1, no 1, cor. 1 of prop. 2. $\square$

Proposition (3.1.6). — *Let X be a locally Noetherian prescheme and $\mathcal{F}$ a coherent $\mathcal{O}_X$ -module; then $\text{Ass}(\mathcal{F})$ is locally finite (that is to say, every point of X admits a neighbourhood whose intersection with $\text{Ass}(\mathcal{F})$ is finite).*

Proof. It suffices to consider the case where X is affine, hence Noetherian, and then the proposition follows from (3.1.2) and from Bourbaki, *Alg. comm.*, chap. IV, § 1, no 4, cor. of th. 2. $\square$

Proposition (3.1.7). — *Let X be a prescheme.*

- (i) *Let $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$ be an exact sequence of quasi-coherent $\mathcal{O}_X$ -modules. Then $\text{Ass}(\mathcal{F}') \subset \text{Ass}(\mathcal{F}) \subset \text{Ass}(\mathcal{F}') \cup \text{Ass}(\mathcal{F}'')$.*
- (ii) *Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$ -module, $(\mathcal{F}_\alpha)$ a family of quasi-coherent $\mathcal{O}_X$ -submodules of $\mathcal{F}$ such that $\mathcal{F}$ is the union of the $\mathcal{F}_\alpha$. Then $\text{Ass}(\mathcal{F}) = \bigcup_\alpha \text{Ass}(\mathcal{F}_\alpha)$.*
- (iii) *For every family $(\mathcal{F}_\alpha)$ of quasi-coherent $\mathcal{O}_X$ -modules, we have $\text{Ass}(\bigoplus_\alpha \mathcal{F}_\alpha) = \bigcup_\alpha \text{Ass}(\mathcal{F}_\alpha)$.*

Proof. We immediately reduce to the corresponding propositions for modules (Bourbaki, *loc. cit.*, § 1, no 1, formula (1), prop. 3 and cor. 1 of prop. 3). $\square$

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Proposition (3.1.8). — *Let X be a locally Noetherian prescheme, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -module, U an open subset of X , and $\mathcal{J}$ a coherent ideal of $\mathcal{O}_X$ defining a closed sub-prescheme of X having $X - U$ as underlying space. The following conditions are equivalent:*

- a) $\text{Ass}(\mathcal{F}) \subset U$.
- b) *For every affine open subset V of X , every section of $\mathcal{F}$ over V whose restriction to $V \cap U$ is zero is itself zero.*
- c) *The canonical homomorphism $\mathcal{F} \rightarrow \mathcal{H}om_{\mathcal{O}_X}(\mathcal{J}, \mathcal{F})$ is injective.*

Proof. The question being local, we can suppose $X = \text{Spec}(A)$ affine, A being a Noetherian ring, $\mathcal{F} = \tilde{M}$, where M is an A -module, and $\mathcal{J} = \tilde{\mathfrak{J}}$, where $\mathfrak{J}$ is an ideal of A . The homomorphism $M \rightarrow \text{Hom}_A(\mathfrak{J}, M)$ sends $m \in M$ to the homomorphism $x \mapsto xm$ of $\mathfrak{J}$ into M ; to say that it is not injective means that there exists $m \neq 0$ in M such that $\mathfrak{J}m = 0$.

Let us first show that *c*) implies *a*). If $\text{Ass}(\mathcal{F}) \cap U \neq \emptyset$, there would be a prime ideal $\mathfrak{p} \in \text{Ass}(M)$ containing $\mathfrak{J}$, hence an element $m \neq 0$ of M such that $\mathfrak{p}m = 0$, and therefore $\mathfrak{J}m = 0$, contradicting *c*).

Next, *b*) implies *c*). Indeed, if $\mathfrak{J}m = 0$ for some $m \neq 0$ in M , then for every prime ideal $\mathfrak{q}$ not containing $\mathfrak{J}$ there exists $a \in \mathfrak{J}$ with $a \notin \mathfrak{q}$. The relation $am = 0$ then implies that the canonical image of m in $M_{\mathfrak{q}}$ is zero. Thus m would define a nonzero section of $\mathcal{F}$ over X whose restriction to U is zero, contrary to *b*).

Finally, *a*) implies *b*). The canonical homomorphism

$$M \longrightarrow \prod_{\mathfrak{p} \in \text{Ass}(M)} M_{\mathfrak{p}}$$

is injective: if N is its kernel, then $\text{Ass}(N) \subset \text{Ass}(M)$; if there existed $\mathfrak{p} \in \text{Ass}(N)$, there would be an element $n \in N$ whose annihilator is $\mathfrak{p}$, while the definition of N would give an $s \notin \mathfrak{p}$ such that $sn = 0$, a contradiction. Hence $\text{Ass}(N) = \emptyset$, and therefore $N = 0$. Under *a*), a section $m \in M$ whose restriction to U is zero has zero image in $M_{\mathfrak{p}}$ for every $\mathfrak{p} \in \text{Ass}(M)$; consequently $m = 0$. $\square$

Corollary (3.1.9). — *Let X be a locally Noetherian prescheme, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -module, and f a section of $\mathcal{O}_X$ over X . For f to be $\mathcal{F}$ -regular (0, 15.2.2), it is necessary and sufficient that $\text{Ass}(\mathcal{F}) \subset X_f$.*

Proof. Indeed, it is immediate that to say that f is $\mathcal{F}$ -regular means that the canonical homomorphism $\mathcal{F} \rightarrow \mathcal{H}om_{\mathcal{O}_X}(f\mathcal{O}_X, \mathcal{F})$ is injective, and it suffices to apply (3.1.8) to the ideal $\mathcal{J} = f\mathcal{O}_X$. $\square$

Proposition (3.1.10). — *Let X and Y be locally Noetherian preschemes and $f : X \rightarrow Y$ an integral morphism. Then, for every quasi-coherent $\mathcal{O}_X$ -module $\mathcal{F}$, we have $f(\text{Ass}(\mathcal{F})) = \text{Ass}(f_*(\mathcal{F}))$.*

Proof. The question being local on Y , and the morphism f being affine, we are immediately reduced to the case where Y is affine, in other words, to the following lemma. $\square$

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Lemma (3.1.10.1). — *Let A and B be two Noetherian rings, $\rho : A \rightarrow B$ a ring homomorphism making B an integral A -algebra, and M a B -module. Then the prime ideals $\mathfrak{p} \in \text{Ass}(M_{[\rho]})$ are the inverse images by ρ of the prime ideals $\mathfrak{q} \in \text{Ass}(M)$.*

Proof. Indeed, if $\mathfrak{q} \in \text{Ass}(M)$, then $\mathfrak{q}$ is the annihilator in B of an element $x \in M$, and hence $\rho^{-1}(\mathfrak{q})$ is the annihilator in A of x . Conversely, suppose $\mathfrak{p} \in \text{Ass}(M_{[\rho]})$, so that $\mathfrak{p}$ is the inverse image under ρ of the annihilator $\mathfrak{b}$ in B of an element $x \in M$. It follows from the first Cohen–Seidenberg theorem that there exists a prime ideal $\mathfrak{q}$ of B containing $\mathfrak{b}$ and whose inverse image is $\mathfrak{p}$ (Bourbaki, *Alg. comm.*, chap. V, § 2, no 1, cor. 2 of th. 1). By considering a prime ideal minimal among those contained in $\mathfrak{q}$ and containing $\mathfrak{b}$, we may evidently suppose that $\mathfrak{q}$ itself is one of these minimal primes. Since $B/\mathfrak{q} \subset M$ is isomorphic to $B/\mathfrak{b}$, we then have $\mathfrak{q} \in \text{Ass}(B/\mathfrak{b}) \subset \text{Ass}(M)$ (Bourbaki, *Alg. comm.*, chap. IV, § 1, no 4, th. 2). $\square$

Corollary (3.1.11). — *Under the hypotheses of (3.1.10), for $\mathcal{F}$ to have no embedded associated prime cycle, it suffices that the same be true of $f_*(\mathcal{F})$.*

Proof. Suppose indeed that $f_*(\mathcal{F})$ has no embedded associated prime cycle. Note that if A is an integral algebra over a field k , all prime ideals of A are maximal (Bourbaki, *Alg. comm.*, chap. V, § 2, no 1, prop. 1); it follows from (I, 6.2.2) that the fibres of f are discrete spaces. If x and x' are two distinct points of $\text{Ass}(\mathcal{F})$, neither can be adherent to the other when $f(x) = f(x')$; and if $f(x) \neq f(x')$, then (3.1.10) and the hypothesis imply that neither of the two points $f(x)$ and $f(x')$ can be adherent to the other, so the same is true of x and x' . $\square$

Remark (3.1.12). — Under the hypotheses of (3.1.10), it may happen, on the contrary, that $\mathcal{F}$ has no embedded associated prime cycle whereas $f_*(\mathcal{F})$ does. For example, take $Y = \text{Spec}(k[T])$, where k is a field (an “affine line”), and let X be the disjoint union of $X_1 = Y$ and $X_2 = \text{Spec}(k)$, the morphism $X_2 \rightarrow Y$ corresponding to the canonical homomorphism $k[T] \rightarrow k[T]/\mathfrak{m}$, where $\mathfrak{m}$ is the maximal ideal (T) . It is clear that the morphism $f : X \rightarrow Y$ is finite. If one takes $\mathcal{F} = \mathcal{O}_X$, then $\mathcal{F}$ has no embedded associated prime cycle, but $f_*(\mathcal{F}) = \tilde{M}$, where M is the direct sum of the $k[T]$ -modules $k[T]$ and k ; hence $\text{Ass}(M)$ consists of the generic point (0) of Y and the point $\mathfrak{m}$.

Proposition (3.1.13). — *Let X be a locally Noetherian prescheme, U an open subset of X , and $i : U \rightarrow X$ the canonical inclusion. For every quasi-coherent $\mathcal{O}_U$ -module $\mathcal{F}$, we have $\text{Ass}(i_*(\mathcal{F})) = \text{Ass}(\mathcal{F})$.*

Proof. Recall that $i_*(\mathcal{F})$ is a quasi-coherent $\mathcal{O}_X$ -module (I, 2.2) and (I, 7.4); as $i_*(\mathcal{F})|_U = \mathcal{F}$, we have $\text{Ass}(i_*(\mathcal{F})) \cap U = \text{Ass}(\mathcal{F})$, and it therefore remains to prove that $\text{Ass}(i_*(\mathcal{F})) \subset U$. But by (3.1.8), this relation means that for every affine open subset V of X , every section of $i_*(\mathcal{F})$ over V which is zero on $U \cap V$, is zero, a condition which is trivially satisfied since $\Gamma(V, i_*(\mathcal{F})) = \Gamma(U \cap V, \mathcal{F}) = \Gamma(U \cap V, i_*(\mathcal{F}))$. $\square$

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3.2. Irredundant decompositions

Proposition (3.2.1). — *Let X be a locally Noetherian prescheme and U a dense open subset of X . The following conditions are equivalent:*

- a) X is reduced.
- b) The induced sub-prescheme on U is reduced, and X has no embedded associated prime cycle.
- c) X has no embedded associated prime cycle and, for every generic point x of an irreducible component of X , one has $\text{length}(\mathcal{O}_x) = 1$.

The prime cycles associated with X are then precisely the irreducible components of X .

Proof. It is clear that if X is reduced, then so is the sub-prescheme induced on U . Moreover, since the existence of embedded associated prime cycles is a local question, we may restrict to the case where $X = \text{Spec}(A)$ is affine and A is Noetherian. If A is reduced, its minimal prime ideals form a reduced primary decomposition of (0) (Bourbaki, *Alg. comm.*, chap. IV, § 2, no 5, prop. 10) and are precisely the elements of $\text{Ass}(A)$; hence A has no embedded associated prime ideals. Thus *a)* implies *b)*. It is immediate that *b)* implies *c)*, since a generic point x of an irreducible component belongs to U , and consequently $\mathcal{O}_x$ is a field. Finally, *c)* implies *a)*. Indeed, let $\mathcal{N}$ be the nilradical of $\mathcal{O}_X$, which is a coherent ideal. By hypothesis, $\text{Supp}(\mathcal{N})$ contains none of the generic points of the irreducible components of X . If $\text{Supp}(\mathcal{N})$ were nonempty and x were one of its maximal points, criterion (3.1.3), *c')*, would give $x \in \text{Ass}(\mathcal{O}_X)$; then $\overline{\{x\}}$ would be an embedded associated prime cycle of X , contrary to the hypothesis. Hence $\mathcal{N} = 0$. $\square$

Definition (3.2.2). — Let X be a locally Noetherian prescheme and $\mathcal{F}$ a coherent $\mathcal{O}_X$ -module. We say that $\mathcal{F}$ is *reduced* if it satisfies the following two conditions: 1° $\mathcal{F}$ has no embedded associated prime cycle; 2° for every maximal point x of $\text{Supp}(\mathcal{F})$, one has $\text{length}(\mathcal{F}_x) = 1$.

Condition 1° means that the prime cycles associated with $\mathcal{F}$ are the irreducible components of $\text{Supp}(\mathcal{F})$ (3.1.4), and condition 2° means that for every generic point x of such a component one has $\text{length}(\mathcal{F}_x) = 1$.

For an affine scheme X , this definition gives in particular the notion of a *reduced module* over a Noetherian ring A : a finitely generated A -module M is said to be *reduced* if it has no embedded associated prime ideals and if, for every $\mathfrak{p} \in \text{Ass}(M)$, one has $\text{length}_{A_{\mathfrak{p}}}(M_{\mathfrak{p}}) = 1$. Returning to a locally Noetherian prescheme X and a coherent $\mathcal{O}_X$ -module $\mathcal{F}$, we say that $\mathcal{F}$ is *reduced at a point* $x \in X$ if $\mathcal{F}_x$ is a reduced $\mathcal{O}_x$ -module. Equivalently, on the local scheme $\text{Spec}(\mathcal{O}_x)$, the module associated with $\mathcal{F}_x$ is reduced. Thus x belongs to no embedded associated prime cycle of $\mathcal{F}$ and $\text{length}(\mathcal{F}_z) = 1$ for every maximal point z of $\text{Supp}(\mathcal{F})$ such that $x \in \overline{\{z\}}$. It is clear that if $\mathcal{F}$ is a reduced coherent $\mathcal{O}_X$ -module, then it is reduced at every point of X . Conversely, if $\mathcal{F}$ is *reduced at a point* x , there exists an open neighbourhood U of x such that $\mathcal{F}|_U$ is a reduced $\mathcal{O}_U$ -module: it suffices to take U meeting no embedded associated prime cycle of $\mathcal{F}$ (such a neighbourhood exists because these cycles form a locally finite family of closed subsets of X). To say that $\mathcal{O}_X$ is reduced at x is equivalent to saying that X is *reduced at the point* x .

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Proposition (3.2.3). — *Let X be a locally Noetherian prescheme, U an open subset of X , and $\mathcal{F}$ a coherent $\mathcal{O}_X$ -module such that $U \cap \text{Supp}(\mathcal{F})$ is dense in $\text{Supp}(\mathcal{F})$. The following conditions are equivalent:*

- a) $\mathcal{F}$ is reduced.
- b) $\mathcal{F}|_U$ is reduced, and $\mathcal{F}$ has no embedded associated prime cycle.

- c) There exist a reduced closed sub-prescheme X' of X and a coherent torsion-free $\mathcal{O}_{X'}$ -module $\mathcal{F}'$ of rank 1 on every irreducible component of X' such that, if $j : X' \rightarrow X$ is the canonical inclusion, then $j_*(\mathcal{F}') = \mathcal{F}$.

Moreover, when these conditions hold, X' is defined by the ideal $\mathcal{J}$ of $\mathcal{O}_X$ that annihilates $\mathcal{F}$.

Proof. To see that c) implies a), we may, by (3.1.3), restrict to the case where $X' = X$ is integral, with generic point x . We may further suppose that $X = \text{Spec}(A)$ is affine and $\mathcal{F} = \tilde{M}$, where A is integral and M is a torsion-free A -module of rank 1. The annihilator of every element of M is then zero, so $\text{Ass}(\mathcal{F}) = \{x\}$ and $\mathcal{F}_x$ is isomorphic to $\mathcal{O}_x$, the field of fractions of A ; hence the conditions of Definition (3.2.2) are satisfied. Since the existence of embedded associated prime cycles is a local question, if $\mathcal{F}$ has no such cycles and $U \cap \text{Supp}(\mathcal{F})$ is dense in $\text{Supp}(\mathcal{F})$, then $\text{Ass}(\mathcal{F}) = \text{Ass}(\mathcal{F}|U)$. Thus a) and b) are equivalent. If a) holds, take for X' the reduced closed sub-prescheme of X whose underlying space is $\text{Supp}(\mathcal{F})$ (I, 5.2.1), and put $\mathcal{F}' = j^*(\mathcal{F})$. A point x of $\text{Ass}(\mathcal{F})$ is necessarily a maximal point of X' , and since $\text{length}(\mathcal{F}_x) = 1$, $\mathcal{F}'_x$ is isomorphic to $\mathcal{F}_x$, hence to the residue field $k(x) = \mathcal{O}_{X',x}$. This proves that $\mathcal{F}'$ is torsion-free and of rank 1 (I, 7.4.6) and (I, 7.4.2). Finally, the last assertion is immediate, since for every $y \in X'$ the annihilator of the $\mathcal{O}_{X',y}$ -module $\mathcal{F}'_y$ is zero. $\square$

Definition (3.2.4). — Let X be a locally Noetherian prescheme and $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -module. We say that $\mathcal{F}$ is *irredundant* if $\text{Ass}(\mathcal{F})$ consists of a single element x ; if $\mathcal{F}$ is of finite type, we say that $\mathcal{F}$ is *integral* if, in addition, $\mathcal{F}$ is reduced (in other words, if $\text{length}(\mathcal{F}_x) = 1$). We say that a quasi-coherent $\mathcal{O}_X$ -submodule $\mathcal{G}$ of a quasi-coherent $\mathcal{O}_X$ -module $\mathcal{F}$ is *primary in $\mathcal{F}$* if $\mathcal{F}/\mathcal{G}$ is irredundant.

For an affine scheme X , this definition gives in particular the notion of an *integral module* over a Noetherian ring A : an A -module M is said to be *integral* if it is of finite type, if M is primary (that is, if $\text{Ass}(M)$ consists of a single prime ideal $\mathfrak{p}$), and if, in addition, $\text{length}_{A_{\mathfrak{p}}}(M_{\mathfrak{p}}) = 1$. Returning to an arbitrary locally Noetherian prescheme X and a coherent $\mathcal{O}_X$ -module $\mathcal{F}$, we say that $\mathcal{F}$ is *integral at a point $x \in X$* if $\mathcal{F}_x$ is an integral $\mathcal{O}_x$ -module. Equivalently, x belongs to only one associated prime cycle of $\mathcal{F}$ (necessarily a nonembedded one), and $\text{length}(\mathcal{F}_z) = 1$ at its generic point z . It is clear that if $\mathcal{F}$ is an integral coherent $\mathcal{O}_X$ -module, then it is integral at every point of X . Conversely, if $\mathcal{F}$ is *integral at a point x* , there exists an open neighbourhood U of x such that $\mathcal{F}|U$ is an integral $\mathcal{O}_U$ -module: it suffices to take U such that $\mathcal{F}|U$ is a reduced $\mathcal{O}_U$ -module (3.2.2).

We say that a locally Noetherian prescheme X is *irredundant* if $\mathcal{O}_X$ is irredundant (which implies that X is *irreducible*). For X to be *integral*, it is necessary and sufficient that the $\mathcal{O}_X$ -module $\mathcal{O}_X$ be integral (I, 2.1.8) and (3.2.1). If $\mathcal{O}_X$ is integral at a point x , that is, if the ring $\mathcal{O}_x$ is integral, we say that X is *integral at the point x* . We say that a closed sub-prescheme Y of X is *primary in X* if the ideal $\mathcal{J}$ of $\mathcal{O}_X$ defining Y is primary in $\mathcal{O}_X$.

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Definition (3.2.5). — Let X be a locally Noetherian prescheme and $\mathcal{F}$ a coherent $\mathcal{O}_X$ -module. An *irredundant decomposition* of $\mathcal{F}$ is a family $(\mathcal{F}_{\alpha})_{\alpha \in I}$ of quotient $\mathcal{O}_X$ -modules of $\mathcal{F}$ such that the $\mathcal{F}_{\alpha}$ are irredundant, the family $(\text{Supp}(\mathcal{F}_{\alpha}))_{\alpha \in I}$ is locally finite, and the canonical homomorphism $\mathcal{F} \rightarrow \bigoplus_{\alpha \in I} \mathcal{F}_{\alpha}$ is injective. We say that such a decomposition is *reduced* if the sets $\text{Ass}(\mathcal{F}_{\alpha})$ are pairwise distinct and there is no proper subset $J \subsetneq I$ for which the subfamily $(\mathcal{F}_{\alpha})_{\alpha \in J}$ is also an irredundant decomposition of $\mathcal{F}$.

If $(\mathcal{F}_{\alpha})_{\alpha \in I}$ is an irredundant decomposition (resp. a reduced irredundant decomposition) of $\mathcal{F}$ and $\mathcal{F}_{\alpha} = \mathcal{F}/\mathcal{G}_{\alpha}$, we also say that the family $(\mathcal{G}_{\alpha})_{\alpha \in I}$ of $\mathcal{O}_X$ -submodules of $\mathcal{F}$ is a *primary decomposition of 0 in $\mathcal{F}$* . The injectivity of the canonical homomorphism $\mathcal{F} \rightarrow \bigoplus_{\alpha \in I} \mathcal{F}_{\alpha}$ is equivalent to the condition $\bigcap_{\alpha \in I} \mathcal{G}_{\alpha} = 0$.

If $(\mathcal{F}_{\alpha})$ is an irredundant decomposition of $\mathcal{F}$, to say that it is reduced is equivalent to saying that the sets $\text{Ass}(\mathcal{F}_{\alpha})$ are pairwise distinct and contained in $\text{Ass}(\mathcal{F})$. If $\text{Ass}(\mathcal{F}_{\alpha}) = \{x_{\alpha}\}$ for every $\alpha \in I$, then $\alpha \mapsto x_{\alpha}$ is a bijection from I onto $\text{Ass}(\mathcal{F})$. These properties are local and therefore follow from Bourbaki, *Alg. comm.*, chap. IV, § 2, no 3, prop. 4.

Proposition (3.2.6). — Let X be a locally Noetherian prescheme and $\mathcal{F}$ a coherent $\mathcal{O}_X$ -module. Then there exists a reduced irredundant decomposition $(\mathcal{F}^{(x)})_{x \in \text{Ass}(\mathcal{F})}$ formed of coherent $\mathcal{O}_X$ -modules such that, for

every $x \in \text{Ass}(\mathcal{F})$, one has $\text{Ass}(\mathcal{F}^{(x)}) = \{x\}$. For every $x \in \text{Ass}(\mathcal{F})$ such that $\overline{\{x\}}$ is not embedded, $\mathcal{F}^{(x)}$ is uniquely determined as the image of the canonical homomorphism $\mathcal{F} \rightarrow i_*(i^*(\mathcal{F}))$, where i is the canonical morphism $\text{Spec}(k(x)) \rightarrow X$.

Proof. For every $x \in \text{Ass}(\mathcal{F})$, let U be an affine open neighbourhood of x , with ring A , and write $\mathcal{F}|_U = \tilde{M}$, where M is a finitely generated A -module. We know (Bourbaki, *Alg. comm.*, chap. IV, § 1, no 1, prop. 4) that there exists a submodule N of M such that, on setting $P = M/N$, one has $\text{Ass}(P) = \{x\}$ and $\text{Ass}(N) = \text{Ass}(M) - \{x\}$. Let $\mathcal{G} = \tilde{P}$, a quasi-coherent $\mathcal{O}_U$ -module, and let $j : U \rightarrow X$ be the canonical inclusion. Let $u : j^*(\mathcal{F}) \rightarrow \mathcal{G}$ be the surjective homomorphism corresponding to $M \rightarrow P$. It induces a homomorphism $j_*(u) : j_*(j^*(\mathcal{F})) \rightarrow j_*(\mathcal{G})$, and hence, by composition, a homomorphism

$$v : \mathcal{F} \xrightarrow{\rho_{\mathcal{F}}} j_*(j^*(\mathcal{F})) \xrightarrow{j_*(u)} j_*(\mathcal{G})$$

whose restriction to U is u . We denote by $\mathcal{F}^{(x)}$ the image of $\mathcal{F}$ under this homomorphism; it is a coherent $\mathcal{O}_X$ -module (I, 6.1.1). By (3.1.13), one has $\text{Ass}(j_*(\mathcal{G})) = \{x\}$, and therefore, by (3.1.7), $\text{Ass}(\mathcal{F}^{(x)}) = \{x\}$, since $\mathcal{F}^{(x)} \neq 0$.

Let $\mathcal{N}^{(x)} = \text{Ker}(v)$. Then $\mathcal{N}^{(x)}|_U = \tilde{N}$, and hence $x \notin \text{Ass}(\mathcal{N}^{(x)})$. It follows that the homomorphism $\mathcal{F} \rightarrow \bigoplus_{x \in \text{Ass}(\mathcal{F})} \mathcal{F}^{(x)}$ is injective. Indeed, its kernel $\mathcal{H}$ is contained in every $\mathcal{N}^{(x)}$, so $\text{Ass}(\mathcal{H})$ is contained in the intersection of the sets $\text{Ass}(\mathcal{N}^{(x)})$, which is empty; consequently, by (3.1.5), $\mathcal{H} = 0$. Since $\text{Ass}(\mathcal{F})$ is locally finite (3.1.6), it is clear that $(\mathcal{F}^{(x)})_{x \in \text{Ass}(\mathcal{F})}$ is a reduced irredundant decomposition of $\mathcal{F}$ satisfying the conditions of the statement. Finally, when $\overline{\{x\}}$ is not embedded, the characterization of $\mathcal{F}^{(x)}$ follows from Bourbaki, *Alg. comm.*, chap. IV, § 2, no 3, prop. 5, since the question is local, together with (I, 1.6.7). $\square$

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Corollary (3.2.7). — Under the hypotheses of (3.2.6), if $\mathcal{F}$ has no embedded associated prime cycle, there exists a unique reduced irredundant decomposition of $\mathcal{F}$.

Corollary (3.2.8). — Let X be a Noetherian prescheme and $\mathcal{F}$ a coherent $\mathcal{O}_X$ -module. There exists a finite filtration $(\mathcal{F}_i)_{0 \leq i \leq n}$ of $\mathcal{F}$ such that $\mathcal{F}_0 = \mathcal{F}$ and $\mathcal{F}_n = 0$, formed of coherent $\mathcal{O}_X$ -modules, and such that the quotients $\mathcal{F}_i/\mathcal{F}_{i+1}$ are either zero or irredundant, with $\text{Ass}(\mathcal{F}_i/\mathcal{F}_{i+1}) \subset \text{Ass}(\mathcal{F})$.

Proof. Indeed, by (3.2.6), $\mathcal{F}$ is isomorphic to an $\mathcal{O}_X$ -submodule of a finite direct sum $\bigoplus_{j=1}^n \mathcal{G}_j$, where the $\mathcal{G}_j$ are irredundant and coherent. Since every quasi-coherent $\mathcal{O}_X$ -submodule of $\mathcal{G}_j$ is either zero or irredundant (3.1.7), the modules

$$\mathcal{F}_i = \mathcal{F} \cap \left(\bigoplus_{j=1}^{n-i} \mathcal{G}_j \right)$$

satisfy the requirements, since $\mathcal{F}_i/\mathcal{F}_{i+1}$ is isomorphic to a coherent $\mathcal{O}_X$ -submodule of $\mathcal{G}_{n-i}$. $\square$

3.3. Relations with flatness

Proposition (3.3.1). — Let $f : X \rightarrow Y$ be a morphism, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -module that is f -flat, and $\mathcal{E}$ a quasi-coherent $\mathcal{O}_Y$ -module. If, for every $y \in Y$, we set $\mathcal{F}_y = \mathcal{F} \otimes_{\mathcal{O}_Y} k(y)$, then

$$(3.3.1.1) \quad \text{Ass}(\mathcal{E} \otimes_{\mathcal{O}_Y} \mathcal{F}) \supset \bigcup_{y \in \text{Ass}(\mathcal{E})} \text{Ass}(\mathcal{F}_y)$$

and the two sides are equal if Y is locally Noetherian.

Proof. (Of course, $\mathcal{F}_y$ is a sheaf on the fibre $f^{-1}(y)$, and this fibre is identified with a subspace of X (I, 3.6.1).) The question being local on X and on Y , we reduce to the case where X and Y are affine, and the proposition is then proved in Bourbaki, *Alg. comm.*, chap. IV, § 2, no 6, th. 2. $\square$

Corollary (3.3.2). — Let Y be a locally Noetherian prescheme with no embedded associated prime cycles, $f : X \rightarrow Y$ a morphism, and $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -module that is f -flat. Then, for every $x \in \text{Ass}(\mathcal{F})$, $f(x)$ is a maximal point of Y .

Proof. It suffices to apply (3.3.1) with $\mathcal{E} = \mathcal{O}_Y$, since $\text{Ass}(\mathcal{O}_Y)$ is by hypothesis the set of maximal points of Y . $\square$

Corollary (3.3.3). — *Under the hypotheses of (3.3.1), suppose in addition that X and Y are locally Noetherian and that $\mathcal{E}$ and $\mathcal{F}$ are coherent. Then the following conditions are equivalent:*

- a) $\mathcal{E} \otimes_Y \mathcal{F}$ has no embedded associated prime cycle.
- b) For every point $y \in \text{Ass}(\mathcal{E}) \cap f(\text{Supp}(\mathcal{F}))$, $\overline{\{y\}}$ is a nonembedded associated prime cycle of $\mathcal{E}$ and $\mathcal{E}_y \otimes_{k(y)} \mathcal{F}_y$ has no embedded associated prime cycle.

Proof. Suppose a) is satisfied. The hypotheses imply that $\mathcal{E} \otimes_Y \mathcal{F}$ is a coherent $\mathcal{O}_X$ -module (0, 5.3.11) and (0, 5.3.5); its associated prime cycles are therefore the irreducible components of $\text{Supp}(\mathcal{E} \otimes_Y \mathcal{F})$ (3.1.4), and for every maximal point x of $\text{Supp}(\mathcal{E} \otimes_Y \mathcal{F})$, $f(x) = y$ is a maximal point of $\text{Supp}(\mathcal{E})$ and x is a maximal point of $\text{Supp}(\mathcal{F}_y)$ (2.5.5). As, by virtue of (3.3.1) and the fact that $y \in f(\text{Supp}(\mathcal{F}))$ implies $\mathcal{F}_y \neq 0$ (I, 9.1.13), every point of $\text{Ass}(\mathcal{E}) \cap f(\text{Supp}(\mathcal{F}))$ is the image by f of a maximal point of $\text{Supp}(\mathcal{E} \otimes_Y \mathcal{F})$, we see that condition b) is satisfied.

Conversely, suppose b) is satisfied, and let us show that if z, z' are two distinct points of $\text{Ass}(\mathcal{E} \otimes_Y \mathcal{F})$, neither of the two can be adherent to the other. In the first place, if $f(z) = f(z') = y$, we have $y \in \text{Ass}(\mathcal{E})$ and z and z' belong to $\text{Ass}(\mathcal{F}_y)$ by (3.3.1), whence $y \in f(\text{Supp}(\mathcal{F}))$; as by hypothesis neither of the two points z, z' is adherent to the other within $f^{-1}(y)$, neither can be adherent to the other within X . If $y = f(z)$ and $y' = f(z')$ are distinct, they belong to $\text{Ass}(\mathcal{E}) \cap f(\text{Supp}(\mathcal{F}))$, hence neither of the two can be adherent to the other in Y ; it follows from the continuity of f that neither of the points z, z' can be adherent to the other in X . $\square$

Proposition (3.3.4). — *Let X and Y be locally Noetherian preschemes, $f : X \rightarrow Y$ a morphism, $\mathcal{E}$ a coherent $\mathcal{O}_Y$ -module, and $\mathcal{F}$ a coherent $\mathcal{O}_X$ -module that is f -flat. Then the following conditions are equivalent:*

- a) $\mathcal{E} \otimes_Y \mathcal{F}$ is reduced (3.2.2).
- b) For every point $y \in \text{Ass}(\mathcal{E}) \cap f(\text{Supp}(\mathcal{F}))$, $\overline{\{y\}}$ is a nonembedded associated prime cycle of $\mathcal{E}$, $\text{length}(\mathcal{E}_y) = 1$, and $\mathcal{F}_y$ is reduced.

Proof. Suppose a) is satisfied. We already know from (3.3.3) that, for every $y \in \text{Ass}(\mathcal{E})$ lying in $f(\text{Supp}(\mathcal{F}))$, $\overline{\{y\}}$ is a nonembedded associated prime cycle of $\mathcal{E}$ and $\mathcal{F}_y$ has no embedded associated prime cycle. Furthermore, by (2.5.5), for every $x \in \text{Ass}(\mathcal{E} \otimes_Y \mathcal{F}) \cap f^{-1}(y)$, we have

$$1 = \text{length}((\mathcal{E} \otimes_Y \mathcal{F})_x) = \text{length}(\mathcal{E}_y) \text{length}((\mathcal{F}_y)_x),$$

hence $\text{length}(\mathcal{E}_y) = \text{length}((\mathcal{F}_y)_x) = 1$, which proves b).

Conversely, suppose b) is satisfied; we already know that every point $x \in \text{Ass}(\mathcal{E} \otimes_Y \mathcal{F})$ is a maximal point of $\text{Supp}(\mathcal{E} \otimes_Y \mathcal{F})$, that $y = f(x)$ is a maximal point of $\text{Supp}(\mathcal{E})$ and x a maximal point of $\text{Supp}(\mathcal{F}_y)$ (3.3.1) and (3.3.3); in addition it follows from the hypothesis and from (2.5.5) that $\text{length}((\mathcal{E} \otimes_Y \mathcal{F})_x) = 1$, which proves a). $\square$

Corollary (3.3.5). — *Let X and Y be locally Noetherian preschemes and let $f : X \rightarrow Y$ be a flat morphism. If Y is reduced at the points of $f(X)$ and $f^{-1}(y)$ is a reduced $k(y)$ -prescheme for every $y \in f(X)$, then X is reduced.*

Proof. Since the nilradical $\mathcal{N}_Y$ is coherent, the set of points where Y is reduced is open (0, 5.2.2), and we may restrict to the case where Y is reduced. It suffices then to apply (3.3.4) with $\mathcal{E} = \mathcal{O}_Y$ and $\mathcal{F} = \mathcal{O}_X$. $\square$

Proposition (3.3.6). — *Let $f : X \rightarrow S$ and $g : Y \rightarrow S$ be two morphisms, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -module, and $\mathcal{G}$ a quasi-coherent $\mathcal{O}_Y$ -module. Suppose that: 1° $\mathcal{G}$ is g -flat; 2° X is locally Noetherian, and for every $s \in S$, $g^{-1}(s)$ is locally Noetherian (which will hold if Y is also locally Noetherian). Let $Z = X \times_S Y$; for every pair (x, y) such that $x \in X$, $y \in Y$, and $f(x) = g(y) = s$, let $T_{x,y}$ be the prescheme $\text{Spec}(k(x) \otimes_{k(s)} k(y))$, and let $I_{x,y}$ be the image of $\text{Ass}(\mathcal{O}_{T_{x,y}})$ by the canonical monomorphism $T_{x,y} \rightarrow Z$ (I, 3.4.9). Then*

$$(3.3.6.1) \quad \text{Ass}(\mathcal{F} \otimes_S \mathcal{G}) = \bigcup_{x \in \text{Ass}(\mathcal{F})} \left(\bigcup_{y \in \text{Ass}(\mathcal{G}_{f(x)})} I_{x,y} \right)$$

where for all $s \in S$, $\mathcal{G}_s = \mathcal{G} \otimes_{\mathcal{O}_S} k(s)$.

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Proof. Let $p : Z \rightarrow X$, $q : Z \rightarrow Y$ be the canonical projections, so that we have the commutative diagram

$$\begin{array}{ccc} X & \xleftarrow{p} & Z \\ f \downarrow & & \downarrow q \\ S & \xleftarrow{g} & Y \end{array}$$

Set $\mathcal{G}' = q^*(\mathcal{G})$, so that $\mathcal{F} \otimes_S \mathcal{G} = \mathcal{F} \otimes_X \mathcal{G}'$; as $\mathcal{G}'$ is p -flat (2.1.4), it follows from (3.3.1) that

$$(3.3.6.2) \quad \text{Ass}(\mathcal{F} \otimes_S \mathcal{G}) = \bigcup_{x \in \text{Ass}(\mathcal{F})} \text{Ass}(\mathcal{G}'_x)$$

with $\mathcal{G}'_x = \mathcal{G}' \otimes_{\mathcal{O}_X} k(x)$. If $s = f(x)$, we have $\mathcal{G}'_x = \mathcal{G}_s \otimes_{k(s)} k(x)$, and $p^{-1}(x) = g^{-1}(s) \otimes_{k(s)} k(x)$; furthermore, as the field $k(x)$ is a flat $k(s)$ -module, the morphism $p^{-1}(x) \rightarrow g^{-1}(s)$ is flat; applying (3.3.1) to this morphism, we get

$$(3.3.6.3) \quad \text{Ass}(\mathcal{G}'_x) = \bigcup_{y \in \text{Ass}(\mathcal{G}_s)} \text{Ass}(\mathcal{O}_{T_{x,y}})$$

whence the proposition. $\square$

We note that if, in the statement, we suppress hypothesis 2°, we can still conclude, by virtue of (3.3.1), the relation

$$(3.3.6.4) \quad \text{Ass}(\mathcal{F} \otimes_S \mathcal{G}) \supset \bigcup_{x \in \text{Ass}(\mathcal{F})} \left(\bigcup_{y \in \text{Ass}(\mathcal{G}_{f(x)})} I_{x,y} \right).$$

Corollary (3.3.7). — Under the hypotheses of (3.3.6), suppose in addition that S is also locally Noetherian and that $f(\text{Ass}(\mathcal{F})) \subset \text{Ass}(\mathcal{O}_S)$. Then

$$(3.3.7.1) \quad \text{Ass}(\mathcal{F} \otimes_S \mathcal{G}) = \bigcup_{(x,y) \in C} I_{x,y}$$

where C is the set of pairs (x, y) such that $x \in \text{Ass}(\mathcal{F})$, $y \in \text{Ass}(\mathcal{G})$ and $f(x) = g(y)$.

Proof. Since $\mathcal{G}$ is g -flat, it follows from (3.3.1) that the relation “ $s \in \text{Ass}(\mathcal{O}_S)$ and $y \in \text{Ass}(\mathcal{G}_s)$ ” is equivalent to “ $y \in \text{Ass}(\mathcal{G})$ ”; the conclusion follows from (3.3.6.1). $\square$

Remarks (3.3.8). — We shall see later (4.2.2) that under the hypotheses of (3.3.6), $T_{x,y}$ is a prescheme with no embedded associated prime cycle; it will follow that if $\mathcal{F}$ and the $\mathcal{G}_s$ have no embedded associated prime cycles, the same is true of $\mathcal{F} \otimes_S \mathcal{G}$.

Corollary (3.3.9). — Under the conditions of (3.3.7), we have

$$(3.3.9.1) \quad q(\text{Ass}(\mathcal{F} \otimes_S \mathcal{G})) \subset \text{Ass}(\mathcal{G})$$

(where $q : X \times_S Y \rightarrow Y$ is the canonical projection).

Proof. Indeed, if $(x, y) \in Z$, we have $q(I_{x,y}) = \{y\} \subset \text{Ass}(\mathcal{G})$. $\square$

3.4. Properties of the sheaves $\mathcal{F}/t\mathcal{F}$

Proposition (3.4.1). — Let X be a locally Noetherian prescheme, t a section of $\mathcal{O}_X$ over X , and Y the closed sub-prescheme of X defined by the ideal $t\mathcal{O}_X$. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -module, let S be the reduced closed sub-prescheme of X having $\text{Supp}(\mathcal{F})$ as underlying space, and let (S_i) be the family of reduced closed sub-preschemes of X whose underlying spaces are the irreducible components of S ; denote by s_i the generic point of S_i . Finally, let Z be an irreducible component of $\text{Supp}(\mathcal{F}/t\mathcal{F}) = S \cap Y$, z its generic point.

- (i) For every i such that $Z \subset S_i$, Z is an irreducible component of $S_i \cap Y$.
- (ii) If Z is not equal to any of the S_i , then

$$(3.4.1.1) \quad \text{length}((\mathcal{F}/t\mathcal{F})_z) \geq \sum_i \text{length}(\mathcal{F}_{s_i})$$

where the sum on the right-hand side is extended to all i such that $Z \subset S_i$.

- (iii) Suppose that Z is not equal to any of the S_i . For the two sides of (3.4.1.1) to be equal, it is necessary and sufficient that the following two conditions be satisfied:

- α) t_z is $\mathcal{F}_z$ -regular (0, 15.1.4).
- β) For every i such that $Z \subset S_i$, the canonical image of the germ t_z in $\mathcal{O}_{S_i, z}$ generates the maximal ideal of this ring (which implies that $\mathcal{O}_{S_i, z}$ is a discrete valuation ring and that the image of t_z is a uniformizer of this ring).

Proof. (i) If $j : Y \rightarrow X$ is the canonical injection, then

$$\mathcal{F}/t\mathcal{F} = \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_Y = j^* \mathcal{F},$$

and hence

$$\text{Supp}(\mathcal{F}/t\mathcal{F}) = j^{-1}(S) = S \cap Y$$

by (I, 9.1.13), which proves the assertion.

(ii) and (iii) The points s_i such that $Z \subset S_i$ are precisely those belonging to $\text{Spec}(\mathcal{O}_z)$. Thus, to prove (ii) and (iii), we may replace X by $\text{Spec}(\mathcal{O}_z)$. If $M = \mathcal{F}_z$, we may therefore suppose that $\mathcal{F} = \tilde{M}$, and then $\mathcal{F}_{s_i} = M_{\mathfrak{p}_i}$, where the $\mathfrak{p}_i$ are the minimal prime ideals of $\mathcal{O}_z$. Moreover, since M is an $\mathcal{O}_z$ -module of finite type, we have $S = S'_{\text{red}}$, where

$$S' = \text{Spec}(\mathcal{O}_z/\mathfrak{a})$$

and $\mathfrak{a}$ is the annihilator of M in $\mathcal{O}_z$ (0, 1.7.4). The two sides of (3.4.1.1) have the same values whether M is regarded as an $\mathcal{O}_z$ -module or as an $(\mathcal{O}_z/\mathfrak{a})$ -module. We may therefore finally replace X by $S' = \text{Spec}(A)$, where A is a Noetherian local ring and M is a faithful A -module. Since Z is closed in S , the hypothesis that $Z \neq S_i$ for every i means that $s_i \notin Z$, and hence that $\dim(A) > 0$. Finally, to say that z is the generic point of Z , an irreducible component of $S \cap V(t)$, means that A/tA has dimension 0 (in other words, that it is a local Artinian ring). We are thus reduced to the following lemma. $\square$

Lemma (3.4.1.2). — Let A be a Noetherian local ring of dimension > 0 , let $\mathfrak{p}_i$ be the minimal prime ideals of A , let $\mathfrak{m}$ be its maximal ideal, and let $t \in \mathfrak{m}$ be such that A/tA is Artinian. Then, for every A -module of finite type M ,

$$(3.4.1.3) \quad \text{length}(M/tM) \geq \sum_i \text{length}(M_{\mathfrak{p}_i}).$$

Moreover, equality holds if and only if the following two conditions are satisfied:

- α) t is M -regular;
- β) for every i such that $M_{\mathfrak{p}_i} \neq 0$, the image of t in $A/\mathfrak{p}_i$ generates the maximal ideal of that ring (which implies that $A/\mathfrak{p}_i$ is a discrete valuation ring).

Proof. Since A does not have dimension 0 and A/tA is Artinian, necessarily $\dim(A) = 1$ (0, 16.3.4), and $t \notin \mathfrak{p}_i$ for every i . The principal ideal (t) is therefore an ideal of definition of A , and hence contains a power of the maximal ideal $\mathfrak{m}$. Let N be the submodule of M consisting of the elements annihilated by a power of t (or, equivalently in view of what we have just proved, by a power of $\mathfrak{m}$), and put $P = M/N$. Then t is P -regular: if $tx \in N$ for some $x \in M$, then $t^k(tx) = 0$ for some integer k , and hence $x \in N$. We shall use the following elementary lemma.

Lemma (3.4.1.4). — *Let A be a ring and*

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

an exact sequence of A -modules. If $t \in A$ is M'' -regular, then the sequence

$$0 \longrightarrow M'/tM' \longrightarrow M/tM \longrightarrow M''/tM'' \longrightarrow 0$$

is exact.

Proof. Since $M/tM = M \otimes_A (A/tA)$, it suffices to prove exactness at M'/tM' . Suppose that the image $x \in M$ of an element $x' \in M'$ is of the form $x = ty$, with $y \in M$. For the images x'', y'' of x, y in M'' , we have $x'' = ty''$. But $x'' = 0$, so the hypothesis gives $y'' = 0$. Thus y is the image of an element $y' \in M'$, and the relation $x = ty$ gives $x' = ty'$ because $M' \rightarrow M$ is injective. $\square$

It follows from this lemma that

$$(3.4.1.5) \quad \text{length}(M/tM) = \text{length}(N/tN) + \text{length}(P/tP).$$

On the other hand, $N_{\mathfrak{p}_i} = 0$ for every i , since $t \notin \mathfrak{p}_i$, and hence $M_{\mathfrak{p}_i} = P_{\mathfrak{p}_i}$. Thus, to prove (3.4.1.3), it suffices to prove it with P in place of M . Furthermore, if the two sides of (3.4.1.3) are equal, then the same inequality for P , together with (3.4.1.5), gives $\text{length}(N/tN) = 0$. Consequently $N/tN = 0$, and Nakayama's lemma gives $N = 0$, since N is of finite type. But $N = 0$ means that t is M -regular. We may therefore reduce to the case $M = P$, that is, suppose from the outset that t is M -regular. This implies that $\mathfrak{m} \notin \text{Ass}(M)$, since t cannot annihilate a nonzero element of M . Because $\dim(A) = 1$, it follows that

$$\text{Ass}(M) \subset \{\mathfrak{p}_i\}_i.$$

We now argue by induction on

$$n = \sum_i \text{length}(M_{\mathfrak{p}_i}).$$

If $n = 0$, then $M_{\mathfrak{p}_i} = 0$ for every i , and hence $M = 0$, since none of the $\mathfrak{p}_i$ belongs to $\text{Ass}(M)$. Both sides of (3.4.1.3) are then zero, and condition β) is vacuous. If $n > 0$, the argument at the beginning of the proof of (3.4.1) allows us, moreover, to suppose that M is faithful. It follows that $M_{\mathfrak{p}_i} \neq 0$ for every i (Bourbaki, *Alg. comm.*, chap. II, § 2, no 2, cor. 2 of prop. 4), and consequently

$$\text{Ass}(M) = \{\mathfrak{p}_i\}_i.$$

Suppose first that $n = 1$. There is then only one minimal prime ideal $\mathfrak{p}$ of A , and the assertion **IV-2 | 48** that $M_{\mathfrak{p}}$ has length 1 means that it is isomorphic, as an $A_{\mathfrak{p}}$ -module, to the residue field

$$k = A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}.$$

Thus $M_{\mathfrak{p}}$ is annihilated by $\mathfrak{p}A_{\mathfrak{p}}$, and consequently $\mathfrak{p}$ is the annihilator of M (Bourbaki, *Alg. comm.*, chap. II, § 2, no 4, formula (9)). Since M is faithful, $\mathfrak{p} = 0$, and hence A is integral. Because $M \neq 0$, Nakayama's lemma gives $M/tM \neq 0$, so that $\text{length}(M/tM) \geq 1$; this proves (3.4.1.3) in this case. Moreover, if $\text{length}(M/tM) = 1$, then M is cyclic (Bourbaki, *Alg. comm.*, chap. II, § 3, no 2, cor. 2 of prop. 4), and hence isomorphic to a quotient $A/\mathfrak{b}$. Since M is faithful, necessarily $\mathfrak{b} = 0$, and M is isomorphic to A . As $\text{length}(A/tA) = 1$, the ideal tA is the maximal ideal $\mathfrak{m}$. Since A is a Noetherian integral local ring, it follows that A is a discrete valuation ring (Bourbaki, *Alg. comm.*, chap. VI, § 3, no 6, prop. 9), with uniformizer t . Conversely, if A is a discrete valuation ring, t is its uniformizer, $\text{length}(M_{\mathfrak{p}}) = 1$, and t is M -regular, then M is torsion-free and therefore isomorphic to a submodule of A (as M is of finite type), hence to A itself. Thus

$$\text{length}(M/tM) = \text{length}(A/tA) = 1.$$

Suppose now that $n \geq 2$. There exists an exact sequence

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

with $M' \neq 0$, $M'' \neq 0$, and

$$\text{Ass}(M) = \text{Ass}(M') \cup \text{Ass}(M'').$$

Indeed, if $\text{Ass}(M)$ has more than one element, this follows from Bourbaki, *Alg. comm.*, chap. IV, § 1, no 1, prop. 4. If, on the contrary, $\text{Ass}(M)$ consists of a single prime ideal, that ideal is necessarily the unique minimal prime ideal $\mathfrak{p}$ of A . The hypothesis then gives $\text{length}(M_{\mathfrak{p}}) \geq 2$, and it is enough

to take for M' the inverse image of a submodule of $M_{\mathfrak{p}}$ distinct from both 0 and $M_{\mathfrak{p}}$. Since t is M -regular, it belongs to none of the prime ideals in $\text{Ass}(M)$ (Bourbaki, *Alg. comm.*, chap. IV, § 1, no 1, cor. 2 of prop. 2). For the same reason, t is both M' -regular and M'' -regular. By (3.4.1.4), the sequence

$$0 \longrightarrow M'/tM' \longrightarrow M/tM \longrightarrow M''/tM'' \longrightarrow 0$$

is exact. For every i , so is the sequence

$$0 \longrightarrow M'_{\mathfrak{p}_i} \longrightarrow M_{\mathfrak{p}_i} \longrightarrow M''_{\mathfrak{p}_i} \longrightarrow 0.$$

Consequently

$$\begin{aligned} \text{length}(M/tM) &= \text{length}(M'/tM') + \text{length}(M''/tM''), \\ \text{length}(M_{\mathfrak{p}_i}) &= \text{length}(M'_{\mathfrak{p}_i}) + \text{length}(M''_{\mathfrak{p}_i}). \end{aligned}$$

The induction hypothesis now proves (3.4.1.3). Moreover, the two sides can be equal only if the analogous inequalities for M' and M'' are both equalities. By the induction hypothesis, this is equivalent to condition β) for those $\mathfrak{p}_i$ such that $M'_{\mathfrak{p}_i} \neq 0$ or $M''_{\mathfrak{p}_i} \neq 0$; these are precisely the ideals for which $M_{\mathfrak{p}_i} \neq 0$. $\square$

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Corollary (3.4.2). — *Under the general hypotheses of (3.4.1), suppose that Z is not equal to any of the S_i and that $\text{length}((\mathcal{F}/t\mathcal{F})_Z) = 1$. Then there is a unique S_i containing Z , and for this value of i , $\text{length}(\mathcal{F}_{S_i}) = 1$; moreover, $\mathcal{O}_{S_i, Z}$ is a discrete valuation ring with uniformizer t_Z , and t_Z is $\mathcal{F}_Z$ -regular.*

Proof. This follows from (3.4.1), the two sides of (3.4.1.1) being then equal. $\square$

Proposition (3.4.3). — *Let X be a locally Noetherian prescheme, t a section of $\mathcal{O}_X$ over X , and Y the closed sub-prescheme of X defined by the ideal $t\mathcal{O}_X$. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -module, let T be an associated prime cycle of $\mathcal{F}$, let T' be an irreducible component of $T \cap Y$, and let x be the generic point of T' . Suppose that t_x is $\mathcal{F}_x$ -regular; then $x \in \text{Ass}(\mathcal{F}/t\mathcal{F})$.*

Proof. As in the proof of (3.4.1), we may reduce to the case $X = \text{Spec}(\mathcal{O}_x)$. Taking (3.1.2) into account, the proposition is then a consequence of (0, 16.4.6.3). $\square$

Proposition (3.4.4). — *Let X be a locally Noetherian prescheme, t a section of $\mathcal{O}_X$ over X , and Y the closed sub-prescheme of X defined by the ideal $t\mathcal{O}_X$. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -module, and let (S_i) be the family of irreducible components of $\text{Supp}(\mathcal{F})$. Let y be a point of Y such that t_y is $\mathcal{F}_y$ -regular and no embedded associated prime cycle of $\mathcal{F}/t\mathcal{F}$ contains y . Then the irreducible components of $\text{Supp}(\mathcal{F}/t\mathcal{F})$ containing y are precisely the irreducible components of the $S_i \cap Y$ containing y , and the associated prime cycles of $\mathcal{F}$ containing y are non-embedded.*

Proof. Let us first prove the last assertion. Let $T \supset T_1$ be two associated prime cycles of $\mathcal{F}$ containing y . If x is the generic point of an irreducible component of $T \cap Y$ containing y , then x is a generization of y , and hence belongs to every neighbourhood of y . The hypothesis that t_y is $\mathcal{F}_y$ -regular therefore implies that t_x is $\mathcal{F}_x$ -regular (0, 15.2.4), so (3.4.3) gives $x \in \text{Ass}(\mathcal{F}/t\mathcal{F})$. Let x_1 be the generic point of an irreducible component of $T_1 \cap Y$ containing y , and let x be the generic point of an irreducible component of $T \cap Y$ containing x_1 . The preceding argument shows that x_1 and x both belong to $\text{Ass}(\mathcal{F}/t\mathcal{F})$. Since $x_1 \in \overline{\{x\}}$, the hypothesis implies $x_1 = x$.

Again denote by T and T_1 the integral closed sub-preschemes of X having these spaces as their respective underlying spaces, and set

$$A = \mathcal{O}_{T, x}, \quad A_1 = \mathcal{O}_{T_1, x}.$$

Then $A_1 = A/\mathfrak{p}$ for a prime ideal $\mathfrak{p}$ of A . By the definitions of x and x_1 , the rings A/tA and A_1/tA_1 are Artinian. On the other hand, we saw above that t_x is $\mathcal{F}_x$ -regular, so $x \notin \text{Ass}(\mathcal{F})$, and consequently A_1 is not Artinian. Thus $\dim A = \dim A_1 = 1$. This implies $\mathfrak{p} = 0$ and $A = A_1$ (0, 16.1.2.2). Since $\text{Spec}(A)$ and $\text{Spec}(A_1)$ are respectively dense in T and T_1 , it follows that $T = T_1$.

The S_i containing y are therefore all the associated prime cycles of $\mathcal{F}$ containing y . If x is the generic point of an irreducible component of $S_i \cap Y$ containing y , then (0, 15.2.4) again shows that t_x is $\mathcal{F}_x$ -regular, and (3.4.3) gives $x \in \text{Ass}(\mathcal{F}/t\mathcal{F})$. This proves the first assertion of (3.4.4). $\square$

Proposition (3.4.5). — Let X be a locally Noetherian prescheme, t a section of $\mathcal{O}_X$ over X , and Y the closed sub-prescheme of X defined by the ideal $t\mathcal{O}_X$. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -module and y a point of Y . Suppose that t_y is $\mathcal{F}_y$ -regular and that $\mathcal{F}/t\mathcal{F}$ is integral at y (3.2.4). Then $\mathcal{F}$ is integral at the point y .

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Proof. Taking into account (3.4.4), it suffices to prove that y is contained in a unique irreducible component of $\text{Supp}(\mathcal{F})$, and that if s is the generic point of this component, $\text{length}(\mathcal{F}_s) = 1$. By hypothesis, y belongs to only one irreducible component of $\text{Supp}(\mathcal{F}/t\mathcal{F})$, and if z is the generic point of this component, then $\text{length}((\mathcal{F}/t\mathcal{F})_z) = 1$. The conclusion follows from (3.4.2). $\square$

Proposition (3.4.6). — The hypotheses being those of (3.4.1), let x be a point of Y . Suppose that Y does not contain any of the S_i containing x , and that $\mathcal{F}/t\mathcal{F}$ is reduced at the point x (3.2.2). Then t_x is $\mathcal{F}_x$ -regular and $\mathcal{F}$ is reduced at the point x . Moreover, if z_j is the generic point of an irreducible component of $\text{Supp}(\mathcal{F}/t\mathcal{F})$ containing x , then z_j is contained in a unique S_i , and $\mathcal{O}_{S_i, z_j}$ is a discrete valuation ring with uniformizer t_{z_j} .

Proof. The assertion that t_x is $\mathcal{F}_x$ -regular follows by applying the following lemma to the ring $\mathcal{O}_x$.

Lemma (3.4.6.1). — Let A be a Noetherian ring, M an A -module of finite type, $\mathfrak{p}_i$ the minimal elements of $\text{Supp}(M)$, and $t \in A$. Suppose that t belongs to none of the $\mathfrak{p}_i$ and that M/tM is a reduced A -module (3.2.2). Then t is M -regular.

Proof. Every prime ideal $\mathfrak{p} \in \text{Supp}(M)$ contains one of the $\mathfrak{p}_i$. Since t belongs to none of the $\mathfrak{p}_i$, multiplication by t on $M_{\mathfrak{p}}$ is not nilpotent (Bourbaki, *Alg. comm.*, chap. IV, § 1, no 4, cor. of prop. 9). Let N be the submodule of M consisting of the elements annihilated by a power of t , and set $P = M/N$. We shall prove that $N = 0$. Since t is P -regular, (3.4.1.4) gives an exact sequence

$$0 \longrightarrow N/tN \longrightarrow M/tM \longrightarrow P/tP \longrightarrow 0.$$

Since N is of finite type, it is annihilated by a power of t , and it therefore suffices to prove that $N/tN = 0$. As N/tN is a submodule of M/tM , it suffices to prove that $(N/tN)_{\mathfrak{p}} = 0$ for every $\mathfrak{p} \in \text{Ass}(M/tM)$, or equivalently that

$$u_{\mathfrak{p}} : (M/tM)_{\mathfrak{p}} \longrightarrow (P/tP)_{\mathfrak{p}}$$

is bijective for every $\mathfrak{p} \in \text{Ass}(M/tM)$. Now $(P/tP)_{\mathfrak{p}} \neq 0$. Indeed,

$$\mathfrak{p} \in \text{Supp}(M/tM) = \text{Supp}(M) \cap V(t)$$

by (0, 1.7.5), so the image of t in $A_{\mathfrak{p}}$ lies in the maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$. The equality $P_{\mathfrak{p}} = tP_{\mathfrak{p}}$ would therefore imply $P_{\mathfrak{p}} = 0$ by Nakayama's lemma. We would then have $M_{\mathfrak{p}} = N_{\mathfrak{p}}$, and multiplication by t on $M_{\mathfrak{p}}$ would be nilpotent, contradicting the observation above because $\mathfrak{p} \in \text{Supp}(M)$. Finally, since M/tM is reduced, $\text{length}((M/tM)_{\mathfrak{p}}) = 1$. As $(P/tP)_{\mathfrak{p}} \neq 0$, the map $u_{\mathfrak{p}}$ is necessarily bijective. This proves the lemma. $\square$

By hypothesis, no embedded associated prime cycle of $\mathcal{F}/t\mathcal{F}$ contains x . It follows from (3.4.4) that no embedded associated prime cycle of $\mathcal{F}$ contains x . On the other hand, applying (3.4.2) to an irreducible component of $\text{Supp}(\mathcal{F}/t\mathcal{F})$ containing x , we find that $\text{length}(\mathcal{F}_{S_i}) = 1$ for every S_i containing x . This completes the proof that $\mathcal{F}_x$ is reduced. The final assertions also follow from (3.4.2). $\square$

Corollary (3.4.7). — Let A be a Noetherian local ring, $\mathfrak{m}$ its maximal ideal, M an A -module of finite type, and $(x_i)_{1 \leq i \leq k}$ a family of elements of $\mathfrak{m}$ forming part of a system of parameters for M (0, 16.3.6). If the A -module

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$$N = M / \left(\sum_{i=1}^k x_i M \right)$$

is integral (3.2.4), then M is integral and the sequence $(x_i)_{1 \leq i \leq k}$ is M -regular.

Proof. Induction on k immediately reduces us to the case $k = 1$; write x for x_1 . The hypothesis that x forms part of a system of parameters for M implies

$$\dim(N) = \dim(M) - 1$$

by (0, 16.3.7). Put $n = \dim(N)$. There is therefore a minimal element $\mathfrak{p}$ of $\text{Supp}(M)$ such that

$$\dim(M/\mathfrak{p}M) = n + 1$$

by (0, 16.3.4); for every integer $j > 0$ one also has

$$\dim(M/\mathfrak{p}^j M) = n + 1$$

by (0, 16.3.5). Moreover, x forms part of a system of parameters for $M/\mathfrak{p}^j M$ (0, 16.3.5). Set

$$M' = M/\mathfrak{p}^j M, \quad N' = M'/xM';$$

then $\dim(N') = n$.

There is an evident surjective homomorphism $v : N \rightarrow N'$. We claim that it is bijective. Indeed, if $P = \text{Ker}(v)$, then $\text{Ass}(P) \subset \text{Ass}(N)$. Since N is integral, if $P \neq 0$, then $\text{Ass}(P)$ and $\text{Ass}(N)$ would both consist of the unique point $\mathfrak{q}$ of

$$\text{Ass}(N) = \text{Supp}(N).$$

But $\dim(N') = \dim(N)$ implies $N'_\mathfrak{q} \neq 0$, and $\text{length}(N_\mathfrak{q}) = 1$ implies $\text{length}(N'_\mathfrak{q}) = 1$, since $N_\mathfrak{q} \rightarrow N'_\mathfrak{q}$ is surjective. Thus $P_\mathfrak{q} = 0$, a contradiction. Hence v is bijective.

It follows that N' , being isomorphic to N , is integral. Moreover, the support of N' , which is the intersection of $\text{Supp}(M')$ with $V(x)$, cannot contain $\text{Supp}(M')$, and the latter is irreducible by construction. Since M'/xM' is integral, and hence reduced, (3.4.6) shows that x is M' -regular. Consequently the kernel of multiplication by x on M is contained in

$$\mathfrak{p}^j M \subset \mathfrak{m}^j M$$

for every integer j , and this kernel is therefore zero by (0, 7.3.5). Thus x is M -regular. Applying (3.4.5), we conclude that M is integral. $\square$

Remark (3.4.8). — The analogue of (3.4.7) obtained by replacing “integral” by “reduced” need not be true. For example, let $C = K[X, Y, Z]$ be the polynomial ring over a field K , let

$$B = C/\mathfrak{p}q, \quad \mathfrak{p} = CZ, \quad \mathfrak{q} = CX^2 + CY,$$

and let A be the local ring of B corresponding to the maximal ideal that is the image of $CX + CY + CZ$ in B . If x, z are the canonical images of X, Z in A , then $xz \neq 0$ but $x^2z^2 = 0$. On the other hand, A/xA is isomorphic to $K[Y, Z]/(YZ)$, so $\dim(A/xA) = 1$ whereas $\dim(A) = 2$. Thus x belongs to a system of parameters of A (0, 16.3.4), and A/xA is reduced, but A is not.

Proposition (3.4.9). — *Let A be a Noetherian ring, M an A -module of finite type, and $f \in A$ an M -regular element such that M/fM has no embedded associated prime ideals. If $\mathfrak{p}_i$ ($1 \leq i \leq m$) are the prime ideals associated to M/fM , then, for every integer $n > 0$, $f^n M$ is the intersection of the inverse images of the $f^n M_{\mathfrak{p}_i}$ under the canonical maps $M \rightarrow M_{\mathfrak{p}_i}$ ($1 \leq i \leq m$).*

Proof. It suffices to prove that the $\mathfrak{p}_i$ are also the prime ideals associated to $M/f^n M$, for then the $\mathfrak{p}_i$ -saturation of $f^n M$ are the components of the necessarily unique reduced primary decomposition of $f^n M$ in M . Now, we have

$$\text{Ass}(f^{n-1}M/f^n M) \subset \text{Ass}(M/f^n M) \subset \text{Ass}(M/f^{n-1}M) \cup \text{Ass}(f^{n-1}M/f^n M)$$

by (3.1.7). Since f is M -regular, $f^{n-1}M/f^n M$ is isomorphic to M/fM , and the result follows by induction on n . $\square$

§4. CHANGE OF BASE FIELD FOR ALGEBRAIC PRESCHMES

4.1. Dimension of algebraic preschemes We shall develop in § 5 the general theory of the dimension of preschemes. The theory of the dimension of algebraic preschemes, however, can be developed more elementarily and also has many special features; we therefore give here a rapid account independent of the general theory.

If K is a field and L an extension of K , we write $\text{tr. deg}_K L$ for the transcendence degree of L over K .

Definition (4.1.1). — Let k be a field and X a prescheme locally of finite type over k . The *dimension* of X is the number

$$(4.1.1.1) \quad \dim(X) = \sup_x \text{tr. deg}_k \kappa(x),$$

where x runs through the set of maximal points of X .

We shall see in (5.2.2) that Definition (4.1.1) depends only apparently on the base field k over which X is locally of finite type, and that the number $\dim(X)$ so defined agrees with the dimension of the underlying space of X (0, 14.1.2). Evidently,

$$\dim(X) = \dim(X_{\text{red}}).$$

Each $\kappa(x)$ is an extension of finite type of k (I, 6.3.3), hence has finite transcendence degree over k . If X is of finite type over k and nonempty, it is Noetherian (I, 6.3.7), hence has only finitely many irreducible components; consequently $\dim(X)$ is finite and nonnegative. Definition (4.1.1) gives

$$\dim(\emptyset) = -\infty.$$

If (X_α) denotes the family of reduced closed sub-preschemes of X whose underlying spaces are the irreducible components of X (I, 5.2.1), then clearly

$$(4.1.1.2) \quad \dim(X) = \sup_\alpha \dim(X_\alpha).$$

The calculation of the dimension is thus reduced to the case of integral preschemes locally of finite type over k . Finally,

$$(4.1.1.3) \quad \dim(X) = \dim(U)$$

for every induced sub-prescheme on an everywhere dense open subset U of X . The notion of dimension is thereby reduced to the case of an affine scheme of finite type over k .

Theorem (4.1.2). — Let X and Y be two preschemes locally of finite type over a field k , and let $f : X \rightarrow Y$ be a k -morphism.

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(i) If f is quasi-compact and dominant, then

$$\dim(Y) \leq \dim(X).$$

(ii) If f is quasi-finite, then

$$\dim(X) \leq \dim(Y).$$

(iii) Suppose that X is of finite type over k . For $\dim(X) \geq n$ (respectively $\dim(X) \leq n$, respectively $\dim(X) = n$), it is necessary and sufficient that there exist an open subset U dense in X and a k -morphism

$$g : U \longrightarrow \mathbf{V}_k^n = \text{Spec}(k[T_1, \dots, T_n])$$

that is surjective (respectively finite, respectively finite and surjective). If X is affine, one may take $U = X$.

Proof. (i) Let y be a maximal point of Y . The fibre $f^{-1}(y)$ contains a maximal point x of X (1.1.5); hence $\kappa(x)$ is an extension of $\kappa(y)$, and therefore

$$\text{tr. deg}_k \kappa(y) \leq \text{tr. deg}_k \kappa(x) \leq \dim(X).$$

This proves (i).

(ii) If x is a maximal point of X , then $\kappa(x)$ is a finite extension of $\kappa(f(x))$ (II, 6.2.2), and hence has the same transcendence degree over k . On considering the reduced closed sub-prescheme of Y whose underlying space is $\overline{\{f(x)\}}$, we are reduced to the following corollary. $\square$

Corollary (4.1.2.1). — Let Y be a k -prescheme locally of finite type. For every sub-prescheme Z of Y , one has

$$\dim(Z) \leq \dim(Y).$$

Suppose in addition that all the irreducible components of Y have the same dimension. Then Z is rare in Y if and only if

$$\dim(Z) < \dim(Y).$$

Proof. Let z be a maximal point of Z . By considering one of the reduced closed sub-preschemes of Y whose underlying space is an irreducible component of Y containing z , we reduce to the case in which Y is integral, with generic point y . Then, by taking in Y an affine open neighbourhood of z , we reduce to the case in which Y is affine and of finite type over k . Write

$$Y = \operatorname{Spec}(A), \quad Z = \operatorname{Spec}(A/\mathfrak{a}),$$

where A is an integral ring and $\mathfrak{a}$ is an ideal of A distinct from A .

By the normalization lemma (Bourbaki, *Alg. comm.*, chap. V, § 3, no. 1, th. 1), there exists a finite sequence $(x_i)_{1 \leq i \leq n}$ of elements of A , algebraically independent over k , such that A is integral over

$$B = k[x_1, \dots, x_n]$$

and $\mathfrak{a} \cap B$ is generated by a subfamily $(x_i)_{1 \leq i \leq p}$, possibly empty, of $(x_i)_{1 \leq i \leq n}$. Let

$$g : Y \longrightarrow \operatorname{Spec}(B) = \mathbf{V}_k^n$$

be the finite dominant morphism corresponding to the canonical injection $B \rightarrow A$. Since

$$C = B/(\mathfrak{a} \cap B) \cong k[x_{p+1}, \dots, x_n],$$

g induces on Z a finite dominant morphism

$$Z \longrightarrow \mathbf{V}_k^{n-p}.$$

Thus

$$\operatorname{tr. deg}_k \kappa(y) = n, \quad \operatorname{tr. deg}_k \kappa(z) = n - p,$$

because $\kappa(y)$ and $\kappa(z)$ are finite extensions of $\kappa(g(y))$ and $\kappa(g(z))$, respectively. This proves the first assertion.

If $p = 0$, then necessarily $z = y$, since y is the only point of Y whose image under g is the generic point of $\mathbf{V}_k^n$; in this case Z contains a nonempty open subset of Y . If $p > 0$, then $z \neq y$, and $\{z\}$ is rare in $\overline{\{y\}}$. This completes the proof. $\square$

Proof of (4.1.2), continued. The corollary proves (ii).

(iii) The sufficiency of the stated conditions follows immediately from (i), (ii), and (1.5.4)(v). To prove necessity, choose a dense open subset U of X that is the union of pairwise disjoint irreducible affine open subsets U_i ($1 \leq i \leq m$), each containing a maximal point of X (0, 2.1.6). We may further suppose that X is reduced; if X is affine, we may of course take $U = X$.

The same argument as in (4.1.2.1) shows that, if $\dim(U_i) = n_i$, there exists a k -morphism

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$$g_i : U_i \longrightarrow \mathbf{V}_k^{n_i}$$

that is finite and dominant, hence surjective (II, 6.1.10). Set

$$n = \dim(X) = \sup_i n_i.$$

For every n_i there is a morphism

$$h_i : \mathbf{V}_k^{n_i} \longrightarrow \mathbf{V}_k^n$$

which is the closed immersion corresponding to the canonical homomorphism

$$k[T_1, \dots, T_n] \longrightarrow k[T_1, \dots, T_{n_i}]$$

and is the identity when $n_i = n$. Since U is the disjoint union of the U_i , define g on each U_i to be $h_i \circ g_i$. Then g is finite and surjective.

When $n' \geq n$, composing g with the canonical closed immersion

$$\mathbf{V}_k^n \longrightarrow \mathbf{V}_k^{n'}$$

gives a finite morphism $U \rightarrow \mathbf{V}_k^{n'}$. When $n' \leq n$, compose g with the canonical morphism

$$p : \mathbf{V}_k^n \longrightarrow \mathbf{V}_k^{n'}$$

corresponding to the canonical injection

$$k[T_1, \dots, T_{n'}] \longrightarrow k[T_1, \dots, T_n].$$

The morphism p is faithfully flat, hence surjective. $\square$

Remark (4.1.3). — Corollary (4.1.2.1) shows that in formula (4.1.1.1) one may let x run through the set of all points of X .

Corollary (4.1.4). — *Let X be a prescheme locally of finite type over a field k , and let K be an extension of k . Then*

$$\dim(X \otimes_k K) = \dim(X).$$

Proof. It is enough to consider the case in which X is of finite type over k . The morphism

$$u : X \otimes_k K \longrightarrow X$$

is faithfully flat (2.2.13)(i). Thus, if U is an everywhere dense open subset of X , then

$$u^{-1}(U) = U \otimes_k K$$

is dense in $X \otimes_k K$ (2.3.10). If

$$g : U \longrightarrow \mathbf{V}_k^n$$

is finite and surjective, then so is

$$g_{(K)} : U \otimes_k K \longrightarrow \mathbf{V}_K^n$$

(I, 3.5.2) and (II, 6.1.5). The corollary follows. $\square$

Corollary (4.1.5). — *Let X and Y be two preschemes locally of finite type over a field k . Then*

$$\dim(X \times_k Y) = \dim(X) + \dim(Y).$$

Proof. It is enough to prove the following. Let U and V be affine neighbourhoods of points $x \in X$ and $y \in Y$ such that

$$\text{tr. deg}_k \kappa(x) = m = \dim(U), \quad \text{tr. deg}_k \kappa(y) = n = \dim(V).$$

Then

$$\dim(U \times_k V) = m + n.$$

Equivalently, by (4.1.2), we may suppose that there exist finite surjective k -morphisms

$$f : X \longrightarrow \mathbf{V}_k^m, \quad g : Y \longrightarrow \mathbf{V}_k^n.$$

The product morphism

$$f \times g : X \times_k Y \longrightarrow \mathbf{V}_k^{m+n}$$

is finite and surjective (I, 3.5.2) and (II, 6.1.5), and the result follows. $\square$

4.2. Associated prime cycles on algebraic preschemes

Proposition (4.2.1). — *Let K and L be two extensions of a field k such that $K \otimes_k L$ is Noetherian. Then the prime ideals associated with $K \otimes_k L$ are minimal, and if E is the residue field of the local ring at such an ideal, then*

$$(4.2.1.1) \quad \text{tr. deg}_K E = \text{tr. deg}_k L, \quad \text{tr. deg}_L E = \text{tr. deg}_k K,$$

whence

$$(4.2.1.2) \quad \text{tr. deg}_k E = \text{tr. deg}_k K + \text{tr. deg}_k L.$$

Proof. It is known that K is an algebraic extension of a purely transcendental extension

$$K' = k(\mathbf{t}), \quad \mathbf{t} = (t_\alpha)_{\alpha \in I},$$

where $(t_\alpha)_{\alpha \in I}$ is a family of indeterminates. The ring

$$k[\mathbf{t}] \otimes_k L = L[\mathbf{t}]$$

is integral, and hence so is $K' \otimes_k L$, which is a ring of fractions of it; the field of fractions of $K' \otimes_k L$ is $L(\mathbf{t})$. There is a commutative diagram of canonical homomorphisms

$$(4.2.1.3) \quad \begin{array}{ccccc} K & \longrightarrow & K \otimes_k L & \longrightarrow & K \otimes_{K'} L(\mathbf{t}) \\ \uparrow & & \uparrow & & \uparrow \\ K' & \longrightarrow & K' \otimes_k L & \longrightarrow & L(\mathbf{t}) \\ \uparrow & & \uparrow & & \\ k & \longrightarrow & L & & \end{array}$$

Since K is faithfully flat over K' ,

$$K \otimes_k L = K \otimes_{K'} (K' \otimes_k L)$$

is faithfully flat over $K' \otimes_k L$, and therefore $K' \otimes_k L$ is Noetherian (0_I, 6.5.2). Moreover, $K' \otimes_k L$ identifies with a subring of $K \otimes_k L$. The contraction to $K' \otimes_k L$ of a prime ideal $\mathfrak{p}$ associated with $K \otimes_k L$ is the zero ideal, since no nonzero element of $K' \otimes_k L$ is a zero-divisor in $K \otimes_k L$ (0_I, 6.3.4) and Bourbaki, *Alg. comm.*, chap. IV, § 1, no. 1, cor. 3 to prop. 2.

Since K is algebraic over K' , the ring $K \otimes_k L$ is integral over $K' \otimes_k L$, and the prime ideals of $K \otimes_k L$ lying over the zero ideal of $K' \otimes_k L$ are necessarily pairwise incomparable (Bourbaki, *Alg. comm.*, chap. V, § 2, no. 1, cor. 1 to prop. 1). This proves the first assertion (Bourbaki, *Alg. comm.*, chap. IV, § 1, no. 3, cor. 1 to prop. 7).

Furthermore, the residue field E of $\mathfrak{p}$ is algebraic over the residue field of the zero ideal of $K' \otimes_k L$, namely $L(\mathbf{t})$. Consequently

$$\mathrm{tr. deg}_L E = \mathrm{tr. deg}_L L(\mathbf{t}) = \mathrm{Card}(I) = \mathrm{tr. deg}_k K.$$

This gives the second equality in (4.2.1.1); interchanging the roles of K and L gives the first, and (4.2.1.2) follows. $\square$

Corollary (4.2.2). — *Under the hypotheses of (3.3.6), if the preschemes $T_{x,y}$ are locally Noetherian, they have no immersed associated prime cycle.*

Corollary (4.2.3). — *Under the hypotheses of (3.3.6) (respectively (3.3.7)), if the $T_{x,y}$ are locally Noetherian and if $\mathcal{F}$ and the $\mathcal{G}_s$ ($s \in S$) (respectively $\mathcal{F}$ and $\mathcal{G}$) have no immersed associated prime cycle, then neither does*

$$\mathcal{F} \otimes_{\mathcal{S}} \mathcal{G}.$$

This follows from (4.2.2), (3.3.2), and the proof of (3.3.6). In particular, since every prescheme over a field k is flat over k , we have thereby proved assertion (i) of the following proposition.

Proposition (4.2.4). — *Let k be a field, and let X and Y be two locally Noetherian k -preschemes such that $X \times_k Y$ is locally Noetherian. Suppose in addition that X and Y are integral. Then:*

- (i) $X \times_k Y$ has no immersed associated prime cycle. Each irreducible component of $X \times_k Y$ dominates X and Y , and the set of these components is in bijective correspondence with the set of irreducible components of

$$\mathrm{Spec}(\mathrm{R}(X) \otimes_k \mathrm{R}(Y)),$$

or equivalently with the set of minimal prime ideals of $\mathrm{R}(X) \otimes_k \mathrm{R}(Y)$, where $\mathrm{R}(X)$ and $\mathrm{R}(Y)$ are the fields of rational functions of X and Y , respectively.

- (ii) If a maximal point z of $X \times_k Y$ is identified with a minimal prime ideal $\mathfrak{p}$ of $\mathrm{R}(X) \otimes_k \mathrm{R}(Y)$, then the local ring

$$\mathcal{O}_{X \times_k Y, z}$$

is isomorphic to the ring of fractions

$$(\mathrm{R}(X) \otimes_k \mathrm{R}(Y))_{\mathfrak{p}}.$$

In particular, if $\mathrm{R}(X)$ or $\mathrm{R}(Y)$ is separable over k , then $X \times_k Y$ is reduced.

(iii) If, in addition, X and Y are locally of finite type over k , then every irreducible component of $X \times_k Y$ has dimension

$$\dim(X) + \dim(Y).$$

Proof. Assertion (iii) follows from (4.2.1.2), taking account of (i) and (ii).

To prove (ii), we may plainly reduce to the case in which

$$X = \operatorname{Spec}(A), \quad Y = \operatorname{Spec}(B)$$

are affine, with A and B integral rings having respective fields of fractions

$$K = R(X), \quad L = R(Y).$$

Assertion (i) shows that every minimal prime ideal $\mathfrak{q}$ of $A \otimes_k B$ is the contraction to $A \otimes_k B$ of a minimal prime ideal $\mathfrak{p}$ of $K \otimes_k L$. Since $K \otimes_k L$ is a ring of fractions of $A \otimes_k B$, the isomorphism

$$(A \otimes_k B)_{\mathfrak{q}} \simeq (K \otimes_k L)_{\mathfrak{p}}$$

follows from (0_I, 1.2.5).

Finally, if, for example, $R(Y)$ is separable over k , then $R(X) \otimes_k R(Y)$ is reduced (Bourbaki, Alg., chap. VIII, § 7, no. 3, th. 1). The local rings at the maximal points of $X \times_k Y$ are therefore also reduced, and it follows that $X \times_k Y$ is reduced (3.2.1). $\square$

Proposition (4.2.5). — *Let k be a field, let X and Y be locally Noetherian k -preschemes, and let $\mathcal{F}$ (respectively $\mathcal{G}$) be a quasi-coherent $\mathcal{O}_X$ -Module (respectively $\mathcal{O}_Y$ -Module). Let (Z'_λ) (respectively (Z''_μ)) be the family of prime cycles associated with $\mathcal{F}$ (respectively $\mathcal{G}$), and denote again by Z'_λ (respectively Z''_μ) the reduced sub-prescheme of X (respectively Y) having Z'_λ (respectively Z''_μ) as its underlying space. If $Z'_\lambda \times_k Z''_\mu$ is locally Noetherian, then its irreducible components $Z_{\lambda\mu\nu}$ dominate Z'_λ and Z''_μ , and $(Z_{\lambda\mu\nu})$ is the family of distinct prime cycles associated with $\mathcal{F} \otimes_k \mathcal{G}$.*

Proof. It suffices to apply (4.2.4) to the product

$$\operatorname{Spec}(\kappa(x)) \times_k \operatorname{Spec}(\kappa(y)).$$

$\square$

Corollary (4.2.6). — *Let k be a field, and let X and Y be two locally Noetherian k -preschemes such that $X \times_k Y$ is locally Noetherian. Let (Z'_λ) (respectively (Z''_μ)) be the family of reduced sub-preschemes of X (respectively Y) whose underlying spaces are the irreducible components of X (respectively Y). Then the irreducible components $Z_{\lambda\mu\nu}$ of $Z'_\lambda \times_k Z''_\mu$ dominate Z'_λ and Z''_μ , and $(Z_{\lambda\mu\nu})$ is the family of irreducible components of $X \times_k Y$.*

Proof. We may reduce to the case in which X and Y are reduced (I, 5.1.8). The irreducible components of X (respectively Y) are then the prime cycles associated with $\mathcal{O}_X$ (respectively $\mathcal{O}_Y$) (3.2.1). Apply (4.2.5) with

$$\mathcal{F} = \mathcal{O}_X, \quad \mathcal{G} = \mathcal{O}_Y,$$

noting that, by definition,

$$\mathcal{O}_X \otimes_k \mathcal{O}_Y = \mathcal{O}_{X \times_k Y}.$$

The corollary follows because $\mathcal{O}_X \otimes_k \mathcal{O}_Y$ has no immersed associated prime cycles, since by hypothesis the same is true of $\mathcal{O}_X$ and $\mathcal{O}_Y$ (4.2.3). $\square$

We apply the preceding results to the case in which Y is the spectrum of an extension K of k .

Proposition (4.2.7). — *Let k be a field, let X be a k -prescheme, and let K be an extension of k such that $X \otimes_k K$ is locally Noetherian. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$ -Module, let x' be a point of $X \otimes_k K$, and let x be its image in X .*

- (i) *Let (Z_λ) be the family of reduced sub-preschemes of X whose underlying spaces are the prime cycles associated with $\mathcal{F}$. Then the irreducible components $Z_{\lambda\mu}$ of $Z_\lambda \otimes_k K$ are the prime cycles associated with $\mathcal{F} \otimes_k K$, and $Z_{\lambda\mu}$ dominates Z_λ . Moreover, $Z_{\lambda\mu}$ is immersed if and only if Z_λ is immersed.*
- (ii) *The point x belongs to an immersed prime cycle associated with $\mathcal{F}$ if and only if x' belongs to an immersed prime cycle associated with $\mathcal{F} \otimes_k K$. The Module $\mathcal{F}$ has no immersed associated prime cycle if and only if the same is true of $\mathcal{F} \otimes_k K$.*

- (iii) The indices λ for which $x \in Z_\lambda$ are precisely those for which there exists an index μ such that $x' \in Z_{\lambda\mu}$. In particular, if x' belongs to only one prime cycle associated with $\mathcal{F} \otimes_k K$, then x belongs to only one prime cycle associated with $\mathcal{F}$.
- (iv) If X is locally of finite type over k , then

$$\dim(Z_{\lambda\mu}) = \dim(Z_\lambda).$$

Proof. The hypothesis implies that X itself is locally Noetherian (2.2.13) and (2.2.14). Assertion (i) follows from (4.2.5) and the proof of (4.2.3), with $\mathcal{G} = \mathcal{O}_Y$ and $Y = \text{Spec}(K)$. Assertions (ii) and (iii) follow from (i) and (2.3.5). Finally, (iv) is a special case of (4.2.4), (iii). $\square$

Corollary (4.2.8). — *If X is locally of finite type over k , then the set of dimensions of the prime cycles associated with $\mathcal{F}$ is the same as that of the prime cycles associated with $\mathcal{F} \otimes_k K$. The set of dimensions of the irreducible components is likewise the same for X and $X \otimes_k K$.*

Proposition (4.2.9). — *Suppose that the hypotheses of (4.2.5) hold, and suppose in addition that $\mathcal{F}$ and $\mathcal{G}$ are coherent. Let $(\mathcal{F}_\lambda)_{\lambda \in L}$ and $(\mathcal{G}_\mu)_{\mu \in M}$ be irredundant decompositions of $\mathcal{F}$ and $\mathcal{G}$, respectively. For every pair $(\lambda, \mu) \in L \times M$, let*

$$(\mathcal{H}_{\lambda\mu\nu})_{\nu \in S(\lambda, \mu)}$$

be a reduced irredundant decomposition of $\mathcal{F}_\lambda \otimes_k \mathcal{G}_\mu$, where

$$S(\lambda, \mu) = \text{Ass}(\mathcal{F}_\lambda \otimes_k \mathcal{G}_\mu)$$

(3.2.5). Then $(\mathcal{H}_{\lambda\mu\nu})$, over all triples (λ, μ, ν) , is an irredundant decomposition of $\mathcal{F} \otimes_k \mathcal{G}$. It is reduced if $(\mathcal{F}_\lambda)$ and $(\mathcal{G}_\mu)$ are reduced.

Proof. The Module $\mathcal{F} \otimes_k \mathcal{G}$ and each $\mathcal{F}_\lambda \otimes_k \mathcal{G}_\mu$ are coherent, and each $\mathcal{F}_\lambda \otimes_k \mathcal{G}_\mu$ identifies with a quotient of $\mathcal{F} \otimes_k \mathcal{G}$. By definition, each $\mathcal{H}_{\lambda\mu\nu}$ identifies with a coherent quotient of $\mathcal{F}_\lambda \otimes_k \mathcal{G}_\mu$, and hence also with a coherent quotient of $\mathcal{F} \otimes_k \mathcal{G}$.

The family of supports of the $\mathcal{H}_{\lambda\mu\nu}$ is locally finite. Indeed,

$$\text{Supp}(\mathcal{F}_\lambda \otimes_k \mathcal{G}_\mu) \subset |\text{Supp}(\mathcal{F}_\lambda) \times_k \text{Supp}(\mathcal{G}_\mu)|,$$

where $\text{Supp}(\mathcal{F}_\lambda)$ and $\text{Supp}(\mathcal{G}_\mu)$ denote the corresponding reduced closed sub-preschemes. For fixed λ and μ , the family of supports of the $\mathcal{H}_{\lambda\mu\nu}$ is locally finite by hypothesis; the assertion therefore follows from the local finiteness of the families $(\text{Supp}(\mathcal{F}_\lambda))$ and $(\text{Supp}(\mathcal{G}_\mu))$.

It remains to show that the canonical homomorphism

$$\mathcal{F} \otimes_k \mathcal{G} \longrightarrow \bigoplus_{\lambda, \mu, \nu} \mathcal{H}_{\lambda\mu\nu}$$

is injective. It is the composite

$$\begin{aligned} \mathcal{F} \otimes_k \mathcal{G} &\longrightarrow \left(\bigoplus_{\lambda} \mathcal{F}_\lambda \right) \otimes_k \left(\bigoplus_{\mu} \mathcal{G}_\mu \right) \\ &\simeq \bigoplus_{\lambda, \mu} (\mathcal{F}_\lambda \otimes_k \mathcal{G}_\mu) \longrightarrow \bigoplus_{\lambda, \mu, \nu} \mathcal{H}_{\lambda\mu\nu}. \end{aligned}$$

The last homomorphism is injective by definition, and the first is injective because it is the tensor product of the canonical injections

$$\mathcal{F} \longrightarrow \bigoplus_{\lambda} \mathcal{F}_\lambda, \quad \mathcal{G} \longrightarrow \bigoplus_{\mu} \mathcal{G}_\mu,$$

and $\mathcal{F}$ and $\mathcal{G}$ are flat over k .

Finally, if $(\mathcal{F}_\lambda)$ and $(\mathcal{G}_\mu)$ are reduced, we may suppose that

$$L = \text{Ass}(\mathcal{F}), \quad M = \text{Ass}(\mathcal{G}).$$

The $\mathcal{H}_{\lambda\mu\nu}$ then form a reduced decomposition because, by (4.2.2), the sets

$$\text{Ass}(\mathcal{F}_\lambda \otimes_k \mathcal{G}_\mu)$$

partition $\text{Ass}(\mathcal{F} \otimes_k \mathcal{G})$ (3.2.5). $\square$

Corollary (4.2.10). — Under the hypotheses of (4.2.7), suppose in addition that $\mathcal{F}$ is coherent, and let $(\mathcal{F}_\lambda)_{\lambda \in L}$ be an irredundant decomposition of $\mathcal{F}$. For every $\lambda \in L$, let

$$(\mathcal{F}_{\lambda\mu})_{\mu \in \text{Ass}(\mathcal{F}_\lambda \otimes_k K)}$$

be a reduced irredundant decomposition of $\mathcal{F}_\lambda \otimes_k K$. Then $(\mathcal{F}_{\lambda\mu})$ is an irredundant decomposition of $\mathcal{F} \otimes_k K$. It is reduced if $(\mathcal{F}_\lambda)$ is reduced.

Proof. Apply (4.2.9) with

$$Y = \text{Spec}(K), \quad \mathcal{G} = \mathcal{O}_Y = K,$$

and with M consisting of a single element. $\square$

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4.3. Reminders on tensor products of fields For the reader's convenience, we recall here certain properties of tensor products of fields that we shall use in the following numbers; for the proofs, we refer to [1].

(4.3.1). Recall that an extension L of a field k is called *primary* if the largest separable algebraic extension of k contained in L is k itself.

Proposition (4.3.2). — Let K and L be two extensions of a field k . If L is a primary extension of k , then $\text{Spec}(L \otimes_k K)$ is irreducible and, if ζ is its generic point, $\kappa(\zeta)$ is a primary extension of K . Conversely, if $\text{Spec}(L \otimes_k K)$ is irreducible for every finite separable extension K of k , then L is a primary extension of k .

Proof. See [1], pp. 14-03 to 14-06. $\square$

Corollary (4.3.3). — If k is separably closed (that is, if its algebraic closure is radical over k), then $\text{Spec}(L \otimes_k K)$ is irreducible for every two extensions K and L of k ; conversely, if this conclusion holds for every such pair, then k is separably closed.

Proof. This follows immediately from (4.3.2). $\square$

Corollary (4.3.4). — Let L be an extension of a field k , and let L_s be the largest separable algebraic extension of k contained in L (sometimes called the separable algebraic closure of k in L). Suppose that L_s is of finite degree over k , and let K be a Galois extension of k containing L_s . Then $\text{Spec}(L \otimes_k K)$ has $[L_s : k]$ irreducible components and is their union, and the residue fields at their generic points are primary extensions of K .

Proof. Indeed,

$$L \otimes_k K = L \otimes_{L_s} (L_s \otimes_k K),$$

and $L_s \otimes_k K$ is isomorphic to a product of $[L_s : k]$ fields, each isomorphic to K (Bourbaki, Alg., chap. VIII, § 7, no. 3, cor. 2 to th. 1). Thus $L \otimes_k K$ is isomorphic to a product of $[L_s : k]$ rings, each isomorphic to $L \otimes_{L_s} K$. Since L is a primary extension of L_s , the conclusion follows from (4.3.2). $\square$

Proposition (4.3.5). — Let K and L be two extensions of a field k . If L is a separable extension of k , then the ring $L \otimes_k K$ is reduced. Conversely, if $L \otimes_k K$ is reduced for every finite radical extension K of k , then L is a separable extension of k .

Proof. See Bourbaki, Alg., chap. VIII, § 7, no. 3, th. 1. $\square$

Corollary (4.3.6). — If k is perfect, then $K \otimes_k L$ is reduced for every two extensions K and L of k ; conversely, if this conclusion holds for every such pair, then k is perfect.

Proof. This follows immediately from (4.3.5). $\square$

Corollary (4.3.7). — Let L be a separable extension of k , and let K be an extension of k such that either K or L is finite over k . Then the residue fields of the semilocal ring $L \otimes_k K$ are separable extensions of K .

Proof. The ring $L \otimes_k K$ is the direct product of its residue fields E_i . For every extension K' of K , the ring $L \otimes_k K'$ is therefore isomorphic to the direct product of the rings $E_i \otimes_K K'$. Since $L \otimes_k K'$ is reduced, the fields E_i are separable over K by (4.3.5). $\square$

Corollary (4.3.8). — If K and L are two finite separable extensions of k , then $K \otimes_k L$ is a product of finitely many separable extensions of k .

Proposition (4.3.9). — *If k is algebraically closed, then the ring $L \otimes_k K$ is an integral domain for every two extensions K and L of k ; conversely, if this conclusion holds for every such pair, then k is algebraically closed.*

Proof. This follows from (4.3.3) and (4.3.6), since a field that is both perfect and separably closed is algebraically closed. $\square$

4.4. Irreducible preschemes and connected preschemes over an algebraically closed field

(4.4.1). Let k be a field, let X be a k -prescheme, and let K be an extension of k . Since the morphism

$$\mathrm{Spec}(K) \longrightarrow \mathrm{Spec}(k)$$

is faithfully flat, quasi-compact, and universally open (2.4.9), the same is true of the projection

$$p : X \otimes_k K \longrightarrow X$$

(2.2.13). It therefore follows from (2.3.5) that every irreducible component of $X \otimes_k K$ dominates an irreducible component of X , and, since p is surjective, every irreducible component of X is dominated by an irreducible component of $X \otimes_k K$. Thus p induces a surjective map from the set of irreducible components of $X \otimes_k K$ to the set of irreducible components of X .

Likewise, because p is continuous and surjective, every connected component of X is the image under p of a union of connected components of $X \otimes_k K$. Hence there is a surjective map from the set of connected components of $X \otimes_k K$ to the set of connected components of X . It follows that every irreducible (resp. connected) component Z' of $X \otimes_k K$ is contained in a unique set of the form $p^{-1}(Z)$, where Z is an irreducible (resp. connected) component of X . For every sub-prescheme of X having Z as its underlying space (and again denoted by Z), the space Z' is an irreducible (resp. connected) component of $Z \otimes_k K$.

In particular, suppose that $X \otimes_k K$ has finitely many irreducible (resp. connected) components, say n (resp. n'). This is the case when X is of finite type over k , since then $X \otimes_k K$ is of finite type over K and hence Noetherian. The number of irreducible (resp. connected) components of X is then at most n (resp. at most n'). Equality with n (resp. n') means that, for every reduced sub-prescheme Z of X whose underlying space is an irreducible (resp. connected) component of X , the prescheme $Z \otimes_k K$ is irreducible (resp. connected) (I, 5.1.8). Consequently, the number of irreducible (resp. connected) components of $X \otimes_k K$ is independent of K if and only if, for every irreducible (resp. connected) component Z of X , the prescheme $Z \otimes_k K$ is irreducible (resp. connected) for every extension K of k .

In particular, if $X \otimes_k K$ is irreducible (resp. connected), then so is X , as also follows directly from the surjectivity of p . Similarly, if $X \otimes_k K$ is reduced (resp. integral), then so is X , since p is faithfully flat (2.1.13).

In this number and the next, we shall examine more closely what can be said about the irreducible (resp. connected) components of $X \otimes_k K$ as K varies. The preceding observations already show that the number of these components is an increasing function of K .

Lemma (4.4.2). — *Let X' and X be two topological spaces, and let $f : X' \rightarrow X$ be a continuous map. Suppose that the following conditions hold:*

- (i) *f is open and surjective (resp. the topological space X is canonically identified with the quotient of X' by the equivalence relation defined by f);*

- (ii) *for every $x \in X$, the fibre $f^{-1}(x)$ is irreducible (resp. connected).*

Then X' is irreducible (resp. connected) if and only if X is.

Proof. The assertion concerning connectedness is Bourbaki, *Top. gén.*, chap. I, 3rd ed., § 11, no. 3, prop. 7. We prove the assertion concerning irreducibility. The condition is plainly necessary because f is surjective; let us prove that it is sufficient.

Let X'_1 and X'_2 be two closed subsets of X' such that $X' = X'_1 \cup X'_2$, and, for $i = 1, 2$, let X_i be the set of $x \in X$ such that $f^{-1}(x) \subset X'_i$. Then

$$X - X_i = f(X' - X'_i),$$

and, since f is open, X_i is a closed subset of X . For every $x \in X$, the fibre $f^{-1}(x)$ is irreducible by hypothesis and is the union of the two closed subsets

$$X'_1 \cap f^{-1}(x) \quad \text{and} \quad X'_2 \cap f^{-1}(x).$$

One of them must equal $f^{-1}(x)$, so $x \in X_1$ or $x \in X_2$. Thus $X = X_1 \cup X_2$. Since X is irreducible, $X = X_1$ or $X = X_2$, and therefore $X' = X'_1$ or $X' = X'_2$. $\square$

Remark (4.4.3). — If the topology of X is not the quotient of that of X' by the equivalence relation defined by f , it can happen that X and all the fibres $f^{-1}(x)$ are irreducible without X' being connected. For example, take for X an irreducible scheme—the affine line $\text{Spec}(k[T])$ over an algebraically closed field k , say—choose a closed point x_0 of X , and take for X' the topological sum of $\{x_0\}$ and the open subspace $X - \{x_0\}$ of X .

Likewise, if the hypothesis that f is open is replaced by the hypothesis that f is closed, or even proper, then it can happen that X and all the fibres $f^{-1}(x)$ are irreducible without X' being irreducible, even though this replacement does ensure that the topology of X is the quotient of that of X' by the relation defined by f . Again take X to be the affine line over k , and put

$$S = X \times_k \mathbf{P},$$

where $\mathbf{P}$ is the projective line over k . Let x_0 be a closed point of X , let t_0 be a closed point of $\mathbf{P}$, and let

$$p : S \longrightarrow X, \quad q : S \longrightarrow \mathbf{P}$$

be the projections. Let X' be the reduced sub-prescheme whose underlying space is the closed set

$$p^{-1}(x_0) \cup q^{-1}(t_0),$$

and let f be the restriction of p to X' . Then f is proper, but X' is not irreducible.

Theorem (4.4.4). — *Let k be an algebraically closed field and let X be a k -prescheme. If X is irreducible (resp. connected), then $X \otimes_k K$ is irreducible (resp. connected) for every extension K of k .*

Proof. We saw in (4.4.1) that

$$p : X \otimes_k K \longrightarrow X$$

is faithfully flat and open. To apply (4.4.2), it therefore suffices to verify that every fibre $p^{-1}(x)$ is irreducible. The $\kappa(x)$ -prescheme $p^{-1}(x)$ is isomorphic to

$$\text{Spec}(\kappa(x) \otimes_k K)$$

(I, 3.6.2), and is therefore integral by the hypothesis on k and (4.3.9). $\square$

Corollary (4.4.5). — *Let k be an algebraically closed field, let X be a k -prescheme, let K be an extension of k , and let*

$$p : X \otimes_k K \longrightarrow X$$

be the canonical projection. If Z is an irreducible (resp. connected) subset of X , then $p^{-1}(Z)$ is irreducible (resp. connected). In particular, if X_0 is an irreducible component (resp. the connected component) of X containing Z , then $p^{-1}(X_0)$ is an irreducible (resp. connected) component of $X \otimes_k K$ containing $p^{-1}(Z)$.

Proof. The second assertion follows from (4.4.1) and the first assertion applied after replacing Z by X_0 . To prove the first, let X' be the underlying space of $X \otimes_k K$, and put $Z' = p^{-1}(Z)$.

The equivalence relation R defined by p on X' is open. The induced relation $R_{Z'}$ on the saturated subset Z' is therefore also open, and Z identifies with the quotient space $Z' / R_{Z'}$ (Bourbaki, *Top. gén.*, chap. I, 3rd ed., § 5, no. 2, prop. 4). We may thus apply (4.4.2) to Z and Z' , proving the first assertion of (4.4.5). $\square$

Corollary (4.4.6). — *Under the hypotheses and notation of (4.4.5), the map*

$$Z \longmapsto p^{-1}(Z)$$

is a bijection from the set of irreducible (resp. connected) components of X to the set of irreducible (resp. connected) components of $X \otimes_k K$. Its inverse is

$$Z' \longmapsto p(Z').$$

4.5. Geometrically irreducible and geometrically connected preschemes

Proposition (4.5.1). — *Let k be a field, let X be a k -prescheme, and let Ω be an algebraically closed extension of k . Let n (resp. n') be the cardinality of the set of irreducible components (resp. connected components) of $X \otimes_k \Omega$. This cardinality is independent of the chosen algebraically closed extension Ω . Moreover, for every extension K of k , the cardinality of the set of irreducible components (resp. connected components) of $X \otimes_k K$ is at most n (resp. at most n').*

Proof. Two algebraically closed extensions Ω and Ω' of k may always be regarded as subextensions of a common extension E of k . Applying (4.4.6) to $X \otimes_k \Omega$ and $X \otimes_k \Omega'$, and to the common extensions E/Ω and E/Ω' , proves the first assertion. For the second, take Ω to be an algebraically closed extension of K and use (4.4.1). $\square$

Definition (4.5.2). — The cardinality n (resp. n') of (4.5.1) is called the *geometric number of irreducible components* (resp. *of connected components*) of X relative to k . If $n = 1$ (resp. $n' \leq 1$), the k -prescheme X is said to be *geometrically irreducible* (resp. *geometrically connected*).

This recovers the definition given in the special case (III, 4.3.4). By (4.5.1), the following properties are equivalent:

- (a) X is geometrically irreducible (resp. geometrically connected);
- (b) for one algebraically closed extension Ω of k , $X \otimes_k \Omega$ is irreducible (resp. connected);
- (c) for every extension K of k , $X \otimes_k K$ is irreducible (resp. connected).

(4.5.3). Let X be a k -prescheme and let Z be a locally closed subset of X . If any sub-prescheme of X having Z as underlying space is again denoted by Z (I, 5.2.1), it follows from (I, 5.1.8) that, for an extension K of k , the number of irreducible (resp. connected) components of $Z \otimes_k K$ is independent of the chosen sub-prescheme with underlying space Z . The *geometric number of irreducible components* (resp. *of connected components*) of Z is therefore defined to be that of any sub-prescheme of X having Z as underlying space.

Proposition (4.5.4). — *Let X and Y be two k -preschemes, and let $f : X \rightarrow Y$ be a surjective k -morphism. If X is geometrically irreducible (resp. geometrically connected), then so is Y .*

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Proof. For every extension K of k , the morphism

$$f_K : X \otimes_k K \longrightarrow Y \otimes_k K$$

is surjective (I, 3.5.2). Since $X \otimes_k K$ is irreducible (resp. connected) by hypothesis, the same is true of $Y \otimes_k K$. $\square$

Definition (4.5.5). — A morphism $f : X \rightarrow Y$ of preschemes is said to be *irreducible* (resp. *connected*) if, for every $y \in Y$, the $\kappa(y)$ -prescheme $f^{-1}(y)$ is geometrically irreducible (resp. geometrically connected).

Proposition (4.5.6). — (i) *Let X be a k -prescheme and let K be an extension of k . The geometric number of irreducible components (resp. connected components) of the K -prescheme $X \otimes_k K$, relative to K , is equal to the geometric number of irreducible components (resp. connected components) of X , relative to k . In particular, $X \otimes_k K$ is geometrically irreducible (resp. geometrically connected) as a K -prescheme if and only if X is so as a k -prescheme.*

(ii) *Let $f : X \rightarrow Y$ and $g : Y' \rightarrow Y$ be two morphisms, and put*

$$X' = X_{(Y')} = X \times_Y Y', \quad f' = f_{(Y')} : X' \longrightarrow Y'.$$

If f is irreducible (resp. connected), then so is f' . The converse holds if g is surjective.

Proof. For (i), if Ω is an algebraically closed extension of K , then

$$X \otimes_k \Omega = (X \otimes_k K) \otimes_K \Omega,$$

which proves the assertion.

For (ii), for every $y' \in Y'$, put $y = g(y')$. Then

$$f'^{-1}(y') = f^{-1}(y) \otimes_{\kappa(y)} \kappa(y')$$

(I, 3.6.4), so both assertions follow from (i). $\square$

Proposition (4.5.7). — *Let $f : X \rightarrow Y$ be a surjective irreducible (resp. connected) morphism, let $g : Y' \rightarrow Y$ be a morphism, and put $X' = X \times_Y Y'$. Suppose in addition that Y' is irreducible (resp. connected), and that f is universally open (resp. flat and quasi-compact, or universally open, or universally closed). Then X' is irreducible (resp. connected).*

Proof. All the properties of f under consideration are stable under base change, so it is enough to treat the case $Y' = Y$. The result then follows from (4.4.2), taking (2.3.12) into account. $\square$

Corollary (4.5.8). — *Let X and Y be two k -preschemes.*

- (i) *If X is geometrically irreducible (resp. geometrically connected) and Y is irreducible (resp. connected), then $X \times_k Y$ is irreducible (resp. connected).*
- (ii) *If X and Y are geometrically irreducible (resp. geometrically connected), then so is $X \times_k Y$.*

Proof. For (i), the structure morphism $X \rightarrow \text{Spec}(k)$ is surjective and universally open (2.4.9), and it is also irreducible (resp. connected). It is therefore enough to apply (4.5.7).

For (ii), if Ω is an algebraically closed extension of k , then

$$(X \times_k Y) \otimes_k \Omega = (X \otimes_k \Omega) \times_{\Omega} (Y \otimes_k \Omega)$$

(I, 3.3.10). By hypothesis and (4.5.6), the two factors on the right are geometrically irreducible (resp. geometrically connected) Ω -preschemes, so (i) applies. $\square$

Proposition (4.5.9). — *Let X be a k -prescheme. The following conditions are equivalent:*

- (a) *X is geometrically irreducible; in other words, $X \otimes_k K$ is irreducible for every extension K of k ;*
- (b) *$X \otimes_k K$ is irreducible for every finite separable extension K of k ;*
- (c) *X is irreducible and, if x is its generic point, $\kappa(x)$ is a primary extension of k .*

Proof. It is clear that (a) implies (b). To see that (b) implies (c), let K be a finite separable extension of k and let

$$p : X \otimes_k K \longrightarrow X$$

be the canonical projection. The maximal points of $X \otimes_k K$ are those of the fibre

$$p^{-1}(x) = \text{Spec}(\kappa(x) \otimes_k K)$$

(2.3.4) and (0_I, 2.1.8). Thus, saying that this fibre is irreducible for every finite separable extension K of k is equivalent, by (4.3.2), to saying that $\kappa(x)$ is a primary extension of k . Conversely, if (c) holds, the same argument, again using (4.3.2), shows that $X \otimes_k K$ is irreducible for every extension K of k . Hence (c) implies (a). $\square$

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Corollary (4.5.10). — *Let X be an irreducible k -prescheme, let x be its generic point, let k' be the separable algebraic closure of k in $\kappa(x)$, and let k'' be a Galois extension of k , of finite degree or not, containing k' . Suppose that k' is finite over k . Then the irreducible components of $X \otimes_k k''$ are geometrically irreducible; their number is $[k' : k]$, and this is also the geometric number of irreducible components of X .*

Proof. As in (4.5.9), it is enough to consider the maximal points of

$$\text{Spec}(\kappa(x) \otimes_k k'').$$

By (4.3.4), there are $[k' : k]$ of them, and their residue fields, which are also the residue fields at the maximal points of $X \otimes_k k''$, are primary extensions of k'' . The result follows from (4.5.9). $\square$

Corollary (4.5.11). — *Let X be a k -prescheme. Suppose that X has only finitely many maximal points x_i ($1 \leq i \leq r$) and that, for each i , the separable algebraic closure k'_i of k in $\kappa(x_i)$ has finite degree over k . Then there is a finite separable extension L of k such that the irreducible components of $X \otimes_k L$ are geometrically irreducible. Their number, equal to*

$$\sum_i [k'_i : k],$$

is the geometric number of irreducible components of X .

Proof. The maximal points of $X \otimes_k L$ are those of the various fibres

$$\mathrm{Spec}(\kappa(x_i) \otimes_k L),$$

by (2.3.4) and (0_I, 2.1.8), applied to the irreducible components of X and their inverse images in $X \otimes_k L$. It therefore suffices to take for L a finite Galois extension of k containing all the k'_i and to apply (4.5.10). $\square$

Notice that (4.5.11) applies in particular to every k -prescheme X of finite type over k . Indeed, X is then Noetherian, so it has only finitely many irreducible components; moreover, each $\kappa(x_i)$ is an extension of finite type of k , and therefore so is the algebraic closure of k in $\kappa(x_i)$ ([1], p. 6-06, Lemma 5).

Remarks (4.5.12). — (i) The notions defined in (4.5.2) depend on the base field k . For brevity, when it is necessary to name the base field, we may say “ k -irreducible” (resp. “ k -connected”) instead of “geometrically irreducible (resp. connected) relative to k ”. This terminology is natural in the context of schemes, but is exactly opposite to Weil’s. For Weil, “ k -irreducible” (resp. “ k -connected”, “ k -normal”, and so on) designates an intrinsic property of a prescheme X , independent of the field k chosen as base field (that is, independent of the chosen morphism $X \rightarrow \mathrm{Spec}(k)$); we express these properties simply by saying that X is irreducible (resp. connected, normal, and so on). On the other hand, Weil says “absolutely irreducible”, “absolutely connected”, “absolutely normal”, and so on, for the corresponding notions relative to $X \otimes_k \Omega$, where Ω is a suitable algebraically closed extension of k . This terminology is explained by his different point of view: an “algebraic variety” V is first given as a geometric object over an algebraically closed field Ω ; the datum of a smaller “field of definition” k (that is, a scheme over k which yields V after extension of the base field to Ω) is an additional and, to some extent, secondary structure. Thus his use of “absolute” or “intrinsic” on the one hand and “relative to k ” on the other is opposite to ours.³ We shall therefore avoid the qualifier “absolute”, which may cause confusion; whenever we use the abbreviated terminology introduced above, in conflict with established terminology, we shall refer to (4.5.12) to avoid ambiguity.

(ii) Proposition (4.5.9) gives a “birational” criterion—one depending only on the residue field at the generic point—for a k -prescheme X to be geometrically irreducible. There is no analogous criterion for geometric connectedness. For example, take

$$X = \mathrm{Spec}(\mathbf{R}[[T, U]]/(T^2 + U^2)),$$

where T and U are indeterminates. Then X is an integral $\mathbf{R}$ -scheme, and its field of rational functions $\kappa(x)$ is $\mathbf{C}((T))$, as is readily verified. The scheme $X \otimes_{\mathbf{R}} \mathbf{C}$ is the union of two irreducible components, the two “isotropic lines”, which have a point in common: the maximal ideal in $\mathbf{C}[[T, U]]/(T^2 + U^2)$ induced by the maximal ideal $(T) + (U)$ of $\mathbf{C}[[T, U]]$. Thus $X \otimes_{\mathbf{R}} \mathbf{C}$ is connected. Let X' be the open complement of the closed point of X corresponding to the maximal ideal $(T) + (U)$ of $\mathbf{R}[[T, U]]$. Then $X' \otimes_{\mathbf{R}} \mathbf{C}$ is not connected, although X and X' have the same field of rational functions.

Proposition (4.5.13). — *Let X and Y be two k -preschemes and let $f : Y \rightarrow X$ be a k -morphism. Suppose that Y is nonempty and geometrically connected, and that X is connected. Then X is geometrically connected.*

Proof. Let Ω be an algebraic closure of k , put

$$X' = X \otimes_k \Omega, \quad Y' = Y \otimes_k \Omega,$$

and let $f' = f_{(\Omega)} : Y' \rightarrow X'$. It is enough to prove that X' is connected. Let U' be a nonempty open-and-closed subset of X' . The projections

$$p : X' \longrightarrow X, \quad q : Y' \longrightarrow Y$$

are open (2.4.10), and they are closed because Ω/k is algebraic (II, 6.1.10). Consequently $U = p(U')$ is a nonempty open-and-closed subset of X , and hence $U = X$. Since Y is nonempty, it follows from

³Zariski’s point of view is already closer to ours, and his terminology is not generally in conflict with that introduced here.

(I, 3.4.7) that

$$V' = f'^{-1}(U')$$

is nonempty. It is also open and closed in Y' , which is connected by hypothesis, so $V' = Y'$. We conclude that $U' = X'$: otherwise, applying the same argument to the open-and-closed subset $X' \setminus U'$ would give $f'^{-1}(X' \setminus U') = Y'$, which is impossible because Y' is nonempty. $\square$

Corollary (4.5.13.1). — (i) *Let X and Y be two k -preschemes and let $f : Y \rightarrow X$ be a k -morphism. If Y is nonempty and geometrically connected, then the connected component X_0 of X containing $f(Y)$ is geometrically connected.*

(ii) *Let X be a k -prescheme. If Y is a geometrically irreducible irreducible component of X , then the connected component X_0 of X containing Y is geometrically connected.*

Proof. For (i), we may suppose that Y is reduced, so that f factors as

$$Y \xrightarrow{g} X_0 \xrightarrow{j} X,$$

where X_0 denotes the reduced sub-prescheme of X whose underlying space is the connected component X_0 (I, 5.2.2). Apply (4.5.13) to g .

For (ii), choose a prescheme structure on the underlying space Y . Since a geometrically irreducible prescheme is in particular geometrically connected, (i) applies to the canonical injection $Y \rightarrow X$. $\square$

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Corollary (4.5.14). — *Let X be a k -prescheme. If $x \in X$ is such that $\kappa(x)$ is a primary extension of k (in particular, if x is a rational point of X), then the connected component of x in X is geometrically connected.*

Proof. Apply (4.5.13.1) to $Y = \text{Spec}(\kappa(x))$ and to the canonical morphism $Y \rightarrow X$. By (4.5.9), the prescheme Y is geometrically irreducible. $\square$

Proposition (4.5.15). — *Let X be a k -prescheme, let $x \in X$, and let k' be the separable algebraic closure of k in $\kappa(x)$. Suppose that:*

- (i) X is connected;
- (ii) k' is a finite extension of k .

Then the geometric number of connected components of X is at most $[k' : k]$. Moreover, if k'' is a finite Galois extension of k containing k' , the connected components of $X \otimes_k k''$ are geometrically connected.

Condition (ii) is satisfied in particular when X is locally of finite type over k .

Proof. By (ii), there are finite Galois extensions k'' of k containing k' . Put $X'' = X \otimes_k k''$. The projection $p : X'' \rightarrow X$ is faithfully flat and finite, hence closed (II, 6.1.10). It follows from (2.3.6), (ii), that the image under p of every connected component X''_α of X'' is all of X . Thus X''_α contains a point $x''_\alpha \in p^{-1}(x)$.

The number of points of $p^{-1}(x)$ is the number of irreducible components of

$$\text{Spec}(\kappa(x) \otimes_k k'')$$

(I, 3.4.9), namely $[k' : k]$ by (4.3.4). Therefore the number of connected components of X'' is at most $[k' : k]$. For every $x''_\alpha \in p^{-1}(x)$, the field $\kappa(x''_\alpha)$ is a primary extension of k'' , by (4.3.5) and (I, 3.4.9). Corollary (4.5.14), applied to x''_α and X'' , now shows that every connected component of X'' is geometrically connected. $\square$

Corollary (4.5.16). — *Let X be a k -prescheme, let (X_α) be the family of its connected components, and choose $x_\alpha \in X_\alpha$ for every α . Suppose that:*

- (i) *the family (X_α) is finite;*
- (ii) *for every α , the separable algebraic closure k'_α of k in $\kappa(x_\alpha)$ is a finite extension of k .*

Then the geometric number of connected components of X is at most

$$\sum_{\alpha} [k'_\alpha : k],$$

and there exists a finite separable extension k'' of k such that all the connected components of $X \otimes_k k''$ are geometrically connected.

Proof. By (i), the connected components of X are open. Apply (4.5.15) to each sub-prescheme induced on the open subset X_α . To obtain one extension k'' with the asserted property, take a finite Galois extension of k containing all the k'_α . $\square$

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Corollary (4.5.17). — Suppose that the k -prescheme X contains a point x such that the separable algebraic closure of k in $\kappa(x)$ is finite over k . Then X is geometrically connected if and only if $X \otimes_k K$ is connected for every finite separable extension K of k .

Remark (4.5.18). — We shall see in § 8 (8.4.5) that the conclusion of (4.5.17) remains valid if, instead of the hypothesis in its statement, one assumes that X is quasi-compact.

Proposition (4.5.19). — Let X be a k -prescheme, let Z be a subset of X , let k' be an algebraically closed extension of k , put

$$X' = X \otimes_k k',$$

and let $p : X' \rightarrow X$ be the canonical projection. Suppose that $Z' = p^{-1}(Z)$ is contained in a single irreducible component X'_0 of X' . Then

$$X'_0 = p^{-1}(X_0),$$

where $X_0 = p(X'_0)$ is an irreducible component of X containing Z ; moreover, X_0 is geometrically irreducible. If, in addition, X'_0 is the only irreducible component of X' which meets Z' , then X_0 is the only irreducible component of X which meets Z .

To prove that $X'_0 = p^{-1}(X_0)$, we shall use the following lemma.

Lemma (4.5.19.1). — Let T and T' be preschemes, let $p : T' \rightarrow T$ be a morphism, put

$$T'' = T' \times_T T',$$

and let $p_1, p_2 : T'' \rightarrow T'$ be the canonical projections. A subset U' of T' is of the form $p^{-1}(U)$, for some subset $U \subset T$, if and only if

$$p_1^{-1}(U') = p_2^{-1}(U').$$

Proof. The structure morphism $q : T'' \rightarrow T$ is, by definition, both $p \circ p_1$ and $p \circ p_2$, so the condition is necessary. Conversely, suppose the condition holds. Let $u' \in U'$ and let $t' \in p^{-1}(p(u'))$. Since $p(t') = p(u')$, there is a point $t'' \in T''$ such that

$$p_1(t'') = u', \quad p_2(t'') = t'$$

(I, 3.4.7). Thus $t'' \in p_1^{-1}(U') = p_2^{-1}(U')$, and consequently $t' = p_2(t'') \in U'$. This proves the lemma. $\square$

Proof of (4.5.19). Form the fibre product

$$X'' = X' \times_X X'$$

relative to $p : X' \rightarrow X$, and let $p_1, p_2 : X'' \rightarrow X'$ be the canonical projections. By (4.5.19.1), it is enough to prove that

$$p_1^{-1}(X'_0) = p_2^{-1}(X'_0).$$

Put

$$S = \text{Spec}(k), \quad S' = \text{Spec}(k'), \quad S'' = S' \times_S S' = \text{Spec}(k' \otimes_k k').$$

Then X'' may also be written $X' \times_{S'} S''$. If $\varphi'' : X'' \rightarrow S''$ is the structure morphism, it is enough to show, for every $s'' \in S''$, that

$$p_1^{-1}(X'_0) \cap \varphi''^{-1}(s'') = p_2^{-1}(X'_0) \cap \varphi''^{-1}(s'').$$

Put

$$T = \text{Spec}(\kappa(s'')), \quad U = X'' \times_{S''} T = \varphi''^{-1}(s''),$$

and let $v : U \rightarrow X''$ be the canonical projection. We must prove that

$$U_1 = v^{-1}(p_1^{-1}(X'_0)), \quad U_2 = v^{-1}(p_2^{-1}(X'_0)),$$

which are irreducible components of U by (4.4.5), are equal. The relevant morphisms form the commutative diagram

$$(4.5.19.2) \quad \begin{array}{ccccccc} X & \xleftarrow{p} & X' & \xleftarrow[p_2]{p_1} & X'' & \xleftarrow{v} & U \\ \downarrow & & \downarrow & & \downarrow \varphi'' & & \downarrow \\ S & \xleftarrow{\quad} & S' & \xleftarrow{\quad} & S'' & \xleftarrow{\quad} & T \end{array}$$

By construction, U_1 and U_2 both contain $w^{-1}(Z)$, where

$$w = p \circ p_1 \circ v = p \circ p_2 \circ v.$$

There is a field K which is a common extension of k' and $\kappa(s')$. If $P = \text{Spec}(K)$, the diagram

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$$\begin{array}{ccc} S' & \xleftarrow{\quad} & P \\ \downarrow & & \downarrow \\ S & \xleftarrow{\quad} & T \end{array}$$

is commutative. Putting $Q = X \otimes_k K$, the diagram

$$\begin{array}{ccc} X' & \xleftarrow{q} & Q \\ \downarrow p & & \downarrow r \\ X & \xleftarrow{w} & U \end{array}$$

is likewise commutative. The two irreducible components $r^{-1}(U_1)$ and $r^{-1}(U_2)$ of Q (4.4.5) therefore contain $q^{-1}(Z')$. By (4.4.5), their images

$$q(r^{-1}(U_1)) \quad \text{and} \quad q(r^{-1}(U_2))$$

are irreducible components of X' containing Z' . Both are therefore X'_0 . Again by (4.4.5), $r^{-1}(U_1) = r^{-1}(U_2)$, and hence $U_1 = U_2$.

Since p is surjective and X'_0 dominates an irreducible component of X , the subset $X_0 = p(X'_0)$ is an irreducible component of X containing Z . Since $p^{-1}(X_0) = X'_0$ is an irreducible component of X' , the component X_0 is geometrically irreducible.

For the last assertion, let X_1 be another irreducible component of X meeting Z . Choose $z \in Z \cap X_1$. There is an irreducible component of X' which meets Z' and dominates X_1 (2.3.5). By hypothesis X'_0 is the only such component, so necessarily $X_1 = X_0$. $\square$

Remark (4.5.20). — The fact that Z' is contained in a single irreducible component X'_0 of X' does *not* imply that $X_0 = p(X'_0)$ is the only irreducible component of X containing Z . Indeed, take $k = \mathbf{R}$, $k' = \mathbf{C}$, and let X be the k -scheme obtained by gluing⁴ the following two schemes at a nongeneric point of each:

$$X_1 = \text{Spec}(\mathbf{R}[S, T]/(S^2 + T^2 + 1)), \quad X_2 = \text{Spec}(\mathbf{C}[T]);$$

the local rings of these two curves at each of their closed points have the same residue field, isomorphic to $\mathbf{C}$. The schemes X_1 and X_2 identify with the two irreducible components of X . If x is their common point, then $p^{-1}(x)$ consists of two distinct points y' and z' ; the inverse image $p^{-1}(X_1)$ is irreducible, while $p^{-1}(X_2)$ is the union of two irreducible components Y' and Z' of X' , with $y' \in Y'$ and $z' \in Z'$. Thus $p^{-1}(x)$ is contained in only one irreducible component of X' .

Proposition (4.5.21). — *Let k be a separably closed field and let k' be an algebraic closure of k . For every k -prescheme X , the canonical projection*

$$X \otimes_k k' \longrightarrow X$$

is a universal homeomorphism. In particular, the irreducible components (resp. connected components) of X are geometrically irreducible (resp. geometrically connected).

⁴The technique of gluing preschemes will be developed in detail in Chapter V. For the case considered here, which concerns algebraic curves, see [38], pp. 68–71.

Proof. By definition, k'/k is radical. The morphism $\mathrm{Spec}(k') \rightarrow \mathrm{Spec}(k)$ is integral, surjective, and radical, so the first assertion follows from (2.4.5), (i). The second follows from the first and (4.5.1). $\square$

4.6. Geometrically reduced algebraic preschemes

Proposition (4.6.1). — *Let k be a field, let X be a k -prescheme, and let Ω be a perfect extension of k . The following conditions are equivalent:*

- (a) *for every reduced k -prescheme S , the prescheme $X \times_k S$ is reduced;*
- (b) *for every extension K of k , the prescheme $X \otimes_k K$ is reduced;*
- (c) *the prescheme $X \otimes_k \Omega$ is reduced;*
- (d) *for every finite radical extension k' of k , the prescheme $X \otimes_k k'$ is reduced;*
- (e) *X is reduced and, for every irreducible component X_α of X , with generic point x_α , the field $\kappa(x_\alpha)$ is a separable extension of k .*

Proof. The implications (a) $\Rightarrow$ (b) $\Rightarrow$ (c) are immediate. Since k' may be regarded as a subextension of Ω , (4.4.1) shows that (c) implies (d).

Suppose (d). Taking $k' = k$ first shows that X is reduced. To prove that $\kappa(x_\alpha)$ is separable over k , it suffices to show that $\kappa(x_\alpha) \otimes_k k'$ is reduced for every finite radical extension k' of k (4.3.5). We may reduce to the case $X = \mathrm{Spec}(A)$, where A is a reduced k -algebra. Then $A \otimes_k k'$ is reduced by hypothesis (I, 5.1.4), and $\kappa(x_\alpha) \otimes_k k'$ is a ring of fractions of this ring; hence it is reduced (0_I, 1.2.8). Thus (d) implies (e).

Finally, suppose (e). We may reduce to the affine case $X = \mathrm{Spec}(A)$ and $S = \mathrm{Spec}(B)$, where A and B are k -algebras. The fields $\kappa(x_\alpha)$ are the fields of fractions of the quotient rings $A/\mathfrak{p}_\alpha$, where $\mathfrak{p}_\alpha = \mathfrak{j}_{x_\alpha}$ are the minimal prime ideals of A . Since $\bigcap_\alpha \mathfrak{p}_\alpha = 0$, the algebra A embeds in $\prod_\alpha A/\mathfrak{p}_\alpha$, and hence in $\prod_\alpha \kappa(x_\alpha)$. Likewise, B may be regarded as a subalgebra of a product $\prod_\beta L_\beta$ of extensions of k . It follows that $A \otimes_k B$ embeds in

$$\left(\prod_\alpha \kappa(x_\alpha) \right) \otimes_k \left(\prod_\beta L_\beta \right),$$

and this tensor product itself embeds in

$$\prod_{\alpha, \beta} (\kappa(x_\alpha) \otimes_k L_\beta)$$

(Bourbaki, *Alg.*, chap. II, 3rd ed., § 7, n° 7, prop. 15). By hypothesis the rings $\kappa(x_\alpha) \otimes_k L_\beta$ are reduced (4.3.5); hence $A \otimes_k B$ is reduced. Thus (e) implies (a). $\square$

Definition (4.6.2). — If the equivalent conditions (a)–(e) of (4.6.1) hold, X is said to be *separable* (or *geometrically reduced*, or *universally reduced*) over k . A k -prescheme X is said to be *geometrically integral* over k if, for every extension K of k , the prescheme $X \otimes_k K$ is integral. By (4.6.1), this is equivalent to saying that X is separable and geometrically irreducible over k .

A commutative k -algebra A is said to be *separable* if $\mathrm{Spec}(A)$ is separable over k ; equivalently, for every extension K of k , the ring $A_{(K)} = A \otimes_k K$ is reduced. This definition agrees with that of Bourbaki, *Alg.*, chap. VIII, § 7, n° 5, def. 1, when A has finite rank over k (loc. cit., corollary to prop. 7), but not in general: a k -algebra may have nonzero radical even when it is integral.

Corollary (4.6.3). — *Let X be an integral k -prescheme. Then X is geometrically reduced (resp. geometrically integral) over k if and only if its field of rational functions $R(X)$ is a separable (resp. separable and primary) extension of k .*

Proof. This follows immediately from (4.5.9) and (4.6.1). $\square$

Corollary (4.6.4). — *Let X be a reduced k -prescheme. For every separable extension k' of k , the prescheme $X' = X \otimes_k k'$ is reduced.*

Proof. Apply the equivalence of (a) and (e) in (4.6.1), replacing X there by $\mathrm{Spec}(k')$ and S by X . $\square$

- Proposition (4.6.5).** — (i) Let X be a k -prescheme and let K be an extension of k . Then X is geometrically reduced (resp. geometrically integral) over k if and only if $X \otimes_k K$ is geometrically reduced (resp. geometrically integral) over K .
(ii) Let X and Y be two k -preschemes. If X and Y are geometrically reduced (resp. geometrically integral) over k , then so is $X \times_k Y$.

Proof. Assertion (i) is an immediate consequence of the definitions and (4.4.1). For the reduced assertion in (ii), let Ω be an algebraically closed extension of k . Then

$$(X \times_k Y) \otimes_k \Omega = X \times_k (Y \otimes_k \Omega).$$

Since $Y \otimes_k \Omega$ is reduced, the right-hand side is reduced by (4.6.1.a). The remainder of (ii) follows from this and (4.5.8.ii). $\square$

Proposition (4.6.6). — Let X be a k -prescheme of finite type. There exists a finite radicial extension k' of k such that $(X_{(k')})_{\text{red}}$ is geometrically reduced over k' .

Proof. Since X is covered by finitely many affine open subsets U_i , we may reduce to the affine case. Indeed, suppose for each i that k_i is a finite radicial extension of k such that $(U_i \otimes_k k_i)_{\text{red}}$ is geometrically reduced over k_i . We may choose a common finite radicial extension k' of k containing all the k_i . By (I, 5.1.8),

$$(U_i \otimes_k k')_{\text{red}} \simeq (U_i \otimes_k k_i)_{\text{red}} \otimes_{k_i} k',$$

and this is geometrically reduced over k' . Thus the same holds for $(X \otimes_k k')_{\text{red}}$.

Now let $X = \text{Spec}(A)$, where A is a k -algebra of finite type, and let Ω be an algebraic closure of k . Write $\mathfrak{N}'$ for the nilradical of $A \otimes_k \Omega$. Since $A \otimes_k \Omega$ is Noetherian, $\mathfrak{N}'$ is generated by finitely many elements

$$y_i = \sum_j x_{ij} \otimes \xi_{ij}, \quad x_{ij} \in A, \quad \xi_{ij} \in \Omega.$$

Let K be a finite subextension of Ω containing all the ξ_{ij} , and let $\mathfrak{N}$ be the ideal of $A \otimes_k K$ generated by the y_i , regarding $A \otimes_k K$ as a subring of $A \otimes_k \Omega$. The y_i are nilpotent in $A \otimes_k K$. Moreover, $\mathfrak{N} \otimes_K \Omega = \mathfrak{N}'$, and hence

$$\mathfrak{N}' \cap (A \otimes_k K) = \mathfrak{N}.$$

Thus $\mathfrak{N}$ contains the nilradical of $A \otimes_k K$ and is therefore equal to it. Since

$$(A \otimes_k \Omega) / \mathfrak{N}' = ((A \otimes_k K) / \mathfrak{N}) \otimes_K \Omega,$$

the prescheme $(X \otimes_k K)_{\text{red}} \otimes_K \Omega$ is reduced. Consequently $(X_{(K)})_{\text{red}}$ is geometrically reduced over K by (4.6.1).

Replacing K by the quasi-Galois extension of k that it generates in Ω , we may further suppose, by (I, 5.1.8), that K is quasi-Galois over k . It is then a separable extension of a finite radicial extension k' of k (Bourbaki, *Alg.*, chap. V, § 10, n° 9, prop. 14). By (I, 5.1.8), $(X_{(k')})_{\text{red}} \otimes_{k'} K$ is isomorphic to $(X_{(K)})_{\text{red}}$, and hence is geometrically reduced. It follows from (4.6.5.i) that $(X_{(k')})_{\text{red}}$ is geometrically reduced. $\square$

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Corollary (4.6.7). — If K is an extension of finite type of k , there exists a finite radicial extension k' of k such that the residue fields of the semilocal ring $K \otimes_k k'$ are separable over k' .

Proof. Apply (4.6.6) to $X = \text{Spec}(A)$, where A is a k -algebra of finite type having K as its field of fractions. $\square$

Corollary (4.6.8). — Let X be a k -prescheme of finite type. There exists a finite extension k' of k such that $(X_{(k')})_{\text{red}}$ is geometrically reduced over k' , the irreducible components of $X_{(k')}$ are geometrically irreducible, and its connected components are geometrically connected.

Proof. This follows immediately from (4.6.6), (4.5.10), and (4.5.15). $\square$

Definition (4.6.9). — Let k be a field, let X be a k -prescheme, and let $x \in X$. We say that X is geometrically reduced (or separable) (resp. geometrically punctually integral) at x over k if, for every extension k' of k and every point x' of $X' = X \otimes_k k'$ above x , the prescheme X' is reduced (resp. integral) at x' ; that is, $\mathcal{O}_{X', x'}$ is reduced (resp. integral).

We say that X is *geometrically punctually integral* if it is geometrically punctually integral at all of its points. Equivalently, for every extension k' of k , all local rings of $X \otimes_k k'$ are integral; in that case $X \otimes_k k'$ is also said to be *punctually integral*.

By (4.6.2), X is geometrically reduced over k if and only if it is geometrically reduced over k at every point $x \in X$.

Proposition (4.6.10). — *Let k be a field, let X be a k -prescheme, let k' be an extension of k , and let x' be a point of $X' = X \otimes_k k'$ with image x in X . Then X is geometrically reduced (resp. geometrically punctually integral) at x over k if and only if X' has the corresponding property at x' over k' . In particular, X is geometrically punctually integral over k if and only if X' is geometrically punctually integral over k' .*

Proof. It is enough to prove the first assertion. Necessity is immediate. Suppose conversely that X' is geometrically reduced (resp. geometrically punctually integral) at x' over k' . Let k'' be an extension of k and let x'' be a point of $X'' = X \otimes_k k''$ above x . There is a point z of $X' \times_X X''$ projecting to x' and x'' (I, 3.4.7). Let s be the image of z in $\text{Spec}(k' \otimes_k k'')$ (I, 3.4.9), put $K = \kappa(s)$, and put $Z = X \otimes_k K$. Then K may be regarded as a composite extension of k' and k'' , and z as a point of Z whose images in X' and X'' are, respectively, x' and x'' . By hypothesis, $\mathcal{O}_{Z,z}$ is reduced (resp. integral). Since $\mathcal{O}_{Z,z}$ is a faithfully flat $\mathcal{O}_{X'',x''}$ -module, it follows that $\mathcal{O}_{X'',x''}$ is reduced (resp. integral) (0_I, 6.5.1). $\square$

Corollary (4.6.11). — *Suppose that k is perfect (resp. algebraically closed). Then X is geometrically reduced (resp. geometrically punctually integral) at x over k if and only if $\mathcal{O}_{X,x}$ is reduced (resp. integral).*

Proof. Necessity is immediate. Conversely, for every extension k' of k , the prescheme $\text{Spec}(\mathcal{O}_{X,x}) \otimes_k k'$ is reduced (4.6.4) (resp. integral (4.4.4)). For every point x' of $X' = X \otimes_k k'$ above x , the local ring $\mathcal{O}_{X',x'}$ is isomorphic to a local ring of $\text{Spec}(\mathcal{O}_{X,x}) \otimes_k k'$, and is therefore reduced (resp. integral). $\square$

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Proposition (4.6.12). — *Let k be a field, let X be a k -prescheme, let $x \in X$, and let Ω be a perfect extension of k . The following conditions are equivalent:*

- (a) X is geometrically reduced over k at x ; that is, for every extension k' of k and every point x' of $X' = X \otimes_k k'$ above x , the local ring $\mathcal{O}_{X',x'}$ is reduced;
- (b) the prescheme $X \otimes_k \Omega$ is reduced at one point above x ;
- (c) for every finite radicial extension k' of k , the prescheme $X' = X \otimes_k k'$ is reduced at the unique point x' above x ;
- (d) the prescheme $\text{Spec}(\mathcal{O}_{X,x})$ is geometrically reduced over k ;
- (e) the local ring $\mathcal{O}_{X,x}$ is reduced and, for every irreducible component Z of X containing x , with generic point z , the field $\kappa(z)$ is a separable extension of k .

Proof. The implications (d) $\Rightarrow$ (a) $\Rightarrow$ (b) $\Rightarrow$ (c) are immediate, using for the last implication (4.6.10), (4.6.11), and the fact that every radicial extension of k is isomorphic to a subextension of Ω .

Suppose (c). By (4.6.1), it suffices to show that $\mathcal{O}_{X,x} \otimes_k k'$ is reduced for every finite radicial extension k' of k . This ring is the local ring of $X' = X \otimes_k k'$ at the unique point x' above x (I, 3.5.8) (I, 3.5.7), and is reduced by (c). Thus (c) implies (d).

Finally, the equivalence of (d) and (e) follows from the equivalence of (b) and (e) in (4.6.1), since the irreducible components of X containing x correspond bijectively to the irreducible components of $\text{Spec}(\mathcal{O}_{X,x})$ (I, 2.4.2). $\square$

Corollary (4.6.13). — *Under the hypotheses of (4.6.12), suppose in addition that X is locally Noetherian. Then conditions (a)–(e) of (4.6.12) are also equivalent to:*

- (f) there exists an open neighbourhood U of x that is geometrically reduced over k .

Proof. Condition (f) plainly implies that X is geometrically reduced at x over k . Conversely, suppose that X is geometrically reduced at x . Since $\mathcal{O}_{X,x}$ is reduced, there is a reduced open neighbourhood U of x in X (I, 6.1.13). After shrinking U , we may further suppose that it meets no irreducible component of X that does not contain x . Condition (e) of (4.6.12), together with condition (e) of (4.6.1), then shows that U is geometrically reduced over k . $\square$

It follows from (4.6.13) that, when X is locally Noetherian, the set of points $x \in X$ at which X is geometrically reduced over k is open; it is the largest open subset of X that is geometrically reduced over k .

(4.6.14). It follows from (4.6.10) and (4.6.11) that, if Ω is an algebraically closed extension of k , saying that X is geometrically punctually integral at x is equivalent to saying that $X \otimes_k \Omega$ is integral at one point above x .

For X to be geometrically punctually integral at x , it is necessary that X be integral at x and that, if z is the generic point of the unique irreducible component of X containing x , the field $\kappa(z)$ be a separable extension of k . This follows from (4.6.10). These conditions are not sufficient, as is shown by (4.5.12.ii). A sufficient but not necessary condition is that X be geometrically reduced at x and belong to only one irreducible component of X , and that this component be geometrically irreducible. If in addition X is locally Noetherian, then x has an open neighbourhood in X that is geometrically integral.

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Recall that a locally Noetherian prescheme that is punctually integral is locally integral (I, 6.1.13). Thus, if a k -prescheme X locally of finite type over k is geometrically punctually integral over k , then $X \otimes_k k'$ is locally integral for every extension k' of k . In this case X is also said to be *geometrically locally integral*.

Proposition (4.6.15). — (i) *Let k be a field and let X be a k -prescheme of finite type. Then X is geometrically punctually integral if and only if it is geometrically reduced and its geometric number of irreducible components equals its geometric number of connected components.*

(ii) *Let X be a prescheme locally of finite type over k . If X is geometrically punctually integral at x , then x has an open neighbourhood U that is geometrically punctually integral. Equivalently, the set of points $x \in X$ at which X is geometrically punctually integral is open in X .*

Proof. (i) Let Ω be an algebraically closed extension of k . The prescheme $X_{(\Omega)}$ is punctually integral (equivalently, locally integral, since it is Noetherian) if and only if it is reduced and the number of its connected components equals the number of its irreducible components (I, 6.1.10). Now apply (4.5.1), (4.6.1), and the first observation in (4.6.14).

(ii) The question is local on X , so we may reduce to the case in which X is of finite type over k . By (4.6.13), x has a geometrically reduced open neighbourhood; hence we may suppose that X itself is geometrically reduced. By (4.6.8), there is a finite extension k' of k such that $X' = X \otimes_k k'$ is reduced and its irreducible components are geometrically irreducible. Let $p : X' \rightarrow X$ be the canonical projection, which is finite and surjective, and let x'_j ($1 \leq j \leq r$) be the points of $p^{-1}(x)$. By hypothesis X' is integral at each x'_j , so each x'_j has an integral open neighbourhood V'_j in X' . Since p is closed (II, 6.1.10), there is an open neighbourhood U of x such that $p^{-1}(U)$ is contained in $\bigcup_j V'_j$. Thus $p^{-1}(U)$ is integral at every point, and hence locally integral. Its irreducible components are consequently open and pairwise disjoint; since they are geometrically irreducible (4.5.9), assertion (i) shows that $p^{-1}(U) = U \otimes_k k'$ is geometrically punctually integral. By definition the same is true of U . $\square$

Proposition (4.6.16). — *Let k be a field, let X be a locally Noetherian k -prescheme, and let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -Module. The following conditions are equivalent:*

- (a) *for every extension of finite type k' of k , if $X' = X \otimes_k k'$, then $\mathcal{F}' = \mathcal{F} \otimes_k k'$ is a reduced $\mathcal{O}_{X'}$ -Module (3.2.2);*
- (b) *for every finite radicial extension k' of k , the Module $\mathcal{F}'$ is a reduced $\mathcal{O}_{X'}$ -Module;*
- (c) *$\mathcal{F}$ is reduced and, if $\mathcal{I}$ is the Ideal of $\mathcal{O}_X$ annihilating $\mathcal{F}$, the closed sub-prescheme defined by $\mathcal{I}$ is geometrically reduced over k .*

If, moreover, X is locally of finite type over k , these conditions are also equivalent to:

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- (d) *for every extension k' of k (or for one algebraically closed extension k' of k), the Module $\mathcal{F}'$ is a reduced $\mathcal{O}_{X'}$ -Module.*

Proof. It is clear that (a) implies (b). Condition (b) first implies that $\mathcal{F}$ is reduced. By (3.2.3), this means the following. If Y is the closed sub-prescheme of X defined by the coherent Ideal $\mathcal{I}$ annihilating $\mathcal{F}$, then Y is reduced and there is a coherent torsion-free $\mathcal{O}_Y$ -Module $\mathcal{G}$, of rank 1 on every irreducible component of Y , such that $j_*(\mathcal{G}) = \mathcal{F}$, where $j : Y \rightarrow X$ is the canonical immersion.

For every extension k' of k , the annihilator of $\mathcal{F}'$ is $\mathcal{J}' = \mathcal{J} \otimes_k k'$, since the projection $X' \rightarrow X$ is flat (2.1.11). The closed sub-prescheme of X' defined by $\mathcal{J}'$ is $Y' = Y \otimes_k k'$. If $j' : Y' \rightarrow X'$ is the canonical immersion and $\mathcal{G}' = \mathcal{G} \otimes_k k'$, then $\mathcal{F}' = j'_*(\mathcal{G}')$. For every extension k' for which X' is locally Noetherian, the Module $\mathcal{F}'$ has no embedded associated prime cycle (4.2.7). Moreover, every maximal point y' of Y' lies above a maximal point y of Y ; since $\mathcal{G}_y \simeq \mathcal{O}_{Y,y}$, one has $\mathcal{G}'_{y'} \simeq \mathcal{O}_{Y',y'}$. Consequently, by (3.2.3), when X' is locally Noetherian, $\mathcal{F}'$ is reduced if and only if Y' is reduced.

It now follows from conditions (d) and (b) of (4.6.1) that (b) implies (c) and (c) implies (a). In (a), the finite-type hypothesis on k'/k serves only to ensure that X' is locally Noetherian. When X is locally of finite type over k , the prescheme X' is locally Noetherian for every extension k' of k ; this proves the equivalence of (d) with the other conditions in that case. $\square$

Definition (4.6.17). — When $\mathcal{F}$ satisfies the equivalent conditions of (4.6.16), it is said to be *geometrically reduced over k* , or *separable over k* .

Proposition (4.6.18). — Let k be a field, let X be a locally Noetherian k -prescheme, and let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -Module. The following conditions are equivalent:

- (a) for every extension of finite type k' of k , if $X' = X \otimes_k k'$, then $\mathcal{F}' = \mathcal{F} \otimes_k k'$ is an integral $\mathcal{O}_{X'}$ -Module (3.2.4);
- (b) for every finite extension k' of k , the Module $\mathcal{F}'$ is an integral $\mathcal{O}_{X'}$ -Module;
- (c) $\mathcal{F}$ is reduced (or integral) and, if $\mathcal{J}$ is the Ideal of $\mathcal{O}_X$ annihilating $\mathcal{F}$, the closed sub-prescheme defined by $\mathcal{J}$ is geometrically integral over k .

If, moreover, X is locally of finite type over k , these conditions are also equivalent to:

- (d) for every extension k' of k (or for one algebraically closed extension k' of k), the Module $\mathcal{F}'$ is an integral $\mathcal{O}_{X'}$ -Module.

Proof. Conditions (a), (b), and (c) imply that $\mathcal{F}$ is integral. Hence $\text{Ass}(\mathcal{F})$ consists of one point x , the generic point of $\text{Supp}(\mathcal{F})$. For every extension k' of k for which X' is locally Noetherian, the prime cycles associated to $\mathcal{F}'$ are therefore nonembedded and are the maximal points of $\text{Supp}(\mathcal{F}')$, all lying above x . By (4.6.16), each of conditions (a), (b), and (c) also implies that $\mathcal{F}$ is geometrically reduced. It is immediate that (a) implies (b), and (b) implies that $\text{Supp}(\mathcal{F})$ is geometrically irreducible (4.5.9). Thus (b) implies (c), because the closed sub-prescheme Y defined by $\mathcal{J}$ is geometrically reduced and geometrically irreducible, and hence geometrically integral. Conversely, if Y is geometrically integral, then for every extension k' of k , the support $\text{Supp}(\mathcal{F}')$ is irreducible. Consequently, whenever X' is locally Noetherian, $\mathcal{F}'$ is integral. This proves that (c) implies (a), and also that (c) implies (d) when X is locally of finite type over k . $\square$

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Definition (4.6.19). — When $\mathcal{F}$ satisfies the equivalent conditions of (4.6.18), it is said to be *geometrically integral over k* .

Proposition (4.6.20). — Let k be a field, let X be a locally Noetherian k -prescheme, let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -Module, and let K be an extension of k such that $X \otimes_k K$ is locally Noetherian. If $\mathcal{F}$ is geometrically reduced (resp. geometrically integral) over k , then $\mathcal{F} \otimes_k K$ is geometrically reduced (resp. geometrically integral) over K .

Proof. For every finite extension K' of K ,

$$(X \otimes_k K) \otimes_K K' = X \otimes_k K'$$

is locally Noetherian. The proofs of (4.6.16) and (4.6.18) show that the hypothesis on $\mathcal{F}$ implies that $\mathcal{F} \otimes_k K'$ is reduced (resp. integral), which proves the assertion. $\square$

Proposition (4.6.21). — Let k be a field, let X and Y be two locally Noetherian k -preschemes such that $X \times_k Y$ is locally Noetherian, let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -Module, and let $\mathcal{G}$ be a coherent $\mathcal{O}_Y$ -Module.

- (i) If $\mathcal{F}$ is geometrically reduced (resp. geometrically integral) and $\mathcal{G}$ is reduced (resp. integral), then $\mathcal{F} \otimes_k \mathcal{G}$ is reduced (resp. integral).
- (ii) If $\mathcal{F}$ and $\mathcal{G}$ are both geometrically reduced (resp. geometrically integral), then so is $\mathcal{F} \otimes_k \mathcal{G}$.

Proof. (i) Using (3.2.3), (4.6.16), and (4.6.18), we may reduce to the case in which

$$\mathrm{Supp}(\mathcal{F}) = X, \quad \mathrm{Supp}(\mathcal{G}) = Y,$$

X is geometrically reduced (resp. geometrically integral), Y is reduced (resp. integral), $\mathcal{F}$ and $\mathcal{G}$ have no embedded associated prime cycles, and

$$\mathcal{F}_x \simeq \mathcal{O}_x, \quad \mathcal{G}_y \simeq \mathcal{O}_y$$

at every maximal point x of X and y of Y . Then $\mathcal{F} \otimes_k \mathcal{G}$ has no embedded associated prime cycle (4.2.3), and its associated prime cycles are exactly the irreducible components of $X \times_k Y$ (4.2.5). We may therefore reduce further to the case in which X and Y are irreducible. If X is geometrically reduced and Y is reduced, then $X \times_k Y$ is reduced (4.2.4.ii). If X is geometrically integral and Y is integral, then $X \times_k Y$ is moreover irreducible (4.5.8.i), and hence integral. Finally, every maximal point z of $X \times_k Y$ lies above a maximal point x of X and a maximal point y of Y . The hypotheses on $\mathcal{F}$ and $\mathcal{G}$ therefore imply, by (I, 9.1.12), that $(\mathcal{F} \otimes_k \mathcal{G})_z \simeq \mathcal{O}_z$.

(ii) The argument is analogous. After reducing to $X = \mathrm{Supp}(\mathcal{F})$ and $Y = \mathrm{Supp}(\mathcal{G})$, the prescheme $X \times_k Y$ is geometrically reduced (resp. geometrically integral) by (4.6.5.ii). $\square$

The definitions in (4.6.18) and (4.6.19) localize as follows.

Definition (4.6.22). — Let k be a field, let X be a locally Noetherian k -prescheme, and let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -Module. We say that $\mathcal{F}$ is *geometrically reduced*, or *separable* (resp. *geometrically punctually integral*), at a point $x \in X$ if, for every finite radicial (resp. finite) extension k' of k , the Module $\mathcal{F} \otimes_k k'$ is reduced (resp. integral) at every point x' of $X \otimes_k k'$ above x ; see (3.2.2) and (3.2.4).

If $\mathcal{F}$ is reduced at x , there is an open neighbourhood U of x such that $\mathcal{F}|_U$ is reduced (3.2.2). Arguing as in (4.6.16), one sees that $\mathcal{F}$ is geometrically reduced at x if and only if $\mathcal{F}$ is reduced at x and the closed sub-prescheme Y defined by the annihilator $\mathcal{J}$ of $\mathcal{F}$ is geometrically reduced at x . It follows that x has an open neighbourhood U such that $\mathcal{F}|_U$ is geometrically reduced (4.6.11). The arguments of (4.6.16) and (4.6.18) likewise show that, if X is locally of finite type over k , then $\mathcal{F}$ is geometrically punctually integral at x if and only if $\mathcal{F}$ is reduced at x and Y is geometrically punctually integral at x . IV-2 | 75

4.7. Multiplicities in the primary decomposition on an algebraic prescheme

Lemma (4.7.1). — Let A and B be two local rings, let $\mathfrak{m}$ be the maximal ideal of A , and let $\rho : A \rightarrow B$ be a local homomorphism. Suppose that B is flat as an A -module and that $B/\mathfrak{m}B$ has finite length as a B -module. Then an A -module M has finite length if and only if $M_{(B)}$ has finite length as a B -module, and

$$(4.7.1.1) \quad \mathrm{length}_B(M_{(B)}) = \mathrm{length}_A(M) \mathrm{length}_B(B/\mathfrak{m}B).$$

Proof. This is a special case of (2.5.5.2). $\square$

Corollary (4.7.2). — Let A and B be two Noetherian rings, and let $\rho : A \rightarrow B$ be a ring homomorphism such that B is flat as an A -module. Let $\mathfrak{q}$ be a minimal prime ideal of B , put $\mathfrak{p} = \rho^{-1}(\mathfrak{q})$, and let M be an A -module of finite type. Then $\mathfrak{p}$ is a minimal prime ideal of A , $M_{\mathfrak{p}}$ has finite length as an $A_{\mathfrak{p}}$ -module, and

$$(4.7.2.1) \quad \mathrm{length}_{B_{\mathfrak{q}}}((M_{(B)})_{\mathfrak{q}}) = \mathrm{length}_{A_{\mathfrak{p}}}(M_{\mathfrak{p}}) \mathrm{length}_{B_{\mathfrak{q}}}(B_{\mathfrak{q}}/\mathfrak{p}B_{\mathfrak{q}}).$$

Proof. The ring $B_{\mathfrak{q}}$ is flat over $A_{\mathfrak{p}}$ (0_I, 6.3.2), and the homomorphism $A_{\mathfrak{p}} \rightarrow B_{\mathfrak{q}}$ is local; hence $B_{\mathfrak{q}}$ is faithfully flat over $A_{\mathfrak{p}}$ (0_I, 6.6.2). It follows from (2.3.5) that $\mathfrak{p}$ is minimal in A . The ring $A_{\mathfrak{p}}$ is therefore Artinian (Bourbaki, *Alg. comm.*, chap. IV, § 2, n° 5, prop. 9), so $M_{\mathfrak{p}}$ has finite length over $A_{\mathfrak{p}}$. Formula (4.7.2.1) is now a special case of (4.7.1.1). $\square$

Proposition (4.7.3). — Let k be a field of characteristic exponent p , let X be an integral locally Noetherian prescheme, and let $K = R(X)$ be the field of rational functions on X . Let k' be an extension of k , let (X'_{α}) be the family of irreducible components of $X' = X_{(k')}$, and let x'_{α} be the generic point of X'_{α} .

- (i) Suppose that X' is locally Noetherian, as is the case if X is locally of finite type over k or if k' is an extension of finite type of k . Let k'' be any extension of k' , set $X'' = X \otimes_k k''$, and let (X''_β) be the family of irreducible components of X'' , with generic points x''_β . If x''_β lies above x'_α , set

$$Z'' = \text{Spec}(\kappa(x'_\alpha)) \otimes_{k'} k'',$$

and let z''_β be the point of Z'' corresponding to x''_β . Then

$$(4.7.3.1) \quad \text{length}(\mathcal{O}_{X'', x''_\beta}) = \text{length}(\mathcal{O}_{X', x'_\alpha}) \text{length}(\mathcal{O}_{Z'', z''_\beta}).$$

In particular, if k'' is a separable extension of k' such that X'' is locally Noetherian, then

$$\text{length}(\mathcal{O}_{X'', x''_\beta}) = \text{length}(\mathcal{O}_{X', x'_\alpha}).$$

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- (ii) If X is locally of finite type over k , or if the extension k'/k is of finite type, then each number

$$\text{length}(\mathcal{O}_{X', x'_\alpha})$$

is a power of p .

- (iii) Suppose that X is locally of finite type over k . If k' is perfect, all the numbers $\text{length}(\mathcal{O}_{X', x'_\alpha})$ are equal and depend only on the extension K/k , not on the particular perfect extension k'/k under consideration. Moreover, there exists a finite radicial extension k_1 of k such that, if x_1 is the generic point of $X_1 = X \otimes_k k_1$, which is irreducible, then

$$\text{length}(\mathcal{O}_{X', x'_\alpha}) = \text{length}(\mathcal{O}_{X_1, x_1}) \quad \text{for every } \alpha.$$

Proof. (i) Recall from (4.2.4) that the finitely many irreducible components X'_α correspond bijectively to the minimal prime ideals $\mathfrak{p}_\alpha$ of $K \otimes_k k'$, and that $\mathcal{O}_{X', x'_\alpha}$ is an Artinian ring isomorphic to $(K \otimes_k k')_{\mathfrak{p}_\alpha}$. These rings therefore depend only on the extensions K/k and k'/k ; in particular, with k' fixed, they are the same for every integral k -prescheme having K as its field of rational functions.

Formula (4.7.3.1) follows from (4.7.2.1). Indeed, apply that formula to the Noetherian ring A of an affine neighbourhood of x'_α in X' and to the ring B of a sufficiently small affine neighbourhood of x''_β in X'' , where $\mathfrak{p}$ and $\mathfrak{q}$ are the minimal prime ideals corresponding respectively to x'_α and x''_β , and take $M = A$. Here $A/\mathfrak{p}$ is the ring of an affine neighbourhood of x'_α in X'_α , while $B = A \otimes_{k'} k''$. If $\mathfrak{q}' = \mathfrak{q}/\mathfrak{p}B$, a minimal prime ideal of $B/\mathfrak{p}B$, then

$$\mathcal{O}_{Z'', z''_\beta} = (B/\mathfrak{p}B)_{\mathfrak{q}'} = B_{\mathfrak{q}}/\mathfrak{p}B_{\mathfrak{q}},$$

which proves (4.7.3.1). The last assertion of (i) follows because Z'' is reduced in this case (4.2.4), and hence $\mathfrak{p}B_{\mathfrak{q}} = \mathfrak{q}B_{\mathfrak{q}}$.

- (ii) If k' is of finite type over k , there is a purely transcendental extension k'_0 of k such that k' is finite over k'_0 . Since k'_0/k is separable, (i), applied with k' and k'' replaced by k and k'_0 , shows that we may replace X by $X \otimes_k k'_0$; thus we may assume that k'/k is finite. There is then a finite quasi-Galois extension k''/k containing k' , and (4.7.3.1) shows that it suffices to prove the assertion for k'' . But k'' is separable over a finite radicial extension k_1 of k , so (i) reduces us to the case $k' = k_1$ radicial. The ring $K \otimes_k k_1$ is then Artinian and has a unique minimal prime ideal (4.3.2). If $\bar{K}$ is an algebraic closure of K , the length of $\bar{K} \otimes_k k_1$ divides the length of the Artinian ring $K \otimes_k k_1$ by (4.7.1.1). The Artinian $\bar{K}$ -algebra $\bar{K} \otimes_k k_1$ has a unique prime ideal, necessarily maximal, and its quotient by that ideal is a finite extension of $\bar{K}$, hence is $\bar{K}$. Its length is therefore its rank over $\bar{K}$, namely $[k_1 : k]$, a power of p .

Suppose next that X is locally of finite type over k . Then X'' is locally Noetherian for every extension k'' of k' . Taking k'' algebraically closed, (4.7.3.1) reduces us to the case in which k' is algebraically closed. Since k' is then separable over the algebraic closure $\bar{k}$ of k , another application of (i) reduces us to $k' = \bar{k}$. After replacing X by an affine open of finite type over k , there is a finite radicial extension k_1/k such that $(X \otimes_k k_1)_{\text{red}}$ is separable over k_1 (4.6.6). Consequently, $(X \otimes_k k_1)_{\text{red}} \otimes_{k_1} \bar{k}$ is reduced (4.6.4).

By (4.7.3.1), this again reduces us to the case in which k' is a finite radicial extension of k , and the preceding argument applies. IV-2 | 77

- (iii) Any two extensions k' and k'' of k embed in a common algebraically closed extension of k . To show that the set of numbers $\text{length}(\mathcal{O}_{X',x'_\alpha})$ for perfect k' does not depend on k' , it is therefore enough to compare them with those obtained after an algebraically closed extension k''/k' ; the assertion then follows from (i), since k''/k' is separable. To prove that all these numbers are equal, we may thus assume $k' = \bar{k}$. For the finite radicial extension k_1/k chosen in (ii), all the points x'_α lie above the unique generic point x_1 of $X_1 = X \otimes_k k_1$. The equality

$$\text{length}(\mathcal{O}_{X',x'_\alpha}) = \text{length}(\mathcal{O}_{X_1,x_1})$$

for every α follows once more from (4.7.3.1) and the choice of k_1 . □

Definition (4.7.4). — Let k be a field, let X be an integral k -prescheme, and let K be the field of rational functions on X . The *radicial multiplicity* of K (or of X) over k is the supremum of the lengths of the Artinian rings $K \otimes_k k_1$, as k_1 ranges over the finite radicial extensions of k . The *separable multiplicity* of K (or of X) over k is the geometric number of irreducible components of X if this number is finite, and $+\infty$ otherwise. Finally, the *total multiplicity* of K (or of X) over k is the product of its radicial and separable multiplicities.

If p is the characteristic exponent of k and the radicial multiplicity of K over k is finite, then it is a power $q = p^f$ of p , by (4.7.3.ii). When $p \neq 1$, the well-defined integer f is called the *inseparability exponent* of K over k ; when $p = 1$, the radicial multiplicity is always 1.

To say that the radicial (resp. separable, resp. total) multiplicity of X over k is 1 means that X is geometrically reduced (resp. geometrically irreducible, resp. geometrically integral) over k .

When X is locally of finite type, its radicial multiplicity over k is also the common length of the local rings A_q at the minimal prime ideals of the total ring of fractions A of $K \otimes_k k'$, for every perfect extension k'/k . Its separable multiplicity is the number of minimal prime ideals of $A = K \otimes_k \Omega$, for every algebraically closed extension Ω/k . Finally, its total multiplicity is the sum of the lengths of the local rings A_q of $A = K \otimes_k \Omega$ at its minimal prime ideals. In this case all three numbers are finite.

Definition (4.7.5). — Let k be a field, let X be a k -prescheme locally of finite type over k , let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -Module, and let $x \in X$ be such that $\mathcal{F}_x$ has finite length as an $\mathcal{O}_{X,x}$ -module. The *geometric length* of $\mathcal{F}$ at x relative to k , or the *radicial multiplicity* of x for $\mathcal{F}$, is

$$\lambda_x(\mathcal{F}) = \text{length}_{\mathcal{O}_{X,x}}(\mathcal{F}_x) \mu_r(\kappa(x) | k),$$

where $\mu_r(\kappa(x) | k)$ denotes the radicial multiplicity of $\kappa(x)$ over k .

When X is integral and x is its generic point, $\text{length}(\mathcal{O}_{X,x}) = 1$. Thus the radicial multiplicity of x for $\mathcal{O}_X$ is precisely the radicial multiplicity of X over k defined in (4.7.4). IV-2 | 78

Proposition (4.7.6). — Let

$$0 \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}'' \longrightarrow 0$$

be an exact sequence of coherent $\mathcal{O}_X$ -Modules. At a point $x \in X$, the geometric length $\lambda_x(\mathcal{F})$ is defined if and only if both $\lambda_x(\mathcal{F}')$ and $\lambda_x(\mathcal{F}'')$ are defined; in that case

$$(4.7.6.1) \quad \lambda_x(\mathcal{F}) = \lambda_x(\mathcal{F}') + \lambda_x(\mathcal{F}'').$$

Proof. This follows immediately from the definition: the length of $\mathcal{F}_x$ is finite if and only if the lengths of $\mathcal{F}'_x$ and $\mathcal{F}''_x$ are finite, and then

$$\text{length}(\mathcal{F}_x) = \text{length}(\mathcal{F}'_x) + \text{length}(\mathcal{F}''_x).$$

□

(4.7.7). Under the hypotheses of (4.7.5), the only points $x \in X$ for which $\mathcal{F}_x$ is a nonzero $\mathcal{O}_{X,x}$ -module of finite length are, by (3.1.2), the maximal points of $\text{Supp}(\mathcal{F})$. This follows from Bourbaki, *Alg. comm.*, chap. IV, § 2, n° 5, cor. 2 to prop. 7, since the question is local on X . There is a closed subscheme Y with underlying space $\text{Supp}(\mathcal{F})$ and a coherent $\mathcal{O}_Y$ -Module $\mathcal{G}$ such that $\mathcal{F} = j_*(\mathcal{G})$, where $j : Y \rightarrow X$ is the canonical injection. If x is a maximal point of Y , then the geometric lengths of

$\mathcal{F}$ and $\mathcal{G}$ at x are equal, because $\mathcal{O}_{X,x}$ and $\mathcal{O}_{Y,x}$ have the same residue field. This gives the following characterization of geometric length.

Proposition (4.7.8). — *Under the hypotheses of (4.7.5), let x be a maximal point of $\text{Supp}(\mathcal{F})$. Let k' be a perfect extension of k , and set $X' = X \otimes_k k'$ and $\mathcal{F}' = \mathcal{F} \otimes_k k'$. Then $\lambda_x(\mathcal{F})$ equals $\text{length}_{\mathcal{O}_{X',x'}}(\mathcal{F}'_{x'})$ at every maximal point x' of $\text{Supp}(\mathcal{F}')$ above x . Moreover, there exists a finite radicial extension k_1 of k such that, on setting $X_1 = X \otimes_k k_1$ and $\mathcal{F}_1 = \mathcal{F} \otimes_k k_1$, the number $\lambda_x(\mathcal{F})$ equals $\text{length}_{\mathcal{O}_{X_1,x_1}}((\mathcal{F}_1)_{x_1})$ at every maximal point x_1 of $\text{Supp}(\mathcal{F}_1)$ above x .*

Proof. The remark in (4.7.7) reduces us to the case in which x is a maximal point of X . Since the question is local on X , suppose $X = \text{Spec}(A)$ is affine Noetherian and $\mathcal{F} = \tilde{M}$, where M is an A -module of finite type. Put $B = A \otimes_k k'$, and let $\mathfrak{p}$ and $\mathfrak{q}$ be the minimal prime ideals of A and B corresponding respectively to x and x' . Formula (4.7.2.1) gives

$$\text{length}_{B_{\mathfrak{q}}}((M_{(B)})_{\mathfrak{q}}) = \text{length}_{A_{\mathfrak{p}}}(M_{\mathfrak{p}}) \text{length}_{B_{\mathfrak{q}}}(B_{\mathfrak{q}}/\mathfrak{p}B_{\mathfrak{q}}).$$

Since $\kappa(x) = A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$, the argument of (4.7.3) shows that the last factor is the length of $(B/\mathfrak{p}B)_{\mathfrak{q}'}$, where $\mathfrak{q}' = \mathfrak{q}/\mathfrak{p}B$. This is a local ring of $(A/\mathfrak{p}) \otimes_k k'$ at a minimal prime ideal, and therefore also a local ring of the total ring of fractions of $\kappa(x) \otimes_k k'$. By definition its length is $\mu_r(\kappa(x) \mid k)$, proving the first assertion. The second follows in the same way by replacing k' with k_1 and using (4.7.3.iii). $\square$

When x is a maximal point of $\text{Supp}(\mathcal{F})$, the geometric length of $\mathcal{F}$ at x is also called the *radicial multiplicity of the maximal prime cycle* $\{x\}$ associated with $\mathcal{F}$.

Corollary (4.7.9). — *Under the hypotheses of (4.7.5), let k' be an extension of k , set $X' = X \otimes_k k'$ and $\mathcal{F}' = \mathcal{F} \otimes_k k'$, and let Z be an irreducible component of $\text{Supp}(\mathcal{F})$. If Z' is an irreducible component of $\text{Supp}(\mathcal{F}')$ dominating Z (4.2.7), then the radicial multiplicities of Z with respect to $\mathcal{F}$ and of Z' with respect to $\mathcal{F}'$ are equal.*

Proof. Choose a perfect extension k'' of k' , set $X'' = X \otimes_k k''$ and $\mathcal{F}'' = \mathcal{F} \otimes_k k''$, and choose a maximal point x'' of $\text{Supp}(\mathcal{F}'')$ above the generic point of Z' . It follows from (4.7.8) that $\text{length}(\mathcal{F}''_{x''})$ equals both multiplicities. $\square$

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Proposition (4.7.10). — *Let k be a field, let X be a k -prescheme locally of finite type over k , let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -Module, let $x \in X$, and let ξ_i be the generic points of the irreducible components of $\text{Supp}(\mathcal{F})$ that contain x . Then $\mathcal{F}$ is geometrically reduced at x (4.6.22) if and only if x belongs to no embedded associated prime cycle of $\mathcal{F}$ and $\lambda_{\xi_i}(\mathcal{F}) = 1$ for every i .*

Proof. First suppose the stated conditions hold. Let k'/k be finite, let x' be a point of $X' = X \otimes_k k'$ above x , and put $\mathcal{F}' = \mathcal{F} \otimes_k k'$. By (4.2.7), x' belongs to no embedded associated prime cycle of $\mathcal{F}'$. Moreover, (4.7.8) shows that $\mathcal{F}'$ has geometric length 1 at every generic point ζ' of an irreducible component of $\text{Supp}(\mathcal{F}')$ containing x' ; by (2.3.4), such a point lies above one of the ξ_i . A fortiori, $\text{length}(\mathcal{F}'_{\zeta'}) = 1$, so $\mathcal{F}'$ is reduced at x' .

Conversely, (4.2.7) first shows that if, for one finite extension k'/k , the Module $\mathcal{F}'$ is reduced at every point x' above x , then x belongs to no embedded associated prime cycle of $\mathcal{F}$. For each ξ_i , take for k' the field k_1 supplied by (4.7.8). The hypothesis that $\mathcal{F}$ is geometrically reduced at x then implies $\lambda_{\xi_i}(\mathcal{F}) = 1$. $\square$

Proposition (4.7.11). — *Let k be a field, let X be a prescheme locally of finite type over k , let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -Module, let k' be an extension of k , set $X' = X \otimes_k k'$ and $\mathcal{F}' = \mathcal{F} \otimes_k k'$, and let $x' \in X'$ lie above $x \in X$. Then $\mathcal{F}$ is geometrically reduced (resp. geometrically punctually integral) at x if and only if $\mathcal{F}'$ is geometrically reduced (resp. geometrically punctually integral) at x' .*

Proof. Every irreducible component of $\text{Supp}(\mathcal{F}')$ containing x' dominates an irreducible component of $\text{Supp}(\mathcal{F})$ containing x , and conversely every such component downstairs is dominated by one containing x' ((4.2.7.i) and (2.3.4)). The assertion concerning geometric reducedness therefore follows from (4.2.7.ii), (4.7.9), and (4.7.10).

Let Y be the closed sub-prescheme of X defined by the annihilator Ideal $\mathcal{J}$ of $\mathcal{F}$. Then $Y' = Y \otimes_k k'$ is the closed sub-prescheme of X' defined by the annihilator Ideal $\mathcal{J}' = \mathcal{J} \otimes_k k'$ of $\mathcal{F}'$. To say that $\mathcal{F}$ (resp. $\mathcal{F}'$) is geometrically punctually integral at x (resp. x') is to say that $\mathcal{F}$ (resp. $\mathcal{F}'$) is geometrically reduced at x (resp. x') and that Y (resp. Y') is geometrically punctually integral at x (resp. x'). By the assertion just proved, either of the two geometric pointwise-integrality hypotheses implies that Y has a separable neighbourhood of x . The conclusion follows from (4.6.10). $\square$

Definition (4.7.12). — Let k be a field, let X be a k -prescheme locally of finite type over k , let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -Module, and let x be a maximal point of $\text{Supp}(\mathcal{F})$.

The *total multiplicity* of x (or of the prime cycle $\{x\}$) for $\mathcal{F}$, relative to k , is the product of the radical multiplicity of x for $\mathcal{F}$ and the separable multiplicity of $\kappa(x)$ over k . IV-2 | 80

Equivalently, the total multiplicity of x for $\mathcal{F}$ is the product of $\text{length}_{\mathcal{O}_{X,x}}(\mathcal{F}_x)$ and the total multiplicity of $\kappa(x)$ over k (4.7.4). With the notation of (4.7.11), the total multiplicity of x for $\mathcal{F}$ is the sum of the total multiplicities, for $\mathcal{F}'$, of the maximal points x'_i of $\text{Supp}(\mathcal{F}')$ above x . Moreover, there exists a finite extension k'/k for which the total multiplicity of x for $\mathcal{F}$ is

$$\sum_i \text{length}(\mathcal{F}'_{x'_i}).$$

Proposition (4.7.13). — Under the hypotheses and notation of (4.7.10), it is sufficient, for $\mathcal{F}$ to be geometrically punctually integral at x , that x belong to only one associated prime cycle of $\mathcal{F}$ (necessarily non-embedded), and that the total multiplicity for $\mathcal{F}$ of the generic point ξ of that cycle be 1.

Proof. Let k'/k be finite, let x' be a point of $X' = X \otimes_k k'$ above x , and put $\mathcal{F}' = \mathcal{F} \otimes_k k'$. The point x' can belong to only one associated prime cycle of $\mathcal{F}'$, since by hypothesis there is only one maximal point of $\text{Supp}(\mathcal{F}')$ above ξ . Moreover, $\mathcal{F}'$ is geometrically reduced at x' by (4.7.10). The conclusion follows. $\square$

4.8. Fields of definition

(4.8.1). Given a prescheme X , throughout this section we denote by $\mathfrak{S}(X)$ the set of sub-preschemes of X , and, as usual, by $\mathfrak{P}(X)$ the set of subsets of the underlying space of X . For every quasi-coherent $\mathcal{O}_X$ -Module $\mathcal{F}$, we denote by $\Phi(\mathcal{F})$ the set of quasi-coherent sub- $\mathcal{O}_X$ -Modules of $\mathcal{F}$.

(4.8.2). Let k be a field, let X and Y be two k -preschemes, let $\mathcal{F}$ and $\mathcal{G}$ be two quasi-coherent $\mathcal{O}_X$ -Modules, and let $K' \subset K$ be two extensions of k . Corresponding to the morphism $\text{Spec}(K) \rightarrow \text{Spec}(K')$, there are canonical maps

$$(4.8.2.1) \quad \Phi(\mathcal{F} \otimes_k K') \longrightarrow \Phi(\mathcal{F} \otimes_k K),$$

$$(4.8.2.2) \quad \text{Hom}(\mathcal{F} \otimes_k K', \mathcal{G} \otimes_k K') \longrightarrow \text{Hom}(\mathcal{F} \otimes_k K, \mathcal{G} \otimes_k K),$$

$$(4.8.2.3) \quad \mathfrak{S}(X_{(K')}) \longrightarrow \mathfrak{S}(X_{(K)}),$$

$$(4.8.2.4) \quad \text{Hom}_{K'}(X_{(K')}, Y_{(K')}) \longrightarrow \text{Hom}_K(X_{(K)}, Y_{(K)}),$$

$$(4.8.2.5) \quad \mathfrak{P}(X_{(K')}) \longrightarrow \mathfrak{P}(X_{(K)}).$$

If $p_X : X_{(K)} \rightarrow X_{(K')}$ is the canonical projection, then (4.8.2.5) sends M to $p_X^{-1}(M)$, and similarly (4.8.2.3) sends Z to $p_X^{-1}(Z) = Z \otimes_{K'} K$ (I, 4.4.1). Map (4.8.2.1) sends

$$\mathcal{H} \longmapsto \mathcal{H}_{(K)} = \mathcal{H} \otimes_{\mathcal{O}_{X_{(K')}}} \mathcal{O}_{X_{(K)}};$$

(4.8.2.2) sends u to $p_X^*(u)$; and (4.8.2.4) sends f to $f_{(K)}$.

If $s_{K,K'}$ denotes any one of these canonical maps, and if $K'' \subset K' \subset K$ are three extensions of k , then

$$(4.8.2.6) \quad s_{K,K''} = s_{K,K'} \circ s_{K',K''}.$$

Proposition (4.8.3). — The canonical maps (4.8.2.1)–(4.8.2.5) are injective.

Proof. The morphism p_X is faithfully flat and quasi-compact. The injectivity of (4.8.2.5) follows from the surjectivity of p_X . The injectivity of (4.8.2.1) follows from (2.2.2), that of (4.8.2.2) from (2.2.7), that of (4.8.2.3) from (2.2.15), and that of (4.8.2.4) from (2.2.16). $\square$

Definition (4.8.4). — With the notation of (4.8.2), a quasi-coherent sub- $\mathcal{O}_{X_{(K)}}$ -Module of $\mathcal{F} \otimes_k K$ (resp. a homomorphism $\mathcal{F} \otimes_k K \rightarrow \mathcal{G} \otimes_k K$, a sub-prescheme of $X_{(K)}$, a K -morphism $X_{(K)} \rightarrow Y_{(K)}$, or a subset of $X_{(K)}$) is said to be *defined over K'* if it belongs to the image of (4.8.2.1) (resp. (4.8.2.2), (4.8.2.3), (4.8.2.4), or (4.8.2.5)). The field K' is then called a *field of definition* of the object in question.

Clearly, K itself is a field of definition for every object in any of the five target sets in (4.8.2). One should note, however, that for a K -prescheme Z , to say that a quasi-coherent $\mathcal{O}_Z$ -Module $\mathcal{H}$ is “defined over a subfield K' of K ” has no meaning unless Z has been *given* in the form $X_{(K)}$ for a prescheme X over a subfield k of K , and unless $\mathcal{H}$ is a sub- $\mathcal{O}_Z$ -Module of an $\mathcal{O}_Z$ -Module that has been *given* in the form $\mathcal{F} \otimes_k K$, where $\mathcal{F}$ is a quasi-coherent $\mathcal{O}_X$ -Module. Analogous remarks apply to the other notions.

It follows from (4.8.2.6) that if an object in one of the five target sets in (4.8.2) is defined over K' , then it is also defined over every intermediate field K'' with $K' \subset K'' \subset K$. Finally, if K_1/K is an extension, then an object in a target set of $s_{K_1, K'}$ is defined over K' if and only if its image under $s_{K_1, K}$ is defined over K' ; this follows from

$$s_{K_1, K'} = s_{K_1, K} \circ s_{K, K'}.$$

Proposition (4.8.5). — With the notation of (4.8.2), let (X_α) be an open covering of X . An element

$$\mathcal{H} \in \Phi(\mathcal{F} \otimes_k K) \quad (\text{resp. } u \in \text{Hom}(\mathcal{F} \otimes_k K, \mathcal{G} \otimes_k K), Z \in \mathfrak{S}(X_{(K)}), \text{ or } M \in \mathfrak{P}(X_{(K)}))$$

is defined over K' if and only if, for every α , its restriction

$$\mathcal{H}|_{(X_\alpha)_{(K)}} \quad (\text{resp. } u|_{(X_\alpha)_{(K)}}, Z \cap (X_\alpha)_{(K)}, \text{ or } M \cap (X_\alpha)_{(K)})$$

is defined over K' .

Proof. This follows immediately from the injectivity of (4.8.2.1), (4.8.2.2), (4.8.2.3), and (4.8.2.5). For example, suppose that for every α there is a quasi-coherent sub- $\mathcal{O}_{(X_\alpha)_{(K')}}$ -Module $\mathcal{H}'_\alpha$ such that

$$\mathcal{H}|_{(X_\alpha)_{(K)}} = \mathcal{H}'_\alpha \otimes_{K'} K.$$

For any α, β , injectivity after restriction to $(X_\alpha \cap X_\beta)_{(K')}$ gives

$$\mathcal{H}'_\alpha|_{(X_\alpha \cap X_\beta)_{(K')}} = \mathcal{H}'_\beta|_{(X_\alpha \cap X_\beta)_{(K')}}.$$

The $\mathcal{H}'_\alpha$ therefore glue to a quasi-coherent sub- $\mathcal{O}_{X_{(K')}}$ -Module $\mathcal{H}'$ of $\mathcal{F} \otimes_k K'$ such that $\mathcal{H} = \mathcal{H}' \otimes_{K'} K$. The other cases are treated in the same way. $\square$

Lemma (4.8.6). — Let k be a field, let K/k be an extension, let A be a k -algebra, let M be an A -module, let $\bar{N}$ be an $A_{(K)}$ -submodule of $M_{(K)}$, and let K' be an intermediate field of K/k . The following conditions are equivalent:

- (a) $\bar{N} = N' \otimes_{K'} K$, where N' is an $A_{(K')}$ -submodule of $M_{(K')}$;
- (b) if $X = \text{Spec}(A)$, the quasi-coherent $\mathcal{O}_{X_{(K)}}$ -Module $\bar{N}$ on $X_{(K)} = \text{Spec}(A_{(K)})$ is defined over K' ;
- (c) $\bar{N} = N' \otimes_{K'} K$, where N' is a K' -vector subspace of $M_{(K')}$.

Proof. The equivalence of (a) and (b) follows immediately from (I, 1.6.5), and (a) plainly implies (c). **IV-2 | 82** Conversely, if (c) holds, then, up to the canonical identification,

$$N' = \bar{N} \cap M_{(K')}.$$

Since $\bar{N}$ and $M_{(K')}$ are $A_{(K')}$ -submodules, so is N' . $\square$

Corollary (4.8.7). — Under the hypotheses of (4.8.6), there is a smallest subfield K' of K containing k for which the equivalent conditions of (4.8.6) hold.

Proof. It is enough to use condition (c), for which the assertion is Bourbaki, *Alg.*, chap. II, 3rd ed., § 8, n° 6, prop. 6. $\square$

Lemma (4.8.8). — With the notation of (4.8.2):

(i) Let $\bar{u} : \mathcal{F} \otimes_k K \rightarrow \mathcal{G} \otimes_k K$ be an $\mathcal{O}_{X(K)}$ -homomorphism, and let

$$\mathcal{H} \subset (\mathcal{F} \otimes_k K) \oplus (\mathcal{G} \otimes_k K)$$

be its graph. Then $\bar{u}$ is defined over K' if and only if $\mathcal{H}$ is.

(ii) Let $\bar{Z}$ be a closed sub-prescheme of $X_{(K)}$, and let $\bar{\mathcal{J}}$ be the quasi-coherent Ideal of $\mathcal{O}_{X(K)}$ defining $\bar{Z}$. Then $\bar{Z}$ is defined over K' if and only if $\bar{\mathcal{J}}$ is.

(iii) Let $\bar{f} : X_{(K)} \rightarrow Y_{(K)}$ be a K -morphism, and let $\bar{Z}$ be its graph, regarded as a sub-prescheme of $X_{(K)} \times_K Y_{(K)}$ (I, 5.3.11). Then $\bar{f}$ is defined over K' if and only if $\bar{Z}$ is.

Proof. Assertion (ii) follows immediately from (I, 4.4.5). Necessity in (iii) is clear. Conversely, suppose $\bar{Z} = Z'_{(K)}$, where Z' is a sub-prescheme of $(X \times_k Y)_{(K')}$. If $p' : Z' \rightarrow X_{(K')}$ is the restriction of the first projection, then $p'_{(K)}$ is an isomorphism, and hence so is p' (2.7.1.viii). Thus Z' is the graph of a K' -morphism $f' : X_{(K')} \rightarrow Y_{(K')}$ (I, 5.3.11), and $\bar{f} = f'_{(K)}$ (I, 5.3.12).

Finally, to prove (i), (4.8.5) permits us to assume $X = \operatorname{Spec}(A)$ affine, with $\mathcal{F} = \tilde{M}$, $\mathcal{G} = \tilde{N}$, and $\bar{u} = \tilde{\bar{v}}$ for an $A_{(K)}$ -homomorphism $\bar{v} : M_{(K)} \rightarrow N_{(K)}$. Suppose the graph of $\bar{v}$ is $P'_{(K)}$, where P' is an $A_{(K')}$ -submodule of $M_{(K')} \oplus N_{(K')}$. If $p' : P' \rightarrow M_{(K')}$ is the restriction of the first projection, then $p' \otimes 1_K$ is an isomorphism of $A_{(K)}$ -modules. Faithful flatness (0_I, 6.4.1) implies that p' is an isomorphism of $A_{(K')}$ -modules. Hence P' is the graph of an $A_{(K')}$ -homomorphism $v' : M_{(K')} \rightarrow N_{(K')}$, and plainly $\bar{v} = v' \otimes 1_K$. $\square$

Proposition (4.8.9). — Let k be a field, let K/k be an extension, let X be a k -prescheme, let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$ -Module, and let $\mathcal{H}$ be a quasi-coherent sub- $\mathcal{O}_{X(K)}$ -Module of $\mathcal{F} \otimes_k K$. Then $\mathcal{H}$ has a smallest field of definition.

Proof. When X is affine, this is precisely (4.8.7). In general, let (X_α) be an affine open covering of X . For every α , there is a smallest subfield K'_α of K containing k over which $\mathcal{H}|_{(X_\alpha)(K)}$ is defined. By (4.8.5), the subfield of K generated by the K'_α is the smallest field of definition of $\mathcal{H}$. $\square$

Corollary (4.8.10). — With the notation of (4.8.9), let $\mathcal{G}$ be a second quasi-coherent $\mathcal{O}_X$ -Module and let

$$\bar{u} : \mathcal{F} \otimes_k K \longrightarrow \mathcal{G} \otimes_k K$$

be an $\mathcal{O}_{X(K)}$ -homomorphism. Then $\bar{u}$ has a smallest field of definition.

Proof. By (4.8.8.i), such a field is also the smallest field of definition of the graph of $\bar{u}$, whose existence follows from (4.8.9). $\square$

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Corollary (4.8.11). — Let k be a field, let X be a k -prescheme, let K/k be an extension, and let $\bar{Z}$ be a closed sub-prescheme of $X_{(K)}$. Then $\bar{Z}$ has a smallest field of definition.

Proof. Let $\bar{\mathcal{J}}$ be the quasi-coherent Ideal of $\mathcal{O}_{X(K)}$ defining $\bar{Z}$. By (4.8.8.ii), a field defines $\bar{Z}$ if and only if it defines $\bar{\mathcal{J}}$, and the assertion follows from (4.8.9). $\square$

Corollary (4.8.12). — Let k be a field, let X and Y be two k -preschemes with Y separated, let K/k be an extension, and let $\bar{f} : X_{(K)} \rightarrow Y_{(K)}$ be a K -morphism. Then $\bar{f}$ has a smallest field of definition.

Proof. By (4.8.8.iii), the existence of such a field is equivalent to the existence of a smallest field of definition for the graph $\bar{Z}$ of $\bar{f}$. Since $\bar{Z}$ is a closed sub-prescheme of $(X \times_k Y)_{(K)}$ (I, 5.4.3), the assertion follows from (4.8.11). $\square$

Proposition (4.8.13). — Under the hypotheses of (4.8.11) (resp. (4.8.9), (4.8.10), or (4.8.12)), suppose in addition that X is of finite type over k (resp. that X is of finite type over k and $\mathcal{F}$ is coherent; that X is of finite type over k and both $\mathcal{F}$ and $\mathcal{G}$ are coherent; or that X and Y are of finite type over k). Then the smallest field of definition of $\bar{Z}$ (resp. $\mathcal{H}$, $\bar{u}$, or $\bar{f}$) is an extension of finite type of k .

Proof. It is enough to treat (4.8.9), and we may assume that X is affine because it is a finite union of affine opens of finite type over k . With the notation of (4.8.6), it remains to show that if A is a k -algebra of finite type and M an A -module of finite type, then one of the fields K' satisfying the equivalent conditions of (4.8.6) may be chosen of finite type over k .

Let $(m_i)_{1 \leq i \leq n}$ generate M over A . Since A is Noetherian, so is $A_{(K)}$, and the $A_{(K)}$ -submodule $\bar{N}$ of $M_{(K)}$ has a finite set of generators n_j ($1 \leq j \leq r$). Write

$$n_j = \sum_i (m_i \otimes 1) b_{ij}, \quad b_{ij} \in A_{(K)},$$

and then

$$b_{ij} = \sum_h a_{ijh} \otimes c_{ijh}, \quad a_{ijh} \in A, \quad c_{ijh} \in K.$$

The extension K' of k generated by the finitely many c_{ijh} has the required property. $\square$

Proposition (4.8.14). — *Let k be a field, let X be a k -prescheme of finite type, and let K/k be an extension containing a perfect extension of k . Then the smallest field of definition of the closed sub-prescheme $(X_{(K)})_{\text{red}}$ of $X_{(K)}$ is a finite radical extension of k .*

Proof. If p is the characteristic exponent of k , the hypothesis implies that $k' = k^{p^{-\infty}}$ is contained in K . The prescheme $(X_{(k')})_{\text{red}}$ is geometrically reduced over k' (4.6.1). Thus

$$(X_{(k')})_{\text{red}} \otimes_{k'} K$$

is reduced and is consequently equal to $(X_{(K)})_{\text{red}}$ (I, 5.1.8). In other words, k' is a field of definition of $(X_{(K)})_{\text{red}}$. Since k'/k is algebraic, the conclusion follows from (4.8.13). $\square$

Remarks (4.8.15). — (i) The conclusion of (4.8.14) need not hold if K is not assumed to contain a perfect extension of k .

For example, let k_0 be a field of characteristic $p > 0$, let $k = k_0(s, t)$ be the field of rational functions in two indeterminates, let

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$$L = k(s^{1/p}, t^{1/p}), \quad X = \text{Spec}(L),$$

let u be a third indeterminate, and put

$$K = k(u, s^{1/p} + ut^{1/p}).$$

One readily verifies that k is algebraically closed in K and that $L \otimes_k K$ has a nonzero nilradical, so $X_{(K)}$ is not reduced. But $(X_{(K)})_{\text{red}}$ is not defined over k , while K contains no intermediate extension of finite degree over k other than k itself. Hence the smallest field of definition of $(X_{(K)})_{\text{red}}$ cannot be finite over k .

- (ii) Given an object in one of the target sets in (4.8.2) and an extension K_1/K , the object has a smallest field of definition if and only if its image under $s_{K_1, K}$ does; the two smallest subfields are necessarily the same, as observed in (4.8.4).

4.9. Fields of definition of subsets of a prescheme

Proposition (4.9.1). — *Let k be a field, let X be a k -prescheme, let K/k be an extension, let $\bar{T}$ be a closed subset of $X_{(K)}$, and let K' be an intermediate field of K/k . The following conditions are equivalent:*

- (a) $\bar{T}$ is defined over K' .
- (b) The open subset $\bar{U} = X_{(K)} - \bar{T}$ of $X_{(K)}$ is defined over K' .
- (b') The sub-prescheme of $X_{(K)}$ induced on the open subset $\bar{U}$ is defined over K' .
- (c) There exists a closed sub-prescheme $\bar{Z}$ of $X_{(K)}$, defined over K' , whose underlying space is $\bar{T}$.

Proof. Let $g : X_{(K)} \rightarrow X_{(K')}$ be the canonical projection. To say that $\bar{T}$ is defined over K' means that there is a closed subset T' of $X_{(K')}$ such that $\bar{T} = g^{-1}(T')$. Since

$$g^{-1}(X_{(K')} - T') = X_{(K)} - g^{-1}(T')$$

for every subset T' of $X_{(K')}$, conditions (4.9.1.a), (4.9.1.b), and (4.9.1.bprime) are equivalent.

If $\bar{Z}$, whose underlying space is $\bar{T}$, is defined over K' , then

$$\bar{Z} = Z' \otimes_{K'} K$$

for a closed sub-prescheme Z' of $X_{(K')}$. If T' is the underlying space of Z' , then $\bar{T} = g^{-1}(T')$ (I, 4.4.1); thus (4.9.1.c) implies (4.9.1.a). Conversely, if (4.9.1.a) holds, choose a closed sub-prescheme Z' of $X_{(K')}$ whose underlying space is T' (I, 5.2.1), and take $\bar{Z} = Z' \otimes_{K'} K$. $\square$

Corollary (4.9.2). — *With the notation of (4.9.1), suppose that K' is a field of definition of $\bar{T}$. Every intermediate field K'' with $k \subset K'' \subset K'$ for which K'/K'' is radicial is also a field of definition of $\bar{T}$.*

Proof. The canonical projection

$$g : X_{(K')} \longrightarrow X_{(K'')}$$

is integral, surjective, and radicial, hence is a homeomorphism (2.4.5). $\square$

Remark (4.9.3). — It follows from (4.9.2) that a closed subset $\bar{T}$ of $X_{(K)}$ need not have a smallest field of definition. For example, let $K = k(t)$, where t is an indeterminate and k is a perfect field of characteristic $p > 0$. A closed subset $\bar{F}$ of $X_{(K)}$ can have a smallest field of definition only if it is already defined over k . Indeed, for every subfield K' of K containing k with $K' \neq k$, one has $(K')^p \neq K'$, the field $(K')^p$ contains k , and $K'/(K')^p$ is radicial. Thus, if K' is a field of definition of $\bar{F}$, then so is $(K')^p$. Nevertheless, one has the following result.

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Proposition (4.9.4). — *With the notation of (4.9.1), suppose that K/K' is separable. Then K' is a field of definition of $\bar{T}$ if and only if it is a field of definition of the reduced sub-prescheme of $X_{(K)}$ whose underlying space is $\bar{T}$.*

Proof. The condition is sufficient by (4.9.1). Conversely, suppose that $\bar{T} = g^{-1}(T')$, where T' is a closed subset of $X_{(K')}$, and let Z' be the reduced sub-prescheme of $X_{(K')}$ whose underlying space is T' . Then

$$\bar{Z} = Z' \otimes_{K'} K = \text{Spec}(K) \times_{K'} Z'$$

is reduced by (4.6.1.e) and the separability of K/K' , and its underlying space is $\bar{T}$. $\square$

Corollary (4.9.5). — *Suppose that one of the following hypotheses holds:*

- (a) k has characteristic zero.
- (b) K/k is algebraic and X is of finite type over k .

Then every closed subset $\bar{T}$ of $X_{(K)}$ has a smallest field of definition. This field is a separable extension of k , and under (4.9.5.b) it is finite over k .

Proof. Under (4.9.5.a), K is separable over each of its subfields. It follows from (4.9.4) that the fields of definition of $\bar{T}$ are exactly those of the reduced sub-prescheme of $X_{(K)}$ whose underlying space is $\bar{T}$. The assertion therefore follows from (4.8.11).

Under (4.9.5.b), the extension K is radicial over a separable extension K_1 of k . For every intermediate field K' of K/k , the extension $K'/(K' \cap K_1)$ is radicial. By (4.9.2), K' is a field of definition of $\bar{T}$ if and only if $K' \cap K_1$ is. This reduces the question to the case $K' = K_1$, hence to the case of a separable algebraic extension of k , and the argument finishes as under (4.9.5.a). $\square$

Proposition (4.9.6). — *Let k be a field, let X be a k -prescheme, let K/k be an extension, let $\bar{U}$ be a subset of $X_{(K)}$ that is both open and closed, and let $\bar{V}$ be its complement. For an intermediate field K' of K/k , the following conditions are equivalent:*

- (a) K' is a field of definition of the subset $\bar{U}$ of $X_{(K)}$.
- (a') K' is a field of definition of the closed sub-prescheme of $X_{(K)}$ induced on the open subset $\bar{U}$.
- (b) K' is a field of definition of the subset $\bar{V}$ of $X_{(K)}$.
- (b') K' is a field of definition of the closed sub-prescheme of $X_{(K)}$ induced on the open subset $\bar{V}$.

Proof. It is clear that (a') implies (a), and (a) implies (b') by (4.9.1). Since $\bar{U}$ and $\bar{V}$ play symmetric roles, (b') implies (b), and (b) implies (a'). $\square$

Corollary (4.9.7). — *Under the hypotheses of (4.9.6), there is a smallest field of definition K' for the subset $\bar{U}$ of $X_{(K)}$. It is also the smallest field of definition of the subset $\bar{V}$ and of the closed sub-preschemes of $X_{(K)}$ induced on the open subsets $\bar{U}$ and $\bar{V}$. If, in addition, X is of finite type over k , then K'/k is finite and separable.*

Proof. In view of (4.9.6) and (4.8.11), only the last assertion remains to be proved. By (4.8.15.ii), we may reduce to the case in which K is algebraically closed. Let $\bar{k}$ be the algebraic closure of k contained in K , and let W_i ($1 \leq i \leq m$) be the connected components of $X_{(\bar{k})}$.

If

$$p : X_{(K)} \longrightarrow X_{(\bar{k})}$$

is the canonical projection, then the $p^{-1}(W_i)$ are the connected components of $X_{(K)}$ (4.4.6). Thus $\bar{U}$ is a union of some of these components and is therefore defined over $\bar{k}$. The conclusion follows from (4.9.5.b). $\square$

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§5. DIMENSION, DEPTH, AND REGULARITY IN LOCALLY NOETHERIAN PRESCHMES

This section merely restates, in geometric language and with various technical complements, the notions and results of commutative algebra set out in Chapter 0, §§ 16 and 17.

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5.1. Dimension of preschemes

(5.1.1). Let A be a ring and let $\mathfrak{J}$ be an ideal of A . Recall (0, 16.1) that the definitions in the theory of the dimension of a ring (0, 16.1.1) and those in the theory of the combinatorial dimension of topological spaces (0, 14.1.2) give the following relations:

$$(5.1.1.1) \quad \dim(\operatorname{Spec}(A)) = \dim(A),$$

$$(5.1.1.2) \quad \dim(V(\mathfrak{J})) = \dim(A/\mathfrak{J}),$$

$$(5.1.1.3) \quad \operatorname{codim}(V(\mathfrak{J}), \operatorname{Spec}(A)) = \operatorname{ht}(\mathfrak{J}).$$

(5.1.1.4) $\operatorname{Spec}(A)$ is a catenary space if and only if A is a catenary ring.

(5.1.1.5) $\operatorname{Spec}(A)$ is equidimensional if and only if A is equidimensional, if and only if the minimal prime ideals $\mathfrak{p}_\alpha$ of A are such that the rings $A/\mathfrak{p}_\alpha$ all have the same dimension.

(5.1.1.6) $\operatorname{Spec}(A)$ is equicodimensional if and only if A is equicodimensional, if and only if all maximal ideals of A have the same height.

Recall that a Noetherian ring A is said to be *biequidimensional* if $\operatorname{Spec}(A)$ is biequidimensional; that is, if $\operatorname{Spec}(A)$ is Noetherian and A is equidimensional, equicodimensional, catenary, and of finite dimension.

Proposition (5.1.2). — *Let X be a prescheme, let Y be an irreducible closed subset of X , and let y be the generic point of Y . Then*

$$(5.1.2.1) \quad \operatorname{codim}(Y, X) = \dim(\mathcal{O}_{X,y}).$$

Proof. Indeed, the irreducible closed subsets of X containing y are canonically in bijection with the irreducible closed subsets of $\operatorname{Spec}(\mathcal{O}_{X,y})$ (I, 2.4.2), and hence with the prime ideals of $\mathcal{O}_{X,y}$; formula (5.1.2.1) follows from the definitions.

In particular, this recovers the fact that the generic points of the irreducible components of X are the only points $x \in X$ such that $\dim(\mathcal{O}_x) = 0$ (I, 1.1.14). $\square$

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Corollary (5.1.3). — *For every closed subset Y of a prescheme X , one has*

$$(5.1.3.1) \quad \operatorname{codim}(Y, X) = \inf_{y \in Y} \dim(\mathcal{O}_{X,y}).$$

Moreover, if X is locally Noetherian, then for every $x \in Y$,

$$(5.1.3.2) \quad \operatorname{codim}_x(Y, X) = \inf_{\substack{y \in Y \\ x \in \{y\}}} \dim(\mathcal{O}_{X,y}).$$

Proof. Formula (5.1.3.2) follows from (5.1.3.1) and (0, 14.2.6). This corollary allows us to *define*, for any subset Y of a prescheme X , the *codimension* $\text{codim}(Y, X)$ of Y in X to be the right-hand side of (5.1.3.1). $\square$

Proposition (5.1.4). — *For every prescheme X , one has*

$$(5.1.4.1) \quad \dim(X) = \sup_{x \in X} \dim(\mathcal{O}_x).$$

If every irreducible closed subset of X contains a closed point, then one also has

$$(5.1.4.2) \quad \dim(X) = \sup_{x \in F} \dim(\mathcal{O}_x),$$

where F is the set of closed points of X .

Proof. It suffices to observe, by (I, 2.4.2), that the chains of irreducible closed subsets of X correspond bijectively to the chains of irreducible closed subsets of a local scheme of X . When every irreducible closed subset of X contains a closed point, one may plainly restrict to the local schemes at closed points of X .

Every irreducible closed subset of X contains a closed point when X is quasi-compact (0, 2.1.3); we shall see below (5.1.11) that the same is true when X is locally Noetherian. $\square$

Corollary (5.1.5). — *A prescheme X is catenary if and only if the local ring $\mathcal{O}_x$ is catenary for every $x \in X$. If, in addition, every irreducible closed subset of X contains a closed point, then for X to be catenary it is enough that $\mathcal{O}_x$ be catenary for every closed point x of X .*

Proof. The argument is the same as in (5.1.4), since it is a matter of comparing chains of irreducible closed subsets with the same endpoints. $\square$

(5.1.6). We now examine the more special properties tied to Noetherian hypotheses.

Recall (0, 16.2.3) that a Noetherian local ring $A \neq 0$ has finite dimension, equal to the minimum number of generators of an ideal of definition of A . For every prime ideal $\mathfrak{p}$ of A , its height, equal to $\dim(A_{\mathfrak{p}})$, is therefore also finite. Together with (5.1.2.1), (5.1.3.1), and (5.1.4.1), these facts show the following.

Proposition (5.1.7). — *For every nonempty closed subset Y of a locally Noetherian prescheme X , the number $\text{codim}(Y, X)$ is finite. If X is Noetherian and affine and Y is an irreducible closed subset of X , then $\text{codim}(Y, X)$ is the minimum number of sections s_i of $\mathcal{O}_X$ over X such that Y is an irreducible component of the set of points $x \in X$ for which $s_i(x) = 0$ for every i .*

Corollary (5.1.8). — *Let X be a locally Noetherian prescheme, let $\mathcal{L}$ be an invertible $\mathcal{O}_X$ -module, and let f be a section of $\mathcal{L}$ over X . Every irreducible component of the set Z of points $x \in X$ such that $f(x) = 0$ has codimension at most 1 in X ; it has codimension 1 if Z contains no irreducible component of X .*

Proof. We may restrict to the case $\mathcal{L} = \mathcal{O}_X$. If y is a generic point of an irreducible component Y of Z , the ideal (f_y) of $\mathcal{O}_{X,y}$ must be such that $\mathcal{O}_{X,y}/(f_y)$ has only one prime ideal. Thus f_y generates an ideal of definition of the Noetherian local ring $\mathcal{O}_{X,y}$, and $\text{codim}(Y, X) \leq 1$ by (5.1.7). If Z contains no irreducible component of X , then $\text{codim}(Y, X)$ cannot be 0 by (0, 14.2.1). $\square$

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Proposition (5.1.9). — *Let X be a locally Noetherian prescheme and let Y be a closed subset of X . If $x \in Y$ and $\mathcal{O}_{X,x}$ is a catenary ring, then*

$$(5.1.9.1) \quad \begin{aligned} \text{codim}_x(Y, X) &= \dim(\mathcal{O}_{X,x}) - \text{codim}(\overline{\{x\}}, Y) \\ &= \dim(\mathcal{O}_{X,x}) - \dim(\mathcal{O}_{Y,x}). \end{aligned}$$

Proof. Let Y_i ($1 \leq i \leq n$) be the irreducible components of Y containing x ; there are only finitely many because Y is locally Noetherian. Let y_i be the generic point of Y_i . Put $A = \mathcal{O}_{X,x}$, and let $\mathfrak{p}_i$ be the prime ideal of A corresponding to $Y_i \cap \text{Spec}(A)$. The irreducible closed subsets of Y containing x correspond bijectively to the prime ideals of A containing one of the $\mathfrak{p}_i$, so

$$\dim(\mathcal{O}_{Y,x}) = \sup_i \dim(A/\mathfrak{p}_i).$$

On the other hand, $\mathcal{O}_{X,y_i} = A_{\mathfrak{p}_i}$, and the hypothesis on A gives

$$\dim(A) = \dim(A/\mathfrak{p}_i) + \dim(A_{\mathfrak{p}_i})$$

by (0, 16.1.4). The conclusion follows from

$$\operatorname{codim}_x(Y, X) = \inf_i \dim(\mathcal{O}_{X,y_i})$$

using (5.1.2.1) and (0, 14.2.6). $\square$

Proposition (5.1.10). — (i) *In every prescheme X , every nonempty locally constructible subset E contains a point x such that $\{x\}$ is locally closed in X ; equivalently, x is isolated in $\overline{\{x\}}$.*
(ii) *Let X be a locally Noetherian prescheme and let $x \in X$ be such that $\{x\}$ is locally closed in X . Then $\dim(\overline{\{x\}}) \leq 1$; consequently, every point $y \neq x$ of $\overline{\{x\}}$ is closed in X .*

Proof. (5.1.10.i) is a special case of the following lemma. $\square$

Lemma (5.1.10.1). — *Let X be a topological space with the following property: for every nonempty locally closed subset Z of X , there exist a subset Z' of Z , locally closed in X (or, equivalently, in Z), and a point $x \in Z'$ that is closed in Z' . Then every nonempty locally closed subset Z of X contains a point x isolated in $\overline{\{x\}}$.*

Proof. Let $Z' \subset Z$ be a locally closed subset of Z containing a point x that is closed in Z' . There is an open neighbourhood U of x in X such that x is closed in U ; hence

$$U \cap \overline{\{x\}} = \{x\},$$

which says precisely that x is isolated in $\overline{\{x\}}$.

The lemma applies to the underlying space of every prescheme X . Indeed, Z is then itself the underlying space of a prescheme (I, 5.2.1), and one may take for Z' an affine open subset of Z , which is a quasi-compact Kolmogorov space and therefore contains a closed point (0, 2.1.3). $\square$

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Proof of (5.1.10.ii). Let Z be the reduced sub-prescheme of X whose underlying space is $\overline{\{x\}}$. The hypothesis implies that $\{x\}$ is open in Z . Thus, for every $z \in Z$, the generic point x of $\operatorname{Spec}(\mathcal{O}_{Z,z})$ is isolated in $\operatorname{Spec}(\mathcal{O}_{Z,z})$. The ring $A = \mathcal{O}_{Z,z}$ is a Noetherian local integral domain, and the hypothesis implies that there exists $f \in A$ such that A_f is a field. It follows from (0, 16.3.3) that $\dim(A) \leq 1$. Since $\dim(\mathcal{O}_{Z,z}) \leq 1$ for every $z \in Z$, one has $\dim(Z) \leq 1$. $\square$

Corollary (5.1.11). — *If X is a locally Noetherian prescheme, every nonempty closed subset of X contains a closed point.*

Proof. Every closed subset of X is constructible (0, 9.1.1) and (0, 9.1.5), so it suffices to apply (5.1.10). $\square$

(5.1.12). Let X be a prescheme, let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$ -module of finite type, and let $S = \operatorname{Supp}(\mathcal{F})$ be its support, which is closed in X (0, 5.2.2). For every $x \in X$, regard $\operatorname{Supp}(\mathcal{F}_x)$ as a closed subset of the local scheme $\operatorname{Spec}(\mathcal{O}_x)$. By definition (0, 16.1.7),

$$\dim(\mathcal{F}_x) = \dim(\operatorname{Supp}(\mathcal{F}_x)).$$

But

$$\operatorname{Supp}(\mathcal{F}_x) = S \cap \operatorname{Spec}(\mathcal{O}_{X,x}) = \operatorname{Spec}(\mathcal{O}_{S,x}),$$

where in the last expression S denotes any closed sub-prescheme of X whose underlying space is S . The irreducible components of $\operatorname{Spec}(\mathcal{O}_{S,x})$ are the intersections of $\operatorname{Spec}(\mathcal{O}_{X,x})$ with the irreducible components of S containing x , and correspond bijectively to the latter. Hence, by (5.1.1.1),

$$(5.1.12.1) \quad \dim(\mathcal{F}_x) = \dim(\mathcal{O}_{S,x}).$$

It also follows from (5.1.2.1) that, if X is locally Noetherian,

$$(5.1.12.2) \quad \dim(\mathcal{F}_x) = \operatorname{codim}(\overline{\{x\}}, S).$$

We say that $\mathcal{F}$ is *equidimensional at the point* $x \in X$ if $\mathcal{F}_x$ is an equidimensional $\mathcal{O}_{X,x}$ -module; that is (0, 16.1.7), if $\text{Supp}(\mathcal{F}_x)$ is equidimensional as a closed subset of $\text{Spec}(\mathcal{O}_{X,x})$. Equivalently, the ring $\mathcal{O}_{S,x}$ is equidimensional.

The *dimension* of $\mathcal{F}$, denoted $\dim(\mathcal{F})$, is the dimension of its support $\text{Supp}(\mathcal{F})$. Equations (5.1.4.1) and (5.1.12.1) give

$$(5.1.12.3) \quad \dim(\mathcal{F}) = \sup_{x \in X} \dim(\mathcal{F}_x).$$

If $X = \text{Spec}(A)$ is affine and $\mathcal{F} = \tilde{M}$, where M is a finite A -module, then $\dim(\mathcal{F}) = \dim(M)$ by (0, 16.1.7) and (5.1.4.1).

Proposition (5.1.13). — *Let X be a prescheme, let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$ -module of finite type, let $x \in X$, and let x' be a generization of x in X . Then*

$$(5.1.13.1) \quad \dim(\mathcal{F}_{x'}) \leq \dim(\mathcal{F}_x).$$

Proof. This follows immediately from (5.1.12.1) and the definitions. $\square$

5.2. Dimension of an algebraic prescheme

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Proposition (5.2.1). — *Let k be a field, let X be an irreducible prescheme locally of finite type over k , and let ζ be its generic point. Then X is biequidimensional and*

$$\dim(X) = \text{tr. deg}_k \kappa(\zeta).$$

Proof. Every local ring $\mathcal{O}_x$ is a local ring of a finite-type k -algebra, hence is a quotient of a regular local ring (0, 17.3.9). Such rings are catenary (0, 16.5.12); consequently X is a catenary space (5.1.5). Since X is irreducible, it follows from (5.1.1) that it is enough to prove, for every closed point $x \in X$, that

$$(5.2.1.1) \quad \dim(\mathcal{O}_x) = \text{tr. deg}_k \kappa(\zeta).$$

We may plainly suppose that X is reduced and affine, hence integral, with ring A , where A is a finite-type k -algebra. Put

$$n = \text{tr. deg}_k \kappa(\zeta), \quad K = \kappa(\zeta) = \text{Frac}(A).$$

There exists a k -subalgebra

$$B = k[t_1, \dots, t_n] \subset A,$$

where the t_i are algebraically independent over k , such that A is a finite B -algebra (Bourbaki, *Algèbre commutative*, chap. V, § 3, n° 1, th. 1). Let $\mathfrak{m} = \mathfrak{j}_x$, which is maximal by hypothesis. Then $\mathfrak{n} = B \cap \mathfrak{m}$ is a maximal ideal of B (ibid., chap. V, § 2, n° 1, prop. 1), and $A_{\mathfrak{m}}$ is a local ring of the finite $B_{\mathfrak{n}}$ -algebra $S^{-1}A$, where $S = B - \mathfrak{n}$. Since $B_{\mathfrak{n}}$ is integrally closed and $S^{-1}A$ is integral,

$$\dim(A_{\mathfrak{m}}) = \dim(B_{\mathfrak{n}})$$

by (0, 16.1.6).

We may therefore reduce to $A = k[t_1, \dots, t_n]$. The residue field $k' = \kappa(x)$ is then a finite extension of k (I, 6.4.2). Put $A' = k'[t_1, \dots, t_n]$. By (I, 2.4.6) and (I, 3.3.14), there exists a maximal ideal $\mathfrak{m}'$ of A' lying above $\mathfrak{m}$ such that k' is the residue field of $A'_{\mathfrak{m}'}$. Since A' is integral and finite over A , the preceding argument gives

$$\dim(A'_{\mathfrak{m}'}) = \dim(A_{\mathfrak{m}}),$$

so we are reduced to the case $A/\mathfrak{m} = k$.

In that case $\mathfrak{m}$ is generated by the polynomials $t_i - a_i$, where $a_i \in k$ is the image of t_i in $A/\mathfrak{m}$. Replacing t_i by $t_i - a_i$, we are finally reduced to

$$\mathfrak{m} = \sum_{i=1}^n At_i.$$

The completion of $A_{\mathfrak{m}}$ is then the formal power-series ring $k[[t_1, \dots, t_n]]$. It has the same dimension as $A_{\mathfrak{m}}$ (0, 16.2.4), while $\dim(k[[t_1, \dots, t_n]]) = n$ by (0, 17.1.4). This proves (5.2.1.1). $\square$

Corollary (5.2.2). — *For a prescheme X locally of finite type over a field k , $\dim(X)$ agrees with the number defined in (4.1.1).*

Proof. Let X_α be the reduced sub-preschemes whose underlying spaces are the irreducible components of X . Then

$$\dim(X) = \sup_\alpha \dim(X_\alpha)$$

by (0, 14.1.2.1), and it suffices to apply (5.2.1) to the X_α . $\square$

Corollary (5.2.3). — *Let X be a prescheme locally of finite type over a field k , and let $x \in X$. Then*

$$(5.2.3.1) \quad \dim_x(X) = \dim(\mathcal{O}_x) + \text{tr. deg}_k \kappa(x).$$

Proof. There is an open neighbourhood U of x in X such that $\dim_x(X) = \dim(U)$ (0, 14.1.4.1). We may suppose that the irreducible components of U are the $X_i \cap U$, where the X_i are the irreducible components of X containing x .

Since $U \cap X_i$ is dense in X_i , formula (4.1.1.3) gives $\dim(X_i) = \dim(U \cap X_i)$, and hence

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$$\dim_x(X) = \sup_i \dim(X_i).$$

The minimal prime ideals of $\mathcal{O}_x$ correspond to the generic points of the X_i , so

$$\dim(\mathcal{O}_{X,x}) = \sup_i \dim(\mathcal{O}_{X_i,x})$$

by (0, 16.1.1.1). We are therefore reduced to the case where X is irreducible. Since X is biequidimensional by (5.2.1),

$$\dim(X) = \dim(\overline{\{x\}}) + \text{codim}(\overline{\{x\}}, X)$$

by (0, 14.3.5.1). Finally,

$$\dim(\overline{\{x\}}) = \text{tr. deg}_k \kappa(x)$$

by (5.2.1), while

$$\text{codim}(\overline{\{x\}}, X) = \dim(\mathcal{O}_x)$$

by (5.1.2). $\square$

Corollary (5.2.4). — *Let X be a prescheme locally of finite type over a field k , let $\mathcal{L}$ be an invertible $\mathcal{O}_X$ -module, and let f be a section of $\mathcal{L}$ over X such that the set*

$$Y = \{x \in X \mid f(x) = 0\}$$

is rare in X . Then

$$\dim(Y) \leq \dim(X) - 1.$$

Equality holds if Y meets every irreducible component of X having maximum dimension.

Proof. If (X_α) is the family of irreducible components of X , then

$$\dim(Y) = \sup_\alpha \dim(Y \cap X_\alpha).$$

We are therefore reduced to the case where X is irreducible, because $Y \cap X_\alpha$ is rare in X_α and every X_α has nonempty interior in X . We may also suppose that $Y \neq \emptyset$. For every maximal point x of Y , biequidimensionality gives

$$\dim(\overline{\{x\}}) = \dim(X) - \text{codim}(\overline{\{x\}}, X).$$

Since Y is rare in X ,

$$\text{codim}(\overline{\{x\}}, X) = 1$$

by (5.1.8), and the result follows. $\square$

Remarks (5.2.5). — (i) Unlike an algebraic k -prescheme, a locally Noetherian prescheme X need not be catenary; see (5.6.11). It is catenary, however, if all its local rings $\mathcal{O}_x$ are quotients of regular local rings (0, 16.5.12), and in particular if X is regular (I, 4.1.4).

Nevertheless, even an integral scheme $X = \text{Spec}(A)$ with A regular need not be biequidimensional. In other words, the closed points of X need not all have the same codimension (0, 14.3.3). For example, let B be a discrete valuation ring, let $\mathfrak{m} = B\pi$ be its maximal ideal, let $k = B/\mathfrak{m}$ be its residue field, let K be its fraction field, and put $A = B[T]$. The ring A has maximal ideals of height 2, such as $(\pi) + (T)$, but also maximal ideals of height 1, such as

the principal ideal $(\pi T - 1)$. Indeed, a nonzero principal prime ideal has height 1 (5.1.8), while

$$A/(\pi T - 1) \simeq B_\pi = K$$

by (0, 1.2.3); thus $(\pi T - 1)$ is maximal and has height 1.

- (ii) If X is locally of finite type over a field, then $\dim(U) = \dim(X)$ for every dense open subset U of X (4.1.1.3). This does not extend to regular Noetherian preschemes, even when they are biequidimensional. For example, let B be a discrete valuation ring with maximal ideal $\mathfrak{m}$. Then $X = \operatorname{Spec}(B)$ has two points, (0) and $\mathfrak{m}$, the latter being its only closed point. One has $\dim(X) = 1$, whereas the open subset $U = \{(0)\}$ has dimension 0 (see § 10).⁵

5.3. Dimension of the support of a module and Hilbert polynomial This section uses the results of Chapter III; it will not be used in the remainder of this chapter. IV-2 | 92

Proposition (5.3.1). — *Let A be an Artinian local ring, let X be a projective scheme over $\operatorname{Spec}(A)$, let $\mathcal{L}$ be an invertible $\mathcal{O}_X$ -module that is very ample relative to A , and let $\mathcal{F}$ be a nonzero coherent $\mathcal{O}_X$ -module. Put*

$$\mathcal{F}(n) = \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes n} \quad (n \in \mathbf{Z}).$$

Then the degree of the Hilbert polynomial

$$P(n) = \chi_A(\mathcal{F}(n))$$

of $\mathcal{F}$ relative to A (III, 2.5.3) equals $\dim(\operatorname{Supp}(\mathcal{F}))$.

Proof. We argue by induction on

$$d = \dim(\operatorname{Supp}(\mathcal{F})).$$

There is a closed subprescheme Y of X whose underlying space is $\operatorname{Supp}(\mathcal{F})$, together with a coherent $\mathcal{O}_Y$ -module $\mathcal{G}$ such that $\mathcal{F} = j_*(\mathcal{G})$, where $j : Y \rightarrow X$ is the canonical injection (Err_{III}, 30). The Hilbert polynomials of $\mathcal{F}$ and $\mathcal{G}$ are plainly the same, so we may restrict to the case $X = \operatorname{Supp}(\mathcal{F})$.

Suppose first that $d = 0$. Every point of X is then closed, and X is an Artinian scheme (I, 6.2.2); hence $\mathcal{F}(n) = \mathcal{F}$ for every integer n . Consequently, by (III, 2.5.3),

$$\chi_A(\mathcal{F}(n)) = \operatorname{length}_A(\Gamma(X, \mathcal{F}))$$

for all sufficiently large n , and the Hilbert polynomial therefore has degree 0.

Suppose now that $d > 0$, and put $Z = \operatorname{Ass}(\mathcal{F})$, a finite set (3.1.6). There are an integer $m > 0$ and a section $f \in \Gamma(X, \mathcal{L}^{\otimes m})$ such that X_f is a neighbourhood of Z (II, 4.5.4). Multiplication by f is an injective homomorphism

$$\mu_f : \mathcal{F} \longrightarrow \mathcal{F}(m)$$

by (3.1.8); thus there is an exact sequence

$$(5.3.1.1) \quad 0 \longrightarrow \mathcal{F} \xrightarrow{\mu_f} \mathcal{F}(m) \longrightarrow \mathcal{G} \longrightarrow 0,$$

where $\mathcal{G}$ is coherent. By Nakayama's lemma, the points $x \in \operatorname{Supp}(\mathcal{G})$ are exactly those for which $f(x) = 0$. We shall deduce that

$$(5.3.1.2) \quad \dim(\operatorname{Supp}(\mathcal{G})) = d - 1.$$

This follows from the next lemma. □

Lemma (5.3.1.3). — *Let A be an Artinian local ring, let X be a projective scheme over $\operatorname{Spec}(A)$, and let $\mathcal{L}$ be an invertible $\mathcal{O}_X$ -module that is ample relative to A . For every section g of $\mathcal{L}$ over X , the set*

$$X \setminus X_g = \{x \in X \mid g(x) = 0\}$$

meets every irreducible component of X having nonzero dimension.

Proof. Indeed, X_g is an affine open subset of X (II, 5.5.7). If it contains an irreducible component X' of X , the reduced closed subprescheme of X whose underlying space is X' is both projective and affine over A , hence finite over A (III, 4.4.2). It is therefore an Artinian scheme and has dimension 0. □

⁵The printed French text has $X = \operatorname{Spec}(A)$ here; the ring just specified, and the asserted two-point spectrum, require $X = \operatorname{Spec}(B)$.

Proof of (5.3.1), continued. Since Z contains the maximal points of X , the open subset X_f is dense; hence $\text{Supp}(\mathcal{G})$ is rare in X . Equation (5.3.1.2) now follows from the lemma and (5.2.4).

From the exact sequence (5.3.1.1), for every n we obtain the exact sequence

$$0 \longrightarrow \mathcal{F}(n) \longrightarrow \mathcal{F}(n+m) \longrightarrow \mathcal{G}(n) \longrightarrow 0.$$

For all sufficiently large n , there is therefore also an exact sequence (III, 2.2.3)

$$0 \longrightarrow \Gamma(X, \mathcal{F}(n)) \longrightarrow \Gamma(X, \mathcal{F}(n+m)) \longrightarrow \Gamma(X, \mathcal{G}(n)) \longrightarrow 0.$$

Taking (III, 2.5.3) into account, we have, for all sufficiently large n ,

$$\chi_A(\mathcal{G}(n)) = \chi_A(\mathcal{F}(n+m)) - \chi_A(\mathcal{F}(n)).$$

By (5.3.1.2) and the induction hypothesis, the polynomial $\chi_A(\mathcal{G}(n))$ has degree $d-1$. The preceding relation therefore implies that $\chi_A(\mathcal{F}(n))$ has degree d . $\square$

5.4. Dimension of the image of a morphism

Proposition (5.4.1). — *Let X and Y be locally Noetherian preschemes, and let $f : X \rightarrow Y$ be a morphism.*

(i) *If f is quasi-finite, then*

$$\dim(X) \leq \dim(\overline{f(X)}) \leq \dim(Y).$$

(ii) *If f is surjective and open (respectively, surjective and closed), then*

$$\dim(X) \geq \dim(Y).$$

Proof. (i) We may replace f by f_{red} (II, 6.2.4), and therefore suppose that X and Y are reduced. Let Z be the reduced closed subprescheme of Y whose underlying space is $\overline{f(X)}$. Then $f = j \circ g$, where $j : Z \rightarrow Y$ is the canonical injection and $g : X \rightarrow Z$ is quasi-finite (I, 5.2.2), (II, 6.2.4). We may thus restrict to the case in which $f(X)$ is dense in Y .

For every $x \in X$, the ring $\mathcal{O}_x$ is then a quasi-finite $\mathcal{O}_{f(x)}$ -module (II, 6.2.2); consequently

$$\mathfrak{m}_{f(x)} \mathcal{O}_x$$

is an ideal of definition of $\mathcal{O}_x$ (0, 7.4.4). If $A \rightarrow B$ is a local homomorphism of Noetherian local rings, with maximal ideal $\mathfrak{m}$ of A , and if $\mathfrak{m}B$ is an ideal of definition of B , then

$$\dim(B) \leq \dim(A)$$

by (0, 16.3.10). Part (i) now follows from (5.1.4).

(ii) By the definition of dimension (0, 14.1.2), it is enough to prove the following. For every sequence $(y_i)_{0 \leq i \leq n}$ of distinct points of Y such that

$$y_i \in \overline{\{y_{i+1}\}} \quad (0 \leq i \leq n-1),$$

there is a sequence $(x_i)_{0 \leq i \leq n}$ of points of X such that

$$x_i \in \overline{\{x_{i+1}\}} \quad (0 \leq i \leq n-1)$$

and $f(x_i) = y_i$ for every i .

Suppose first that f is surjective and open. We construct the x_i by induction on i . Surjectivity gives an $x_0 \in X$ such that $f(x_0) = y_0$. Suppose the x_i have been chosen for $i \leq m$ so that $f(x_i) = y_i$ for $i \leq m$ and $x_i \in \overline{\{x_{i+1}\}}$ for $i < m$. Since f is open and y_{m+1} is a generization of y_m , there is an $x_{m+1} \in f^{-1}(y_{m+1})$ that is a generization of x_m (1.10.3). The induction therefore continues.

Suppose now that f is surjective and closed. We construct the x_i by descending induction on i . Surjectivity gives an x_n such that $f(x_n) = y_n$. Suppose the x_i have been chosen with the required properties for $i > m$. Because f is closed,

$$f(\overline{\{x_{m+1}\}}) = \overline{\{f(x_{m+1})\}} = \overline{\{y_{m+1}\}}$$

(Bourbaki, *Topologie générale*, chap. I, 3rd ed., § 5, n° 4, prop. 9). There is therefore an $x_m \in \overline{\{x_{m+1}\}}$ such that $f(x_m) = y_m$, and the descending induction continues. $\square$

Corollary (5.4.2). — *Let X and Y be locally Noetherian preschemes, and let $f : X \rightarrow Y$ be a finite morphism, hence a closed morphism. Then*

$$\dim(X) = \dim(\overline{f(X)}).$$

If f is moreover surjective, then $\dim(X) = \dim(Y)$.

Remarks (5.4.3). — (i) We saw in (4.1.2) that if X and Y are preschemes locally of finite type over a field k , and f is a k -morphism, the inequality

$$\dim(Y) \leq \dim(X)$$

already holds when f is quasi-compact and dominant. By contrast, if one assumes only that X and Y are locally Noetherian, it is possible to have $\dim(Y) > \dim(X)$ even when f is of finite type, bijective, and a local immersion.

For example, let A be a discrete valuation ring, let K be its fraction field, and let k be its residue field. Put $Y = \operatorname{Spec}(A)$, and denote by a the generic point of Y and by b its closed point; thus $\kappa(a) = K$ and $\kappa(b) = k$. Let

$$X = \operatorname{Spec}(K) \amalg \operatorname{Spec}(k),$$

and let $f : X \rightarrow Y$ agree on the two summands with the respective canonical morphisms (I, 2.4.5). The morphism f is plainly a bijective local immersion, since $\{a\}$ is open in Y . It is also of finite type: if π is a uniformizer of A , then $K = A[\pi^{-1}]$, so K is a finitely generated A -algebra. Nevertheless,

$$\dim(X) = 0 \quad \text{and} \quad \dim(Y) = 1.$$

(ii) Let $A \subset B$ be Noetherian rings, with B a finite A -algebra. Corollary (5.4.2) again gives

$$\dim(B) = \dim(A)$$

(0, 16.1.5). Suppose in addition that A is a Noetherian local ring. Then B is a Noetherian semilocal ring (Bourbaki, *Algèbre commutative*, chap. IV, § 2, n° 5, cor. 3 of prop. 9). If $\mathfrak{n}_i$ ($1 \leq i \leq r$) are the maximal ideals of B , then

$$(5.4.3.1) \quad \dim(A) = \sup_i \dim(B_{\mathfrak{n}_i}).$$

Under these hypotheses one need not have

$$\dim(B_{\mathfrak{n}_i}) = \dim(A)$$

for every i ; replacing B by the direct product of B with the residue field k of A gives an immediate example. There are even examples in which A and B are integral domains of dimension 2 and some of the $B_{\mathfrak{n}_i}$ have dimension strictly less than $\dim(A)$ (5.6.11). We shall see later, in (5.6.4) and (5.6.10), that this last phenomenon cannot occur when A is a quotient of a regular local ring.

5.5. Dimension formula for a morphism of finite type

(5.5.1).

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Recall (0, 16.3.9) that if A and B are Noetherian local rings, k is the residue field of A , and $\varphi : A \rightarrow B$ is a local homomorphism, then

$$(5.5.1.1) \quad \dim(B) \leq \dim(A) + \dim(B \otimes_A k).$$

We deduce the following.

Proposition (5.5.2). — *Let X and Y be locally Noetherian preschemes, let $f : X \rightarrow Y$ be a morphism, let $x \in X$, and put $y = f(x)$. Then*

$$(5.5.2.1) \quad \dim(\mathcal{O}_x) \leq \dim(\mathcal{O}_y) + \dim(\mathcal{O}_x \otimes_{\mathcal{O}_y} \kappa(y)).$$

In particular, if x is a maximal point of the fibre $f^{-1}(y)$, then

$$(5.5.2.2) \quad \dim(\mathcal{O}_x) \leq \dim(\mathcal{O}_y),$$

because

$$\mathcal{O}_x \otimes_{\mathcal{O}_y} \kappa(y) = \mathcal{O}_x / \mathfrak{m}_y \mathcal{O}_x$$

is the local ring of x in the prescheme $f^{-1}(y)$ and therefore has dimension 0.

We shall now obtain a formula more precise than (5.5.2.1) when f is assumed to be of finite type. **IV-2 | 95**

Proposition (5.5.3). — *Let A be a Noetherian ring, let T be an indeterminate, and let $\mathfrak{p}$ be a prime ideal of A . Then*

$$\mathfrak{p}' = \mathfrak{p}A[T]$$

is a prime ideal of $B = A[T]$, and $\mathfrak{p}' \cap A = \mathfrak{p}$. There are infinitely many prime ideals of B , distinct from $\mathfrak{p}'$, whose intersection with A is $\mathfrak{p}$; these prime ideals are pairwise incomparable. Moreover, if $\mathfrak{q}$ is any such ideal, then

$$(5.5.3.1) \quad \dim(B_{\mathfrak{q}}) = \dim(B_{\mathfrak{p}'}) + 1 = \dim(A_{\mathfrak{p}}) + 1.$$

Proof. For the first assertions, replacing A by $A/\mathfrak{p}$ and using

$$A[T]/\mathfrak{p}A[T] = (A/\mathfrak{p})[T]$$

reduces us immediately to the case $\mathfrak{p} = 0$. The prime ideals of $B = A[T]$ whose intersection with A is 0 are exactly those disjoint from the multiplicative set $S = A - \{0\}$ of the domain A . There is an inclusion-preserving bijection from this set of prime ideals to the prime ideals of

$$S^{-1}A[T] = K[T],$$

where K is the fraction field of A (Bourbaki, *Algèbre commutative*, chap. II, § 2, n° 5, prop. 11). The assertions follow.

Furthermore, by (5.5.1.1),⁶

$$\dim(B_{\mathfrak{q}}) \leq \dim(A_{\mathfrak{p}}) + \dim(B_{\mathfrak{q}}/\mathfrak{p}B_{\mathfrak{q}}).$$

Let k be the fraction field of $A/\mathfrak{p}$. The ring $B_{\mathfrak{q}}/\mathfrak{p}B_{\mathfrak{q}}$ is canonically identified with

$$(k[T])_{\bar{\mathfrak{q}}},$$

where $\bar{\mathfrak{q}}$ is a nonzero prime ideal of $k[T]$. It is therefore a discrete valuation ring and has dimension 1. Thus

$$\dim(B_{\mathfrak{q}}) \leq \dim(A_{\mathfrak{p}}) + 1.$$

Let

$$\mathfrak{p} = \mathfrak{p}_0 \supset \mathfrak{p}_1 \supset \cdots \supset \mathfrak{p}_m$$

be a chain of prime ideals of A of maximum length. The ideals $\mathfrak{p}_j B$ ($0 \leq j \leq m$) are prime in B , pairwise distinct, and contained in $\mathfrak{q}$. Hence

$$\dim(B_{\mathfrak{q}}) \geq m + 1 = \dim(A_{\mathfrak{p}}) + 1.$$

It follows that

$$\mathrm{ht}(\mathfrak{q}) = \mathrm{ht}(\mathfrak{p}) + 1.$$

Since $\mathfrak{q} \neq \mathfrak{p}'$, we also have

$$\mathrm{ht}(\mathfrak{q}) \geq \mathrm{ht}(\mathfrak{p}') + 1 \geq \mathrm{ht}(\mathfrak{p}) + 1.$$

All these inequalities are equalities, and (5.5.3.1) follows. □

Corollary (5.5.4). — *For every Noetherian ring A ,*

$$(5.5.4.1) \quad \dim(A[T_1, \dots, T_r]) = \dim(A) + r,$$

where the T_i are indeterminates.

By contrast, there are examples of non-Noetherian local rings A such that

$$\dim(A) = 1 \quad \text{and} \quad \dim(A[T]) = 3$$

[30].

Corollary (5.5.5). — *For every locally Noetherian prescheme X , the dimension of*

$$X[T_1, \dots, T_r] = X \otimes_{\mathbf{Z}} \mathbf{Z}[T_1, \dots, T_r],$$

where the T_i are indeterminates, is $\dim(X) + r$.

⁶The printed French text cites (5.5.1.2), which does not exist. The displayed inequality used here is (5.5.1.1).

Proof. This follows from (5.5.4) and (0, 14.1.7). $\square$

Corollary (5.5.6). — *Under the hypotheses of (5.5.3), let $\mathfrak{q}$ be a prime ideal of $B = A[T]$ such that $\mathfrak{q} \cap A = \mathfrak{p}$. If k and k' are the residue fields of $A_{\mathfrak{p}}$ and $B_{\mathfrak{q}}$, respectively, then*

$$(5.5.6.1) \quad \dim(A_{\mathfrak{p}}) + 1 = \dim(B_{\mathfrak{q}}) + \text{tr. deg}_k k'.$$

Proof.

If $\mathfrak{q} = \mathfrak{p}B$, then

$$\dim(B_{\mathfrak{q}}) = \dim(A_{\mathfrak{p}})$$

by (5.5.3.1), while $k' = k(T)$. Thus (5.5.6.1) holds. If $\mathfrak{q} \neq \mathfrak{p}B$, then

$$\dim(B_{\mathfrak{q}}) = \dim(A_{\mathfrak{p}}) + 1,$$

and $\mathfrak{q}$ corresponds to a nonzero prime ideal of $k[T]$. Consequently k' is algebraic over k , and (5.5.6.1) again holds. $\square$

Lemma (5.5.7). — *Let A be a Noetherian local domain, let $\mathfrak{m}$ be its maximal ideal, and let k be its residue field.*

- (i) *Let B be a domain containing A such that $B = A[x]$ for some $x \in B$, and let $\mathfrak{q}$ be a prime ideal of B such that $\mathfrak{q} \cap A = \mathfrak{m}$. Then*

$$(5.5.7.1) \quad \dim(A) + \text{tr. deg}_A B \geq \dim(B_{\mathfrak{q}}) + \text{tr. deg}_k k',$$

where k' is the residue field of $B_{\mathfrak{q}}$ and $\text{tr. deg}_A B$ denotes the transcendence degree of the fraction field of B over that of A .

- (ii) *Suppose that for every maximal ideal $\mathfrak{n}$ of $A[T]$ such that $\mathfrak{n} \cap A = \mathfrak{m}$, the ring $(A[T])_{\mathfrak{n}}$ is catenary. Then the two sides of (5.5.7.1) are equal.*

Proof. (i) If x is transcendental over the fraction field of A , then $\text{tr. deg}_A B = 1$, and the two sides of (5.5.7.1) are equal by (5.5.6).

Suppose instead that x is algebraic over the fraction field of A . Then

$$B = A[T]/\mathfrak{p},$$

where $\mathfrak{p}$ is a nonzero prime ideal of $A[T]$ such that $\mathfrak{p} \cap A = 0$, because B contains A . Thus

$$\text{ht}(\mathfrak{p}) = 1$$

by (5.5.3). The ideal $\mathfrak{q}$ is of the form $\mathfrak{n}/\mathfrak{p}$, where $\mathfrak{n} \supset \mathfrak{p}$ is a prime ideal of $A[T]$ satisfying $\mathfrak{n} \cap A = \mathfrak{m}$, and

$$B_{\mathfrak{q}} = (A[T])_{\mathfrak{n}}/\mathfrak{p}(A[T])_{\mathfrak{n}}.$$

Formula (0, 16.1.4.1), applied to

$$X = \text{Spec}((A[T])_{\mathfrak{n}}) \quad \text{and} \quad Y = \text{Spec}(B_{\mathfrak{q}}),$$

gives

$$(5.5.7.2) \quad \begin{aligned} \dim((A[T])_{\mathfrak{n}}) &\geq \text{ht}(\mathfrak{p}(A[T])_{\mathfrak{n}}) + \dim(B_{\mathfrak{q}}) \\ &= 1 + \dim(B_{\mathfrak{q}}). \end{aligned}$$

Indeed,

$$\text{ht}(\mathfrak{p}(A[T])_{\mathfrak{n}}) = \text{ht}(\mathfrak{p}) = 1$$

by the bijection between prime ideals of $A[T]$ contained in $\mathfrak{n}$ and prime ideals of $(A[T])_{\mathfrak{n}}$. Finally, (5.5.6.1) gives

$$(5.5.7.3) \quad \dim((A[T])_{\mathfrak{n}}) = \dim(A) + 1 - \text{tr. deg}_k k',$$

because $B_{\mathfrak{q}}$ and $(A[T])_{\mathfrak{n}}$ have the same residue field. Here $\text{tr. deg}_A B = 0$, and (5.5.7.1) follows.

- (ii) If $(A[T])_{\mathfrak{n}}$ is catenary, the two sides of (5.5.7.2) are equal (0, 16.1.4); hence the two sides of (5.5.7.1) are equal as well. $\square$

Theorem (5.5.8 (dimension formula)). — *Let A be a Noetherian local domain, let B be a domain containing A and finitely generated as an A -algebra, and let $\mathfrak{q}$ be a prime ideal of B such that $\mathfrak{q} \cap A$ is the maximal ideal $\mathfrak{m}$ of A . Let k and k' be the residue fields of A and $B_{\mathfrak{q}}$, respectively. Then*

$$(5.5.8.1) \quad \dim(A) + \text{tr. deg}_A B \geq \dim(B_{\mathfrak{q}}) + \text{tr. deg}_k k'.$$

Moreover, the two sides are equal if, for every finitely generated A -subalgebra A' of $B[T]$ and every maximal ideal $\mathfrak{m}'$ of A' such that $\mathfrak{m}' \cap A = \mathfrak{m}$, the local ring $A'_{\mathfrak{m}'}$ is catenary.

Proof.

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Write

$$B = A[x_1, \dots, x_n]$$

and argue by induction on n . Put

$$C = A[x_1, \dots, x_{n-1}], \quad \mathfrak{r} = \mathfrak{q} \cap C.$$

The ring $C_{\mathfrak{r}}$ is a Noetherian local domain. Let

$$S = C - \mathfrak{r}, \quad B' = S^{-1}B, \quad \mathfrak{q}' = S^{-1}\mathfrak{q}.$$

Then

$$B_{\mathfrak{q}} = B'_{\mathfrak{q}'}, \quad B' = C_{\mathfrak{r}}[x_n], \quad \mathfrak{q}' \cap C_{\mathfrak{r}} = \mathfrak{r}C_{\mathfrak{r}}.$$

Moreover, the fraction fields of B' and $C_{\mathfrak{r}}$ are those of B and C , respectively. If k_1 is the residue field of $C_{\mathfrak{r}}$, then (5.5.7.1) gives

$$(5.5.8.2) \quad \dim(C_{\mathfrak{r}}) + \text{tr. deg}_C B \geq \dim(B_{\mathfrak{q}}) + \text{tr. deg}_{k_1} k'.$$

The induction hypothesis gives

$$(5.5.8.3) \quad \dim(A) + \text{tr. deg}_A C \geq \dim(C_{\mathfrak{r}}) + \text{tr. deg}_{k_1} k_1.$$

Adding (5.5.8.2) and (5.5.8.3) term by term yields (5.5.8.1).

For the second assertion, first note that every finitely generated A -subalgebra of $C[T]$ is also a finitely generated A -subalgebra of $B[T]$. By induction on n , we may therefore suppose that the two sides of (5.5.8.3) are equal.

By (5.5.7), to prove equality in (5.5.8.2), it is enough to show that whenever $\mathfrak{n}$ is a maximal ideal of $C_{\mathfrak{r}}[T]$ satisfying

$$\mathfrak{n} \cap C_{\mathfrak{r}} = \mathfrak{r}C_{\mathfrak{r}},$$

the ring $(C_{\mathfrak{r}}[T])_{\mathfrak{n}}$ is catenary. We have

$$C_{\mathfrak{r}}[T] = S^{-1}C[T].$$

Thus $\mathfrak{n} = S^{-1}\mathfrak{n}'$ for a prime ideal $\mathfrak{n}'$ of $C[T]$ such that $\mathfrak{n}' \cap C = \mathfrak{r}$, and

$$(C_{\mathfrak{r}}[T])_{\mathfrak{n}} = (C[T])_{\mathfrak{n}'}$$

Choose a maximal ideal $\mathfrak{m}'$ of $C[T]$ containing $\mathfrak{n}'$. Then $\mathfrak{m}' \cap A = \mathfrak{m}$, and $(C[T])_{\mathfrak{m}'}$ is catenary by hypothesis. Since $(C[T])_{\mathfrak{n}'}$ is a localization of this ring, it too is catenary (0, 16.1.4). $\square$

5.6. Dimension formula and universally catenary rings

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Proposition (5.6.1). — *Let A be a Noetherian ring. The following properties are equivalent:*

- (a) *Every polynomial ring $A[T_1, \dots, T_n]$ is catenary.*
- (b) *Every A -algebra of finite type is catenary.*
- (c) *The ring A is catenary, and the following dimension formula holds. Let B' be any integral local A -algebra essentially of finite type (1.3.8); let $\mathfrak{q}$ be its maximal ideal and $\mathfrak{p}$ the inverse image of $\mathfrak{q}$ in A ; let A' be the image of $A_{\mathfrak{p}}$ in B' ; and let k and k' be the residue fields of A' and B' , respectively. Then*

$$(5.6.1.1) \quad \dim(A') + \text{tr. deg}_{A'} B' = \dim(B') + \text{tr. deg}_k k'.$$

Proof. Suppose first that condition (b) holds. Then B' is a local A' -algebra essentially of finite type (1.3.10); hence

$$B' = B''_{\mathfrak{q}''},$$

where B'' is a finitely generated A' -subalgebra of B' and $\mathfrak{q}''$ is a prime ideal of B'' lying over the maximal ideal of A' . Moreover, B' and B'' have the same fraction field, so

$$\text{tr. deg}_{A'} B' = \text{tr. deg}_{A'} B''.$$

To prove (5.6.1.1), it is enough to show that every finitely generated A' -subalgebra A_1 of $B'[T]$ is catenary. Indeed, the two sides of (5.5.8.1), applied to A' , B'' , and $\mathfrak{q}''$, are then equal. But Condition (b) implies that every A -algebra essentially of finite type is catenary (0, 16.1.4), and A_1 is such an A -algebra (1.3.9). Thus (b) implies (c).

It is immediate that (b) implies (a). Conversely, (a) implies (b), because every A -algebra of finite type is a quotient of a polynomial algebra (0, 16.1.4).

It remains to prove that (c) implies (b). Since every quotient by a prime ideal of an A -algebra of finite type is an integral A -algebra of finite type, it is enough to prove the following. Let B be an integral A -algebra of finite type, and let $\mathfrak{q}' \subset \mathfrak{q}$ be prime ideals of B . Then

$$(5.6.1.2) \quad \dim(B_{\mathfrak{q}}/\mathfrak{q}'B_{\mathfrak{q}}) + \dim(B_{\mathfrak{q}'}) = \dim(B_{\mathfrak{q}}).$$

By (0, 16.1.4.2), this is precisely the required catenarity condition.

Let $\mathfrak{p}$ and $\mathfrak{p}'$ be the inverse images in A of $\mathfrak{q}$ and $\mathfrak{q}'$, respectively, and let $\mathfrak{n}$ be the kernel of

$$A_{\mathfrak{p}} \longrightarrow B_{\mathfrak{q}}.$$

The image of $A_{\mathfrak{p}}$ in $B_{\mathfrak{q}}$ is isomorphic to $A_{\mathfrak{p}}/\mathfrak{n}$. Applying (5.6.1.1) to this image and to $B_{\mathfrak{q}}$ gives

$$(5.6.1.3) \quad \begin{aligned} \dim(A_{\mathfrak{p}}/\mathfrak{n}) + \text{tr. deg}_{A_{\mathfrak{p}}/\mathfrak{n}} B_{\mathfrak{q}} \\ = \dim(B_{\mathfrak{q}}) + \text{tr. deg}_{\kappa(\mathfrak{p})} \kappa(\mathfrak{q}). \end{aligned}$$

The kernel of $A_{\mathfrak{p}'} \rightarrow B_{\mathfrak{q}'}$ is $\mathfrak{n}A_{\mathfrak{p}'}$. Applying (5.6.1.1) again gives

$$(5.6.1.4) \quad \begin{aligned} \dim(A_{\mathfrak{p}'}/\mathfrak{n}A_{\mathfrak{p}'}) + \text{tr. deg}_{A_{\mathfrak{p}'}/\mathfrak{n}A_{\mathfrak{p}'}} B_{\mathfrak{q}'} \\ = \dim(B_{\mathfrak{q}'}) + \text{tr. deg}_{\kappa(\mathfrak{p}')} \kappa(\mathfrak{q}'). \end{aligned}$$

Finally, $B/\mathfrak{q}'$ is an integral A -algebra of finite type. The inverse image of $\mathfrak{q}/\mathfrak{q}'$ in A is $\mathfrak{p}$, and the kernel of

$$A_{\mathfrak{p}} \longrightarrow B_{\mathfrak{q}}/\mathfrak{q}'B_{\mathfrak{q}}$$

is $\mathfrak{p}'A_{\mathfrak{p}}$. Hence

$$(5.6.1.5) \quad \begin{aligned} \dim(A_{\mathfrak{p}}/\mathfrak{p}'A_{\mathfrak{p}}) + \text{tr. deg}_{\kappa(\mathfrak{p}')} (B_{\mathfrak{q}}/\mathfrak{q}'B_{\mathfrak{q}}) \\ = \dim(B_{\mathfrak{q}}/\mathfrak{q}'B_{\mathfrak{q}}) + \text{tr. deg}_{\kappa(\mathfrak{p})} \kappa(\mathfrak{q}). \end{aligned}$$

Add (5.6.1.4) and (5.6.1.5) term by term. The fields $\kappa(\mathfrak{p}')$ and $\kappa(\mathfrak{q}')$ are the fraction fields of $A_{\mathfrak{p}}/\mathfrak{p}'A_{\mathfrak{p}}$ and $B_{\mathfrak{q}}/\mathfrak{q}'B_{\mathfrak{q}}$, respectively. Furthermore, $A_{\mathfrak{p}'}/\mathfrak{n}A_{\mathfrak{p}'}$ and $A_{\mathfrak{p}}/\mathfrak{n}$ have the same fraction field, as do $B_{\mathfrak{q}'}$ and $B_{\mathfrak{q}}$. Since A is catenary, (0, 16.1.4.2) gives

$$\dim(A_{\mathfrak{p}}/\mathfrak{p}'A_{\mathfrak{p}}) + \dim(A_{\mathfrak{p}'}/\mathfrak{n}A_{\mathfrak{p}'}) = \dim(A_{\mathfrak{p}}/\mathfrak{n}).$$

Taking these observations and (5.6.1.3) into account yields (5.6.1.2). $\square$

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Definition (5.6.2). — A Noetherian ring is said to be *universally catenary* if it satisfies the equivalent conditions of (5.6.1).

Remarks (5.6.3). — (i) If A is universally catenary, then so is $S^{-1}A$ for every multiplicative subset S of A . This follows at once from 5.6.1(a) and the fact that every ring of fractions of a catenary ring is catenary.

Conversely, suppose that $A_{\mathfrak{p}}$ is universally catenary for every prime ideal $\mathfrak{p}$ of A . Then A is universally catenary. Indeed, apply 5.6.1(c); if $S = A - \mathfrak{p}$, then $S^{-1}B$ is an $A_{\mathfrak{p}}$ -algebra of finite type and

$$B_{\mathfrak{q}} = (S^{-1}B)_{S^{-1}\mathfrak{q}}.$$

- (ii) A locally Noetherian prescheme X is said to be *universally catenary* if $\mathcal{O}_x$ is universally catenary for every $x \in X$. By (i), a Noetherian ring A is universally catenary if and only if $\text{Spec}(A)$ is universally catenary.
- (iii) The criterion 5.6.1(c) involves only the images of A in integral A -algebras of finite type, hence only integral quotient rings of A .

It follows that if $\mathfrak{p}_i$ ($1 \leq i \leq n$) are the minimal prime ideals of A , then A is universally catenary if and only if every $A/\mathfrak{p}_i$ is universally catenary. Likewise, a locally Noetherian prescheme X is universally catenary if and only if X_{red} is universally catenary.

- (iv) The criterion 5.6.1(b) and (i) show that every A -algebra essentially of finite type over a universally catenary ring A is universally catenary (1.3.8).

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Proposition (5.6.4). — *Every quotient ring of a regular ring is universally catenary.*

Proof. By 5.6.3(iv), it is enough to show that every regular ring A is universally catenary. Each polynomial ring $A[T_1, \dots, T_n]$ is regular (0, 17.3.7), hence catenary (0, 16.5.12), and the assertion follows from 5.6.1(a). $\square$

In particular, Cohen's theorem (0, 19.8.8) implies that every complete Noetherian local ring is universally catenary. Likewise, every algebra of finite type over a field is universally catenary, and every prescheme locally of finite type over a field is universally catenary.

We shall see in (6.3.7) that “regular ring” may be replaced by “Cohen–Macaulay ring” in (5.6.4).

Proposition (5.6.5). — *Let Y be an irreducible locally Noetherian prescheme, let X be an irreducible prescheme, and let $f : X \rightarrow Y$ be a dominant morphism locally of finite type. Let ξ and η be the generic points of X and Y , respectively, and put*

$$e = \dim(f^{-1}(\eta)) = \text{tr. deg}_{\kappa(\eta)} \kappa(\xi),$$

the dimension of the generic fibre (0, 2.1.8), (4.1.1). For every $x \in X$, with $y = f(x)$, one has

$$(5.6.5.1) \quad e + \dim(\mathcal{O}_y) \geq \text{tr. deg}_{\kappa(y)} \kappa(x) + \dim(\mathcal{O}_x)$$

and

$$(5.6.5.2) \quad \dim(\mathcal{O}_x) \leq \dim(\mathcal{O}_y) + \dim(\mathcal{O}_x \otimes_{\mathcal{O}_y} \kappa(y)) - \delta(x),$$

where

$$\delta(x) = \dim_x(f^{-1}(y)) - e.$$

If Y is universally catenary, equality holds in (5.6.5.1) and (5.6.5.2). If, in addition, x is closed in $f^{-1}(y)$, then

$$(5.6.5.3) \quad \dim(\mathcal{O}_x) = \dim(\mathcal{O}_y) + e.$$

Proof. We may restrict to the case in which X and Y are affine, and hence f is of finite type, and then replace both preschemes by their reductions (1.5.4). Thus X and Y may be assumed integral.

Since f is dominant, $\mathcal{O}_y$ identifies with a subring of $\mathcal{O}_x$, and their fraction fields are $\kappa(\eta)$ and $\kappa(\xi)$, respectively (I, 8.2.7). Moreover, $\mathcal{O}_x$ is the localization at a prime ideal $\mathfrak{q}$ of an integral $\mathcal{O}_y$ -algebra B of finite type (I, 3.6.5). Applying (5.5.8.1) to $A = \mathcal{O}_y$, B , and $\mathfrak{q}$ gives (5.6.5.1).

Furthermore,

$$\mathcal{O}_x \otimes_{\mathcal{O}_y} \kappa(y) = \mathcal{O}_x / \mathfrak{m}_y \mathcal{O}_x$$

is the local ring of the fibre $f^{-1}(y)$ at x (I, 3.6.1). Since the fibre is of finite type over $\kappa(y)$, (5.2.3.1) gives

$$\dim(\mathcal{O}_x \otimes_{\mathcal{O}_y} \kappa(y)) = \dim_x(f^{-1}(y)) - \text{tr. deg}_{\kappa(y)} \kappa(x).$$

Thus (5.6.5.2) is simply another form of (5.6.5.1).

When Y is universally catenary, equality in (5.6.5.1) is exactly (5.6.1.1), applied to $A = \mathcal{O}_y$ and $B' = \mathcal{O}_x$. Finally, if x is closed in the fibre, then in general IV-2 | 100

$$\text{tr. deg}_{\kappa(y)} \kappa(x) = \dim(\overline{\{x\}} \cap f^{-1}(y));$$

this follows from (5.2.1), applied to the reduced closed sub-prescheme of $f^{-1}(y)$ whose underlying space is the displayed subspace. Equation (5.6.5.3) follows. $\square$

We shall prove in (13.1.1) that under the hypotheses of (5.6.5) one always has $\delta(x) \geq 0$. Thus in this case (5.6.5.2) sharpens (5.5.2.1).

Corollary (5.6.6). — *Under the general hypotheses of (5.6.5),*

$$(5.6.6.1) \quad \dim(X) \leq \dim(Y) + e.$$

If Y is universally catenary, equality holds in (5.6.6.1) if and only if

$$(5.6.6.2) \quad \dim(Y) = \sup_{y \in f(X)} \dim(\mathcal{O}_y).$$

In particular, equality holds in (5.6.6.1) when Y is locally of finite type over a field.

Proof. Apply (5.6.5.1) at a closed point x of X , and use (5.1.4.2) and (5.1.11); this gives (5.6.6.1). Conversely, every nonempty fibre $f^{-1}(y)$ contains a point closed in that fibre (5.1.11). If Y is universally catenary, (5.6.5.3) holds at such a point. Taking suprema in (5.6.5.3) as x ranges over X , and observing that every point closed in X is *a fortiori* closed in $f^{-1}(f(x))$, yields the second assertion by (5.1.4.1) and (5.1.4.2).

For the last assertion, choose an affine open neighbourhood U of the generic point of X such that $f|_U$ is of finite type. Then $f(U)$ is constructible (1.8.5) and dense in Y , and therefore contains a nonempty open, hence dense, subset of Y (0, 9.2.2). Since Y is locally of finite type over a field, it is universally catenary (5.6.4); moreover, the two sides of (5.6.6.2) are equal by (5.2.2) and (4.1.1.3).

Condition (5.6.6.2) is trivially satisfied when f is surjective. $\square$

Corollary (5.6.7). — *Let Y be a locally Noetherian prescheme, let $f : X \rightarrow Y$ be a morphism locally of finite type, and let n be an integer. If*

$$\dim(f^{-1}(y)) \leq n$$

for every $y \in Y$, then

$$(5.6.7.1) \quad \dim(X) \leq \dim(Y) + n.$$

Proof. By (0, 14.1.7) and (1.5.4), we may assume that X and Y are affine and reduced, so that f is of finite type. Let X_i ($1 \leq i \leq m$) be the integral closed sub-preschemes whose underlying spaces are the irreducible components of X . Then

$$\dim(X) = \sup_i \dim(X_i).$$

Let Z_i be the reduced closed sub-prescheme of Y whose underlying space is $\overline{f(X_i)}$. It is integral (0, 2.1.5), and the restriction of f factors as

$$X_i \xrightarrow{g_i} Z_i \xrightarrow{j_i} Y,$$

where j_i is the canonical injection (I, 5.2.2), and g_i is dominant and of finite type (1.5.4).

We have $\dim(Z_i) \leq \dim(Y)$. If ζ_i is the generic point of Z_i , then IV-2 | 101

$$\dim(g_i^{-1}(\zeta_i)) \leq n$$

by hypothesis. Thus it is enough to show

$$\dim(X_i) \leq \dim(Z_i) + n$$

for every i , and this is (5.6.6.1). $\square$

Corollary (5.6.8). — *Let Y be a locally Noetherian prescheme, let X be irreducible, let $f : X \rightarrow Y$ be a morphism locally of finite type, and let $n \geq 0$ be an integer. Suppose that Y is universally catenary and that*

$$\dim(f^{-1}(y)) \geq n \quad \text{for every } y \in Y.$$

Then

$$(5.6.8.1) \quad \dim(X) \geq \dim(Y) + n.$$

If instead every fibre has dimension exactly n , then

$$(5.6.8.2) \quad \dim(X) = \dim(Y) + n.$$

Proof. Since $n \geq 0$, either hypothesis implies that f is surjective, and therefore Y is irreducible. If η is the generic point of Y , then $\dim(f^{-1}(\eta)) \geq n$, respectively $\dim(f^{-1}(\eta)) = n$. The conclusion follows from (5.6.6). $\square$

Remarks (5.6.9). — (i) Even if Y is regular, X is irreducible, and $f : X \rightarrow Y$ is dominant and of finite type, the two sides of (5.6.6.1) need not be equal. Take $Y = \operatorname{Spec}(A)$ for a discrete valuation ring A , take $X = \operatorname{Spec}(K)$ for its fraction field K , and let f be the canonical morphism.

(ii) Example 5.4.3(i) shows that the hypothesis that X is irreducible cannot be omitted from (5.6.6) or (5.6.8), even when all the other hypotheses hold. We shall see in (10.6.1) that such phenomena cannot occur under additional hypotheses on Y , satisfied for example when Y is locally of finite type over a field or over a Dedekind ring having infinitely many prime ideals.

Proposition (5.6.10). — *Let A be an integral universally catenary Noetherian local ring, and let B be an integral finite A -algebra containing A . For every maximal ideal $\mathfrak{n}$ of B ,*

$$\dim(B_{\mathfrak{n}}) = \dim(A).$$

Proof. We have $\operatorname{tr. deg}_A B = 0$, and the residue field of $B_{\mathfrak{n}}$ is algebraic over the residue field of A .⁷ Every maximal ideal of B lies over the maximal ideal of A , so the assertion follows from (5.6.1.1). $\square$

Example (5.6.11). — Let A be an integral, integrally closed Noetherian local ring. Then (0, 16.1.6) shows that the conclusion of (5.6.10) holds for every integral finite A -algebra B containing A . We now construct, by contrast, an integral catenary Noetherian local ring A for which that conclusion fails.

Use the construction of 5.2.5(i). Let k_0 be a field, let

$$k = k_0(S_i)_{i \in \mathbb{N}}$$

be a purely transcendental extension of infinite transcendence degree, and let $V = k[S]_{(S)}$ be the discrete valuation ring obtained by localizing $k[S]$ at (S) . Put $E = V[T]$. In E , the maximal ideal

$$\mathfrak{m} = (S) + (T)$$

has height 2, whereas the maximal ideal

$$\mathfrak{m}' = (ST - 1)$$

has height 1.

The corresponding residue fields are

$$\kappa(\mathfrak{m}) = k, \quad \kappa(\mathfrak{m}') = k(S),$$

the latter being the fraction field of V . By the choice of k , these fields are isomorphic. Let

$$\varepsilon : E \longrightarrow E/\mathfrak{m}, \quad \varepsilon' : E \longrightarrow E/\mathfrak{m}'$$

be the canonical homomorphisms, choose an isomorphism

$$\sigma : \kappa(\mathfrak{m}) \xrightarrow{\sim} \kappa(\mathfrak{m}'),$$

⁷The printed French text says “the fraction field of A ”. Here the field required by the dimension formula is the residue field of A .

and let C be the subring of E formed by the elements x such that

$$\varepsilon'(x) = \sigma(\varepsilon(x)).$$

This is a special case of the “gluing procedures” to be studied systematically in Chapter V.

It is immediate that

$$\mathfrak{n} = \mathfrak{m}\mathfrak{m}' = \mathfrak{m} \cap \mathfrak{m}'$$

is a maximal ideal of C , and that $C/\mathfrak{n}$ identifies with the subfield of $(E/\mathfrak{m}) \times (E/\mathfrak{m}')$ formed by the pairs $(z, \sigma(z))$. Moreover,

$$E = C + C(ST - 1),$$

so E is a finite C -algebra; it is plainly the integral closure of C . We next prove that C is Noetherian.

Lemma (5.6.11.1). — *Let R be a ring, let S be a subring, and let*

$$\mathfrak{K} = \text{Ann}_S(R/S)$$

be the conductor of S in R , that is, the largest ideal of S which is also an ideal of R .

- (i) *For every ideal $\mathfrak{J} \subset \mathfrak{K}$ of R , there is a strictly increasing map from the set of ideals $\mathfrak{a}$ of S such that $R\mathfrak{a} = \mathfrak{J}$ to the set of $(S/\mathfrak{K})$ -submodules of $\mathfrak{J}/\mathfrak{K}\mathfrak{J}$.*
- (ii) *If $S/\mathfrak{K}$ and R are Noetherian and R is a finite S -module, then every increasing sequence of ideals of S contained in $\mathfrak{K}$ is stationary.*

Proof. For (i), if $R\mathfrak{a} = \mathfrak{J}$, then

$$\mathfrak{K}\mathfrak{J} = \mathfrak{K}(R\mathfrak{a}) = \mathfrak{K}\mathfrak{a} \subset \mathfrak{a}.$$

Thus $\mathfrak{a} \mapsto \mathfrak{a}/\mathfrak{K}\mathfrak{J}$ gives the asserted strictly increasing map.

For (ii), let $(\mathfrak{L}_r)$ be an increasing sequence of ideals of S contained in $\mathfrak{K}$. Since R is Noetherian, the sequence $(R\mathfrak{L}_r)$ is stationary; after discarding finitely many terms, suppose that all its members are the same ideal $\mathfrak{J}$ of R . The module $\mathfrak{J}/\mathfrak{K}\mathfrak{J}$ is finite over R , hence over S , and therefore finite over the Noetherian ring $S/\mathfrak{K}$. It is consequently Noetherian, and (i) shows that $(\mathfrak{L}_r)$ is stationary. $\square$

Apply the lemma with $R = E$ and $S = C$. The ideal $\mathfrak{n}$ is the conductor of C in E . For every ideal $\mathfrak{a}$ of C ,

$$\mathfrak{a}/(\mathfrak{n} \cap \mathfrak{a}) \simeq (\mathfrak{a} + \mathfrak{n})/\mathfrak{n}$$

is a submodule of the simple C -module $C/\mathfrak{n}$. It is therefore enough to prove that every ideal $\mathfrak{a} \cap \mathfrak{n}$ is finitely generated; this follows from (5.6.11.1). Thus C is Noetherian.

By (0, 16.1.5),

$$\dim(C) = \dim(E) = 2.$$

Set $A = C_{\mathfrak{n}}$ and $U = C - \mathfrak{n}$. Then

$$B = U^{-1}E$$

is the integral closure of A and is a finite A -module. The ideals $\mathfrak{m}$ and $\mathfrak{m}'$ are the only prime ideals of E containing $\mathfrak{n}$. Hence B is semilocal, and its local rings at its two maximal ideals are isomorphic to $E_{\mathfrak{m}}$ and $E_{\mathfrak{m}'}$, of dimensions 2 and 1, respectively. Thus $\dim(B) = 2$ and, by (0, 16.1.5), $\dim(A) = 2$.

Since A is an integral local ring of dimension 2, it is catenary: every prime ideal other than 0 and the maximal ideal has height 1. Nevertheless it does not satisfy the conclusion of (5.6.10), and therefore is not universally catenary.

The ideal $\mathfrak{n}$ has height 2 in C . If $\mathfrak{p}$ is a height-1 prime ideal of C , there is a unique prime ideal $\mathfrak{p}'$ of E above $\mathfrak{p}$; it has height 1, and

$$C_{\mathfrak{p}} = E_{\mathfrak{p}'}$$

Existence follows from integrality, and the Cohen–Seidenberg theorem shows that every prime ideal of E above $\mathfrak{p}$ has height 1.

Moreover, E is a finite C -algebra and $C_{\mathfrak{p}}$ is integrally closed (Bourbaki, *Algèbre commutative*, IV-2 | 103 chap. V, § 1, n° 5, cor. 5 of prop. 16). This proves uniqueness and the displayed equality.

It would be interesting to know whether every integral Noetherian local ring satisfying the conclusion of (5.6.10) is universally catenary. Nagata asserted this [33], but his proof does not appear to be complete.

5.7. Depth and property (S_k)

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Definition (5.7.1). — Let X be a locally Noetherian prescheme and let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -module. The *depth* (respectively, *codepth*) of $\mathcal{F}$ at a point $x \in X$ is the number $\text{prof}(\mathcal{F}_x)$ (respectively, $\text{coprof}(\mathcal{F}_x)$) (0, 16.4.5), (0, 16.4.9). The *codepth* of $\mathcal{F}$ is the number

$$(5.7.1.1) \quad \text{coprof}(\mathcal{F}) = \sup_{x \in X} \text{coprof}(\mathcal{F}_x).$$

The module $\mathcal{F}$ is said to be a *Cohen–Macaulay $\mathcal{O}_X$ -module at x* if $\mathcal{F}_x$ is a Cohen–Macaulay $\mathcal{O}_x$ -module, that is (0, 16.5.1), if $\text{coprof}(\mathcal{F}_x) = 0$. It is said to be a *Cohen–Macaulay $\mathcal{O}_X$ -module* if it is so at every point, or equivalently if $\text{coprof}(\mathcal{F}) = 0$. A point $x \in X$ such that $\mathcal{O}_x$ is a Cohen–Macaulay ring is also called a *Cohen–Macaulay point* of X .

The *codepth* of X , denoted by $\text{coprof}(X)$, is the number $\text{coprof}(\mathcal{O}_X)$. The prescheme X is said to be *Cohen–Macaulay* if $\mathcal{O}_X$ is a Cohen–Macaulay $\mathcal{O}_X$ -module, or equivalently if $\text{coprof}(X) = 0$. Every locally Noetherian prescheme of dimension 0 is evidently Cohen–Macaulay. To say that $\text{Spec}(A)$ is Cohen–Macaulay means that A is a Cohen–Macaulay ring (0, 16.5.13).

Definition (5.7.2). — Let X be a locally Noetherian prescheme, let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -module, and let k be an integer, positive or negative. The module $\mathcal{F}$ is said to satisfy property (S_k) if, for every $x \in X$,

$$(5.7.2.1) \quad \text{prof}(\mathcal{F}_x) \geq \inf\{k, \dim(\mathcal{F}_x)\}.$$

It is said to satisfy property (S_k) at a point $x \in X$ if, for every generization x' of x in X ,

$$(5.7.2.2) \quad \text{prof}(\mathcal{F}_{x'}) \geq \inf\{k, \dim(\mathcal{F}_{x'})\}.$$

The prescheme X is said to satisfy property (S_k) (respectively, property (S_k) at x) if $\mathcal{O}_X$ does so (respectively, does so at x).

The module $\mathcal{F}$ satisfies (S_k) if and only if it satisfies (S_k) at every point of X . If U is an open subset of X and $\mathcal{F}$ satisfies (S_k) , then $\mathcal{F}|_U$ satisfies (S_k) . Conversely, if (U_α) is an open covering of X and $\mathcal{F}|_{U_\alpha}$ satisfies (S_k) for every α , then $\mathcal{F}$ satisfies (S_k) .

Remarks (5.7.3). — (i) Recall that $\text{prof}(\mathcal{F}_x) \leq \dim(\mathcal{F}_x)$ (0, 16.4.5.1) whenever $\mathcal{F}_x \neq 0$. Thus $\mathcal{F}$ satisfies (S_k) precisely when $\text{prof}(\mathcal{F}_x) \geq k$ except at points x for which $\dim(\mathcal{F}_x) < k$, while at those latter points

$$\dim(\mathcal{F}_x) = \text{prof}(\mathcal{F}_x).$$

Equivalently (0, 16.5.1), $\mathcal{F}$ is Cohen–Macaulay at those points. At points where $\dim(\mathcal{F}_x) = k$, one likewise has $\text{prof}(\mathcal{F}_x) = k$.

Hence $\mathcal{F}$ is also Cohen–Macaulay at those points. To say that $\mathcal{F}$ satisfies (S_k) for every k is therefore the same as to say that $\mathcal{F}$ is a Cohen–Macaulay $\mathcal{O}_X$ -module. If $k' \geq k$, then $(S_{k'})$ implies (S_k) ; if $k \leq 0$, every coherent $\mathcal{O}_X$ -module satisfies (S_k) .

- (ii) To verify (5.7.2.1), it is enough to consider $x \in \text{Supp}(\mathcal{F})$; otherwise $\dim(\mathcal{F}_x) = -\infty$ (0, 14.1.2).
- (iii) If $X = \text{Spec}(A)$, where A is Noetherian, and $\mathcal{F} = \tilde{M}$, where M is a finite A -module, then M is said to satisfy (S_k) when $\mathcal{F}$ does.

For an arbitrary locally Noetherian prescheme X , saying that $\mathcal{F}$ satisfies (S_k) at x means the following. Put $Y = \text{Spec}(\mathcal{O}_x)$; then the $\mathcal{O}_Y$ -module $\tilde{\mathcal{F}}_x$ satisfies (S_k) . Notice that (5.7.2.1) does not in general imply (5.7.2.2) for every generization x' of x , since there is no general inequality between $\text{prof}(\mathcal{F}_x)$ and $\text{prof}(\mathcal{F}_{x'})$ (0, 16.4.6).

- (iv) It follows immediately from the definition that if $\mathcal{F}$ satisfies (S_k) at x , then it also satisfies (S_k) at every generization of x .
- (v) Property (S_k) is especially important for $k = 1$ and $k = 2$. Serre introduced it for $k = 2$ in order to express his normality criterion; see (5.8.5).
- (vi) Let Y be a closed sub-prescheme of X , let $j : Y \rightarrow X$ be the canonical immersion, and let $\mathcal{G}$ be a coherent $\mathcal{O}_Y$ -module. For every $x \in Y$,

$$\dim(\mathcal{G}_x) = \dim((j_*\mathcal{G})_x), \quad \text{prof}(\mathcal{G}_x) = \text{prof}((j_*\mathcal{G})_x),$$

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and consequently

$$\operatorname{coprof}(\mathcal{G}_x) = \operatorname{coprof}((j_*\mathcal{G})_x).$$

Thus $\mathcal{G}$ satisfies (S_k) if and only if $j_*\mathcal{G}$ does.

Proposition (5.7.4). — *Let X be a locally Noetherian prescheme and let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -module. Put $S = \operatorname{Supp}(\mathcal{F})$, and for every $n \geq 0$ let Z_n be the set of points $z \in X$ such that $\operatorname{coprof}(\mathcal{F}_z) > n$; thus $Z_n \subset S$.*

(i) *The module $\mathcal{F}$ satisfies (S_k) if and only if, for every $n \geq 0$,*

$$(5.7.4.1) \quad \operatorname{codim}(Z_n, S) > n + k.$$

(ii) *Suppose in addition that every Z_n is closed in X . Then $\mathcal{F}$ satisfies (S_k) at a point $x \in S$ if and only if, for every $n \geq 0$,*

$$(5.7.4.2) \quad \operatorname{codim}_x(Z_n, S) > n + k.$$

Proof. (i) By definition (5.1.3),

$$\operatorname{codim}(Z_n, S) = \inf_{z \in Z_n} \dim(\mathcal{O}_{S,z}),$$

and (5.7.4.1) therefore says (5.1.12.2) that, for every $z \in X$ and every $n \geq 0$,

$$\dim(\mathcal{F}_z) - \operatorname{prof}(\mathcal{F}_z) \geq n + 1$$

implies

$$\dim(\mathcal{F}_z) \geq n + k + 1.$$

If $a = \dim(\mathcal{F}_z)$ and $b = \operatorname{prof}(\mathcal{F}_z)$, then $b \leq a$; requiring, for every $n \geq 0$, that $b \leq a - n - 1$ imply $k \leq a - n - 1$ is equivalent to $b \geq \inf\{k, a\}$. This proves (i).

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(ii) The argument is the same as in (i), except that one restricts to the points z which are generizations of x and uses (5.1.3.2). □

Proposition (5.7.5). — *Let X be a locally Noetherian prescheme and let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -module. The module $\mathcal{F}$ satisfies (S_k) , where $k \geq 1$, if and only if the following condition holds: for every integer r with $0 \leq r < k$, every open subset U of X , and every $(\mathcal{F}|U)$ -regular sequence $(f_i)_{1 \leq i \leq r}$ of sections of $\mathcal{O}_X$ over U , the module*

$$(\mathcal{F}|U) / \left(\sum_{i=1}^r f_i(\mathcal{F}|U) \right)$$

has no embedded associated prime cycle.

Proof. For $k = 1$, the assertion says that $\mathcal{F}$ satisfies (S_1) if and only if $\mathcal{F}$ has no embedded associated prime cycle. Indeed, (S_1) says that $\operatorname{prof}(\mathcal{F}_x) \geq 1$ at precisely the points $x \in \operatorname{Supp}(\mathcal{F})$ for which $\dim(\mathcal{F}_x) > 0$; at the other points of the support, $\dim(\mathcal{F}_x) = \operatorname{prof}(\mathcal{F}_x) = 0$. Now $\operatorname{prof}(\mathcal{F}_x) \geq 1$ means that the maximal ideal $\mathfrak{m}_x$ is not associated to $\mathcal{F}_x$ (0, 16.4.6(i)), or equivalently that x is not associated to $\mathcal{F}$ (3.1.1). If S' is a sub-prescheme of X whose underlying space is $\operatorname{Supp}(\mathcal{F})$, then

$$\dim(\mathcal{F}_x) = \dim(\mathcal{O}_{S',x})$$

(5.1.12.1); saying that this dimension is positive is the same as saying that x is not a maximal point of $\operatorname{Supp}(\mathcal{F})$ (5.1.2). This proves the assertion for $k = 1$.

Let $k > 1$, and first suppose that $\mathcal{F}$ satisfies (S_k) . By induction on k and the case just proved, it is enough to show that if f is an $\mathcal{F}$ -regular section of $\mathcal{O}_X$, then $\mathcal{F}/f\mathcal{F}$ satisfies (S_{k-1}) . At each $x \in X$, the element f_x is $\mathcal{F}_x$ -regular. If it is invertible, the quotient stalk is zero. Otherwise $f_x \in \mathfrak{m}_x$, and

$$\operatorname{prof}(\mathcal{F}_x/f_x\mathcal{F}_x) = \operatorname{prof}(\mathcal{F}_x) - 1, \quad \dim(\mathcal{F}_x/f_x\mathcal{F}_x) = \dim(\mathcal{F}_x) - 1$$

(0, 16.4.6(i)), (0, 16.3.4); the assertion follows. Thus the condition is necessary.

Conversely, suppose the stated condition holds. We argue by induction on k . Let $x \in X$, and first suppose that $\dim(\mathcal{F}_x) \geq k$. The induction hypothesis gives (S_{k-1}) , so $\operatorname{prof}(\mathcal{F}_x) \geq k - 1$. By

(0, 15.2.4), after replacing X by an open neighbourhood U of x , there is an $(\mathcal{F}|U)$ -regular sequence $(f_i)_{1 \leq i \leq k-1}$ with $(f_i)_x \in \mathfrak{m}_x$. Put

$$\mathcal{G} = (\mathcal{F}|U) / \left(\sum_{i=1}^{k-1} f_i (\mathcal{F}|U) \right).$$

By hypothesis $\mathcal{G}$ has no embedded associated prime cycle, while

$$\dim(\mathcal{G}_x) = \dim(\mathcal{F}_x) - (k-1) \geq 1$$

(0, 16.3.4). Hence x is not associated to $\mathcal{G}$, so $\text{prof}(\mathcal{G}_x) \geq 1$ (0, 16.4.6); since

$$\text{prof}(\mathcal{G}_x) = \text{prof}(\mathcal{F}_x) - (k-1),$$

we obtain $\text{prof}(\mathcal{F}_x) \geq k$.

If instead $\dim(\mathcal{F}_x) = r < k$, the induction hypothesis gives

$$\text{prof}(\mathcal{F}_x) \geq \inf\{k-1, r\} = r.$$

Thus $\mathcal{F}$ satisfies (S_k) , completing the proof. $\square$

Corollary (5.7.6). — Suppose that $\mathcal{F}$ satisfies (S_k) , where $k \geq 1$. If $(f_i)_{1 \leq i \leq r}$ is an $\mathcal{F}$ -regular sequence of sections of $\mathcal{O}_X$ over X and $r < k$, then

$$\mathcal{F} / \left(\sum_{i=1}^r f_i \mathcal{F} \right)$$

satisfies (S_{k-r}) .

Proof. This follows immediately from (5.7.5). $\square$

Corollary (5.7.7). — The module $\mathcal{F}$ satisfies (S_2) if and only if $\mathcal{F}$ has no embedded associated prime cycle and, for every open subset U of X and every $(\mathcal{F}|U)$ -regular section f of $\mathcal{O}_X$ over U , the module $(\mathcal{F}|U) / f(\mathcal{F}|U)$ has no embedded associated prime cycle.

Proof. This follows immediately from (5.7.5). $\square$

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Remarks (5.7.8). — Let X be a locally Noetherian prescheme of dimension 1. The following conditions are equivalent: X is Cohen–Macaulay; X satisfies (S_1) ; X satisfies (S_n) for some $n \geq 1$. This follows from Definitions (5.7.1) and (5.7.2). By (5.7.5), these conditions are also equivalent to the absence of an embedded associated prime cycle. In particular, every reduced locally Noetherian prescheme of dimension 1 is Cohen–Macaulay.

Proposition (5.7.9). — Let A and B be Noetherian rings, let $\rho : A \rightarrow B$ be a ring homomorphism, and let M be a B -module such that the A -module $M_{[\rho]}$ is finite. Let $\mathfrak{p}$ be a prime ideal of A . The prime ideals of B lying over $\mathfrak{p}$ and belonging to $\text{Supp}(M)$ are finite in number. If $(\mathfrak{q}_i)_{1 \leq i \leq n}$ is the family of these ideals, then

$$(5.7.9.1) \quad \dim_{A_{\mathfrak{p}}}(M_{\mathfrak{p}}) = \sup_i \dim_{B_{\mathfrak{q}_i}}(M_{\mathfrak{q}_i}),$$

$$(5.7.9.2) \quad \text{prof}_{A_{\mathfrak{p}}}(M_{\mathfrak{p}}) = \inf_i \text{prof}_{B_{\mathfrak{q}_i}}(M_{\mathfrak{q}_i}).$$

Proof. Let $\mathfrak{b}$ be the annihilator of M . The ring $B/\mathfrak{b}$ identifies with an A -submodule of $\text{End}_A(M_{[\rho]})$, and is therefore finite over A . Hence only finitely many prime ideals of B lie over $\mathfrak{p}$ and contain $\mathfrak{b}$; these are precisely the ones belonging to $\text{Supp}(M)$.

Replacing B by $B/\mathfrak{b}$ does not change the right-hand sides of (5.7.9.1) and (5.7.9.2) (0, 16.1.9), (0, 16.4.8); we may therefore suppose that B is finite over A . Put $S = A - \mathfrak{p}$ and $B' = S^{-1}B$. Then B' is a Noetherian semilocal ring whose maximal ideals are $S^{-1}\mathfrak{q}_i$, and

$$\dim_{A_{\mathfrak{p}}}(M_{\mathfrak{p}}) = \dim_{B'}(M_{\mathfrak{p}})$$

(0, 16.1.9). Formula (5.7.9.1) is now the semilocal dimension formula (0, 16.1.7.4).

For (5.7.9.2), as in (0, 16.4.8), it is enough to treat the case $\text{prof}_{A_p}(M_p) = 0$. The same argument shows that M_p contains a B' -submodule of finite length, and hence a simple B' -submodule. Such a module is isomorphic to the residue field of one of the local rings B_{q_i} . Thus

$$\text{prof}_{B_{q_i}}(M_{q_i}) = 0$$

for at least one i (0, 16.4.6), proving the formula. $\square$

Corollary (5.7.10). — Assume the hypotheses of (5.7.9), and suppose in addition that A is local. The A -module $M_{[\rho]}$ is Cohen–Macaulay if and only if every M_{q_i} , for q_i lying over the maximal ideal of A , is a Cohen–Macaulay B_{q_i} -module and all the numbers $\dim_{B_{q_i}}(M_{q_i})$ are equal.

Proof. By (5.7.9.1) and (5.7.9.2), these conditions are equivalent to

$$\dim_A(M_{[\rho]}) = \text{prof}_A(M_{[\rho]}).$$

$\square$

Corollary (5.7.11). — Assume the hypotheses of (5.7.9).

- (i) If $M_{[\rho]}$ satisfies (S_k) , then M satisfies (S_k) .
- (ii) Suppose that for every pair of prime ideals q, q' of B satisfying $\rho^{-1}(q) = \rho^{-1}(q')$, one has

$$\dim_{B_q}(M_q) = \dim_{B_{q'}}(M_{q'}).$$

If M satisfies (S_k) , then $M_{[\rho]}$ satisfies (S_k) .

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Proof. This follows immediately from (5.7.9.1), (5.7.9.2), and the definition of (S_k) . $\square$

(5.7.12). In accordance with (5.7.1), if A is any Noetherian ring and M is a finite A -module, define

$$\text{coprof}_A(M) = \text{coprof}(\tilde{M}) = \sup_{x \in X} \text{coprof}_{A_x}(M_x), \quad X = \text{Spec}(A).$$

We shall see in (6.11.5) that this definition agrees with that of (0, 16.4.9) when A is a Noetherian local ring.

Corollary (5.7.13). — Let A and B be Noetherian rings, let $\rho : A \rightarrow B$ be a ring homomorphism, and let M be a B -module such that $M_{[\rho]}$ is a finite A -module. Then

$$(5.7.13.1) \quad \text{coprof}_A(M_{[\rho]}) \geq \text{coprof}_B(M).$$

Proof. This follows from (5.7.12) and (5.7.9.1), (5.7.9.2). $\square$

5.8. Regular preschemes and property (R_k) . Serre's normality criterion

(5.8.1). Recall (0, 4.1.4) that a ringed space $(X, \mathcal{O}_X)$ is said to be *regular* at a point $x \in X$ if $\mathcal{O}_x$ is a regular local ring. When dealing with preschemes, we shall use this terminology in this chapter only when X is locally Noetherian.

Definition (5.8.2). — A locally Noetherian prescheme X is said to be *regular in codimension $\leq k$* , or to satisfy property (R_k) , if the set of points where X is not regular has codimension $> k$. In other words (5.1.3), for every $x \in X$,

$$\dim(\mathcal{O}_x) \leq k \implies \mathcal{O}_x \text{ is regular.}$$

To say that X is regular means that X satisfies (R_k) for every k .

If $X = \text{Spec}(A)$, where A is a Noetherian ring, we say that A satisfies (R_k) if X does; to say that X is regular is to say that the ring A is regular (0, 17.3.6). For an arbitrary locally Noetherian prescheme X , we say that X satisfies (R_k) at a point $x \in X$ if the local ring $\mathcal{O}_x$ satisfies (R_k) . Thus, for every generization x' of x in X , the relation $\dim(\mathcal{O}_{x'}) \leq k$ must imply that $\mathcal{O}_{x'}$ is a regular local ring. By (0, 17.3.6), saying that X is regular at x is equivalent to saying that X satisfies (R_n) at x for every $n \geq 0$.

Proposition (5.8.3). — Let k be a field and let X be a prescheme locally of finite type over k . For every $x \in X$, there is an open neighbourhood of x in X which is isomorphic to a subscheme of a regular k -scheme.

Proof. There is an affine open neighbourhood U of x isomorphic to a k -scheme $\text{Spec}(A)$, where A is a k -algebra of finite type. The algebra A is therefore isomorphic to a quotient of a polynomial algebra $k[T_1, \dots, T_n]$, and the latter is a regular ring (0, 17.3.7). $\square$

(5.8.4). By (5.8.2), saying that X satisfies (R_0) means that, at every maximal point x of X , the ring $\mathcal{O}_x$ is a field (0, 17.1.4); in other words, X is reduced at x . The set U of points where X is reduced is open, because the nilradical of X is a coherent $\mathcal{O}_X$ -module (I, 6.1.1), (0, 5.2.2). Consequently, saying that X satisfies (R_0) is equivalent to saying that U is everywhere dense.

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Proposition (5.8.5). — *A locally Noetherian prescheme X is reduced if and only if it satisfies (S_1) and (R_0) .*

Proof. Taking (5.7.5) into account, this follows from the preceding remark and (3.2.1). $\square$

Theorem (5.8.6). — (Serre's criterion). *Let X be a locally Noetherian prescheme. Then X is normal if and only if it satisfies (S_2) and (R_1) . Equivalently, for every $x \in X$, the following conditions hold:*

- (i) *If $\dim(\mathcal{O}_x) \leq 1$, then $\mathcal{O}_x$ is regular; that is, it is a field or a discrete valuation ring (0, 17.1.4).*
- (ii) *If $\dim(\mathcal{O}_x) \geq 2$, then $\text{prof}(\mathcal{O}_x) \geq 2$.*

Proof. The conditions are necessary. Indeed, saying that X is normal means that, for every $x \in X$, the ring $\mathcal{O}_x$ is an integrally closed Noetherian local ring. If $\dim(\mathcal{O}_x) = 0$, then $\mathcal{O}_x$ is a field because it is a domain; if $\dim(\mathcal{O}_x) = 1$, then $\mathcal{O}_x$ is a discrete valuation ring by (II, 7.1.6). Furthermore, for every nonzero element $f_x \in \mathcal{O}_x$, the associated prime ideals of $\mathcal{O}_x / f_x \mathcal{O}_x$ are non-embedded (Bourbaki, *Alg. comm.*, chap. VII, § 1, no. 4, prop. 8). Hence $\mathcal{O}_x$ satisfies (S_2) by (5.7.7).

Conversely, suppose that the two conditions hold. It follows first from (5.8.5) that X is reduced. Since the question is local, we may suppose that $X = \text{Spec}(A)$, where A is a reduced Noetherian ring (I, 5.1.4). If R is the total ring of fractions of A , then R is a direct product of finitely many fields; taking (II, 6.3.6) into account, it is enough to prove that A is integrally closed in R .

Let $h = f/g \in R$ be integral over A , where $f, g \in A$ and g is a non-zero-divisor. There is a relation

$$(5.8.6.1) \quad f^n + \sum_{i=1}^n a_i f^{n-i} g^i = 0, \quad a_i \in A \quad (1 \leq i \leq n).$$

Let $\mathfrak{p}$ be a prime ideal of A such that $\dim(A_{\mathfrak{p}}) = 1$, and write $f_{\mathfrak{p}}$ and $g_{\mathfrak{p}}$ for the images of f and g in $A_{\mathfrak{p}}$. It follows from (5.8.6.1) that $f_{\mathfrak{p}}/g_{\mathfrak{p}}$ is integral over $A_{\mathfrak{p}}$. This quotient belongs to the total ring of fractions of $A_{\mathfrak{p}}$, since flatness implies that $g_{\mathfrak{p}}$ is still a non-zero-divisor (0, 5.3.1). But $A_{\mathfrak{p}}$ is regular and hence integrally closed, so $f_{\mathfrak{p}}/g_{\mathfrak{p}} \in A_{\mathfrak{p}}$. Equivalently,

$$(fA)_{\mathfrak{p}} \subset (gA)_{\mathfrak{p}}.$$

Since g is a non-zero-divisor in A , condition (S_2) and (5.7.7) imply that A/gA has only non-embedded associated prime ideals $\mathfrak{p}_i$ ($1 \leq i \leq n$). The ideal gA is therefore an intersection of primary ideals $\mathfrak{q}_i$ corresponding to the $\mathfrak{p}_i$; moreover, each $\mathfrak{q}_i$ is the inverse image in A , under $A \rightarrow A_{\mathfrak{p}_i}$, of $(gA)_{\mathfrak{p}_i}$ (Bourbaki, *Alg. comm.*, chap. IV, § 2, no. 3, prop. 5). By the Hauptidealsatz (0, 16.3.2), $\dim(A_{\mathfrak{p}_i}) = 1$ for every i , so the preceding argument gives

$$(fA)_{\mathfrak{p}_i} \subset (gA)_{\mathfrak{p}_i} \quad (1 \leq i \leq n).$$

Now fA is contained in the intersection of the inverse images of the $(fA)_{\mathfrak{p}_i}$. It is therefore contained in the intersection of the inverse images of the $(gA)_{\mathfrak{p}_i}$, which is gA . Thus $fA \subset gA$, and hence $f/g \in A$. $\square$

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5.9. Z-pure and Z-closed modules Some of the notions and results in this section and the next are special cases of notions and results developed in Chapter III in the theory of local cohomology. For the reader's convenience, we give an independent account here.

(5.9.1). Let X be a locally Noetherian prescheme and let Z be a subset of X *stable under specialization*. This means that the closure of every finite subset M of Z is contained in Z . It follows that Z is the union of a filtered increasing family (Z_α) of closed subsets of X ; conversely, such a union is plainly stable under specialization.

Put $U_\alpha = X - Z_\alpha$. Then $X - Z$ is the intersection of the filtered decreasing family of open subsets U_α . Let $i_\alpha : U_\alpha \rightarrow X$ be the canonical injection, and, whenever $U_\alpha \supset U_\beta$, let $i_{\alpha\beta} : U_\beta \rightarrow U_\alpha$ be the canonical injection, so that $i_\beta = i_\alpha \circ i_{\alpha\beta}$.

Let $\mathcal{F}$ be an $\mathcal{O}_X$ -module, not necessarily quasi-coherent. We have

$$(i_\beta)_*(\mathcal{F}|_{U_\beta}) = (i_\alpha)_*((i_{\alpha\beta})_*(\mathcal{F}|_{U_\beta})).$$

The canonical homomorphism (0, 4.4.3.2)

$$\mathcal{F}|_{U_\alpha} \longrightarrow (i_{\alpha\beta})_*(\mathcal{F}|_{U_\beta})$$

therefore gives, after applying $(i_\alpha)_*$, a homomorphism

$$\rho_{\beta\alpha} : (i_\alpha)_*(\mathcal{F}|_{U_\alpha}) \longrightarrow (i_\beta)_*(\mathcal{F}|_{U_\beta}).$$

For $U_\alpha \supset U_\beta \supset U_\gamma$, one immediately checks that $\rho_{\gamma\alpha} = \rho_{\gamma\beta} \circ \rho_{\beta\alpha}$. Thus the modules $(i_\alpha)_*(\mathcal{F}|_{U_\alpha})$ form an inductive system under the homomorphisms $\rho_{\beta\alpha}$. We put

$$(5.9.1.1) \quad \mathcal{H}_{X/Z}^0(\mathcal{F}) = \varinjlim_\alpha (i_\alpha)_*(\mathcal{F}|_{U_\alpha}).$$

This $\mathcal{O}_X$ -module does not depend on the increasing family (Z_α) of closed subsets having union Z . Indeed, let V be a Noetherian open subset of X . In the category of $\mathcal{O}_V$ -modules, the functor $\mathcal{G} \mapsto \Gamma(V, \mathcal{G})$ commutes with inductive limits (G, II.3.10.1); hence (5.9.1.1) gives

$$(5.9.1.2) \quad \Gamma(V, \mathcal{H}_{X/Z}^0(\mathcal{F})) = \varinjlim_\alpha \Gamma(V \cap U_\alpha, \mathcal{F}).$$

Let (Z'_λ) be another filtered increasing family of closed subsets of X with union Z . For fixed α , the set $V \cap Z_\alpha$ is the union of the $V \cap Z_\alpha \cap Z'_\lambda$. Now $V \cap Z_\alpha$ is locally closed in X , and every closed irreducible subset of it has a generic point. Since the $V \cap Z_\alpha \cap Z'_\lambda$ form a filtered increasing family of closed subsets of $V \cap Z_\alpha$, there is a λ such that

$$V \cap Z_\alpha \cap Z'_\lambda = V \cap Z_\alpha \quad (0, 9.2.4).$$

Equivalently, $V \cap Z_\alpha \subset V \cap Z'_\lambda$. The two filtered decreasing families $V \cap U_\alpha$ and $V \cap U'_\lambda$, where $U'_\lambda = X - Z'_\lambda$, are therefore cofinal. The asserted independence follows from (5.9.1.2).

The subset Z need not be constructible. For example, take $X = \text{Spec}(A)$, where A is a Noetherian domain having infinitely many maximal ideals, and let Z be the complement of the generic point of X . IV-2 | 110

If Z is closed and $i : X - Z \rightarrow X$ is the canonical injection, then

$$(5.9.1.3) \quad \mathcal{H}_{X/Z}^0(\mathcal{F}) = i_*(\mathcal{F}|_{X-Z}).$$

In particular, when $Z = \emptyset$, $\mathcal{H}_{X/Z}^0(\mathcal{F}) = \mathcal{F}$.

Proposition (5.9.2). — (i) The functor $\mathcal{F} \mapsto \mathcal{H}_{X/Z}^0(\mathcal{F})$ is left exact.

(ii) If $\mathcal{F}$ is quasi-coherent, so is $\mathcal{H}_{X/Z}^0(\mathcal{F})$.

Proof. Assertion (i) follows from (5.9.1.1), from the left exactness of $(i_\alpha)_*$, and from the fact that inductive limits preserve exactness in the category of $\mathcal{O}_X$ -modules. Assertion (ii) follows from (I, 9.2.2) and from the fact that an inductive limit of quasi-coherent $\mathcal{O}_X$ -modules is quasi-coherent (I, 1.3.9). □

Remark (5.9.3). — If $\mathcal{F}$ is an $\mathcal{O}_X$ -algebra, then $\mathcal{H}_{X/Z}^0(\mathcal{F})$ is one as well (0, 4.2.4). In particular, $\mathcal{H}_{X/Z}^0(\mathcal{O}_X)$ is a quasi-coherent $\mathcal{O}_X$ -algebra, and, for every $\mathcal{O}_X$ -module $\mathcal{F}$, $\mathcal{H}_{X/Z}^0(\mathcal{F})$ is a module over that algebra; it is quasi-coherent when $\mathcal{F}$ is quasi-coherent (I, 9.6.1).

More specifically, suppose that $X = \operatorname{Spec}(A)$, where A is a Noetherian domain. Then $\mathcal{H}_{X/Z}^0(\mathcal{O}_X)$ is the $\mathcal{O}_X$ -algebra $\tilde{B}$, where

$$(5.9.3.1) \quad B = \bigcap_{\mathfrak{p} \in X-Z} A_{\mathfrak{p}}.$$

This follows at once from (5.9.1.2) and (I, 8.2.1.1).

Proposition (5.9.4). — Let X be a locally Noetherian prescheme, let $Z \subset X$ be stable under specialization, let X' be a locally Noetherian prescheme, and let $f : X' \rightarrow X$ be flat. Then $Z' = f^{-1}(Z)$ is stable under specialization, and, for every quasi-coherent $\mathcal{O}_X$ -module $\mathcal{F}$, there is a canonical isomorphism

$$(5.9.4.1) \quad f^*(\mathcal{H}_{X/Z}^0(\mathcal{F})) \xrightarrow{\sim} \mathcal{H}_{X'/Z'}^0(f^*\mathcal{F}).$$

Proof. With the notation of (5.9.1), $Z'_\alpha = f^{-1}(Z_\alpha)$ is closed in X' , and Z' is the union of the Z'_α . The base change of i_α to X' is the canonical injection $i'_\alpha : U'_\alpha \rightarrow X'$, where $U'_\alpha = X' - Z'_\alpha = f^{-1}(U_\alpha)$. Since f is flat, the canonical homomorphism

$$f^*((i_\alpha)_*(\mathcal{F}|_{U_\alpha})) \longrightarrow (i'_\alpha)_*(f^*\mathcal{F}|_{U'_\alpha})$$

is bijective (2.3.1). For $\alpha \leq \beta$, the diagram

$$\begin{array}{ccc} f^*((i_\alpha)_*(\mathcal{F}|_{U_\alpha})) & \xrightarrow{\sim} & (i'_\alpha)_*(f^*\mathcal{F}|_{U'_\alpha}) \\ \downarrow & & \downarrow \\ f^*((i_\beta)_*(\mathcal{F}|_{U_\beta})) & \xrightarrow{\sim} & (i'_\beta)_*(f^*\mathcal{F}|_{U'_\beta}) \end{array}$$

is commutative. Passing to the limit gives a canonical isomorphism

$$\varinjlim_\alpha f^*((i_\alpha)_*(\mathcal{F}|_{U_\alpha})) \xrightarrow{\sim} \mathcal{H}_{X'/Z'}^0(f^*\mathcal{F}).$$

Since f^* commutes with inductive limits (0, 4.3.2), this is precisely (5.9.4.1). □ IV-2 | 111

Corollary (5.9.5). — Under the hypotheses of (5.9.4), if $\mathcal{H}_{X/Z}^0(\mathcal{F})$ is coherent, then $\mathcal{H}_{X'/Z'}^0(f^*\mathcal{F})$ is coherent. The converse holds when f is faithfully flat and quasi-compact.

Proof. The first assertion follows from (5.9.4.1) and (0, 5.3.11). The second amounts to saying that if $\mathcal{H}_{X'/Z'}^0(f^*\mathcal{F})$ is of finite type, then so is $\mathcal{H}_{X/Z}^0(\mathcal{F})$; this follows from (5.9.4.1) and (2.5.2). □

Corollary (5.9.6). — Let X be a locally Noetherian prescheme, let $Z \subset X$ be stable under specialization, and let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$ -module. For every $x \in X$, put $X_x = \operatorname{Spec}(\mathcal{O}_x)$ and $Z_x = Z \cap X_x$. There is a functorial isomorphism

$$(5.9.6.1) \quad (\mathcal{H}_{X/Z}^0(\mathcal{F}))_x \xrightarrow{\sim} \mathcal{H}_{X_x/Z_x}^0(\tilde{\mathcal{F}}_x).$$

Proof. Apply (5.9.4) to the canonical flat morphism $X_x \rightarrow X$ and use (I, 1.6.5). □

(5.9.7). With the notation of (5.9.1), for every α there is a canonical functorial homomorphism

$$\mathcal{F} \longrightarrow (i_\alpha)_*(\mathcal{F}|_{U_\alpha}) \quad (0, 4.4.3.2).$$

These homomorphisms form an inductive system, and passage to the limit gives a canonical functorial homomorphism

$$(5.9.7.1) \quad \rho_{X/Z} : \mathcal{F} \longrightarrow \mathcal{H}_{X/Z}^0(\mathcal{F}).$$

Proposition (5.9.8). — Let X be a locally Noetherian prescheme, let $Z \subset X$ be stable under specialization, and let $\mathcal{F}$ be an $\mathcal{O}_X$ -module. The following conditions are equivalent:

- a) The homomorphism $\rho_{X/Z}$ in (5.9.7.1) is injective (respectively bijective).

b) For every Noetherian open subset V of X , the homomorphism

$$(\rho_{X/Z})_V : \Gamma(V, \mathcal{F}) \longrightarrow \varinjlim_{\alpha} \Gamma(V \cap U_{\alpha}, \mathcal{F})$$

is injective (respectively bijective).

They are also equivalent to:

a') For every closed subset $T \subset Z$ of X , the canonical homomorphism (0, 4.4.3.2)

$$\mathcal{F} \longrightarrow i_*(\mathcal{F}|_{X-T}),$$

where $i : X - T \rightarrow X$ is the canonical injection, is injective (respectively bijective).

b') For every closed subset $T \subset Z$ and every Noetherian open subset $V \subset X$, the restriction homomorphism

$$\Gamma(V, \mathcal{F}) \longrightarrow \Gamma(V \cap (X - T), \mathcal{F})$$

is injective (respectively bijective).

Proof. Taking (5.9.1.2) into account, the equivalence of a) and b), and likewise that of a') and b'), follows from the definition of the functor Γ and its left exactness.

For every closed subset $T \subset Z$, the homomorphism $(\rho_{X/Z})_V$ is the composite

$$(5.9.8.1) \quad \Gamma(V, \mathcal{F}) \longrightarrow \Gamma(V \cap (X - T), \mathcal{F}) \longrightarrow \varinjlim_{\alpha} \Gamma(V \cap U_{\alpha}, \mathcal{F}).$$

Thus b) implies b') in the injective case, while b') implies b) directly from the definition of an inductive limit.

It remains to show that, if $\rho_{X/Z}$ is bijective, then so is $\Gamma(V, \mathcal{F}) \rightarrow \Gamma(V \cap (X - T), \mathcal{F})$. By (5.9.8.1), it is enough to observe that whenever $U' \subset U$ are open subsets of V containing $V \cap (X - Z)$, the restriction homomorphism $\Gamma(U, \mathcal{F}) \rightarrow \Gamma(U', \mathcal{F})$ is injective. This follows from the injectivity of $\rho_{X/Z}$ by replacing V with U and $V \cap (X - T)$ with U' . $\square$

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Definition (5.9.9). — Under the hypotheses of (5.9.8), the module $\mathcal{F}$ is called *Z-pure* (respectively *Z-closed*) if $\rho_{X/Z}$ is injective (respectively bijective).

If $X = \text{Spec}(A)$ is affine and $\mathcal{F} = \tilde{M}$, where M is an A -module, we call M *Z-pure* (respectively *Z-closed*) when $\mathcal{F}$ is.

We say that $\mathcal{F}$ is *Z-pure* (respectively *Z-closed*) at a point $x \in X$ if, with the notation of (5.9.6), $\widetilde{\mathcal{F}}_x$ is Z_x -pure (respectively Z_x -closed). By (5.9.6), this is equivalent to saying that

$$\mathcal{F}_x \longrightarrow (\mathcal{H}_{X/Z}^0(\mathcal{F}))_x$$

is injective (respectively bijective). At every point $x \in X - Z$, $\mathcal{F}$ is *Z-closed* by (5.9.8).

Corollary (5.9.10). — (i) Let (V_{λ}) be an open covering of X . Then $\mathcal{F}$ is *Z-pure* (respectively *Z-closed*) if and only if, for every λ , $\mathcal{F}|_{V_{\lambda}}$ is $(Z \cap V_{\lambda})$ -pure (respectively $(Z \cap V_{\lambda})$ -closed).
(ii) Let $Z' \subset Z$ be stable under specialization. If $\mathcal{F}$ is *Z-pure* (respectively *Z-closed*), then it is Z' -pure (respectively Z' -closed).

Proof. This follows immediately from condition b') of (5.9.8). $\square$

Proposition (5.9.11). — Under the hypotheses of (5.9.8), the supports of $\ker(\rho_{X/Z})$ and $\text{coker}(\rho_{X/Z})$ are contained in Z , and $\mathcal{H}_{X/Z}^0(\mathcal{F})$ is *Z-closed*.

Furthermore, if $u : \mathcal{F} \rightarrow \mathcal{F}'$ is an $\mathcal{O}_X$ -module homomorphism and $\mathcal{F}'$ is *Z-closed*, then u factors uniquely as

$$\mathcal{F} \xrightarrow{\rho_{X/Z}} \mathcal{H}_{X/Z}^0(\mathcal{F}) \xrightarrow{v} \mathcal{F}'.$$

If, in addition, the supports of $\ker(u)$ and $\text{coker}(u)$ are contained in Z , then v is an isomorphism.

Proof. The first assertion says that, for every $x \in X - Z$,

$$\ker(\rho_{X/Z})_x = \text{coker}(\rho_{X/Z})_x = 0.$$

Equivalently, $(\rho_{X/Z})_x$ is bijective, or $\mathcal{F}$ is *Z-closed* at x , as noted after (5.9.9).

To prove that $\mathcal{H}_{X/Z}^0(\mathcal{F})$ is Z -closed, it is enough to show, for every Noetherian open $V \subset X$, that

$$\varinjlim_{\beta} \Gamma(V \cap U_{\beta}, \mathcal{H}_{X/Z}^0(\mathcal{F})) = \Gamma(V, \mathcal{H}_{X/Z}^0(\mathcal{F})).$$

By definition,

$$\Gamma(V \cap U_{\beta}, \mathcal{H}_{X/Z}^0(\mathcal{F})) = \varinjlim_{\alpha} \Gamma(V \cap U_{\alpha} \cap U_{\beta}, \mathcal{F}).$$

The double family $(U_{\alpha} \cap U_{\beta})$ is filtered decreasing, so the assertion follows from (5.9.1) and the theorem on double inductive limits.

We now prove the second part. For every α , $(i_{\alpha})_*(u)$ is the unique homomorphism making the

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$$\begin{array}{ccc} \mathcal{F} & \xrightarrow{u} & \mathcal{F}' \\ \downarrow & & \downarrow \\ (i_{\alpha})_*(\mathcal{F}|_{U_{\alpha}}) & \xrightarrow{(i_{\alpha})_*(u)} & (i_{\alpha})_*(\mathcal{F}'|_{U_{\alpha}}) \end{array}$$

commute. Hence there is a unique homomorphism w making all diagrams

$$\begin{array}{ccc} (i_{\alpha})_*(\mathcal{F}|_{U_{\alpha}}) & \xrightarrow{(i_{\alpha})_*(u)} & (i_{\alpha})_*(\mathcal{F}'|_{U_{\alpha}}) \\ \downarrow & & \downarrow \\ \mathcal{H}_{X/Z}^0(\mathcal{F}) & \xrightarrow{w} & \mathcal{H}_{X/Z}^0(\mathcal{F}') \end{array}$$

commute. Since $\mathcal{F}'$ is Z -closed, it identifies canonically with $\mathcal{H}_{X/Z}^0(\mathcal{F}')$. This proves existence and uniqueness of v .

Suppose finally that the supports of $\ker(u)$ and $\operatorname{coker}(u)$ are contained in Z . We show that, for every Noetherian open $V \subset X$, the homomorphism

$$\Gamma(V, \mathcal{H}_{X/Z}^0(\mathcal{F})) \longrightarrow \Gamma(V, \mathcal{F}')$$

is an isomorphism. If a section $t \in \Gamma(V, \mathcal{H}_{X/Z}^0(\mathcal{F}))$ maps to zero, then, for some α , it is represented by a section on $V \cap U_{\alpha}$. By the hypothesis on u , its germ vanishes at every point of $V \cap (X - Z)$. It therefore vanishes on an open neighbourhood of $V \cap (X - Z)$, and so represents zero in $\Gamma(V, \mathcal{H}_{X/Z}^0(\mathcal{F}))$.

Conversely, let $s' \in \Gamma(V, \mathcal{F}')$. For every $x \in V \cap (X - Z)$, there is an open neighbourhood $W^{(x)}$ of x and a section $s^{(x)}$ of $\mathcal{F}$ on $W^{(x)}$ whose image under u is $s'|W^{(x)}$. Hence $s'|W^{(x)}$ is also the image under v of a section $t^{(x)}$ of $\mathcal{H}_{X/Z}^0(\mathcal{F})$. Since v is injective, these sections agree on overlaps. They therefore glue to a section t over an open subset $U \subset V$ containing $V \cap (X - Z)$. As $\mathcal{H}_{X/Z}^0(\mathcal{F})$ is Z -closed, t extends uniquely over V ; its image under v agrees with s' on U , and hence on V by the same argument.

The module $\mathcal{H}_{X/Z}^0(\mathcal{F})$ is called the Z -closure of $\mathcal{F}$. □

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Remarks (5.9.12). — (i) Let $\mathcal{C}(X)$ be the category of $\mathcal{O}_X$ -modules, and let $\mathcal{C}_Z(X)$ be its full subcategory of modules whose support is contained in Z . This is a localizing subcategory in the sense of Gabriel, and $\mathcal{H}_{X/Z}^0$ is precisely Gabriel's localization functor (cf. [27]); this gives another proof of (5.9.11). When Z is closed, the functor $i^* : \mathcal{C}(X) \rightarrow \mathcal{C}(X - Z)$, where $i : X - Z \rightarrow X$, defines an equivalence

$$\mathcal{C}(X)/\mathcal{C}_Z(X) \simeq \mathcal{C}(X - Z).$$

(ii) By (5.9.11), $\mathcal{H}_{X/Z}^0(\mathcal{F}) = 0$ if and only if $\operatorname{Supp}(\mathcal{F}) \subset Z$. Indeed, the vanishing implies that $\ker(\rho_{X/Z}) = \mathcal{F}$. Conversely, if $\operatorname{Supp}(\mathcal{F}) \subset Z$, apply the second part of (5.9.11) to the unique map $u : \mathcal{F} \rightarrow 0$; the corresponding map $v : \mathcal{H}_{X/Z}^0(\mathcal{F}) \rightarrow 0$ is an isomorphism.

- (iii) The preceding constructions make sense for every locally Noetherian ringed space in which every closed irreducible subset has a unique generic point. In particular, on a locally Noetherian prescheme they apply to arbitrary sheaves of abelian groups, regarded as modules over the sheaf associated with the constant presheaf $\mathbf{Z}$. For every $x \in X$ one still has the canonical isomorphism (5.9.6.1), in which $\widetilde{\mathcal{F}}_x$ denotes the sheaf induced by $\mathcal{F}$ on the subspace X_x of X . A direct proof follows from (5.9.1.2) and the theorem on double inductive limits.

5.10. Property (S₂) and Z-closure

(5.10.1). Let X be a locally Noetherian prescheme and let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -module. For every subset T of X , put

$$(5.10.1.1) \quad \text{prof}_T(\mathcal{F}) = \inf_{z \in T} \text{prof}(\mathcal{F}_z).$$

Proposition (5.10.2). — *Let X be a locally Noetherian prescheme, let Z be a subset of X stable under specialization, and let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$ -module. The following conditions are equivalent:*

- a) $\mathcal{F}$ is Z -pure.
- b) $\text{Ass}(\mathcal{F})$ does not meet Z .

If, in addition, $\mathcal{F}$ is coherent, these conditions are also equivalent to:

- c) $\text{prof}_Z(\mathcal{F}) \geq 1$.

Proof. To say that $\mathcal{F}$ is Z -pure means that, for every Noetherian open subset V of X and every open subset $U \supset X - Z$, the restriction homomorphism

$$\Gamma(V, \mathcal{F}) \longrightarrow \Gamma(V \cap U, \mathcal{F})$$

is injective (5.9.8). By (3.1.8), this is equivalent to $V \cap \text{Ass}(\mathcal{F}) \subset U$, which proves the equivalence of a) and b). Further, to say that $x \in \text{Ass}(\mathcal{F})$ means that no element of $\mathfrak{m}_x$ is $\mathcal{F}_x$ -regular (3.1.2); when $\mathcal{F}$ is coherent, this is equivalent to $\text{prof}(\mathcal{F}_x) = 0$. The equivalence of b) and c) follows immediately. $\square$

Corollary (5.10.3). — *Let*

$$0 \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}'' \longrightarrow 0$$

be an exact sequence of quasi-coherent $\mathcal{O}_X$ -modules. If $\mathcal{F}$ is Z -pure, then so is $\mathcal{F}'$. Conversely, if $\mathcal{F}'$ and $\mathcal{F}''$ are Z -pure, then so is $\mathcal{F}$.

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Proof. This follows from the form (5.10.2), b), of the condition for a quasi-coherent $\mathcal{O}_X$ -module to be Z -pure, and from (3.1.7). $\square$

Corollary (5.10.4). — *Suppose that $\mathcal{F}$ is coherent. For $\mathcal{F}$ to be Z -pure at a point $x \in X$, it is necessary and sufficient that*

$$\text{prof}_{Z_x}(\widetilde{\mathcal{F}}_x) \geq 1,$$

with the notation of (5.9.6).

Proof. This follows immediately from (5.10.2) and (5.9.6). $\square$

Theorem (5.10.5). — *Let X be a locally Noetherian prescheme, let Z be a subset of X stable under specialization, and let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -module. For $\mathcal{F}$ to be Z -closed, it is necessary and sufficient that*

$$\text{prof}_Z(\mathcal{F}) \geq 2.$$

Proof. By (5.10.2), we may restrict ourselves to the case in which $\mathcal{F}$ is Z -pure and $\text{prof}_Z(\mathcal{F}) \geq 1$. Moreover, to say that $\text{prof}_Z(\mathcal{F}) \geq 2$ is equivalent to saying that

$$\text{prof}_{Z_\alpha}(\mathcal{F}) \geq 2$$

for every closed subset Z_α of Z ; similarly, by (5.9.8), to say that $\mathcal{F}$ is Z -closed is equivalent to saying that it is Z_α -closed for every α . We may therefore first reduce to the case in which Z is closed.

The question is local. For each $x \in Z$, it is enough to prove the theorem for $\mathcal{F}|_U$, where U is an affine open neighbourhood of x ; we may thus suppose that $X = U$ is affine. Then $\text{Ass}(\mathcal{F})$ is finite (3.1.6), and, since $\text{Ass}(\mathcal{F}) \subset X - Z$, there is a section f of $\mathcal{O}_X$ over X such that

$$\text{Ass}(\mathcal{F}) \subset X_f \subset X - Z \quad (\text{II}, 4.5.4).$$

It follows that f is $\mathcal{F}$ -regular (3.1.9), and for every $y \in Z$ one has $f_y \in \mathfrak{m}_y$, hence

$$\text{prof}(\mathcal{F}_y) = 1 + \text{prof}(\mathcal{F}_y / f_y \mathcal{F}_y) \quad (0, 16.4.6).$$

Thus $\text{prof}_Z(\mathcal{F}) \geq 2$ is equivalent to $\text{prof}_Z(\mathcal{F} / f \mathcal{F}) \geq 1$, or, by (5.10.2), to $\mathcal{F} / f \mathcal{F}$ being Z -pure. It remains to show that this last property is equivalent to $\mathcal{F}$ being Z -closed.

Consider the exact sequence

$$0 \longrightarrow \mathcal{F} \xrightarrow{f} \mathcal{F} \longrightarrow \mathcal{F} / f \mathcal{F} \longrightarrow 0,$$

where multiplication by f is injective by hypothesis. Put $W = X - Z$. We have the commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & \Gamma(X, \mathcal{F}) & \xrightarrow{f} & \Gamma(X, \mathcal{F}) & \longrightarrow & \Gamma(X, \mathcal{F} / f \mathcal{F}) \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \Gamma(W, \mathcal{F}) & \xrightarrow{f} & \Gamma(W, \mathcal{F}) & \longrightarrow & \Gamma(W, \mathcal{F} / f \mathcal{F}) \end{array}$$

whose two rows are exact, since X is affine. If the restriction homomorphism

$$\Gamma(X, \mathcal{F}) \longrightarrow \Gamma(W, \mathcal{F})$$

is bijective, the diagram shows that

$$\Gamma(X, \mathcal{F} / f \mathcal{F}) \longrightarrow \Gamma(W, \mathcal{F} / f \mathcal{F})$$

is injective; hence $\mathcal{F} / f \mathcal{F}$ is Z -pure (5.9.8).

Conversely, suppose that $\mathcal{F} / f \mathcal{F}$ is Z -pure, and let s be a section of $\mathcal{F}$ over W . Since $X_f \subset W$, there is an integer $n > 0$ such that $f^n(s|_{X_f})$ extends to a section t of $\mathcal{F}$ over X (I, 1.4.1). The restrictions of t and $f^n s$ to X_f are equal; the relation $\text{Ass}(\mathcal{F}) \subset X_f$ therefore implies, by (5.10.2), that $t|_W = f^n s$. Since f is $\mathcal{F}$ -regular, it is enough to show that $t = f^n t'$ for some $t' \in \Gamma(X, \mathcal{F})$.

This is equivalent to the image of t in $\Gamma(X, \mathcal{F} / f^n \mathcal{F})$ being zero. For every k , $f^k \mathcal{F} / f^{k+1} \mathcal{F}$ is isomorphic to $\mathcal{F} / f \mathcal{F}$, hence is Z -pure. By induction on n , using (5.10.3), it follows that $\mathcal{F} / f^n \mathcal{F}$ is Z -pure. But the image of $t|_W = f^n s$ in $\Gamma(W, \mathcal{F} / f^n \mathcal{F})$ is zero; the assertion follows from Z -purity. $\square$

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Corollary (5.10.6). — *Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -module. For $\mathcal{F}$ to be Z -closed at a point x , it is necessary and sufficient that*

$$\text{prof}_{Z_x}(\widetilde{\mathcal{F}}_x) \geq 2.$$

Proof. This follows from (5.9.6) and (5.10.5). $\square$

Theorem (5.10.7). — (Hartshorne). *Let X be a locally Noetherian prescheme and let Y be a closed subset of X . Suppose that $\text{prof}(\mathcal{O}_{X,y}) \geq 2$ for every $y \in Y$. Then, for every connected component C of X , $C - (C \cap Y)$ is connected.*

Proof. We may restrict ourselves to the case in which X is connected. By (5.10.5), the canonical homomorphism

$$\mathcal{O}_X \longrightarrow i_*(\mathcal{O}_X|_{X-Y}),$$

where $i : X - Y \rightarrow X$ is the canonical injection, is bijective. Consequently the restriction homomorphism

$$\Gamma(X, \mathcal{O}_X) \longrightarrow \Gamma(X - Y, \mathcal{O}_X)$$

is also bijective. It remains to apply (III, 7.8.6.1). $\square$

Corollary (5.10.8). — *Let X be a locally Noetherian prescheme and let d be an integer such that, for every $x \in X$,*

$$\dim(\mathcal{O}_x) \geq d \implies \text{prof}(\mathcal{O}_x) \geq 2.$$

Suppose that X is connected. If X' and X'' are two distinct irreducible components of X , there is a sequence $(X_i)_{0 \leq i \leq n}$ of irreducible components of X such that $X_0 = X'$, $X_n = X''$, and

$$\text{codim}(X_{i-1} \cap X_i, X) \leq d - 1 \quad (1 \leq i \leq n).$$

In this case one says that X is connected in codimension $\leq d - 1$.

Proof. If Y is a closed subset of X such that $\text{codim}(Y, X) \geq d$, then $\dim(\mathcal{O}_{X,y}) \geq d$ for every $y \in Y$ (5.1.3); hence $\text{prof}(\mathcal{O}_{X,y}) \geq 2$ for every $y \in Y$, and (5.10.7) shows that $X - Y$ is connected.

For $\text{codim}(Y, X) \geq d$, it is necessary and sufficient that, for every $y \in Y$, there be an open neighbourhood V of y such that $\text{codim}(Y \cap V, V) \geq d$ (0, 14.2.3). Let $\mathfrak{Y}$ be the set of closed subsets Y of X of codimension at least d . The union of two members of $\mathfrak{Y}$ belongs to $\mathfrak{Y}$ (0, 14.2.5), and every closed subset of a member of $\mathfrak{Y}$ also belongs to $\mathfrak{Y}$. In other words, $\mathfrak{Y}$ is an *antifilter* of closed subsets of X . The corollary now follows from the following topological lemma. $\square$

Lemma (5.10.8.1). — *Let X be a connected locally Noetherian topological space and let $\mathfrak{Y}$ be an antifilter of closed subsets of X . Suppose that a closed subset Y belongs to $\mathfrak{Y}$ whenever, for every $y \in Y$, there exist an open neighbourhood V of y in X and a $Y_y \in \mathfrak{Y}$ such that $V \cap Y = V \cap Y_y$. The following conditions are equivalent:*

- a) *For every $Y \in \mathfrak{Y}$, the space $X - Y$ is connected.*
- b) *If X' and X'' are two distinct irreducible components of X , there is a sequence $(X_i)_{0 \leq i \leq n}$ of irreducible components of X such that $X_0 = X'$, $X_n = X''$, and*

$$X_{i-1} \cap X_i \notin \mathfrak{Y} \quad (1 \leq i \leq n).$$

Proof. Suppose b) holds and let $Y \in \mathfrak{Y}$. Put $U = X - Y$. If U' and U'' are two distinct irreducible components of U , there are irreducible components X' and X'' of X such that $X' \cap U = U'$ and $X'' \cap U = U''$ (0, 2.1.6).

Choose a sequence (X_i) with the property in b), and put $U_i = X_i \cap U$ for $0 \leq i \leq n$. Each U_i is an irreducible component of U (0, 2.1.6), and $U_i \cap U_{i-1} \neq \emptyset$ for $1 \leq i \leq n$: otherwise $X_i \cap X_{i-1} \subset Y$, so $X_i \cap X_{i-1} \in \mathfrak{Y}$, contrary to the choice of the X_i . Thus U is connected. IV-2 | 117

Conversely, suppose a) holds. Let Y be the union of the sets $X_\alpha \cap X_\beta$, where (X_α, X_β) ranges over the pairs of distinct irreducible components of X for which $X_\alpha \cap X_\beta \in \mathfrak{Y}$. Every $y \in Y$ has an open neighbourhood V meeting only finitely many irreducible components of X . It follows both that Y is closed and that $V \cap Y$ is the intersection of V with a member of $\mathfrak{Y}$. By hypothesis, $Y \in \mathfrak{Y}$; hence $U = X - Y$ is connected. Moreover, Y is nowhere dense in X .

Let X' and X'' be distinct irreducible components of X , and let U' and U'' be their respective traces on U . They are distinct irreducible components of U (0, 2.1.6). The union of those irreducible components W of U for which there is a sequence $(U_i)_{0 \leq i \leq n}$ of irreducible components of U satisfying

$$U_0 = U', \quad U_{i-1} \cap U_i \neq \emptyset \quad (1 \leq i \leq n), \quad U_n = W,$$

is both open and closed in U , because U is locally Noetherian and its irreducible components form a locally finite family of closed subsets. Since U is connected, there is such a sequence ending at U'' . Let X_i be the irreducible component of X such that $X_i \cap U = U_i$ (0, 2.1.6). Since $U_{i-1} \neq U_i$, one has $X_{i-1} \neq X_i$. If $X_{i-1} \cap X_i \in \mathfrak{Y}$ for some i , then $X_{i-1} \cap X_i \subset Y$, contrary to $U_{i-1} \cap U_i \neq \emptyset$. This proves b). $\square$

The hypothesis on X in (5.10.8) is satisfied when X is a Cohen–Macaulay prescheme and $d \geq 2$.

Corollary (5.10.9). — *If a Noetherian local ring A satisfies (S_2) and is catenary, then it is equidimensional.*

Proof. The hypothesis of (5.10.8) is satisfied by $X = \text{Spec}(A)$ with $d = 2$. By that corollary, to show that all irreducible components of X have the same dimension, it is enough to consider two such components X' and X'' with

$$\text{codim}(X' \cap X'', X) = 1.$$

There is then an irreducible component Z of $X' \cap X''$ such that $\text{codim}(Z, X) = 1$. Since

$$1 = \text{codim}(Z, X) \geq \text{codim}(Z, X') \geq 1,$$

one has $\text{codim}(Z, X') = 1$; similarly $\text{codim}(Z, X'') = 1$. Since X is catenary, this implies $\dim(X') = \dim(X'')$. $\square$

Proposition (5.10.10). — *Let X be a locally Noetherian prescheme, let Z be a subset of X stable under specialization, and let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -module. Suppose that the $\mathcal{O}_X$ -module $\mathcal{H}_{X/Z}^0(\mathcal{F})$ is coherent. Then:*

(i)

$$\text{prof}_Z(\mathcal{H}_{X/Z}^0(\mathcal{F})) \geq 2.$$

(ii) For every $x \in \text{Ass}(\mathcal{F}) \cap (X - Z)$,

$$\text{codim}(Z \cap \overline{\{x\}}, \overline{\{x\}}) \geq 2.$$

(iii) The set U of points $x \in X$ such that

$$\text{prof}_{Z_x}(\widetilde{\mathcal{F}}_x) \geq 2$$

(with the notation of (5.9.6)) is open in X ; moreover, $X - U \subset Z$, and U is the largest open subset of X such that $\mathcal{F}|_U$ is $(Z \cap U)$ -closed.

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Proof. Put, for brevity,

$$\mathcal{F}' = \mathcal{H}_{X/Z}^0(\mathcal{F}).$$

By (5.9.11), $\mathcal{F}'$ is Z -closed, so assertion (i) follows from (5.10.5).

Let $x \in \text{Ass}(\mathcal{F}) \cap (X - Z)$. The restrictions of $\mathcal{F}$ and $\mathcal{F}'$ to $X - Z$ are canonically isomorphic (5.9.11); hence $x \in \text{Ass}(\mathcal{F}')$. For $y \in Z \cap \overline{\{x\}}$, let $\mathfrak{p}$ be the prime ideal of $\mathcal{O}_y$ corresponding to x . It is associated to the $\mathcal{O}_y$ -module $\mathcal{F}'_y$. Consequently, by (i) and (0, 16.4.6.2),

$$2 \leq \text{prof}(\mathcal{F}'_y) \leq \dim(\mathcal{O}_y/\mathfrak{p}) = \text{codim}(\overline{\{y\}}, \overline{\{x\}}),$$

which proves (ii).

Finally, by (5.10.5), U is the set of points x for which $\widetilde{\mathcal{F}}_x$ is Z_x -closed; by (5.9.6), it is equivalently the set of points at which the canonical homomorphism

$$\mathcal{F}_x \longrightarrow \mathcal{F}'_x$$

is bijective. Thus $X - U$ is the union of the supports of $\text{Ker}(\rho_{X/Z})$ and $\text{Coker}(\rho_{X/Z})$. These are coherent $\mathcal{O}_X$ -modules by the hypothesis and (0, 5.3.4), so their supports are closed (0, 5.2.2). Hence U is open. It is plainly the largest open subset on which $\mathcal{F}$ is $(Z \cap U)$ -closed, and $X - U \subset Z$ follows from (5.9.11). $\square$

We shall see in (5.11.1) that, in the most important cases, assertion (ii) conversely implies that $\mathcal{H}_{X/Z}^0(\mathcal{F})$ is coherent.

(5.10.11). Let X be a locally Noetherian prescheme and let Z be a subset of X stable under specialization. We saw in (5.9.3) that

$$\mathcal{A} = \mathcal{H}_{X/Z}^0(\mathcal{O}_X)$$

is a quasi-coherent $\mathcal{O}_X$ -algebra. The X -scheme

$$X' = \text{Spec}(\mathcal{A})$$

(II, 1.3.1) is called the Z -closure of X .

For every $\mathcal{O}_X$ -module $\mathcal{F}$, $\mathcal{H}_{X/Z}^0(\mathcal{F})$ is an $\mathcal{A}$ -module, and it is quasi-coherent when $\mathcal{F}$ is quasi-coherent. In that case there is a unique quasi-coherent $\mathcal{O}_{X'}$ -module $\mathcal{F}'$ such that, if $g : X' \rightarrow X$ denotes the structural morphism, then

$$(5.10.11.1) \quad \mathcal{H}_{X/Z}^0(\mathcal{F}) = g_*(\mathcal{F}').$$

Proposition (5.10.12). — *Use the notation of (5.10.11).*

(i) Let $x \in X$. The morphism obtained from g by localization,

$$X' \times_X X_x \longrightarrow X_x = \operatorname{Spec}(\mathcal{O}_{X,x}),$$

is an isomorphism if and only if $\mathcal{O}_X$ is Z -closed at x . This condition holds for every $x \in X - Z$.

(ii) Put $Z' = g^{-1}(Z)$ and suppose that X' is locally Noetherian. Then $\mathcal{F}'$ is Z' -closed. If, in addition, $\mathcal{F}'$ is a coherent $\mathcal{O}_{X'}$ -module, then

$$\operatorname{prof}_{Z'}(\mathcal{F}') \geq 2.$$

(iii) Suppose that both $\mathcal{H}_{X/Z}^0(\mathcal{O}_X)$ and $\mathcal{H}_{X/Z}^0(\mathcal{F})$ are coherent. Then $g : X' \rightarrow X$ is finite. The set U of points $x \in X$ such that

$$\operatorname{prof}_{Z_x}(\tilde{\mathcal{O}}_x) \geq 2 \quad \text{and} \quad \operatorname{prof}_{Z_x}(\tilde{\mathcal{F}}_x) \geq 2$$

is open in X and satisfies $X - U \subset Z$. Moreover, U is the largest open subset of X such that $g^{-1}(U) \rightarrow U$ is an isomorphism and the restriction to U of the canonical g -morphism from $\mathcal{F}$ to $\mathcal{F}'$ is an isomorphism.

Proof. Assertion (i) follows from the definitions, and (iii) is an immediate consequence of (5.10.10), (iii).

For (ii), let V be an open subset of X containing $X - Z$, put $V' = g^{-1}(V)$, and let $i : V \rightarrow X$ and $i' : V' \rightarrow X'$ be the canonical injections.

The canonical homomorphism

$$\rho_{X'/Z'} : \mathcal{F}' \longrightarrow i'_*(\mathcal{F}'|V')$$

has the property that $g_*(\rho_{X'/Z'})$ is the canonical homomorphism

$$\rho_{X/Z} : \mathcal{H}_{X/Z}^0(\mathcal{F}) \longrightarrow i_*(\mathcal{H}_{X/Z}^0(\mathcal{F})|V),$$

by (II, 1.4.2) and (5.10.11.1). Since $\mathcal{H}_{X/Z}^0(\mathcal{F})$ is Z -closed (5.9.11), $\rho_{X/Z}$ is an isomorphism, and therefore so is $\rho_{X'/Z'}$. The open subset $X' - Z'$ is the intersection of the filtered family of inverse images $g^{-1}(V)$, as V ranges over the open subsets of X containing $X - Z$. Since X' is locally Noetherian, (5.9.1) now shows that $\mathcal{F}'$ is Z' -closed. The final assertion follows from (5.10.5). $\square$

(5.10.13). We shall now apply the preceding results when Z is one of the sets $Z^{(n)}(X)$ (or simply $Z^{(n)}$), defined as the set of points $x \in X$ such that

$$\dim(\mathcal{O}_x) \geq n.$$

It is clear that $Z^{(n)}$ is stable under specialization. A closed subset T of X is contained in $Z^{(n)}$ if and only if

$$\operatorname{codim}(T, X) \geq n.$$

We shall be concerned here with the case $n = 2$.

Proposition (5.10.14). — *Let X be a locally Noetherian prescheme, and let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -module whose support is X .*

- (i) *The module $\mathcal{F}$ satisfies (S_1) if and only if it is $Z^{(1)}$ -pure.*
- (ii) *The module $\mathcal{F}$ satisfies (S_2) if and only if it is $Z^{(2)}$ -closed and $Z^{(1)}$ -pure, or equivalently, if it is $Z^{(2)}$ -closed and has no associated prime cycle of codimension 1.*

Proof. To say that $\mathcal{F}$ has property (S_1) means that it has no embedded associated prime cycle (5.7.5), or equivalently that

$$\dim(\mathcal{F}_x) = 0 \quad \text{for every } x \in \operatorname{Ass}(\mathcal{F})$$

(3.1.4). Since $\operatorname{Supp}(\mathcal{F}) = X$, this is equivalent to $\dim(\mathcal{O}_x) = 0$ (5.1.12.1); in other words, $\operatorname{Ass}(\mathcal{F})$ does not meet $Z^{(1)}$. Assertion (i) therefore follows from (5.10.2).

To say that $\mathcal{F}$ is $Z^{(2)}$ -closed means that

$$\operatorname{prof}_{Z^{(2)}}(\mathcal{F}) \geq 2,$$

or equivalently that, for every $x \in X$,

$$\dim(\mathcal{O}_x) \geq 2 \implies \operatorname{prof}(\mathcal{F}_x) \geq 2.$$

Thus (S_2) implies that $\mathcal{F}$ is $Z^{(2)}$ -closed. It also implies (S_1) , so $\mathcal{F}$ has no embedded associated prime cycle (5.7.5); since its support is X , this says that all its associated prime cycles have codimension 0.

Conversely, suppose that $\mathcal{F}$ is $Z^{(2)}$ -closed and has no associated prime cycle of codimension 1. It remains only to consider a point x for which $\dim(\mathcal{O}_x) = 1$. By hypothesis $x \notin \text{Ass}(\mathcal{F})$, and hence $\text{prof}(\mathcal{F}_x) > 0$. Since $\dim(\mathcal{F}_x) = \dim(\mathcal{O}_x) = 1$, it follows that $\text{prof}(\mathcal{F}_x) = 1$. Therefore $\mathcal{F}$ satisfies (S_2) . Finally, if $\mathcal{F}$ is $Z^{(1)}$ -pure, it satisfies (S_1) by (i); as observed above, all its associated prime cycles then have codimension 0, so the same argument applies.

Notice that a module may be $Z^{(2)}$ -closed without satisfying (S_1) . For example, if $\dim X = 1$, then $Z^{(2)} = \emptyset$ and every $\mathcal{O}_X$ -module is $Z^{(2)}$ -closed, even though a module may have embedded associated prime cycles.

We recall that Chapter III gives, by means of local cohomology, a characterization of (S_n) for every $n \geq 1$ that generalizes (5.10.14). $\square$

Corollary (5.10.15). — *Let X be a locally Noetherian prescheme, and let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -module with support X . Suppose that $\mathcal{F}$ has no associated prime cycle of codimension 1 and that*

$$\mathcal{F}' = \mathcal{H}_{X/Z^{(2)}}^0(\mathcal{F})$$

is coherent. Then:

(i) $\mathcal{F}'$ satisfies (S_2) .

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(ii) *The set U of points $x \in X$ at which $\mathcal{F}$ satisfies (S_2) (5.7.2) is open in X , and*

$$\text{codim}(X - U, X) \geq 2.$$

Proof. By (5.9.11), $\mathcal{F}'$ is $Z^{(2)}$ -closed. Moreover, $\text{Supp}(\mathcal{F}') = X$: every maximal point of X lies outside $Z^{(2)}$, and at such a point $\mathcal{F}'_x = \mathcal{F}_x \neq 0$. Thus the support of $\mathcal{F}'$ is dense in X ; it is also closed because $\mathcal{F}'$ is coherent, and hence it is all of X .

It remains to see that $\mathcal{F}'$ has no associated prime cycle of codimension 1. If $x \in \text{Ass}(\mathcal{F}')$ and

$$\dim(\mathcal{F}'_x) = \dim(\mathcal{O}_x) = 1,$$

then $x \notin Z^{(2)}$, so $\mathcal{F}'_x = \mathcal{F}_x$ and consequently $x \in \text{Ass}(\mathcal{F})$, contrary to the hypothesis. Assertion (i) now follows from (5.10.14).

For (ii), one has

$$Z^{(n)}(X_x) = Z^{(n)}(X) \cap X_x$$

with the notation of (5.9.6), by (I, 2.4.2). The absence of associated prime cycles of codimension 1 for $\mathcal{F}$ implies the same property for $\widetilde{\mathcal{F}}_x$. By (5.10.14), $\widetilde{\mathcal{F}}_x$ satisfies (S_2) if and only if it is $Z^{(2)}(X_x)$ -closed. The assertion therefore follows from (5.10.6) and (5.10.10), (iii). $\square$

Proposition (5.10.16). — *Let X be a locally Noetherian prescheme, let*

$$X' = \text{Spec}(\mathcal{H}_{X/Z^{(2)}}^0(\mathcal{O}_X))$$

be its $Z^{(2)}$ -closure, and let $g : X' \rightarrow X$ be the structural morphism. Suppose that X has no associated prime cycle of codimension 1.

(i) *At a point $x \in X$, the prescheme X satisfies (S_2) if and only if the morphism*

$$X'_x \longrightarrow X_x$$

deduced from g (with the notation of (5.10.12)) is an isomorphism. This condition always holds if

$$\text{codim}(\overline{\{x\}}, X) \leq 1.$$

(ii) *Suppose in addition that g is finite (see (5.11.2) for sufficient conditions). Then the set U of points at which X satisfies (S_2) is open,*

$$\text{codim}(X - U, X) \geq 2,$$

and U is the largest open subset of X for which $g^{-1}(U) \rightarrow U$ is an isomorphism.

(iii) Under the hypotheses of (ii), X' satisfies (S_2) . Moreover, for every $x' \in X'$ such that

$$\text{codim}(\overline{\{x'\}}, X') \leq 1,$$

the point $x = g(x')$ satisfies

$$\text{codim}(\overline{\{x\}}, X) = \text{codim}(\overline{\{x'\}}, X').$$

(iv) Under the hypotheses of (ii), let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -module with support X , and suppose that $\mathcal{H}_{X/Z^{(2)}}^0(\mathcal{F})$ is coherent. The $\mathcal{O}_{X'}$ -module $\mathcal{F}'$ defined by

$$g_*(\mathcal{F}') = \mathcal{H}_{X/Z^{(2)}}^0(\mathcal{F})$$

is coherent and satisfies (S_2) , and its support is a union of irreducible components of X' .

Proof. Assertions (i) and (ii) have already been proved in substance: (i) follows from (5.10.12), (i), and (5.10.14), while (ii) is the special case of (5.10.15), (ii), obtained by taking $\mathcal{F} = \mathcal{O}_X$.

We prove (iii). Put $x = g(x')$. Since g is finite, so is $X'_x \rightarrow X_x$, and therefore

$$\dim(\mathcal{O}_{X',x'}) \leq \dim(X'_x) \leq \dim(X_x) = \dim(\mathcal{O}_{X,x})$$

by (5.4.1). Suppose first that $\dim(\mathcal{O}_{X',x'}) \leq 1$. Then $\dim(\mathcal{O}_{X,x}) \leq 1$ as well: otherwise $x \in Z^{(2)}$ and (5.10.12), (ii), applied to $\mathcal{O}_X$, would give

$$\text{prof}(\mathcal{O}_{X',x'}) \geq 2,$$

which is impossible. Hence $x \notin Z^{(2)}$, and (5.10.12), (i), gives an isomorphism

$$\mathcal{O}_{X,x} \simeq \mathcal{O}_{X',x'}.$$

In particular their dimensions are equal. Since X has no associated prime cycle of codimension 1, the equality $\dim(\mathcal{O}_{X,x}) = 1$ implies $x \notin \text{Ass}(\mathcal{O}_X)$, so

$$\text{prof}(\mathcal{O}_{X,x}) = \text{prof}(\mathcal{O}_{X',x'}) = 1.$$

If instead $\dim(\mathcal{O}_{X',x'}) \geq 2$, then $\dim(\mathcal{O}_{X,x}) \geq 2$, so $x \in Z^{(2)}$ and (5.10.12), (ii), gives

$$\text{prof}(\mathcal{O}_{X',x'}) \geq 2.$$

This proves all the assertions in (iii).

For (iv), replace $\mathcal{O}_X$ by $\mathcal{F}$ in the preceding argument. Since g is finite and $g_*(\mathcal{F}')$ is coherent, $\mathcal{F}'$ is coherent. The same argument proves that $\mathcal{F}'$ satisfies (S_2) and that, if $\dim(\mathcal{F}'_{x'}) \leq 1$, then $\mathcal{F}_x$ and $\mathcal{F}'_{x'}$ are di-isomorphic. In particular,

$$\dim(\mathcal{F}'_{x'}) = 0 \implies \dim(\mathcal{F}_x) = 0 \implies \dim(\mathcal{O}_{X,x}) = 0,$$

because $\text{Supp}(\mathcal{F}) = X$, and hence also $\dim(\mathcal{O}_{X',x'}) = 0$. Every irreducible component of $\text{Supp}(\mathcal{F}')$ is therefore an irreducible component of X' . Finally, $\text{Supp}(\mathcal{F}')$ is closed because $\mathcal{F}'$ is coherent. $\square$

Proposition (5.10.17). — Let A be a Noetherian integral domain, and denote by $A^{(1)}$ the intersection of the local rings $A_{\mathfrak{p}}$, where $\mathfrak{p}$ ranges over the prime ideals of A of height 1. Suppose that $A^{(1)}$ is a finite A -algebra. Then:

- (i) The ring $A^{(1)}$ satisfies (S_2) .
- (ii) The set U of primes $\mathfrak{p} \in \text{Spec}(A)$ for which the canonical homomorphism

$$A_{\mathfrak{p}} \longrightarrow (A^{(1)})_{\mathfrak{p}}$$

is bijective is precisely the set of primes for which $A_{\mathfrak{p}}$ satisfies (S_2) . The set U is open in $X = \text{Spec}(A)$ and

$$\text{codim}(X - U, X) \geq 2.$$

- (iii) For every multiplicative subset S of A , $(S^{-1}A)^{(1)}$ is a finite $S^{-1}A$ -algebra.
- (iv) Let B be a finite integral A -algebra that is an integral domain and contains A . Then $B^{(1)}$ is a finite B -algebra. Moreover, if $\mathfrak{q}$ is a prime ideal of B of height 1, then $\mathfrak{q} \cap A$ has height 1 in A .

Proof. Formula (5.9.3.1) shows that

$$X' = \operatorname{Spec}(A^{(1)})$$

is the $Z^{(2)}$ -closure of $X = \operatorname{Spec}(A)$. Since A has no embedded associated prime ideals, assertions (i) and (ii) are special cases of (5.10.16), (i)–(iii).

For (iii), observe that

$$(S^{-1}A)^{(1)} = S^{-1}A^{(1)},$$

which is a special case of (5.9.4). Indeed, $S^{-1}A$ is flat over A , its prime ideals are the ideals $S^{-1}\mathfrak{p}$ with $\mathfrak{p} \in \operatorname{Spec}(A)$ disjoint from S , and

$$\operatorname{ht}(S^{-1}\mathfrak{p}) = \operatorname{ht}(\mathfrak{p}).$$

Since $A^{(1)}$ is finite over A , its localization $S^{-1}A^{(1)}$ is finite over $S^{-1}A$.

To prove (iv), put $Y = \operatorname{Spec}(B)$ and let $f : Y \rightarrow X$ be the structural morphism. Since f is finite, (5.4.1) gives, for every $y \in Y$,

$$\dim(\mathcal{O}_{Y,y}) \leq \dim(\mathcal{O}_{X,f(y)}).$$

Consequently, if

$$T = f^{-1}(Z^{(2)}(X)),$$

then $T \supset Z^{(2)}(Y)$.

We claim that

$$\mathcal{G} = \mathcal{H}_{X/Z^{(2)}(X)}^0(f_*(\mathcal{O}_Y))$$

is coherent. The module $f_*(\mathcal{O}_Y)$ corresponds to B viewed as an A -module. Since B is a finite integral A -algebra and an integral domain, its field of fractions is finite over the field of fractions of A ; hence B is contained in a finite free A -module. By (5.9.2), (i), $\mathcal{G}$ is therefore a quasi-coherent $\mathcal{O}_X$ -submodule of

$$(\mathcal{H}_{X/Z^{(2)}(X)}^0(\mathcal{O}_X))^n$$

for a suitable n . The latter module is coherent by hypothesis, and hence so is $\mathcal{G}$.

By the definition (5.9.1.2), there is an isomorphism

$$\mathcal{G} \simeq f_*(\mathcal{H}_{Y/T}^0(\mathcal{O}_Y)).$$

It follows, in particular, that $\mathcal{H}_{Y/T}^0(\mathcal{O}_Y)$ is a coherent $\mathcal{O}_Y$ -module. Applying (5.10.10), (ii), to $\mathcal{O}_Y$ and the generic point of Y , we obtain

$$\operatorname{codim}(T, Y) \geq 2,$$

that is, $T \subset Z^{(2)}(Y)$. Thus $T = Z^{(2)}(Y)$. Formula (5.9.3.1) now identifies the corresponding algebra with $B^{(1)}$, proving that $B^{(1)}$ is finite over B . Finally, if $\mathfrak{q}$ has height 1, then $\mathfrak{q} \notin T$, so $\mathfrak{q} \cap A \notin Z^{(2)}(X)$. Incomparability for the integral extension $A \subset B$ excludes $\mathfrak{q} \cap A = (0)$; hence

$$\operatorname{ht}(\mathfrak{q} \cap A) = 1.$$

□

5.11. Coherence criteria for the modules $\mathcal{H}_{X/Z}^0(\mathcal{F})$

Proposition (5.11.1). — *Let X be a locally Noetherian prescheme, let Z be a subset of X stable under specialization, and let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -module. Denote by (x_α) the family of points of*

$$\text{Ass}(\mathcal{F}) \cap (X - Z),$$

and, for every α , let Y_α be the reduced closed subprescheme of X whose underlying space is $\overline{\{x_\alpha\}}$, and put $Z_\alpha = Z \cap \overline{\{x_\alpha\}}$. The following two conditions are equivalent:

- a) $\mathcal{H}_{X/Z}^0(\mathcal{F})$ is a coherent $\mathcal{O}_X$ -module.
- b) For every α , $\mathcal{H}_{Y_\alpha/Z_\alpha}^0(\mathcal{O}_{Y_\alpha})$ is a coherent $\mathcal{O}_{Y_\alpha}$ -module.

These conditions imply:

- c) For every α , $\text{codim}(Z_\alpha, Y_\alpha) \geq 2$.

Moreover, conditions a), b), and c) are equivalent if, in addition, either of the following properties holds:

- (i) *Every point of X has an open neighbourhood isomorphic to a subscheme of a regular scheme; one also says in this case that X is locally immersible in a regular scheme.*
- (ii) *For every α , Y_α is universally catenary (5.6.2), and its normalization $\tilde{Y}_\alpha$ (II, 6.3.8) is finite over Y_α .*

Proof. All the properties under consideration are local on X . We may therefore restrict ourselves to the case in which $X = \text{Spec}(A)$ is affine, A is Noetherian, and $\mathcal{F} = \tilde{M}$, where M is a finite A -module. For every α , let $h_\alpha : Y_\alpha \rightarrow X$ be the canonical injection. Then

$$(h_\alpha)_*(\mathcal{O}_{Y_\alpha}) = \mathcal{G}_\alpha$$

is the $\mathcal{O}_X$ -module corresponding to the quotient $A/\mathfrak{j}_{x_\alpha}$; by the definition of $\text{Ass}(\mathcal{F})$, this quotient is isomorphic to an A -submodule of M . Since $\mathcal{H}_{X/Z}^0(\mathcal{G}_\alpha)$ is a quasi-coherent $\mathcal{O}_X$ -submodule of $\mathcal{H}_{X/Z}^0(\mathcal{F})$ (5.9.2), condition a) implies that it is coherent. On the other hand, the definition (5.9.1.2) gives

$$\mathcal{H}_{X/Z}^0(\mathcal{G}_\alpha) \simeq (h_\alpha)_*(\mathcal{H}_{Y_\alpha/Z_\alpha}^0(\mathcal{O}_{Y_\alpha})).$$

Thus a) implies b).

To prove the converse, it is enough to find a finite filtration $(\mathcal{F}_i)_{0 \leq i \leq n}$ by coherent $\mathcal{O}_X$ -modules, with $\mathcal{F}_0 = \mathcal{F}$ and $\mathcal{F}_n = 0$, such that

$$\mathcal{H}_{X/Z}^0(\mathcal{F}_i/\mathcal{F}_{i+1})$$

is coherent for every i . Indeed, the assertion then follows by descending induction from the exact sequences

$$0 \longrightarrow \mathcal{F}_{i+1} \longrightarrow \mathcal{F}_i \longrightarrow \mathcal{F}_i/\mathcal{F}_{i+1},$$

the left exactness of $\mathcal{H}_{X/Z}^0$ (5.9.2), and (0I, 5.3.3) and (I, 6.1.1). In view of (3.2.8), it is therefore enough to treat the case in which $\mathcal{F}$ is irredundant, that is, $\text{Ass}(M) = \{\mathfrak{p}\}$ consists of one element. We use the following lemma.

Lemma (5.11.1.1). — *Let A be a Noetherian ring and let M be a finite A -module such that $\text{Ass}(M) = \{\mathfrak{p}\}$. There is a finite filtration $(M_i)_{0 \leq i \leq r}$ of M , with $M_0 = M$ and $M_r = 0$, such that every quotient M_i/M_{i+1} is isomorphic to a submodule of $A/\mathfrak{p}$.*

Proof. The canonical homomorphism $M \rightarrow M_{\mathfrak{p}} = N$ is injective (Bourbaki, *Algèbre commutative*, Chapter IV, §1, no. 2, Proposition 6). Put

$$B = A_{\mathfrak{p}}, \quad \mathfrak{m} = \mathfrak{p}A_{\mathfrak{p}}.$$

Then $\mathfrak{m}$ is the maximal ideal of B and $\text{Ass}(N) = \{\mathfrak{m}\}$ (*loc. cit.*, Proposition 5).

Since N is a finite B -module, there is an integer s such that $\mathfrak{m}^s N = 0$. The filtration by the powers $\mathfrak{m}^j N$ may be refined to a finite filtration (N_i) of N whose successive quotients are isomorphic to

$$B/\mathfrak{m} = \text{Frac}(A/\mathfrak{p}).$$

Intersecting it with M gives $M_i = M \cap N_i$. Each quotient M_i/M_{i+1} is then isomorphic to a finite $(A/\mathfrak{p})$ -submodule of $\text{Frac}(A/\mathfrak{p})$, and every such module is isomorphic to a submodule of $A/\mathfrak{p}$. $\square$

Completion of the proof of Proposition 5.11.1. The lemma and the preceding filtration argument reduce the proof, under condition b), to showing that $\mathcal{H}_{X/Z}^0(\mathcal{G})$ is coherent when

$$\mathcal{G} = (A/\mathfrak{p})^\sim$$

and $\mathfrak{p}$ is an associated prime of M . If $\mathfrak{p} = \mathfrak{j}_y$ with $y \in Z$, the support of $\mathcal{G}$ is a closed subset contained in Z , because Z is stable under specialization. Hence $\mathcal{H}_{X/Z}^0(\mathcal{G}) = 0$ by (5.9.1.2). Otherwise, $\mathfrak{p} = \mathfrak{j}_{x_\alpha}$ for some α , and

$$\mathcal{G} = (h_\alpha)_*(\mathcal{O}_{Y_\alpha}).$$

The displayed isomorphism above and condition b) then show that $\mathcal{H}_{X/Z}^0(\mathcal{G})$ is coherent. This proves the equivalence of a) and b).

Condition a) implies c) by (5.10.10, ii). It remains to prove, under either hypothesis (i) or (ii), that c) implies b). If X satisfies (i), each Y_α does as well. Consequently, in view of (5.9.3.1), the required assertion follows from Corollary 5.11.2 below. $\square$

Corollary (5.11.2). — *Let A be a Noetherian domain satisfying either of the following hypotheses:*

- (i) *A is a quotient of a regular Noetherian ring.*
- (ii) *A is universally catenary, and its integral closure A' is a finite A -algebra.*

Then

$$A^{(1)} = \bigcap_{\substack{\mathfrak{p} \in \text{Spec}(A) \\ \text{ht}(\mathfrak{p})=1}} A_{\mathfrak{p}}$$

is a finite A -algebra.

Proof. Suppose first that (i) holds, and put $X = \text{Spec}(A)$. The set U of points at which X satisfies (S_2) is open by (6.11.2).⁸ Put $Z = X - U$. Then $\text{codim}(Z, X) \geq 2$. Indeed, if $x \in X$ and $\dim(\mathcal{O}_x) \leq 1$, then $\dim(\mathcal{O}_{x'}) \leq 1$ for every generization x' of x . Since $\mathcal{O}_{x'}$ is a domain, it has positive depth whenever its dimension is positive, and therefore X satisfies (S_2) at x .

Thus $Z \subset Z^{(2)}(X)$. If $j : U \rightarrow X$ is the canonical injection, (5.9.1.2) gives

$$\mathcal{H}_{X/Z^{(2)}(X)}^0(\mathcal{O}_X) \simeq j_*(\mathcal{H}_{U/Z^{(2)}(U)}^0(\mathcal{O}_U)).$$

The prescheme U satisfies (S_2) , so (5.10.14) identifies the module inside the parentheses with $\mathcal{O}_U$. Since $\text{codim}(Z, X) \geq 2$, Chapter III, §9 shows that $j_*(\mathcal{O}_U)$ is coherent. Finally, (5.9.3.1) identifies the corresponding A -module with $A^{(1)}$, proving the assertion under (i).

Now suppose that (ii) holds. The ring A' is a Noetherian normal domain, hence it is the intersection of its local rings $A'_{\mathfrak{p}'}$ as $\mathfrak{p}'$ ranges over the height-one prime ideals of A' (Bourbaki, *Algèbre commutative*, Chapter VII, §1, no. 6, Theorem 4). For such a prime $\mathfrak{p}'$, put

$$\mathfrak{p} = \mathfrak{p}' \cap A, \quad S = A - \mathfrak{p}.$$

The ring $A'_{\mathfrak{p}'}$ is the localization of the finite $A_{\mathfrak{p}}$ -algebra $S^{-1}A'$ at $S^{-1}\mathfrak{p}'$. The latter prime lies above the maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$ and is therefore maximal. Moreover, $A_{\mathfrak{p}}$ is universally catenary by (5.6.3, i). Formula (5.6.10) consequently gives

$$\dim(A_{\mathfrak{p}}) = \dim(A'_{\mathfrak{p}'}) = 1.$$

It follows that $A^{(1)} \subset A'$. Since A' is a finite A -module and A is Noetherian, its submodule $A^{(1)}$ is finite as well. $\square$

Remark (5.11.3). — The conclusion that $A^{(1)}$ is a finite A -algebra need no longer hold if, in hypothesis (ii) of (5.11.2), one assumes only that A is catenary.

Indeed, consider the catenary local ring A constructed in (5.6.11); its integral closure A' (denoted there by B) is a finite A -algebra. If $A^{(1)}$ were also finite over A , then, being contained in the fraction field of A , it would be contained in A' . On the other hand, in the notation of (5.6.11), every height-one prime ideal of A is of the form $\mathfrak{p}A$, where $\mathfrak{p}$ is a height-one prime of C , and

$$A_{\mathfrak{p}A} = C_{\mathfrak{p}}.$$

⁸The reader may verify that Corollary 5.11.2 is not used in the proof of (6.11.2).

Moreover, $C_p = E_{p'}$, where p' is the unique prime of E above p . Thus A_{pA} is integrally closed and consequently contains A' . By definition this gives $A' \subset A^{(1)}$, so finiteness of $A^{(1)}$ would force $A^{(1)} = A'$. This is impossible: one of the two prime ideals of A' above the maximal ideal nA of A has height 1, whereas nA has height 2, contradicting (5.10.17, iv).

Corollary (5.11.4). — *Let X be a locally Noetherian prescheme, let Z be a closed subset of X , put $U = X - Z$, and let $i : U \rightarrow X$ be the canonical injection. Let $\mathcal{F}$ be a coherent $\mathcal{O}_U$ -module. A necessary condition for $i_*(\mathcal{F})$ to be a coherent $\mathcal{O}_X$ -module is that*

$$\text{codim}(\overline{\{x\}} \cap Z, \overline{\{x\}}) \geq 2 \quad \text{for every } x \in \text{Ass}(\mathcal{F}).$$

This condition is sufficient in either of the following cases:

- (i) X is locally immersible in a regular scheme.
- (ii) X is universally catenary and universally Japanese; by definition, the latter means that every point $x \in X$ has an affine open neighbourhood whose ring is universally Japanese (0, 23.1.1).

Proof. By (I, 9.4.7), there is a coherent $\mathcal{O}_X$ -submodule $\mathcal{G}$ of $i_*(\mathcal{F})$ such that $\mathcal{G}|_U = \mathcal{F}$. Plainly,

$$\text{Ass}(\mathcal{G}) \subset \text{Ass}(\mathcal{F}) \subset \text{Ass}(i_*(\mathcal{F})),$$

and

$$\text{Ass}(i_*(\mathcal{F})) = \text{Ass}(\mathcal{F})$$

by (3.1.13). Hence $\text{Ass}(\mathcal{G}) = \text{Ass}(\mathcal{F})$. It remains to apply (5.11.1) to $\mathcal{G}$, observing that

$$i_*(\mathcal{F}) = \mathcal{H}_{X/Z}^0(\mathcal{G}),$$

and that, when X is universally catenary and universally Japanese, hypothesis (ii) of (5.11.1) holds by definition. $\square$

Corollary (5.11.5). — *Let X be a locally Noetherian prescheme, let Z be a subset of X stable under specialization, and let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -module such that $\text{Ass}(\mathcal{F}) \subset X - Z$. Consider the conditions:*

- a) $\mathcal{H}_{X/Z}^0(\mathcal{F})$ is a coherent $\mathcal{O}_X$ -module.
- d) For every subset T of Z closed in X , $i_*(\mathcal{F}|_{X-T})$ is coherent, where $i : X - T \rightarrow X$ is the canonical injection. It is enough here to consider an increasing filtered family of such closed subsets whose union is Z .

Then a) implies d). If X satisfies either hypothesis (i) or (ii) of (5.11.1), the two conditions are equivalent.

Proof. The hypothesis and (3.1.13) give

$$\text{Ass}(i_*(\mathcal{F}|_{X-T})) = \text{Ass}(\mathcal{F}).$$

It follows from (5.10.2) that the canonical maps

$$i_*(\mathcal{F}|_{X-T}) \longrightarrow \mathcal{H}_{X/Z}^0(\mathcal{F})$$

are injective. Condition a) therefore implies d).

Conversely, condition d) and (5.11.1) imply, in the notation of that proposition, that

$$\text{codim}(T \cap Y_\alpha, Y_\alpha) \geq 2.$$

Since Z is the union of its subsets closed in X , it follows that

$$\text{codim}(Z \cap Y_\alpha, Y_\alpha) \geq 2.$$

The last assertion now follows from (5.11.1). $\square$

Corollary (5.11.6). — *Let A be a Noetherian ring and put $X = \text{Spec}(A)$. Consider the following properties:*

- a) For every quotient domain B of A , the ring $B^{(1)}$ (5.11.2) is a finite B -algebra.
- b) For every coherent $\mathcal{O}_X$ -module $\mathcal{F}$ and every subset Z of X stable under specialization such that

$$\text{codim}(\overline{\{x\}} \cap Z, \overline{\{x\}}) \geq 2$$

for every $x \in \text{Ass}(\mathcal{F}) \cap (X - Z)$, the module $\mathcal{H}_{X/Z}^0(\mathcal{F})$ is coherent.

c) For every closed subset T of X and every coherent $\mathcal{O}_U$ -module $\mathcal{G}$, where $U = X - T$, such that

$$\operatorname{codim}(\overline{\{x\}} \cap T, \overline{\{x\}}) \geq 2 \quad (x \in \operatorname{Ass}(\mathcal{G})),$$

the module $i_*(\mathcal{G})$ is coherent, where $i : U \rightarrow X$ is the canonical injection.

d) For every quotient domain B of A and every ideal $\mathfrak{J}$ of B of height at least 2, the ring

$$\bigcap_{\mathfrak{p} \not\supset \mathfrak{J}} B_{\mathfrak{p}}$$

is a finite B -algebra.

Then

$$a) \iff b) \implies c) \iff d).$$

Moreover, all four properties hold in either of the following cases:

- (i) A is a quotient of a regular ring.
- (ii) A is universally catenary and universally Japanese.

Proof. Let $B = A/\mathfrak{q}$, where $\mathfrak{q}$ is a prime ideal of A , so that $Y = \operatorname{Spec}(B) = V(\mathfrak{q})$ is a closed subset of X . Put $Z = Z^{(2)}(Y)$, viewed as a subset of X stable under specialization. Since B is a domain, $\operatorname{Ass}(A/\mathfrak{q})$ consists only of the generic point $\mathfrak{q}$ of Y . If b) holds, apply it to

$$\mathcal{F} = (A/\mathfrak{q})^\sim$$

and to Z . Formula (5.9.3.1) then shows that $B^{(1)}$ is a finite A -module, and hence a finite B -module. Thus b) implies a).

Conversely, suppose a) holds. Let $\mathcal{F}$ and Z satisfy the assumptions in b), and use the notation of (5.11.1). Each $Y_\alpha = \operatorname{Spec}(B_\alpha)$, where B_α is a quotient domain of A . Since $Z_\alpha \subset Z^{(2)}(Y_\alpha)$, condition a), (5.10.2), and the fact that $\operatorname{Ass}(\mathcal{O}_{Y_\alpha})$ consists of the generic point show that

$$\mathcal{H}_{Y_\alpha/Z_\alpha}^0(\mathcal{O}_{Y_\alpha})$$

is coherent. Property b) follows from (5.11.1).

To prove that c) implies d), argue as above with $\mathcal{G} = (A/\mathfrak{q})^\sim|_U$ and $T = V(\mathfrak{J})$. Conversely, d) implies c) by applying the same componentwise argument and (5.11.1). Property c) is plainly a special case of b). Finally, (5.11.2), together with the definitions of universally catenary and universally Japanese rings, proves all four properties under either hypothesis (i) or (ii). $\square$

Remarks (5.11.7). — (i) It is not known whether condition d) in (5.11.5) implies condition a) without an additional hypothesis on X . We shall see in (7.2.4) that it does when X is a local scheme. Likewise, when A is a Noetherian local ring, the four properties a), b), c), and d) of (5.11.6) are equivalent (7.2.4). It is not known whether this result extends to every Noetherian ring.

- (ii) If A satisfies property a) of (5.11.6), then so does every localization $S^{-1}A$ and every finite A -algebra C . Indeed, every quotient domain of $S^{-1}A$ has the form

$$S^{-1}(A/\mathfrak{q}),$$

where $\mathfrak{q}$ is a prime ideal of A disjoint from S . On the other hand, if $\mathfrak{r}$ is a prime ideal of C and $\mathfrak{p}$ is its inverse image in A , then $C/\mathfrak{r}$ is a finite integral $(A/\mathfrak{p})$ -algebra containing $A/\mathfrak{p}$. The assertions therefore follow from (5.10.17, iii) and (5.10.17, iv).

5.12. Relations between the properties of a Noetherian local ring A and of a quotient ring A/tA IV-2 | 126

We have already seen in (3.4) relations between the properties of A and those of A/tA concerning associated prime ideals, as well as the properties of being a domain or being reduced. In this subsection we give further relations between the properties of these rings, involving the notions of dimension and depth.

Proposition (5.12.1). — *Let A be a Noetherian local ring, put $X = \text{Spec}(A)$, and let t be an element of A belonging to a system of parameters (0, 16.3.6). Let X_0 be the closed subspace $V(t)$ of X , and let $X_1 = X - X_0$. Let Z be a subset of X stable under specialization. Suppose that X is biequidimensional (as is the case when A is equidimensional and is a quotient of a regular local ring (0, 16.5.12)). If $Z_0 = Z \cap X_0$ and $Z_1 = Z \cap X_1$, then*

$$(5.12.1.1) \quad \text{codim}(Z_0, X_0) \leq \text{codim}(Z, X) \leq \text{codim}(Z_1, X_1).$$

Proof. The second inequality follows from the definition (5.1.3). For the first, it is enough to treat the case in which Z is closed. Indeed, by hypothesis Z is a union of closed subsets Z_α , and inequalities

$$\text{codim}(X_0 \cap Z_\alpha, X_0) \leq \text{codim}(Z_\alpha, X)$$

for all α give

$$\begin{aligned} \text{codim}(Z_0, X_0) &= \inf_{\alpha} \text{codim}(X_0 \cap Z_\alpha, X_0) \\ &\leq \inf_{\alpha} \text{codim}(Z_\alpha, X) = \text{codim}(Z, X). \end{aligned}$$

Suppose therefore that Z is closed. Then

$$\text{codim}(Z, X) = \dim(X) - \dim(Z)$$

by (0, 14.3.5.1).

Moreover, X_0 is catenary and equidimensional, and $\dim(X_0) = \dim(X) - 1$ by (0, 16.3.4); hence IV-2 | 127

$$\text{codim}(Z_0, X_0) = \dim(X) - 1 - \dim(Z_0).$$

Since $\dim(Z_0) \geq \dim(Z) - 1$ (0, 16.3.4), the first inequality in (5.12.1.1) follows. $\square$

Proposition (5.12.2). — *Let A be a catenary Noetherian local ring, let M be a finite A -module, let t be an M -regular element belonging to the maximal ideal $\mathfrak{m}$ of A , and let $k \geq 1$. If M/tM is equidimensional and satisfies (S_k) , then the same is true of M .*

Proof. In view of (Err_{III}, 30) and (5.7.3, vi), we may reduce to the case

$$\text{Supp}(M) = \text{Spec}(A) = X.$$

Put $X_0 = V(t) = \text{Spec}(A/tA)$. Since M/tM satisfies (S_1) , it has no embedded associated prime ideals (5.7.5). Applying (3.4.4) at the closed point of X (and of every closed subscheme of X), we see that the irreducible components of $\text{Supp}(M/tM)$ are exactly the irreducible components of $Y_i \cap X_0$, where Y_i ($1 \leq i \leq r$) are the irreducible components of X .

Because t is M -regular, $V(t)$ contains no maximal point of X (3.1.8). Every irreducible component of $Y_i \cap X_0$ therefore has codimension 1 in Y_i (5.1.8). Since X is catenary and all irreducible components of $\text{Supp}(M/tM)$ have the same dimension, the Y_i all have the same dimension. Thus X is equidimensional, and hence biequidimensional because A is local and catenary.

It remains to prove (S_k) . By the criterion (5.7.4), with its notation, we must show that

$$\text{codim}(Z_n, X) > n + k$$

for every integer $n \geq 0$. The hypothesis on M/tM and the same criterion give

$$\text{codim}(Z_n \cap X_0, X_0) > n + k.$$

Every specialization of a point of Z_n again belongs to Z_n , as will be proved in (6.11.5).⁹ Finally, t belongs to a system of parameters for M (0, 16.4.1), and hence also to one for A , because the annihilator of M is nilpotent under the hypothesis $\text{Supp}(M) = \text{Spec}(A)$. The required inequality now follows from (5.12.1). $\square$

⁹The reader may verify that Proposition 5.12.2 is not used in the proof of (6.11.5).

Remark (5.12.3). — If in Proposition 5.12.2 one assumes only that M/tM is equidimensional, it need not follow that M is equidimensional. For example, let k be a field, let $B = k[T, U, V]$, let C be the local ring of B at the maximal ideal $BT + BU + BV$, and put

$$\mathfrak{p} = CU, \quad \mathfrak{q} = CV + C(T + U), \quad A = C/\mathfrak{p}\mathfrak{q}.$$

Geometrically, A is the local ring, at their intersection point x , of the closed subscheme of affine 3-space formed by a plane and a line meeting that plane at x . Let t, u, v be the images of T, U, V in A . The element t is not a zero divisor in A , and

$$A/tA \simeq C_0/\mathfrak{p}_0\mathfrak{q}_0,$$

where C_0 is the local ring of $B_0 = k[U, V]$ at the maximal ideal $B_0U + B_0V$, while

$$\mathfrak{p}_0 = C_0U, \quad \mathfrak{q}_0 = C_0U + C_0V.$$

Thus $\mathfrak{q}_0$ is the maximal ideal of C_0 . It is immediate that $\text{Spec}(A/tA)$ is irreducible of dimension 1, that A/tA does not satisfy (S_1) , and that $\text{Spec}(A)$ is not equidimensional.

Corollary (5.12.4). — *Let A be a catenary Noetherian local ring, let t be a regular element belonging to the maximal ideal of A , and let $k \geq 1$. If A/tA satisfies (S_k) , then so does A .*

Proof. For $k = 1$ this follows from (3.4.4), in view of the interpretation of (S_1) in (5.7.5). For $k \geq 2$, Hartshorne's criterion (5.10.9) shows that A/tA is equidimensional, and Proposition 5.12.2 applies. $\square$

Proposition (5.12.5). — *Let A be a catenary Noetherian local ring, let t be a regular element of A belonging to its maximal ideal, and let $k \geq 0$. If A/tA is reduced and equidimensional and satisfies (R_k) , then A has these same three properties.*

Proof. We already know from (3.4.6) that A is reduced, and Proposition 5.12.2, applied with $k = 1$, shows that A is equidimensional. Put

$$X = \text{Spec}(A), \quad X_0 = V(t) = \text{Spec}(A/tA),$$

and let Z be the set of points at which X is not regular. Every specialization of a point of Z belongs to Z (0, 17.3.2). On the other hand, if $x \in X_0$ and X_0 is regular at x , then X is regular at x (0, 17.1.8). Consequently the nonregular locus Z'_0 of X_0 contains $Z_0 = Z \cap X_0$. The hypothesis gives

$$k < \text{codim}(Z'_0, X_0) \leq \text{codim}(Z_0, X_0).$$

Since A is equidimensional and catenary, Proposition 5.12.1 gives

$$\text{codim}(Z_0, X_0) \leq \text{codim}(Z, X).$$

Thus X satisfies (R_k) . $\square$

Remark (5.12.6). — If in Proposition 5.12.5 one assumes only that A/tA is reduced and satisfies (R_k) , it need not follow that A satisfies (R_k) . For example, let k be a field and let $P(U, V) \in k[U, V]$ be irreducible, with the curve Γ defined by (P) in $\text{Spec}(k[U, V])$ singular at the maximal ideal $(U) + (V)$. One may take, for example,

$$P(U, V) = U(U^2 + V^2) + (U^2 - V^2),$$

which defines a cubic with a double point. Let $B = k[T, U, V, W]$, let C be the local ring of B at the maximal ideal $BT + BU + BV + BW$, and put

$$\mathfrak{p} = CW + CP(U - T, V), \quad \mathfrak{q} = CU.$$

Consider the local ring $A = C/\mathfrak{p}\mathfrak{q}$. Geometrically, the corresponding closed subscheme X of affine 4-space is the union of a hyperplane H and a 2-dimensional “cylinder” L with base Γ ; the “singular line” Y of L is not contained in H , and A is the local ring of X at the point $x = Y \cap H$.

Let t be the image of T in A . The scheme $\text{Spec}(A/tA)$ has x as its only singular point; its local ring there is A/tA , which is reduced and has dimension 2. By contrast, the generic point y of Y , defined by the image in A of

$$C(U - T) + CV + CW,$$

is a singular point of X , and $\mathcal{O}_{X,y}$ has dimension 1. Thus A/tA is reduced and satisfies (R_1) , whereas A does not satisfy (R_1) .

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Corollary (5.12.7). — *Let A be a catenary Noetherian local ring and let t be a regular element of A belonging to its maximal ideal. If A/tA is a domain and is integrally closed, then so is A .*

Proof. By Serre's criterion (5.8.6) and the fact that A is local, it is enough to prove that $\text{Spec}(A)$ satisfies (S_2) and (R_1) . The hypothesis on A/tA implies (R_1) for A by Proposition 5.12.5 and (S_2) for A by Corollary 5.12.4. $\square$

Proposition (5.12.8). — (Hironaka's lemma). *Let A be a reduced, equidimensional, catenary Noetherian local ring. Suppose in addition that, for every minimal prime ideal $\mathfrak{q}_i$ of A , the ring $B_i = A/\mathfrak{q}_i$ has the property that $B_i^{(1)}$ (notation of (5.10.17)) is a finite B_i -algebra. This last condition holds, for example, when A has either property (i) or property (ii) of (5.11.6).*

Let $t \in A$ be a non-zero-divisor satisfying the following conditions:

- (i) The A -module A/tA has a unique nonembedded associated prime ideal $\mathfrak{p}$. (It necessarily has height 1 by the Hauptidealsatz (0, 16.3.2), but it is not assumed to be the only associated prime of A/tA .)
- (ii) The ideal $\mathfrak{p}A_{\mathfrak{p}}$ is generated by $t/1$.
- (iii) The ring $A/\mathfrak{p}$ is integrally closed.

Then A is a domain and is integrally closed, and $\mathfrak{p} = tA$.

Proof. Put $X = \text{Spec}(A)$ and $Z = V(tA)$. Condition (i) says that Z is irreducible, with generic point corresponding to $\mathfrak{p}$. By (ii), the maximal ideal of the Noetherian local ring $A_{\mathfrak{p}}$ is generated by one element. Thus $A_{\mathfrak{p}}$ is a discrete valuation ring with uniformizer $t/1$ (Bourbaki, *Algèbre commutative*, chap. VI, §3, no. 6, prop. 9), and $A_{\mathfrak{p}}/tA_{\mathfrak{p}}$, its residue field, has length 1 as an $A_{\mathfrak{p}}$ -module.

It follows from (3.4.2) that Z is contained in only one irreducible component, say X_i , of X . For every other irreducible component X_j one consequently has

$$\dim(X_j \cap Z) < \dim(Z) = \dim(X_i \cap Z).$$

Since t is not a zero divisor, it belongs to none of the minimal prime ideals $\mathfrak{q}_j$, and hence

$$\text{codim}(X_i \cap Z, X_i) = \text{codim}(X_j \cap Z, X_j) = 1$$

for every $j \neq i$ (5.1.8). But A is biequidimensional; the displayed strict inequality would therefore force $\dim(X_i) \neq \dim(X_j)$ (0, 14.3.3.1), a contradiction. Hence A has only one minimal prime. Since it is reduced, A is a domain.

Let $A^{(1)}$ have the meaning of (5.10.17). It is a finite A -algebra and therefore a Noetherian semilocal ring. Every nonembedded associated prime $\mathfrak{r}$ of $A^{(1)}/tA^{(1)}$ has height 1 by the Hauptidealsatz. Its contraction $\mathfrak{r} \cap A$ also has height 1 (5.10.17, iv), and, since it contains tA , condition (i) forces

$$\mathfrak{r} \cap A = \mathfrak{p}.$$

Let A' be the integral closure of A and put $S = A - \mathfrak{p}$. The integral closure of $A_{\mathfrak{p}}$ is $S^{-1}A'$ (Bourbaki, *Algèbre commutative*, chap. V, §1, no. 5, prop. 17); it equals $A_{\mathfrak{p}}$ because this ring is a discrete valuation ring. Consequently there is only one prime $\mathfrak{p}'$ of A' above $\mathfrak{p}$ (*loc. cit.*, §2, no. 1, lemma 1), and therefore only one prime $\mathfrak{p}^{(1)}$ of $A^{(1)}$ above $\mathfrak{p}$. Moreover,

$$A_{\mathfrak{p}} = A'_{\mathfrak{p}'} = A_{\mathfrak{p}^{(1)}}^{(1)}.$$

The ring $A^{(1)}$ satisfies (S_2) (5.10.17, i). Since t is a non-zero-divisor on $A^{(1)}$, the quotient

$$A^{(1)}/tA^{(1)}$$

has no embedded associated primes (5.7.7). Thus $\mathfrak{p}^{(1)}$ is its only associated prime, so $tA^{(1)}$ is $\mathfrak{p}^{(1)}$ -primary. Equivalently, $tA^{(1)}$ is the inverse image in $A^{(1)}$ of

$$tA_{\mathfrak{p}^{(1)}}^{(1)} = tA_{\mathfrak{p}},$$

which is the maximal ideal of $A_{\mathfrak{p}}$ by (ii). It follows that

$$tA^{(1)} = \mathfrak{p}^{(1)}$$

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is prime in $A^{(1)}$.

The domain $A^{(1)}/\mathfrak{p}^{(1)}$ is finite over $A/\mathfrak{p}$ and has the same fraction field, namely the residue field of $A_{\mathfrak{p}}^{(1)} = A_{\mathfrak{p}}$. Condition (iii) therefore gives

$$A^{(1)}/\mathfrak{p}^{(1)} = A/\mathfrak{p}.$$

Hence

$$A^{(1)} = A + \mathfrak{p}^{(1)} = A + tA^{(1)}.$$

Since t lies in the Jacobson radical of the Noetherian semilocal ring $A^{(1)}$, Nakayama's lemma yields $A^{(1)} = A$. Thus

$$\mathfrak{p}^{(1)} = \mathfrak{p} = tA.$$

Finally, A is catenary and $A/tA = A/\mathfrak{p}$ is a domain and integrally closed by (iii). Corollary 5.12.7 shows that A is integrally closed. $\square$

Remark (5.12.9). — The conclusion of Hironaka's lemma fails without the hypothesis that A is equidimensional. Let k be a field, put

$$B = k[X, Y, Z], \quad \mathfrak{r}_1 = BZ, \quad \mathfrak{r}_2 = BX + BY, \quad C = B/\mathfrak{r}_1\mathfrak{r}_2.$$

Let $\mathfrak{m} = BX + BY + BZ$, let $\mathfrak{n} = \mathfrak{m}/\mathfrak{r}_1\mathfrak{r}_2$ be its image in C , and let $A = C_{\mathfrak{n}}$. Finally, let t_0 be the image of $Y + Z$ in C , and let t be its image in A . Thus $\text{Spec}(C)$ is the union of a plane P and a line D not contained in P , while $\text{Spec}(C/t_0C)$ is a line D' contained in P and passing through $D \cap P$.

The ring A is reduced and catenary, since it is a quotient of a regular ring, and its minimal prime ideals $\mathfrak{q}_1$ and $\mathfrak{q}_2$ are the images of $\mathfrak{r}_1$ and $\mathfrak{r}_2$. The A -module A/tA has a unique nonembedded associated prime $\mathfrak{p}$, the image of $BY + BZ$; the ideal $\mathfrak{p}A_{\mathfrak{p}}$ is generated by $t/1$, and $A/\mathfrak{p} \simeq k[X]$. Nevertheless, A is not a domain.

Corollary (5.12.10). — *Let A be a Noetherian local domain satisfying one of the following hypotheses:*

- a) *A is a quotient of a regular ring;*
- b) *A is universally catenary and universally Japanese.*

Let $(x_i)_{1 \leq i \leq n}$ be a sequence of n elements of A , and put

$$\mathfrak{J} = x_1A + \cdots + x_nA.$$

Suppose that:

- (i) *$A/\mathfrak{J}$ has a unique nonembedded associated prime $\mathfrak{p}$, and $\mathfrak{p}$ has height n ;*
- (ii) *the maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$ is generated by the $x_i/1$ (so $A_{\mathfrak{p}}$ is a regular local ring of dimension n);*
- (iii) *$A/\mathfrak{p}$ is integrally closed.*

Then, for every i with $0 \leq i \leq n$, the quotient

$$A/(x_1A + \cdots + x_iA)$$

is integrally closed and has dimension $\dim(A) - i$. In particular, $\mathfrak{J}$ is prime and equals $\mathfrak{p}$, the ring A is integrally closed, and $(x_i)_{1 \leq i \leq n}$ is an A -regular sequence.

Proof. We argue by induction on n . The assertion is trivial for $n = 0$ and is Hironaka's lemma for $n = 1$, so suppose $n \geq 2$. Put

$$\mathfrak{J}' = x_1A + \cdots + x_{n-1}A,$$

and let $\mathfrak{q}$ be a prime minimal among those containing $\mathfrak{J}'$. Then $\text{ht}(\mathfrak{q}) \leq n - 1$ (0, 16.3.1). We first show that $\mathfrak{q} \subset \mathfrak{p}$. If $\mathfrak{p}'$ is minimal among the primes containing $\mathfrak{q} + x_nA$, then $\mathfrak{p}'/\mathfrak{q}$ has height at most 1 by the Hauptidealsatz (0, 16.3.2); because A is catenary, $\text{ht}(\mathfrak{p}') \leq n$. But $\mathfrak{p}'$ contains $\mathfrak{J}$, and the unique minimal prime of $A/\mathfrak{J}$ is $\mathfrak{p}$. Hence $\mathfrak{p}' = \mathfrak{p}$, and $\mathfrak{q} \subset \mathfrak{p}$.

The prime $\mathfrak{q}$ is therefore induced from a prime of $A_{\mathfrak{p}}$ minimal over $\mathfrak{J}'A_{\mathfrak{p}}$. By (ii) and (0, 17.1.7), **IV-2** | 131 the ideal $\mathfrak{J}'A_{\mathfrak{p}}$ is prime. Consequently

$$\mathfrak{q} = A \cap \mathfrak{J}'A_{\mathfrak{p}}$$

is the unique nonembedded associated prime of $A/\mathfrak{J}'$, and has height $n - 1$. Moreover,

$$\mathfrak{q}A_{\mathfrak{q}} = \mathfrak{J}'A_{\mathfrak{q}},$$

and, since $A_q = (A_p)_{qA_p}$, the maximal ideal of A_q is generated by the images of $x_1, \dots, x_{n-1}$. Thus the shortened sequence already satisfies conditions (i) and (ii).

We verify condition (iii) for it. Put $B = A/q$, let $\mathfrak{n} = \mathfrak{p}/q$, and let t be the image of x_n in B . Property a) (respectively b)) passes from A to B (5.6.3, iv), and $t \neq 0$ because $x_n \notin q$. A prime of B minimal over t comes from a prime of A minimal over $q + x_n A$; the only such prime is $\mathfrak{p}$. Hence $\mathfrak{n}$ is the unique nonembedded associated prime of B/tB . Furthermore,

$$B_{\mathfrak{n}} = A_{\mathfrak{p}}/qA_{\mathfrak{p}}, \quad \mathfrak{n}B_{\mathfrak{n}} = \mathfrak{p}A_{\mathfrak{p}}/qA_{\mathfrak{p}}.$$

Since $qA_{\mathfrak{p}} = \mathfrak{J}'A_{\mathfrak{p}}$ and $\mathfrak{p}A_{\mathfrak{p}}$ is generated by $x_1, \dots, x_n$, the ideal $\mathfrak{n}B_{\mathfrak{n}}$ is generated by t . Finally, $B/\mathfrak{n} = A/\mathfrak{p}$ is integrally closed. Hironaka's lemma therefore gives

$$B = A/q \text{ integrally closed,} \quad \mathfrak{p} = q + x_n A.$$

The induction hypothesis applied to $x_1, \dots, x_{n-1}$ now shows that $A/(x_1 A + \dots + x_i A)$ is integrally closed and has dimension $\dim(A) - i$ for $0 \leq i \leq n-1$, and that $q = \mathfrak{J}'$. Hence

$$\mathfrak{p} = \mathfrak{J}' + x_n A = \mathfrak{J},$$

so $A/\mathfrak{J} = A/\mathfrak{p}$ is integrally closed. Finally, $\dim(A_{\mathfrak{p}}) = n$, and biequidimensionality of A gives

$$\dim(A/\mathfrak{p}) = \dim(A) - \dim(A_{\mathfrak{p}}) = \dim(A) - n.$$

This completes the induction. □

5.13. Properties preserved under passage to a direct limit Throughout this subsection, $(A_\alpha, \varphi_{\beta\alpha})$ IV-2 | 131 denotes a direct system of rings whose index set is filtered and preordered. We put

$$A = \varinjlim_{\alpha} A_{\alpha}$$

and denote by $\varphi_{\alpha} : A_{\alpha} \rightarrow A$ the canonical homomorphism.

Proposition (5.13.1). — (i) For every index α , let $\mathfrak{a}_{\alpha}$ be an ideal of A_{α} such that

$$\varphi_{\beta\alpha}(\mathfrak{a}_{\alpha}) \subset \mathfrak{a}_{\beta} \quad (\alpha \leq \beta).$$

Then the $\mathfrak{a}_{\alpha}$, with the restrictions of the $\varphi_{\beta\alpha}$, form a direct system; $\mathfrak{a} = \varinjlim_{\alpha} \mathfrak{a}_{\alpha}$ is canonically identified with an ideal of A , and

$$A/\mathfrak{a} \simeq \varinjlim_{\alpha} (A_{\alpha}/\mathfrak{a}_{\alpha}).$$

If, in addition,

$$\mathfrak{a}_{\alpha} = \varphi_{\beta\alpha}^{-1}(\mathfrak{a}_{\beta}) \quad (\alpha \leq \beta),$$

then

$$\mathfrak{a}_{\alpha} = \varphi_{\alpha}^{-1}(\mathfrak{a}) \quad \text{for every } \alpha.$$

(ii) Conversely, for every ideal $\mathfrak{a}$ of A , if

$$\mathfrak{a}_{\alpha} = \varphi_{\alpha}^{-1}(\mathfrak{a}) \quad \text{for every } \alpha,$$

then the $\mathfrak{a}_{\alpha}$ form a direct system and

$$\mathfrak{a} = \varinjlim_{\alpha} \mathfrak{a}_{\alpha}.$$

Proof. Assertion (ii) is plainly a special case of (i). In (i), the canonical identification of $\mathfrak{a}$ with an ideal of A , and that of $A/\mathfrak{a}$ with $\varinjlim_{\alpha} (A_{\alpha}/\mathfrak{a}_{\alpha})$, follow from the exactness of the direct-limit functor in the category of abelian groups.

Moreover, $\mathfrak{a}$ is the union of the increasing family of images $\varphi_{\alpha}(\mathfrak{a}_{\alpha})$. Thus, if $x_{\alpha} \in A_{\alpha}$ and $\varphi_{\alpha}(x_{\alpha}) \in \mathfrak{a}$, there are $\beta \geq \alpha$ and $y_{\beta} \in \mathfrak{a}_{\beta}$ such that

$$\varphi_{\alpha}(x_{\alpha}) = \varphi_{\beta}(y_{\beta}).$$

For some $\gamma \geq \beta$ one then has

$$\varphi_{\gamma\alpha}(x_{\alpha}) = \varphi_{\gamma\beta}(y_{\beta}) \in \mathfrak{a}_{\gamma}.$$

Under the additional inverse-image hypothesis this implies $x_{\alpha} \in \mathfrak{a}_{\alpha}$, and proves the last assertion of (i). □

Corollary (5.13.2). — Let $\mathfrak{N}_\alpha$ be the nilradical of A_α . The nilradical of A is canonically identified with $\varinjlim \mathfrak{N}_\alpha$, and

$$A_{\text{red}} \simeq \varinjlim_{\alpha} (A_\alpha)_{\text{red}}.$$

In particular, if every A_α is reduced, then A is reduced.

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Proof. It is clear that

$$\varphi_{\beta\alpha}(\mathfrak{N}_\alpha) \subset \mathfrak{N}_\beta \quad (\alpha \leq \beta),$$

and that $\varinjlim \mathfrak{N}_\alpha$ is a nil ideal of A , hence is contained in the nilradical $\mathfrak{N}$ of A . Conversely, if $x \in \mathfrak{N}$, then $x^h = 0$ for some integer h . Write $x = \varphi_\alpha(x_\alpha)$ with $x_\alpha \in A_\alpha$. The relation $\varphi_\alpha(x_\alpha^h) = 0$ implies that for some $\beta \geq \alpha$,

$$\varphi_{\beta\alpha}(x_\alpha^h) = 0.$$

Putting $x_\beta = \varphi_{\beta\alpha}(x_\alpha)$, we obtain $x_\beta^h = 0$, hence $x_\beta \in \mathfrak{N}_\beta$, while $x = \varphi_\beta(x_\beta)$. This proves the assertion, in view of (5.13.1). $\square$

Proposition (5.13.3). — (i) For every index α , let $\mathfrak{p}_\alpha$ be a prime ideal of A_α such that

$$\varphi_{\beta\alpha}(\mathfrak{p}_\alpha) \subset \mathfrak{p}_\beta \quad (\alpha \leq \beta).$$

Then $\mathfrak{p} = \varinjlim \mathfrak{p}_\alpha$ is a prime ideal of A . In particular, if all the A_α are domains, then A is a domain.

(ii) If, in addition,

$$\mathfrak{p}_\alpha = \varphi_{\beta\alpha}^{-1}(\mathfrak{p}_\beta) \quad (\alpha \leq \beta),$$

then there is a canonical identification

$$A_{\mathfrak{p}} \simeq \varinjlim_{\alpha} (A_\alpha)_{\mathfrak{p}_\alpha}.$$

(iii) Suppose that A is local, with maximal ideal $\mathfrak{p}$, and that the homomorphisms $\varphi_\alpha : A_\alpha \rightarrow A$ are injective. If $\mathfrak{p}_\alpha = \varphi_\alpha^{-1}(\mathfrak{p})$, then

$$A \simeq \varinjlim_{\alpha} (A_\alpha)_{\mathfrak{p}_\alpha},$$

and every canonical homomorphism $(A_\alpha)_{\mathfrak{p}_\alpha} \rightarrow A$ is injective.

Proof. The two assertions in (i) are equivalent, since

$$A/\mathfrak{p} \simeq \varinjlim_{\alpha} (A_\alpha/\mathfrak{p}_\alpha).$$

Suppose therefore that all the A_α are domains, and let $x, y \in A$ satisfy $xy = 0$. For some α one may write

$$x = \varphi_\alpha(x_\alpha), \quad y = \varphi_\alpha(y_\alpha),$$

with $x_\alpha, y_\alpha \in A_\alpha$ and $\varphi_\alpha(x_\alpha y_\alpha) = 0$. Hence there is $\beta \geq \alpha$ such that, on putting

$$x_\beta = \varphi_{\beta\alpha}(x_\alpha), \quad y_\beta = \varphi_{\beta\alpha}(y_\alpha),$$

one has $x_\beta y_\beta = 0$. Since A_β is a domain, $x_\beta = 0$ or $y_\beta = 0$, and therefore $x = 0$ or $y = 0$.

For (ii), the hypothesis says that the multiplicative subsets $A_\alpha - \mathfrak{p}_\alpha$ form a direct system whose direct limit is $A - \mathfrak{p}$. The local rings $(A_\alpha)_{\mathfrak{p}_\alpha}$ therefore form a direct system with local transition maps, and the canonical homomorphisms to $A_{\mathfrak{p}}$ induce a surjection

$$\psi : \varinjlim_{\alpha} (A_\alpha)_{\mathfrak{p}_\alpha} \longrightarrow A_{\mathfrak{p}}.$$

To prove injectivity, let $x_\alpha \in A_\alpha$ and $s_\alpha \in A_\alpha - \mathfrak{p}_\alpha$, and suppose that

$$\frac{\varphi_\alpha(x_\alpha)}{\varphi_\alpha(s_\alpha)} = 0 \quad \text{in } A_{\mathfrak{p}}.$$

There is $s' \in A - \mathfrak{p}$ such that $s' \varphi_\alpha(x_\alpha) = 0$. After replacing α by a larger index, we may write $s' = \varphi_\alpha(s'_\alpha)$ with $s'_\alpha \in A_\alpha - \mathfrak{p}_\alpha$. Then $\varphi_\alpha(s'_\alpha x_\alpha) = 0$, so for some $\beta \geq \alpha$,

$$\varphi_{\beta\alpha}(s'_\alpha x_\alpha) = 0 \quad \text{in } A_\beta.$$

It follows that the image of x_α/s_α in the direct limit is zero.

Finally, the first assertion of (iii) follows from (ii) and $A_{\mathfrak{p}} = A$. Every element of $A_{\alpha} - \mathfrak{p}_{\alpha}$ maps to a unit of A , and the injectivity of $(A_{\alpha})_{\mathfrak{p}_{\alpha}} \rightarrow A$ follows from that of φ_{α} . $\square$

Corollary (5.13.4). — Suppose that all the A_{α} are domains and that all the transition homomorphisms $\varphi_{\beta\alpha}$ are injective. If A'_{α} denotes the integral closure of A_{α} , then

$$A' = \varinjlim_{\alpha} A'_{\alpha}$$

is the integral closure of A . In particular, if every A_{α} is integrally closed, then A is integrally closed.

If the A_{α} are unibranch local rings (respectively, geometrically unibranch local rings) (0, 23.2.1), and the $\varphi_{\beta\alpha}$ are injective local homomorphisms, then A is unibranch (respectively, geometrically unibranch).

Proof. Let K_{α} be the fraction field of A_{α} . By (5.13.3, ii), applied to $\mathfrak{p}_{\alpha} = (0)$,

$$K = \varinjlim_{\alpha} K_{\alpha}$$

is the fraction field of A . Thus $A' = \varinjlim_{\alpha} A'_{\alpha}$ is identified with a subring of K which is plainly integral over A .

Conversely, let $z \in K$ satisfy an equation of integral dependence

$$z^n + \sum_{i=1}^n c_i z^{n-i} = 0, \quad c_i \in A.$$

There are an index α , an element $z_{\alpha} \in K_{\alpha}$, and elements $c_{i\alpha} \in A_{\alpha}$ whose images are z and the c_i , respectively, and such that the same equation holds in K_{α} . Hence $z_{\alpha} \in A'_{\alpha}$, so $z \in A'$. This also proves at once that a direct limit of the integrally closed rings occurring here is integrally closed.

Suppose now that the A_{α} are local and that the transition maps are local. Then A is local and, if k_{α} denotes the residue field of A_{α} ,

$$k = \varinjlim_{\alpha} k_{\alpha}$$

is the residue field of A (0, 10.3.1.3). If every A'_{α} is local, then A' is local, and hence A is unibranch. If, moreover, the residue field k'_{α} of A'_{α} is a radical extension of k_{α} for every α , then

$$k' = \varinjlim_{\alpha} k'_{\alpha},$$

the residue field of A' , is a radical extension of k . Thus A is geometrically unibranch. $\square$

(5.13.5). A ring A is called *normal* if $\text{Spec}(A)$ is normal; in other words, if for every prime ideal $\mathfrak{p}$ of A , the ring $A_{\mathfrak{p}}$ is a domain and is integrally closed.

This implies that A is reduced and that $\mathfrak{p}$ can contain only one minimal prime ideal $\mathfrak{q}$ of A ; in other words, $\mathfrak{p}$ belongs to only one irreducible component of $\text{Spec}(A)$. Moreover, $\mathfrak{q}$ is the inverse image of (0) under the canonical homomorphism $A \rightarrow A_{\mathfrak{p}}$, and hence

$$A_{\mathfrak{p}} \simeq (A/\mathfrak{q})_{\mathfrak{p}/\mathfrak{q}}.$$

It follows that the domain $A/\mathfrak{q}$ is integrally closed, being the intersection of integrally closed rings (I, 8.2.1.1).

If A is Noetherian and normal, it is the direct product of finitely many integrally closed domains (I, 5.1.4).

Corollary (5.13.6). — Suppose that all the A_{α} are normal and that, for $\alpha \leq \beta$, every irreducible component of $\text{Spec}(A_{\beta})$ dominates an irreducible component of $\text{Spec}(A_{\alpha})$. Then

$$A = \varinjlim_{\alpha} A_{\alpha}$$

is normal.

Proof. Let $\mathfrak{p}$ be a prime ideal of A , and put

$$\mathfrak{p}_{\alpha} = \varphi_{\alpha}^{-1}(\mathfrak{p})$$

for every α . By hypothesis, $\mathfrak{p}_{\alpha}$ contains a unique minimal prime ideal $\mathfrak{q}_{\alpha}$ of A_{α} . If $\alpha \leq \beta$, then $\varphi_{\beta\alpha}^{-1}(\mathfrak{q}_{\beta})$ is a minimal prime ideal of A_{α} contained in $\mathfrak{p}_{\alpha}$, hence equals $\mathfrak{q}_{\alpha}$.

Put

$$B_\alpha = A_\alpha / \mathfrak{q}_\alpha.$$

The domains B_α form a direct system under the injective homomorphisms

$$\varphi'_{\beta\alpha} : B_\alpha \longrightarrow B_\beta$$

induced by the $\varphi_{\beta\alpha}$. Each B_α is integrally closed, so

$$B = \varinjlim_\alpha B_\alpha$$

is integrally closed by (5.13.4). If

$$\mathfrak{p}'_\alpha = \mathfrak{p}_\alpha / \mathfrak{q}_\alpha,$$

then

$$(B_\alpha)_{\mathfrak{p}'_\alpha} = (A_\alpha)_{\mathfrak{p}_\alpha}.$$

Consequently,

$$\varinjlim_\alpha (B_\alpha)_{\mathfrak{p}'_\alpha} = \varinjlim_\alpha (A_\alpha)_{\mathfrak{p}_\alpha} = A_{\mathfrak{p}}$$

is a localization of B at a prime ideal (5.13.3). It is therefore a domain and is integrally closed, proving the corollary. $\square$

Proposition (5.13.7). — *Suppose that every A_α is regular and that, for $\alpha \leq \beta$, the homomorphism $\varphi_{\beta\alpha}$ makes A_β a flat A_α -module. Suppose in addition that*

$$A = \varinjlim_\alpha A_\alpha$$

is Noetherian. Then A is regular.

Proof. By definition, it is enough to prove that the Noetherian local ring $A_{\mathfrak{p}}$ is regular for every prime ideal $\mathfrak{p}$ of A . Put

$$\mathfrak{p}_\alpha = \varphi_\alpha^{-1}(\mathfrak{p}).$$

By (5.13.3),

$$\mathfrak{p} = \varinjlim_\alpha \mathfrak{p}_\alpha, \quad A_{\mathfrak{p}} = \varinjlim_\alpha (A_\alpha)_{\mathfrak{p}_\alpha}.$$

The rings $(A_\alpha)_{\mathfrak{p}_\alpha}$ are regular, and for $\alpha \leq \beta$ the ring $(A_\beta)_{\mathfrak{p}_\beta}$ is flat over $(A_\alpha)_{\mathfrak{p}_\alpha}$ (0, 6.3.2). We are therefore reduced to the case in which all the A_α , as well as A , are local.

To prove that A is regular, it is enough to show that for any two finite A -modules M and N , there is an integer i_0 such that

$$\mathrm{Tor}_i^A(M, N) = 0 \quad (i \geq i_0)$$

(0, 17.3.1) and (0, 17.2.6). We use the following lemma, in which the rings A_α are again arbitrary. $\square$

Lemma (5.13.7.1). — *For every finitely presented A -module M , there are an index α and a finitely presented A_α -module M_α such that*

$$M \simeq M_\alpha \otimes_{A_\alpha} A.$$

Proof. We may write $M = A^r / P$, where P is a finitely generated A -submodule of A^r . Since

$$A^r = \varinjlim_\alpha A_\alpha^r$$

and P is finitely generated, there are an index α and a finitely generated A_α -submodule $P_\alpha \subset A_\alpha^r$ whose canonical image generates P . Right exactness of the tensor product then gives

$$M \simeq (A_\alpha^r / P_\alpha) \otimes_{A_\alpha} A.$$

$\square$

Completion of the proof of Proposition 5.13.7. Since A is Noetherian, the finite modules M and N are finitely presented. After increasing α if necessary, the lemma gives

$$M = M_\alpha \otimes_{A_\alpha} A, \quad N = N_\alpha \otimes_{A_\alpha} A.$$

The A_α -module A is flat (0, 6.1.2), and base change for Tor gives

$$\mathrm{Tor}_i^A(M, N) \simeq \mathrm{Tor}_i^{A_\alpha}(M_\alpha, N_\alpha) \otimes_{A_\alpha} A$$

(III, 6.3.9). Since A_α is regular, the right-hand side vanishes for all sufficiently large i . This proves the proposition. $\square$

Corollary (5.13.8). — *Let $k \geq 0$. Suppose that the A_α are Noetherian and satisfy (R_k) . Suppose also that, for $\alpha \leq \beta$, the homomorphism $\varphi_{\beta\alpha}$ makes A_β a flat A_α -module, that $A = \varinjlim A_\alpha$ is Noetherian, and that for every α the induced map*

$$\varphi_\alpha : \mathrm{Spec}(A) \longrightarrow \mathrm{Spec}(A_\alpha)$$

is a homeomorphism. Then A satisfies (R_k) .

Proof. Let $\mathfrak{p}$ be a prime ideal of A , and put

$$\mathfrak{p}_\alpha = \varphi_\alpha^{-1}(\mathfrak{p}).$$

Then

$$\mathfrak{p} = \varinjlim_\alpha \mathfrak{p}_\alpha, \quad A_\mathfrak{p} = \varinjlim_\alpha (A_\alpha)_{\mathfrak{p}_\alpha}$$

by (5.13.3). It follows from (I, 2.4.2) and the hypothesis that

$$\mathrm{Spec}(A_\mathfrak{p}) \longrightarrow \mathrm{Spec}((A_\alpha)_{\mathfrak{p}_\alpha})$$

is a surjective homeomorphism for every α . Hence

$$\dim(A_\mathfrak{p}) = \dim((A_\alpha)_{\mathfrak{p}_\alpha}) \quad \text{for every } \alpha.$$

If $\dim(A_\mathfrak{p}) \leq k$, then each $(A_\alpha)_{\mathfrak{p}_\alpha}$ is regular. Moreover, for $\alpha \leq \beta$, $(A_\beta)_{\mathfrak{p}_\beta}$ is flat over $(A_\alpha)_{\mathfrak{p}_\alpha}$ (0, 6.3.2). Proposition (5.13.7) therefore shows that $A_\mathfrak{p}$ is regular. Thus A satisfies (R_k) . $\square$

§6. FLAT MORPHISMS OF LOCALLY NOETHERIAN PRESCHEMES

Let X, Y be two locally Noetherian preschemes, $f : X \rightarrow Y$ a morphism. For every $y \in Y$, the fibre $f^{-1}(y) = X \times_Y \mathrm{Spec}(k(y))$ is also a locally Noetherian prescheme, since it suffices to see this when $Y = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$ are spectra of Noetherian rings, and then $B \otimes_A k(y)$ is a ring of fractions of the quotient ring $B/\mathfrak{I}_y B$, hence is Noetherian. We propose first of all in this paragraph to relate the properties of X , of Y , and of the fibres $f^{-1}(y)$ (where y ranges over Y), under the hypothesis that the morphism f is *flat*. The questions treated will reduce to the study of relations between the Noetherian local rings $\mathcal{O}_y$, $\mathcal{O}_x$, and $\mathcal{O}_x \otimes_{\mathcal{O}_y} k(y)$, the homomorphism $\mathcal{O}_y \rightarrow \mathcal{O}_x$ being local and making $\mathcal{O}_x$ a flat $\mathcal{O}_y$ -module. In nos 6.11 to 6.13 (rather separate from the rest of the paragraph, by their “absolute” rather than “relative” nature), we apply certain preceding results to find criteria allowing us to affirm that the *singular locus* (or certain analogous sets) of certain preschemes are closed sets; these criteria will play an important role in §7.

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6.1. Flatness and dimension

Proposition (6.1.1). — *Let A, B be two local Noetherian rings, $\mathfrak{m}$ the maximal ideal of A , $k = A/\mathfrak{m}$ its residue field, $\varphi : A \rightarrow B$ a local homomorphism. Suppose that for every prime ideal $\mathfrak{p}$ of A , distinct from $\mathfrak{m}$, and for every minimal element $\mathfrak{q}$ of the set of prime ideals of B containing $\mathfrak{p}B$, $\varphi^{-1}(\mathfrak{q})$ is distinct from $\mathfrak{m}$ (in other words, no irreducible component of $\text{Spec}(B/\mathfrak{p}B)$ is contained in the inverse image of the closed point $\mathfrak{m}$ of $\text{Spec}(A)$ in $\text{Spec}(B)$). Then we have*

$$(6.1.1.1) \quad \dim(B) = \dim(A) + \dim(B \otimes_A k).$$

Proof. We argue by induction on $n = \dim(A)$; the assertion is trivial for $n = 0$, $\mathfrak{m}B$ being then contained in the nilradical of B . We may therefore suppose $n > 0$. Let $\mathfrak{q}_i$ ($1 \leq i \leq m$) be the minimal prime ideals of B , and let $\mathfrak{p}_i = \varphi^{-1}(\mathfrak{q}_i)$; for every i , we have $\mathfrak{p}_i \neq \mathfrak{m}$; otherwise (since $n > 0$) there would exist a prime ideal $\mathfrak{p} \neq \mathfrak{m}$ contained in $\mathfrak{p}_i$ and as $\mathfrak{q}_i$ is minimal among the prime ideals of B containing $\mathfrak{p}B$, we would arrive at a contradiction with the hypothesis. Therefore $\mathfrak{m}$ is distinct from the union of the $\mathfrak{p}_i$ and of the minimal prime ideals $\mathfrak{p}'_j$ ($1 \leq j \leq r$) of A (Bourbaki, *Alg. comm.*, chap. II, §1, n° 1, prop. 2) and there exists an $a \in \mathfrak{m}$ not belonging to any of the $\mathfrak{p}_i$ nor any of the $\mathfrak{p}'_j$. Set $A' = A/aA$, $B' = B/aB$; we have (0, 16.3.4)

$$\dim(A') = \dim(A) - 1, \quad \dim(B') = \dim(B) - 1$$

by construction of a , and on the other hand $B' \otimes_{A'} k = B \otimes_A k$, hence

$$\dim(B' \otimes_{A'} k) = \dim(B \otimes_A k)$$

and it suffices to prove (6.1.1) for A' and B' ; now, by virtue of the bijective correspondence between ideals of A (resp. B) containing a and ideals of A' (resp. B'), the hypotheses of the statement are also valid for A' and B' ; we may therefore apply the induction hypothesis, which completes the proof. $\square$

Corollary (6.1.2). — *Let A, B be two local Noetherian rings, k the residue field of A , $\varphi : A \rightarrow B$ a local homomorphism, $M \neq 0$ an A -module of finite type, $N \neq 0$ a B -module of finite type. If N is a flat A -module (resp. if B is a flat A -module), then we have*

$$(6.1.2.1) \quad \dim_B(M \otimes_A N) = \dim_A(M) + \dim_{B \otimes_A k}(N \otimes_A k)$$

(resp. (6.1.1)).

Proof. It suffices to prove the assertion concerning N ; on the other hand, if $\mathfrak{b}$ is the annihilator of N , we may replace B by $B/\mathfrak{b}$, and therefore suppose that $\text{Supp}(N) = \text{Spec}(B)$; the hypothesis then means that the morphism $f : \text{Spec}(B) \rightarrow \text{Spec}(A)$ corresponding to φ is *quasi-flat* (2.3.3); we conclude (2.3.4) that for every prime ideal $\mathfrak{p} \neq \mathfrak{m}$ of A , every irreducible component of $f^{-1}(V(\mathfrak{p}))$ dominates $V(\mathfrak{p})$, and therefore the condition of (6.1.1) is satisfied, whence the conclusion. $\square$

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Corollary (6.1.3). — *Let A, B be two local Noetherian rings, $\mathfrak{m}$ the maximal ideal of A , k its residue field, $\varphi : A \rightarrow B$ a local homomorphism. Suppose that $\dim(B \otimes_A k) = 0$ (that is to say (0, 16.2.1) that $\mathfrak{m}B$ is an ideal of definition of B). Then we have $\dim(B) \leq \dim(A)$. If in addition there exists a B -module of finite type N that is a flat A -module and has support equal to $\text{Spec}(B)$ (which holds in particular if B is a flat A -module), then $\dim(B) = \dim(A)$.*

Proof. The first assertion follows from (5.5.1) and the second from (6.1.2). $\square$

Corollary (6.1.4). — *Let X, Y be two locally Noetherian preschemes, $f : X \rightarrow Y$ a surjective morphism. For every closed subset Z of Y , we have*

$$(6.1.4.1) \quad \text{codim}(f^{-1}(Z), X) \leq \text{codim}(Z, Y)$$

Moreover, if f is *quasi-flat* (2.3.3), the two sides of (6.1.4) are equal.

Proof. Indeed, if y is a maximal point of Z and x a maximal point of the fibre $f^{-1}(y)$, we have $\dim(\mathcal{O}_{X,x}) \leq \dim(\mathcal{O}_{Y,y})$ by virtue of (5.5.2); the inequality (6.1.4) then follows from (5.1.2.1) and from (5.1.3.1) and from the fact that f is surjective. If in addition f is *quasi-flat*, we know (2.3.4) that every irreducible component of $f^{-1}(Z)$ dominates an irreducible component of Z ; the maximal points of $f^{-1}(Z)$ are therefore the maximal points of the fibres $f^{-1}(y)$, where y ranges over the set of

maximal points of Z (0, 2.1.8), and for each of these maximal points x we have $\dim(\mathcal{O}_x \otimes_{\mathcal{O}_y} k(x)) = 0$; as in addition $\mathcal{O}_x$ is a flat $\mathcal{O}_y$ -module, we have $\dim(\mathcal{O}_x) = \dim(\mathcal{O}_y)$ by virtue of (6.1.1), whence the equality in (6.1.4), taking into account that f is surjective. $\square$

The proposition (6.1.1) admits the following partial converse:

Proposition (6.1.5). — *Let A, B be two local Noetherian rings, k the residue field of A , $\varphi : A \rightarrow B$ a local homomorphism, M a B -module of finite type. Suppose that:*

- 1° A is a regular ring.
- 2° M is a Cohen-Macaulay B -module.
- 3° $\dim_B(M) = \dim(A) + \dim_{B \otimes_A k}(M \otimes_A k)$.

Then M is a flat A -module.

Proof. We argue by induction on $n = \dim(A)$. If $n = 0$, A is a field since it is regular (0, 17.1.4) and the assertion is trivial. Suppose $n > 0$; let $\mathfrak{m}$ be the maximal ideal of A , and let $x \in \mathfrak{m}$ be an element not belonging to $\mathfrak{m}^2$; we know then (0, 17.1.8) that $A' = A/xA$ is regular and $\dim(A') = \dim(A) - 1$. Set

$$B' = B/xB, \quad M' = M/xM = M \otimes_A A';$$

then

$$B' \otimes_{A'} k = B \otimes_A k, \quad M' \otimes_{A'} k = M \otimes_A k$$

and by virtue of (5.5.1.2) we have

$$\dim_{B'}(M') \leq \dim(A') + \dim_{B' \otimes_{A'} k}(M' \otimes_{A'} k);$$

we conclude therefore from (0, 16.3.4) that we have

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$$\dim_B(M) \leq \dim_{B'}(M') + 1 \leq (\dim(A') + \dim_{B \otimes_A k}(M \otimes_A k)) + 1 = \dim(A) + \dim_{B \otimes_A k}(M \otimes_A k).$$

Since by hypothesis the extreme terms of these inequalities are equal, we have necessarily: (i) $\dim_{B'}(M') = \dim_B(M) - 1$, and as M is by hypothesis a Cohen-Macaulay B -module, this means that x is M -regular (0, 16.1.9) and (0, 16.5.5); (ii) $\dim_{B'}(M') = \dim(A') + \dim_{B' \otimes_{A'} k}(M' \otimes_{A'} k)$; as M' is a Cohen-Macaulay B' -module by virtue of (i) and of (0, 16.1.9) and (0, 16.5.5) and A' is regular, the induction hypothesis proves that M' is a flat A' -module; we deduce then from (0, 10.2.7) that M is a flat A -module, since x is M -regular by (i). $\square$

6.2. Flatness and projective dimension

Proposition (6.2.1). —

- (i) *Let A, B be two rings, $\varphi : A \rightarrow B$ a homomorphism such that B is a flat A -module. Then, for every A -module M , we have*

$$(6.2.1.1) \quad \dim \cdot \operatorname{proj}_B(M \otimes_A B) \leq \dim \cdot \operatorname{proj}_A(M).$$

- (ii) *Suppose in addition that A is a Noetherian ring, B a faithfully flat A -module, and M an A -module of finite type; then*

$$(6.2.1.2) \quad \dim \cdot \operatorname{proj}_B(M \otimes_A B) = \dim \cdot \operatorname{proj}_A(M).$$

Proof.

- (i) We may restrict to the case where $n = \dim \cdot \operatorname{proj}_A(M)$ is finite; there therefore exists a left resolution

$$0 \longrightarrow P_n \longrightarrow P_{n-1} \longrightarrow \dots \longrightarrow P_0 \longrightarrow M \longrightarrow 0$$

of M by projective A -modules; as B is a flat A -module, the sequence

$$0 \longrightarrow P_n \otimes_A B \longrightarrow P_{n-1} \otimes_A B \longrightarrow \dots \longrightarrow P_0 \otimes_A B \longrightarrow M \otimes_A B \longrightarrow 0$$

is exact; on the other hand, the $P_i \otimes_A B$ are projective B -modules; whence the conclusion.

(ii) Suppose that $\dim . \operatorname{proj}_B(M \otimes_A B) = m$, and consider an exact sequence

$$0 \longrightarrow R \longrightarrow P_{m-1} \longrightarrow P_{m-2} \longrightarrow \dots \longrightarrow P_0 \longrightarrow M \longrightarrow 0$$

where the P_i ($0 \leq i \leq m-1$) are projective A -modules of finite type; as A is Noetherian, R is also an A -module of finite type. As B is a flat A -module, we also have an exact sequence

$$0 \longrightarrow R \otimes_A B \longrightarrow P_{m-1} \otimes_A B \longrightarrow \dots \longrightarrow P_0 \otimes_A B \longrightarrow M \otimes_A B \longrightarrow 0$$

and the hypothesis on $M \otimes_A B$ implies that $R \otimes_A B$ is a projective B -module (0, 17.2.1). As B is a faithfully flat A -module, we conclude that R is a projective A -module (Bourbaki, *Alg. comm.*, chap. I^{er}, § 3, n° 6, prop. 12); hence $\dim . \operatorname{proj}_A(M) \leq m$, which completes the proof. $\square$

Corollary (6.2.2). — *Let $f : X \rightarrow Y$ be a flat morphism of preschemes, $\mathcal{E}$ a quasi-coherent $\mathcal{O}_Y$ -Module. If $\dim . \operatorname{proj}(\mathcal{E}) \leq n$ (0, 17.2.14), then $\dim . \operatorname{proj}(f^*(\mathcal{E})) \leq n$.*

Proposition (6.2.3). — *Let A, B be two local Noetherian rings, k the residue field of A , $\varphi : A \rightarrow B$ a local homomorphism, N a B -module of finite type. Suppose that B and N are flat A -modules. Then we have*

$$(6.2.3.1) \quad \dim . \operatorname{proj}_B(N) = \dim . \operatorname{proj}_{B \otimes_A k}(N \otimes_A k).$$

Proof. Consider a left resolution of N by free B -modules of finite type

$$\dots \longrightarrow L_i \longrightarrow L_{i-1} \longrightarrow \dots \longrightarrow L_0 \longrightarrow N \longrightarrow 0$$

As the L_i and N are flat A -modules (B being a flat A -module) it follows from (2.1.10) that the $Z_i(L_\bullet)$ are flat A -modules, that $L_\bullet \otimes_A k$ is a resolution of $N \otimes_A k$ and that $Z_i(L_\bullet \otimes_A k) = Z_i(L_\bullet) \otimes_A k$ for all i . Note that since B is Noetherian, the $Z_i(L_\bullet)$ are B -modules of finite type, hence the $Z_i(L_\bullet \otimes_A k)$ are $(B \otimes_A k)$ -modules of finite type. Now, to say that a $(B \otimes_A k)$ -module of finite type is flat is equivalent to saying that it is *free* or even *projective* (0, 10.1.3). As (where we set $C = B$) the $Z_i(L_\bullet)$ are flat B -modules, it follows on the other hand from (0, 10.2.5) that, for $Z_i(L_\bullet)$ to be a flat B -module, it is necessary and sufficient that $Z_i(L_\bullet) \otimes_A k$ be a $(B \otimes_A k)$ -module that is flat. The smallest integer n such that $Z_{n-1}(L_\bullet)$ is a free B -module is therefore also the smallest integer n such that $Z_{n-1}(L_\bullet \otimes_A k)$ is a free $(B \otimes_A k)$ -module, which proves the proposition (0, 17.2.1). $\square$

6.3. Flatness and depth

Proposition (6.3.1). — *Let A, B be two local Noetherian rings, k the residue field of A , $\varphi : A \rightarrow B$ a local homomorphism, M an A -module of finite type, N a B -module of finite type. If N is a flat A -module $\neq 0$, then we have*

$$(6.3.1.1) \quad \operatorname{prof}_B(M \otimes_A N) = \operatorname{prof}_A(M) + \operatorname{prof}_{B \otimes_A k}(N \otimes_A k).$$

Proof. We may restrict to the case where $M \neq 0$, without which both sides of (6.3.1) are all equal to $+\infty$. We will argue by induction on the integer n equal to the second member of (6.3.1.1) (which is finite by virtue of the hypotheses, of (0, 16.4.6.2) and of the lemma of Nakayama applied to N). Suppose first $n = 0$; then, if $\mathfrak{m}$ and $\mathfrak{n}$ denote the maximal ideals of A and B respectively, $\mathfrak{m}$ is a prime ideal associated to M and $\mathfrak{n}/\mathfrak{m}B$ a prime ideal of $B/\mathfrak{m}B = B \otimes_A k$ associated to $N \otimes_A k$. We conclude then from (3.3.1) that $\mathfrak{n}$ is a prime ideal associated to $M \otimes_A N$, hence (0, 16.4.6), (i) that $\operatorname{prof}_B(M \otimes_A N) = 0$, and we distinguish two cases:

a) Suppose $\operatorname{prof}_A(M) > 0$. Let $x \in \mathfrak{m}$ be an M -regular element, and set

$$A' = A/xA, \quad B' = B/xB, \quad M' = M/xM, \quad N' = N/xN;$$

then

$$B' \otimes_{A'} k = B \otimes_A k, \quad N' \otimes_{A'} k = N \otimes_A k$$

since $x \in \mathfrak{m}$, and

$$M' \otimes_{A'} N' = (M \otimes_A N)/x(M \otimes_A N);$$

in addition, as N is a flat A -module, the hypothesis that x is M -regular implies that x is also $(M \otimes_A N)$ -regular. We thus have (0, 16.4.6), (ii) and (0, 16.4.8))

$$\operatorname{prof}_{A'}(M') = \operatorname{prof}_A(M) - 1, \quad \operatorname{prof}_{B'}(M' \otimes_{A'} N') = \operatorname{prof}_B(M \otimes_A N) - 1$$

and

$$\text{prof}_{B' \otimes_{A'} k}(N' \otimes_{A'} k) = \text{prof}_{B \otimes_A k}(N \otimes_A k).$$

The equality (6.3.1.1) is therefore a consequence of the same relation for A' , B' , M' and N' ; but as N is a flat A -module, $N' = N \otimes_A A'$ is a flat A' -module (0, 6.2.1); we may therefore apply the induction hypothesis, which proves (6.3.1.1) in this case.

b) Suppose $\text{prof}_{B \otimes_A k}(N \otimes_A k) > 0$. Consider an element $y \in \mathfrak{n}$ which is $(N \otimes_A k)$ -regular; it follows from (0, 10.2.4) that y is N -regular and that we then have

$$N' = N/yN$$

is a flat A -module, since N is supposed to be a flat A -module. Applying (0, 6.1.2) to the exact sequence of A -modules

$$0 \longrightarrow N \xrightarrow{y} N \longrightarrow N' \longrightarrow 0$$

we conclude that we have isomorphisms

$$(M \otimes_A N)/y(M \otimes_A N) \xrightarrow{\sim} M \otimes_A N' \quad \text{and} \quad N' \otimes_A k \xrightarrow{\sim} (N \otimes_A k)/y(N \otimes_A k)$$

and in addition that y is $(M \otimes_A N)$ -regular. We thus have

$$\text{prof}_{B \otimes_A k}(N' \otimes_A k) = \text{prof}_{B \otimes_A k}(N \otimes_A k) - 1$$

and

$$\text{prof}_B(M \otimes_A N') = \text{prof}_B(M \otimes_A N) - 1;$$

the induction hypothesis shows that the relation (6.3.1.1) is valid for A , B , M and N' , and from what precedes, we deduce (6.3.1.1) for A , B , M and N . $\square$

Corollary (6.3.2). — *Under the hypotheses of (6.3.1), suppose in addition $M \neq 0$ and $N \neq 0$; then we have*

$$(6.3.2.1) \quad \text{coprof}_B(M \otimes_A N) = \text{coprof}_A(M) + \text{coprof}_{B \otimes_A k}(N \otimes_A k).$$

Proof. This follows immediately from (6.3.1.1) and (6.1.2) and from the definition of the codepth (0, 16.4.9). $\square$

Corollary (6.3.3). — *Under the hypotheses of (6.3.1), suppose in addition $M \neq 0$ and $N \neq 0$; for $M \otimes_A N$ to be a Cohen-Macaulay B -module, it is necessary and sufficient that M be a Cohen-Macaulay A -module and that $N \otimes_A k$ be a $(B \otimes_A k)$ -module of Cohen-Macaulay.*

Proof. This follows from the corollary (6.3.2) and from the definition of Cohen-Macaulay modules by the fact that their codepth is zero (0, 16.5.1), taking into account that the condition $N \neq 0$ is equivalent to $N \otimes_A k \neq 0$ by virtue of the lemma of Nakayama. $\square$

Corollary (6.3.4). — *Under the hypotheses of (6.3.1), suppose in addition that $N \otimes_A k$ is a B -module of finite length; then we have*

$$(6.3.4.1) \quad \text{prof}_B(M \otimes_A N) = \text{prof}_A(M).$$

If in addition $M \neq 0$ and $N \neq 0$, then we have

$$(6.3.4.2) \quad \text{coprof}_B(M \otimes_A N) = \text{coprof}_A(M).$$

Proof. It amounts to the same to say that $N \otimes_A k$ is a B -module of finite length or a $(B \otimes_A k)$ -module of finite length, and we know (0, 16.2.3) that modules of finite length $\neq 0$ are of dimension 0 and depth 0.

The corollary (6.3.4) is applicable in particular when $B \otimes_A k$ is an Artinian ring, that is to say when $\mathfrak{m}B$ is an ideal of definition of the ring B . $\square$

Corollary (6.3.5). — *Let X, Y be two locally Noetherian preschemes, $f : X \rightarrow Y$ a flat morphism.*

- (i) *If, at a point $x \in X$, $\text{coprof}(\mathcal{O}_x) \leq n$, then, setting $y = f(x)$, we have $\text{coprof}(\mathcal{O}_y) \leq n$ and $\text{coprof}(\mathcal{O}_x \otimes_{\mathcal{O}_y} k(y)) \leq n$; in particular, if $\mathcal{O}_x$ is a Cohen-Macaulay ring, the same is true of $\mathcal{O}_y$ and of $\mathcal{O}_x \otimes_{\mathcal{O}_y} k(y)$.*
- (ii) *Conversely, suppose that $\mathcal{O}_x \otimes_{\mathcal{O}_y} k(y)$ is a Cohen-Macaulay ring. Then, if $\mathcal{O}_y$ is a Cohen-Macaulay ring, we have $\text{coprof}(\mathcal{O}_x) \leq n$ (resp. $\mathcal{O}_x$ is a Cohen-Macaulay ring).*

Proof. It suffices to apply (6.3.2) for $M = \mathcal{O}_y$ and $N = B = \mathcal{O}_x$. $\square$

Corollary (6.3.6). — *Let A be a Cohen-Macaulay ring. Then $A' = A[T_1, \dots, T_n]$ (T_i indeterminates) is a Cohen-Macaulay ring.*

Proof. Indeed, if we set $Y = \text{Spec}(A)$, $X = \text{Spec}(A')$ and if $f : X \rightarrow Y$ is the canonical morphism, f is flat (since A' is a free A -module (2.1.2)) and for every $y \in Y$, $k(y)[T_1, \dots, T_n]$ is a regular ring (0, 17.3.7), hence a fortiori a Cohen-Macaulay ring; it therefore suffices to apply (6.3.5). $\square$

Corollary (6.3.7). — *Every quotient ring of a Cohen-Macaulay ring is universally catenary.*

Proof. This follows from (6.3.6), from (5.6.1) and from (0, 16.5.12). $\square$

Proposition (6.3.8). — *Let A be a local Noetherian ring; suppose that there exists a Cohen-Macaulay A -module M of finite type whose support is equal to $\text{Spec}(A)$. Let B be a quotient ring of A , and let $f : \text{Spec}(\widehat{B}) \rightarrow \text{Spec}(B)$ be the canonical morphism; then, for every $x \in \text{Spec}(B)$, $f^{-1}(x)$ is a Cohen-Macaulay prescheme.*

Proof. As B is an A -module of finite type, one has $\widehat{B} = B \otimes_A \widehat{A}$ (0, 7.3.3); hence f is the restriction to $\text{Spec}(\widehat{B})$ of the canonical morphism $\text{Spec}(\widehat{A}) \rightarrow \text{Spec}(A)$, and the fibres of these two morphisms at a point of $\text{Spec}(B)$ are the same. We are therefore reduced to proving the proposition when $B = A$. Let $\mathfrak{p}$ be a prime ideal minimal of A . Set, to simplify, $A' = \widehat{A}$, $M' = \widehat{M} = M \otimes_A A'$ (0, 7.3.3); we know (0, 16.5.2) that M' is a Cohen-Macaulay A' -module, which implies that for every prime ideal $\mathfrak{p}'$ of A' , $M'_{\mathfrak{p}'}$ is a Cohen-Macaulay $A'_{\mathfrak{p}'}$ -module (0, 16.5.10). Taking into account (I, 3.6.5), one sees that if we set $A'' = A' \otimes_A A_{\mathfrak{p}}$ and $M'' = M' \otimes_A A_{\mathfrak{p}}$, M'' is a Cohen-Macaulay A'' -module. For every prime ideal $\mathfrak{p}''$ of A'' above $\mathfrak{p}$, $M''_{\mathfrak{p}''} = M_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} A''_{\mathfrak{p}''}$ is a Cohen-Macaulay $A''_{\mathfrak{p}''}$ -module (0, 16.5.10); on the other hand, $M_{\mathfrak{p}}$ is by hypothesis a Cohen-Macaulay $A_{\mathfrak{p}}$ -module and $A''_{\mathfrak{p}''}$ is a flat $A_{\mathfrak{p}}$ -module since A' is a flat A -module (0, 7.3.3) and (6.3.2). We conclude from (6.3.3) that, if k is the residue field of $A_{\mathfrak{p}}$, $k \otimes_{A_{\mathfrak{p}}} A''_{\mathfrak{p}''}$ is a Cohen-Macaulay ring. But since $A_{\mathfrak{p}}$ is of dimension 0, the prime ideals of A' correspond bijectively to those of $k \otimes_{A_{\mathfrak{p}}} A''$ (I, 3.5.7), and if $\mathfrak{q}''$ is the prime ideal of $k \otimes_{A_{\mathfrak{p}}} A''$ corresponding to $\mathfrak{p}''$, the local rings $(k \otimes_{A_{\mathfrak{p}}} A'')_{\mathfrak{q}''}$ and $k \otimes_{A_{\mathfrak{p}}} A''_{\mathfrak{p}''}$ are isomorphic. Therefore (0, 16.5.13), the ring $k \otimes_{A_{\mathfrak{p}}} A''$ is a Cohen-Macaulay ring. $\square$

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6.4. Flatness and property (S_k)

Proposition (6.4.1). — *Let X, Y be two locally Noetherian preschemes, $f : X \rightarrow Y$ a morphism, $\mathcal{E}$ a coherent $\mathcal{O}_Y$ -Module, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module and f -flat.*

- (i) *Let $x \in \text{Supp}(\mathcal{F})$ be such that $\mathcal{E} \otimes_{\mathcal{O}_Y} \mathcal{F}$ has property (S_k) at the point x ; then $\mathcal{E}$ has property (S_k) at the point $y = f(x)$.*
- (ii) *Suppose that for every $y \in f(\text{Supp}(\mathcal{F}))$, $\mathcal{F}_y = \mathcal{F} \otimes_{\mathcal{O}_y} k(y)$ has property (S_k) ; then, if for a point $y \in f(\text{Supp}(\mathcal{F}))$, $\mathcal{E}$ has property (S_k) at the point y , $\mathcal{E} \otimes_{\mathcal{O}_Y} \mathcal{F}$ has property (S_k) at every point of $f^{-1}(y)$.*

Proof.

- (i) We know (Err_{III}, 30) that there exists a closed sub-prescheme X' of X having for underlying space $\text{Supp}(\mathcal{F})$ and a coherent $\mathcal{O}_{X'}$ -Module $\mathcal{F}'$ such that $\mathcal{F} = j_*(\mathcal{F}')$, where j is the canonical injection $X' \hookrightarrow X$; we may replace X by X' and $\mathcal{F}$ by $\mathcal{F}'$, in other words suppose that $\text{Supp}(\mathcal{F}) = X$. Let y' be a generization of y in Y ; we know (2.3.4) that there is a generization x' of x in X such that $y' = f(x')$; we may in addition suppose that x' is the generic point of an irreducible component of $f^{-1}(y')$; we then have $\dim(\mathcal{F}_{x'} \otimes_{\mathcal{O}_y} k(y')) = 0$. By virtue of (6.1.2), we then have, setting $\mathcal{G} = \mathcal{E} \otimes_{\mathcal{O}_Y} \mathcal{F}$,

$$\dim(\mathcal{G}_{x'}) = \dim(\mathcal{E}_{y'})$$

and by virtue of (6.3.1)

$$\text{prof}(\mathcal{G}_{x'}) = \text{prof}(\mathcal{E}_{y'})$$

taking into account that the depth is always at most equal to the dimension. By hypothesis, **IV-2** | 142 we have

$$\text{prof}(\mathcal{G}_{x'}) \geq \inf(k, \dim(\mathcal{G}_{x'}))$$

hence

$$\text{prof}(\mathcal{E}_{y'}) \geq \inf(k, \dim(\mathcal{E}_{y'}))$$

which proves the first assertion.

- (ii) As for every generization x' of $x \in f^{-1}(y)$, $y' = f(x')$ is a generization of y , we may restrict to verifying that if $x \in \text{Supp}(\mathcal{G})$ and $f(x) = y$, we have $\text{prof}(\mathcal{G}_x) \geq \inf(k, \dim(\mathcal{G}_x))$; it follows from (6.1.2) and (6.3.1) that we have

$$\dim(\mathcal{G}_x) = \dim(\mathcal{E}_y) + \dim(\mathcal{F}_x \otimes_{\mathcal{O}_y} k(y))$$

$$\text{prof}(\mathcal{G}_x) = \text{prof}(\mathcal{E}_y) + \text{prof}(\mathcal{F}_x \otimes_{\mathcal{O}_y} k(y)).$$

By hypothesis, we have

$$\text{prof}(\mathcal{E}_y) \geq \inf(k, \dim(\mathcal{E}_y))$$

$$\text{prof}(\mathcal{F}_x \otimes_{\mathcal{O}_y} k(y)) \geq \inf(k, \dim(\mathcal{F}_x \otimes_{\mathcal{O}_y} k(y)))$$

whence, adding term by term,

$$\begin{aligned} \text{prof}(\mathcal{G}_x) &\geq \inf(k, \dim(\mathcal{E}_y)) + \inf(k, \dim(\mathcal{F}_x \otimes_{\mathcal{O}_y} k(y))) \geq \\ &\geq \inf(k, \dim(\mathcal{E}_y) + \dim(\mathcal{F}_x \otimes_{\mathcal{O}_y} k(y))) = \inf(k, \dim(\mathcal{G}_x)) \end{aligned}$$

which proves (ii). □

Corollary (6.4.2). — *Let X, Y be two locally Noetherian preschemes, $f : X \rightarrow Y$ a flat morphism, $\mathcal{E}$ a coherent $\mathcal{O}_Y$ -Module. Suppose that for every $y \in Y$, the prescheme $f^{-1}(y)$ has property (S_k) . Then, for $f^*(\mathcal{E})$ to have property (S_k) at a point x , it is necessary and sufficient that $\mathcal{E}$ have property (S_k) at the point $f(x)$.*

Remark (6.4.3). — Let A, B be two local Noetherian rings, k the residue field of A , $\varphi : A \rightarrow B$ a local homomorphism, M a B -module of finite type that is a flat A -module, and suppose in addition that the A -module A and the $(B \otimes_A k)$ -module $M \otimes_A k$ have property (S_n) ; can one then conclude that the B -module M has property (S_n) ? We do not know even when $n = 1$, $M = B$ and $B = \hat{A}$, which is to say we do not know whether in general the completion of a local Noetherian ring satisfying (S_n) also satisfies (S_n) , even for $n = 1$. Setting $Y = \text{Spec}(A)$, $X = \text{Spec}(B)$ and $\mathcal{F} = \hat{M}$, it would suffice, by virtue of (6.4.1), (ii), to show that for every $y \in Y$, $\mathcal{F}_y$ has property (S_k) (and not only when y is the closed point of Y); it would also suffice to show that the set U of $x \in X$ such that $\mathcal{F}_x \otimes_{\mathcal{O}_{f(x)}} k(f(x))$ satisfies (S_k) is open in X (since it contains the closed point of X by hypothesis). We will show further on (12.1.4) that U is open when B is a local ring of a finite type A -algebra. On the other hand, for $B = \hat{A}$ and $M = B$, we have seen in (6.3.8) that the preschemes $f^{-1}(y)$ are Cohen-Macaulay preschemes (in other words satisfy (S_n)) for every n when we suppose that A is a quotient of a local Cohen-Macaulay ring (or more generally of a local ring satisfying the hypotheses of (6.3.8)), for A to satisfy (S_n) , it is necessary and sufficient that its completion $\hat{A}$ satisfy (S_n) . It would remain to see whether this property holds without restriction on A .

6.5. Flatness and property (R_k)

Proposition (6.5.1). — *Let A, B be two local Noetherian rings, k the residue field of A , $\varphi : A \rightarrow B$ a local homomorphism, for which B is a flat A -module. Then:*

- (i) *If B is regular, A is regular.*
- (ii) *If A and $B \otimes_A k$ are regular, B is regular.*

This proposition is given here for the record, having already been proved in (0, 17.3.3).

Corollary (6.5.2). — *Let X, Y be two locally Noetherian preschemes, $f : X \rightarrow Y$ a flat morphism.*

- (i) *If X is regular at a point x , Y is regular at the point $f(x)$.*
- (ii) *If for a $y \in f(X)$, Y is regular at the point y and if $f^{-1}(y)$ is a regular prescheme at a point x , then X is regular at the point x .*

Proposition (6.5.3). — *Let X, Y be two locally Noetherian preschemes, $f : X \rightarrow Y$ a flat morphism.*

- (i) *If X has property (R_k) at a point x (resp. has property (R_k)), Y has property (R_k) at the point $f(x)$ (resp. Y has property (R_k) at all points of $f(X)$).*
- (ii) *Suppose that for every $y \in f(X)$, the prescheme $f^{-1}(y)$ has property (R_k) ; then, if for a point $y \in f(X)$, Y has property (R_k) at the point y , X has property (R_k) at every point of $f^{-1}(y)$.*

Proof.

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- (i) Set $y = f(x)$ and let y' be a generization of y ; as in the proof of (6.4.1), there is a generic point x' of an irreducible component of $f^{-1}(y')$ which is a generization of x ; hence $\dim(\mathcal{O}_{x'} \otimes_{\mathcal{O}_{y'}} k(y')) = 0$ and by virtue of the hypothesis and of (6.1.2), $\dim(\mathcal{O}_{x'}) = \dim(\mathcal{O}_{y'})$. The hypothesis implies, provided that $\dim(\mathcal{O}_{x'}) > k$, that $\mathcal{O}_{x'}$ is a regular ring, and then $\mathcal{O}_{y'}$ is regular by virtue of (6.5.1).
- (ii) As for every generization x' of $x \in f^{-1}(y)$, $y' = f(x')$ is a generization of y , we may restrict to verifying that if $\dim(\mathcal{O}_x) \leq k$, then $\mathcal{O}_x$ is a regular ring. Now, by virtue of (6.1.2)

$$\dim(\mathcal{O}_x) = \dim(\mathcal{O}_y) + \dim(\mathcal{O}_x \otimes_{\mathcal{O}_y} k(y))$$

hence if $\dim(\mathcal{O}_x) \leq k$, we have *a fortiori* $\dim(\mathcal{O}_y) \leq k$ and $\dim(\mathcal{O}_x \otimes_{\mathcal{O}_y} k(y)) \leq k$ and the hypothesis implies that $\mathcal{O}_y$ and $\mathcal{O}_x \otimes_{\mathcal{O}_y} k(y)$ are regular rings. We conclude then from (6.5.1) that $\mathcal{O}_x$ is a regular ring. □

Corollary (6.5.4). — *Let X, Y be two locally Noetherian preschemes, $f : X \rightarrow Y$ a flat morphism.*

- (i) *If X is normal at a point x , Y is normal at the point $f(x)$.*
- (ii) *If, for every $y \in f(X)$, $f^{-1}(y)$ is a normal prescheme and if Y is normal at a point $y \in f(X)$, then X is a normal prescheme at every point of $f^{-1}(y)$.*

Proof. This follows immediately from the normality criterion of Serre (5.8.6) and from (6.4.1) applied for $k = 2$ and (6.5.3) applied for $k = 1$. □

Remarks (6.5.5). —

- (i) Let A, B be two local Noetherian rings, $\varphi : A \rightarrow B$ a local homomorphism such that B is a flat A -module. Let k be the residue field of A , and suppose that the two rings A and $B \otimes_A k$ satisfy property (R_i) (5.8.2); then it does not necessarily follow that B satisfies (R_i) , even in the particular case where $i = 0$ or $i = 1$ and where B is the completion $\hat{A}$ of A . Nagata has in fact given an example where A is normal (hence satisfies (R_1)) but where $\hat{A}$ is not even reduced (hence does not satisfy (R_0)) [30]. We cannot apply the proposition (6.5.3) to this case because the fibres $f^{-1}(y)$ do not necessarily satisfy property (R_i) at the points distinct from the closed point of $Y = \text{Spec}(A)$. We will show further on (7.8.3), (v)) that such phenomena do not arise for the local Noetherian rings that one encounters most often in applications.

(ii) The property for a prescheme of being *integral* does not at all behave like the properties we have just examined and the preceding ones: it can happen that $f : X \rightarrow Y$ is a flat morphism of finite type, that all the fibres $f^{-1}(y)$ are geometrically regular (and even geometrically regular (6.7.6)) and that Y is integral, without X being even locally integral. For example, let k be an algebraically closed field of characteristic 0, and let A be the integral ring $k[U, V]/(UV - (U + V)^3)$ (whose spectrum is therefore a “cubic with a double point”); A is not integrally closed, and if u, v are the canonical images of U, V in A , the integral closure of A is the ring $C = A[t]$, with $t = (u - v)/(u + v)$, which satisfies the equation $t^2 = 1 + u + v$, whence $u = \frac{1}{2}(t^3 + t^2 - t)$, $v = \frac{1}{2}(-t^3 + t^2 + t)$ and therefore $C = k[t]$, isomorphic to a polynomial ring in one indeterminate over k . If $\mathfrak{m} = Au + Av$, the maximal ideal of A (corresponding to the “double point” of the cubic), there are two maximal ideals $\mathfrak{n}_1, \mathfrak{n}_2$ of C , generated respectively by $t - 1$ and $t + 1$. Let B be the subring of the product $C \times C$ formed by the pairs of polynomials (f, g) such that $f(1) = g(-1)$ and $f(-1) = g(1)$ ($\text{Spec}(B)$ is the prescheme obtained by “gluing” two copies of $\text{Spec}(C)$, the point $\mathfrak{n}_1$ (resp. $\mathfrak{n}_2$) of one of the two copies being “glued” to the point $\mathfrak{n}_2$ (resp. $\mathfrak{n}_1$) of the other; cf. chap. V, where this operation will be discussed in a general way). There are therefore two maximal ideals $\mathfrak{r}_1, \mathfrak{r}_2$ of B above the maximal ideal $\mathfrak{m}$ of A . Moreover, as the “gluing” process commutes with localisation and completion, one easily verifies that the canonical homomorphisms $\hat{A}_{\mathfrak{m}} \rightarrow \hat{B}_{\mathfrak{r}_1}$ and $\hat{A}_{\mathfrak{m}} \rightarrow \hat{B}_{\mathfrak{r}_2}$ are bijective; consequently $B_{\mathfrak{r}_1}$ and $B_{\mathfrak{r}_2}$ are flat $A_{\mathfrak{m}}$ -modules (Bourbaki, *Alg. comm.*, chap. III, §3, n°5, prop. 10). For every other maximal ideal $\mathfrak{p}$ of A , there is a single maximal ideal $\mathfrak{q}_1, \mathfrak{q}_2$ of B above $\mathfrak{p}$, and the homomorphisms $A_{\mathfrak{p}} \rightarrow B_{\mathfrak{q}_1}$ and $A_{\mathfrak{p}} \rightarrow B_{\mathfrak{q}_2}$ are bijective. We see thus that the morphism $\text{Spec}(B) \rightarrow \text{Spec}(A)$ is flat and finite, and that all its fibres are geometrically regular (it is even *étale*, as we will see later (17.6.3)); however it is immediate that B is not integral.

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6.6. Properties of transitivity

Proposition (6.6.1). — *For a locally Noetherian prescheme Z , let us denote by $\mathbf{P}(Z)$ any one of the following properties:*

- a) Z is a Cohen-Macaulay prescheme.
- b) Z satisfies (S_n) .
- c) Z is regular.
- d) Z satisfies (R_n) .
- e) Z is normal.
- f) Z is reduced.

Let then X, Y, Z be three locally Noetherian preschemes, $f : X \rightarrow Y, g : Y \rightarrow Z$ two morphisms.

- (i) Suppose that f and g are flat and that for every $y \in f(X)$ (resp. every $z \in g(Y)$), the prescheme $f^{-1}(y)$ (resp. $g^{-1}(z)$) satisfies $\mathbf{P}$. Then $h = g \circ f$ is flat, $h^{-1}(z)$ satisfies $\mathbf{P}$ for every $z \in h(X)$, and for every $z \in g(Y)$ and every $t \in h(X)$, $h^{-1}(z)$ satisfies $\mathbf{P}$.
- (ii) Suppose the following conditions are satisfied: f is faithfully flat, $h = g \circ f$ is flat, for every $y \in Y$, the prescheme $f^{-1}(y)$ satisfies $\mathbf{P}$, and for every $z \in g(Y)$, the prescheme $h^{-1}(z)$ satisfies $\mathbf{P}$. Then g is flat, and for every $z \in g(Y)$, $g^{-1}(z)$ satisfies $\mathbf{P}$.

Proof. (i) We already know (0, 2.1.6) that h is flat; furthermore, for every $z \in Z$, $f_z = f \otimes \mathbf{1}_{k(z)} : h^{-1}(z) \rightarrow g^{-1}(z)$ is flat (0, 2.1.4), and for every $y \in g^{-1}(z)$, $f_z^{-1}(y)$ is isomorphic to $f^{-1}(y)$ (I, 3.6.4). The conclusion follows then respectively from (6.3.5), (ii), (6.4.2), (6.5.2), (ii), (6.5.3), (ii), (6.5.4), (ii) and (3.3.5), (ii).

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(ii) We already know that g is flat (2.2.13); in addition, for every $z \in Z$, f_z is faithfully flat (2.2.13). The conclusion follows then respectively from (6.3.5), (i), (6.4.2), (6.5.2), (i), (6.5.3), (i), (6.5.4), (ii) and (2.1.13). $\square$

Remark (6.6.2). — Suppose h flat, f faithfully flat, and suppose that for every $z \in Z$, $h^{-1}(z)$ is of coprofundity $\leq n$ (5.7.1); then it follows from the reasoning of (6.6.1), (ii) and (6.3.2) that $g^{-1}(z)$ is of coprofundity $\leq n$ for every $z \in g(Y)$ and that for every $y \in Y$, $f^{-1}(y)$ is of coprofundity $\leq n$.

6.7. Application to base changes in algebraic preschemes

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Proposition (6.7.1). — Let k be a field, X a k -prescheme locally Noetherian, $\mathcal{F}$ an $\mathcal{O}_X$ -Module coherent. Let k' be an extension of k ; set $X' = X \otimes_k k'$, $\mathcal{F}' = \mathcal{F} \otimes_k k'$ and let $p : X' \rightarrow X$ be the canonical projection. We suppose either that X is locally of finite type over k , or that k' is an extension of finite type of k , so that in both cases X' is locally Noetherian. Let x' be a point of X' , $x = p(x')$. Then:

- (i) We have $\text{coprof}(\mathcal{F}_x) = \text{coprof}(\mathcal{F}'_{x'})$; in particular, for $\mathcal{F}_x$ to be an $\mathcal{O}_x$ -module of Cohen-Macaulay, it is necessary and sufficient that $\mathcal{F}'_{x'}$ be an $\mathcal{O}_{x'}$ -module of Cohen-Macaulay.
- (ii) For $\mathcal{F}$ to satisfy the property (S_n) at the point x , it is necessary and sufficient that $\mathcal{F}'$ satisfy (S_n) at the point x' .

Proof. We know that p is faithfully flat (2.2.13); the two assertions are then consequences respectively of (6.3.2) and (6.4.2), provided that one proves that $p^{-1}(x) = \text{Spec}(k(x) \otimes_k k')$ is a Cohen-Macaulay prescheme; as one of the two fields $k(x)$, k' is an extension of finite type of k by hypothesis, we are reduced to proving the following: $\square$

Lemma (6.7.1.1). — Let K, L be two extensions of a field k , one of which is of finite type (so that the ring $K \otimes_k L$ is Noetherian). Then $K \otimes_k L$ is a Cohen-Macaulay ring.

Proof. Suppose for example that L is an extension of finite type of k , so that L is extension of a pure extension $k(\mathbf{t})$ of k ($\mathbf{t}$ being a system $(t_i)_{1 \leq i \leq m}$ of indeterminates). We then set $A = K \otimes_k k(\mathbf{t})$, $B = K \otimes_k L$, B is an A -module flat (0, 6.2.1) and of finite type; the morphism $h : \text{Spec}(B) \rightarrow \text{Spec}(A)$ is closed and finite, and as every Artinian prescheme is of Cohen-Macaulay, to prove that B is a Cohen-Macaulay ring, it suffices to prove that A is a Cohen-Macaulay ring (6.3.5). Now, if S is the set of nonzero elements of $k[\mathbf{t}]$, then $A = S^{-1}A'$, where $A' = K \otimes_k k[\mathbf{t}] = K[\mathbf{t}]$; by (0, 16.5.13), it suffices to prove that A' is regular (0, 17.3.7).

B) Let (L_λ) be the filtering family of sub-extensions of L which are of finite type over k ; by virtue of A), each of the rings $C_\lambda = K \otimes_k L_\lambda$ is regular; furthermore, for $L_\lambda \subset L_\mu$, L_μ is an L_λ -module flat, so C_μ is an C_λ -module flat (0, 6.2.1). Since $C = K \otimes_k L = \varinjlim (K \otimes_k L_\lambda)$ is Noetherian, we can apply (5.13.7), and C is therefore regular. $\square$

Remarks (6.7.5). — (i) In the proof of the fact that $P(X, x)$ implies $P(X', x')$, one cannot dispense with the hypothesis that k' is a *separable* extension of k . We have indeed already seen (4.6.1) when P is the property d ; but even when X is *geometrically integral* (4.6.2), it can happen that X is *regular* without X' being *normal*.

(ii) As recalled in (i), it follows from (4.6.1) that when X is not a geometrically reduced algebraic prescheme over k , there exist finite radical extensions k' of k such that $X' = X \otimes_k k'$ is not reduced. It is interesting to give an example where X is a regular scheme over k , such that k is *algebraically closed* in the field K of rational functions of X . Let k be a field of characteristic $p > 0$, in which there exist two elements a, b forming a p -free family over k^p . Denoting again by B the ring $k[S, T]$, consider this time the polynomial $P(S, T) = T^p - aS^p - b$; as $aS^p + b$ is not a p -th power in $k(S)$, P is irreducible as a polynomial of $k(S)[T]$, and the scheme $X_0 = \text{Spec}(B/PB)$ is therefore an integral affine curve over k , whose field of rational functions is $K = k(S)[t]$, where t is a root of P ; let us show that k is *algebraically closed* in K . Suppose indeed that K contains an element z algebraic over k and not in k ; one would then also have $z \notin k(S)$, so $[k(S)[z] : k(S)] > 1$; as $[K : k(S)] = p$, and as K is radical over $k(S)$, one would have $z^p \in k(S)$, so $c = z^p \in k$ since z is algebraic over k ; but then one would have $t^p = aS^p + b \in k^p(S)$ hence a and b would belong to k^p , which is absurd. So X is the normalisation of the curve X_0 in K , which is therefore a normal curve (and hence *regular*) over k . If $k' = k(a^{1/p}, b^{1/p})$, it is clear that $aS^p + b$ is a p -th power in $k'(S)$, hence $K \otimes_k k'$ is *not reduced*, nor a fortiori the scheme $X \otimes_k k'$.

Definition (6.7.6). — Let k be a field, X a k -prescheme locally Noetherian. We say that X is *geometrically regular* at a point x (resp. has the property (R_n) *geometric*) at a point x , resp. is geometrically normal at a point x if, for every finite extension k' of k , $X' = X_{(k')} = X \otimes_k k'$ is regular (resp. has the property (R_n) , resp. is normal) at every point x' of X' whose projection in X is x . We say that X is *geometrically regular* (resp. has the property (R_n) *geometric*, or equivalently is geometrically regular in codimension $\leq n$), resp. is *geometrically normal* if X is geometrically regular (resp. has the property (R_n) *geometric*, resp. is geometrically normal) at every point.

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We say that an algebra A over k is a *geometrically regular ring* (resp. geometrically normal, resp. geometrically reduced, resp. having the property (R_n) geometric) if $\text{Spec}(A)$ has the same property.

If $X = \text{Spec}(K)$, where K is an extension of k , then saying that X is geometrically regular, or geometrically normal, or *geometrically reduced*, amounts to saying that K is a *separable* extension of k : this follows from (4.6.1) and the fact that if K is a separable extension of k and k' a finite extension of k , $K \otimes_k k'$ is composed of a direct product of a finite number of fields (Bourbaki, *Alg.*, chap. VIII, § 7, no 3, cor. 1 of th. 1).

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Proposition (6.7.7). — *Let k be a field, X a k -prescheme locally Noetherian, x a point of X ; denote by $Q(k')$ the relation “ k' is an extension of k , and $P(X_{(k')}, x')$ is true for every point x' of $X_{(k')}$ whose projection in X is x ”, where $P(Z, z)$ is one of the properties a), b), c) of the statement (6.7.4). Then the following properties are equivalent:*

- a) $Q(k')$ is true for every finite extension k' of k .
- b) $Q(k')$ is true for every finite radicial extension k' of k .
- c) $Q(k')$ is true for every extension of finite type k' of k .

Suppose in addition that X is locally of finite type over k ; then the three preceding properties are also equivalent to the following:

- d) $Q(k')$ is true for every extension k' of k .
- e) $Q(k')$ is true for a perfect extension k' of k .
- f) $Q(k')$ is true for every extension $k' = k^{p^{-s}}$, where p is the characteristic exponent of k , and $s > 0$.

Proof. To prove the equivalence of a), b) and c), it clearly suffices to show that b) implies c). Let K be then an extension of finite type of k , which is then the field of fractions of a k -algebra of finite type A ; set $Y = \text{Spec}(A)$. We know (4.6.6) that there exists a finite and radicial extension k' of k such that $Y' = (Y \otimes_k k')_{\text{red}}$ is separable over k' ; if η' is the generic point of Y' , $K' = k(\eta')$ is a separable extension of k' by (4.6.1). This being so, $P(X_{(k')}, x')$ is true by hypothesis for every point x' of $X_{(k')}$ above x , so it follows from (6.7.4) that $P(X_{(K')}, x'')$ is true for every point x'' of $X_{(K')}$ above x , since x'' is above a point x' of $X_{(k')}$, itself above x , and that K' is separable over k' . But we also have $X_{(K')} = (X_{(K)})_{(K')}$ and for every $x_0 \in X_{(K)}$ above x , there exists $x'' \in X_{(K')}$ above x_0 ; it therefore follows from (6.7.4) that $P(X_{(K)}, x_0)$ is true.

Suppose now that X is locally of finite type over k , so that for every extension K of k , $X_{(K)}$ is locally Noetherian. As a radicial extension of k is isomorphic to a sub-extension of a perfect extension of k , it follows immediately from (6.7.4) that e) implies f); also, every finite radicial extension of k is contained in an extension $k^{p^{-s}}$, so f) implies b); d) trivially implies e), and finally e) implies d). Indeed, if K is any extension whatever of k , there exists an extension K' of k containing (up to k -isomorphisms) k' and K ; since K' is a separable extension of k' , one deduces from (6.7.4) that $Q(k')$ is true, then that $Q(K)$ is true, reasoning as in the first part of the proof. It therefore suffices to prove that b) implies e). We shall take $k' = k^{p^{-\infty}}$ in e).

The question being local on X , one can furthermore limit oneself to the case where X is affine and of finite type over k , by replacing X by a neighbourhood of x ; let then $X = \text{Spec}(B)$, where B is a k -algebra of finite type; furthermore, by definition of the property P in the cases considered, one can also replace X by the local scheme of X at the point x , hence suppose that B is a *local* Noetherian ring. Set $B' = B \otimes_k k'$; k' is the inductive limit of its finite sub-extensions k'_λ and if one sets $B'_\lambda = B \otimes_k k'_\lambda$, the morphism $f_\lambda : \text{Spec}(B') \rightarrow \text{Spec}(B'_\lambda)$ is a *homeomorphism* over $\text{Spec}(B'_\lambda)$ for every λ ;

indeed f_λ is bijective by virtue of (I, 3.5.2), (3.5.7) and (3.5.8); on the other hand, it is closed by (II, 6.1.10). One can now apply (5.13.6) which proves (by virtue of the hypothesis b)) that b) implies e) when $P(Z, z)$ is the property (R_n) at the point z . The case where P is the property of being regular (i.e. of having the property (R_n) for every n) follows trivially. Finally, taking into account Serre's normality criterion (5.8.6), b) implies e) further when $P(Z, z)$ is the property of being normal at the point z , since this is equivalent to saying that Z has at the point z the properties (S_2) and (R_1) : one applies what precedes for $n = 1$, and (6.7.2), (ii) for $n = 2$. $\square$

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Corollary (6.7.8). — *Let k be a field; for a k -prescheme locally Noetherian X and a point $x \in X$, denote by $P(X, x)$ any of the following properties:*

- (i) $\text{coprof}(\mathcal{O}_x) \leq n$.
- (ii) $\mathcal{O}_x$ is a Cohen-Macaulay ring.
- (iii) X satisfies the property (S_n) at the point x .
- (iv) X is geometrically regular at the point x .
- (v) X has the property (R_n) geometric at the point x .
- (vi) X is geometrically normal at the point x .
- (vii) X is geometrically reduced (i.e. separable) at the point x .

Let k' be an extension of k ; one supposes either that k' is of finite type, or that X is locally of finite type over k , so that $X' = X \otimes_k k'$ is locally Noetherian. Let x' be a point of X' whose projection in X is x . Then, for $P(X, x)$ to be true, it is necessary and sufficient that $P(X', x')$ be so.

Proof. This has already been seen for the property (vii) (4.6.11), and for the properties (i), (ii) and (iii) (6.7.1). For (iv), (v) and (vi), this follows from (6.7.7): indeed, the fact that the condition is necessary follows from the equivalence of the criteria $a)$ and $c)$, and the equivalence of $c)$ and $d)$ when X is locally of finite type over k . To see that the condition is sufficient, let k'' be a finite radicial extension of k ; one can always consider k' and k'' as sub-extensions of an extension K of k ; set $X'' = X \otimes_k k''$, $X_0 = X \otimes_k K$, and note that since k'' is a radicial extension of k , there is only one point x'' of X'' above x (I, 3.5.7) and (3.5.8). If then x_0 is any point of X_0 above x , it is of the same kind as $P(X_0, x_0)$ by virtue of (6.7.7), $c)$ and $d)$, and if k' is an extension of k of finite type, one can suppose that it is also locally of finite type over k' , X' is also locally of finite type. One deduces then from (6.7.4) that the property $Q(k'')$ is true (with the notations of (6.7.7)), hence $P(X, x)$ is true by (6.7.7), $b)$). $\square$

6.8. Regular, normal, reduced, smooth morphisms

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Definition (6.8.1). — Let X, Y be two preschemes, $f : X \rightarrow Y$ a morphism such that the fibre $f^{-1}(y)$ is a locally Noetherian prescheme for every $y \in Y$, x a point of X . We say that f is respectively a morphism:

- (i) of coprofundity $\leq n$ at the point x ;
- (ii) of Cohen-Macaulay at the point x ;
- (iii) (S_n) at the point x ;
- (iv) regular at the point x ;
- (v) (R_n) at the point x ;
- (vi) normal at the point x ;
- (vii) reduced at the point x ;

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if f is flat at the point x , and if in addition the corresponding property $P(f^{-1}(f(x)), x)$ (notation of (6.7.8)) is true.

We say that f is *smooth* at the point x if it is locally of finite presentation in a neighbourhood of x and if it is regular at the point x . We say respectively that f is a morphism: of coprofundity $\leq n$, of Cohen-Macaulay, (S_n) , regular, (R_n) , normal, reduced, smooth, if it has the corresponding property at every point of X .

Proposition (6.8.2). — Let X, Y be two locally Noetherian preschemes, $f : X \rightarrow Y$ a morphism, x a point of X . Denote by $\mathbf{M}(f, x)$ one of the properties (i) to (vii) of the definition (6.8.1), or the property of being smooth at the point x . Let Y' be a locally Noetherian prescheme, $g : Y' \rightarrow Y$ a morphism, $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \rightarrow Y'$. Suppose that f or g is locally of finite type. Then, for every $x' \in X'$ above x , the property $\mathbf{M}(f, x)$ implies $\mathbf{M}(f', x')$.

Proof. Set $y = f(x)$, $y' = f'(x')$; by transitivity of fibres (I, 3.6.4), since either $f^{-1}(y)$ or $k(y')$ is locally of finite type over $k(y)$ (I, 6.4.11), it follows from (6.7.8) that the properties $P(f^{-1}(y), x)$ and $P(f'^{-1}(y'), x')$ are equivalent; furthermore, since f is flat at the point x , f' is flat at the point x' (2.1.4), which proves the proposition, taking into account (1.4.3), (iii). $\square$

Proposition (6.8.3). — For a morphism f of locally Noetherian preschemes, let $\mathbf{M}(f)$ be one of the following properties: being Cohen-Macaulay, (S_n) , regular, (R_n) , normal, reduced.

- (i) Let X, Y, Z be three locally Noetherian preschemes, $f : X \rightarrow Y$, $g : Y \rightarrow Z$ two morphisms. If $\mathbf{M}(f)$ and $\mathbf{M}(g)$ are true, $\mathbf{M}(g \circ f)$ is true.
- (ii) Conversely, if f is surjective and if $\mathbf{M}(f)$ and $\mathbf{M}(g \circ f)$ are true, then $\mathbf{M}(g)$ is true.

- (iii) Let X, Y, Y' be three locally Noetherian preschemes, $f : X \rightarrow Y, h : Y' \rightarrow Y$ two morphisms; set $X' = X \times_Y Y', f' = f_{(Y')} : X' \rightarrow Y'$. Suppose that f or h is locally of finite type. If $M(f)$ is true, then so is $M(f')$; the converse is true when h is faithfully flat.

Proof. The conclusions of (i) and (iii) are also true when M is the property of being smooth and (in (iii)) h is quasi-compact.

(i) We already know that if f and g are flat, so is $u = g \circ f$, and that if f and $g \circ f$ are flat and f is surjective for every $z \in Z$, the morphism $f_z = f \otimes \mathbf{1}_{k(z)} : u^{-1}(z) \rightarrow g^{-1}(z)$ is flat (resp. faithfully flat if f is) and for every $y \in g^{-1}(z)$, $f_z^{-1}(y)$ is isomorphic to $f^{-1}(y)$ (I, 3.6.4). If $M(f)$ and $M(g)$ are true, $M(g \circ f)$ is then true for the cases where M is the property of being Cohen-Macaulay or (S_n) , by virtue of (6.6.1). On the other hand, let k' be a finite extension of $k(z)$; setting $Y'_z = g^{-1}(z) \otimes_{k(z)} k', X'_z = u^{-1}(z) \otimes_{k(z)} k'$, and $f'_z = f_z \otimes \mathbf{1}_{k'} : X'_z \rightarrow Y'_z$; the morphism f'_z

is flat (resp. faithfully flat) and for every $y' \in Y'_z$, the fibre $f'^{-1}_z(y')$ is isomorphic to $f^{-1}(y) \otimes_{k(y)} k(y')$, denoting by y the image of y' in $g^{-1}(z)$ (I, 3.6.4). When M is the property of being regular, (R_n) , normal or reduced, the hypothesis that $M(f)$ and $M(g)$ are true implies that Y'_z and each of the fibres $f'^{-1}_z(y')$ for $y' \in Y'_z$ have the corresponding property among the properties $c), d), e), f)$ of (6.6.1); one deduces from (6.6.1), (i) that X'_z has the same property, hence $M(g \circ f)$ is true.

(ii) Conversely, the hypothesis that $M(g \circ f)$ and $M(f)$ are true and that f is surjective implies from (6.6.1), (ii), f'_z being surjective for every z ; hence $M(g)$ is true.

(iii) The first assertion follows immediately from (6.8.2). On the other hand, if h is faithfully flat and f' flat, f is flat (2.4.1); since, with the notations of (6.8.2), the properties $P(f^{-1}(y), x)$ and $P(f'^{-1}(y'), x')$ are equivalent (6.7.8), one sees that $M(f)$ and $M(f')$ are then equivalent.

Finally, the last assertion of the proposition follows from (1.4.3), (iii) and (2.7.1), (iv). $\square$

Remarks (6.8.4). — (i) If f is faithfully flat, g flat and if $g \circ f$ is of coprofundity $\leq n$, then g is of coprofundity $\leq n$, as follows from the reasoning of (6.6.2).

(ii) When, in (6.8.1), one takes for Y the spectrum of a field k , the notions (iv), (v) and (vi) reduce to those defined in (6.7.5). It is clear that these latter are relative to the base field k . The definition (6.8.1) thus leads to saying that a prescheme X is “regular (resp. (R_n) , resp. normal) over k ” instead of saying that it is “geometrically regular (resp. (R_n) , resp. normal) relative to k ”; one will avoid confusing this notion with the property of being regular (resp. (R_n) , resp. normal) which is independent of k . One can make in this connection the same remarks as in (4.5.12).

Proposition (6.8.5). — Let k be a field, X, Y two k -preschemes locally Noetherian, at least one of which is locally of finite type over k . For a k -prescheme Z , denote by $P(Z)$ one of the properties $c)$ to $f)$ of (6.6.1); the property “ $P(Z)$ geometric” is then defined in (6.7.6) (resp. (4.6.1)). Then:

- (i) If X has the property P , and Y the property P geometric, $X \times_k Y$ has the property P .
- (ii) If X and Y have the property P geometric, so does $X \times_k Y$.

Proof. Indeed, let $f : X \times_k Y \rightarrow X, g : X \rightarrow \text{Spec}(k)$ be the structural morphisms, which are faithfully flat (2.2.13). The hypothesis that Y has the property P geometric implies by virtue of (6.8.2) that $M(f)$ is true, M being the property (6.8.3) which corresponds to P ; under the hypothesis (ii), $M(g)$ is also true, hence the assertion (ii) follows from (6.8.3), (i). As for the assertion (i), it follows directly from (6.6.1), (i). $\square$

Theorem (6.8.6). — Let X, Y be two locally Noetherian preschemes, $f : X \rightarrow Y$ a morphism locally of finite type, x a point of $X, y = f(x)$. The following properties are equivalent:

- a) f is smooth at the point x .
- b) f is regular at the point x .
- c) $\mathcal{O}_x$ is a formally smooth $\mathcal{O}_y$ -algebra (0, 19.3.1) for the adic topologies on $\mathcal{O}_x$ and $\mathcal{O}_y$.
- c') $\mathcal{O}_x$ is a formally smooth $\mathcal{O}_y$ -algebra (0, 19.3.1) for the discrete topologies on $\mathcal{O}_x$ and $\mathcal{O}_y$.

Proof. The equivalence of a) and b) follows from the definitions (6.8.1) and from the fact that for locally Noetherian preschemes, morphisms locally of finite type are locally of finite presentation (1.4.2).

Secondly, for $\mathcal{O}_x$ to be a formally smooth $\mathcal{O}_y$ -algebra for the adic topologies, it is necessary and sufficient that $\mathcal{O}_x$ be flat over $\mathcal{O}_y$ and that $\mathcal{O}_x \otimes_{\mathcal{O}_y} k(y)$ be formally smooth over $k(y)$ for its adic

topology (0, 19.7.1). The latter condition is equivalent to geometric regularity over $k(y)$ (0, 19.6.6); this proves the equivalence of $b)$ and $c)$. Finally, to prove the equivalence of $c)$ and $c')$, one may reduce to the affine case $Y = \text{Spec}(A)$, $X = \text{Spec}(C)$, where A is Noetherian and C is a finite-type Noetherian A -algebra, say $C = A[T_1, \dots, T_n]/\mathfrak{J}$. Since $\mathfrak{J}/\mathfrak{J}^2$ is a finitely presented C -module, the equivalence follows from (0, 22.6.4). $\square$

Corollary (6.8.7). — *Let X, Y be two locally Noetherian preschemes, $f : X \rightarrow Y$ a morphism locally of finite type. Then the set of points $x \in X$ where f is smooth (or regular) is open in X .*

Proof. Indeed it follows from (0, 22.6.5) that the set of $x \in X$ satisfying the condition $c')$ of (6.8.6) is open in X , and we conclude by (6.8.6). $\square$

Remark (6.8.8). — In (17.5.1), we shall show that the equivalence of $b)$ and $c')$ in (6.8.6), as well as the corollary (6.8.7), remain valid without the Noetherian hypothesis on X and Y , provided that one restricts to morphisms *locally of finite presentation*.

6.9. The theorem of generic flatness

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Theorem (6.9.1). — *Let Y be a locally Noetherian and integral prescheme, $u : X \rightarrow Y$ a morphism of finite type, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module. There then exists a non-empty open subset U of Y such that $\mathcal{F}|_{u^{-1}(U)}$ is flat over U .*

Proof. One can clearly limit oneself to the case where $Y = \text{Spec}(A)$ is affine; then X is a finite union of affine opens X_i of finite type over Y ; if, for each i , there exists a non-empty open U_i in Y such that $\mathcal{F}|_{(X_i \cap u^{-1}(U_i))}$ is flat over U_i , it is clear that taking for U the intersection of the U_i , we shall have answered the question; one can therefore suppose that $X = \text{Spec}(B)$, where B is an A -algebra of finite type. We then have $\mathcal{F} = \bar{M}$, where M is a B -module of finite type; the theorem will follow from the following: $\square$

Lemma (6.9.2). — *Let A be an integral Noetherian ring, B a finite-type A -algebra, and M a finite B -module. Then there exists $f \neq 0$ in A such that M_f is a free A_f -module.*

Proof. Denote by K the field of fractions of A ; then $B \otimes_A K$ is an algebra of finite type over K and $M \otimes_A K$ is a $(B \otimes_A K)$ -module of finite type. We shall reason by induction on the *dimension* n of the support of $M \otimes_A K$, which is either $-\infty$ or finite and ≥ 0 . Suppose first $n = -\infty$, i.e. $M \otimes_A K = 0$; if $(m_i)_{1 \leq i \leq r}$ is a system of generators of the B -module M , there therefore exists $f \neq 0$ in A such that $f m_i = 0$ for $1 \leq i \leq r$; hence $M_f = 0$ and the lemma is true in this case. Suppose now $n \geq 0$. We know that there exists a composition series $M = M_1 \supset M_2 \supset \dots \supset M_q = 0$ of the B -module M such that each of the quotients $N_i = M_i/M_{i+1}$ is isomorphic to a B -module of the form $B/\mathfrak{p}_i$, where $\mathfrak{p}_i$ is a prime ideal of B (Bourbaki, *Alg. comm.*, chap. IV, § 1, no 4, th. 1). If the theorem is true for each of the N_i , there exists for each i an $f_i \neq 0$ in A such that $(N_i)_{f_i}$ is an A_{f_i} -module free for $1 \leq i \leq q-1$. But $(N_i)_f = (M_i)_f / (M_{i+1})_f$ (0, 1.3.2) and since an extension of free modules is free, we deduce that M_f is free, setting $f = f_1 f_2 \dots f_{q-1}$, and it therefore suffices (in the case where $M = B$ and B is integral) to show that there exists $g \neq 0$ in A and elements t_i ($1 \leq i \leq m$) algebraically independent over A and such that B_g is integral over $A_g[t_1, \dots, t_m]$. One can replace A by A_g , B by B_g and suppose that B is integral over $C = A[t_1, \dots, t_m]$, hence a C -module of finite type and *without torsion*. We know furthermore (4.1.2) that the dimension of $\text{Spec}(B \otimes_A K)$ is equal to m , hence $m = n$.

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This being so, if h is the rank of the C -module without torsion B , there exists a short exact sequence of C -modules

$$0 \longrightarrow C^h \longrightarrow B \longrightarrow M' \longrightarrow 0$$

where M' is a torsion C -module of finite type; the support of M' does not therefore contain the generic point of $\text{Spec}(C)$ (I, 7.4.6) and hence the support of $M' \otimes_A K$ does not contain the generic point of $\text{Spec}(C \otimes_A K)$ (I, 9.1.13.1); one concludes (4.1.2.1) that its dimension is $< n$. By the induction hypothesis, there exists $f \neq 0$ in A such that M'_f is a free A_f -module; moreover C_f^h is a free A_f -module, and therefore so is B_f , which by (0, 1.3.2) is an extension of free modules. $\square$

Corollary (6.9.3). — *Let S be a Noetherian prescheme, $u : X \rightarrow S$ a morphism of finite type, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module. There then exists a partition $(S_i)_{1 \leq i \leq n}$ of S into a finite number of locally closed subsets of S ,*

such that, if one still denotes by S_i the reduced sub-prescheme having S_i for underlying space, and if one sets $X_i = X \times_S S_i$, the $\mathcal{O}_{X_i}$ -Module $\mathcal{F}_i = \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S_i}$ is flat over S_i .

Proof. We proceed by Noetherian induction (0, 2.2.2) on the set of closed subsets T of S such that the reduced closed sub-prescheme with underlying space T satisfies the conclusion of (6.9.3). One can therefore limit oneself to proving the corollary for S by supposing that it is true for every reduced closed sub-prescheme whose underlying space is a closed subset distinct from S . Since the morphism $S_{\text{red}} \rightarrow S$ is of finite type and surjective, one can replace S by S_{red} , hence suppose S reduced and non-empty. Since S is Noetherian, the interior T of an irreducible component of S is non-empty, and the induced prescheme on T is integral; there therefore exists by virtue of (6.9.1) a non-empty open $U \subset T$

such that $\mathcal{F}|_{U^{-1}(U)}$ is flat over U . If Y is then the reduced closed sub-prescheme of S with underlying space $S - U$, there exists by hypothesis a partition (Y_i) of Y into locally closed subsets of Y (hence of S) such that $\mathcal{F}_i = \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{Y_i}$ is flat over Y_i for every i ; it is clear that the Y_i and U form a partition answering the question. $\square$

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6.10. Dimension and depth of a module normally flat along a closed sub-prescheme

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(6.10.1). Let X be a locally Noetherian prescheme, $\mathcal{J}$ a quasi-coherent Ideal of $\mathcal{O}_X$, Y the closed sub-prescheme of X defined by $\mathcal{J}$, $j : Y \rightarrow X$ the canonical injection. For every integer $k \geq 0$, the $\mathcal{O}_X$ -Module $\mathcal{J}^k \mathcal{F} / \mathcal{J}^{k+1} \mathcal{F}$ is annihilated by $\mathcal{J}$, hence of the form $j_*(\mathcal{G}_k)$, where $\mathcal{G}_k = j^*(\mathcal{J}^k \mathcal{F} / \mathcal{J}^{k+1} \mathcal{F}) = j^*(\mathcal{J}^k \mathcal{F})$ is a coherent $\mathcal{O}_Y$ -Module. By abuse of language, we shall denote by $\text{gr}_{\mathcal{J}}^{\bullet}(\mathcal{F})$ the graded $\mathcal{O}_Y$ -Module equal to the direct sum

$$\bigoplus_{k=0}^{\infty} \mathcal{G}_k = j^* \left(\bigoplus_{k=0}^{\infty} \mathcal{J}^k \mathcal{F} / \mathcal{J}^{k+1} \mathcal{F} \right);$$

in particular, we have $\text{gr}_{\mathcal{J}}^0(\mathcal{F}) = \mathcal{G}_0 = \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_Y = j^*(\mathcal{F})$. We shall say (following Hironaka) that $\mathcal{F}$ is *normally flat along* Y if $\text{gr}_{\mathcal{J}}^{\bullet}(\mathcal{F})$ is a flat $\mathcal{O}_Y$ -Module. This amounts to the same (2.1.2) and (0, 6.1.2) as saying that each of the $\mathcal{O}_Y$ -Modules $\mathcal{G}_k = j^*(\mathcal{J}^k \mathcal{F})$ is *locally free*.

Proposition (6.10.2). — *Let X be a locally Noetherian prescheme, Y a closed integral sub-prescheme of X . For every coherent $\mathcal{O}_X$ -Module $\mathcal{F}$, there exists an open U of X such that $Y \cap U \neq \emptyset$ and that $\mathcal{F}|_U$ is normally flat along $Y \cap U$.*

Proof. Indeed, let $\mathcal{J}$ be the coherent Ideal of $\mathcal{O}_X$ defining Y ; the $\mathcal{O}_Y$ -Algebra $\mathcal{B} = \text{gr}_{\mathcal{J}}^{\bullet}(\mathcal{O}_X)$ is quasi-coherent and of finite type car elle est engendrée par $\text{gr}_{\mathcal{J}}^1(\mathcal{O}_X)$, inverse image of $\mathcal{J} / \mathcal{J}^2$; as $\text{gr}_{\mathcal{J}}^{\bullet}(\mathcal{F})$ is a $\mathcal{B}$ -module quasi-coherent generated by $\text{gr}_{\mathcal{J}}^{\bullet}(\mathcal{F})$, it is a $\mathcal{B}$ -Module of finite type. If one sets $Z = \text{Spec}(\mathcal{B})$, the structural morphism $u : Z \rightarrow Y$ is of finite type, and if $\mathcal{H}$ is the coherent $\mathcal{O}_Z$ -Module such that $u_*(\mathcal{H}) = \text{gr}_{\mathcal{J}}^{\bullet}(\mathcal{F})$, it suffices to apply to u and to $\mathcal{H}$ the theorem of generic flatness (6.9.1) to prove the proposition. $\square$

Proposition (6.10.3). — *The notations being those of (6.10.1), suppose that $\mathcal{F}$ is normally flat along Y . Then:*

- (i) $Y \cap \text{Supp}(\mathcal{F})$ is a part both open and closed of Y (in other words, a union of connected components of Y).
- (ii) If $\mathcal{J}$ is locally nilpotent, $\text{Supp}(\mathcal{F})$ is an open and closed part of X , and for every $x \in \text{Supp}(\mathcal{F})$, we have

$$(6.10.3.1) \quad \dim(\mathcal{F}_x) = \dim(\mathcal{O}_{Y,x})$$

$$(6.10.3.2) \quad \text{prof}(\mathcal{F}_x) = \text{prof}(\mathcal{O}_{Y,x})$$

$$(6.10.3.3) \quad \text{coprof}(\mathcal{F}_x) = \text{coprof}(\mathcal{O}_{Y,x}).$$

Proof. (i) With the notations of (6.10.1), we have $Y \cap \text{Supp}(\mathcal{F}) = \text{Supp}(\mathcal{G}_0)$ (I, 9.1.13), and as by hypothesis $\mathcal{G}_0$ is a locally free $\mathcal{O}_Y$ -Module of finite type, its support is at the same time open and closed in Y .

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(ii) The hypothesis that $\mathcal{J}$ is locally nilpotent implies that the underlying spaces of X and Y are the same, whence the two first assertions of (ii), taking into account (i); it remains to prove (6.10.3.2) and (6.10.3.3) by difference. Let $(f_i)_{1 \leq i \leq p}$ be a finite family of elements of the maximal ideal of $\mathcal{O}_{X,x}$ whose images in $\mathcal{O}_{Y,x} = \mathcal{O}_{X,x}/\mathcal{J}_x$ form a maximal $\mathcal{O}_{Y,x}$ -regular sequence; the hypothesis on $\mathcal{F}$ and $\mathcal{J}$ implies that the $\mathcal{O}_{Y,x}$ -module $\mathrm{gr}_{\mathcal{J}_x}^\bullet(\mathcal{F}_x)$ is free of finite type, hence the suite (f_i) is $\mathrm{gr}_{\mathcal{J}_x}^\bullet(\mathcal{F}_x)$ -regular; one deduces that this suite is also $\mathcal{F}_x$ -regular (0, 15.1.19). On the other hand, let n be the largest integer such that $\mathcal{F}_x^{(n)} = \mathcal{J}_x^n \mathcal{F}_x \neq 0$. The hypothesis implies also that $\mathrm{gr}_{\mathcal{J}_x}^\bullet(\mathcal{F}_x/\mathcal{F}_x^{(n)})$ is free of finite type, hence the suite (f_i) is also $(\mathcal{F}_x/\mathcal{F}_x^{(n)})$ -regular (*loc. cit.*). Applying the lemma (3.4.1.4), one concludes by induction on i that there is an exact sequence

$$0 \longrightarrow \mathcal{F}_x^{(n)} / \left(\sum_{i=1}^p f_i \mathcal{F}_x^{(n)} \right) \longrightarrow \mathcal{F}_x / \left(\sum_{i=1}^p f_i \mathcal{F}_x \right).$$

But the hypothesis implies that $\mathcal{F}_x^{(n)}$ is a free $\mathcal{O}_{Y,x}$ -module of finite type and $\neq 0$, hence $\mathcal{F}_x^{(n)} / (\sum_{i=1}^p f_i \mathcal{F}_x^{(n)})$ is isomorphic to a module of the form $(\mathcal{O}_{Y,x} / (\sum_{i=1}^p f_i \mathcal{O}_{Y,x}))^m$ with $m > 0$; as $\mathrm{prof}(\mathcal{O}_{Y,x} / (\sum_{i=1}^p f_i \mathcal{O}_{Y,x})) = 0$ (0, 16.4.6) one also has, by the characterisation (0, 16.4.6), (i) of modules of zero depth, $\mathrm{prof}(\mathcal{F}_x^{(n)} / (\sum_{i=1}^p f_i \mathcal{F}_x^{(n)})) = 0$, then $\mathrm{prof}(\mathcal{F}_x / (\sum_{i=1}^p f_i \mathcal{F}_x)) = 0$; the $(f_i)_x$ belonging to the maximal ideal of $\mathcal{O}_{Y,x}$ and forming a maximal $\mathcal{F}_x$ -regular suite, this shows that $\mathrm{prof}(\mathcal{F}_x) = p$ (0, 16.4.6), (ii). $\square$

Corollary (6.10.4). — Let $U_{S_n}(\mathcal{F})$ (resp. $U_{C_n}(\mathcal{F})$) be the set of $x \in X$ such that $\mathcal{F}$ satisfies (S_n) (resp. such that $\mathrm{coprof}(\mathcal{F}_x) \leq n$). If $\mathcal{F}$ is normally flat along Y , and if $\mathcal{J}$ is locally nilpotent, then $U_{S_n}(\mathcal{F}) = U_{S_n}(\mathcal{O}_Y)$ and $U_{C_n}(\mathcal{F}) = U_{C_n}(\mathcal{O}_Y)$.

Proposition (6.10.5). — The notations being those of (6.10.1), suppose that Y is connected and non-empty, and that $\mathcal{F}$ is normally flat along Y . For every integer $n \geq 0$, set

$$(6.10.5.1) \quad r(n) = \mathrm{rg}_{\mathcal{O}_Y}(\mathrm{gr}_{\mathcal{J}}^n(\mathcal{F}))$$

(the locally free $\mathcal{O}_Y$ -Module $\mathrm{gr}_{\mathcal{J}}^n(\mathcal{F})$ being necessarily of constant rank). Then:

- (i) There exists a polynomial $P \in \mathbf{Q}[T]$ such that $P(n) = r(n)$ for all sufficiently large n .
- (ii) We have $Y \cap \mathrm{Supp}(\mathcal{F}) = \emptyset$ (in other words $\mathcal{J}|_{\mathcal{F}} = 0$), or $Y \cap \mathrm{Supp}(\mathcal{F}) = Y$ (in other words $Y \subset \mathrm{Supp}(\mathcal{F})$).

In the second case, denote $d - 1$ the degree of P ; for every maximal point y of Y , we have

$$(6.10.5.2) \quad \dim(\mathcal{F}_y) = d$$

and in particular

$$(6.10.5.3) \quad \mathrm{codim}(Y, \mathrm{Supp}(\mathcal{F})) = d.$$

(iii) Suppose $Y \subset \mathrm{Supp}(\mathcal{F})$. For every $x \in Y$, we have

$$(6.10.5.4) \quad \dim(\mathcal{F}_x) = \dim(\mathcal{O}_{Y,x}) + d.$$

More precisely, there exist in $\mathcal{J}_x$ elements f_i ($1 \leq i \leq d$) forming part of a system of parameters for $\mathcal{F}_x$ (0, 16.3.6) and such that, in $\mathrm{Spec}(\mathcal{O}_{X,x})$, we have $V(\mathcal{J}_x) = V(\sum_{i=1}^d f_i \mathcal{O}_{X,x}) \cap \mathrm{Supp}(\mathcal{F})$.

Proof. As Y is assumed connected, the first assertion of (ii) follows from (6.10.3), (i). If $Y \cap \mathrm{Supp}(\mathcal{F}) = \emptyset$, assertion (i) is trivial with $P = 0$. Suppose now that $\mathrm{Supp}(\mathcal{F}) \supset Y$, and let y be a maximal point of Y . Since then $Y_y = \mathrm{Supp}(\mathcal{O}_{Y,y})$, the module $\mathcal{F}_y/\mathcal{J}_y \mathcal{F}_y$ has finite length (3.1.4). Setting $s(n) = \mathrm{length}(\mathrm{gr}_{\mathcal{J}_y}^n(\mathcal{F}_y))$, we know that there exists a polynomial $R \in \mathbf{Q}[T]$ such that $s(n) = R(n)$ for n large enough, namely $R(n) = H(n+1) - H(n)$, where H is the Hilbert–Samuel polynomial of $\mathcal{F}_y$ for the $\mathcal{J}_y$ -adic filtration (0, 16.2.1). Since the $\mathcal{O}_{Y,y}$ -modules $\mathrm{gr}_{\mathcal{J}_y}^n(\mathcal{F}_y)$ are free by hypothesis, we have $r(n) = s(n)/m$, denoting by m the length of the $\mathcal{O}_{X,y}$ -module $\mathcal{O}_{Y,y} = \mathcal{O}_{X,y}/\mathcal{J}_y$; thus (i) holds with $P = \frac{1}{m}R$. We know furthermore (0, 16.2.3) that $\deg(H) = \dim(\mathcal{F}_y)$, whence the relation (6.10.5.2); the relation (6.10.5.3) follows from (5.1.12.2).

It remains to prove (iii) for any point $x \in Y$. Set $A = \mathcal{O}_{X,x}$, $\mathfrak{J} = \mathcal{J}_x$, and $M = \mathcal{F}_x$, so that $\mathcal{O}_{Y,x} = A/\mathfrak{J}$; let $S = \mathrm{gr}_{\mathfrak{J}}^\bullet(A)$, which is a graded $(A/\mathfrak{J})$ -algebra of finite type, with positive degrees,

such that $S_0 = A/\mathfrak{J}$, and generated by its homogeneous elements of degree 1; let finally $N = \text{gr}_{\mathfrak{J}}^{\bullet}(M)$, which is a graded S -module of finite type, each homogeneous component N_n being by hypothesis a free $(A/\mathfrak{J})$ -module of rank $r(n)$. Let $\mathfrak{m}$ be the maximal ideal of A , $k = A/\mathfrak{m}$ its residue field; $B = S \otimes_A k$ is a graded k -algebra of finite type, with positive degrees, generated by its homogeneous elements of degree 1 and such that $B_0 = k$, so that $\mathfrak{q} = B^+ = \bigoplus_{n \geq 1} B_n$ is a maximal ideal of B ; $E = N \otimes_A k$ is a graded B -module of finite type such that $\text{rg}_k(E_n) = r(n)$. Applying (0, 16.2.7) to B and E , choose homogeneous elements $t_i \in B_{n_i}$ ($1 \leq i \leq d$), and lift each t_i to an element $f_i \in \mathfrak{J}^{n_i}$. For $n \geq \sup(n_i)$, consider the A -submodule $\sum_{i=1}^d f_i \mathfrak{J}^{n-n_i} M$ of M ; as the homogeneous component of degree n of the submodule $\sum_i t_i E$ of E is equal to E_n as soon as n is large enough, one sees that, for n large enough, we have

$$\mathfrak{J}^n M = \sum_{i=1}^d f_i \mathfrak{J}^{n-n_i} M + \mathfrak{m} \mathfrak{J}^n M$$

and as $\mathfrak{J}^n M$ is an A -module of finite type, this implies, by Nakayama's lemma,

$$\mathfrak{J}^n M = \sum_{i=1}^d f_i \mathfrak{J}^{n-n_i} M \subset \sum_{i=1}^d f_i M.$$

If $\mathfrak{a}$ is the annihilator of M , we have therefore (0, 1.7.5) in $\text{Spec}(A)$

$$V\left(\sum_{i=1}^d f_i A\right) \cap V(\mathfrak{a}) \subset V(\mathfrak{J}^n) \cap V(\mathfrak{a}) = V(\mathfrak{J}) \cap V(\mathfrak{a}) = V(\mathfrak{J})$$

since by hypothesis $Y \cap \text{Spec}(A) = V(\mathfrak{J}) \cap \text{Supp}(M) = V(\mathfrak{J})$. Since on the other hand the f_i belong to S , we have $V(\mathfrak{J}) \subset V(\sum_{i=1}^d f_i A)$, which proves the last relation of (iii). It remains to show that the f_i form part of a system of parameters for M . Replacing A by $A/\mathfrak{a}$ and the f_i by their images in $A/\mathfrak{a}$, one can limit oneself to the case where $\mathfrak{a} = 0$; one then has to show that we have $\dim(A/(\sum_{i=1}^d f_i A)) \leq \dim(A) - d$ and it remains therefore to prove (0, 16.3.7) that we have

$$\dim(A) \geq \dim(A/\mathfrak{J}) + d.$$

Now, let $\mathfrak{p}$ be a prime ideal of A containing $\mathfrak{J}$ and such that $\dim(A/\mathfrak{J}) = \dim(A/\mathfrak{p})$, so that $\mathfrak{p}$ is minimal among the prime ideals containing $\mathfrak{J}$. We have therefore $\mathfrak{p} = \mathfrak{i}_y$ where $y \in \text{Spec}(A)$ is a maximal point of Y . But by virtue of (6.10.5.2) and the hypothesis $\text{Spec}(A) = \text{Supp}(M)$, we have $\dim(A_{\mathfrak{p}}) = d$; the inequality $\dim(A/\mathfrak{p}) + \dim(A_{\mathfrak{p}}) \leq \dim(A)$ (0, 16.1.4) completes the proof. $\square$

Proposition (6.10.6). — *Let X be a locally Noetherian prescheme, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module, Y a closed irreducible sub-prescheme of X , of generic point $y \in \text{Supp}(\mathcal{F})$. There then exists a neighbourhood U open and non-empty of y in X such that, for every $x \in U \cap Y$, we have*

$$(6.10.6.1) \quad \dim(\mathcal{F}_x) = \dim(\mathcal{F}_y) + \dim(\mathcal{O}_{Y,x})$$

$$(6.10.6.2) \quad \text{prof}(\mathcal{F}_x) = \text{prof}(\mathcal{F}_y) + \text{prof}(\mathcal{O}_{Y,x})$$

$$(6.10.6.3) \quad \text{coprof}(\mathcal{F}_x) = \text{coprof}(\mathcal{F}_y) + \text{coprof}(\mathcal{O}_{Y,x}).$$

Proof. Let $Y' = Y_{\text{red}}$, which is a closed integral sub-prescheme of Y having the same underlying space, and defined by a locally nilpotent Ideal $\mathcal{J}$ of $\mathcal{O}_Y$ (I, 6.1.6). It follows that $\text{gr}_{\mathcal{J}}^{\bullet}(\mathcal{O}_v)$ is a coherent $\mathcal{O}_{v'}$ -Module, and as Y' is integral, there exists an open neighbourhood V of y in Y' such that $\text{gr}_{\mathcal{J}}^{\bullet}(\mathcal{O}_v)|_V$ is locally free (0, 5.2.7); otherwise said, replacing X by a neighbourhood of y , one can suppose that $\mathcal{O}_v$ is normally flat along Y' ; one then deduces from (6.10.3) that we have

$$\dim(\mathcal{O}_{Y,x}) = \dim(\mathcal{O}_{Y',x}), \quad \text{prof}(\mathcal{O}_{Y,x}) = \text{prof}(\mathcal{O}_{Y',x})$$

for every $x \in Y$; this allows us to limit ourselves to the case where the closed sub-prescheme Y is integral.

This being so, by virtue of (6.10.2), one can, replacing X by an open neighbourhood of y , suppose that $\mathcal{F}$ is normally flat along Y . Suppose now $p = \text{prof}(\mathcal{F}_y)$; replacing at need X by an open neighbourhood of y , one can further suppose that there exist p sections f_i ($1 \leq i \leq p$) of $\mathcal{O}_X$ above X , forming an $\mathcal{F}$ -regular suite, such that the $(f_i)_y$ belong to the maximal ideal $\mathfrak{m}_y$ of $\mathcal{O}_X$ at the point y and that $\text{prof}(\mathcal{F}_y/(\sum_{i=1}^p (f_i)_y \mathcal{F}_y)) = 0$ (0, 16.4.6). If $\mathfrak{J}$ is the Ideal of $\mathcal{O}_X$ defining Y , the hypothesis $\text{prof}(\mathcal{G}_y) = 0$ implies that $\mathcal{G}_y$ contains a sub-module isomorphic to $\mathcal{O}_{X,y}/\mathfrak{m}_y$ (0, 16.4.6); reasoning

as above (taking into account (0, 5.3.8) and (5.2.2)) shows that one can suppose that there exists a sub- $\mathcal{O}_X$ -Module $\mathcal{G}'$ of $\mathcal{G}$ isomorphic to $\mathcal{O}_X/\mathfrak{J}$ so that $\mathcal{G}'_x = \mathcal{O}_{Y,x}$ for every $x \in Y$. Setting $\mathcal{G}'' = \mathcal{G}/\mathcal{G}'$, replacing X by an open neighbourhood of y , one can, by virtue of (6.10.2), suppose that $\mathcal{G}'$ and $\mathcal{G}''$ are normally flat along Y . Let then x be any point of Y , and set $q = \text{prof}(\mathcal{O}_{Y,x}) = \text{prof}(\mathcal{G}'_x)$; let $(g_i)_{1 \leq i \leq q}$ be a maximal $\mathcal{G}''_x$ -regular suite of elements of the maximal ideal $\mathfrak{m}_x$ of $\mathcal{O}_{Y,x}$. Each of the homogeneous components of $\text{gr}_{\mathcal{G}''_x}^\bullet(\mathcal{G}''_x)$ is a flat $\mathcal{O}_{Y,x}$ -module of finite type by hypothesis, hence a free $\mathcal{O}_{Y,x}$ -module, and it is therefore also $\text{gr}_{\mathcal{G}''_x}^\bullet(\mathcal{G}''_x)$ -regular, hence it is also $\text{gr}_{\mathcal{G}''_x}^\bullet(\mathcal{G}''_x)$ -regular, and whence the result $\text{gr}_{\mathcal{G}''_x}^\bullet(\mathcal{G}''_x)$ is $\mathcal{G}''_x$ -regular (0, 15.1.19). Applying the lemma (0, 15.1.18) to the exact sequence

$$0 \longrightarrow \mathcal{G}'_x \longrightarrow \mathcal{G}_x \longrightarrow \mathcal{G}''_x \longrightarrow 0$$

by induction on j , one concludes an exact sequence

$$0 \longrightarrow \mathcal{G}'_x / \left(\sum_{j=1}^q g_j \mathcal{G}'_x \right) \longrightarrow \mathcal{G}_x / \left(\sum_{j=1}^q g_j \mathcal{G}_x \right).$$

But by hypothesis $\text{prof}(\mathcal{G}'_x / (\sum_{j=1}^q g_j \mathcal{G}'_x)) = 0$ (0, 16.4.6); by the characterisation of modules of zero depth, one concludes that $\text{prof}(\mathcal{G}_x / (\sum_{j=1}^q g_j \mathcal{G}_x)) = 0 = \text{prof}(\mathcal{F}_x / (\sum_{i=1}^p f_i \mathcal{F}_x + \sum_{j=1}^q g_j \mathcal{F}_x))$; the suite (g_i) being $\mathcal{G}''_x$ -regular and $\mathcal{G}''_x$ being $\mathcal{F}_x$ -regular, the suite $((f_i)_x)$ is $\mathcal{F}_x$ -regular by hypothesis and is formed of elements of the maximal ideal; one deduces (0, 16.4.6) that $\text{prof}(\mathcal{F}_x) = p + q$, which completes the proof. $\square$

6.11. Criteria for the sets $U_{S_n}(\mathcal{F})$ or $U_{C_n}(\mathcal{F})$ to be open

Lemma (6.11.1). — *Let X be a locally Noetherian prescheme, $\mathcal{F}$ an $\mathcal{O}_X$ -Module coherent; then the function $x \mapsto \dim. \text{proj}(\mathcal{F}_x)$ is upper semi-continuous in X .*

Proof. One can restrict to the case where $X = \text{Spec}(A)$ is the spectrum of a Noetherian ring and $\mathcal{F} = \tilde{M}$, where M is an A -module of finite type. Suppose that for an $x \in X$, we have $\dim. \text{proj}(M_x) = n < +\infty$ (if $n = +\infty$ there is nothing to prove); there exists a resolution

$$L_{n-1} \longrightarrow L_{n-2} \longrightarrow \cdots \longrightarrow L_0 \longrightarrow M \longrightarrow 0$$

where the L_i are free A -modules of finite type (A being Noetherian), whence an exact sequence

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$$0 \longrightarrow R \longrightarrow L_{n-1} \longrightarrow L_{n-2} \longrightarrow \cdots \longrightarrow L_0 \longrightarrow M \longrightarrow 0$$

where R is an A -module of finite type; one deduces an exact sequence

$$0 \longrightarrow R_x \longrightarrow (L_{n-1})_x \longrightarrow \cdots \longrightarrow (L_0)_x \longrightarrow M_x \longrightarrow 0$$

where the $(L_i)_x$ are free A_x -modules of finite type; as by hypothesis $\dim. \text{proj}(M_x) = n$, this implies that R_x is a projective A_x -module of finite type (M, VI, 2.1) and hence a free A_x -module of finite type (0III, 10.1.3). We conclude that there exists an open neighbourhood U of x in X such that, for every $x' \in U$, $R_{x'}$ is a free $A_{x'}$ -module (0I, 5.2.7), whence $M_{x'}$ admits a projective resolution of length n , in other words $\dim. \text{proj}(M_{x'}) \leq n$, which proves the lemma. $\square$

Proposition (6.11.2). — *Let X be a locally Noetherian prescheme and locally immersible in a regular scheme (5.11.1); let $\mathcal{F}$ be an $\mathcal{O}_X$ -Module coherent. Then:*

- (i) (M. Auslander) *The function $x \mapsto \text{coprof}(\mathcal{F}_x)$ is upper semi-continuous in X (in other words, for every integer n , the set $U_{C_n}(\mathcal{F})$ of $x \in X$ such that $\text{coprof}(\mathcal{F}_x) \leq n$ is open).*
- (ii) *For every integer n , the set $U_{S_n}(\mathcal{F})$ of $x \in X$ such that $\mathcal{F}$ has the property (S_n) at the point x is open.*

Proof. (i) The question being local on X , one can, by virtue of the hypothesis, restrict to the case where X is a closed sub-prescheme of an affine regular scheme Y . If $j : X \rightarrow Y$ is the canonical injection, and $\mathcal{G} = j_*(\mathcal{F})$, one knows that one has then, for every $x \in X$, $\dim(\mathcal{F}_x) = \dim(\mathcal{G}_x)$ and $\text{prof}(\mathcal{F}_x) = \text{prof}(\mathcal{G}_x)$, hence $\text{coprof}(\mathcal{F}_x) = \text{coprof}(\mathcal{G}_x)$, and as $\text{coprof}(\mathcal{G}_x) = 0$ for $y \notin X \rightarrow Y$, one is reduced to proving the property for $\mathcal{G}$; in other words, one can restrict to the case where X is an affine regular scheme. One knows then (0, 17.3.4) that one has

$$\text{prof}(\mathcal{F}_x) = \dim(\mathcal{O}_{X,x}) - \dim. \text{proj}(\mathcal{F}_x).$$

On the other hand, if S is the unique closed sub-prescheme of X having for underlying space $\text{Supp}(\mathcal{F})$, one has $\dim(\mathcal{F}_x) = \text{codim}(\{x\}, S)$ (5.1.12.1), and, by virtue of (5.1.9)

$$\dim(\mathcal{F}_x) = \dim(\mathcal{O}_{X,x}) - \text{codim}_x(S, X)$$

since $\mathcal{O}_{X,x}$ is a regular ring, and *a fortiori* equidimensional (0, 16.5.12). We can therefore write

$$(6.11.2.1) \quad \text{coprof}(\mathcal{F}_x) = \dim. \text{proj}(\mathcal{F}_x) - \text{codim}_x(S, X)$$

and the proposition follows then from (6.11.1) and from (0, 14.2.6).

(ii) As the $U_{C_n}(\mathcal{F})$ are open for every n by (i), the $Z_n = X - U_{C_n}(\mathcal{F})$ are closed in X ; furthermore it is clear that $Z_{n+1} \subset Z_n$, and as $\dim(\mathcal{F}_x)$ is finite for every $x \in X$ and $\text{coprof}(\mathcal{F}_x) \leq \dim(\mathcal{F}_x)$, the intersection of the Z_n is *empty*; as one can restrict to the case where X is affine, hence quasi-compact, one can suppose that there exists an m □

such that $Z_m = \emptyset$. Now, it follows from (5.7.4) that the relation $x \in U_{S_k}(\mathcal{F})$ is equivalent to the set of relations

$$(6.11.2.2) \quad \text{codim}_x(Z_n, S) > n + k$$

for every $n \geq 0$; but for $n \geq m$ this relation is automatically satisfied, hence one need only consider the relations (6.11.2.2) for $0 \leq n < m$. Now (0, 14.2.6) the set $V_{n,k}$ of x satisfying (6.11.2.2) is open, and $U_{S_k}(\mathcal{F})$, intersection of the $V_{n,k}$ for $0 \leq n < m$, is also open.

Let us recall that the hypothesis that X is locally immersible in a regular scheme is always satisfied when X is a prescheme *locally of finite type over a field k* (5.8.3).

Corollary (6.11.3). — *Under the hypotheses of (6.11.2), the set $\text{CM}(\mathcal{F})$ of points $x \in X$ such that $\mathcal{F}_x$ is a Cohen-Macaulay module is open in X .*

Proof. Indeed, it is the set $U_{C_0}(\mathcal{F})$. □

Remarks (6.11.4). — (i) The reasoning of (6.11.2), (ii) proves that (without any hypothesis on X) when $U_{C_n}(\mathcal{F})$ is open for *every* integer n , then $U_{S_n}(\mathcal{F})$ is open for every integer n .

(ii) We have $\text{CM}(\mathcal{F}) = \bigcap U_{S_n}(\mathcal{F})$. If every point $x \in X$ admits an open neighbourhood V of *finite* dimension, the sequence of intersections $V \cap U_{S_n}(\mathcal{F})$ is *stationary* since there exists an m such that $\dim(\mathcal{F}_x) \leq m$ for every $x \in V$; consequently, if the $U_{S_n}(\mathcal{F})$ are open in X , then $\text{CM}(\mathcal{F})$ is also open in X .

(iii) We shall write $U_{S_n}(X)$, $U_{C_n}(X)$, $\text{CM}(X)$ instead of $U_{S_n}(\mathcal{O}_X)$, $U_{C_n}(\mathcal{O}_X)$, $\text{CM}(\mathcal{O}_X)$.

Proposition (6.11.5). — *Let A be a local Noetherian ring, M an A -module of finite type. For every prime ideal $\mathfrak{p}$ of A , we have*

$$(6.11.5.1) \quad \text{coprof}_{A_{\mathfrak{p}}}(M_{\mathfrak{p}}) \leq \text{coprof}_A(M).$$

Proof. One can restrict to the case where $M_{\mathfrak{p}} \neq 0$. As $\hat{A}$ is a faithfully flat A -module (0_I, 7.3.5), there exists an ideal $\mathfrak{q}$ of $\hat{A}$ lying over $\mathfrak{p}$ (0_I, 6.5.1); as $\hat{M}_{\mathfrak{q}} = (M \otimes_A \hat{A})_{\mathfrak{q}} = M \otimes_A \hat{A}_{\mathfrak{q}}$ and $\hat{A}_{\mathfrak{q}}$ is a flat A -module, hence also a flat $A_{\mathfrak{p}}$ -module (0_I, 6.2.1), it follows from (6.3.2), applied to the local homomorphism $A_{\mathfrak{p}} \rightarrow \hat{A}_{\mathfrak{q}}$, to $M_{\mathfrak{p}}$ and $\hat{M}_{\mathfrak{q}}$, that $\text{coprof}_{A_{\mathfrak{p}}}(M_{\mathfrak{p}}) \leq \text{coprof}_{\hat{A}_{\mathfrak{q}}}(\hat{M}_{\mathfrak{q}})$. On the other part, $X = \text{Spec}(\hat{A})$ is isomorphic to a sub-prescheme of a regular scheme by virtue of the Cohen structure theorem (0, 19.8.8, (i)). One deduces therefore from (6.11.2) that $\text{coprof}_{\hat{A}_{\mathfrak{q}}}(\hat{M}_{\mathfrak{q}}) \leq \text{coprof}_{\hat{A}}(\hat{M})$; finally, one knows that $\text{coprof}_{\hat{A}}(\hat{M}) = \text{coprof}_A(M)$ (0, 16.4.10), which finishes the proof of (6.11.5). This proposition justifies the definition of the coprofundity of an A -module when A is not a local ring, given in (5.7.12). □

Proposition (6.11.6). — *Let X be a locally Noetherian prescheme, n an integer > 0 , $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module. Suppose that for every closed integral sub-prescheme Y of X , there exists an open non-empty part W of Y such that the sub-prescheme of X induced on W satisfies (S_n) . Then the set $U_{S_n}(\mathcal{F})$ is open in X .*

Proof. The question being local on X , one can restrict to the case where X is Noetherian. We shall reason by induction on n : for $n = 1$, the set $U_{S_1}(\mathcal{F})$ is open in X , for the set of $x \in X$ where $\mathcal{F}$ does not satisfy (S_1) is the set of x such that $\mathcal{F}_x$ admits immersed associated prime ideals (5.7.5); if (Z_j) is the finite family of the prime cycles associated to $\mathcal{F}$ which are immersed, then $U_{S_1}(\mathcal{F}) = X - \bigcup_j Z_j$,

whence our assertion since the Z_j are closed. We shall therefore suppose henceforth that $n > 1$. In the second place, one can restrict to the case where $\text{Supp}(\mathcal{F}) = X$, for there is a closed sub-prescheme T of X having $\text{Supp}(\mathcal{F})$ as underlying space, and an $\mathcal{O}_T$ -Module coherent $\mathcal{G}$ such that $\mathcal{F} = j_*(\mathcal{G})$, $j : T \rightarrow X$ being the canonical injection (**ErrIII**, 30); as it amounts to the same to prove that $\mathcal{F}$ or $\mathcal{G}$ satisfies (S_n) at a point $x \in \text{Supp}(\mathcal{F})$, one can restrict to considering the case where $T = X$. Let us note finally that by definition (5.7.2), $U_{S_n}(\mathcal{F})$ is stable by generization. We shall now use the following lemma:

Lemma (6.11.6.1). — *Let X be a Noetherian space in which every irreducible closed subset admits a generic point. For E to be an open subset of X , it is necessary and it suffices that E be stable by generization, and that, for every open part V of X and every irreducible closed subset Y of V , such that $V - Y \subset E$ and that the generic point of Y belongs to E , $E \cap Y$ contains an open non-empty part of Y .*

Proof. We observe that this criterion implies that of (**0III**, 9.2.6) when every irreducible closed subset admits a generic point. There is obviously nothing other than to prove the sufficiency of the conditions stated.

Consider the interior U of E ; the closed set $X - U$ is a union of its irreducible components, which are finite in number and closed in X . If $E \neq U$, the hypothesis that E is stable by generization would imply that the generic point z of one of the irreducible components of $X - U$ would belong to E . Now, z belongs to only one of the irreducible components of $X - U$; if T is the union of the other irreducible components of $X - U$, $V = X - T$ is open in X , union of U and of the set $Y = \{\bar{z}\} \cap V$ closed in V and irreducible. By hypothesis $E \cap Y$ contains an open part W of Y ; we conclude that $U \cup W$ is open in V , hence in X , which is absurd since U is supposed to be the interior of E .

By virtue of this lemma, one can suppose that there exists in X an integral sub-prescheme Y of generic point y such that $\mathcal{F}$ satisfies (S_n) at the point y and at every point of $X - Y$, and restrict to verifying that there exists in X an open neighbourhood of y such that $\mathcal{F}$ satisfies (S_n) in this neighbourhood. We distinguish two cases:

1° y is a maximal point of X ; as there exists an open neighbourhood of y meeting no irreducible component of X other than $\{\bar{y}\}$, one can suppose that X is irreducible, hence *a fortiori* has the same underlying space as Y , so that Y is defined by the Nilradical of $\mathcal{O}_X$, which is *nilpotent*. On the other hand, one can, replacing X by an open neighbourhood of y , suppose that $\mathcal{F}$ is *normally flat* along Y (6.9.1); it follows then from (6.10.4) that $U_{S_n}(\mathcal{F}) = U_{S_n}(\mathcal{O}_Y)$, and as the latter is by hypothesis an open neighbourhood of y in X , this proves the proposition in this case.

2° y is not a maximal point of X , in other words $\text{Supp}(\mathcal{F}) = X$, $\dim(\mathcal{F}_y) \geq 1$, hence also $\text{prof}(\mathcal{F}_y) \geq 1$ since $\mathcal{F}$ satisfies by hypothesis (S_n) (and *a fortiori* (S_1)) at the point y . Replacing X by an open neighbourhood of y if necessary, one can therefore suppose that there exists a section f of $\mathcal{O}_X$ over X , $\mathcal{F}$ -regular and such that $f_y \in \mathfrak{m}_y$, or equivalently $f(y) = 0$ (**0**, 15.2.4); one has then also $f(x) = 0$ for every $x \in Y$ (**0I**, 5.5.2). One knows that $\mathcal{F}/f\mathcal{F}$ satisfies (S_{n-1}) at the point y (5.7.6). Applying the induction hypothesis, and replacing X by an open neighbourhood of y , one can therefore suppose that $\mathcal{F}/f\mathcal{F}$ satisfies (S_{n-1}) at every point of X . But for every $x \in Y$, the relation $f(x) = 0$ implies that $\text{prof}(\mathcal{F}_x|_{f_x\mathcal{F}_x}) = \text{prof}(\mathcal{F}_x) - 1$ and $\dim(\mathcal{F}_x|_{f_x\mathcal{F}_x}) = \dim(\mathcal{F}_x) - 1$ (**0**, 16.3.4 and 16.4.6); the relation

$$\text{prof}(\mathcal{F}_x|_{f_x\mathcal{F}_x}) \geq \inf(n-1, \dim(\mathcal{F}_x|_{f_x\mathcal{F}_x}))$$

is therefore equivalent to

$$\text{prof}(\mathcal{F}_x) \geq \inf(n, \dim(\mathcal{F}_x)).$$

As we supposed that $\mathcal{F}$ satisfies (S_{n-1}) at every point of $X - Y$, this finishes the proof. $\square$

Corollary (6.11.7). — *With the notations being those of (6.11.6):*

- (i) *The set $U_{S_1}(\mathcal{F})$ is open in X .*
- (ii) *For the set $U_{S_n}(\mathcal{F})$ to be open, it suffices that every maximal point x of $\text{Supp}(\mathcal{F})$, belonging to $U_{S_n}(\mathcal{F})$, is interior to $U_{S_n}(\mathcal{F})$.*

Proof. Assertion (i) was proved in the course of the proof of (6.11.6); on the other hand, for $n = 2$ the case 2° of the proof of (6.11.6) is valid without any hypothesis on X , since (with the same notations) $U_{S_1}(\mathcal{F})$ and $U_{S_1}(\mathcal{F}/f\mathcal{F})$ are open in X . As for the case 1° of this proof, the hypothesis ensures precisely that it is unnecessary to consider it. $\square$

Proposition (6.11.8). — *Let X be a locally Noetherian prescheme satisfying the following property:*

(CMU) *Every closed integral sub-prescheme Y of X contains an open non-empty W such that the prescheme induced by X on W is a Cohen-Macaulay prescheme.*

Then, for every coherent $\mathcal{O}_X$ -Module $\mathcal{F}$, the function $x \mapsto \text{coprof}(\mathcal{F}_x)$ is locally constructible and upper semi-continuous; the sets $U_{C_n}(\mathcal{F})$ and $U_{S_n}(\mathcal{F})$ are open in X .

Proof. Indeed, let Y be a closed integral sub-prescheme of X , of generic point y ; by virtue of (6.10.6), there is a neighbourhood V of y in Y such that, for every $x \in V \cap Y$, we have

$$(6.11.8.1) \quad \text{coprof}(\mathcal{F}_x) = \text{coprof}(\mathcal{F}_y) + \text{coprof}(\mathcal{O}_{Y,x}).$$

But by hypothesis there exists an open non-empty W of Y such that, for $x \in V \cap W$, $\text{coprof}(\mathcal{O}_{Y,x}) = 0$, hence $\text{coprof}(\mathcal{F}_x)$ is constant in a neighbourhood of y in Y , which proves that the function $x \mapsto \text{coprof}(\mathcal{F}_x)$ is locally constructible (0_{III}, 9.3.2); it follows furthermore from (6.11.5) and from (0_{III}, 9.3.4) that this function is upper semi-continuous. The last assertion follows from (6.11.4), (i), or from (6.11.6). $\square$

Remarks (6.11.9). — (i) If X satisfies (CMU), then every prescheme X' locally of finite type over X also satisfies (CMU). Let Y' be a closed integral subprescheme of X' , let y' be its generic point, put $y = u(y')$, and let Y be the integral closed subprescheme of X with underlying space $\overline{\{y\}}$. Then $u|_{Y'}$ factors as

$$Y' \xrightarrow{v} Y \xrightarrow{j} X,$$

where j is the canonical injection and v is locally of finite type.

Restricting to affine open neighbourhoods of y and y' , we may suppose that $X = \text{Spec}(A)$, where A is an integral Cohen-Macaulay ring, and $X' = \text{Spec}(A')$, where A' is an integral finite-type A -algebra containing A . After replacing A by A_f for some $f \neq 0$, we may suppose that A' contains a polynomial ring $A[T_1, \dots, T_n] = A''$ and is finite over A'' . The ring A'' is Cohen-Macaulay (6.3.6); hence we may reduce to the case where A' is finite over A . After one further localisation, A' is a finite free A -module. It is therefore a Cohen-Macaulay A -module (0, 16.5.1); since it is finite over A , it is also a Cohen-Macaulay A' -module (0, 16.5.3), and hence a Cohen-Macaulay ring.

(ii) Suppose that there exists a coherent $\mathcal{O}_X$ -Module $\mathcal{F}$ such that $\text{Supp}(\mathcal{F}) = X$ and that $\mathcal{F}$ is an $\mathcal{O}_X$ -Module of Cohen-Macaulay. Then X satisfies the condition (CMU): indeed, with the notations of the proof of (6.11.8), the relation (6.11.8.1) shows that $\text{coprof}(\mathcal{O}_{Y,x}) = 0$ in a neighbourhood (with respect to Y) of the generic point y of Y .

It is unknown whether there exist locally Noetherian preschemes of dimension ≥ 2 that do not satisfy (CMU) (if $\dim(X) = 1$, it is immediate that every maximal point of X_{red} admits an open neighbourhood of dimension 1, hence a Cohen-Macaulay ring).

6.12. Nagata's criteria for $\text{Reg}(X)$ to be open

(6.12.1). Given a locally Noetherian prescheme X , we call the *singular locus* of X and denote by $\text{Sing}(X)$ the set of points $x \in X$ such that X is not regular at the point x (in other words, such that the local ring $\mathcal{O}_x$ is not regular); the complement $X - \text{Sing}(X)$, set of $x \in X$ where X is regular, is denoted $\text{Reg}(X)$. We propose in this subsection to give conditions ensuring that $\text{Sing}(X)$ is *closed* (i.e. $\text{Reg}(X)$ *open*).

Proposition (6.12.2). — *Let X be a locally Noetherian prescheme. The following conditions are equivalent:*

- a) $\text{Reg}(X)$ is open in X .
- b) For every $x \in \text{Reg}(X)$, there exists an open non-empty part of $\overline{\{x\}}$ contained in $\text{Reg}(X)$.

Moreover, these conditions imply the following:

- c) For every $x \in \text{Reg}(X)$, if we denote by Y the reduced closed sub-prescheme of X having $\overline{\{x\}}$ as underlying space, $\text{Reg}(Y)$ is a neighbourhood of x in Y .

Proof. The equivalence of a) and b) follows from the fact that $\text{Reg}(X)$ is stable by generization (0, 17.3.2) and from (0_{III}, 9.2.6). To prove that c) implies b), one can restrict to the case where $X = \text{Spec}(A)$ is the spectrum of an affine ring of x ; if $(t_i)_{1 \leq i \leq n}$ is a regular system of parameters

(0, 17.1.6) of the regular local ring A_x , one can suppose (replacing X by an open neighbourhood of x if necessary) that $t_i = (s_i)_x$ for every i , where $s_i \in A$ and where the family $(s_i)_{1 \leq i \leq n}$ is A -regular (0, 15.2.4). One has then $Y = \text{Spec}(A/\mathfrak{p})$, where $\mathfrak{p} = \mathfrak{j}_x$; as the t_i generate the maximal ideal $\mathfrak{m} = \mathfrak{p}A_x$ of A_x , one can further suppose (replacing X by a smaller neighbourhood of x) that the s_i generate $\mathfrak{p}$. For every $y \in Y$, the $(s_i)_y$ generate therefore $\mathfrak{p}_y$; as they form a suite A_y -regular, they also form a suite $\text{gr}_{\mathfrak{p}_y}^0(\mathcal{O}_y^Y)$ -regular, so they form a $\text{gr}_{\mathfrak{p}_y}(\mathcal{O}_y^Y)$ -regular sequence and it is thus also $\mathcal{O}_y^Y$ -regular, whence the conclusion. $\square$

Corollary (6.12.3). — *Let X be a locally Noetherian prescheme. The following conditions are equivalent:*

- a) *For every sub-prescheme Y of X , $\text{Reg}(Y)$ is open in Y .*
- b) *For every closed integral sub-prescheme Y of X , $\text{Reg}(Y)$ contains an open non-empty part of Y .*

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Proof. It is clear that a) implies b). Conversely, to see that b) implies a), consider a closed integral sub-prescheme Z of Y with generic point z ; if Y' is the integral closed sub-prescheme of X whose underlying space is the closure $Y' = \overline{\{z\}}$ of z in X , Z is open in Y' and the sub-prescheme Z of Y' , being reduced, is induced by Y' on the open subset of Z in Y' . Now the hypothesis implies that $\text{Reg}(Y')$ is a neighbourhood of z in Y' , hence $\text{Reg}(Z)$ is a neighbourhood of z in Z ; it suffices then to apply (6.12.2) by replacing X by Y and Y by Z . $\square$

Theorem (6.12.4). — (Nagata). — *Let A be a Noetherian ring, $X = \text{Spec}(A)$. The following conditions are equivalent:*

- a) *For every prescheme X' locally of finite type over X , $\text{Reg}(X')$ is open in X' .*
- b) *For every A -algebra A' finite and integral, there exists an open non-empty subset of $X' = \text{Spec}(A')$ contained in $\text{Reg}(X')$.*
- c) *For every prime ideal $\mathfrak{p}$ of A and every finite radicial extension K' of the field of fractions K of $A/\mathfrak{p}$, there exists a sub- A -algebra A' of K' , finite over A , having K' as field of fractions, and such that there exists in $X' = \text{Spec}(A')$ an open non-empty subset contained in $\text{Reg}(X')$.*

Proof. It is clear that a) implies b). To see that b) implies c), it suffices to note that one can take as generators of the extension K' of K elements integral over $A/\mathfrak{p}$ (and *a fortiori* over A), and as these elements are finite in number, they generate a finite A -algebra A' whose field of fractions is K' ; one can then apply b) to A' . It remains therefore to prove that c) implies a). The question being local on X' , one can first suppose that $X' = \text{Spec}(A')$, where A' is an A -algebra of finite type; in view of (6.12.2), it suffices to prove that for every closed integral sub-prescheme Y' of X' , $\text{Reg}(Y')$ contains an open non-empty part of Y' . In other words, one can restrict to proving that if A' is an A -algebra integral of finite type and $X' = \text{Spec}(A')$, $\text{Reg}(X')$ contains an open non-empty part. Let K' be the field of fractions of A' ; if $\mathfrak{p}$ is the canonical image in X of the generic point of X' , K' is an extension of the field of fractions K of $A/\mathfrak{p}$, and $A/\mathfrak{p}$ is identified with a subring of A' , A' being a finite $(A/\mathfrak{p})$ -algebra of finite type. One can obviously restrict in what follows to the proof when $\mathfrak{p} = 0$. We distinguish two cases:

I) K' is a separable extension of K . — One is then reduced to proving the

Lemma (6.12.4.1). — *Let A be a Noetherian integral ring, A' an A -algebra integral of finite type, such that the field of fractions K' of A' is a separable extension of the field of fractions K of A . If $\text{Spec}(A)$ contains an open non-empty subset consisting of regular points, then so does $\text{Spec}(A')$.*

Proof. Replacing A by a ring of fractions A_f , so that $D(f) \subset \text{Reg}(\text{Spec}(A))$, one can already suppose the ring A regular. By hypothesis there exists in K' a system $(t_i)_{1 \leq i \leq n}$ of elements algebraically independent over K and such that K' is a finite separable algebraic extension of $K(t_1, \dots, t_n)$; considering a finite system of generators t'_j ($1 \leq j \leq m$) of K' over $K(t_1, \dots, t_n)$, which one can suppose to be integral over $A_1 = A[t_1, \dots, t_n]$, one sees that $A'_1 = A_1[t'_1, \dots, t'_m]$ is finite over A_1 and has K' as field of fractions. If one sets $X' = \text{Spec}(A')$, $X'_1 = \text{Spec}(A'_1)$, it follows from the fact that the fields of rational functions $R(X')$ and $R(X'_1)$ are both isomorphic to K' and from the fact that X' and X'_1 are A -preschemes of finite type, that there exists an open subset $U' \subset X'$ and an open $U'_1 \subset X'_1$ which are A -isomorphic (I, 6.5.5). One is therefore reduced to proving that $\text{Reg}(X'_1)$ contains a non-empty open subset, in other words one can suppose that A' is an A_1 -algebra finite. Now, one knows (0, 17.3.7) that A_1 is a regular ring, and one can therefore restrict to the case where

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A' is an A -algebra *finite* and K' an extension *separable* and *finite* of K . Let ξ be the generic point of X , $A'_\xi = K'$ is then a free module over $A_\xi = K$, hence (Bourbaki, *Alg., comm.*, chap. II, § 5, no 1, cor. of prop. 2) one can, replacing A by a ring of fractions A_f , suppose that A' is a free A -module of finite type. Let then $(x_h)_{1 \leq h \leq r}$ be a basis of this A -module, and set

$$d = \det(\mathrm{Tr}_{A'/A}(x_h x_k)) = \det(\mathrm{Tr}_{K'/K}(x_h x_k)) \in A.$$

As K' is separable over K , one knows (Bourbaki, *Alg.*, chap. IX, § 2, prop. 5) that $d \neq 0$; replacing A by a ring of fractions A_z , one can therefore suppose d *invertible* in A . But then, for every $z \in \mathrm{Spec}(A)$, if one denotes by $\bar{x}_h$ ($1 \leq h \leq r$) the canonical image of x_h in $A'(z) = A' \otimes_A k(z)$, one has $\det(\mathrm{Tr}_{A'(z)/k(z)}(\bar{x}_h \bar{x}_k)) = \bar{d}$, canonical image of d in $k(z) = A_z/\mathfrak{m}_z$, and as $\bar{d}$ is invertible (hence $\neq 0$) in $k(z)$, one knows (*loc. cit.*) that $A'(z)$ is a $k(z)$ -algebra *separable*, hence a direct product of finite separable field extensions of $k(z)$. Since such an algebra is a *regular* ring, one sees that the morphism $g : X' \rightarrow X$ is *flat* and that its fibres $g^{-1}(z)$ are *regular* for every $z \in X$; one concludes then from (6.5.2, (ii)) that X' is *regular*, which finishes the proof in this case. $\square$

II) *General case.* — As A' is an A -module without torsion, $A' \otimes_A K$ is identified with a subring of K' , hence $X'' = \mathrm{Spec}(A' \otimes_A K)$ is an integral K -prescheme whose field of rational functions is K' . One knows (4.6.6) that there exists a finite radicial extension K_1 of K such that $X'_1 = \mathrm{Spec}(A' \otimes_A K_1) = X'' \otimes_K K_1$, $(X'_1)_{\mathrm{red}}$ is a K_1 -prescheme *geometrically reduced* of finite type; furthermore, the morphism $\mathrm{Spec}(K_1) \rightarrow \mathrm{Spec}(K)$ being radicial, finite and surjective, it is a universal homeomorphism (2.4.5), hence X'_1 is homeomorphic to X'' , and consequently $(X'_1)_{\mathrm{red}}$ is integral; its field of rational functions K'_1 is a finite radicial extension of K' and a *separable* extension of finite type of K_1 (4.6.1). By virtue of hypothesis *c*) of the statement, there is a sub- A -algebra *finite* A_1 of K_1 , having K_1 as field of fractions and such that if one sets $X_1 = \mathrm{Spec}(A_1)$, $\mathrm{Reg}(X_1)$ contains an open non-empty subset. Let A'_1 be the image of the canonical homomorphism $A' \otimes_A A_1 \rightarrow K'_1$ and let $X'_1 = \mathrm{Spec}(A'_1)$; A'_1 is an integral ring which is an A' -algebra *finite* and whose field of fractions is K'_1 ; furthermore, as the composite homomorphism $A' \rightarrow A' \otimes_A A_1 \rightarrow A'_1 \rightarrow K'_1$ is identical to $A' \rightarrow K' \rightarrow K'_1$, hence injective, the homomorphism $A' \rightarrow A'_1$ is injective; the morphism $g : X'_1 \rightarrow X'$ is therefore finite and surjective (II, 6.1.10). This being the hypothesis on A_1 and part I) of the proof imply that $\mathrm{Reg}(X'_1)$ contains an open non-empty V'_1 ; as g is closed (II, 6.1.10), one can suppose that $V'_1 = g^{-1}(V')$, where V' is an open affine of X' ; replacing A' by the ring of V' and A'_1 by that of V'_1 , one can therefore suppose that X'_1 is *regular*. Furthermore, the same reasoning as in I) applied to the generic point ξ' of X' allows (replacing X' by a neighbourhood of ξ') to suppose that A'_1 is an A' -module *free*, and hence that the morphism g is *flat*. But it follows then from (6.5.2, (i)) that X' is *regular*. $\square$

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Corollary (6.12.5). — (Zariski). — *Let k be a field; for every k -prescheme X locally of finite type over k , the set of $x \in X$ where X is regular (resp. geometrically regular over k) is open in X .*

Proof. The assertion of the corollary relative to the property of being regular follows from (6.12.4) by taking $A = k$. The assertion relative to the property of being geometrically regular follows already from (6.8.7); one can also deduce it from (6.12.4) in the following way: set $k' = k^{p^{-\infty}}$ (p being the characteristic exponent of k); as the morphism $\mathrm{Spec}(k') \rightarrow \mathrm{Spec}(k)$ is radicial, integral and surjective, it is a universal homeomorphism (2.4.5), hence the morphism projection $X \otimes_k k' \rightarrow X$ is a universal homeomorphism. The projection in X of $\mathrm{Reg}(X \otimes_k k')$ is the set of points of X where X is geometrically regular over k , by virtue of (6.7.7, e)); hence this set is open from what precedes. $\square$

Corollary (6.12.6). — *Let A be a ring having one of the following properties:*

- (i) *A is a Dedekind ring and its field of fractions K is of characteristic 0.*
- (ii) *A is a semi-local Noetherian ring of dimension ≤ 1 .*

Then, for every prescheme X locally of finite type over A , $\mathrm{Reg}(X)$ is open in X .

Proof. Let us verify in both cases the condition *c*) of (6.12.4). In both cases, a prime ideal $\mathfrak{p}$ of A is maximal or minimal; if $\mathfrak{p}$ is maximal, an $(A/\mathfrak{p})$ -algebra finite and integral is a field and the condition *c*) of (6.12.4) is trivially satisfied. Suppose therefore $\mathfrak{p}$ non-maximal, and distinguish the two cases of the statement.

(i) If K is of characteristic 0, there is no radicial extension of K other than K itself; as a Dedekind ring is regular (0, 17.1.4), the condition *c*) of (6.12.4) is trivially satisfied.

(ii) One can then suppose A integral (6.12.2), let K be its field of fractions; if K' is a finite radical extension of K , A' a sub- A -algebra of K' generated by a finite system of generators of K' over K , integral over A , A' is a semi-local integral ring of dimension 1 (0, 16.1.5), and consequently, in $X' = \text{Spec}(A')$, the reduced set at the generic point is open and evidently contained in $\text{Reg}(X')$, which proves the condition c) of (6.12.4) in this case. $\square$

This corollary applies notably when $A = \mathbb{Z}$.

Theorem (6.12.7). — (Nagata). — *Let A be a local Noetherian complete ring, $X = \text{Spec}(A)$. Then $\text{Reg}(X)$ is open in X .*

Proof. In view of (6.12.2), one is reduced to the case where moreover A is integral, and to proving that $\text{Reg}(X)$ contains an open non-empty part of X . We distinguish two cases:

I) *The field of fractions K of A is of characteristic 0.* — One knows then (0, 19.8.8) that there exists a complete discrete valuation ring C and a subring B of A such that A is a B -algebra finite and that B is isomorphic to a formal power series ring $C[[T_1, \dots, T_r]]$. As C is regular (II, 7.1.6), so is B (0, 17.3.8); furthermore, the field of fractions L of B being of characteristic 0, K is a separable extension finite of L , hence one can apply (6.12.4.1) to B , and this proves the proposition.

II) *The field of fractions K of A is of characteristic $p > 0$.* — Then A contains the prime field $\mathbb{F}_p$, and the theorem has been proved in this case (0, 22.7.6). $\square$

Corollary (6.12.8). — *Let A be a local Noetherian complete ring. Then, for every prescheme X locally of finite type over A , $\text{Reg}(X)$ is open in X .*

Proof. Let us verify the condition c) of (6.12.4). If $\mathfrak{p}$ is prime in A , $A/\mathfrak{p}$ is again a local Noetherian complete ring; if K' is a finite extension of the field of fractions K of $A/\mathfrak{p}$, K' is the field of fractions of a sub- A -algebra finite A' of K' , generated by a finite system of generators of K' over K , integral over A . One knows then that A' is a semi-local complete ring (Bourbaki, *Alg. comm.*, chap. IV, § 2, no 5, cor. 3 of prop. 9), hence a product of local complete rings, and as A' is integral, it is a local complete ring; and by virtue of (6.12.7), if $X' = \text{Spec}(A')$, $\text{Reg}(X')$ is open and non-empty, whence the conclusion. $\square$

Proposition (6.12.9). — *Let X be a locally Noetherian prescheme such that $\text{Reg}(X)$ is open; then, for every $n \geq 0$, the set $U_{R_n}(X)$ of points $x \in X$ where X satisfies (R_n) is open.*

Proof. Indeed (5.8.2), $U_{R_n}(X)$ is the union of $\text{Reg}(X)$ and of the set of $x \in \text{Sing}(X)$ such that the generic points z_i of the irreducible components of $\text{Sing}(X)$ containing x satisfy the relation $\dim(\mathcal{O}_{X,z_i}) \geq n+1$ (5.1.2); in other words, these points $x \in \text{Sing}(X)$ are those for which $\text{codim}_x(\text{Sing}(X), X) \geq n+1$; as the function $x \mapsto \text{codim}_x(\text{Sing}(X), X)$ is lower semi-continuous (0, 14.2.6), the set of these points is open in $\text{Sing}(X)$, which finishes the proof. $\square$

Let us also note the following elementary result:

Proposition (6.12.10). — *Let X be a locally Noetherian prescheme. The set $U_{R_0}(X)$ is open in X . For $U_{R_1}(X)$ to be open in X , it is necessary and it suffices that every maximal $x \in X$ such that $x \in U_{R_1}(X)$ (which means that X is reduced at x) be interior to $U_{R_1}(X)$.*

Proof. By definition (5.8.2), $U_{R_0}(X)$ is the set $X - \bigcup_{\alpha} X'_{\alpha}$, where X'_{α} ranges over the set of irreducible components of X such that X is not reduced at their generic point; as the set of irreducible components of X is locally finite, $U_{R_0}(X)$ is open. As for $U_{R_1}(X)$, the stated condition is trivially necessary; let us prove that it is sufficient. Let X''_{β} be the irreducible components of X such that, at their generic point x''_{β} , the prescheme X is reduced; by hypothesis there exists an open $U \subset U_{R_1}(X)$ containing all the x''_{β} . Denote then by Z_{λ} the irreducible components of the closed set $Z = X - U$ whose generic point z_{λ} is such that $\dim(\mathcal{O}_{X,z_{\lambda}}) \leq 1$ and that $\mathcal{O}_{X,z_{\lambda}}$ is non-regular. No point belonging to one of the Z_{λ} can belong to $U_{R_1}(X)$; but conversely, if $x \in Z$ does not belong to any of the Z_{λ} , then, for every generization x' of x , either x' belongs to U , or $\dim(\mathcal{O}_{X,x'}) \geq 2$, or $\dim(\mathcal{O}_{X,x'}) = 1$, and as x' does not belong to any Z_{λ} , $\overline{\{x'\}}$ is necessarily one of the irreducible components of Z , of codimension 1 in X , distinct from the Z_{λ} , hence $\mathcal{O}_{X,x'}$ is regular by definition. One concludes that $U_{R_1}(X) = X - \bigcup_{\lambda} Z_{\lambda}$; as the set of Z_{λ} is locally finite in Z , $\bigcup_{\lambda} Z_{\lambda}$ is closed, which finishes the proof that $U_{R_1}(X)$ is open in X . $\square$

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6.13. Criteria for $\text{Nor}(X)$ to be open

(6.13.1). Given a locally Noetherian prescheme X , we shall denote by $\text{Nor}(X)$ the set of $x \in X$ where X is *normal*; this set contains $\text{Reg}(X)$ and is contained in the (open) set of points x such that $\mathcal{O}_{X,x}$ is integral (i.e. the set of points x where X is reduced, and which belong to only one irreducible component of X).

Proposition (6.13.2). — *Let X be a locally Noetherian prescheme reduced, X' its normalization (II, 6.3.8). If the canonical morphism $f : X' \rightarrow X$ is of finite type, $\text{Nor}(X)$ is open in X .*

Proof. Indeed, on $X' = \text{Spec}(f_*(\mathcal{O}_{X'}))$ and $f_*(\mathcal{O}_{X'})$ is a coherent $\mathcal{O}_X$ -Algebra by hypothesis (II, 6.1.3). To say that X is normal at a point x means that the canonical homomorphism $\mathcal{O}_x \rightarrow (f_*(\mathcal{O}_{X'}))_x$ is bijective (II, 6.3.4); but the set of these points is open since $\mathcal{O}_X$ and $f_*(\mathcal{O}_{X'})$ are coherent (0_I, 5.2.7). $\square$

Corollary (6.13.3). — *If A is an integral Noetherian Japanese ring, $\text{Nor}(\text{Spec}(A))$ is open in $\text{Spec}(A)$.*

Proposition (6.13.4). — *Let X be a locally Noetherian prescheme. For $\text{Nor}(X)$ to be open, it is necessary and it suffices that every maximal point x of X such that $x \in \text{Nor}(X)$ (which means that X is reduced at the point x) be interior to $\text{Nor}(X)$.*

Proof. There is only to prove the sufficiency of the condition. By virtue of the remark made in (6.13.1), one can restrict (replacing X by an open of X in which X is reduced), i.e. to suppose that the maximal points of X belong to $U_{S_1}(X)$ and to $U_{R_1}(X)$; as $\text{Nor}(X) = U_{S_1}(X) \cap U_{R_1}(X)$ by virtue of Serre's criterion (5.8.6), one deduces from (6.11.7) and (6.12.10) that these two sets are open; hence $\text{Nor}(X)$ is open in X . $\square$

Corollary (6.13.5). — *If X is such that $\text{Reg}(X)$ is open in X , then $\text{Nor}(X)$ is open in X .*

Proof. Indeed, every maximal point x where X is reduced (hence $\mathcal{O}_{X,x}$ is a field) belongs to $\text{Reg}(X)$, hence is interior to $\text{Reg}(X)$ by hypothesis, and *a fortiori* is interior to $\text{Nor}(X)$. $\square$

Proposition (6.13.6). — (Nagata). — *Let A be a Noetherian integral ring, K its field of fractions, K' a finite extension of K , A' the integral closure of A in K' . For A' to be a finite A -algebra, it is necessary and it suffices that the two following conditions be satisfied:*

- (i) *There exists $f \neq 0$ in A such that the integral closure A'_f of the ring of fractions A_f in K' is a finite A_f -algebra.*
- (ii) *For every prime ideal $\mathfrak{p} \in \text{Spec}(A)$, the integral closure $A'_\mathfrak{p}$ of the local ring $A_\mathfrak{p}$ in K' is a finite $A_\mathfrak{p}$ -algebra.*

Proof. The conditions are necessary, for for every multiplicative subset S of A , $S^{-1}A'$ is the integral closure of $S^{-1}A$ in K' and is by hypothesis a $S^{-1}A$ -module of finite type. To see that the conditions are sufficient, consider the filtered increasing family (B_λ) of sub- A -algebras of K' which are A -algebras finite and have K' as field of fractions. Set $Y = \text{Spec}(A)$, $X_\lambda = \text{Spec}(B_\lambda)$, denote by u_λ the morphism $X_\lambda \rightarrow Y$ and let $S_\lambda = X_\lambda - \text{Nor}(X_\lambda)$, $T_\lambda = u_\lambda(S_\lambda)$. One can restrict to the B_λ which contain a finite set whose image in A'_f is a system of generators of the A_f -module A'_f ; this implies, by virtue of hypothesis (i), that $(B_\lambda)_f = A'_f$ for every λ , or equivalently that $u_\lambda^{-1}(D(f))$ is contained in $\text{Nor}(X_\lambda)$. By virtue of (6.13.2), S_λ is therefore closed in Y . But for every $\mathfrak{p} \in \text{Spec}(A)$, it exists by virtue of (ii) a λ such that $(B_\lambda)_\mathfrak{p} = A'_\mathfrak{p}$, and consequently all points of X_λ lying over $\mathfrak{p}$ belong to $\text{Nor}(X_\lambda)$; in other words T_λ is closed in Y , $\bigcup_\lambda T_\lambda$ is closed, and as u_λ is a finite morphism, T_λ is closed in Y . But for every $\mathfrak{p} \in \text{Spec}(A)$, it exists by virtue of (ii) a λ such that $(B_\lambda)_\mathfrak{p} = A'_\mathfrak{p}$, and consequently all points of X_λ lying over $\mathfrak{p}$ belong to $\text{Nor}(X_\lambda)$; in other words $\mathfrak{p} \notin T_\lambda$; as Y is Noetherian and the T_λ are closed, there exists a λ_0 such that $T_{\lambda_0} = \emptyset$; hence B_{λ_0} is integrally closed, and as its field of fractions is K' , one has $B_{\lambda_0} = A'$. $\square$

Proposition (6.13.7). — *Let A be a Noetherian ring, $X = \text{Spec}(A)$. The following conditions are equivalent:*

- a) *For every prescheme X' locally of finite type over X , $\text{Nor}(X')$ is open in X' .*
- b) *For every A -algebra finite and integral A' , $\text{Nor}(\text{Spec}(A'))$ is open in $\text{Spec}(A')$.*
- c) *For every prime ideal $\mathfrak{p}$ of A and every finite radicial extension K' of the field of fractions K of $A/\mathfrak{p}$, there exists a sub- A -algebra A' of K' , finite over A , having K' as field of fractions and such that $\text{Nor}(\text{Spec}(A'))$ is open in $\text{Spec}(A')$.*

Proof. The fact that $a)$ implies $b)$ and that $b)$ implies $c)$ is proved as in (6.12.4). To show that $c)$ implies $a)$, one reduces as in (6.12.4) to proving that if A' is an A -algebra integral and of finite type, the generic point of $\text{Spec}(A')$ is interior to $\text{Nor}(\text{Spec}(A'))$. One distinguishes two cases as in (6.12.4) by proving first the

Lemma (6.13.7.1). — *Let A be a Noetherian integral ring, A' an A -algebra integral of finite type, containing A and such that the field of fractions K' of A' is a separable extension of the field of fractions K of A . If $\text{Nor}(\text{Spec}(A))$ is open in $\text{Spec}(A)$, $\text{Nor}(\text{Spec}(A'))$ is open in $\text{Spec}(A')$.*

Proof. It suffices to prove (in view of (6.13.2)) that the generic point of $\text{Spec}(A')$ is interior to $\text{Nor}(\text{Spec}(A'))$. The proof follows the same approach as that of (6.12.4.1), from which we conserve the notations. We note that when A is integrally closed, and then one takes $A_1 = A[t_1, \dots, t_n]$ and is integrally closed (Bourbaki, *Alg. comm.*, chap. V, § 1, no 3, cor. 2 of prop. 13); one reduces next to the case where A' is a free A -module of finite type; the reasoning then applies (replacing A by an appropriate ring of fractions A_f with $f \neq 0$) the reasoning from case I) of (6.12.4) to see that the fibres $g^{-1}(z)$ of the morphism $g : X' \rightarrow X$ are regular and *a fortiori* normal. Furthermore g is flat and X is normal, hence (6.5.4, (ii)) X' is normal. One then passes to the general case as in (6.12.4; II)), from which we conserve further the notations; applying hypothesis $c)$, one sees this time that $\text{Nor}(X'_1)$ is open and one reduces thus to the case where X'_1 is normal and $g : X' \rightarrow X'$ is flat and surjective; one concludes by applying (6.5.4, (i)). $\square$

This lemma being proved, one passes to the general case as in (6.12.4; II)), from which we conserve the notations; applying hypothesis $c)$, one sees this time that $\text{Nor}(X'_1)$ is open and one reduces thus to the case where X'_1 is normal and $g : X' \rightarrow X'$ flat and surjective; one concludes by applying (6.5.4, (i)). $\square$

6.14. Base change and integral closure

Proposition (6.14.1). — *Let Y, Y' be two locally Noetherian preschemes, $g : Y' \rightarrow Y$ a normal morphism (6.8.1). Then, for every normal Y -prescheme X , the prescheme $X' = X \times_Y Y'$ is normal.*

Proof. We note that in this statement one does not suppose X locally Noetherian. $\square$

Lemma (6.14.1.1). — *Let R be a ring that is a direct product of finitely many fields.*

- (i) *For a subring A of R having R as total ring of fractions to be normal, it is necessary and it suffices that it be integrally closed in R .*
- (ii) *Let (A_λ) be a family of normal subrings of R ; if $A = \bigcap_\lambda A_\lambda$ admits R as total ring of fractions, A is normal.*

Proof. (i) As A is a subring of R , A_p is a subring of R_p for every prime ideal p of A ; furthermore R_p is a ring of fractions of A_p ; furthermore R_p is necessarily a direct product of finitely many fields, hence every non-zero divisor of 0 in R_p is invertible; this proves that R_p is the total ring of fractions of A_p . If A is integrally closed in R , A_p is therefore integrally closed in R_p ; but if R_p is a direct product of at least two rings, the integral closure of A_p in R_p is a direct product of at least two nonzero rings, which is absurd since A_p is a local ring; hence R_p is necessarily a field and A_p is integral and integrally closed, which by definition means that A is normal. Conversely, if A is normal, R_p , total ring of fractions of an integral ring A_p , is a field and A_p is integrally closed in R_p ; hence R is an integral element over A , its image in each R_p is integral over A_p , hence belongs to A_p ; we conclude that $x \in A$ (Bourbaki, *Alg. comm.*, chap. II, § 3, no 3, cor. 1 of th. 1), and hence A is integrally closed in R .

(ii) As $A \subset A_\lambda \subset R$ for every λ , R is also the total ring of fractions of each A_λ . In view of the characterisation of normal rings having R as total ring of fractions, the assertion (ii) follows from Bourbaki, *Alg. comm.*, chap. V, § 1, no 3, prop. 12. $\square$

This lemma being proved, the proof of (6.14.1) proceeds in several steps.

I) *Reduction to the case where $Y = \text{Spec}(A)$, $Y' = \text{Spec}(A')$, $X = \text{Spec}(B)$, A, A' being local Noetherian rings, A integral, B the integral closure of A .* — It suffices to prove that for every $x' \in X'$, the local ring $\mathcal{O}_{X', x'}$ is integral and integrally closed; let x, y, y' be the canonical images of x' in X, Y, Y' respectively; if one sets $A = \mathcal{O}_{Y, y}$, $A' = \mathcal{O}_{Y', y'}$, $B = \mathcal{O}_{X, x}$, the rings $\mathcal{O}_{X, x}, \mathcal{O}_{Y, y}, \mathcal{O}_{Y', y'}$ for x, y, y' fixed, are the local

rings at the prime ideals of the ring $B' = B \otimes_A A'$ (I, 3.6.5), and so it suffices to prove that B' is a *normal* ring. Let us note on the other hand that $\text{Spec}(A') \rightarrow \text{Spec}(A)$ is a normal morphism by definition (6.8.1 and I, 3.6.5); one is reduced to the case where Y and Y' are local schemes, that is to say to the case where A is an integral and integrally closed local ring (0_1 , 6.8.1 and I, 3.6.5); one is reduced to the case where Y and Y' are local schemes. Let us designate by (B_α) the family of *integral closures* of the sub- A -algebras of *finite type* of B ; it is clear that B is the union of the filtered increasing family of the B_α . As $\varinjlim$ commutes with the tensor product, B' is therefore also the union of the filtered increasing family $\overline{B'_\alpha} = B_\alpha \otimes_A A'$. To prove that B' is normal, it suffices, by virtue of (5.13.6), to show that the rings B'_α are *normal* and that, for $\alpha \leq \beta$, every irreducible component of $\text{Spec}(B'_\beta)$ *dominates* one irreducible component of $\text{Spec}(B'_\alpha)$. But this last property follows from the hypothesis that A' is an A -module flat and from (2.3.7, (ii)).

One can therefore restrict to proving that $B' = B \otimes_A A'$ is normal when B is the integral closure of a A -algebra integral of *finite type* C ; in this case, if one sets $C' = C \otimes_A A'$, the morphism $\text{Spec}(C') \rightarrow \text{Spec}(C)$ is normal (6.8.2); as $B' = B \otimes_C C'$, one sees that one can replace A and A' by C and C' respectively, hence suppose that A is integral and that B is the *integral closure* of A . Finally, the procedure from the beginning allows to restrict to the case where A is *local* (taking into account the fact that, if B is the integral closure of A , $B_{\mathfrak{p}}$ is the integral closure of $A_{\mathfrak{p}}$ for every prime ideal $\mathfrak{p}$ of A).

II) *Reduction to the case where A is a local integral ring of dimension 1, B a discrete valuation ring, integral closure of A and the morphism $\text{Spec}(B) \rightarrow \text{Spec}(A)$ radicial.* — Let K be the field of fractions of A . One knows (0, 23.2.7) that B is intersection of a family (V_λ) of discrete valuation rings such that, for every $x \in K$, one has $x \in V_\lambda$ except for a finite number of indices λ . One therefore has an exact sequence of A -modules

$$0 \longrightarrow B \longrightarrow K \longrightarrow \bigoplus_{\lambda} (K/V_{\lambda}).$$

Set $K' = K \otimes_A A'$, $V'_\lambda = V_\lambda \otimes_A A'$; by flatness, one deduces from the preceding exact sequence a new exact sequence

$$0 \longrightarrow B' \longrightarrow K' \longrightarrow \bigoplus_{\lambda} (K'_\lambda/V'_\lambda)$$

and hence $B' = \bigcap_{\lambda} V'_\lambda$. Furthermore $\text{Spec}(K')$ is the fibre of the morphism $\text{Spec}(A') \rightarrow \text{Spec}(A)$ at the generic point of $\text{Spec}(A)$, hence K' is a Noetherian ring, *normal*, and hence a direct product of finitely many integrally closed integral rings; the direct product L' of the fields of fractions of these latter is the total ring of fractions of A' , hence also that of B' . One can therefore apply the lemma (6.14.1.1), and if one proves that each of the V'_λ is a *normal* ring, then so is B' .

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One knows on the other hand (0, 23.2.7) that one can take the V_λ such that there is a sub- A -algebra finite C of K such that V_λ is the integral closure of $C_{\mathfrak{p}_\lambda}$ where $\mathfrak{p}_\lambda$ is a prime ideal of height 1 in C . If one sets $C'_{\mathfrak{p}_\lambda} = C_{\mathfrak{p}_\lambda} \otimes_A A'$, the morphism $\text{Spec}(C'_{\mathfrak{p}_\lambda}) \rightarrow \text{Spec}(C_{\mathfrak{p}_\lambda})$ is normal (6.8.2) and one has $V'_\lambda = V_\lambda \otimes_{C_{\mathfrak{p}_\lambda}} C'_{\mathfrak{p}_\lambda}$; one can therefore replace B by V_λ and A by $C_{\mathfrak{p}_\lambda}$, hence suppose that A is local, integral and of *dimension* 1, B its integral closure and a discrete valuation ring. There is a sub- A -algebra *finite* A_1 of B such that the morphism $\text{Spec}(B) \rightarrow \text{Spec}(A_1)$ is radicial (0, 23.2.5), which implies in particular that A_1 is also a local ring (obviously of dimension 1); furthermore one can suppose that B and A_1 have the same residue field (*loc. cit.*). One can therefore by the same method replace A by A_1 . Applying if necessary the procedure from the beginning of I), one can suppose finally that A' is also a local ring and that the homomorphism $A \rightarrow A'$ is *local*.

III) *End of the proof.* — We shall first establish the following lemma: □

Lemma (6.14.1.2). — *Let A be a local Noetherian integral ring of dimension 1, A' a local Noetherian integral ring, $A \rightarrow A'$ a local homomorphism such that the corresponding morphism $\text{Spec}(A') \rightarrow \text{Spec}(A)$ is normal. Let K be the field of fractions of A , $\mathfrak{m}$ the maximal ideal of A , $k = A/\mathfrak{m}$ its residue field.*

- (i) If $\mathfrak{q}'_j$ ($1 \leq j \leq r$) are the minimal ideals of A' , $K' = K \otimes_A A'$ is a direct product of integral rings and integrally closed $K'_j \supset A'/\mathfrak{q}'_j$ ($1 \leq j \leq r$), K'_j having as field of fractions the field of fractions L'_j of $A'/\mathfrak{q}'_j$, so that the total ring of fractions L' of A' is identified with the direct product of the L'_j and A' is a subring of K' .
- (ii) The ideal $\mathfrak{p}' = \mathfrak{m}A'$ of A' is prime; the ring $A'_{\mathfrak{p}'}$ is of dimension 1 and is identified with a sub-product of the fields L'_j for the indices j such that $\mathfrak{q}'_j \subset \mathfrak{p}'$.

(iii) If $\overline{A'_{p'}}$ denotes the subring of L' , product of $A'_{p'}$ and of the L'_j such that $q'_j \not\subset p'$, one has

$$(6.14.1.3) \quad A' = K' \cap \overline{A'_{p'}}.$$

Proof. Assertion (i) has already been seen in the course of the proof, independent of the hypothesis on the dimension of A . The hypothesis that the morphism $\text{Spec}(A') \rightarrow \text{Spec}(A)$ is normal implies that $A' \otimes_A k = A'/mA'$ is a normal local ring and *integral*, which shows already that $p' = mA'$ is prime. One has, by (6.1.2), $\dim(A'_{p'}) = \dim(A) + \dim(A'_{p'}/mA'_{p'})$. But $A'_{p'}/mA'_{p'} = A'_{p'}/p'A'_{p'}$ is the residue field of $A'_{p'}$, hence $\dim(A'_{p'}) = 1$. The fact that $A'_{p'}$ is contained in the composed direct of the L'_j for these indices j such that $q'_j \subset p'$, for these indices j are the minimal prime ideals of $A'_{p'}$.

It remains to prove (6.14.1.3). Conversely, let y' be an element of this intersection; let a be a "parameter" for A , i.e. an element such that Aa is m -primary, and let a' be the image of a in A' ; every element of K can be written x/a^n for $x \in A$ and $n > 0$ since Aa contains a power of m ; hence one can write $y' = x'/a'^n$ with $x' \in A'$. Note now that p' is the *only* prime ideal associated to $A'/a'^n A' = (A/a^n A) \otimes_A A'$: as A' is a flat A -module, this follows from (3.3.1), m being the only prime ideal associated to $A/a^n A$ since A is integral. Hence, $a'^n A'$ is the inverse image in A' of $a'^n A'_{p'}$; as by hypothesis $y' \in \widehat{A'_{p'}}$, the image of x' in $A'_{p'}$ belongs to $a'^n A'_{p'}$; whence $x' \in a'^n A'$ and $y' \in A'$, which finishes the proof.

In the situation in which we have arrived at the end of II), B' is *radicial* over A' (I, 3.5.7) and is hence also a local ring; furthermore B' is integer over A' , hence, if q' is the unique prime ideal of B' lying over p' , $q'B'_{p'}$ is the unique maximal ideal of $B'_{p'}$ (Bourbaki, *Alg. comm.*, chap. V, § 2, no 1, prop. 1), hence $B'_{q'} = B'_{p'} = B \otimes_{A_p} A'_{p'}$. We shall first prove that $B'_{p'}$ is *Noetherian* and *normal*. $B'_{p'}$ contains $A'_{p'}$, and is contained in $K'_{p'}$, hence in the product L_1 of the L'_j such that $q'_j \subset p'$. The composed direct L'' of the fields of fractions of these latter is also the total ring of fractions of A' , hence also that of $B'_{p'}$. For every index j such that $q'_j \subset p'$, let α'_j be the product of the L'_h for these indices h such that $h \neq j$, so that $\alpha'_j \cap A'_{p'} = q'_j A'_{p'}$; as every element of $A' - p'$ is regular in L_1 , one has also

$$\alpha'_j \cap A'_{p'} = q'_j A'_{p'}.$$

Let $\tau'_j = B'_{p'} \cap \alpha'_j$, so that $B'_{p'}/\tau'_j$ is isomorphic to the projection of $B'_{p'}$ in L'_j ; hence $B'_{p'}/\tau'_j$ contains the local integral ring of dimension 1, $A'_{p'}/q'_j A'_{p'}$, and is contained in its field of fractions L'_j ; it is hence a Noetherian ring by virtue of the theorem of Krull-Akizuki (Bourbaki, *Alg. comm.*, chap. VII, § 2, no 5, prop. 5).

As the intersection of the τ'_j is reduced to 0, one deduces that $B'_{p'}$ itself is Noetherian, by virtue of the classical lemma:

Lemma (6.14.1.4). — *Let R be a ring, a and b two ideals of R ; if R/a and R/b are Noetherian, so is $R/(a \cap b)$.*

Proof. Indeed, let c be an ideal of R such that $a \cap b \subset c$; one has therefore $a \cap b \subset a \cap c \subset a$; now $c/(a \cap c)$ is an R -module isomorphic to $(a + c)/a$, hence an ideal of R/a , and hence is of finite type; on the other hand $(a \cap c)/(a \cap b)$ is a sub- R -module of $a/(a \cap b)$, and this latter is isomorphic to $(a + b)/b$, hence is of finite type as an ideal of R/b ; hence $(a \cap c)/(a \cap b)$ is also an R -module of finite type, and it follows finally that $c/(a \cap b)$ is also an R -module of finite type. $\square$

Let us note on the other hand that $\text{Spec}(A)$ is made up of the closed point m and of the generic point (o) , of residue fields k and K respectively; as in the case where we are placed, the fields of fractions of B and its residue field are respectively isomorphic to K and k , the fibres of the morphism $u : \text{Spec}(B'_{p'}) \rightarrow \text{Spec}(B)$ at the closed point and at the generic point of $\text{Spec}(B)$ are respectively *isomorphic* to the fibres of the morphism $\text{Spec}(A'_{p'}) \rightarrow \text{Spec}(A)$ at the closed point and at the generic point of $\text{Spec}(A)$ (I, 3.6.4), hence are *geometrically normal*; as on the other hand the morphism u is flat, we conclude that it is *normal* (6.8.1). But B and $B'_{p'}$ are Noetherian and B is normal, hence (6.5.4) $B'_{p'}$ is *normal*.

This being so, B is union of the filtered increasing family of its sub- A -algebras of finite type A_α ; by flatness, B' is union of the filtered increasing family of the $A'_\alpha = A_\alpha \otimes_A A'$; if one sets $p'_\alpha = q' \cap A'_\alpha$,

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$B'_{q'}$ is also union of the filtered increasing family of the $(A'_\alpha)_{p'_\alpha}$ (5.13.3). Denote by L'' the direct product of the fields L'_j such that $q'_j \not\subset p'$; then for every α , $(A'_\alpha)_{p'_\alpha}$ is contained in $K'_{p'}$, and the ring $\overline{B'_{q'}} = L'' \times B'_{q'}$ is therefore union of the rings $L'' \times (A'_\alpha)_{p'_\alpha} = (\overline{A'_\alpha})_{p'_\alpha}$. But each of the A_α is local, Noetherian, integral and of dimension 1, and the morphism $\text{Spec}(A'_\alpha) \rightarrow \text{Spec}(A_\alpha)$ is normal (6.8.2), hence one can apply (6.14.1.2) and one has

$$A'_\alpha = K' \cap (\overline{A'_\alpha})_{p'_\alpha}$$

for every α ; taking the inductive limit of each of the two members, it follows that

$$B' = K' \cap \overline{B'_{q'}}.$$

But $\overline{B'_{q'}}$, being the direct product of the normal rings L'' and $B'_{q'}$, is normal, and as it has the same total ring of fractions as K' , the lemma (6.14.1.1) shows that B' is normal. $\square$

Corollary (6.14.2). — *Let k be a field, X a k -prescheme normal (not necessarily locally Noetherian). Then, for every separable extension k' of k , $X_{(k')} = X \otimes_k k'$ is normal.*

Proof. One knows indeed (6.7.6) that the morphism $\text{Spec}(k') \rightarrow \text{Spec}(k)$ is then normal. $\square$

Corollary (6.14.3). — *Let k be a field, X, Y two k -preschemes integral and normal, whose fields of rational functions $K = R(X)$, $L = R(Y)$ are separable extensions of k . Then the prescheme $X \times_k Y$ is normal.*

Proof. The question being local on X and Y , one can suppose that $X = \text{Spec}(A)$, $Y = \text{Spec}(B)$, where A and B are two integral rings and integrally closed, of fields of fractions K and L respectively; suppose first that A is a k -algebra of finite type, hence K is a finite type extension of k . By flatness, $A \otimes_k B$ is a sub- k -algebra of $K \otimes_k L$; now $K \otimes_k L$ is a Noetherian normal ring (by virtue of (6.7.1) or (6.14.2)), hence a direct product of finitely many integral rings C_i , so that if E_i is the field of fractions of C_i , the direct product E of the E_i is the total ring of fractions of $K \otimes_k L$; it is also the total ring of fractions of $A \otimes_k B$, since $K \otimes_k L$ is a ring of fractions of $A \otimes_k B$. This being so, one can write $A \otimes_k B = (A \otimes_k L) \cap (K \otimes_k B)$, and it follows from (6.14.2) that $A \otimes_k L$ and $K \otimes_k B$ are normal, hence $A \otimes_k B$ is normal by virtue of (6.14.1.1).

Consider now the general case; A is then union of the filtered increasing family of its sub- k -algebras of finite type A_α , hence $A \otimes_k B$ is the inductive limit of the normal rings $A_\alpha \otimes_k B$. To prove that $A \otimes_k B$ is normal, it suffices hence (5.13.6) to prove that every irreducible component of $\text{Spec}(A_\beta \otimes_k B)$ dominates one irreducible component of $\text{Spec}(A_\alpha \otimes_k B)$ for $A_\alpha \subset A_\beta$, which follows immediately from the fact that B is a k -module flat, from (2.3.7, (ii)) and from the fact that A_α and A_β are integral rings. $\square$

Proposition (6.14.4). — *Let A be a Noetherian ring, A' a Noetherian A -algebra such that the morphism $\text{Spec}(A') \rightarrow \text{Spec}(A)$ is normal. Let B be an A -algebra, C the integral closure of A in B ; set $B' = B \otimes_A A'$, $C' = C \otimes_A A'$, C' being identified with a subring of B' ; then C' is the integral closure of A' in B' .*

Proof. We shall proceed in several steps.

I) *Reduction to the case where B is reduced.* — Set $B_0 = B_{\text{red}} = B/\mathfrak{N}$ where $\mathfrak{N}$ is the nilradical of B , and let C_0 be the integral closure of A in B_0 . One has the following lemma:

Lemma (6.14.4.1). — *Let A be a ring, B an A -algebra, $B_0 = B/\mathfrak{n}$ the quotient of B by a nilideal. If C_0 is the integral closure of A in B_0 , the inverse image C of C_0 by the canonical homomorphism $\varphi : B \rightarrow B_0$ is the integral closure of A in B .*

Proof. Indeed, if $x \in B$ is such that $\varphi(x)$ satisfies an integral dependence equation over A , one deduces that x satisfies a relation of the form $x^n + a_1 x^{n-1} + \dots + a_n \in \mathfrak{n}$ with $a_k \in A$, whence, by raising to a sufficiently large power, an integral dependence equation for x , with coefficients in A . $\square$

One has therefore the exact sequence of A -modules

$$0 \longrightarrow C \longrightarrow B \longrightarrow B_0/C_0$$

whence, by flatness, the exact sequence

$$0 \longrightarrow C' \longrightarrow B' \longrightarrow B'_0/C'_0$$

where one has set $B'_0 = B_0 \otimes_A A'$, $C'_0 = C_0 \otimes_A A'$. If one proves that C'_0 is the integral closure of A' in B'_0 , the lemma (6.14.4.1) will prove that C' is the integral closure of A' in B' . One can therefore restrict to the case where B is an A -algebra of finite type; let q_i ($1 \leq i \leq n$) be its minimal prime ideals; as B is supposed *reduced*, it is identified with a *subring* of the product B_0 of the B/q_i ; if C_0 is the integral closure of A in B_0 , one has $C = B \cap C_0$; if one sets $B'_0 = B_0 \otimes_A A'$, $C'_0 = C_0 \otimes_A A'$, one has, by flatness, $C' = B' \cap C'_0$ (0_I, 6.1.3); it suffices therefore to prove that C'_0 is the integral closure of A' in B'_0 . But C_0 is the direct product of the C_i , integral closures of A in the $B_i = B/q_i$; hence C'_0 is the direct product of the $C'_i = C_i \otimes_A A'$ and it suffices to show that C'_i is the integral closure of A' in $B'_i = B_i \otimes_A A'$. One is thus reduced to the case where B is integral and an A -algebra of finite type; if $\mathfrak{p}$ is the kernel of the homomorphism $A \rightarrow B$, one has also $B' = B \otimes_{A/\mathfrak{p}} (A'/\mathfrak{p}A')$; as the morphism $\text{Spec}(A'/\mathfrak{p}A') \rightarrow \text{Spec}(A/\mathfrak{p})$ is normal, one can replace A by $A/\mathfrak{p}$ and A' by $A'/\mathfrak{p}A'$, and suppose hence that $A \subset B$.

II) *Reduction to the case where B is an integral A -algebra of finite type containing A .* — Let (B_α) be the filtered increasing family of sub- A -algebras of finite type of B , and let C_α be the integral closure of A in B_α . It follows immediately from the definition of the integral closure that C is union of the C_α ; if one sets $B'_\alpha = B_\alpha \otimes_A A'$, $C'_\alpha = C_\alpha \otimes_A A'$, C'_α is contained in B'_α by flatness, and for the same reason B' is union of the filtered increasing family of the B'_α and C' union of the filtered increasing family of the C'_α . If one proves that C'_α is the integral closure of A' in B'_α , it will follow immediately that C' is the integral closure of A' in B' . One can therefore restrict to the case where B is an A -algebra of finite type; let q_i ($1 \leq i \leq n$) be its minimal prime ideals; as B is supposed *reduced*, it is identified with a *subring* of the product B_0 of the B/q_i ; if C_0 is the integral closure of A in B_0 , one has $C = B \cap C_0$; if one sets $B'_0 = B_0 \otimes_A A'$, $C'_0 = C_0 \otimes_A A'$, one has, by flatness, $C' = B' \cap C'_0$ (0_I, 6.1.3); it suffices therefore to prove that C'_0 is the integral closure of A' in B'_0 . But C_0 is the direct product of the C_i , integral closures of A in the $B_i = B/q_i$; hence C'_0 is the direct product of the $C'_i = C_i \otimes_A A'$ and it suffices to show that C'_i is the integral closure of A' in $B'_i = B_i \otimes_A A'$. One is thus reduced to the case where B is integral and A -algebra of finite type; if $\mathfrak{p}$ is the kernel of the homomorphism $A \rightarrow B$, one has also $B' = B \otimes_{A/\mathfrak{p}} (A'/\mathfrak{p}A')$; as the morphism $\text{Spec}(A'/\mathfrak{p}A') \rightarrow \text{Spec}(A/\mathfrak{p})$ is normal, one can replace A by $A/\mathfrak{p}$ and A' by $A'/\mathfrak{p}A'$, and suppose hence that $A \subset B$.

III) *Case where A is a field, B a field, finite extension of A and such that A is algebraically closed in B .* — One has then $C = A$, and A' is a Noetherian geometrically normal A -algebra, hence a direct product of integrally closed integral rings A'_i , the fields of fractions K'_i of the A'_i being *separable* extensions of A (4.6.1). The A' -algebra B' is therefore the direct product of the $B \otimes_A A'_i = B'_i$. One can therefore restrict to the case where A' is integral, its field of fractions K' being a separable extension of A . Now, as K' is a separable extension of A , $B \otimes_A K'$ is reduced (4.3.7), and as furthermore B is a primary extension of A , $B \otimes_A K'$ is integral (4.3.2); it is hence of the same type as B' . Let L' be the field of fractions of $B \otimes_A K'$ (which is also that of B'); it is obviously a composed field of $B(K')$ of its sub-fields B and K' ; as A is algebraically closed in B , that K' is a separable extension of A , and that B and K' are linearly disjoint over A , K' is algebraically closed in $B(K') = L'$ (Bourbaki, *Alg.*, chap. V, § 9, exerc. 2); *a fortiori*, A' , being integrally closed, is integrally closed in L' , hence in B' , which proves the proposition in this case.

IV) *Case where A is integral, B a field, finite extension of the field of fractions K of A .* — Then $B' = (A' \otimes_A K) \otimes_K B$ is a B -algebra Noetherian geometrically normal, and hence B' is a direct product of integral rings; the total ring of fractions L' of B' is then a direct product of finitely many fields. This being so, as B is the field of fractions of C , B' is a ring of fractions of C' and L' is therefore also the total ring of fractions of C' . Now (0, 6.14.1) as C is a normal ring and $\text{Spec}(A') \rightarrow \text{Spec}(A)$ a normal morphism of Noetherian rings, C' is a normal ring; but it follows from (6.14.1.1) that C' is then integrally closed in L' , and *a fortiori* in B' , and hence is the integral closure of A' in B' .

V) *End of the proof.* — By part II), one can suppose that B is an A -algebra integral of finite type containing A . Let K be the field of fractions of A , L that of B , which is a finite type extension of K . Let M be the algebraic closure of K in L , which is a finite algebraic extension of K ; let C_0 be the integral closure of A in M , which is also the integral closure of A in L ; if one sets $C'_0 = C_0 \otimes_A A'$, one has hence $C' = B' \cap C'_0$ by flatness (0_I, 6.1.3). Now, it follows from IV) that C'_0 is the integral closure of A' in $M' = M \otimes_A A'$; furthermore, M' is a Noetherian ring and the morphism $\text{Spec}(M') \rightarrow \text{Spec}(M)$ is normal (as we have seen in the course of IV)); applying III) to M and M' in place of A and A' and

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to L in place of B , one deduces that C'_0 is the integral closure of A' in L' , and $C' = C'_0 \cap B'$ is the integral closure of A' in L' , and hence C' is the integral closure of A' in B' . $\square$

Corollary (6.14.5). — *Let A be a Noetherian ring, A' a Noetherian A -algebra such that the morphism $\text{Spec}(A') \rightarrow \text{Spec}(A)$ is normal. Let B, C be two A -algebras, $\varphi : B \rightarrow C$ an A -homomorphism, D the integral closure of B in C . If one sets $B' = B \otimes_A A'$, $C' = C \otimes_A A'$, $D' = D \otimes_A A'$, D' is the integral closure of B' in C' .*

Proof. Let (B_λ) be the filtered increasing family of sub- A -algebras of finite type of B , and set $B'_\lambda = B_\lambda \otimes_A A'$, so that B' is union of the filtered increasing family of the sub- A' -algebras of finite type B'_λ . Let D_λ be the integral closure of B_λ in C , and set $D'_\lambda = D_\lambda \otimes_A A'$, so that D is union of the D_λ and D' union of the D'_λ ; if we prove that D'_λ is the integral closure of B'_λ in C' , it will follow that D' is the integral closure of B' in C' . Now B_λ and B'_λ are Noetherian and the morphism $\text{Spec}(B'_\lambda) \rightarrow \text{Spec}(B_\lambda)$ is normal (6.8.2); one can therefore apply (6.14.4) by replacing A, A' and B by B_λ, B'_λ and C respectively, whence the corollary. $\square$

One can for example apply (6.14.5) when A is its local ring *excellent* and A' its completion $\hat{A}$, since in this case $\text{Spec}(A') \rightarrow \text{Spec}(A)$ is a regular morphism (7.8.2).

6.15. Geometrically unibranch preschemes

(6.15.1). We shall say that a prescheme X is *unibranch* (resp. *geometrically unibranch*) at a point x , or that the point x is *unibranch* (resp. *geometrically unibranch*) in X if the local ring $\mathcal{O}_{X,x}$ is unibranch (resp. geometrically unibranch) (0, 23.2.1); we say that X is *unibranch* (resp. *geometrically unibranch*) if it is so at every point. It follows from this definition that, for X to be unibranch (resp. geometrically unibranch) at a point, it is necessary and it suffices that X_{red} be so at this point.

(6.15.2). One must take care that with the definitions of (6.15.1), a local ring A can be unibranch (resp. geometrically unibranch) *without* $\text{Spec}(A)$ being so; in other terms, it can happen that there are prime ideals $\mathfrak{p}$ of A such that $A_{\mathfrak{p}}$ is not unibranch; it amounts to the same to say that on a prescheme, the notion of geometrically unibranch point is *not stable by generization*. We shall see this on the following example.

Let K be an algebraically closed field of characteristic 0, B the integral ring $K[U, V, W]/(U^2(U - W) - V^2(U + W))$ (U, V, W indeterminates) so that $Y = \text{Spec}(B)$ is a “cone with vertex the origin, having a double generatrix”. We shall designate by u, v, w the images of U, V, W in B . Let R be the field of fractions of B , and consider in R the element $t = v(u + w)/u$, which does not belong to B ; let us show that $C = B[t]$ is the *integral closure* of B . One has indeed $t^2 = u^2 - w^2$, hence t is integral over B , and $v = tu/(u + w)$; the ring $C_1 = K[t, u, w]$ is integrally closed, for it is isomorphic to $K[T, U, W]/(T^2 - U^2 + W^2)$ and is therefore the integral closure of the integrally closed ring $K[U, W]$ in the quadratic extension $K(U, W)(\sqrt{U^2 + W^2})$ of its field of fractions (Bourbaki, *Alg. comm.*, chap. V, § 1, no 6, prop. 18). The ring of fractions $K[t, u, w, 1/(u + w)]$ of C_1 is hence integrally closed. In the same way, one sees that the ring $C_2 = K[t, v, w]$ is integrally closed, for t is a root of the equation $t(t - v) - w^2(2v - t) = 0$, and hence $K[t, v, w, 1/(t - v)] = K[t, v, w, \frac{tu}{u+w}, w, \frac{u+w}{tw}]$ is integrally closed. Finally, taking into account the fact that u and w are algebraically independent over K , one proves easily that $C = K[t, u, v, w] = K[t, u, w, 1/(u + w)] \cap K[t, v, w, 1/(t - v)]$, which finishes showing that C is the integral closure of B . It is immediate that if $\mathfrak{m}_0$ is the maximal ideal of C lying over $\mathfrak{n}_0$, the ideal generated by u, v, w (“vertex of the cone”), there is only one maximal ideal of C lying over $\mathfrak{m}_0$, namely the ideal generated by t, u, v, w . If one sets $A = B_{\mathfrak{m}_0}$, one deduces easily that $A' = C_{\mathfrak{m}_0}$ is the integral closure of A , which is hence unibranch, and hence also geometrically unibranch since its residue field K is algebraically closed. But in A the prime ideal $\mathfrak{p}$ generated by u and v is such that the integral closure $A_{\mathfrak{p}}[t]$ of $A_{\mathfrak{p}}$ is not a local ring.

We shall see further below (9.7.10) that when X is a locally Noetherian prescheme such that, if X' is the normalization of X_{red} , the canonical morphism $X' \rightarrow X$ is *finite* (which will be the case if X is such that the rings of its affine open subsets are *universally Japanese* (0, 23.1.1)), then the set of points $x \in X$ where X is geometrically unibranch is *locally constructible*.

(6.15.3). We shall say for short that a morphism $f : X \rightarrow Y$ is *radicial at a point* $y \in Y$, if $f^{-1}(y)$ is empty or reduced to a single point x and if $k(x)$ is a radicial extension of $k(y)$. It amounts to the same

to say that f is radicial at every point of Y or that f is radicial (I, 3.5.8). If $g : f^{-1}(y) \rightarrow \operatorname{Spec}(k(y))$ is the base change inverse image of f by the base change $\operatorname{Spec}(k(y)) \rightarrow Y$, it amounts to the same to say that f is radicial at the point $y \in Y$, or that g is radicial.

Lemma (6.15.3.1). — (i) Let $f : X \rightarrow Y$, $g : Y \rightarrow Z$ be two morphisms. If f is radicial at a point y and g radicial at the point $z = g(y)$, then $g \circ f$ is radicial at the point $g(y)$. The converse is true if f is surjective.

(ii) Let $f : X \rightarrow Y$, $h : Y' \rightarrow Y$ be two morphisms, and let $f' = f|_{Y'} : X \times_Y Y' \rightarrow Y'$. Let $y' \in Y'$ and $y = h(y')$; then for f to be radicial at the point y , it is necessary and it suffices that f' be radicial at the point y' .

Proof. (i) If f is radicial at the point y and g radicial at the point z , $g^{-1}(z)$ is reduced to y and $f^{-1}(g^{-1}(z)) = f^{-1}(y)$ is empty or reduced to a single point x ; furthermore $k(x)$ is a radicial extension of $k(y)$ and $k(y)$ a radicial extension of $k(z)$, hence $k(x)$ is a radicial extension of $k(z)$. Conversely, suppose f surjective; if $g \circ f$ is radicial at the point $z = g(y)$, $g^{-1}(z)$ is reduced to the point y , for otherwise $f^{-1}(g^{-1}(z))$ would have at least two distinct points; furthermore, $f^{-1}(y) = f^{-1}(g^{-1}(z))$ is reduced to a single point x , and by hypothesis one has $k(z) \subset k(y) \subset k(x)$, and $k(x)$ is a radicial extension of $k(z)$, hence $k(y)$ is a radicial extension of $k(z)$ and $k(x)$ radicial over $k(y)$.

(ii) Let $g : f^{-1}(y) \rightarrow \operatorname{Spec}(k(y))$, $g' : f'^{-1}(y') \rightarrow \operatorname{Spec}(k(y'))$ be the morphisms inverse images of f and f' respectively; as $f'^{-1}(y') = f^{-1}(y) \otimes_{k(y)} k(y')$ (I, 3.6.4), g' is the inverse image of g and the assertion reduces to (2.6.1, (v)). $\square$

Suppose then X a prescheme *reduced* not having more than a *finite* number of irreducible components, and let X' be its *normalization* (II, 6.3.8); we say (*loc. cit.*) that the canonical morphism $f : X' \rightarrow X$ is surjective. The definition (6.15.1) shows then that for X to be geometrically unibranch at a point, it is necessary and it suffices that f be *radicial at this point*; for X to be geometrically unibranch, it is necessary and it suffices that f be *radicial*.

(6.15.4). Generalizing the definition given in (I, 2.2.9), we shall say, for two reduced preschemes X, Y , a morphism $f : X \rightarrow Y$ is *birational* if the restriction of f to the set of maximal points of X is a *bijection* of this set onto the set of maximal points of Y , and if, for every maximal point y of Y , the morphism $X \times_Y \operatorname{Spec}(\mathcal{O}_y) \rightarrow \operatorname{Spec}(\mathcal{O}_y)$ induced by f is an isomorphism (in other words, if for the fibre $f^{-1}(y)$ reduced to a single point x (maximal in X) and if the homomorphism $k(y) \rightarrow k(x)$ induced by f is bijective); this notion reduces to that of (I, 2.2.9) when X and Y have only a *finite* number of irreducible components.

Lemma (6.15.4.1). — Let $f : X \rightarrow Y$ be a morphism, $g : Y' \rightarrow Y$ a flat morphism, $X' = X \times_Y Y'$, $f' = f|_{Y'} : X' \rightarrow Y'$. If f_{red} is birational, so is f'_{red} .

Proof. Indeed, if y' is a maximal point of Y' , one knows (2.3.4, (ii)) that $y = g(y')$ is a maximal point of Y ; there is a unique point x of X in $f^{-1}(y)$ by hypothesis; since moreover $k(x) = k(y)$. It follows therefore from (I, 3.4.9) that $f'^{-1}(y')$ is reduced to a single point x' and that $k(x') = k(y')$. We conclude from the reasoning that, noting (2.3.7, (ii)), every irreducible component of X' dominates one irreducible component of Y' . $\square$

Proposition (6.15.5). — Let $f : X \rightarrow Y$ be a morphism such that f_{red} is entire and birational (6.15.4), hence surjective.

- (i) For Y to be geometrically unibranch at a point y , it is necessary and it suffices that f be radicial at the point y and that X be geometrically unibranch at the unique point x of $f^{-1}(y)$.
- (ii) For Y to be geometrically unibranch, it is necessary and it suffices that X be geometrically unibranch and that f be radicial.

Proof. One can obviously restrict to the case where X and Y are *reduced*; the fact that f is surjective follows from the fact that f is closed (II, 6.1.10) and that $f(X)$ contains all the maximal points of Y . This also shows immediately that (i) implies (ii). Using (I, 3.6.5) and (II, 6.1.5), one can suppose, to prove (i), that $Y = \operatorname{Spec}(\mathcal{O}_{Y,y})$, in other words $Y = \operatorname{Spec}(A)$ (with $A = \mathcal{O}_{X,x}$) is a local scheme. Note that $A' = \mathcal{O}_{X',x'}$ is by hypothesis a faithfully flat A -module, hence containing A , and hence the hypothesis that X is geometrically unibranch at the point x , or the hypothesis that X' is geometrically

unibranch at the point x' , both imply that A is a local ring *integral* (since A , being reduced, is also isomorphic to a subring of A'_{red}). Let K be the field of fractions of A , B the integral closure of A , and set $Y = \text{Spec}(B)$; it amounts to the same to say that X is geometrically unibranch at the point x or that the morphism $g : Y \rightarrow X$ is radicial at the point x (6.15.3). Let $Y' = Y \otimes_k k' = Y \times_X X'$ so that one has the commutative diagram

$$\begin{array}{ccc} Y & \xrightarrow{h} & Y' \\ g \uparrow & & \uparrow g' \\ X & \xrightarrow{f} & X' \end{array}$$

Note that f is a flat morphism; hence (6.15.4.1) the entire morphism g'_{red} is birational. As Y is normal, Y' is geometrically unibranch (6.15.6). For X to be geometrically unibranch at the point x , it is necessary and it suffices therefore that g' be radicial at the point x' (6.15.5). Now, x is the projection of x' in X and it is equivalent to say that g is radicial at the point x or that g' is radicial at the point x' (6.15.3.1); finally, for X to be geometrically unibranch at this point, it is necessary and it suffices that g be radicial at this point, whence the proposition. $\square$

Proposition (6.15.6). — *Let k be a field, X a k -prescheme. If X is normal, then, for every extension k' of k , $X' = X \otimes_k k'$ is geometrically unibranch.*

Proof. One knows that k' is an algebraic extension of a pure extension k_0 of k , and if k'' is the largest separable extension of k_0 contained in k' , k' is a radicial extension of k'' and k'' a separable extension of k . One knows (6.14.2) that $X'' = X \otimes_k k''$ is normal; as $X \otimes_k k' = X'' \otimes_{k''} k'$, one sees that one can restrict to the case where the extension k' of k is *radicial*. Furthermore (I, 3.6.5), one can suppose that $X = \text{Spec}(A)$, where A is a *local integral and integrally closed* ring (since X is normal). The projection morphism $f : X' \rightarrow X$ is a universal homeomorphism, since $\text{Spec}(k') \rightarrow \text{Spec}(k)$ is a universal homeomorphism (2.4.5); as $X' = \text{Spec}(A')$ where $A' = A \otimes_k k'$, one sees therefore that A' is a local ring whose nilradical $\mathfrak{N}'$ is the unique minimal prime ideal, whence $X'_{\text{red}} = \text{Spec}(A'_0)$, where $A'_0 = A' / \mathfrak{N}'$ is a local integral ring; furthermore, if K is the field of fractions of A , the field of fractions K'_0 of A'_0 is radicial over K , since the morphism f is radicial. But A'_0 is integer over A , its integral closure B is also the integral closure of A , and one knows (Bourbaki, *Alg. comm.*, chap. V, § 2, no 3, lemma 3) that B is the set of $x \in K'_0$ of which a power p^m -th (for m large enough) belongs to A (p being the characteristic exponent of K); furthermore there exists only a single prime ideal of B lying over every prime ideal of A ; in particular B is a local ring and its residue field is a radicial extension of that of A , and *a fortiori* of the same type for A'_0 , which proves that A'_0 is geometrically unibranch, and hence so is X' as well. $\square$

Proposition (6.15.7). — *Let k be a field, X a k -prescheme, k' an extension of k , $X' = X \otimes_k k'$. Let x' be a point of X' , x its projection in X . For X to be geometrically unibranch at the point x , it is necessary and it suffices that X' be geometrically unibranch at the point x' . For X to be geometrically unibranch, it is necessary and it suffices that X' be so.*

Proof. The second assertion follows from the first and from the fact that the projection $f : X' \rightarrow X$ is a surjective morphism. To prove the first assertion, one can, by virtue of (I, 5.1.8), restrict to the case where X is reduced, and by virtue of (I, 3.6.5), to the case $X = \text{Spec}(A)$ (with $A = \mathcal{O}_{X,x}$) is a local scheme. Note that $A' = \mathcal{O}_{X',x'}$ is by hypothesis a faithfully flat A -module, hence containing A , and hence the hypothesis that X is geometrically unibranch at the point x , or the hypothesis that X' is geometrically unibranch at the point x' , both imply that A is a local ring *integral* (since A , being reduced, is also isomorphic to a subring of A'_{red}). Let K be the field of fractions of A , B the integral closure of A , and set $Y = \text{Spec}(B)$; it amounts to the same to say that X is geometrically unibranch at the point x or that the morphism $g : Y \rightarrow X$ is radicial at the point x (6.15.3). Let

$Y' = Y \otimes_k k' = Y \times_X X'$ so that one has the commutative diagram

$$\begin{array}{ccc} Y & \xrightarrow{h} & Y' \\ g \uparrow & & \uparrow g' \\ X & \xrightarrow{f} & X' \end{array}$$

Note that f is a flat morphism; hence (6.15.4.1) the entire morphism g'_{red} is birational. As Y is normal, Y' is normal (and *a fortiori* geometrically unibranch) (6.14.1). For X' to be geometrically unibranch at x' , it is necessary and it suffices therefore that g' be radicial at this point (6.15.5); we conclude hence from (6.15.3.1) that f' is radicial at every point $x' \in p^{-1}(x)$, and it follows then from (6.15.5) that X' is geometrically unibranch (hence integral since it is reduced) at each of these points. $\square$

Lemma (6.15.8). — *Let k be a separably closed field (in other words, such that the algebraic closure of k is radicial over k), X a k -prescheme locally of finite type over k , x a closed point of X . If X is unibranch at the point x , it is geometrically unibranch at this point.*

Proof. Indeed, one knows (I, 6.4.2) that $k(x)$ is an algebraic extension of k ; as the residue field of the integral closure $(\mathcal{O}_x)_{\text{red}}$ (integral closure which is by hypothesis a local ring) is an algebraic extension of $k(x)$, it is a radicial extension of k by hypothesis, hence also of $k(x)$. $\square$

Corollary (6.15.9). — *Let k be a field, X a k -prescheme locally of finite type over k , k' a separable extension close of k (in other words, such that the algebraic closure of k is radicial over k'); then, for X to be geometrically unibranch, it is necessary and it suffices that $X' = X \otimes_k k'$ be unibranch.*

Proof. In view of (6.15.7), one is reduced to proving that if X is unibranch and k separably closed, then X is geometrically unibranch. Now, X is geometrically unibranch at its closed points (6.15.8); we conclude from (10.4.9) that X is geometrically unibranch at all its points, relying on the fact (pointed out at the end of (6.15.2)) that the set of points where X is geometrically unibranch is constructible (of course, the corollary (6.15.9) will not be used before that point). $\square$

This result justifies, to a certain extent, the terminology of “geometrically unibranch”.

Proposition (6.15.10). — *Let Y, Y' be two locally Noetherian preschemes, $g : Y' \rightarrow Y$ a normal morphism (6.8.1), $f : X \rightarrow Y$ a morphism; set $X' = X \times_Y Y'$ and let $p : X' \rightarrow X$ be the canonical projection, x' a point of X' . Then, if X is reduced (resp. geometrically unibranch, resp. integral and geometrically unibranch) at the point $x = p(x')$, X' is reduced (resp. geometrically unibranch, resp. integral and geometrically unibranch) at the point x' .*

Proof. By virtue of (I, 3.6.5), one can restrict to the case where $Y = \text{Spec}(A)$, $Y' = \text{Spec}(A')$, $X = \text{Spec}(B)$, where A, A', B are local rings, the homomorphisms $A \rightarrow A'$ and $A \rightarrow B$ being local, A, A' Noetherian rings, and x being the closed point of X ; it suffices to prove that if B is reduced (resp. geometrically unibranch, resp. integral and geometrically unibranch) then $B' = B \otimes_A A'$ is also geometrically unibranch at the points of $p^{-1}(x)$, resp. $\text{Spec}(B')$ is integral and geometrically unibranch at these points). Suppose first B reduced; B is union of the filtered increasing family of its sub- A -algebras of finite type B_λ , and by flatness, B' is union of the filtered increasing family of the sub- A' -algebras $B'_\lambda = B_\lambda \otimes_A A'$. Now, the morphism $\text{Spec}(B'_\lambda) \rightarrow \text{Spec}(B_\lambda)$ is normal (6.8.2), and *a fortiori* reduced; the fact that B'_λ is reduced is consequence of (3.3.5); one concludes that B' itself is reduced by (5.13.2).

Suppose now B geometrically unibranch; in view of (I, 5.1.8), one can furthermore suppose B reduced, hence *integral*. Let C be its integral closure, which is by hypothesis a local ring; set $Z = \text{Spec}(C)$, $C' = C \otimes_A A'$, $Z' = \text{Spec}(C') = Z \times_X X'$, so that one has the commutative diagram

$$\begin{array}{ccc} Z & \longrightarrow & Z' \\ l \uparrow & & \uparrow l' \\ X & \xrightarrow{p} & X' \end{array}$$

By virtue of the first part of the reasoning, X' is *reduced*; on the other hand, as $Z' = Z \times_Y Y'$, it follows from (6.14.1) that Z' is normal (and *a fortiori* geometrically unibranch). As f is entire and birational, so is f' by (6.15.4.1), since p is flat. Finally, the hypothesis that X is geometrically unibranch at the point x implies that f is radicial at this point (6.15.5); we conclude hence from (6.15.3.1) that f' is radicial at every point $x' \in p^{-1}(x)$, and it follows then from (6.15.5) that X' is geometrically unibranch (hence integral since it is reduced) at each of these points. $\square$

Remarks (6.15.11). — (i) In the proof of (6.15.10), one cannot dispense with appealing to (6.14.1), even when $X = Y$, for in that case the integral closure is the ring B , which is not necessarily a Noetherian ring even if B is.

(ii) The example (6.5.5, (ii)) shows that in (6.15.10), one cannot replace “geometrically unibranch” by “integral” in the statement, even if the residue field $k(x)$ is algebraically closed and the morphism g is étale.

§7. FORMAL PROPERTIES OF MORPHISMS; NOETHERIAN APPROXIMATION

7.1. Formal equidimensionality and formally catenary rings

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Definition (7.1.1). — We say that a local Noetherian ring A is *formally equidimensional* if its completion $\hat{A}$ is equidimensional (0, 16.1.4).

Proposition (7.1.2). — Let A be a local Noetherian ring, $\mathfrak{p}_i$ ($1 \leq i \leq n$) its minimal prime ideals. For A to be formally equidimensional, it is necessary and sufficient that the $A/\mathfrak{p}_i$ be equidimensional (in other words, that the $A/\mathfrak{p}_i$ have the same dimension).

Proof. Indeed, this follows from the fact that for every prime ideal $\mathfrak{p}'$ of $\hat{A}$, $\mathfrak{p}' \cap A$ contains one of the $\mathfrak{p}_i$, hence $\mathfrak{p}'$ contains one of the $\mathfrak{p}_i \hat{A}$, and $\hat{A}/\mathfrak{p}_i \hat{A} = (A/\mathfrak{p}_i) \otimes_A \hat{A}$ is the completion of the local ring $A/\mathfrak{p}_i$ (0, 7.3.3), hence has the same dimension. Every maximal chain of prime ideals of $\hat{A}$ therefore identifies canonically with a maximal chain of prime ideals of one of the $\hat{A}/\mathfrak{p}_i \hat{A}$, and conversely, whence the conclusion. $\square$

Proposition (7.1.3). — Let A, A' be two local Noetherian rings, $\mathfrak{m}$ the maximal ideal of A , $\varphi : A \rightarrow A'$ a local homomorphism; suppose that A' is a flat A -module, and that A' is equidimensional and catenary. Then:

- i) A is equidimensional and catenary.
- ii) Suppose in addition that $\mathfrak{m}A'$ is an ideal of definition of A' . Then, if $\mathfrak{a}$ is an ideal of A , for $A'/\mathfrak{a}A'$ to be equidimensional, it is necessary and sufficient that $A/\mathfrak{a}A$ be so. In particular, for every prime ideal $\mathfrak{p}$ of A , $A'/\mathfrak{p}A'$ is equidimensional.

Proof. Let us first prove the second assertion of (ii). Set $X = \text{Spec}(A)$, $X' = \text{Spec}(A')$, $Y = V(\mathfrak{p}) = \text{Spec}(A/\mathfrak{p})$, $Y' = V(\mathfrak{p}A') = \text{Spec}(A'/\mathfrak{p}A')$; if $f : X' \rightarrow X$ is the morphism corresponding to φ , Y' is also the closed sub-prescheme $f^{-1}(Y)$ of X' . As $\mathfrak{m}A'$ is an ideal of definition of A' , we have

$$(7.1.3.1) \quad \dim(X) = \dim(X')$$

by virtue of (6.1.3); similarly $\mathfrak{m}A'/\mathfrak{p}A'$ is an ideal of definition of $A'/\mathfrak{p}A'$ and Y' is flat over Y (2.1.4), hence (6.1.3)

$$(7.1.3.2) \quad \dim(Y) = \dim(Y').$$

Let y be the generic point of Y , Y'_i ($1 \leq i \leq r$) the irreducible components of Y' , y'_i the generic point of Y'_i ($1 \leq i \leq r$); we have $f(y'_i) = y$ for all i (2.3.4). On the other hand, y'_i is the maximal point of the fibre $f^{-1}(y)$ (0, 2.1.8), hence $\mathcal{O}_{X',y'_i} \otimes_{\mathcal{O}_{X,y}} k(y) = \mathcal{O}_{X',y'_i}/\mathfrak{m}_y \mathcal{O}_{X',y'_i}$ is of dimension 0, in other words $\mathfrak{m}_y \mathcal{O}_{X',y'_i}$ is an ideal of definition of $\mathcal{O}_{X',y'_i}$; we can again apply (6.1.3), which gives

$$(7.1.3.3) \quad \dim(\mathcal{O}_{X',y'_i}) = \dim(\mathcal{O}_{X,y})$$

that is to say (5.1.2)

$$(7.1.3.4) \quad \text{codim}(Y'_i, X') = \text{codim}(Y, X)$$

for all i . As A' is equidimensional and catenary, we have, by virtue of (0, 14.3.5)

$$(7.1.3.5) \quad \dim(X') = \dim(Y'_i) + \text{codim}(Y'_i, X')$$

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for all i ; by virtue of (7.1.3.1), (7.1.3.2) and (7.1.3.4) and the inequality $\dim(Y'_i) \leq \dim(Y')$, we deduce

$$(7.1.3.6) \quad \dim(X) \leq \dim(Y) + \operatorname{codim}(Y, X)$$

hence the two sides of this inequality are equal by (0, 14.2.2.2); moreover this equality implies

$$(7.1.3.7) \quad \dim(Y'_i) = \dim(Y') = \dim(Y)$$

for all i , which proves the second assertion of (ii).

Let us now prove the first assertion of (ii). Let $\mathfrak{p}_j$ ($1 \leq j \leq n$) be the minimal prime ideals of A among those containing $\mathfrak{a}$, and let $\mathfrak{p}'_{jh}$ be the prime ideals of A' minimal among those containing $\mathfrak{p}_j A'$ ($1 \leq h \leq m_j$); we know then (2.3.4) that the $\mathfrak{p}'_{jh}$ ($1 \leq j \leq n$, $1 \leq h \leq m_j$ for all j) are also the minimal prime ideals among those containing $\mathfrak{a} A'$. From what we have just seen, for all j , the dimensions of the rings $A'/\mathfrak{p}'_{jh}$ ($1 \leq h \leq m_j$) are all equal and are equal to $\dim(A'/\mathfrak{p}_j A')$, itself equal to $\dim(A/\mathfrak{p}_j)$ by (7.1.3.2). To say that $A'/\mathfrak{a} A'$ is equidimensional therefore means that the dimensions of the $A/\mathfrak{p}_j$ are all equal, that is to say that $A/\mathfrak{a}$ is equidimensional.

Let us finally prove (i). Let $\mathfrak{r}'$ be a prime ideal of A' minimal among those containing $\mathfrak{m} A'$, and set $A'' = A'_{\mathfrak{r}'}$; as A'' is a flat A' -module, it is also a flat A -module; moreover A'' is equidimensional and catenary (0, 16.1.4), and $\mathfrak{m} A''$ is an ideal of definition of A'' . We can therefore restrict ourselves to the case where $\mathfrak{m} A'$ is an ideal of definition of A' . If then $\mathfrak{p}$ and $\mathfrak{q} \subset \mathfrak{p}$ are two prime ideals of A , we can apply (ii) to $A/\mathfrak{q}$ and $A'/\mathfrak{q} A'$, which is equidimensional and flat over $A/\mathfrak{q}$; if $Z = \operatorname{Spec}(A/\mathfrak{q})$ and $Y = \operatorname{Spec}(A/\mathfrak{p})$, we have $\dim(Z) = \dim(Y) + \operatorname{codim}(Y, Z)$; moreover one can apply (7.1.3.6), which shows that $\operatorname{codim}(Y, X) = \dim X - \dim Y$; as X is equicodimensional, these relations show that it is biequidimensional by virtue of (0, 14.3.3). $\square$

Corollary (7.1.4). — *Let A be a local Noetherian ring that is formally equidimensional. Then:*

- i) *A is equidimensional and catenary (in other words, biequidimensional).*
- ii) *Let $\mathfrak{a}$ be an ideal of A ; for $A/\mathfrak{a}$ to be equidimensional, it is necessary and sufficient that $A/\mathfrak{a}$ be formally equidimensional. In particular, for every prime ideal $\mathfrak{p}$ of A , $A/\mathfrak{p}$ is formally equidimensional.*

Proof. If $\mathfrak{m}$ is the maximal ideal of A , $\widehat{\mathfrak{m} A}$ is an ideal of definition of $\widehat{A}$. By hypothesis $\widehat{A}$ is equidimensional, and on the other hand it is known to be catenary (5.6.4); it suffices then to apply (7.1.3) with $A' = \widehat{A}$. $\square$

Corollary (7.1.5). — *Let A be a local Noetherian ring such that there exists a finite type A -module M that is a Cohen-Macaulay A -module and for which $\operatorname{Supp}(M) = \operatorname{Spec}(A)$ (this holds for example if A is a Cohen-Macaulay ring). Then A is formally equidimensional, hence (7.1.4) for a quotient ring B of A to be formally equidimensional, it is necessary and sufficient that it be equidimensional.*

Proof. Indeed, $\widehat{M} = M \otimes_A \widehat{A}$ is a Cohen-Macaulay $\widehat{A}$ -module (0, 16.5.2), with support equal to $\operatorname{Spec}(\widehat{A})$; hence (0, 16.5.4), $\widehat{A}$ is equidimensional. $\square$

Remark (7.1.6). — We have seen (6.3.8) that under the hypotheses of (7.1.5), for every integral quotient ring B of A , the fibres of the canonical morphism $\operatorname{Spec}(\widehat{B}) \rightarrow \operatorname{Spec}(B)$ are preschemes of Cohen-Macaulay; hence (6.4.3) and (5.7.5), for B to have no embedded associated prime ideals, it is necessary and sufficient that the same hold for $\widehat{B}$.

Lemma (7.1.7). — *Let A be a local Noetherian ring, $\mathfrak{p}_i$ ($1 \leq i \leq r$) its minimal prime ideals. Suppose that each of the rings $A/\mathfrak{p}_i$ is formally equidimensional. Then, for every prime ideal $\mathfrak{q}$ of A and every i such that $\mathfrak{p}_i \subset \mathfrak{q}$, the ring $A_{\mathfrak{q}}/\mathfrak{p}_i A_{\mathfrak{q}}$ is formally equidimensional.*

Proof. As $A_{\mathfrak{q}}/\mathfrak{p}_i A_{\mathfrak{q}}$ is the local ring of $A/\mathfrak{p}_i$ at the prime ideal $\mathfrak{q}/\mathfrak{p}_i$, we can restrict ourselves to the case where A is integral and formally equidimensional. Set $A' = \widehat{A}$, and let $\mathfrak{q}'$ be one of the prime ideals of A' minimal among those containing $\mathfrak{q} A'$; if we set $B = A_{\mathfrak{q}}$, $B' = A'_{\mathfrak{q}'}$, then B' is a flat B -module (0, 6.3.2). Set $C = \widehat{B}$, $C' = \widehat{B}'$; as B' is a flat B -module, we know that C' is a flat C -module (Bourbaki, Alg. comm., chap. III, §5, n° 4, prop. 4). As A' is catenary (5.6.4) and equidimensional by hypothesis, the same holds for $B' = A'_{\mathfrak{q}'}$ (0, 16.1.4); moreover, as A' is isomorphic to a quotient of a regular ring by the theorem of Cohen (0, 19.8.8), the same holds for B' (0, 17.3.9); we conclude therefore from (7.1.5) that C' is equidimensional. On the other hand, C' is catenary (5.6.4), hence C is equidimensional by virtue of (7.1.3), (i). $\square$

Theorem (7.1.8). — *Let A be a local Noetherian ring. The following conditions are equivalent:*

- a) *Every integral quotient ring of A is formally equidimensional.*
- b) *The quotient rings of A by its minimal prime ideals are formally equidimensional.*
- c) *Every local ring B that is an A -algebra essentially of finite type (1.3.8) and that is equidimensional, is formally equidimensional.*

Proof. It is trivial that c) implies a) and that a) implies b). On the other hand, as every prime ideal of A contains a minimal prime ideal of A , b) implies a) by virtue of (7.1.4), (ii). It remains to see that a) implies c).

It suffices to prove that the quotients of B by its minimal prime ideals are formally equidimensional (7.1.2); as every quotient ring of B is also an A -algebra essentially of finite type (1.3.9), we can first suppose B integral. If $\mathfrak{q}$ is the inverse image in A of the maximal ideal of B , we know that B is also an $A_{\mathfrak{q}}$ -algebra essentially of finite type (1.3.10), and it follows from a) and from (7.1.7) that every integral quotient of $A_{\mathfrak{q}}$ is formally equidimensional; we can therefore suppose that the homomorphism $A \rightarrow B$ is a local homomorphism. The kernel $\mathfrak{r}$ of this homomorphism is then a prime ideal of A , and by virtue of a), every integral quotient ring of $A/\mathfrak{r}$ is formally equidimensional; as B is an $(A/\mathfrak{r})$ -algebra essentially of finite type, we see that we can also suppose that A is integral and is a subring of B . We know (1.3.11) that B is a quotient of a local ring of the form $C_{\mathfrak{p}}$, where $C = A[T_1, \dots, T_n]$ is a polynomial algebra and $\mathfrak{p}$ is a prime ideal of C above the maximal ideal $\mathfrak{m}$ of A . By virtue of (7.1.7), it suffices to prove that $C_{\mathfrak{p}}$ is formally equidimensional, in other words one can restrict to the case where $B = C_{\mathfrak{p}}$.

Set $A' = \hat{A}$, and $C' = A'[T_1, \dots, T_n] = C \otimes_A A'$; there is a unique prime ideal $\mathfrak{p}'$ of C' above the maximal ideal $\mathfrak{m}A'$ of A' , hence above $\mathfrak{p}$; set $B' = C'_{\mathfrak{p}'}$; the homomorphism $B \rightarrow B'$ is local and makes B' a flat B -module. We know that $\hat{B}'$ is a flat $\hat{B}$ -module (Bourbaki, *Alg. comm.*, chap. III, §5, n° 4, prop. 4); as $\hat{B}'$ is also (7.1.3), (i)), it will suffice to prove that $\hat{B}'$ is equidimensional, in order to deduce that $\hat{B}$ is so as well, which will complete the proof.

Now, A' is a quotient of a regular ring by the theorem of Cohen (0, 19.8.8), hence by virtue of (7.1.5), to show that B' is formally equidimensional, it suffices to prove that B' is equidimensional, since C' is a quotient of a regular ring, hence catenary (5.6.4). But the minimal prime ideals of C' are the ideals $\mathfrak{q}'C'$, where $\mathfrak{q}'$ ranges over the minimal prime ideals of A' (5.5.3), and we have $C'/\mathfrak{q}'C' = (A'/\mathfrak{q}')[T_1, \dots, T_n]$. As A' is equidimensional by hypothesis (since A is integral), the same holds for C' by (5.5.4). $\square$

Definition (7.1.9). — When the equivalent conditions of (7.1.8) are satisfied, we say that A is a *formally catenary ring*.

For a local Noetherian ring, it is therefore equivalent to say that it is formally equidimensional or formally catenary and equidimensional, by virtue of (7.1.2).

Proposition (7.1.10). — *Every local quotient ring of a local Cohen-Macaulay ring is formally catenary.*

Proof. This follows immediately from (7.1.8) and (7.1.5) applied to the integral quotient rings of A . $\square$

Proposition (7.1.11). —
 i) *A local Noetherian ring that is formally catenary is universally catenary.*
 ii) *If A is a local Noetherian ring that is formally catenary, every local ring B that is an A -algebra essentially of finite type is formally catenary.*

Proof. Assertion (ii) follows immediately from condition c) of (7.1.8) and from the fact that if C is a B -algebra essentially of finite type, C is also an A -algebra essentially of finite type (1.3.9). For the proof of (i), let us first note that by virtue of condition b) of (7.1.8), a local Noetherian ring that is formally catenary A is catenary, since the quotients of A by its minimal prime ideals are so (7.1.4). If now E is an A -algebra of finite type, it follows from what precedes and from (ii) that every local ring of E at a prime ideal is catenary, hence E is catenary. Therefore A is universally catenary. $\square$

Remarks (7.1.12). —
 i) We do not know if, conversely, a local Noetherian ring that is universally catenary is always formally catenary.

- ii) A local Noetherian ring A of dimension 1 is necessarily equidimensional (since in this case A would have dimension 0, being its maximal ideal minimal, a case where A is of dimension 0). As $\dim(\hat{A}) = 1$, this shows that A is even *formally equidimensional*, and *a fortiori* formally catenary (hence *universally catenary*). By contrast, the local ring of dimension 2 defined in (5.6.11), which is catenary but not universally catenary, is *a fortiori* not formally catenary.

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Corollary (7.1.13). — *Let Y be an irreducible prescheme locally Noetherian of dimension 1, X an irreducible prescheme, $f : X \rightarrow Y$ a dominant morphism of finite type, ξ, η_i the generic points of X and Y respectively. Then, for all $y \in f(X)$, the dimensions of the irreducible components of $f^{-1}(y)$ are all equal to $\deg \cdot \mathrm{tr}_{k(\eta)} k(\xi)$.*

Proof. By hypothesis, $f^{-1}(\eta)$ is irreducible of generic point ξ and of dimension $\deg \cdot \mathrm{tr}_{k(\eta)} k(\xi)$ (5.2.1). As Y is irreducible and of dimension 1, we have $\dim(\mathcal{O}_y) = 1$ for all $y \neq \eta$, and $\mathcal{O}_y$ is universally catenary by virtue of (7.1.12), (ii). If $y \in f(X)$ and z is a generic point of an irreducible component Z of $f^{-1}(y)$, we have therefore, by (5.6.5)

$$\dim(Z) = 1 + \dim(f^{-1}(\eta)) - \dim(\mathcal{O}_{X,z}).$$

But we have $\dim(\mathcal{O}_{X,z}) \leq \dim(\mathcal{O}_y)$ (0, 16.3.9), and on the other hand, as z is not the generic point of X , $\dim(\mathcal{O}_{X,z}) > 0$; hence $\dim(\mathcal{O}_{X,z}) = 1$ and $\dim(Z) = \dim(f^{-1}(\eta))$. $\square$

7.2. Strictly equidimensional rings and strictly formally catenary rings

(7.2.1). Notations. — For an integral Noetherian ring A , we denote by $A^{(i)}$ the intersection of the local rings $A_{\mathfrak{p}}$, where $\mathfrak{p}$ ranges over the set of prime ideals of A of height 1 (5.10.17). If A is a local Noetherian integral ring, we denote by $A^{(a)}$ the intersection of the $A_{\mathfrak{p}}$ where this time $\mathfrak{p}$ ranges over the set of all the prime ideals of A distinct from the maximal ideal.

We will say that a local Noetherian ring A is *strictly equidimensional* if, for every prime ideal $\mathfrak{p} \in \mathrm{Ass}(A)$, we have $\dim(A/\mathfrak{p}) = \dim(A)$; this amounts to saying that A is equidimensional and without embedded associated prime ideals.

(7.2.1.1). Example. — Let A be a local Noetherian ring of dimension 1. Then the following conditions are equivalent:

- a) A has no embedded associated prime ideals.
- a') $\hat{A}$ has no embedded associated prime ideals.
- b) A is strictly equidimensional.
- b') $\hat{A}$ is strictly equidimensional.
- c) A is a Cohen-Macaulay ring.

Indeed, $a)$ means that the maximal ideal $\mathfrak{m}$ of A is not associated to A , hence the associated prime ideals to A are the minimal ideals of A , all distinct from $\mathfrak{m}$, and for such an ideal $\mathfrak{p}$, we necessarily have $\dim(A/\mathfrak{p}) = 1$; hence $a)$ implies $b)$. Conversely, $b)$ implies that every associated prime ideal $\mathfrak{p} \in \mathrm{Ass}(A)$ is distinct from $\mathfrak{m}$, hence minimal, and therefore $b)$ implies $a)$. We already know that $a)$ and $c)$ are equivalent for a local ring of dimension 1 (5.7.8). As it comes to the same thing to say that A is a Cohen-Macaulay ring or that $\hat{A}$ is a Cohen-Macaulay ring (0, 16.5.2), and since $\dim(\hat{A}) = 1$, we see finally that $a')$ and $b')$ are equivalent to $c)$.

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Proposition (7.2.2). — *Let A be a local Noetherian integral ring; set $A' = \hat{A}$, $X = \mathrm{Spec}(A)$, $X' = \mathrm{Spec}(A')$ and denote by $f : X' \rightarrow X$ the canonical morphism. Let a be the closed point of X , $j : X - \{a\} \rightarrow X$ the canonical injection. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -Module, $\mathcal{F}' = f^*(\mathcal{F}) = \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_{X'}$; then the following conditions are equivalent:*

- a) The $\mathcal{O}_X$ -Module $j_*(\mathcal{F}|_{X - \{a\}})$ is coherent.
- b) For all $x' \in \mathrm{Ass}(\mathcal{F}')$, we have $\dim(\{x'\}) \geq 2$.

Proof. Let a' be the closed point of X' , which is the unique point of the fibre $f^{-1}(a)$, and let $j' : X' - \{a'\} \rightarrow X'$ be the canonical injection. As the morphism f is faithfully flat and quasi-compact, it is sufficient to show that $j_*(\mathcal{F}|_{X - \{a\}})$ is coherent or that $j'_*(\mathcal{F}'|_{X' - \{a'\}})$ is coherent. On

the other hand, as A' is isomorphic to a quotient of a regular local ring by the theorem of Cohen (0, 19.8.8), it follows from (5.11.4) that condition b) of the statement was equivalent to the fact that $j'_*(\mathcal{F}'|X' - \{a'\})$ is coherent. Whence the proposition.

Instead of applying (5.11.4) and using the theorem of Cohen, which (by (5.11.2)) implicitly appeals to the cohomological results of chap. III, one can also use the fact that A' is universally catenary (5.6.4) and universally Japanese by virtue of (7.6.5) (the proof of this last result not using (7.2.2)). $\square$

Proposition (7.2.3). — *Let A be a local Noetherian integral ring. The notations being as in (7.2.2), the following conditions are equivalent:*

- a) $A^{(1)}$ is a finite A -algebra.
- b) For every closed subset T of X of codimension ≥ 2 , if we denote by $i : X - T \rightarrow X$ the canonical injection, $i_*(\mathcal{O}_{X-T})$ is a coherent $\mathcal{O}_X$ -Module.
- c) For all $x' \in \text{Ass}(\mathcal{O}_{X'})$ and every closed subset T of X of codimension ≥ 2 , we have $\text{codim}(f^{-1}(T) \cap \overline{\{x'\}}, \overline{\{x'\}}) \geq 2$.

Proof. Set $Z = Z^{(2)}(X)$ (5.10.13) and $Z' = f^{-1}(Z)$, which are stable parts under specialisation; the conditions a) and b) are equivalent respectively to the following properties: $a_1)$ $\mathcal{H}_{X/Z}^0(\mathcal{O}_X)$ is coherent; $b_1)$ $\mathcal{H}_{X/T}^0(\mathcal{O}_X)$ is coherent for every closed subset T of X of codimension ≥ 2 . Taking into account (5.9.5), these two properties are equivalent respectively to: $a'_1)$ $\mathcal{H}_{X'/Z'}^0(\mathcal{O}_{X'})$ is coherent; $b'_1)$ setting $T' = f^{-1}(T)$, $\mathcal{H}_{X'/T'}^0(\mathcal{O}_{X'})$ is coherent for every closed subset T of X of codimension ≥ 2 . Now, every point of $\text{Ass}(\mathcal{O}_{X'})$ projects to the generic point of X since f is a flat morphism (3.3.2); as by definition this point does not belong to Z , we see that $\text{Ass}(\mathcal{O}_{X'})$ does not meet Z' ; the equivalence of $a'_1)$ and $b'_1)$ therefore follows from (5.11.5), since A' is isomorphic to a quotient of a regular local ring by the theorem of Cohen (0, 19.8.8) (or alternatively because A' is universally Japanese (7.6.5)). **IV-2 | 189**
For this reason, conditions $b'_1)$ and c) are also equivalent by virtue of (5.11.4). $\square$

Proposition (7.2.4). — *Let A be a local Noetherian ring and set $X = \text{Spec}(A)$. The following conditions are equivalent:*

- a) For every integral quotient ring B of A , the ring $B^{(1)}$ is a finite B -algebra.
- b) For every coherent $\mathcal{O}_X$ -Module $\mathcal{F}$ and every closed subset Z , stable under specialisation, and such that for all $x \in \text{Ass}(\mathcal{F}) \cap (X - Z)$, we have $\text{codim}(\overline{\{x\}} \cap Z, \overline{\{x\}}) \geq 2$, the $\mathcal{O}_X$ -Module $\mathcal{H}_{X/Z}^0(\mathcal{F})$ is coherent.
- c) For every closed subset T of X and every coherent $\mathcal{O}_U$ -Module $\mathcal{G}$ (where $U = X - T$) such that, for all $x \in \text{Ass}(\mathcal{G})$, we have $\text{codim}(\overline{\{x\}} \cap T, \overline{\{x\}}) \geq 2$, $i_*(\mathcal{G})$ (where $i : U \rightarrow X$ is the canonical injection) is a coherent $\mathcal{O}_X$ -Module.
- d) For every integral quotient ring B of A , and every ideal $\mathfrak{J}$ of height ≥ 2 in B , the ring $\bigcap_{\mathfrak{p} \not\supset \mathfrak{J}} B_{\mathfrak{p}}$ is a finite B -algebra.
- e) For every integral quotient ring B of A and every prime ideal $\mathfrak{q}$ of B , with $C = B_{\mathfrak{q}}$ in a prime ideal of B , the ring $C^{(a)}$ is a finite C -algebra (or, what amounts to the same, if $Y = \text{Spec}(C)$ and U is the complement in Y of the closed point of C and $j : U \rightarrow Y$ the canonical injection, $j_*(\mathcal{O}_U)$ is a coherent $\mathcal{O}_Y$ -Module).
- f) For every integral quotient ring B of A , every local ring $C = B_{\mathfrak{q}}$ in a prime ideal $\mathfrak{q}$ of B , with $\dim(C) \geq 2$, and for every ideal $\mathfrak{t} \in \text{Ass}(\widehat{C})$, we have $\dim(\widehat{C}/\mathfrak{t}) \geq 2$.

Proof. We already know (5.11.6) that a) and b) are equivalent, as well as that c) and a) imply d), and that a) implies d). The equivalence of a) and d) in the present case follows from (7.2.3) applied to the condition (7.2.3, b)), by virtue of (5.9.3.1). The equivalence of e) and f) is a consequence of (7.2.2) applied to the $\mathcal{O}_Y$ -Module coherent $\mathcal{O}_Y$ itself. We have already remarked (5.11.7), (ii) that if A satisfies a), the same holds for every A -algebra of finite type and for every ring of fractions of A . The condition a) for A therefore implies (with the notations of e)) that the ring C satisfies a), hence also c); but as C is integral and $\dim(C) \geq 2$, one can apply condition c) by taking T the closed point of $Y = \text{Spec}(C)$, which proves that a) implies e); on the other hand, $\dim(C) \geq 2$, one can apply c) the condition c) by taking T the closed point of Y , which gives that a) implies a); we can obviously restrict ourselves to the case where A is integral and $B = A$, and it amounts to showing that condition e) implies the condition (7.2.3, c)). Let us consider the notations of (7.2.3, c)), and

let $y' \in f^{-1}(T) \cap \overline{\{x'\}}$, and set $y = f(y')$ and $C = \mathcal{O}_{X,y}$; as $y \in T$, $\dim(C) \geq 2$, and hence (with the notations of e)), $j_*(\mathcal{O}_U)$ is a coherent $\mathcal{O}_Y$ -Module. As the morphism f is flat, $f_z : Y' \rightarrow Y$; the morphism $f : X' \rightarrow X$ is flat, being the base change $g = f_{\{Y\}} : Y' \rightarrow Y$. The underlying space is identified with $f^{-1}(Y)$ (I, 3.6.5), and as A is integral and f flat, $\text{Ass}(\mathcal{O}_{X'})$ is contained in the fibre of the generic point of X (3.3.2); this last being contained in Y , we have $x' \in Y'$. Let $U' = g^{-1}(U)$, and let j' be the canonical injection $U' \rightarrow Y'$; as $j_*(\mathcal{O}_U)$ is a coherent $\mathcal{O}_Y$ -Module and g is flat, it follows from (5.9.4) that $j'_*(\mathcal{O}_{U'})$ is a coherent $\mathcal{O}_{Y'}$ -Module. But we have $x' \in \text{Ass}(\mathcal{O}_{U'})$, hence we conclude from (5.10.10) that, in Y' , $\text{codim}(f^{-1}(T) \cap \overline{\{x'\}}, \overline{\{x'\}}) \geq 2$ in X' . $\square$

Theorem (7.2.5). — *Let A be a local Noetherian ring. The following conditions are equivalent:*

- a) *For every integral quotient ring B of A , the completion $\widehat{B}$ is strictly equidimensional (7.2.1).*
- b) *A is formally catenary (7.1.9) and the fibres of the canonical morphism $\text{Spec}(\widehat{A}) \rightarrow \text{Spec}(A)$ satisfy (S_1) (in other words, have no embedded associated prime cycle).*
- c) *A is universally catenary (5.6.2) and for every integral quotient ring B of A , the ring $B^{(1)}$ is a finite B -algebra (cf. (7.2.4)).*
- d) *A is universally catenary and for every integral quotient ring B of A and every local ring $C = B_{\mathfrak{q}}$ in a prime ideal $\mathfrak{q}$ of B , with $\dim(C) \geq 2$, the completed ring $C^{(a)}$ is a finite C -algebra.*
- e) *A is universally catenary and for every integral quotient ring B of A and every local ring $C = B_{\mathfrak{q}}$ in a prime ideal $\mathfrak{q}$ of B , with $\dim(C) \geq 2$, the completed ring $\widehat{C}$ is such that $\text{Spec}(\widehat{C})$ has no associated prime cycle of dimension 1.*

Moreover, when these conditions are satisfied, for every integral quotient ring B of A that is strictly equidimensional, the completion $\widehat{B}$ is strictly equidimensional.

Proof. The equivalence of c), d) and e) was proved in (7.2.4). To show that a) and b) are equivalent, recall (7.1.9) that to say that A is formally catenary means that for every integral quotient ring B of A , $\widehat{B}$ is equidimensional. On the other hand, for every fibre of the morphism $\text{Spec}(\widehat{A}) \rightarrow \text{Spec}(A)$ at a point $x \in \text{Spec}(A)$, this fibre of the morphism $\text{Spec}(\widehat{B}) \rightarrow \text{Spec}(B)$ at the generic point x of $\text{Spec}(B)$, where $B = A/\mathfrak{J}_x$; to say that this fibre has no embedded associated prime cycle is equivalent to saying that $\widehat{B}$ has no embedded associated prime ideals, by virtue of (3.3.3); this proves the equivalence of a) and b). The same reasoning shows that if a) is satisfied, then, for every integral quotient ring B of A that has no embedded associated prime ideals, the completion $\widehat{B}$ is also without embedded associated prime ideals; on the other hand, the hypothesis that A is formally catenary implies that if B is equidimensional, it is also the case for $\widehat{B}$ (7.1.8); this establishes the last assertion of the theorem.

Let us show now that a) implies c). Condition a) implies that A is universally catenary (7.1.11); let us show on the other hand that for every integral quotient ring B of A , $B^{(1)}$ is then a finite B -algebra. Taking into account (7.2.3) applied to B , it amounts to showing that if $X = \text{Spec}(B)$, T is a closed subset of codimension ≥ 2 in X , $X' = \text{Spec}(\widehat{B})$, and $g : X' \rightarrow X$ is the canonical morphism, then for all $x' \in \text{Ass}(\mathcal{O}_{X'})$, $\text{codim}(g^{-1}(T) \cap \overline{\{x'\}}, \overline{\{x'\}}) \geq 2$. But by hypothesis X' has no associated prime cycle of dimension 1, hence $\inf(\text{codim}(g^{-1}(T) \cap \overline{\{x'\}}, \overline{\{x'\}}))$ when x' ranges over $\text{Ass}(\mathcal{O}_{X'})$, is nothing other than $\text{codim}(g^{-1}(T), X')$. Now, as g is a faithfully flat morphism, we have (6.1.4)

$$\text{codim}(g^{-1}(T), X') = \text{codim}(T, X) \geq 2.$$

It remains to prove that c) implies a). Let us reason by induction on $n = \dim(A)$, the theorem being trivial for $n = 0$; we can moreover restrict to the case where $A = B$ is integral. We proceed in two steps, n being ≥ 1 .

I) Suppose first that A satisfies (S_2) . — Set $X = \text{Spec}(A)$, $A' = \widehat{A}$, $X' = \text{Spec}(A')$ and let $u : X' \rightarrow X$ be the canonical morphism; it amounts to showing that for all $x' \in \text{Ass}(\mathcal{O}_{X'})$, we have $\dim(\overline{\{x'\}}) = n$. Let $f \neq 0$ be an element of the maximal ideal of A , and set $C = A/fA$; we know (5.7.6) that A/fA satisfies (S_1) ; moreover, as the prime ideals of A among those containing f are of height 1 by virtue of the Hauptidealsatz, $C = A/fA$ is strictly equidimensional and $\dim(C) = n - 1$. We have $\widehat{C} = C \otimes_A A' = A'/fA'$, and f is A' -regular by flatness (0, 6.3.4); if we set $Y' = V(fA') = \text{Spec}(\widehat{C})$, then, for every maximal point y' of $Y' \cap \overline{\{x'\}}$, we have $y' \in \text{Ass}(\mathcal{O}_{Y'})$ by virtue of (3.4.3); on the other

hand, we have $y' \neq x'$ since $u(x')$ is the generic point of X (3.3.2) and we conclude (5.1.8) that

$$(7.2.5.1) \quad \dim(\overline{\{y'\}}) = \dim(\overline{\{x'\}}) - 1.$$

But the quotient ring $C = A/fA$ also satisfies hypothesis $c)$ of the statement (5.6.1) and (5.11.7), (ii), hence by the induction hypothesis, we have $\dim(\overline{\{y'\}}) = \dim(C) = n - 1$; the conclusion therefore follows in this case from (7.2.5.1).

II) *General case.* — As by hypothesis $A^{(1)}$ is a finite A -algebra, it is a semi-local ring, and each of its local rings $(A^{(1)})_{\mathfrak{n}}$ at a maximal ideal $\mathfrak{n}$ and moreover, as A is universally catenary, the rings $(A^{(1)})_{\mathfrak{n}}$ (in finite number) are all of *dimension n* (5.6.10); moreover, these rings satisfy hypothesis $c)$ of the statement (5.6.1) and (5.11.7), (ii). We know that the completion of $A^{(1)}$, equal to $\hat{A} \otimes_A A^{(1)}$, is composed of the completions of the local rings $(A^{(1)})_{\mathfrak{n}}$. Set $X_1 = \text{Spec}(A^{(1)})$, $X'_1 = \text{Spec}((A^{(1)})^\wedge) = X' \times_X X_1$; for all $x'_1 \in \text{Ass}(\mathcal{O}_{X'_1})$, it follows from what precedes and from case I) that $\dim(\overline{\{x'_1\}}) = n$. Let $u_1 = u_{(X_1)} : X'_1 \rightarrow X_1$ be the canonical morphism; as A and $A^{(1)}$ have the same field of fractions, the inverse image of the generic point x of X by the projection $X_1 \rightarrow X$ reduces to the generic point x_1 of X_1 , the inverse image by the projection $X'_1 \rightarrow X'$ of the generic point x' of X' induces an isomorphism of the prescheme $u_1^{-1}(x_1)$ to the prescheme $u^{-1}(x)$. This being the case, the points of $\text{Ass}(\mathcal{O}_{X'_1})$ (resp. $\text{Ass}(\mathcal{O}_{X'_1})$) are the generic points of the associated prime cycles to $u^{-1}(x)$ (resp. $u_1^{-1}(x_1)$) (3.3.1). For all $x' \in \text{Ass}(\mathcal{O}_{X'})$, and if Z' (resp. Z'_1) is the closed sub-prescheme of X' (resp. X'_1) having $\overline{\{x'\}}$ (resp. $\overline{\{x'_1\}}$) as underlying space, the projection $Z'_1 \rightarrow Z'$ is a finite and surjective morphism; we conclude (5.4.2) that $\dim(\overline{\{x'\}}) = \dim(\overline{\{x'_1\}}) = n$. $\square$

Definition (7.2.6). — When a local Noetherian ring A satisfies the equivalent conditions of (7.2.5), we say that it is *strictly formally catenary*.

Corollary (7.2.7). — *Let A be a local Noetherian ring such that there exists an A -module M of finite type that is a Cohen-Macaulay A -module and for which $\text{Supp}(M) = \text{Spec}(A)$; then A is strictly formally catenary.*

Proof. Indeed, A is formally catenary (7.1.5) and (7.1.9), and on the other hand the fibres of the canonical morphism $\text{Spec}(\hat{A}) \rightarrow \text{Spec}(A)$ are preschemes of Cohen-Macaulay (6.3.8), hence *a fortiori* satisfy (S_1) . $\square$

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Corollary (7.2.8). — *If A is a local Noetherian ring that is strictly formally catenary, every local ring B that is an A -algebra essentially of finite type is strictly formally catenary.*

Proof. Indeed, B is formally catenary (7.1.11) and it follows from (7.4.4)¹⁰ that the fibres of $\text{Spec}(\hat{B}) \rightarrow \text{Spec}(B)$ satisfy (S_1) , whence the conclusion. $\square$

Corollary (7.2.9). — *Every local Noetherian ring of dimension 1 is strictly formally catenary.*

Proof. Indeed, every integral quotient ring B of such a ring A is of dimension 0 or 1, hence its completion is strictly equidimensional (7.2.1.1). $\square$

Remarks (7.2.10). — *i)* It follows from (7.2.7) and (7.2.8) that every quotient ring of a Cohen-Macaulay local ring is strictly formally catenary. A local Noetherian ring of dimension 2 satisfying (S_2) is strictly formally catenary, since it is a Cohen-Macaulay ring. Recall however that there exist integral local Noetherian rings of dimension 2 that are not universally catenary, nor *a fortiori* strictly formally catenary (5.6.11).

ii) We do not know if a ring that is formally catenary (7.1.9) is strictly formally catenary; this is because we do not know if for a local Noetherian integral ring B , $\hat{B}$ has no embedded associated prime ideals (6.4.3). We also do not know if, in the equivalent conditions $c)$, $d)$, $e)$ of (7.2.5), one can replace the hypothesis that A is universally catenary by the hypothesis that A is catenary; one can show that this is the case when A is Henselian (18.9.6).

¹⁰The corollary (7.2.8) is not used to prove (7.4.4).

7.3. Formal fibres of Noetherian local rings

(7.3.1). We will consider in this number and the two following properties $P(Z, k)$ of the following form:

“ Z is a locally Noetherian prescheme over a field k ,
and for all $z \in Z$, we have $Q(\mathcal{O}_z, k)$ ”

where $Q(A, k)$ is a property of a local Noetherian ring A that is a k -algebra; one assumes that if k' is a field isomorphic to k , and if A' is a local Noetherian ring that is a k' -algebra di-isomorphic to A , the properties $Q(A, k)$ and $Q(A', k')$ are equivalent.

If X, Y are two locally Noetherian preschemes, we will say that a morphism $f : X \rightarrow Y$ is a P -morphism if:

- 1° f is flat;
- 2° for all $y \in Y$, the property $P(f^{-1}(y), k(y))$ is true.

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Lemma (7.3.2). — *Let X, Y be two locally Noetherian preschemes, $f : X \rightarrow Y$ a morphism. The following properties are equivalent:*

- a) f is a P -morphism.
- b) For all $x \in X$, if we set $y = f(x)$, then $\mathcal{O}_x$ is a flat $\mathcal{O}_y$ -module and the property $Q(\mathcal{O}_x \otimes_{\mathcal{O}_y} k(y), k(y))$ is true.
- c) For all $x \in X$, if we set $y = f(x)$, the morphism $\text{Spec}(\mathcal{O}_x) \rightarrow \text{Spec}(\mathcal{O}_y)$ corresponding to the homomorphism $\mathcal{O}_y \rightarrow \mathcal{O}_x$ is a P -morphism.
- c') For every closed point $x \in X$, the morphism $\text{Spec}(\mathcal{O}_x) \rightarrow \text{Spec}(\mathcal{O}_y)$ is a P -morphism.

Proof. The equivalence of a) and b) follows from the definitions; the same holds for b) and c), taking into account (I, 2.4.2); finally the equivalence of b) and c') follows from the fact that, by (5.1.11), $\overline{\{x\}}$ contains a closed point. $\square$

Corollary (7.3.3). — *Let A, B be two Noetherian rings, $\varphi : A \rightarrow B$ a homomorphism such that the corresponding morphism $\text{Spec}(B) \rightarrow \text{Spec}(A)$ is a P -morphism. Then, for every multiplicative subset S of A , the morphism $\text{Spec}(S^{-1}B) \rightarrow \text{Spec}(S^{-1}A)$ is a P -morphism.*

Proof. This follows immediately from (7.3.2) and (I, 1.6.2). $\square$

(7.3.4). We will suppose throughout what follows that the property Q is such that the following three conditions are satisfied:

- (P_I) (transitivity). — If $f : X \rightarrow Y$ is a regular morphism (6.8.1) and $g : Y \rightarrow Z$ is a P -morphism, then $g \circ f$ is a P -morphism.
- (P_{II}) (descent). — If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are morphisms of locally Noetherian preschemes such that f is faithfully flat and that $g \circ f$ is a P -morphism, then g is a P -morphism.
- (P_{III}) — For every field k , the property $P(\text{Spec}(k), k)$ is true.

Remarks (7.3.5). — i) Let us note that the conditions (P_I) and (P_{III}) imply that every regular morphism is a P -morphism.

ii) Let us note that the hypotheses of (P_I) (resp. (P_{II})) imply that $h = g \circ f$ (resp. g) is flat (2.2.13); the hypotheses of (P_I) or of (P_{II}) imply on the other hand that for all $z \in Z$, $f_z : h^{-1}(z) \rightarrow g^{-1}(z)$ is flat (2.1.4); as, for all $y \in g^{-1}(z)$, $f_z^{-1}(y)$ is isomorphic to $f^{-1}(y)$ (I, 3.6.4), this shows that it suffices, to verify conditions (P_I) and (P_{II}), to do it only when Z is the spectrum of a field.

iii) In certain cases, the property Q will be such that the following condition is also satisfied:

(P'_I) — If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are two P -morphisms, then $g \circ f$ is a P -morphism.

(7.3.6). We will say that the property P is *geometric* if it satisfies (in addition to (P_I), (P_{II}) and (P_{III})) the condition:

(P_{IV}) (finite extension of the base field). — If $P(Z, k)$ is true, then, for every finite extension k' of k , $P(Z \otimes_k k', k')$ is true.

Lemma (7.3.7). — Let $f : X \rightarrow Y$ be a P -morphism of locally Noetherian preschemes, $g : Y' \rightarrow Y$ a morphism locally of finite type. Then, if P satisfies the condition (P_{IV}) , the morphism $f' = f_{(Y')} : X \times_Y Y' \rightarrow Y'$ is a P -morphism.

Proof. Indeed, for all $y' \in Y'$, if we set $y = g(y')$, $k(y')$ is an extension of finite type of $k(y)$ (I, 6.4.11) and $f'^{-1}(y') = f^{-1}(y) \otimes_{k(y)} k(y')$ (I, 3.6.4); it suffices therefore to apply (P_{IV}) . $\square$

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(7.3.8). Examples. — The following properties $P(Z, k)$ satisfy the conditions (P_I) , (P_{II}) and (P_{III}) :

- (i) (also denoted (i_n)) Z is of cofundity $\leq n$.
- (ii) Z is a prescheme of Cohen-Macaulay.
- (iii) (also denoted (iii_n)) Z satisfies (S_n) .
- (iv) Z is regular.
- (v) (also denoted (v_n)) Z satisfies (R_n) .
- (vi) Z is reduced.
- (vii) Z is normal.

For properties (ii) to (vii), this in fact proves the stronger condition (P'_I) which also implies the property (iv) implies all the properties (i) to (vii), and the property (iv') implies all the properties enumerated in (7.3.8). Recall also that (i_0) is equivalent to (ii); the conjunction of (iii_1) and of (v_0) (resp. of (iii_1) and of (v'_0)) is equivalent to (vi) (resp. (vi')); the conjunction of (iii_2) and of (v_1) (resp. of (iii_2) and of (v'_1)) is equivalent to (vii) (resp. (vii')) (5.8.6).

Moreover, the following are also geometric:

- (iv') Z is geometrically regular.
- (v') (also denoted (v'_n)) Z possesses the property (R_n) geometric.
- (vi') Z is geometrically reduced.
- (vii') Z is geometrically normal.

Remarks (7.3.9). — *i)* Each of the properties (iv') , (v_n) , (vi') , (vii') implies respectively the corresponding property (iv), (v_n) , (vi), (vii). The property (iv) implies all the properties (i) to (vii), and the property (iv') implies all the properties enumerated in (7.3.8). Recall also that (i_0) is equivalent to (ii); the conjunction of (iii_1) and of (v_0) (resp. of (iii_1) and of (v'_0)) is equivalent to (vi) (resp. (vi')); the conjunction of (iii_2) and of (v_1) (resp. of (iii_2) and of (v'_1)) is equivalent to (vii) (resp. (vii')) (5.8.6).

ii) In all the examples of (7.3.8), the property $Q(\mathcal{O}_z, k)$ which serves to define P is such that, for every *generalization* z' of z in Z , $Q(\mathcal{O}_z, k)$ implies $Q(\mathcal{O}_{z'}, k)$: by virtue of (2.3.4), it suffices to verify this for properties (i) to (vii) by (7.3.9), (i) and even (i) to (v) from the remark (7.3.9), (i): but this is included in the definition for (iii) and (v) (5.7.2) and (5.8.2), this follows from (6.3.9) for (i), and finally from (0, 16.5.10) and (17.3.2) for (ii) and (iv).

(7.3.10). We will say that a property $P(Z, k)$ is of the *first type* if the property $Q(A, k)$ which serves to define it defines a property $R(A)$ *not involving* k and is true when A is a *field*; we will say that $P(Z, k)$ is of the *second type* if the property $Q(A, k)$ is of the form

“For every finite extension k' of k and every point z' of $\text{Spec}(A \otimes_k k')$ above the closed point of $\text{Spec}(A)$, we have $R(\mathcal{O}_{z'})$ ”

$R(A)$ being assumed true when A is a field. It is clear that in the examples of (7.3.8), the properties (i) to (vii) are of the first type, the properties (iv') to (vii') of the second type. IV-2 | 195

This being the case, if we take up the reasoning of (6.6.1) and (6.8.3), we observe that when P is of the first or second type for a property $R(A)$, the conditions (P_I) and (P_{II}) have the following consequences on R :

(R_I) Let X, Y be two locally Noetherian preschemes, $f : X \rightarrow Y$ a *regular* morphism; then, for all $x \in X$, the property $R(\mathcal{O}_{f(x)})$ implies the property $R(\mathcal{O}_x)$.

(R_{II}) Let $\varphi : A \rightarrow B$ be a homomorphism of local Noetherian rings making B a flat A -module; then $R(B)$ implies $R(A)$.

Moreover, the property (P_{III}) results from the fact that $R(k)$ is true for every field k ; finally, when P is of the second type, the condition (P_{IV}) is a consequence of the definition of P and of the transitivity of finite extensions.

Remark (7.3.11). — We leave to the reader the task of formulating the property R corresponding to each of the examples of (7.3.8), and of verifying the conditions (R_I) and (R_{II}) in each case, with the help of the results of § 6. In fact, except for example (i) of (7.3.8), the property R satisfies in the other cases the following condition:

(R'_I) Let X, Y be two locally Noetherian preschemes, $f : X \rightarrow Y$ a P -morphism (for the property P of the first or second type defined by R); then, for all $x \in X$, the property $R(\mathcal{O}_{f(x)})$ implies the property $R(\mathcal{O}_x)$.

The reasoning of (6.6.1) and (6.8.3) proves that if R satisfies the conditions (R'_I) (7.3.4), (iii) and (P_{II}) , then P satisfies the conditions (P'_I) (7.3.5), (iii) and (P_{II}) .

Proposition (7.3.12). — Let P be a property of the first or the second type defined from a property R verifying the conditions (R_I) and (R_{II}) (resp. (R'_I) and (R_{II})). If for every locally Noetherian prescheme X , $U_R(X)$ denotes the set of $x \in X$ such that $R(\mathcal{O}_x)$ is true, then, for every regular morphism (resp. every P -morphism) $f : X \rightarrow Y$ of locally Noetherian preschemes, we have

$$(7.3.12.1) \quad U_R(X) = f^{-1}(U_R(Y)).$$

Proof. This is an immediate consequence of the definitions. $\square$

(7.3.13). Given a semi-local Noetherian ring A , we will call *formal fibres* of A the fibres of the canonical morphism $f : \text{Spec}(\hat{A}) \rightarrow \text{Spec}(A)$; for every prime ideal $\mathfrak{p}$ of A , the formal fibre at $\mathfrak{p}$ is therefore the scheme $\text{Spec}(\hat{A} \otimes_A k(\mathfrak{p}))$; as the completion of the local ring $A/\mathfrak{p}$ is $\hat{A}/\mathfrak{p}\hat{A} = (A/\mathfrak{p}) \otimes_A \hat{A}$, we see that the formal fibre of A at $\mathfrak{p}$ is also the *formal fibre* of $A/\mathfrak{p}$ at the generic point (o) of $\text{Spec}(A/\mathfrak{p})$.

The property P being defined as in (7.3.1), we will say that *the formal fibres of A have the property P* , or that A is a *P -ring*, if, for all $x \in \text{Spec}(A)$, $P(f^{-1}(x), k(x))$ is true. As the morphism f is flat, this amounts to saying that f is a P -morphism. IV-2 | 196

Proposition (7.3.14). — Let A be a semi-local Noetherian ring, $\mathfrak{m}_i$ ($1 \leq i \leq r$) its maximal ideals, and set $A_i = A_{\mathfrak{m}_i}$; for A to be a P -ring, it is necessary and sufficient that each of the A_i be so.

Proof. Indeed, $\hat{A}$ is the direct product of the $\hat{A}_i$, hence the formal fibre of A at a point $x \in \text{Spec}(A)$ is the sum of the formal fibres at x of those A_i such that $x \in \text{Spec}(A_i)$. $\square$

Proposition (7.3.15). — Let A be a semi-local Noetherian ring.

- i) If A is a P -ring, every quotient ring of A is a P -ring.
- ii) If in addition P satisfies the condition (P_{IV}) (7.3.6), then every finite A -algebra is a P -ring.

Proof. For every ideal $\mathfrak{a}$ of A , $\hat{A} \otimes_A (A/\mathfrak{a})$ is the completion of $A/\mathfrak{a}$, and the formal fibres of $A/\mathfrak{a}$ are therefore the formal fibres of A at the points of $V(\mathfrak{a})$, whence (i). On the other hand, if B is a finite A -algebra, B is a semi-local ring, we have $\hat{B} = B \otimes_A \hat{A}$, and (ii) follows from (7.3.7). $\square$

Proposition (7.3.16). — Suppose that the property P is of the first (resp. second) type, defined from a property R (7.3.10). When P is of the second type, suppose moreover that the relation

“for every finite extension k' of k , and every $z' \in Z \otimes_k k'$, $R(\mathcal{O}_{z'})$ is true”

implies $P(Z, k)$ (this is verified for the examples (iv') to (vii') of (7.3.8), by virtue of (6.7.7) and (4.6.1)).

Let A be a semi-local Noetherian ring. If the property R satisfies the conditions (R_I) and (R_{II}) , the following properties are equivalent:

- a) A is a P -ring.
- b) For every integral quotient ring B of A (resp. every finite integral A -algebra B) and every prime ideal $\mathfrak{q}$ of $\hat{B}$ whose inverse image in B is 0, $R(\hat{B}_{\mathfrak{q}})$ is true.

If in addition R satisfies the condition (R'_I) , the properties a) and b) are also equivalent to:

- c) For every quotient ring B of A (resp. every finite A -algebra B), if we set $Y = \text{Spec}(B)$, $Y' = \text{Spec}(\hat{B})$, and if $g : Y' \rightarrow Y$ is the canonical morphism, we have (with the notations of (7.3.12))

$$(7.3.16.1) \quad U_R(Y') = g^{-1}(U_R(Y)).$$

Proof. The equivalence of *a*) and *b*) is immediate when P is of the first type: indeed, for every prime ideal $\mathfrak{p}$ of A , the formal fibre of $B = A/\mathfrak{p}$ at the generic point $\mathfrak{p} \in \operatorname{Spec}(A)$ is nothing other than the formal fibre of A at the point $\mathfrak{p}$ (7.3.13). On the other hand, if we specialise *c*) to the case where B is integral and y the generic point of $Y = \operatorname{Spec}(B)$, we have $\mathcal{O}_{X,y} = k(y)$, hence $y \in U_R(Y)$, hence $R(k)$ is true for every field k ; by expressing that every point $y' \in g^{-1}(y)$ belongs to $U_R(Y')$, we obtain the statement *b*), hence *c*) implies *b*).

When P is of the second type, the equivalence of *a*) and *b*) follows from the more general lemma below. $\square$

Lemma (7.3.16.2). — *Let A, A' be two rings, $\varphi : A \rightarrow A'$ a homomorphism, $f : \operatorname{Spec}(A') \rightarrow \operatorname{Spec}(A)$ the corresponding morphism. In order that for all $x \in \operatorname{Spec}(A)$, every finite extension k of $k(x)$ and every point z of $f^{-1}(x) \otimes_{k(x)} k = Z$, the property $R(\mathcal{O}_{Z,z})$ be true, it is necessary and sufficient that: for every integral finite A -algebra B , if $g : \operatorname{Spec}(A' \otimes_A B) \rightarrow \operatorname{Spec}(B)$ is the morphism deduced from f by base extension to B , and if $T = g^{-1}(\xi)$ is the fibre of the generic point ξ of $\operatorname{Spec}(B)$, then, for all $t \in T$, $R(\mathcal{O}_{T,t})$ is true.*

Proof. The condition is trivially necessary since x is the point of $\operatorname{Spec}(A)$ above which ξ lies, $k(\xi)$ is a finite extension of $k(x)$. Conversely, let us consider a point $x \in \operatorname{Spec}(A)$ and let k be a finite extension of $k(x)$. If we set $\mathfrak{p} = \mathfrak{I}_x$, $k(x)$ is the field of fractions of $A/\mathfrak{p}$, and if there is a base of k over $k(x)$ formed of integral elements over $A/\mathfrak{p}$; if B is the subring of k generated by these elements, B is an integral finite A -algebra, and k is the field $k(\xi)$ at the generic point ξ of $\operatorname{Spec}(B)$; as x is the image of ξ in $\operatorname{Spec}(A)$, the fibre $g^{-1}(\xi)$ is nothing other than $f^{-1}(x) \otimes_{k(x)} k$, which completes the proof of the lemma. $\square$

Proposition (7.3.17). — *Suppose that the property R satisfies the conditions (R'_I) and (R_{II}) , and that P is the property of the first (resp. second) type defined from R (7.3.10). Then, if A is a P -ring, the properties $R(A)$ and $R(\widehat{A})$ are equivalent.*

Proof. This follows from (7.3.12.1) applied to $Y = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(\widehat{A})$. $\square$

Proposition (7.3.18). — *Suppose that P is a property of the first (resp. second) type defined from a property R satisfying the conditions (R'_I) and (R_{II}) , and also the condition following:*

(R_{III}) *For every local Noetherian complete ring C , if we set $Z = \operatorname{Spec}(C)$, the set $U_R(Z)$ (7.3.12) is open in Z .*

Then, if A is a P -ring and if we set $X = \operatorname{Spec}(A)$, the set $U_R(X)$ is open in X .

Proof. Indeed, if $X' = \operatorname{Spec}(\widehat{A})$ and if $f : X' \rightarrow X$ is the canonical morphism, $U_R(X') = f^{-1}(U_R(X))$ (7.3.12). As f is faithfully flat and quasi-compact (2.3.12), and as by hypothesis $U_R(X')$ is open in X' , the conclusion follows from (2.3.12). $\square$

Remarks (7.3.19). — *i)* The property R which serves to define P satisfies the condition (R_{III}) in all the examples enumerated in (7.3.8). For (i), (ii) and (iii), this follows from (6.11.2) and the theorem of Cohen (0, 19.8.8); when P is one of the properties (iv), (iv'), (v), (v'), this follows from (6.12.7) and (6.12.9), and when P is one of the properties (vi) and (vi'), the assertion of (7.3.18) is trivial, being true for every locally Noetherian prescheme.

ii) We have already noted (6.4.3) that when P is one of the properties (ii) or (iii₁) of (7.3.8), we do not know if every local Noetherian ring is a P -ring. We also do not know if, in the equivalent conditions *c*), *d*), *e*) of (7.2.5), one can replace the hypothesis that A is universally catenary by the hypothesis that A is catenary; one can show that this is the case when A is Henselian (18.9.6).

7.4. Permanence of properties of formal fibres

(7.4.1). Throughout the rest of this number, we suppose that the property P is of the form defined in (7.3.1), and satisfies the conditions (P_I) , (P_{II}) and (P_{III}) of (7.3.4). Moreover, we suppose that the property Q which serves to define P is such that, for every generization z' of z in Z , $Q(\mathcal{O}_z, k)$ implies $Q(\mathcal{O}_{z'}, k)$.

Lemma (7.4.2). — *Let A, A' be two local Noetherian rings, $\varphi : A \rightarrow A'$ a local homomorphism such that $f = {}^a\varphi : \text{Spec}(A') \rightarrow \text{Spec}(A)$ is a P -morphism. Then, if the formal fibres of A' are geometrically regular, A is a P -ring.*

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Proof. Consider the completed homomorphism $\widehat{\varphi} : \widehat{A} \rightarrow \widehat{A'}$ and the corresponding morphism $\widehat{f} = {}^a\widehat{\varphi} : \text{Spec}(\widehat{A'}) \rightarrow \text{Spec}(\widehat{A})$; we have the commutative diagram

$$\begin{array}{ccc} \text{Spec}(\widehat{A}) & \xleftarrow{\widehat{f}} & \text{Spec}(\widehat{A'}) \\ \uparrow g & & \uparrow g' \\ \text{Spec}(A) & \xleftarrow{f} & \text{Spec}(A') \end{array}$$

where g and g' are the canonical morphisms. As by hypothesis f is a P -morphism and g' a regular morphism, it follows from (P_I) that $f \circ g' = g \circ \widehat{f}$ is a P -morphism. On the other hand, the hypothesis that f is a P -morphism implies that f is flat, hence it is the same for $\widehat{f}$ (Bourbaki, *Alg. comm.*, chap. III, §5, n°4, cor. of prop. 3), which is moreover a local homomorphism, hence faithfully flat (0, 6.6.2); it follows then from (P_{II}) that g is a P -morphism. $\square$

Corollary (7.4.3). — i) *Let A be a P -ring local Noetherian, $A' = \widehat{A}$ its completion. Suppose that for every prime ideal $\mathfrak{p}'$ of A' , the formal fibres of $A'_{\mathfrak{p}'}$ are geometrically regular; then, for every prime ideal $\mathfrak{p}$ of A , $A_{\mathfrak{p}}$ is a P -ring.*

ii) *Suppose that P satisfies the condition (P_{IV}) (7.3.6). Let A be a P -ring local Noetherian, B an A -algebra locally essentially of finite type, and such that the homomorphism $A \rightarrow B$ is local. Set $A' = \widehat{A}$, and let $\mathfrak{n}'$ be the unique prime ideal of $B' = B \otimes_A A'$ above the maximal ideal of A' . If the formal fibres of $B'_{\mathfrak{n}'}$ are geometrically regular, then B is a P -ring.*

Proof. (i) As by hypothesis $\text{Spec}(A') \rightarrow \text{Spec}(A)$ is a P -morphism, the same holds for $\text{Spec}(A'_{\mathfrak{p}'}) \rightarrow \text{Spec}(A_{\mathfrak{p}'})$ by virtue of (7.3.2, b)), for every prime ideal $\mathfrak{p}'$ of A' above $\mathfrak{p}$ and every prime ideal $\mathfrak{p}$ of A . It suffices then to apply the lemma (7.4.2) to this morphism, noting that the morphism $\text{Spec}(A') \rightarrow \text{Spec}(A)$ is surjective.

(ii) By virtue of (7.4.2), it suffices to prove that the morphism $\text{Spec}(B'_{\mathfrak{n}'}) \rightarrow \text{Spec}(B)$ is a P -morphism. Now we have $B = C_{\mathfrak{n}}$, where C is a finite type A -algebra and $\mathfrak{n}$ is a prime ideal of C above the maximal ideal of A . If we set $C' = C \otimes_A A'$, it follows from the hypotheses and from (7.3.7) that $\text{Spec}(C') \rightarrow \text{Spec}(C)$ is a P -morphism; as $B'_{\mathfrak{n}'}$ is a local ring of C' at a prime ideal of C' above $\mathfrak{n}$, the conclusion follows from (7.3.2, b)). $\square$

Theorem (7.4.4). — *The hypotheses on P being those of (7.4.1), let A be a P -ring local Noetherian.*

- i) *For every prime ideal $\mathfrak{p}$ of A , $A_{\mathfrak{p}}$ is a P -ring.*
- ii) *Suppose in addition that P satisfies the condition (P_{IV}) . Then every local ring that is an A -algebra essentially of finite type is a P -ring.*

Proof. (i) Applying (7.4.3), (i), everything reduces to seeing that for every prime ideal $\mathfrak{p}'$ of $A' = \widehat{A}$, $A'_{\mathfrak{p}'}$ has its formal fibres geometrically regular. Now, this has been proved in (0, 22.3.3) and (0, 22.5.8).

(ii) Let B be an A -algebra essentially of finite type that is a local ring; if $\mathfrak{p}$ is the prime ideal of A above which is the maximal ideal of B , B is also an $A_{\mathfrak{p}}$ -algebra essentially of finite type (1.3.8); by virtue of (i), we can therefore suppose that $\mathfrak{p}$ is the maximal ideal $\mathfrak{m}$ of A . We have

therefore $B = C_q$, where C is a finite type A -algebra, q a prime ideal of C (necessarily above m); by virtue of (i), one can therefore suppose that $B = C_q$.

Let B be an A -algebra essentially of finite type that is a local ring; if p is the prime ideal of A above which the maximal ideal of B lies, B is also an A_p -algebra essentially of finite type (1.3.8); by virtue of (i), one can therefore suppose that the homomorphism $A \rightarrow B$ is local.

Applying (7.4.3), (ii) with $A' = \hat{A}$, $B' = B \otimes_A A'$ is a local ring of $C \otimes_A A'$ at a prime ideal above the maximal ideal of A' ; as A and A' have the same residue field k , the residue field of $C \otimes_A A'$ at this prime ideal is equal to k' . Moreover $C \otimes_A A'$ is a finite type A' -algebra generated by a single element. Hence (7.4.3), (ii) one can restrict to proving (7.4.4.1) when P is the property (iv') of (7.3.8), that A is complete and C is generated by a single element t . $\square$

Lemma (7.4.4.1). — *Let A be a P -ring local Noetherian, k its residue field, C a finite type A -algebra, B a local ring in a prime ideal n of C , such that: 1° the homomorphism $A \rightarrow B$ is local; 2° the residue field k' of B is a finite extension of k . If P satisfies (P_{IV}) , then B is a P -ring.*

Proof. Let $(x_i)_{1 \leq i \leq m}$ be a system of generators of the A -algebra C ; let us show first that we can argue by induction on m . Let C' be the sub- A -algebra of C generated by $x_1, \dots, x_{m-1}$, and let $n' = n \cap C'$. The homomorphism $A \rightarrow C_n$ factors through the local homomorphisms $A \rightarrow C'_{n'} \rightarrow C_n$; if k'' is the residue field of $C'_{n'}$, the extension $k \rightarrow k'$ factors through k'' , hence k'' is a finite extension of k and k' a finite extension of k'' . The induction hypothesis implies that $C'_{n'}$ is a P -ring. If $S' = C' - n'$, then C_n is a local ring of $S'^{-1}C = C'_{n'}[x_m/1]$; the induction hypothesis applied once more shows that C_n is a P -ring. Thus the problem reduces to the case where C is an A -algebra generated by a single element t .

Applying (7.4.3), (ii), with $A' = \hat{A}$, the ring $B' = B \otimes_A A'$ is a local ring of $C \otimes_A A'$ at a prime ideal above n ; as A and A' have the same residue field k , the residue field of $C \otimes_A A'$ at this prime ideal is equal to k' . Moreover $C \otimes_A A'$ is generated by a single element over A' . It follows therefore from (7.4.3), (ii) that one can restrict to proving (7.4.4.1) when P is the property (iv') of (7.3.8), that A is complete and C generated by a single element t .

To show that the formal fibres of $B = C_n$ are then geometrically regular, let us apply the criterion (7.3.16), b)). Let B_1 be a finite integral B -algebra, hence generated by a finite number of elements integral over B . By multiplying these elements by an element of B , one can suppose that they are integral over C , and we can therefore write $B_1 = S^{-1}C_1$, where C_1 is a sub- C -algebra of B_1 generated by a finite number of integral elements over C , hence a finite integral C -algebra. On the other hand, B_1 is a semi-local ring, and every local ring B_2 of B_1 at a maximal ideal is a local ring of C_1 , such that $A \rightarrow B_2$ is a local homomorphism; moreover, the residue field of B_2 is a finite extension of k' , hence of k . We see therefore (taking into account (7.3.14) and (i)) that we are reduced to the following question: let C be a Noetherian integral ring containing a subring C_0 which is an A -algebra generated by a single element t and such that C is a finite C_0 -algebra; if n is a maximal ideal of C above the maximal ideal of A , and $B = C_n$, is the completion $\hat{B}_q$ regular for every prime ideal q of $\hat{B}$ such that $q \cap A = 0$?

lying over the maximal ideal of A , and $B = C_n$, the question is to show that for every prime ideal q of $\hat{B}$ whose inverse image in B is 0, the ring $\hat{B}_q$ is regular. One may moreover replace A by its image in C , which is a complete local ring (as a quotient of A) and integral. The desired conclusion follows from the more general lemma below. $\square$

Lemma (7.4.4.2). — *Let A be a complete Noetherian local integral ring, k its residue field, C an integral ring containing A , such that there exists $t \in C$ for which C is a finite $A[t]$ -algebra. Let n be a maximal ideal of C above the maximal ideal m of A ; set $B = C_n$, $X = \text{Spec}(B)$, $B' = \hat{B}$, $X' = \text{Spec}(\hat{B})$; if $U = \text{Reg}(X)$, $U' = \text{Reg}(X')$, and if $f : X' \rightarrow X$ is the canonical morphism, then $f^{-1}(U) \subset U'$.*

Proof. The assertion needed for (7.4.4.1) follows from this lemma because, since B is integral, the generic point of X belongs to U . Since C is an A -algebra of finite type and n is a maximal ideal of C , the residue field k' of $C_n = B$ (hence also of B') is a finite extension of k (I, 6.4.11) and (6.4.2).

Set $Y = \text{Spec}(C)$. It follows from (6.12.8) that $\text{Reg}(Y)$ is open in Y ; since the local rings of X are local rings of Y (I, 2.4.2), we have $U = X \cap \text{Reg}(Y)$, hence U is open in X . On the other hand, (6.12.7)

shows that U' is open in X' , hence $S' = X' - U'$ is closed. Consequently $S' \cap f^{-1}(U)$ is locally closed in X' , and the whole question is to prove that this set is empty. If it were non-empty, (5.1.10) would give a prime ideal $\mathfrak{p}'$ in $S' \cap f^{-1}(U)$ such that $\dim(B'/\mathfrak{p}') \leq 1$. The prime ideal $\mathfrak{p}'$ cannot be the maximal ideal $\mathfrak{m}B'$ of B' , where $\mathfrak{m}$ is the maximal ideal of B . Indeed, this would mean that $B = B_{\mathfrak{m}}$ is regular, hence so is $B' = \widehat{B}$ (0, 17.1.5), and we would have $\mathfrak{m}B' \in U'$, contrary to the hypothesis. We must therefore have $\dim(B'/\mathfrak{p}') = 1$.

Set $\mathfrak{p} = B \cap \mathfrak{p}'$. By hypothesis $B_{\mathfrak{p}}$ is regular, whereas $B'_{\mathfrak{p}'}$ is not. Since $B'_{\mathfrak{p}'}$ is a flat $B_{\mathfrak{p}}$ -module, (6.5.2) shows that the fibre Z of f at $\mathfrak{p}$ is not regular at $\mathfrak{p}'$. We may reduce to the case $\mathfrak{p} = 0$. Indeed, in general, set $\mathfrak{q} = \mathfrak{p} \cap C$ and $\mathfrak{r} = \mathfrak{p} \cap A$. Then $C/\mathfrak{q}$ is a finite $(A/\mathfrak{r})[\bar{t}]$ -algebra, where $\bar{t}$ is the class of t modulo $\mathfrak{q}$; since $\mathfrak{p} = \mathfrak{q}B$, the ring $B/\mathfrak{p}$ is equal to $(C/\mathfrak{q})_{\mathfrak{n}/\mathfrak{q}}$, and $\mathfrak{n}/\mathfrak{q}$ is a maximal ideal of $C/\mathfrak{q}$ lying over the maximal ideal $\mathfrak{m}/\mathfrak{r}$ of $A/\mathfrak{r}$. Thus the hypotheses of (7.4.4.2) are still satisfied by $A/\mathfrak{r}$, $C/\mathfrak{q}$, and $B/\mathfrak{p}$; since the completion of $B/\mathfrak{p}$ is $B'/\mathfrak{p}'$, this proves the reduction.

Suppose therefore that $\mathfrak{p} = 0$, so that Z is the generic fibre of f and the homomorphism $B \rightarrow B'/\mathfrak{p}'$ is injective. Set $V = B'/\mathfrak{p}'$, and distinguish two cases.

I) V is a finite A -algebra. Since $B \subset V$, B is a fortiori a finite A -algebra; as A is complete, so is B (Bourbaki, *Alg. comm.*, chap. IV, § 2, n° 5, cor. 3 of prop. 9). Hence $B' = B$ and $\mathfrak{p}' = 0$, so $B'_{\mathfrak{p}'}$ is a field, contrary to the hypothesis.

II) V is not a finite A -algebra. Since the local ring A is complete, V is not a quasi-finite A -algebra (0, 7.4.1) and (0, 7.4.2); but the residue field k' of V is a finite extension of the residue field k of A , hence (0, 7.4.4) the ideal $\mathfrak{m}V$ is not an ideal of definition of V . Since V is a one-dimensional Noetherian local integral ring, 0 is the only ideal of V that is not an ideal of definition, hence $\mathfrak{m}V = 0$. But $A \subset V$ and V is integral, so $\mathfrak{m} = 0$ and $A = k$ is a field. It follows first that $\dim(C) \leq 1$ (0, 16.1.5); but since $\dim(V) = 1$, the inequalities

$$\dim(V) \leq \dim(B') = \dim(B) \leq \dim(C)$$

show that $\dim(C) = \dim(B) = \dim(B') = \dim(B'/\mathfrak{p}') = 1$, and consequently $\mathfrak{p}'$ is a minimal prime ideal of B' . We therefore obtain a contradiction once we prove that $B'_{\mathfrak{p}'}$ is a field, or equivalently that B' is reduced. Since C is a finite-type k -algebra, its integral closure C_1 is a finite C -algebra (Bourbaki, *Alg. comm.*, chap. V, § 3, n° 2, th. 2). If $S = C - \mathfrak{n}$, then $B_1 = S^{-1}C_1$ is the integral closure of B , hence a finite B -algebra and therefore a one-dimensional semi-local Noetherian integral and integrally closed ring (0, 16.1.5). If $\mathfrak{m}_j$ ($1 \leq j \leq h$) are its maximal ideals, the rings $(B_1)_{\mathfrak{m}_j}$ are discrete valuation rings (II, 7.1.6), and the completion B'_1 of B_1 is the direct product of the completed discrete valuation rings $(B_1)_{\mathfrak{m}_j}$ (Bourbaki, *Alg. comm.*, chap. III, § 2, n° 13, prop. 18). Thus B'_1 is reduced; since the completion B' of B is a subring of B'_1 (Bourbaki, *Alg. comm.*, chap. IV, § 2, n° 5, cor. 3 of prop. 9), B' is reduced as well. $\square$

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Corollary (7.4.5). — Suppose that the property P satisfies the conditions (P_I) , (P_{II}) , (P_{III}) . Let A be a Noetherian ring. The following conditions are equivalent:

- a) For every prime ideal $\mathfrak{p}$ of A , $A_{\mathfrak{p}}$ is a P -ring.
- b) For every maximal ideal $\mathfrak{m}$ of A , $A_{\mathfrak{m}}$ is a P -ring.

If in addition P satisfies (P_{IV}) , the properties a) and b) are also equivalent to:

- c) For every finite type A -algebra B and every prime ideal $\mathfrak{q}$ of B , $B_{\mathfrak{q}}$ is a P -ring.

Proof. The equivalence of a) and b) follows from (7.4.4), (i), and that of a) and c) from (7.4.4), (ii). When condition a) of (7.4.5) is satisfied, we say that A is a P -ring; for semi-local Noetherian rings, this definition coincides with that of (7.3.13), when the conditions (P_I) , (P_{II}) and (P_{III}) are satisfied. The ring $\mathbb{Z}$ is a P -ring (7.3.19), (iv); every local Noetherian complete ring is a P -ring. $\square$

Proposition (7.4.6). — Suppose that the property P satisfies the conditions (P_I) , (P_{II}) and (P_{III}) . Let A be a Noetherian ring, $\mathfrak{J}$ an ideal of A , $\widehat{A}$ the separated completion of A for the $\mathfrak{J}$ -adic topology. Then, if A is a P -ring (7.4.5), the canonical morphism $\text{Spec}(\widehat{A}) \rightarrow \text{Spec}(A)$ is a P -morphism.

Proof. Using (7.3.2), c'), it suffices to prove that for every maximal ideal $\mathfrak{n}$ of $\widehat{A}$ with inverse image $\mathfrak{m}$ in A , the morphism $\text{Spec}((\widehat{A})_{\mathfrak{n}}) \rightarrow \text{Spec}(A_{\mathfrak{m}})$ is a P -morphism. We know (Bourbaki, *Alg. comm.*,

chap. III, §3, n°4, prop. 8) that the canonical homomorphism $A_m \rightarrow (\hat{A})_n$ is injective, that the mA_m -adic topology on A_m is induced by the $n(\hat{A})_n$ -adic topology, and that A_m is dense in $(\hat{A})_n$ for this topology, so that the completion of A_m for the mA_m -adic topology identifies with that of $(\hat{A})_n$ for the $n(\hat{A})_n$ -adic topology. We have therefore two morphisms

$$\mathrm{Spec}((A_m)^\wedge) \xrightarrow{f} \mathrm{Spec}((\hat{A})_n) \xrightarrow{g} \mathrm{Spec}(A_m)$$

□

Corollary (7.4.7). — Suppose that P satisfies the conditions (P_I) , (P'_I) , (P_{II}) , (P_{III}) and (P_{IV}) . Then, if A is a P -ring (7.4.5), the canonical morphism $\mathrm{Spec}(A[[T_1, \dots, T_r]]) \rightarrow \mathrm{Spec}(A)$ is a P -morphism. In particular, if in addition A is integral, K its field of fractions, and if $\mathfrak{p}$ is a prime ideal of $B = A[[T_1, \dots, T_r]]$ such that $\mathfrak{p} \cap A = 0$, then the property $P(B_{\mathfrak{p}}, K)$ is true.

Proof. Indeed, $\mathrm{Spec}(B) \rightarrow \mathrm{Spec}(A)$ factorises as

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$$\mathrm{Spec}(A[[T_1, \dots, T_r]]) \xrightarrow{f} \mathrm{Spec}(A[T_1, \dots, T_r]) \xrightarrow{g} \mathrm{Spec}(A).$$

It is clear that the morphism g is regular (0.17.3.7); by virtue of (7.4.5), $A[T_1, \dots, T_r]$ is a P -ring, hence it follows from (7.4.6) that f is a P -morphism; as g is regular, hence also a P -morphism (7.3.5), (i), it follows from (P'_I) that $g \circ f$ is a P -morphism.

One will note that the conclusion is still valid if instead of supposing that P satisfies (P'_I) , one only supposes that a composite $g \circ f$ is a P -morphism when g is regular and f a P -morphism (symmetric condition to that of (P'_I) in some sense). □

Remark (7.4.8). — The preceding results pose the following problems:

- A) Let A be a Zariski complete ring, and let $\mathfrak{J}$ be an ideal of definition of A ; if the ring $A/\mathfrak{J}$ is a P -ring, is A also a P -ring? It would follow that for every Noetherian P -ring A and every ideal $\mathfrak{J}$ of A , the separated completion $\hat{A}$ for the $\mathfrak{J}$ -adic topology would also be a P -ring.
- B) Let k be a valued field, complete and non-discrete; one calls *ring of restricted formal power series* $k\{T_1, \dots, T_n\}$ the subring of the formal power series ring $k[[T_1, \dots, T_n]]$ formed of the series whose coefficients *tend to 0*. Is such a ring a P -ring?
- C) If A is a linearly topologised P -ring, S a multiplicative subset of A , are the rings $A\{S^{-1}\}$ (0, 7.6.1) and $A_{\{S\}}$ (0, 7.6.15) P -rings?

7.5. A criterion for P -morphisms In what follows of this section, we will apply (7.5.1) to the study of completed tensor products $A \hat{\otimes}_k B$ of local Noetherian rings which are algebras over a field k . IV-2 | 207

Lemma (7.5.5). — Let k be a field, A, B two local Noetherian complete rings containing k , the residue field of A being a finite extension of k . Let C be the completed tensor product $A \hat{\otimes}_k B$ (0, 7.7.5). Then:

- (i) C is a semi-local Noetherian complete ring.
- (ii) C is a flat A -module and a flat B -module.
- (iii) If $\mathfrak{m}$ is the maximal ideal of A , $\mathfrak{m}C$ is contained in the radical of C , and $C/\mathfrak{m}C$ is isomorphic to $(A/\mathfrak{m}) \otimes_k B$.
- (iv) The residue fields of the components of C are finite extensions of the residue field of B .

The properties (i), (iii) and the first assertion of (ii) are particular cases of (0, 19.7.1.2). To prove that C is a flat B -module, note that for every $h > 0$,

$$C/\mathfrak{m}^h C = (A/\mathfrak{m}^h) \otimes_k B$$

is a flat B -module, since k is a field; it therefore suffices to apply (0, 10.2.6) to each local component of C . Finally, if $\mathfrak{n}$ is the maximal ideal of B , the residue fields of the local components of C are also those of the local components of the Artinian ring $(A/\mathfrak{m}) \otimes_k (B/\mathfrak{n})$, and hence are finite extensions of $B/\mathfrak{n}$.

Proposition (7.5.6). — Let k be a field, and let A and B be complete Noetherian local rings containing k , whose residue fields are finite extensions of k . Let R be a property satisfying (R'_I) , (R_{III}) and (R_{IV}) , with P defined from R as in (7.5.0). Suppose that $R(A)$ and $P(\mathrm{Spec}(B), k)$ hold; the latter means that, for every finite extension k' of k and each complete local ring B_h occurring as a component of $B \otimes_k k'$, one has $R(B_h)$. Then $R(C_j)$ holds for every complete local component C_j of $A \hat{\otimes}_k B$.

Proof. By (7.5.5), (iii), each homomorphism $A \rightarrow C_j$ is local; by (7.5.5), (ii), each C_j is flat over A ; and by (7.5.5), (iv), its residue field is a finite extension of $A/\mathfrak{m}$. Moreover,

$$C/\mathfrak{m}C = (A/\mathfrak{m}) \otimes_k B,$$

so the definition of P and the fact that $A/\mathfrak{m}$ is finite over k imply

$$P(\mathrm{Spec}(C_j \otimes_A A/\mathfrak{m}), A/\mathfrak{m}).$$

Thus all the hypotheses of (7.5.1) hold for the local homomorphism $A \rightarrow C_j$, so $\mathrm{Spec}(C_j) \rightarrow \mathrm{Spec}(A)$ is a P -morphism. The conclusion follows from (R'_1) and $R(A)$. $\square$

Corollary (7.5.7). — (Chevalley). — *Let k be a perfect field (resp. algebraically closed), A, B two local Noetherian complete rings containing k and whose residue fields are finite extensions of k . Then, if A and B are reduced (resp. integral), the completed tensor product $A \hat{\otimes}_k B$ is reduced (resp. integral).*

Proof. I) Suppose k is perfect and A and B are reduced. Let A_i (resp. B_j) be the quotients of A (resp. B) by its minimal prime ideals. Then A and B identify with subrings of $\prod_i A_i$ and $\prod_j B_j$. Since the tensor products are taken over a field, $A \otimes_k B$ identifies with a subring of $\prod_{i,j} (A_i \otimes_k B_j)$; the tensor-product topology is induced by the product of the corresponding topologies. Hence $A \hat{\otimes}_k B$ embeds in $\prod_{i,j} (A_i \hat{\otimes}_k B_j)$, and we are reduced to the case where A and B are integral.

Let A' and B' be their integral closures. By Nagata's finiteness theorem (0, 23.1.6), A' (resp. B') is a finite A -module (resp. B -module) and a complete local ring. It suffices to show that $A \hat{\otimes}_k B$ embeds in $A' \hat{\otimes}_k B$ and that the latter embeds in $A' \hat{\otimes}_k B'$. This follows from the lemma below.

Lemma (7.5.7.1). — *Let A, B be two local Noetherian complete rings containing a field k and whose residue fields are finite extensions of k . Let $\mathfrak{m}$ be the maximal ideal of A . For every A -module M of finite type (equipped with the $\mathfrak{m}$ -adic topology), the completed tensor product $M \hat{\otimes}_k B$ (0, 7.7.1) identifies canonically with $M \otimes_A (A \hat{\otimes}_k B)$.*

Proof. Indeed, one has a canonical isomorphism $M \otimes_k B \xrightarrow{\sim} M \otimes_A (A \otimes_k B)$, hence a canonical composite homomorphism

$$\varphi : M \otimes_k B \longrightarrow M \otimes_A (A \otimes_k B) \longrightarrow M \otimes_A (A \hat{\otimes}_k B)$$

and it is immediate that this homomorphism is continuous for the tensor product topologies; furthermore, $M \otimes_A (A \hat{\otimes}_k B)$ is separated and complete ((7.5.5) and (0, 7.7.8)), hence one also has by completion a continuous homomorphism

$$\hat{\varphi} : M \hat{\otimes}_k B \longrightarrow M \otimes_A (A \hat{\otimes}_k B)$$

It is immediate that this homomorphism is bijective when M is free of finite type; as one has an exact sequence $L' \rightarrow L \rightarrow M \rightarrow 0$, where L and L' are free A -modules of finite type, one deduces a commutative diagram

$$\begin{array}{ccccccc} L' \hat{\otimes}_k B & \longrightarrow & L \hat{\otimes}_k B & \longrightarrow & M \hat{\otimes}_k B & \longrightarrow & 0 \\ \downarrow & & \downarrow & & \downarrow & & \\ L' \otimes_A (A \hat{\otimes}_k B) & \longrightarrow & L \otimes_A (A \hat{\otimes}_k B) & \longrightarrow & M \otimes_A (A \hat{\otimes}_k B) & \longrightarrow & 0 \end{array}$$

where the rows are exact (the first by virtue of the definition of the completed tensor product and of (0, 13.2.2)); the first two vertical arrows being bijections, it follows that so is the third. $\square$

The fact that $A \hat{\otimes}_k B$ identifies with a subring of $A' \otimes_A (A \hat{\otimes}_k B)$ follows then from the fact that $A \hat{\otimes}_k B$ is a flat A -module (7.5.5). One can thus further suppose that A and B are *integrally closed*; as moreover k is supposed *perfect*, $\mathrm{Spec}(A)$ and $\mathrm{Spec}(B)$ are *geometrically normal* over k , by virtue of (6.7.7, b)); one can then apply (7.5.6) taking for R the property (vi) of (7.5.3).

II) Suppose k is algebraically closed and A and B are integral. The argument in I) again reduces us to the case where A and B are integrally closed. By (7.5.6), $\mathrm{Spec}(A \hat{\otimes}_k B)$ is normal, so $A \hat{\otimes}_k B$ is a finite product of integrally closed complete local rings. It remains only to show that it is local. If $\mathfrak{m}$ and $\mathfrak{n}$ are the maximal ideals of A and B , it suffices to show that $(A/\mathfrak{m}) \otimes_k (B/\mathfrak{n})$ is local. Since both residue fields are finite extensions of the algebraically closed field k , they are equal to k , and the conclusion follows. $\square$

Remark (7.5.8). — It would be interesting to determine whether, in (7.5.7), one can replace the hypothesis that k is perfect or algebraically closed by the hypothesis that $\text{Spec}(A)$ or $\text{Spec}(B)$ is *geometrically integral* over k , or at least by the hypothesis that A (for example) is integral and contains a subring A_0 isomorphic to a formal power series ring $k[[T_1, \dots, T_n]]$, such that the field of fractions K of A is *separable* over the field of fractions K_0 of A_0 (one can show that this condition entails that A is geometrically integral over k , and that the two properties are equivalent when $[k : k^p]$ is finite (p being the characteristic exponent of k)). Likewise, it would be desirable to develop variants of (7.5.6) and (7.5.7) where one would weaken the hypothesis of finiteness of the residue fields of A and B , by supposing for example that only one of them is a finite extension of k , the other being arbitrary.

7.6. Applications: I. Japanese local rings

Proposition (7.6.1). — *Let A be a reduced local Noetherian ring whose formal fibres are geometrically normal. Then the completion $\widehat{A}$ is reduced, the integral closure A' of A in its total ring of fractions is a finite A -algebra (hence a semi-local Noetherian ring), and its completion $\widehat{A'}$ is isomorphic to the integral closure of $\widehat{A}$ in its total ring of fractions.*

Proof.

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The formal fibres of A are *a fortiori* geometrically reduced, so the reducedness of A implies that of $\widehat{A}$ by (7.3.17). Let $\mathfrak{p}_i$ be the minimal prime ideals of A and put $B_i = A/\mathfrak{p}_i$. The formal fibres of the local rings B_i are geometrically normal (7.3.15); hence $\widehat{B}_i$ is reduced, and (0, 23.1.7), (i), shows that the integral closure B'_i of B_i in its field of fractions is finite over B_i , and therefore over A . Since $A' = \prod_i B'_i$ (II, 6.3.8), A' is a finite A -module.

Let $\mathfrak{m}_j$ be the maximal ideals of the semi-local ring A' . Its completion $\widehat{A'}$ identifies with the product of the completions $\widehat{A'_{\mathfrak{m}_j}}$. By the hypothesis and (7.3.15), the formal fibres of each $A'_{\mathfrak{m}_j}$ are geometrically normal. Since A' is normal, so is each $A'_{\mathfrak{m}_j}$, and (7.3.17) then shows that each $\widehat{A'_{\mathfrak{m}_j}}$, and hence $\widehat{A'}$, is normal.

Since A' is finite over A , one has $\widehat{A'} = A' \otimes_A \widehat{A}$. If R denotes the total ring of fractions of A , then

$$\widehat{A} \subset \widehat{A'} \subset R' = R \otimes_A \widehat{A}.$$

Flatness of $\widehat{A}$ over A implies that every regular element of A remains regular in $\widehat{A}$; hence R' embeds canonically in the total ring of fractions R'' of $\widehat{A}$. Since $\widehat{A'}$ is normal and finite over $\widehat{A}$, it is precisely the integral closure of $\widehat{A}$ in R'' . $\square$

Corollary (7.6.2). — *Under the hypotheses of (7.6.1), there is a bijective correspondence between the maximal ideals $\mathfrak{m}_j$ of A' (in other words, the set of closed points of $\text{Spec}(A')$ above the closed point of A) and the set of minimal prime ideals $\mathfrak{q}_j$ of $\widehat{A}$ (in other words, the set of maximal points of $\text{Spec}(\widehat{A})$); in this correspondence, the completion $\widehat{A'_{\mathfrak{m}_j}}$ of $A'_{\mathfrak{m}_j}$ is isomorphic to the integral closure of $\widehat{A}/\mathfrak{q}_j$.*

Proof. Indeed, one knows that the integral closure of $\widehat{A}$ in its total ring of fractions is the direct product of the integral closures of the $\widehat{A}/\mathfrak{q}_j$, which are complete local rings (0, 23.1.6). $\square$

Corollary (7.6.3). — *Under the hypotheses of (7.6.1), for $\widehat{A}$ to be integral, it is necessary and sufficient that A be unibranch; for $\widehat{A}$ to be geometrically unibranch, it is necessary and sufficient that A be.*

Proof. This is a particular case of (7.6.2). $\square$

Theorem (7.6.4). — (Zariski-Nagata). — *Let A be a semi-local Noetherian ring. The following conditions are equivalent:*

- a) *For every finite reduced A -algebra C , the completion $\widehat{C}$ is a reduced ring.*
- a') *For every integral quotient ring B of A , with field of fractions K , the completion $\widehat{B}$ is reduced, and the fields L_i of the total ring of fractions of $\widehat{B}$ are separable extensions of K .*
- a'') *The formal fibres are geometrically reduced (in other words, for every integral quotient ring B of A , with field of fractions K , the K -algebra $\widehat{B} \otimes_B K$ is separable (4.6.2)).*
- b) *Every integral quotient ring B of A is a Japanese ring.*

Proof.

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To show that $a)$ implies $a')$, it suffices to verify that the L_i are separable extensions of K , or equivalently (4.6.1) that for every finite extension K' of K , the ring $\widehat{B} \otimes_B K'$ is reduced. The field K' is generated by finitely many elements integral over B ; these generate a finite B -subalgebra B' of K' whose field of fractions is K' . One has

$$\widehat{B'} = \widehat{B} \otimes_B B'$$

(0, 7.3.3), and therefore, by associativity of the tensor product,

$$\widehat{B} \otimes_B K' = \widehat{B'} \otimes_{B'} K'.$$

Now B' is a finite integral A -algebra, so $\widehat{B'}$ is reduced by $a)$. Since $\widehat{B'}$ is flat over B' , the nonzero elements of B' are $\widehat{B'}$ -regular; hence $\widehat{B'} \otimes_{B'} K'$ identifies with a subring of the total ring of fractions of $\widehat{B'}$, and is therefore reduced.

To see that $a')$ implies $a'')$, consider any point x of $X = \text{Spec}(A)$, and let Y be the integral closed subscheme of X having $\{\overline{x}\}$ as its underlying space. Then $Y = \text{Spec}(B)$, where B is an integral quotient ring of A , and $\text{Spec}(\widehat{B}) = Y \times_X \text{Spec}(\widehat{A})$. Thus the formal fibre of A at x is the same as that of B at its generic point, namely $\text{Spec}(\widehat{B} \otimes_B K)$. Its local rings are those of $\text{Spec}(\widehat{B})$ at the points of this fibre, so the hypothesis implies that $\widehat{B} \otimes_B K$ is reduced; the remaining condition in $a')$ then says that it is a separable K -algebra (4.6.1).

The condition $a'')$ implies $a)$, for it follows from (7.3.15) that the formal fibres of C are geometrically reduced, and if C is reduced, it follows hence that so is $\widehat{C}$ (particular case of (7.3.17)).

The fact that $a')$ implies $b)$ is a particular case of (0, 23.1.7). It remains to prove that $b)$ implies $a)$. If C is a finite A -algebra and $\mathfrak{q}$ is a prime ideal of C , then the inverse image $\mathfrak{p}$ of $\mathfrak{q}$ in A is prime and $C/\mathfrak{q}$ is a finite $(A/\mathfrak{p})$ -algebra. Thus $b)$ implies that every integral quotient ring of C is Japanese. We are therefore reduced to proving that, under $b)$, the completion of every integral quotient ring of A is reduced.

We argue by induction on $n = \dim(A)$, the assertion being trivial for $n = 0$. Replacing A by the quotients by its minimal prime ideals $\mathfrak{p}_i$, we may restrict to the case where A is integral. For every nonzero prime ideal $\mathfrak{p}$, the induction hypothesis already shows that the completion of $A/\mathfrak{p}$ is reduced; it therefore suffices to prove that $\widehat{A}$ is reduced.

Moreover, the integral closure A' of A is, by hypothesis, a finite A -module, hence a semi-local Noetherian ring, and $\widehat{A}$ identifies with a subring of $\widehat{A'}$ (0, 7.3.3). It is therefore enough to prove that $\widehat{A'}$ is reduced. As observed above, $b)$ also holds for A' , and $\dim(A') = n$ (0, 16.1.5); hence we may suppose that A is *integrally closed*. Let $t \neq 0$ belong to the Jacobson radical of A , and let $\mathfrak{q}_j$ ($1 \leq j \leq r$) be the prime ideals minimal among those containing tA ; then:

- (i) t is regular.
- (ii) A/tA has no immersed associated prime ideals.
- (iii) The $A_{\mathfrak{q}_j}$ are discrete valuation rings.
- (iv) The completions of the $A/\mathfrak{q}_j$ are reduced.

Indeed, (i) is trivial since A is integral and $t \neq 0$. As A is integrally closed, A/tA satisfies (S_1) , that is to say (5.7.7) is without immersed associated prime ideals (Bourbaki, *Alg. comm.*, chap. VII, § 1, no 4, prop. 8). Since A is integrally

closed, the $A_{\mathfrak{q}_j}$ are, and one knows (*loc. cit.*) that these rings are of dimension 1, hence they are discrete valuation rings, whence (iii). Finally, (iv) follows from the hypothesis of induction. The proof will be completed if we prove the

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Lemma (7.6.4.1). — (Zariski). — *Let A be a semi-local Noetherian ring, t an element of its radical satisfying the conditions (i) to (iv) above; then $\widehat{A}$ is reduced.*

Proof. Set for simplicity $A' = \widehat{A}$, and consider t as an element of A' ; one has $A'/tA' = (A/tA) \otimes_A A'$, and as A' is a flat A -module, it follows from (3.3.1) that the prime ideals of A' associated to the A' -module A'/tA' are the prime ideals $\mathfrak{q}'_{jh}$ such that $\mathfrak{q}'_{jh}$ lies above $\mathfrak{q}_j$ and is associated to the k_j -module $(A'/tA') \otimes_A k_j$, where k_j is the field of fractions of $A/\mathfrak{q}_j$. Now, $\text{Spec}((A'/tA') \otimes_A k_j)$ is the formal fibre of A at the point $\mathfrak{q}_j$, or equivalently that of $A/\mathfrak{q}_j$ at its generic point. By (iv), the completion $A'/\mathfrak{q}_j A'$ of $A/\mathfrak{q}_j$ is reduced; hence so is its formal fibre at the generic point. This formal

fibre therefore has no embedded associated prime, and its irreducible components are fields (3.2.1). It follows from (3.3.3) and (ii) that $\mathfrak{q}_j A'_{\mathfrak{q}_{jh}}$ is the maximal ideal of $A'_{\mathfrak{q}_{jh}}$ for every h . Since $A_{\mathfrak{q}_j}$ is a discrete valuation ring, its maximal ideal is principal; therefore the maximal ideal of the Noetherian local ring $A'_{\mathfrak{q}_{jh}}$ is principal, and this ring is a discrete valuation ring. We have thus verified the relevant hypotheses for the complete semi-local ring A' . It therefore suffices to show that if A is complete and satisfies (i) to (iii), then A is reduced.

The hypotheses (i) and (ii) imply that, if $\varphi : A \rightarrow \prod_{j=1}^r A_{\mathfrak{q}_j}$ is the canonical homomorphism, then for every $m \geq 0$ one has $t^m A = \varphi^{-1}(\varphi(t^m A))$ (3.4.9). Since the image of tA in each discrete valuation ring $A_{\mathfrak{q}_j}$ is a power of its maximal ideal, the tA -adic topology on A is the inverse image under φ of the product topology on $\prod_{j=1}^r A_{\mathfrak{q}_j}$. As t belongs to the Jacobson radical of A , the tA -adic topology is separated, hence φ is injective. By (iii), the rings $A_{\mathfrak{q}_j}$ are integral; their product is reduced, and therefore so is A . $\square$

 $\square$

Corollary (7.6.5). — *Let A be a semi-local Noetherian ring satisfying the equivalent conditions of (7.6.4); every semi-local A -algebra essentially of finite type also satisfies the conditions of (7.6.4).*

Proof. The condition a'') of (7.6.4) means indeed that A is a P -ring, where $P(Z, k)$ is the property “ Z is geometrically reduced over k ”; the corollary is then a consequence of the general theorem (7.4.4). $\square$

Corollary (7.6.6). — *Let A be a semi-local Noetherian integral ring of dimension 1, K its field of fractions. For A to be a Japanese ring, it is necessary and sufficient that the completion $\hat{A}$*

be reduced and that, if $\mathfrak{q}_j$ ($1 \leq j \leq n$) are the minimal prime ideals of $\hat{A}$, the fields of fractions of the integral rings $\hat{A}/\mathfrak{q}_j$ be separable extensions of K .

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Proof. The integral quotients of A are themselves fields, hence the condition b) of (7.6.4) is equivalent to the hypothesis that A is a Japanese ring, and the hypothesis a') to the conditions on $\hat{A}$, whence the conclusion. $\square$

Remarks (7.6.7). — (i) The equivalent conditions of (7.6.6) mean again that the formal fibres of A are geometrically regular (7.3.19), (iv)). One has already observed (*loc. cit.*) that this condition is satisfied when A is a discrete valuation ring whose field of fractions is of characteristic 0.

(ii) The same reasoning as in the first part of the proof of (7.6.4) shows that for a semi-local Noetherian ring A , the two following conditions are equivalent:

- a) For every integral quotient ring B of A , the completion $\hat{B}$ is reduced.
- a') The formal fibres of A are reduced.

Taking into account (0, 23.1.7), (i), these two conditions imply the following:

- b) For every integral quotient ring B of A , the integral closure of B is a finite B -algebra.

When A is a universally catenary ring, one can prove that the condition b) conversely implies a); we will not need to use this result.

7.7. Applications: II. Universally Japanese rings

(7.7.1). Recall (0, 23.1.1) that one calls a ring *universally Japanese* a ring A such that every integral A -algebra of finite type is a Japanese ring. This amounts to saying that for every integral A -algebra B of finite type, the integral closure of B is a finite B -algebra.

Theorem (7.7.2). — (Nagata). — *Let A be a Noetherian ring. The following conditions are equivalent:*

- a) A is a universally Japanese ring.
- b) Every integral quotient ring of A is a Japanese ring.
- c) For every maximal ideal $\mathfrak{m}$ of A , every integral quotient ring of $A_{\mathfrak{m}}$ is a Japanese ring, and for every integral quotient ring B of A , with field of fractions K , every finite radical extension K' of K and every sub- B -algebra finite B' of K' , with field of fractions K' , there exists $f' \neq 0$ in B' such that $B'_{f'}$ is integrally closed (cf. (6.13.7)).

Moreover, if A satisfies these conditions, so does every ring of fractions $S^{-1}A$ of A and every A -algebra of finite type.

Proof. Let us first show that $b)$ implies $c)$; every integral quotient ring of $A_{\mathfrak{m}}$ is a ring of fractions of an integral quotient ring of A , hence a Japanese ring (0, 23.1.1). On the other hand, every quotient B of A by one of its prime ideals is a Japanese ring (0, 23.1.1). One deduces that the set $\text{Nor}(\text{Spec}(B'))$ is open (6.13.3), hence contains a non-empty open set $D(f') = \text{Spec}(B'_{f'})$.

Let us show secondly that, if A satisfies $c)$, so does every A -algebra of finite type B . In the first instance, for every prime ideal $\mathfrak{q}$ of B , $B_{\mathfrak{q}}$ is a local ring in $S^{-1}B$, where $S = A - \mathfrak{p}$, $\mathfrak{p}$ being the inverse image of $\mathfrak{q}$ in A ; as by hypothesis every integral quotient ring of $A_{\mathfrak{p}}$ is a Japanese ring, so is every integral quotient ring of $B_{\mathfrak{q}}$, since $S^{-1}B$ is an $A_{\mathfrak{p}}$ -algebra of finite type (7.6.5). On the other hand, if C is an integral quotient ring of B , with field of fractions K , K' a finite radical extension of K and C' a sub- C -algebra finite of K' , with field of fractions K' , C' is an A -algebra integral of finite type, and in virtue of (6.13.7), the hypothesis $c)$ on A entails that the spectrum of C' contains a non-empty open part whose points are all normal; in other words, there is a $g' \neq 0$ in C' such that $C'_{g'}$ is integrally closed, which proves our assertion.

This result shows that, to prove the equivalence of $a)$, $b)$ and $c)$, it suffices to show that $c)$ implies $b)$, and even that for every integral quotient ring B of A , with field of fractions K , the integral closure of B is a B -module of finite type. On the other hand, if A satisfies $b)$, so does every integral quotient ring of A , hence is a Japanese ring, and we have seen above that A verifies the conditions $a)$, $b)$, $c)$; it follows that the same holds for every A -algebra of finite type. Moreover, if A satisfies $b)$, every integral quotient ring of $S^{-1}A$ is an integral quotient ring of A , hence is a Japanese ring by hypothesis, and as $A/\mathfrak{p}$ is by hypothesis a Japanese ring, so is $S^{-1}(A/\mathfrak{p})$ (0, 23.1.1). This completes the proof of the last assertion of the statement. $\square$

Corollary (7.7.3). — *Let A be a Noetherian ring satisfying the following conditions:*

- (i) For every maximal ideal $\mathfrak{m}$ of A , the formal fibres of $A_{\mathfrak{m}}$ are geometrically reduced.
- (ii) The equivalent conditions of (6.12.4) are satisfied.

Then A is universally Japanese.

Proof. Indeed, it follows then from (7.6.4) that every integral quotient ring of $A_{\mathfrak{m}}$ is a Japanese ring; whence the conclusion, by virtue of (7.7.2). $\square$

Corollary (7.7.4). — *A Dedekind ring whose field of fractions is of characteristic 0 (in particular $\mathbb{Z}$) is a universally Japanese ring. A local Noetherian ring whose formal fibres are geometrically reduced (in particular a local Noetherian complete ring) is universally Japanese.*

Proof. The first assertion follows from (7.7.3), (7.6.7), (i) and (6.12.6). The second follows from (7.6.4) and (7.7.2). $\square$

7.8. Excellent rings

(7.8.1). The previous paragraphs of this section (as well as (6.11), (6.12) and (6.13)) have been devoted to the study of problems frequently encountered in the use of rings and *Noetherian* preschemes, and which can be grouped into the following types:

A) For a *local* Noetherian ring A , do the properties of A “transfer” to its *completion* $\hat{A}$? For example, if A is reduced (resp. integral, resp. integral and integrally closed), is the same true of $\hat{A}$? Most of these questions are related to the properties of the local *formal fibres* of A (recall that these are the fibres of the canonical morphism $\text{Spec}(\hat{A}) \rightarrow \text{Spec}(A)$). An exception is formed by the properties related to the notion of dimension, in particular the notion of being equidimensional; it is then the condition of *chains* and its various refinements that play the essential role.

B) For a locally Noetherian prescheme X (in particular for an affine scheme $\text{Spec}(A)$), does the set of $x \in X$ at which the local ring $\mathcal{O}_x$ possesses a certain property (for example being integrally closed, or being a Cohen-Macaulay ring, or being regular) form an *open* set?

C) For an *integral* ring A , is the integral closure of A a finite extension of its field of fractions? Of course, this question can be reformulated for Noetherian preschemes (II, 6.3).

The problems of type B) or C) may also be posed for *local* rings, but it does not suffice in general that they be solved affirmatively for every local ring $A_{\mathfrak{p}}$ of a ring A for them to be solved for A (cf. (6.13.6)).

Let us also emphasise that in the study of these problems, we have been systematically concerned with whether the affirmative answer is *stable* for one or the other of the two most important operations in Commutative Algebra: localisation and passage to an algebra of finite type.

The results obtained in the study of these problems lead to identifying a class of Noetherian rings whose behaviour in this regard is the best possible:

Definition (7.8.2). — A ring A is called *excellent* if it is Noetherian and satisfies the following conditions:

- (i) A is universally catenary (or, which amounts to the same (5.6.3), (i)), for every prime ideal $\mathfrak{p}$ of A , $A_{\mathfrak{p}}$ is universally catenary).
- (ii) For every prime ideal $\mathfrak{p}$ of A , the formal fibres of $A_{\mathfrak{p}}$ are geometrically regular.
- (iii) For every integral quotient ring B of A and every finite radical extension K' of the field of fractions K of B , there exists a sub- B -algebra finite B' of K' , containing B , having K' as field of fractions, and such that the set of regular points of $\text{Spec}(B')$ contains a non-empty open set.

With this terminology, the essential content of the results of § 7 and of part of § 6 is summarised as follows:

(7.8.3). (i) In the conditions (i) and (ii) of the definition (7.8.2) one may restrict to the maximal ideals $\mathfrak{p}$ in A . For a local Noetherian ring to be excellent, it is necessary and

sufficient that it be universally catenary and that its formal fibres be geometrically regular.

(ii) If A is an excellent ring, so is every ring of fractions $S^{-1}A$ and every A -algebra of finite type.

(iii) A local complete ring (in particular a field or a ring of formal power series over a field) is excellent. A Dedekind ring whose field of fractions is of characteristic 0 (in particular $\mathbf{Z}$) is excellent.

(iv) Let A be an excellent ring, $X = \text{Spec}(A)$; the set $\text{Reg}(X)$ (resp. $\text{Nor}(X)$, $U_{R_n}(X)$) of points where X is regular (resp. normal, resp. satisfies (R_n)) is open in X . For every coherent $\mathcal{O}_X$ -Module $\mathcal{F}$, the set $U_{C_n}(\mathcal{F})$ (resp. $U_{S_n}(\mathcal{F})$, $\text{CM}(\mathcal{F})$) of $x \in X$ where $\text{coprof}(\mathcal{F}_x) \leq n$ (resp. the points where $\mathcal{F}$ satisfies (S_n) , resp. the points where $\mathcal{F}_x$ is a Cohen-Macaulay $\mathcal{O}_x$ -Module) is open in X .

(v) Let A be an excellent ring, $\mathfrak{J}$ an ideal of A , and $\hat{A}$ the separated completion of A for the $\mathfrak{J}$ -adic topology. Then the canonical morphism $f : \text{Spec}(\hat{A}) \rightarrow \text{Spec}(A)$ is regular (in other words, flat and with geometrically regular fibres). If we set $X = \text{Spec}(A)$, $X' = \text{Spec}(\hat{A})$, use the notation of (iv), and put $\mathcal{F}' = \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_{X'}$, then

(7.8.3.1).

$$\begin{aligned} \text{Reg}(X') &= f^{-1}(\text{Reg}(X)), & \text{Nor}(X') &= f^{-1}(\text{Nor}(X)), & U_{R_n}(X') &= f^{-1}(U_{R_n}(X)) \\ U_{C_n}(\mathcal{F}') &= f^{-1}(U_{C_n}(\mathcal{F})), & U_{S_n}(\mathcal{F}') &= f^{-1}(U_{S_n}(\mathcal{F})), & \text{CM}(\mathcal{F}') &= f^{-1}(\text{CM}(\mathcal{F})) \end{aligned}$$

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In particular, if $\mathfrak{J}$ is contained in the radical of A (for example if A is local and $\mathfrak{J}$ its maximal ideal), for A to be regular (resp. normal, resp. reduced, resp. to satisfy (R_n) , resp. its codepth to be $\leq n$, resp. to satisfy (S_n) , resp. to be a Cohen-Macaulay ring), it is necessary and sufficient that the same hold for $\hat{A}$; in particular, for A to have no immersed associated primes, it is necessary and sufficient that the same hold for $\hat{A}$.

(vi) An excellent ring is universally Japanese; in particular, if A is integral, its integral closure in every finite extension of its field of fractions is a finite A -algebra.

(vii) Let A be a reduced excellent local ring. Then the completion $\hat{A}$ is reduced, the integral closure A' of A in its total ring of fractions is a finite A -algebra (hence a semi-local Noetherian ring), and the integral closure of $\hat{A}$ in its total ring of fractions is isomorphic to the completion $\hat{A}'$ of A' . Furthermore, there is a bijective canonical correspondence between the set of maximal points of $\text{Spec}(\hat{A})$ (in other words, the set of prime minimal ideals of $\hat{A}$) and the set of closed points of $\text{Spec}(A')$ (in other words, the set of maximal ideals of A'). For the local ring A to be unibranch, it is necessary and sufficient that $\hat{A}$ be integral; for A to be geometrically unibranch, it is necessary and sufficient that $\hat{A}$ be.

(viii) If A is an excellent integral ring, the integral closure of A is the intersection of the integral closures of the $A_{\mathfrak{p}}$, where $\mathfrak{p}$ ranges over the set of prime ideals of A of height 1.

(ix) Let A be an excellent ring, $X = \text{Spec}(A)$, Z a closed part of X , $U = X - Z$, $i : U \rightarrow X$ the canonical injection, $\mathcal{F}$ an $\mathcal{O}_U$ -Module coherent. For the $\mathcal{O}_X$ -Module $i_*(\mathcal{F})$ to be coherent, it is necessary and sufficient that, for every $x \in \text{Ass}(\mathcal{F})$, one has $\text{codim}(\overline{\{x\}} \cap Z, \overline{\{x\}}) \geq 2$. In particular, if A is integral and $\mathcal{F}$ torsion-free, for $i_*(\mathcal{F})$ to be coherent, it is necessary and sufficient that $\text{codim}(Z, X) \geq 2$.

(x) Let A be an excellent local ring. For every integral quotient ring B of A , $\hat{B}$ is equidimensional. **IV-2 | 216**
For A to be equidimensional, it is necessary and sufficient that $\hat{A}$ be.

Proof. Most of these results have already been proved.

(i) The first assertion follows from (5.6.3), (i) and (7.4.4), (i); the second from (7.4.4), (7.3.18), (6.12.7) and (6.12.4).

Assertion (ii) follows from (5.6.1), (5.6.3), (i), (7.4.4) and (6.12.4).

(iii) The first assertion has been seen in (5.6.4) and (7.4.4), taking into account (i). The second has been seen in (5.6.4) and (7.3.19), (iv), taking into account (i).

Assertion (iv) follows from (6.12.4), (6.13.5), (6.12.9) and (6.11.8) (taking into account (ii)).

The first assertion of (v) is a consequence of (7.4.6); the relations follow, taking into account respectively (6.5.1), (6.5.4), (6.5.3), (6.3.5) and (6.4.2). The last assertion of (v) is a particular case corresponding to the property (S_n) for $n = 1$ (5.7.5).

Assertion (vi) follows from (7.7.3), and assertion (vii) from (7.6.1), (7.6.2) and (7.6.3).

For assertion (viii), let us note that the ring $A^{(1)}$, intersection of the $A_{\mathfrak{p}}$ for all prime ideals $\mathfrak{p}$ of height 1 of A , is a finite A -algebra, as it follows from (vi), from (7.8.2), (i) and (5.11.2). As the integral closure A' of A is a finite A -algebra, the prime ideals of height 1 of A' are exactly those which lie above prime ideals of height 1 of A (5.10.17), (iv); the conclusion follows hence from (0, 23.2.9).

The first assertion of (ix) is a consequence of (5.6.3), (i) and (5.11.4); the second is a consequence of (vi) and (5.11.4), if one observes that when $\mathcal{F}$ is torsion-free, $\text{Ass}(\mathcal{F})$ is reduced to the generic point of $\text{Spec}(A)$. Finally, to prove (x), it suffices to observe that B is universally catenary and universally Japanese by (ii); as B is an excellent local ring by (ii), the ring $B^{(1)}$ is a finite B -algebra (5.11.2), hence $A_{\mathfrak{m}}$ is formally catenary, and the same holds for every $A_{\mathfrak{p}}$ for every prime ideal $\mathfrak{p}$ of A , hence also A . This shows that A is strictly formally catenary by (7.2.5), c) and completes the proof of (x). $\square$

Remarks (7.8.4). — (i) If one considers only the Noetherian rings A that satisfy the conditions (ii) and (iii) of the definition (7.8.2), one verifies by inspection of the preceding proofs that the assertions (ii), (iii), (iv), (v), (vi) and (vii) of (7.8.3) are still valid, replacing “excellent” by “satisfying (7.8.2), (ii) and (iii)”.

(ii) The catenary local ring A constructed in (5.6.11) satisfies the conditions (ii) and (iii) (but not condition (i) of the definition (7.8.2)). Indeed, the formal fibre at the maximal ideal nC_n of A is geometrically regular, and that at the other prime ideals $\mathfrak{p}$ of A coincides with that of the ring E at these prime ideals;

hence the condition (6.12.4), a)) is also satisfied for A by virtue of (7.8.3), (v); this proves hence that A satisfies the condition (7.8.2), (ii). On the other hand, the complement of the closed point in $\text{Spec}(A)$ is isomorphic to an open affine, whose ring of affine functions is the ring of a

polynomial ring; by taking the body k_0 of characteristic 0 (7.8.3), (iii); consequently E is an excellent ring by (7.8.3), (ii) and this proves that A satisfies the condition (7.8.2), (ii).

(iii) One can replace the condition (i) of the definition (7.8.2) by the following (weaker in appearance, taking into account (5.6.10)):

(i') For every minimal prime ideal $\mathfrak{p}_i$ of A ($1 \leq i \leq r$), let A'_i be the integral closure of the integral ring $A_i = A/\mathfrak{p}_i$; then, for every maximal ideal $\mathfrak{m}'$ of A'_i , denoting by $\mathfrak{m}$ the inverse image of $\mathfrak{m}'$ in A ,

(7.8.4.1).

$$\dim(A_{\mathfrak{m}}) = \dim(A'_{i, \mathfrak{m}'}).$$

Indeed, one has seen in a Remark (i) that under only the hypotheses (ii) and (iii) of (7.8.2), the analogues of the properties (ii), (v), (vi) and (vii) of (7.8.3) are still valid. One deduces hence from (vi) that the A'_i satisfy (ii) and (iii), and that $((A'_i)_{\mathfrak{m}'})^\wedge$ are integral and integrally closed. Let now $\mathfrak{m}$ be an arbitrary maximal ideal of A ; for every i such that $\mathfrak{p}_i \subset \mathfrak{m}$, the integral closure of $A_{\mathfrak{m}}/\mathfrak{p}_i A_{\mathfrak{m}}$ is an $(A_{\mathfrak{m}}/\mathfrak{p}_i A_{\mathfrak{m}})$ -algebra finite by (vi), hence semi-local, and its local components are of the form $(A'_i)_{\mathfrak{m}'}$, where $\mathfrak{m}'$ is a prime ideal (not necessarily maximal) of A'_i above $\mathfrak{m}/\mathfrak{p}_i$; it follows then from (vii), from what precedes and from (7.8.4.1) that the quotients of the completion $(A_{\mathfrak{m}}/\mathfrak{p}_i A_{\mathfrak{m}})^\wedge$ by its prime minimal ideals all have the same dimension equal to $\dim(A_{\mathfrak{m}})$. Consequently ((7.1.9) and (7.1.8), b)), $A_{\mathfrak{m}}$ is formally catenary, and *a fortiori* (7.1.11) universally catenary; it follows that so is $A_{\mathfrak{p}}$ for every prime ideal $\mathfrak{p}$ of A (5.6.3), (i), hence also A .

In particular, if A is normal, or more generally if $A_{\mathfrak{m}}$ is unibranch for every maximal ideal $\mathfrak{m}$ of A , the condition (i') is always satisfied, since $\mathfrak{m}$ can contain only a single prime ideal $\mathfrak{p}_i$ (0, 16.1.5); one can then omit (i) from the definition (7.8.2). More particularly, if A is a Noetherian ring, *local unibranch*, it is excellent if and only if its formal fibres are geometrically regular.

(iv) Recall that a quotient ring of a local Cohen-Macaulay ring is also regular (and *a fortiori* a quotient ring of a regular local ring) also satisfies the properties (7.8.3), (ix) and (x) by virtue of (7.2.7); but the interest of quotient rings of regular rings lies above all in their cohomological properties (chap. III, 3^e Partie).

(v) We will see further on (§ 18) that if k is a field with a complete non-discrete valuation, the ring $k\{\{T_1, \dots, T_n\}\}$ of convergent power series (in a neighbourhood of 0 in k^n) is an excellent ring.

(7.8.5). One says that a locally Noetherian prescheme X is *excellent* if for some covering (U_α) of X by affine opens, each of the rings A_α of the U_α is excellent; this property is then true for *every* covering (U_α) of X by affine opens.

Proposition (7.8.6). — *Let X be an excellent prescheme.*

(i) *If $f : X' \rightarrow X$ is a morphism locally of finite type, X' is excellent.*

(ii) *If X is reduced, its normalised X' (II, 6.3.8) is finite over X .*

(iii) *The sets $\text{Reg}(X)$, $\text{Nor}(X)$, $U_{R_n}(X)$, as well as $U_{C_n}(\mathcal{F})$, $U_{S_n}(\mathcal{F})$ and $\text{CM}(\mathcal{F})$ for every coherent $\mathcal{O}_X$ -Module $\mathcal{F}$, are open in X .*

Proof. This follows immediately from (7.8.3), (ii), (vi) and (iv). Let us also note that (7.8.3), (ix) is also valid without modification of the statement when X is an excellent prescheme. $\square$

7.9. Excellent rings and resolution of singularities

(7.9.1). Given a *locally Noetherian reduced* prescheme X , one calls a *resolving morphism* for X a morphism $f : X' \rightarrow X$ such that X' is *regular* and that f is *proper* and *birational*. When such a morphism exists, one says that one can *resolve the singularities* of X (or more simply that one can *resolve* X). For example, if A is a *Japanese ring* of dimension 1, one can resolve $X = \operatorname{Spec}(A)$ since the morphism $X' \rightarrow X$ (where X' is the *normalised* of X) is finite (hence proper) and birational, and the local rings of X' are integrally closed rings of dimension 1, hence discrete valuation rings, hence regular.

(7.9.2). It is clear that if one can resolve X , one can also resolve every *open* of X and every local prescheme $\operatorname{Spec}(\mathcal{O}_x)$ of X (II, 5.4.2). On the other hand, if one can resolve X , it is clear that there exists in X an open subset *everywhere* dense contained in $\operatorname{Reg}(X)$. These remarks show immediately that if, for every closed *integral* sub-prescheme Y of X and every Y -prescheme Y' *integral, finite* and *radicial* over Y , one can resolve Y' , then the affine opens of X satisfy condition (iii) of (7.8.2).

On the other hand, for local schemes, one has the following proposition:

Proposition (7.9.3). — *Let A be a local Noetherian reduced ring, and suppose that one can resolve $\operatorname{Spec}(A)$. Then the fibres of the canonical morphism $\operatorname{Spec}(\hat{A}) \rightarrow \operatorname{Spec}(A)$ at the maximal points of $\operatorname{Spec}(A)$ are regular.*

Proof. Set $X = \operatorname{Spec}(A)$, $X' = \operatorname{Spec}(\hat{A})$, and let $g : X' \rightarrow X$ be the canonical morphism. Let $f : Y \rightarrow X$ be a resolving morphism; set $Y' = X' \times_X Y$, and let $f' : Y' \rightarrow X'$ and $g' : Y' \rightarrow Y$ be the canonical projections, so that one has a commutative diagram

$$\begin{array}{ccc} X' & \xleftarrow{f'} & Y' \\ g \downarrow & & \downarrow g' \\ X & \xleftarrow{f} & Y \end{array}$$

It will suffice to prove that the prescheme Y' is *regular*: indeed, as f is birational and of finite type, there is an open U dense in X such that the restriction $f^{-1}(U) \rightarrow U$ of f is an isomorphism and that $f^{-1}(U)$ is everywhere dense in Y (I, 6.5.5); the preschemes induced respectively on the open $g^{-1}(f^{-1}(U))$ of X' and the open $g'^{-1}(f^{-1}(U))$ of Y' are therefore isomorphic; this proves that $g^{-1}(U)$ is regular, and *a fortiori* so are the fibres of g at the maximal points of X , which are contained in U .

Let us note that the morphism f' is proper (II, 5.4.2), and in particular of finite type, hence Y' is Noetherian, since it is thus of X' . Let a' be the closed point of X' ; to see that Y' is regular, it will suffice to prove that *for every $y' \in f'^{-1}(a')$, the local ring $\mathcal{O}_{Y',y'}$ is regular*. Indeed, one will then have $f'^{-1}(a') \subset \operatorname{Reg}(Y')$. But as X' is the spectrum of a complete local ring and f' is of finite type, $\operatorname{Reg}(Y')$ is open in Y' (6.12.8), and as the morphism f' is closed and $f'^{-1}(a') \subset \operatorname{Reg}(Y')$, there exists a neighbourhood V' of a' in X' such that $f'^{-1}(V') \subset \operatorname{Reg}(Y')$. As $\hat{A}$ is an integral local ring, on necessarily has $V' = X'$, hence $\operatorname{Reg}(Y') = Y'$ and the proposition will be proved.

Now, if a is the closed point of X , one has $g^{-1}(a) = \{a'\}$ and the residue fields $k(a)$ and $k(a')$ are isomorphic; hence $g'^{-1}(f^{-1}(a)) = f'^{-1}(a')$ and the restriction $f'^{-1}(a') \rightarrow f^{-1}(a)$ of g' is an isomorphism of these two fibres. Let $y = g'(y')$ and set $B = \mathcal{O}_{X,y}$; if $\mathfrak{n}$ is the maximal ideal of B , $B' = B \otimes_A \hat{A}$ has a single maximal ideal $\mathfrak{n}'$ above $\mathfrak{n}$ of B and of the maximal ideal of $\hat{A}$, and one has $\mathcal{O}_{Y',y'} = B'_{\mathfrak{n}'}$. To show that the local ring $B'_{\mathfrak{n}'}$ is regular, it suffices to establish that its completion is regular (0, 17.1.5), that is, or by hypothesis B is regular, hence so is its completion $\hat{B}$. The conclusion will follow from the following lemma:

Lemma (7.9.3.1). — *Let A, B be two local Noetherian rings, $\rho : A \rightarrow B$ a local homomorphism such that the ring $B' = B \otimes_A \hat{A}$ is Noetherian. Then, for every maximal ideal $\mathfrak{n}'$ of B' above the maximal ideal $\mathfrak{n}$ of B and the maximal ideal of $\hat{A}$, the completions of B and of $B'_{\mathfrak{n}'}$ are isomorphic.*

Proof. Let $\mathfrak{m}$ be the maximal ideal of A and set $B'' = B'_{\mathfrak{n}'}$. For every integer $h > 0$, one has $B'/\mathfrak{m}^h B' = B \otimes_A (\hat{A}/\mathfrak{m}^h \hat{A})$; but $\hat{A}/\mathfrak{m}^h \hat{A}$ is isomorphic to $A/\mathfrak{m}^h$, hence $B'/\mathfrak{m}^h B'$ is isomorphic to $B/\mathfrak{m}^h B$, and in particular it is a local ring whose maximal ideal is $\mathfrak{n}'/\mathfrak{m}^h B'$; consequently $B''/\mathfrak{m}^h B''$, isomorphic

to $(B'/m^h B')_{n'}$, is B -isomorphic to $B'/m^h B'$, and finally isomorphic to $B/m^h B$; in particular, the maximal ideal of B'' is equal to nB'' , and $B''/n^h B''$ is isomorphic to $B/n^h B$ from what precedes. As $\widehat{B''} = \varprojlim_h (B''/n^h B'')$, the lemma is proved, as was (7.9.3). $\square$

$\square$

Corollary (7.9.4). — *Let A be a local Noetherian ring.*

- (i) *If, for every integral quotient ring B of A , one can resolve $\text{Spec}(B)$, then the formal fibres of A are regular.*
- (ii) *Suppose that, for every integral quotient ring B of A , and every integral ring B' containing B , which is a finite B -algebra and whose field of fractions is radical over that of B , one can resolve $\text{Spec}(B')$; then the formal fibres of A are geometrically regular.*

Proof. Every formal fibre of A being a formal fibre of a ring at the generic point of its spectrum, it is clear that (i) is an immediate consequence of (7.9.3), and (ii) follows from (i) and (6.7.7). $\square$

Proposition (7.9.5). — *Let X be a locally Noetherian prescheme, such that, for every integral closed subprescheme Y , finite and radical over X , one can resolve Y . Then the rings of the affine opens of X satisfy the conditions (ii) and (iii) of (7.8.2); if moreover X is universally catenary (5.6.4),*

in particular if X is locally immersible in a regular prescheme (5.6.4), X is an excellent prescheme. IV-2 | 220

Proof. This follows immediately from (7.9.4) and (7.9.2). $\square$

Remark (7.9.6). — It is possible that the converse of (7.9.5) be true, that is to say that X is reduced and that the rings of the affine opens of X satisfy the conditions (ii) and (iii) of (7.8.2) implies the possibility of resolving X (and consequently also the possibility of resolving every reduced prescheme locally of finite type over X , by virtue of (7.4.4) and (6.12.4)). This is in any case true when one limits oneself to reduced Noetherian preschemes whose residue fields are of characteristic 0, as shown by the recent results of Hironaka [35] (the latter states his results under hypotheses that are too restrictive, the reasoning being in fact valid under the only hypotheses (ii) and (iii) of (7.8.2) for the rings of the affine opens of X , combined with the fact that the residue fields are of characteristic 0). On the contrary, in the general case, it is known so far only how to resolve algebraic schemes of dimension 2 over a perfect field [36]. Given the importance that the resolution of singularities is acquiring in the topological study of schemes (particularly concerning their homological and homotopical properties), this gives a particular interest to the category of excellent rings and excellent preschemes.

One may also note in this regard that the less delicate part of the reasoning of Hironaka (*loc. cit.*) shows independently of all hypothesis on the characteristics of the residue fields, and using only the properties (ii) and (iii) of (7.8.2) for the local rings of X , the problem of resolution of singularities of a prescheme X reduces to the case where $X = \text{Spec}(A)$, A being a local Noetherian integral ring and complete. Consequently, if subsequent results were to disprove the more advanced conjecture above, and were to lead to formulating more restrictive conditions on the local rings of X , these could only be through the intermediary of restrictive conditions imposed on the local complete rings (probably conditions on their residue fields, perhaps the condition $[k : k^p] < +\infty$ (p being the characteristic exponent of k)).

(7.9.7). Consider on the one hand a full subcategory $\mathcal{C}$ of the category of locally Noetherian preschemes, and on the other hand a property $R(A)$, subject to the following conditions:

1° For every $X \in \mathcal{C}$, every prescheme locally of finite type over X belongs to $\mathcal{C}$. For every local Noetherian ring A such that $\text{Spec}(A) \in \mathcal{C}$ and every multiplicative subset S of A , one has $\text{Spec}(S^{-1}A) \in \mathcal{C}$.

2° For every $X \in \mathcal{C}$, the set $U_R(X)$ of $x \in X$ such that $R(\mathcal{O}_{X,x})$ is true is open in X .

3° For every local Noetherian ring A such that $\text{Spec}(A) \in \mathcal{C}$ and every regular element t of the maximal ideal of A , $R(A/tA)$ implies $R(A)$.

Let us denote as in (7.5.0) by $P(Z, k)$, for a field k and a k -prescheme $Z \in \mathcal{C}$, the following property:

“For every finite extension k' of k , all the local rings of $Z \otimes_k k'$ satisfy the property R .”

We will further suppose that R satisfies the following condition:

4° If $P(Z, k)$ is true, then, for every extension of finite type k'' of k , all the local rings of $Z \otimes_k k''$ satisfy the property R .

Let us note that these conditions are satisfied when one takes for $\mathcal{C}$ the category of *excellent* preschemes and for R one of the properties (i) to (viii) of (7.5.3); for condition 1°, this follows from (7.8.3), (ii)); for condition 2°, from the reasoning of (7.5.3), and for conditions 3° and 4°, from the reasoning of (7.5.3), taking into account that every excellent ring is catenary. Finally, for condition 4°, it follows, for (i), (ii) and (iii) of (6.7.1); for (iv), (v), (vi) of (6.7.7), and for (viii) of (4.6.1); as concerns property (vii) of (7.5.3), the corresponding property $P(Z, k)$ implies that Z is locally integral (being locally Noetherian) and that each of the integral sub-preschemes of Z whose sum is geometrically integral, one also concludes the property 4° in this case.

With these notations and hypotheses:

Proposition (7.9.8). — *Let A be a local Noetherian ring, k its residue field, B an integral quotient ring of A such that $\text{Spec}(B) \in \mathcal{C}$, $\varphi : A \rightarrow B$ a local homomorphism making B a flat A -module; set $Y = \text{Spec}(A)$, $X = \text{Spec}(B)$ and let $f : X \rightarrow Y$ be the morphism corresponding to φ . Suppose that:*

1° *the property $P(B \otimes_A k, k)$ is true;*

2° *for every finite morphism $Y_1 \rightarrow Y$, one can resolve $(Y_1)_{\text{red}}$.*

Then, for every $y \in Y$, the property $P(f^{-1}(y), k(y))$ is true.

Proof. Let us note that condition 2° of (7.9.8) is still satisfied when one replaces Y by the spectrum of a local ring in a maximal ideal of a finite A -algebra (7.9.2); on the other hand, by virtue of condition 1° of (7.9.7), the spectrum of a ring of fractions of $B \otimes_A A'$ also belongs to $\mathcal{C}$. The lemma (7.3.16.2) shows then (as in part I) of the reasoning of (7.5.2)) that it suffices to prove that, if Y is integral and if y is the generic point of Y , the local rings of the fibre $f^{-1}(y)$ satisfy R .

This being so, there exists by hypothesis a regular prescheme Y' and a morphism proper and birational $g : Y' \rightarrow Y$; as Y' is Noetherian and locally integral, the hypothesis that g is birational implies that Y' is integral. As the morphism $g' : X' = X \times_Y Y' \rightarrow X$ is also birational, and let $f' : Y' \rightarrow X'$ and $g' : X' \rightarrow X$ be the canonical projections, so that one has a commutative diagram

(7.9.8.1).

$$\begin{array}{ccc} X & \xleftarrow{g'} & X' \\ f \downarrow & & \downarrow f' \\ Y & \xleftarrow{g} & Y' \end{array}$$

where f' and g' are the canonical projections. Taking into account condition 2° of (7.9.7), the same reasoning as at the beginning of (7.9.3) shows that it suffices to prove

that $U_R(X') = X'$, then, designating by a the closed point of X , that $g'^{-1}(a) \subset U_R(X')$. Now, let $x' \in g'^{-1}(a)$, and set $z' = f'(x')$, so that $b = g(z')$ is the closed point of Y . One has therefore $f'^{-1}(z') = f^{-1}(b) \otimes_{k(b)} k(z')$, and as $k(z')$ is an extension of finite type of k (since g is proper, hence of finite type), the hypothesis implies, by virtue of condition 4° of (7.9.7), that it follows that $P(f'^{-1}(z'), k(z'))$ is true, and as $\mathcal{O}_{x'}$ is a regular ring and that $\text{Spec}(\mathcal{O}_{x'}) \in \mathcal{C}$ by virtue of condition 1° of (7.9.7), the lemma (7.5.1.1) proves that $R(\mathcal{O}_{X', x'})$ is true, which completes the proof. $\square$

Corollary (7.9.9). — *Let A, B be two local Noetherian rings, $\varphi : A \rightarrow B$ a local homomorphism making B a flat A -module. Suppose that A is integral and geometrically unibranch, and that for every A -algebra finite integral A' one can resolve $\text{Spec}(A')$ (this will be the case if the formal fibres of A are geometrically regular and the residue field of A is of characteristic 0, by virtue of the results of Hironaka [35]). Let k be the residue field of A and suppose that $\text{Spec}(B \otimes_A k)$ is geometrically punctually integral (4.6.9). Then B is integral.*

Proof. We will apply (7.9.8) taking for $\mathcal{C}$ the category of all locally Noetherian preschemes, for R the property of being integral; the reasoning of (7.5.3) and (7.9.7) show that conditions 1°, 2°, 3° and 4° of (7.9.7) are satisfied in this case. The hypothesis on $B \otimes_A k$ and the proposition (7.9.8) show hence (with the notations of (7.9.8)) that the fibres $f^{-1}(y)$ are geometrically punctually integral for $y \in Y$; *a fortiori* the $f^{-1}(y)$ for $y \in Y$ are reduced preschemes, and as A is integral (and *a fortiori* reduced) (3.3.5). It remains to prove that $X = \text{Spec}(B)$ is irreducible. Now, the proof of (7.9.8) proves that X' is locally integral (being locally Noetherian). As the morphism $g' : X' \rightarrow X$ is surjective, it

suffices hence to see that X' is irreducible, or (since X' is locally integral), that X' is *connected*. But it suffices for this to prove that the fibre $g'^{-1}(a)$ is *connected*. Indeed, if this point is established, X' cannot be the sum of two open non-empty preschemes X'_1, X'_2 , for one would then have for example $g'^{-1}(a) \cap X'_2 = \emptyset$; but the restriction $X'_2 \rightarrow X$ of g' being proper (since X'_2 is a closed sub-prescheme of X'), $g'(X'_2)$ would be a closed non-empty part of X not containing the closed point a , which is absurd. Now, one has $g'^{-1}(a) = g^{-1}(b) \otimes_{k(b)} k(a)$; but $g : Y' \rightarrow Y$ is proper and birational, hence, in the Stein factorisation $Y' \xrightarrow{v} Y'' \xrightarrow{u} Y$ of the morphism g (III, 4.3.3), the morphism finite u is also birational and consequently Y'' is integral. As A is supposed geometrically unibranch, it follows then from (III, 4.3.4) that $g^{-1}(b)$ is *geometrically connected*, which completes the proof. $\square$

Remarks (7.9.10). — (i) We will show further on (18.8.11) that for a local Noetherian reduced ring B to be geometrically unibranch, it is necessary and sufficient that for every étale morphism $h : X' \rightarrow X = \text{Spec}(B)$ and every point $x' \in X'$ above the closed point of X , $\mathcal{O}_{X',x'}$ is integral. One deduces that if, in (7.9.9) one supposes, not only that $\text{Spec}(B \otimes_A k)$ is geometrically punctually integral, but also geometrically unibranch, then one can conclude that B is *geometrically unibranch*.

(ii) Let X, Y be two excellent preschemes, $f : X \rightarrow Y$ a flat morphism, and suppose that for every finite morphism $Y_1 \rightarrow Y$, one can resolve $(Y_1)_{\text{red}}$. It follows from (7.9.8), taking for R the property of being *regular*, that the set U of $x \in X$ such that the fibre at the point $f(x)$ of the morphism

$$\text{Spec}(\mathcal{O}_x \otimes_{\mathcal{O}_{f(x)}} k(f(x))) \longrightarrow \text{Spec}(k(f(x)))$$

is *geometrically regular*, is *stable by generisation*. We do not know if this set is *open* (or, which amounts to the same (0, 9.2.5), if it is *constructible*) even in the particular case where Y is the spectrum of $\mathbf{Z}$ or of a polynomial ring in one indeterminate $k[T]$ over a field k , and where consequently the condition of “resolvability” imposed on Y is trivially satisfied.

§8. PROJECTIVE LIMITS OF PRESCHMES

8.1. Introduction

(8.1.1). In this paragraph, we will systematically study the following situation. Let I be a filtered preordered increasing set, $(A_\lambda, \varphi_{\mu\lambda})$ an inductive system of rings having I for index set, $A = \varinjlim A_\lambda$ its inductive limit. For every $\alpha \in I$ and every A_α -prescheme X_α , consider the A_λ -preschemes $X_\lambda = X_\alpha \otimes_{A_\alpha} A_\lambda$ for $\lambda \geq \alpha$, and the A -prescheme $X = X_\alpha \otimes_{A_\alpha} A$; it is clear that the preschemes X_λ (for $\lambda \geq \alpha$) form a *projective system*, and we will see (8.2.5) that X is a *projective limit* of this system in the category of preschemes. We propose to find conditions on X_α or on the A_λ allowing us to prove properties of the following type: for X to have a property P (for example, the property of being proper over $S = \text{Spec}(A)$, or irreducible, or connected, etc.), it is necessary and sufficient that there exist an index $\mu \geq \alpha$ such that, for every $\lambda \geq \mu$, X_λ has (with respect to $S_\lambda = \text{Spec}(A_\lambda)$, as the case may be) the same property P . We will obtain analogous statements for properties of $\mathcal{O}_X$ -Modules, of A -morphisms of A -preschemes, etc. We will also show (8.9.1) that the datum of an A -prescheme of finite presentation (1.6.1) is essentially equivalent to the datum of an A_λ -prescheme of finite presentation X_λ for some sufficiently large λ , X being then *isomorphic* to $X_\lambda \otimes_{A_\lambda} A$. There are also analogous statements for $\mathcal{O}_X$ -Modules of finite presentation, their homomorphisms, the A -morphisms of A -preschemes of finite presentation, etc.

(8.1.2). The utility of such results will appear, for example, in the following questions:

- a) Let Y be a prescheme, y a point of Y , (U_λ) the projective system of filtered decreasing open affine neighbourhoods of y in Y ; if A_λ is the ring of U_λ , the A_λ form a filtered increasing inductive system, whose inductive limit A is the local ring $\mathcal{O}_y$; moreover, if we denote by $\mathfrak{p}_\lambda$ the prime ideal $\mathfrak{i}_y$ in the ring A_λ , the inductive system (A_λ) is cofinal in every inductive system $(A_\alpha)_f$, where f ranges over $A_\alpha - \mathfrak{p}_\alpha$ (for a fixed α), since the $D(f)$ form a fundamental system of neighbourhoods of y in U_α , hence in Y . The results of the present paragraph will therefore imply that the algebraic geometry of $\mathcal{O}_y$ -preschemes of finite presentation (and the theory of Modules of finite presentation on these preschemes) is essentially equivalent to the algebraic geometry of preschemes of finite presentation on “sufficiently small” open neighbourhoods of y . Thus, the statement (8.10.5), (xii) implies that if a morphism of finite

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presentation $f : X \rightarrow Y$ is such that $X \times_Y \operatorname{Spec}(\mathcal{O}_y)$ is proper over $\operatorname{Spec}(\mathcal{O}_y)$, it is necessary and sufficient that there exist an open neighbourhood U of y in Y such that $f^{-1}(U)$ is proper over U .

An important particular case, classical in a certain sense, is the one where Y is integral and where y is its generic point, so that $\mathcal{O}_y$ is nothing other than the field $R(Y) = K$ of rational functions on Y . The results of the present paragraph then amount to interpreting the algebraic geometry over K in terms of algebraic geometry over “sufficiently small” non-empty open subsets of Y , that is to say, intuitively, in terms of “families” of geometric objects indexed by the points of such an open set. This point of view has moreover been commonly used for a long time, not only in Algebraic Geometry over algebraically closed fields, but also in the arithmetic study of varieties defined over a number field K (a finite extension of $\mathbf{Q}$), considering the latter as the field of fractions of its ring of integers A (the “theory of reduction modulo $\mathfrak{p}$ ” ($\mathfrak{p}$ a prime ideal of A); cf. (I, 3.7)). The results of §§ 8 and 9 thus provide, among other things, some of the foundations of the language of “reduction modulo $\mathfrak{p}$ ” in arithmetic.

We note that in the example considered here, the morphisms $S_\mu \rightarrow S_\lambda$ (for $\lambda \leq \mu$) are the canonical open immersions $U_\mu \hookrightarrow U_\lambda$, and *a fortiori* are flat morphisms (but not faithfully flat in general), which explains the interest of the statements that appeal to such a restriction.

- b) Suppose that the A_λ are fields, so that $A = \varinjlim A_\lambda$ is also a field. This case arises generally when one starts from geometric data over an arbitrary field K , which one considers as an extension of a field k (for example the prime subfield of K). There is then interest in considering K as an inductive limit of its sub-extensions that are of finite type over k , which allows us to reduce many questions to the case where K is a finite type extension of k . Using also the method sketched in a), one can then reduce generally to the case of a base ring A which is an integral algebra of finite type over k .

We note that in this example, the morphisms $S_\mu \rightarrow S_\lambda$ are faithfully flat.

- c) Suppose that we are interested in properties, local on Y , of preschemes of finite presentation over an arbitrary prescheme Y , which we can from now on suppose to be an affine scheme of ring A . There is then interest in considering A as an inductive limit of its subrings that are $\mathbf{Z}$ -algebras of finite type, which allows us to reduce many questions to the case where Y is the spectrum of such an algebra. This is the explication of the “Kroneckerian point of view” according to which Algebraic Geometry reduces to the Algebraic Geometry of preschemes of finite type over $\mathbf{Z}$ (which one sometimes qualifies as “Absolute Algebraic Geometry”). This example shows us in particular that in most questions “relative” to a prescheme of base Y , one can reduce to the case where Y is Noetherian.

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We note that in this example, in contrast to the previous ones, the morphisms $S_\mu \rightarrow S_\lambda$ have in general no particular regularity property.

In what follows, when we apply the results that follow to any one of the three particular situations that have just been described, we will not go into detail about the procedure that allows us to make these applications, being content to refer to what precedes.

(8.1.3). In example a) of (8.1.2), we saw that if Y is an integral prescheme with generic point y , $f : X \rightarrow Y$ a morphism of finite presentation, and if the generic fibre $f^{-1}(y)$ is proper over $k(y)$, then there exists an open neighbourhood U of y in Y such that $f^{-1}(U)$ is proper over U ; it follows *a fortiori* that for every $s \in U$, $f^{-1}(s)$ is proper over $k(s)$. It turns out that one needs a converse, ensuring that if $f^{-1}(s)$ is proper over $k(s)$, then $f^{-1}(y)$ is proper over $k(y)$. For example, suppose that X and Y are algebraic preschemes over an algebraically closed field k (one can take for k the field $\mathbf{C}$ of complex numbers, to fix ideas); one sometimes needs to know that, for every $s \in Y$, rational over k , the fibre $f^{-1}(s)$ is proper over $k(s)$, then $f^{-1}(y)$ is proper over $k(y)$, and consequently $f^{-1}(U)$ is proper over U for a neighbourhood U of y ¹¹. Now this statement will follow easily from the following: the set E of points $s \in Y$ such that $f^{-1}(s)$ is proper over $k(s)$ is constructible (and therefore identical to all of

¹¹We note that such a statement is ultimately purely geometric in nature, in the sense that it only appeals to rational points over k , and not to generic points; for example, when $k = \mathbf{C}$, this statement has an obvious topological significance for the analyst, interpreting “proper” in the topological sense of the term, for the underlying analytic spaces formed by the points of X and Y rational over $\mathbf{C}$.

Y if it contains the closed points of Y, by the theorem of zeros of Hilbert (I, 10.4.8)); this amounts again to saying that if $f^{-1}(y)$ is not proper over $k(y)$, then there exists an open neighbourhood U of y such that $f^{-1}(s)$ is not proper over $k(s)$ for every $s \in U$ (cf. (9.6.1), (iv))). This example illustrates the interest of systematically developing *criteria of constructibility*: this is what will be done in §9.

8.2. Projective limits of preschemes

(8.2.1). Let S_0 be a ringed space, L a filtered preordered increasing set, $(\mathcal{A}_\lambda, \varphi_{\mu\lambda})$ an inductive system of $\mathcal{O}_{S_0}$ -Algebras (not necessarily commutative), having L for index set. We know that, considered as an inductive system of $\mathcal{O}_{S_0}$ -Modules, $(\mathcal{A}_\lambda, \varphi_{\mu\lambda})$ admits an inductive limit $\mathcal{A}$; let us denote by $\varphi_\lambda : \mathcal{A}_\lambda \rightarrow \mathcal{A}$ the canonical homomorphism of $\mathcal{O}_{S_0}$ -Modules. Let $m_\lambda : \mathcal{A}_\lambda \otimes \mathcal{A}_\lambda \rightarrow \mathcal{A}_\lambda$ be the homomorphism of $\mathcal{O}_{S_0}$ -Modules that defines multiplication in the $\mathcal{O}_{S_0}$ -Algebra $\mathcal{A}_\lambda$; the hypothesis on the $\varphi_{\mu\lambda}$ implies that the m_λ form an inductive system of homomorphisms, and since the functor $\varinjlim$ commutes with tensor products, $m = \varinjlim m_\lambda$ is a homomorphism $\mathcal{A} \otimes \mathcal{A} \rightarrow \mathcal{A}$ of $\mathcal{O}_{S_0}$ -Modules; by passing to the limit on the commutative diagrams expressing the associativity of m_λ and the existence of the identity section in $\mathcal{A}_\lambda$, one sees that m defines on $\mathcal{A}$ a structure of $\mathcal{O}_{S_0}$ -Algebra and that φ_λ is a homomorphism of $\mathcal{O}_{S_0}$ -Algebras for every $\lambda \in L$. Furthermore, $\mathcal{A}$ is the inductive limit of the system $(\mathcal{A}_\lambda, \varphi_{\mu\lambda})$ in the category of $\mathcal{O}_{S_0}$ -Algebras, in other words, for every $\mathcal{O}_{S_0}$ -Algebra $\mathcal{B}$, the canonical map

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$$(8.2.1.1) \quad \text{Hom}_{\mathcal{O}_{S_0}\text{-Alg.}}(\mathcal{A}, \mathcal{B}) \longrightarrow \varprojlim \text{Hom}_{\mathcal{O}_{S_0}\text{-Alg.}}(\mathcal{A}_\lambda, \mathcal{B})$$

which to every homomorphism $f : \mathcal{A} \rightarrow \mathcal{B}$ of $\mathcal{O}_{S_0}$ -Algebras associates the family $(f \circ \varphi_\lambda)$, is *bijjective*. Indeed, we already know that it is injective and identifies $\text{Hom}_{\mathcal{O}_{S_0}\text{-Alg.}}(\mathcal{A}, \mathcal{B})$ with a subset of $\varprojlim \text{Hom}_{\mathcal{O}_{S_0}\text{-Mod.}}(\mathcal{A}_\lambda, \mathcal{B})$; it all comes down to seeing that if (f_λ) is an inductive system of $\mathcal{O}_{S_0}$ -Algebra homomorphisms, $f_\lambda : \mathcal{A}_\lambda \rightarrow \mathcal{B}$, its inductive limit $f : \mathcal{A} \rightarrow \mathcal{B}$, which is by definition a homomorphism of $\mathcal{O}_{S_0}$ -Modules, is also a homomorphism of $\mathcal{O}_{S_0}$ -Algebras; but this results from passage to the inductive limit in the commutative diagram of $\mathcal{O}_{S_0}$ -Modules

$$\begin{array}{ccc} \mathcal{A}_\lambda \otimes \mathcal{A}_\lambda & \xrightarrow{f_\lambda \otimes f_\lambda} & \mathcal{B} \otimes \mathcal{B} \\ m_\lambda \downarrow & & \downarrow \\ \mathcal{A}_\lambda & \xrightarrow{f_\lambda} & \mathcal{B} \end{array}$$

and from the fact that the functor $\varinjlim$ commutes with tensor products. We note finally that if the $\mathcal{A}_\lambda$ are commutative $\mathcal{O}_{S_0}$ -Algebras, the same holds for $\mathcal{A}$.

(8.2.2). Suppose now that S_0 is a *prescheme*, and that the $\mathcal{A}_\lambda$ are (commutative) *quasi-coherent* $\mathcal{O}_{S_0}$ -Algebras; we know then that $\mathcal{A} = \varinjlim \mathcal{A}_\lambda$ is a quasi-coherent $\mathcal{O}_{S_0}$ -Algebra (I, 4.1.1). Let us denote by S_λ (resp. S) the spectrum of the $\mathcal{O}_{S_0}$ -Algebra $\mathcal{A}_\lambda$ (resp. $\mathcal{A}$) (II, 1.3.1), and let $u_{\lambda\mu} : S_\mu \rightarrow S_\lambda$ (for $\lambda \leq \mu$) and $u_\lambda : S \rightarrow S_\lambda$ be the S_0 -morphisms corresponding to the homomorphisms $\varphi_{\mu\lambda}$ and φ_λ respectively (II, 1.2.7); it is clear that $(S_\lambda, u_{\lambda\mu})$ is a *projective system* in the category of S_0 -preschemes. We note that the $u_{\lambda\mu}$ and u_λ are *affine morphisms* (II, 1.6.2), hence *quasi-compact* and *separated*.

Proposition (8.2.3). — *With the notations of (8.2.2), the morphisms $u_\lambda : S \rightarrow S_\lambda$ make S a projective limit of the projective system $(S_\lambda, u_{\lambda\mu})$ in the category of preschemes. Furthermore, if $h : S_0 \rightarrow T$ is a morphism, making every S_0 -prescheme a T-prescheme, S is also a projective limit of the system $(S_\lambda, u_{\lambda\mu})$ in the category of T-preschemes.*

Proof. Let us first prove the second assertion of the statement in the case $T = S_0$. It all comes down to proving that if X is any S_0 -prescheme, the canonical map

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$$(8.2.3.1) \quad \text{Hom}_{S_0}(X, S) \longrightarrow \varprojlim \text{Hom}_{S_0}(X, S_\lambda)$$

which to every S_0 -morphism $v : X \rightarrow S$ associates the family $(u_\lambda \circ v)$, is *bijjective*. Now, if $g : X \rightarrow S_0$ is the structural morphism and if we set $\mathcal{B} = g_*(\mathcal{O}_X)$, which is an $\mathcal{O}_{S_0}$ -Algebra, the map (8.2.3.1) identifies canonically with (8.2.1.1) (II, 1.2.7), and the conclusion follows from what we have seen in (8.2.1).

The other assertions of (8.2.3) are consequences of the following general lemma: □

Lemma (8.2.4). — *Let $\mathbf{C}$ be a category, T an object of $\mathbf{C}$, $\mathbf{C}_T$ the subcategory of objects of $\mathbf{C}$ over T . Let $(S_\lambda, u_{\lambda\mu})$ be a projective system in $\mathbf{C}_T$; then every projective limit of this system in $\mathbf{C}_T$ is also a projective limit in $\mathbf{C}$, and conversely.*

Proof. Let $f_\lambda : S_\lambda \rightarrow T$ be the structural morphism. Suppose that S is a projective limit of $(S_\lambda, u_{\lambda\mu})$ in $\mathbf{C}$, and let us denote by $u_\lambda : S \rightarrow S_\lambda$ the canonical corresponding morphisms. Consider a projective system of T -morphisms (in $\mathbf{C}$) $w_\lambda : Y \rightarrow S_\lambda$, where $Y \in \mathbf{C}_T$. There exists by hypothesis a unique morphism (in $\mathbf{C}$) $w : Y \rightarrow S$ such that $w_\lambda = u_\lambda \circ w$ for every λ . The hypothesis that the u_λ are T -morphisms implies that the morphisms $f_\lambda \circ u_\lambda : S \rightarrow T$ are all the same, and this composite f makes S a T -object. If $g : Y \rightarrow T$ is the structural morphism of Y , one has $f \circ w = f_\lambda \circ u_\lambda \circ w = f_\lambda \circ w_\lambda = g$ for every λ , which proves that w is a T -morphism. Conversely, suppose (with the same notations) that S is a projective limit of $(S_\lambda, u_{\lambda\mu})$ in $\mathbf{C}_T$, and consider now a projective system of morphisms (in $\mathbf{C}$) $w_\lambda : Y \rightarrow S_\lambda$. The composites $f_\lambda \circ w_\lambda : Y \rightarrow T$ are then all the same: indeed, for any two indices λ, μ , there is an index ν such that $\lambda \leq \nu$ and $\mu \leq \nu$, whence $f_\nu = f_\lambda \circ u_{\lambda\nu} = f_\mu \circ u_{\mu\nu}$; since $w_\lambda = u_{\lambda\nu} \circ w_\nu$ and $w_\mu = u_{\mu\nu} \circ w_\nu$, one has $f_\lambda \circ w_\lambda = f_\lambda \circ u_{\lambda\nu} \circ w_\nu = f_\nu \circ w_\nu$ and $f_\mu \circ w_\mu = f_\mu \circ u_{\mu\nu} \circ w_\nu = f_\nu \circ w_\nu$. If $g : Y \rightarrow T$ is the unique morphism thus defined, g makes Y a T -object, and the w_λ are T -morphisms; they hence have a projective limit $w : Y \rightarrow S$ which is a T -morphism, and *a fortiori* a morphism of $\mathbf{C}$; moreover, the first part of the argument shows that any projective limit w' (in $\mathbf{C}$) of the projective system (w_λ) is also a T -morphism, and as $w' : Y \rightarrow S'$ in $\mathbf{C}_T$ gives $p \circ w' = w$ and $p_\lambda \circ w_\lambda = p_\lambda \circ w'_\lambda$, the definition of S'_λ as a fibre product $S_\lambda \times_T T'$ gives $u'_\lambda \circ w' = w'_\lambda$, and it is immediate that w' is the unique T' -morphism satisfying these relations, whence the lemma. $\square$

Proposition (8.2.5). — *Under the conditions of (8.2.2), let α be an element of L , X_α an S_α -prescheme. For every $\lambda \geq \alpha$, set $X_\lambda = X_\alpha \times_{S_\alpha} S_\lambda$, and for $\alpha \leq \lambda \leq \mu$, set $v_{\lambda\mu} = 1_{X_\alpha} \times u_{\lambda\mu}$, so that $(X_\lambda, v_{\lambda\mu})$ is a projective system of X_α -preschemes, whose index set is formed of the $\lambda \geq \alpha$ in L . Set likewise $X = X_\alpha \times_{S_\alpha} S$ and $v_\lambda = 1_{X_\alpha} \times u_\lambda$. Then the X_α -morphisms $v_\lambda : X \rightarrow X_\lambda$ make X a projective limit of the projective system $(X_\lambda, v_{\lambda\mu})$ in the category of X_α -preschemes, or in the category of all preschemes.*

Proof. This will also follow from the following general lemma: $\square$

Lemma (8.2.6). — *Let $\mathbf{C}$ be a category in which fibre products exist, $q : T' \rightarrow T$ a morphism of $\mathbf{C}$, $\mathbf{C}_T$ (resp. $\mathbf{C}_{T'}$) the category of objects of $\mathbf{C}$ over T (resp. T'). Let $(S_\lambda, u_{\lambda\mu})$ be a projective system (not necessarily filtered) in $\mathbf{C}_T$, and set $S'_\lambda = S_\lambda \times_T T'$, $u'_{\lambda\mu} = u_{\lambda\mu} \times 1_{T'}$, so that $(S'_\lambda, u'_{\lambda\mu})$ is a projective system in $\mathbf{C}_{T'}$. Then, if (S, u_λ) is a projective limit of $(S_\lambda, u_{\lambda\mu})$ in $\mathbf{C}_T$, $(S \times_T T', u_\lambda \times 1_{T'})$ is a projective limit of $(S'_\lambda, u'_{\lambda\mu})$ in $\mathbf{C}_{T'}$.*

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Proof. By hypothesis, for every λ , one has a commutative diagram

$$\begin{array}{ccccc} S' & \xrightarrow{u'_\lambda} & S'_\lambda & \xrightarrow{h'_\lambda} & T' \\ p \downarrow & & & & \downarrow q \\ S & \xrightarrow{u_\lambda} & S_\lambda & \xrightarrow{h_\lambda} & T \end{array}$$

where we have set $S' = S \times_T T'$, $u'_\lambda = u_\lambda \times 1_{T'}$, $h'_\lambda = h_\lambda \times 1_{T'}$. Let Y be a T' -object, $g' : Y \rightarrow T'$, and consider a projective system of T' -morphisms $w'_\lambda : Y \rightarrow S'_\lambda$. Then Y is a T -object for the morphism $g = q \circ g'$, and the $w_\lambda = p_\lambda \circ w'_\lambda$ are T -morphisms since $h_\lambda \circ w_\lambda = h_\lambda \circ p_\lambda \circ w'_\lambda = q \circ h'_\lambda \circ w'_\lambda = q \circ g'$ by hypothesis. Furthermore, one verifies immediately that (w_λ) is a projective system. There thus exists by hypothesis a unique T -morphism $w : Y \rightarrow S$ such that $u_\lambda \circ w = w_\lambda$ for every λ . By the definition of the fibre product, there is a unique T' -morphism $w' : Y \rightarrow S'$ such that $p \circ w' = w$. Then $u'_\lambda \circ w' = u_\lambda \circ w' = w_\lambda = p_\lambda \circ w'_\lambda$, and since $h'_\lambda \circ u'_\lambda \circ w' = g' = h'_\lambda \circ w'_\lambda$, the definition of S'_λ as the fibre product $S_\lambda \times_T T'$ gives $u'_\lambda \circ w' = w'_\lambda$, and it is immediate that w' is the unique T' -morphism satisfying these relations, whence the lemma. $\square$

Remark (8.2.7). — Given a ringed space S , the inductive limits with respect to a preordered set L (not necessarily filtered) exist in the category of $\mathcal{O}_S$ -Algebras, commutative or not, since the inductive limit exists by (8.2.1) and, on the other hand, for any two homomorphisms of $\mathcal{O}_S$ -Algebras $\mathcal{A} \rightarrow \mathcal{B}$, $\mathcal{A} \rightarrow \mathcal{C}$, the tensor product $\mathcal{B} \otimes_{\mathcal{A}} \mathcal{C}$ is the corresponding “amalgamated sum” in this category.

When S is a prescheme, we say that the tensor product $\mathcal{B} \otimes_{\mathcal{A}} \mathcal{C}$ is a quasi-coherent $\mathcal{O}_S$ -Algebra when this is so for $\mathcal{A}$, $\mathcal{B}$, and $\mathcal{C}$ (I, 1.3.13); from this one concludes that, in the category of quasi-coherent $\mathcal{O}_S$ -Algebras, inductive limits for any preordered index set always exist. This allows us to generalise the definition of a projective limit of preschemes and propositions (8.2.3) and (8.2.5) to the case where the preordered set L is not necessarily filtered.

(8.2.8). The notations being those of (8.2.2), set $u_{\lambda\mu} = (\psi_{\lambda\mu}, \theta_{\lambda\mu})$ and $u_{\lambda} = (\psi_{\lambda}, \theta_{\lambda})$; $(S_{\lambda}, \psi_{\lambda\mu})$ is thus a projective system of topological spaces and (ψ_{λ}) a projective system of continuous maps $S \rightarrow S_{\lambda}$.

Proposition (8.2.9). — *With the notations of (8.2.8), the projective limit of the projective system (ψ_{λ}) of continuous maps is a homeomorphism of the underlying space of S onto the projective limit of the projective system $(S_{\lambda}, \psi_{\lambda\mu})$ of topological spaces.*

Proof. Let T be the topological limit space of the system $(S_{\lambda}, \psi_{\lambda\mu})$ and set $\psi = \varprojlim \psi_{\lambda} : S \rightarrow T$. We can restrict to the case where $S_0 = S_{\alpha}$ for some $\alpha \in L$, and $\lambda \geq \alpha$. IV-3 | 11

(i) Let us show first that the topology of S is the inverse image by ψ of the topology of T ; in other words, if $\pi_{\lambda} : T \rightarrow S_{\lambda}$ is the canonical map, it amounts to showing that every open subset of S is a union of open subsets of the form $\psi^{-1}(\pi_{\lambda}^{-1}(U_{\lambda}))$, where U_{λ} is open in S_{λ} . Now, every open subset of S is by definition a union of open subsets obtained in the following way: one considers an open affine U_0 of S_0 , of ring A_0 , so that $\psi_{\alpha}^{-1}(U_0)$ is the open affine subset of S of ring $A = \Gamma(U_0, \mathcal{A})$, then one takes an element $f \in A$ and considers $\psi_{\alpha}^{-1}(U_0)$, i.e., the open set $D(f)$. These are the open subsets $D(f)$ which form a base of the topology of S (II, 1.3.1). Now, if one sets $A_{\lambda} = \Gamma(U_0, \mathcal{A}_{\lambda})$, one has $A = \varinjlim A_{\lambda}$ (I, 1.3.9), hence there exists an index λ such that f is the canonical image of an element $f_{\lambda} \in A_{\lambda}$; one has then $D(f) = \psi_{\lambda}^{-1}(D(f_{\lambda}))$ (I, 1.2.2), and since $\psi_{\lambda} = \pi_{\lambda} \circ \psi$, our assertion is proved.

(ii) Let us now show that ψ is *bijective*, which will complete the proof. Since S is a Kolmogoroff space, it already follows from (i) that ψ is injective, and it remains to show that ψ is surjective. We can evidently replace T by an open subset $\pi_{\alpha}^{-1}(U_0)$, where U_0 is an open affine in $S_{\alpha} = S_0$, hence we can restrict to the case where the S_{λ} and S are affine, in other words $\mathcal{A}_{\lambda}$ is the sheaf of algebras associated to an A_0 -algebra A_{λ} , and $\mathcal{A}$ the sheaf of algebras associated to $A = \varinjlim A_{\lambda}$; we will still denote $\varphi_{\mu\lambda} : A_{\lambda} \rightarrow A_{\mu}$ and $\varphi_{\lambda} : A_{\lambda} \rightarrow A$ the canonical homomorphisms. By definition, an element of T is a family $(\mathfrak{p}_{\lambda})_{\lambda \in L}$, where $\mathfrak{p}_{\lambda}$ is a prime ideal of A_{λ} and where $\mathfrak{p}_{\lambda} = \varphi_{\mu\lambda}^{-1}(\mathfrak{p}_{\mu})$ for $\lambda \leq \mu$. We know ((0, 5.13.3) and (5.13.1)) that there exists a prime ideal $\mathfrak{p}$ of A such that $\mathfrak{p}_{\lambda} = \varphi_{\lambda}^{-1}(\mathfrak{p})$ for every $\lambda \in L$, which completes the proof. $\square$

In particular, we have thus proved the

Corollary (8.2.10). — *Let $(A_{\lambda})_{\lambda \in L}$ be a filtered inductive system of rings, and let $\varphi_{\lambda} : A_{\lambda} \rightarrow A$ be the canonical homomorphisms. The canonical map $\mathfrak{p} \mapsto (\varphi_{\lambda}^{-1}(\mathfrak{p}))$ is a homeomorphism of $\text{Spec}(A)$ onto the topological space $\varprojlim \text{Spec}(A_{\lambda})$.*

Corollary (8.2.11). — *With the notations of (8.2.8), for every quasi-compact open subset U of S , there exist an index λ and a quasi-compact open subset U_{λ} of S_{λ} such that $U = \psi_{\lambda}^{-1}(U_{\lambda})$.*

Proof. This follows from the fact that, by definition of the projective limit topology, the $\psi_{\lambda}^{-1}(U_{\lambda})$ (U_{λ} a quasi-compact open subset of S_{λ}) form a base of the topology of S , and from the fact that the index set L is filtered. $\square$

Corollary (8.2.12). — *With the notations of (8.2.8), the inductive limit of the inductive system of homomorphisms $\theta_{\lambda}^* : \psi_{\lambda}^*(\mathcal{O}_{S_{\lambda}}) \rightarrow \mathcal{O}_S$ of sheaves of rings on S is an isomorphism*

$$(8.2.12.1) \quad \varinjlim_{\lambda} \psi_{\lambda}^*(\mathcal{O}_{S_{\lambda}}) \xrightarrow{\sim} \mathcal{O}_S.$$

Proof. We can evidently suppose the S_{λ} affine; with the notations of the proof of (8.2.9), it all comes down to seeing that the inductive limit of the inductive system of canonical maps $(A_{\lambda})_{\mathfrak{p}_{\lambda}} \rightarrow A_{\mathfrak{p}}$ is an isomorphism, which is nothing other than (0, 5.13.3), (ii). $\square$

Proposition (8.2.13). — Suppose that the morphisms $u_{\lambda\mu} : S_\mu \rightarrow S_\lambda$ are open immersions, so that S_μ identifies with an open subset of S_λ for $\lambda \leq \mu$. Then, for every $\alpha \in L$, the underlying space of the prescheme S identifies with the intersection of the S_λ for $\lambda \geq \alpha$, and the structural sheaf $\mathcal{O}_S$ of the sheaf induced (G, II, 1.5) by $\mathcal{O}_{S_\alpha}$ on this intersection; more generally, for every $\mathcal{O}_{S_\alpha}$ -Module $\mathcal{F}_\alpha$, $u_\alpha^*(\mathcal{F}_\alpha)$ identifies with the $\mathcal{O}_S$ -Module induced by $\mathcal{F}_\alpha$ on S .

Proof. The first assertion follows from (8.2.9), in view of the definition of a projective limit of topological spaces; furthermore, all the $\psi_\lambda^*(\mathcal{O}_{S_\lambda})$ are equal to the sheaf induced by $\mathcal{O}_{S_\alpha}$ on S by definition (0, 1.3.7.1), and with the notations of the proof of (8.2.9), the homomorphisms $(A_\lambda)_{\mathfrak{p}_\lambda} \rightarrow (A_\mu)_{\mathfrak{p}_\mu}$ are bijective for a system $(\mathfrak{p}_\lambda)$ of ideals of primes corresponding to a single point of S ; the assertion relative to $\mathcal{O}_S$ is then deduced from (8.2.12). The last assertion then follows from the definition of $u_\alpha^*(\mathcal{F}_\alpha)$ (0, 1.4.3.1). $\square$

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Remark (8.2.14). — The results of (8.2.9) and (8.2.12) show that S is a projective limit of the projective system $(S_\lambda, u_{\lambda\mu})$ in the category of all ringed spaces (or of all ringed spaces with a local ringed space structure). Indeed, let Y be a ringed space, and consider a projective system of morphisms of ringed spaces $w_\lambda : Y \rightarrow S_\lambda$. If one sets $w_\lambda = (\rho_\lambda, \omega_\lambda)$, the ρ_λ form a projective system of continuous maps and $\rho = \varprojlim \rho_\lambda$ a continuous map $Y \rightarrow S$, by virtue of (8.2.9) and (8.2.12). The unique homomorphism $\omega^\# : \rho^*(\mathcal{O}_S) \rightarrow \mathcal{O}_Y$ such that $\omega_\lambda^\# = \omega^\# \circ \rho^*(\theta_\lambda^*)$ for every λ , and there is thus a unique morphism of ringed spaces $w : Y \rightarrow S$ satisfying these relations.

8.3. Constructible subsets of a projective limit of preschemes

(8.3.1). In what follows throughout this paragraph, we suppose the conditions of (8.2.2) are satisfied, and we keep the notations.

Theorem (8.3.2). — For every λ , let E_λ, F_λ be two subsets of S_λ . Set

$$(8.3.2.1) \quad E = \bigcap_{\lambda} u_\lambda^{-1}(E_\lambda), \quad F = \bigcup_{\lambda} u_\lambda^{-1}(F_\lambda).$$

Suppose the following conditions are satisfied:

- (i) For every λ , E_λ is pro-constructible and F_λ is ind-constructible (1.9.4).
- (ii) For $\lambda \leq \mu$, we have

$$E_\mu \subset u_{\lambda\mu}^{-1}(E_\lambda), \quad F_\mu \supset u_{\lambda\mu}^{-1}(F_\lambda).$$

- (iii) There exists $\alpha \in L$ such that S_α is quasi-compact (which implies that S_λ is quasi-compact for $\lambda \geq \alpha$).

Then the following properties are equivalent:

- a) $E \subset F$.
- b) There exists $\lambda \geq \alpha$ such that $u_\lambda^{-1}(E_\lambda) \subset u_\lambda^{-1}(F_\lambda)$ (and then we also have $u_\mu^{-1}(E_\mu) \subset u_\mu^{-1}(F_\mu)$ for $\mu \geq \lambda$).
- c) There exists $\lambda \geq \alpha$ such that $E_\lambda \subset F_\lambda$ (and then we also have $E_\mu \subset F_\mu$ for $\mu \geq \lambda$).

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The parenthetical remarks in b) and c) follow from (ii). Set

$$G_\lambda = E_\lambda \cap (S_\lambda - F_\lambda), \quad G = E \cap (S - F).$$

Then G_λ is a pro-constructible subset of S_λ (1.9.5), (i), and by virtue of (8.3.2.1) we have

$$G_\mu \subset u_{\lambda\mu}^{-1}(G_\lambda) \quad \text{for } \lambda \leq \mu$$

$$G = \bigcap_{\lambda} u_\lambda^{-1}(G_\lambda).$$

We are thus reduced to proving the particular case of (8.3.2) corresponding to $F_\lambda = \emptyset$ for every λ :

Corollary (8.3.3). — For every λ , let E_λ be a pro-constructible subset of S_λ such that $E_\mu \subset u_{\lambda\mu}^{-1}(E_\lambda)$. Suppose that there exists $\alpha \in L$ such that S_α is quasi-compact. Then the following conditions are equivalent:

- a) $E = \bigcap_{\lambda} u_\lambda^{-1}(E_\lambda) = \emptyset$.
- b) There exists λ such that $u_\lambda^{-1}(E_\lambda) = \emptyset$ (and then $u_\mu^{-1}(E_\mu) = \emptyset$ for $\mu \geq \lambda$).
- c) There exists λ such that $E_\lambda = \emptyset$ (and then $E_\mu = \emptyset$ for $\mu \geq \lambda$).

Proof. It is clear that *c*) implies *a*). Let us prove that *a*) implies *b*): since S_λ is quasi-compact, E_λ , being pro-constructible, is also quasi-compact, and it is the same for $u_\lambda^{-1}(E_\lambda)$ (1.9.5), (vi); then the filtered decreasing family of pro-constructible subsets $u_\lambda^{-1}(E_\lambda)$ has empty intersection, hence (1.9.9) one of them is empty.

Finally, let us show that *b*) implies *c*). Since S_α is quasi-compact and L is filtered, we can replace S_α by an open affine, hence suppose (8.2.2) that S and the S_λ for $\lambda \geq \alpha$ are affine; one has then (1.9.2.1), for $\lambda \geq \alpha$,

$$u_\lambda(S) = \bigcap_{\mu \geq \lambda} u_{\lambda\mu}(S_\mu)$$

whence

$$E_\lambda \cap u_\lambda(S) = \bigcap_{\mu \geq \lambda} (E_\lambda \cap u_{\lambda\mu}(S_\mu)).$$

Now, since u_λ and the $u_{\lambda\mu}$ are quasi-compact, $u_\lambda(S)$ and $u_{\lambda\mu}(S_\mu)$ are pro-constructible in S_λ (1.9.5), (vii), hence the sets $E_\lambda \cap u_{\lambda\mu}(S_\mu)$ for $\mu \geq \lambda$ form a filtered decreasing family of pro-constructible subsets of S_λ . Since S_λ is quasi-compact, the hypothesis *b*) implies that the intersection of this family is empty, hence (1.9.9) one of the sets $E_\lambda \cap u_{\lambda\mu}(S_\mu)$ is empty, hence $E_\mu \subset u_{\lambda\mu}^{-1}(E_\lambda)$ is empty. $\square$

Corollary (8.3.4). — For every λ , let F_λ be an ind-constructible subset of S_λ such that for $\lambda \leq \mu$ one has $u_{\lambda\mu}^{-1}(F_\lambda) \subset F_\mu$. Suppose that there exists $\alpha \in L$ such that S_α is quasi-compact. Then the following conditions are equivalent:

- a) The set $F = \bigcup_\lambda u_\lambda^{-1}(F_\lambda)$ is equal to S .
- b) There exists λ such that $u_\lambda^{-1}(F_\lambda) = S$ (and then $u_\mu^{-1}(F_\mu) = S$ for $\mu \geq \lambda$).
- c) There exists λ such that $F_\lambda = S_\lambda$ (and then $F_\mu = S_\mu$ for $\mu \geq \lambda$).

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This follows immediately from (8.3.3) by passing to complements.

Corollary (8.3.5). — For every λ , let E_λ, F_λ be two constructible subsets of S_λ such that, for $\mu \geq \lambda$, one has $E_\mu \subset u_{\lambda\mu}^{-1}(E_\lambda)$ and $F_\mu \supset u_{\lambda\mu}^{-1}(F_\lambda)$. Suppose that there exists α such that S_α is quasi-compact. Then, for $E \subset F$ (resp. $E = F$), it is necessary and sufficient that there exist λ such that $E_\lambda \subset F_\lambda$ (resp. $E_\lambda = F_\lambda$), in which case one also has $E_\mu \subset F_\mu$ (resp. $E_\mu = F_\mu$) for $\mu \geq \lambda$.

This is a particular case of (8.3.2).

Corollary (8.3.6). — Suppose that there exists α such that S_α is quasi-compact. For $S = \emptyset$, it is necessary and sufficient that there exist λ such that $S_\lambda = \emptyset$.

Corollary (8.3.7). — On *a*, for every λ ,

$$(8.3.7.1) \quad u_\lambda(S) = \bigcap_{\mu \geq \lambda} u_{\lambda\mu}(S_\mu).$$

Proof. It is clear that the first member of (8.3.7.1) is contained in the second. Let s be a point of S_λ and set $X_\lambda = \text{Spec}(k(s))$; consider the projective system $(X_\mu, v_{\mu\nu})$ where $X_\mu = X_\lambda \times_{S_\lambda} S_\mu$ and $v_{\mu\nu} = 1 \times u_{\mu\nu}$ for $\lambda \leq \mu \leq \nu$; its projective limit is $X = X_\lambda \times_{S_\lambda} S$ (8.2.5). If $s \in \bigcap_{\mu \geq \lambda} u_{\lambda\mu}(S_\mu)$, this implies that $X_\mu \neq \emptyset$ for every $\mu \geq \lambda$ (I, 3.4.8); it follows then from (8.3.6) that $X \neq \emptyset$, hence $s \in u_\lambda(S)$ by (I, 3.4.8). $\square$

Proposition (8.3.8). — (i) For the morphism $u_\lambda : S \rightarrow S_\lambda$ to be dominant (resp. surjective), it is necessary and sufficient that for $\mu \geq \lambda$ the morphism $u_{\lambda\mu} : S_\mu \rightarrow S_\lambda$ be dominant (resp. surjective).
(ii) If, for some index λ , the morphisms $u_{\lambda\mu} : S_\mu \rightarrow S_\lambda$ are flat (resp. faithfully flat), then for every $\mu \geq \lambda$, the morphism $u_\lambda : S \rightarrow S_\lambda$ is flat (resp. faithfully flat).
(iii) Suppose that the morphisms $u_\mu : S \rightarrow S_\mu$ are surjective for $\mu \geq \lambda$. For u_λ to be an open morphism (resp. universally open), it is necessary and sufficient that, for every $\mu \geq \lambda$, $u_{\lambda\mu}$ be an open morphism (resp. universally open).

Proof. (i) Since $u_\lambda(S) \subset u_{\lambda\mu}(S_\mu)$ for every $\mu \geq \lambda$, the necessity of the conditions is trivial, and it follows immediately from (8.3.7.1) that if the $u_{\lambda\mu}$ are surjective, then so is u_λ . Suppose now that the $u_{\lambda\mu}$ are dominant for $\mu \geq \lambda$, and consider in S_λ an open quasi-compact non-empty

subset U_λ ; then the $U_\mu = u_{\lambda\mu}^{-1}(U_\lambda)$ for $\mu \geq \lambda$ form a projective system whose projective limit is $U = u_\lambda^{-1}(U_\lambda)$ (8.2.5). By hypothesis the U_μ are all non-empty, hence it is the same for U by (8.3.6), which proves that u_λ is dominant.

- (ii) By virtue of (i), it suffices to consider the case where the $u_{\lambda\mu}$ are flat; one can then restrict to the case where S_λ is affine, hence also the S_μ and S , and the assertion follows from (2.1.2) and (0, 6.2.3).
- (iii) By virtue of (8.2.5) and of (I, 3.5.2), it suffices to treat the case of open morphisms. Since $u_\lambda = u_{\lambda\mu} \circ u_\mu$ and u_μ is surjective, we know that if u_λ is open then so is $u_{\lambda\mu}$ (Bourbaki, *Top. g  n.*, chap. I^{er}, 3^e ed., §5, n^o 1, prop. 1). Conversely, to show that u_λ is open when all the $u_{\lambda\mu}$ are open for $\mu \geq \lambda$, it suffices to see that for every quasi-compact open subset V of S , $u_\lambda(V)$ is open in S_λ ; but there exists then $\mu \geq \lambda$ such that $V = u_\mu^{-1}(V_\mu)$, where V_μ is open in S_μ (8.2.11); since u_μ is surjective, one has $V_\mu = u_\mu(V)$ and $u_\lambda(V) = u_{\lambda\mu}(u_\mu(V))$ is therefore open by hypothesis.

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One notes that we can arrange for u_λ to be open without the $u_{\lambda\mu}$ being open and without u_λ being surjective either. Indeed, one has an example by considering an integral ring A which is not a field, and its field of fractions K , which is the inductive limit of the A_f , where f ranges over $A - \{0\}$; setting $S_1 = \text{Spec}(A)$, $S_f = \text{Spec}(A_f)$, and $S = \varinjlim S_f = \text{Spec}(K)$, the morphism $S \rightarrow S_1$ is not open, even though the morphisms $S_f \rightarrow S_1$ are open. $\square$

(8.3.9). For every prescheme X , we will denote as usual by $\mathfrak{P}(X)$ the set of all subsets of the underlying set of X , by $\mathfrak{C}(X)$ (resp. $\mathfrak{Dc}(X)$, $\mathfrak{F}\mathfrak{C}(X)$, $\mathfrak{L}\mathfrak{F}\mathfrak{C}(X)$) the set of constructible (resp. constructible and open, constructible and closed, and locally closed) subsets of X . It is clear that $(\mathfrak{P}(S_\lambda), u_{\lambda\mu}^{-1})$ is an *inductive system of sets* and that the maps $u_\lambda^{-1} : \mathfrak{P}(S_\lambda) \rightarrow \mathfrak{P}(S)$ form an inductive system of maps, whence, by passing to the inductive limit, a canonical map

$$(8.3.9.1) \quad u_{\mathfrak{P}} : \varinjlim \mathfrak{P}(S_\lambda) \longrightarrow \mathfrak{P}(S).$$

Furthermore, it follows from (1.8.2) that $u_{\lambda\mu}^{-1}$ sends $\mathfrak{C}(S_\lambda)$ (resp. $\mathfrak{Dc}(S_\lambda)$, $\mathfrak{F}\mathfrak{C}(S_\lambda)$, $\mathfrak{L}\mathfrak{F}\mathfrak{C}(S_\lambda)$) into $\mathfrak{C}(S_\mu)$ (resp. $\mathfrak{Dc}(S_\mu)$, $\mathfrak{F}\mathfrak{C}(S_\mu)$, $\mathfrak{L}\mathfrak{F}\mathfrak{C}(S_\mu)$) and that u_λ^{-1} sends $\mathfrak{C}(S_\lambda)$ (resp. $\mathfrak{Dc}(S_\lambda)$, $\mathfrak{F}\mathfrak{C}(S_\lambda)$, $\mathfrak{L}\mathfrak{F}\mathfrak{C}(S_\lambda)$) into $\mathfrak{C}(S)$ (resp. $\mathfrak{Dc}(S)$, $\mathfrak{F}\mathfrak{C}(S)$, $\mathfrak{L}\mathfrak{F}\mathfrak{C}(S)$). We thus have, by restriction of (8.3.9.1), canonical maps

$$(8.3.9.2) \quad \varinjlim \mathfrak{C}(S_\lambda) \longrightarrow \mathfrak{C}(S)$$

$$(8.3.9.3) \quad \varinjlim \mathfrak{Dc}(S_\lambda) \longrightarrow \mathfrak{Dc}(S)$$

$$(8.3.9.4) \quad \varinjlim \mathfrak{F}\mathfrak{C}(S_\lambda) \longrightarrow \mathfrak{F}\mathfrak{C}(S)$$

$$(8.3.9.5) \quad \varinjlim \mathfrak{L}\mathfrak{F}\mathfrak{C}(S_\lambda) \longrightarrow \mathfrak{L}\mathfrak{F}\mathfrak{C}(S).$$

(8.3.10). Let $g_\alpha : X_\alpha \rightarrow S_\alpha$ be a morphism; with the notations of (8.2.5) one has, on the one hand, a canonical map $v_{\mathfrak{P}} : \varinjlim \mathfrak{P}(S_\lambda) \rightarrow \mathfrak{P}(X)$; on the other hand, from the morphisms of projection $g_\lambda : X_\lambda \rightarrow S_\lambda$ for every $\lambda \geq \alpha$ and a morphism of projection $g : X \rightarrow S$. It is clear that the $g_\lambda^{-1} : \mathfrak{P}(S_\lambda) \rightarrow \mathfrak{P}(X_\lambda)$ form an inductive system of maps, and that the diagrams

$$\begin{array}{ccc} \mathfrak{P}(S_\lambda) & \xrightarrow{u_\lambda^{-1}} & \mathfrak{P}(S) \\ g_\lambda^{-1} \downarrow & & \downarrow g^{-1} \\ \mathfrak{P}(X_\lambda) & \xrightarrow{v_\lambda^{-1}} & \mathfrak{P}(X) \end{array}$$

are commutative; one deduces from this by passing to the inductive limit a commutative diagram

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$$\begin{array}{ccc} \varinjlim \mathfrak{P}(S_\lambda) & \xrightarrow{u_{\mathfrak{P}}} & \mathfrak{P}(S) \\ \downarrow & & \downarrow g^{-1} \\ \varinjlim \mathfrak{P}(X_\lambda) & \xrightarrow{v_{\mathfrak{P}}} & \mathfrak{P}(X) \end{array}$$

and it follows from (1.8.2) that one has analogous diagrams replacing $\mathfrak{P}$ by $\mathfrak{C}$, $\mathfrak{Dc}$, $\mathfrak{F}\mathfrak{C}$, or $\mathfrak{L}\mathfrak{F}\mathfrak{C}$.

(8.3.10.1)

It follows from (8.3.5) that under the hypothesis that for some $\alpha \in L$, S_α is *quasi-compact*, the canonical map (8.3.9.2) is *injective*. Furthermore:

Theorem (8.3.11). — *Suppose that there exists α such that S_α is quasi-compact and quasi-separated. Then the canonical maps (8.3.9.2), (8.3.9.3), (8.3.9.4), and (8.3.9.5) are bijective.*

Proof. By virtue of the preceding remark, it remains to prove that these maps are *surjective*; as every constructible subset of S is a finite union of sets of the form $U \cap \mathbb{C}V$, where U and V are open and constructible, it will suffice to prove that (8.3.9.3) is surjective for it to be the same for (8.3.9.2) (and also for (8.3.9.4), by passing to complements). Now, since the morphisms $S_\lambda \rightarrow S_\alpha$ are affine and separated, the S_λ for $\lambda \geq \alpha$ and S are quasi-compact and quasi-separated, and one knows that the constructible open subsets of such a prescheme are none other than the quasi-compact open subsets (1.8.1). The conclusion then follows from (8.2.11) except for the map (8.3.9.5). To prove that this last map is surjective, consider a locally closed and constructible subset Z of S ; Z is therefore quasi-compact (0_{III}, 9.1.4). As every point $x \in Z$ admits by hypothesis an open quasi-compact neighbourhood V_x in S such that $Z \cap V_x$ is closed in V_x ; we can cover Z by a finite number of the V_x , hence there is a quasi-compact open subset U containing Z such that Z is closed in U ; in other words, Z is also constructible in U (0_{III}, 9.1.8). We know (8.2.11) that there exist λ and a quasi-compact open subset U_λ of S_λ such that $U = u_\lambda^{-1}(U_\lambda)$. Applying to U (which is a projective limit of $U_\lambda = u_{\lambda\mu}^{-1}(U_\lambda)$) the map (8.3.9.4), one sees that there exist $\mu \geq \lambda$ and a closed constructible subset Z_μ in U_μ such that $Z = u_\mu^{-1}(Z_\mu)$. But since the canonical immersion $U_\mu \rightarrow S_\mu$ is quasi-compact by hypothesis (1.2.7), she is of finite presentation (1.6.2), (i), and Z_μ is also a constructible subset of S_μ by virtue of (1.8.4) and (1.8.1); since Z_μ is evidently locally closed in S_μ , this completes the proof. $\square$

Corollary (8.3.12). — *Suppose that there exists α such that S_α is quasi-compact, and let, for every λ , Z_λ be a constructible subset of S_λ such that $Z_\mu = u_{\lambda\mu}^{-1}(Z_\lambda)$ for $\mu \geq \lambda$. If $Z = u_\lambda^{-1}(Z_\lambda)$, then, for Z to be open (resp. closed, resp. locally closed) in S , it is necessary and sufficient that there exist $\lambda \geq \alpha$ such that Z_λ be so in S_λ .*

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Proof. Cover S_α by a finite number of open affine subsets $U_\alpha^{(i)}$; then the $U^{(i)} = u_\alpha^{-1}(U_\alpha^{(i)})$ form an open covering of S , and for Z to be open (resp. closed, resp. locally closed) in S , it is necessary and sufficient that each of the $Z \cap U^{(i)}$ be so in $U^{(i)}$. Since L is filtered and each $Z \cap U^{(i)}$ is constructible in $U^{(i)}$ (0_{III}, 9.1.8), one can restrict to proving the corollary when S_α is affine, hence quasi-compact and quasi-separated; but then it follows from (8.3.11). $\square$

Proposition (8.3.13). — *Suppose that the morphisms $u_{\lambda\mu} : S_\mu \rightarrow S_\lambda$ are flat and that there exists α such that S_α is quasi-compact. For every λ , let Z'_λ, Z''_λ be two pro-constructible subsets of S_λ , such that $Z'_\mu = u_{\lambda\mu}^{-1}(Z'_\lambda)$ and $Z''_\mu = u_{\lambda\mu}^{-1}(Z''_\lambda)$ for $\mu \geq \lambda$; suppose further that $\overline{Z'_\alpha}$ is constructible in S_α . Set $Z' = u_\lambda^{-1}(Z'_\lambda)$, $Z'' = u_\lambda^{-1}(Z''_\lambda)$; for $Z'' \subset \overline{Z'}$, it is necessary and sufficient that there exist $\lambda \geq \alpha$ such that $Z''_\lambda \subset \overline{Z'_\lambda}$.*

Proof. Indeed, one knows that $u_\lambda : S \rightarrow S_\lambda$ is also a flat morphism for every λ (8.3.8), (i); as Z'_λ is pro-constructible, the closure of Z'_μ for $\mu \geq \lambda$ (resp. of Z') in S_μ (resp. S) is equal to $u_{\lambda\mu}^{-1}(\overline{Z'_\lambda})$ (resp. $u_\lambda^{-1}(\overline{Z'_\lambda})$) (2.3.10). Since the $u_{\lambda\mu}^{-1}(\overline{Z'_\lambda})$ and $u_\lambda^{-1}(\overline{Z'_\lambda})$ are constructible (1.8.2), the conclusion follows from (8.3.2). $\square$

8.4. Criteria of irreducibility and connectedness for projective limits of preschemes

Proposition (8.4.1). — *Suppose that there exists an index α such that S_α is quasi-compact.*

- (i) *If S is not irreducible and if in addition the underlying space of S is Noetherian and S_α is quasi-separated, there exists $\lambda \geq \alpha$ such that, for $\mu \geq \lambda$, S_μ is not irreducible.*
- (ii) *If S is not connected, there exists $\lambda \geq \alpha$ such that, for $\mu \geq \lambda$, S_μ is not connected.*

Proof. Suppose that S is a union of two distinct closed subsets S', S'' of S (resp. disjoint non-empty closed subsets). In case (i), S' and S'' are constructible since the space S is Noetherian. By virtue of (8.3.11), there exist $\lambda \geq \alpha$ and two constructible closed subsets S'_λ, S''_λ of S_λ such that $S' = u_\lambda^{-1}(S'_\lambda)$, $S'' = u_\lambda^{-1}(S''_\lambda)$; since $S = S' \cup S''$, it follows from (8.3.11) that one can suppose $S_\lambda = S'_\lambda \cup S''_\lambda$; since S'_λ and S''_λ are distinct from S_λ , this proves that S_λ is not irreducible.

In case (ii), S' and S'' are open and quasi-compact, hence, by virtue of (8.2.11), there exist $\lambda \geq \alpha$ and two open quasi-compact subsets S'_λ, S''_λ of S_λ such that $S' = u_\lambda^{-1}(S'_\lambda)$, $S'' = u_\lambda^{-1}(S''_\lambda)$. Moreover, since S' and S'' are open and closed in S , they are both pro-constructible and ind-constructible (1.9.6), hence constructible (1.9.5), and it follows from (8.3.5) that one can suppose λ taken such that $S_\lambda = S'_\lambda \cup S''_\lambda$ and $S'_\lambda \cap S''_\lambda = \emptyset$, which shows that S_λ is not connected. $\square$

Proposition (8.4.2). — *Suppose that the underlying space of S is Noetherian and that one of the two following conditions is satisfied:*

- a) *For $\lambda \leq \mu$, $u_{\lambda\mu} : S_\mu \rightarrow S_\lambda$ is dominant, and there exists α such that S_α is quasi-compact.*
- b) *There exists α such that the underlying space of S_α is Noetherian, and for $\mu \geq \lambda$, $u_{\lambda\mu}$ is a homeomorphism of S_μ onto a subspace of S_λ .*

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Under these conditions, there exists λ such that, for every $\mu \geq \lambda$:

- (i) *For every irreducible component Y_i of S ($1 \leq i \leq m$), $\overline{u_\mu(Y_i)}$ is an irreducible component of S_μ , and the map $Y_i \mapsto \overline{u_\mu(Y_i)}$ is a bijection of the set of irreducible components of S onto the set of irreducible components of S_μ .*
- (ii) *For every connected component C_j of S ($1 \leq j \leq n$), $\overline{u_\mu(C_j)}$ is a connected component of S_μ , and the map $C_j \mapsto \overline{u_\mu(C_j)}$ is a bijection of the set of connected components of S onto the set of connected components of S_μ .*

Proof. Let us first establish the

Lemma (8.4.2.1). — *Under condition a) or b) of (8.4.2), there exists λ such that, for $\mu \geq \lambda$, $u_\mu : S \rightarrow S_\mu$ is dominant.*

In case a), this has already been proved without supposing the space S Noetherian (8.3.8), (i). In case b), set $Z_\alpha = \overline{u_\alpha(S)}$; as a closed subset of the Noetherian space S_α , Z_α is constructible, and since $u_\alpha^{-1}(Z_\alpha) = S$, it follows from (8.3.5) that there exists $\lambda \geq \alpha$ such that $u_{\lambda\mu}^{-1}(Z_\lambda) = S_\mu$. But since $u_{\alpha\mu}$ is a homeomorphism and $u_\alpha(S)$ is dense in S_α , it follows from $S \rightarrow Z_\mu = S_\mu$.

This lemma being proved, one can suppose that for every λ , u_λ is a dominant morphism.

- (i) Each of the S_λ is a union of the $\overline{u_\lambda(Y_i)}$, which are irreducible. On the other hand, if U_i is the open subset of S complementary to the union of the Y_j of index $j \neq i$ ($1 \leq i \leq m$), the U_i are pairwise disjoint and $Y_i = \overline{U_i}$ (0, 2.1.6). Since the underlying space of S is Noetherian, the U_i are quasi-compact, hence there exist an index λ and open subsets $U_{i\lambda}$ of S_λ such that $U_i = u_\lambda^{-1}(U_{i\lambda})$ for $1 \leq i \leq m$ (8.2.11). From this one concludes that if one sets $U_{i\mu} = u_{\lambda\mu}^{-1}(U_{i\lambda})$ for $\lambda \leq \mu$, the $U_{i\mu}$ are pairwise disjoint, the $U_{i\mu}$ are open and $u_\mu(U_i)$ is dense in $U_{i\mu}$ since u_μ is dominant. By this, none of the closures $\overline{U_{i\mu}}$ is contained in another and $u_\mu(U_i)$ is dense in $U_{i\mu}$ since u_μ is dominant; hence $\overline{U_{i\mu}} = \overline{u_\mu(Y_i)}$, which proves that the $\overline{u_\mu(Y_i)}$ are the irreducible components of S_μ (0, 2.1.7) and completes the proof.
- (ii) Since the underlying space S is Noetherian, the C_j are open and closed in S and quasi-compact; the same reasoning as in (i) shows that there exist λ and open subsets $V_{j\lambda}$ of S_λ such that $C_j = u_\lambda^{-1}(V_{j\lambda})$ for $1 \leq j \leq n$. One sees also, as in (i), that if one sets $V_{j\mu} = u_{\lambda\mu}^{-1}(V_{j\lambda})$ for $\mu \geq \lambda$, the $V_{j\mu}$ are pairwise disjoint, and $u_\mu(C_j)$ is dense in $V_{j\mu}$. Furthermore, it follows from (8.3.4) that for μ sufficiently large, the union of the $V_{j\mu}$ is S_μ , since every open subset

of a Noetherian prescheme is ind-constructible (1.9.6). The $V_{j\mu}$ are thus the connected components of S_μ , this completes the proof. $\square$

We note that if the morphisms $u_{\lambda\mu}$ are *immersions*, they satisfy in particular condition b) of (8.4.2).

Corollary (8.4.3). — Suppose one of the conditions a), b) of (8.4.2) is satisfied, the underlying space of S being Noetherian; then, for S to be irreducible, it is necessary and sufficient that there exist λ such that S_μ be so for every $\mu \geq \lambda$.

Proposition (8.4.4). — Suppose that there exists α such that S_α is quasi-compact and that, for $\lambda \leq \mu$, $u_{\lambda\mu} : S_\mu \rightarrow S_\lambda$ is dominant. Then, for S to be connected, it is necessary and sufficient that there exist λ such that S_μ be so for every $\mu \geq \lambda$. IV-3 | 19

Proof. The condition is sufficient by virtue of (8.4.1), (i); on the other hand, we have seen (8.3.8), (i) that $u_\lambda : S \rightarrow S_\lambda$ is dominant for λ sufficiently large, hence, if S is connected, it is the same for S_λ , since $u_\lambda(S)$ is dense in S_λ and is connected. $\square$

Corollary (8.4.5). — Let k be a field, X a quasi-compact k -prescheme. For X to be geometrically connected (4.5.2), it is necessary and sufficient that, for every finite and separable extension K of k , $X \otimes_k K$ be connected.

Proof. The condition is trivially necessary. To see that it is sufficient, we must prove that if Ω is an algebraic closure of k , $X \otimes_k \Omega$ is connected (4.5.1). Now, Ω is a filtered inductive limit of the finite sub-extensions of k , and for $k' \subset k'' \subset \Omega$, the morphism $X \otimes_k k'' \rightarrow X \otimes_k k'$ is surjective. We are thus reduced, by virtue of (8.4.4), to proving that $X \otimes_k k'$ is connected for every finite extension k' of k . But if K is the largest separable extension contained in k' , the morphism $X \otimes_k k' \rightarrow X \otimes_k K$ is finite, surjective, and radicial, hence (2.4.5) a homeomorphism, and since $X \otimes_k K$ is connected by hypothesis, so is $X \otimes_k k'$. $\square$

Remark (8.4.6). — (i) The proof of (8.4.2) shows that the conclusion of this proposition is valid if one supposes that the underlying space of S is Noetherian, that there exists α such that S_α is quasi-compact, and finally that the $u_\lambda : S \rightarrow S_\lambda$ are dominant.

(ii) On the other hand, the conclusion of (8.4.2) can be at fault when the u_λ are not dominant for λ sufficiently large, even when the S_λ and S are Noetherian, as shown by the following example. Take $\mathbf{N}$ for index set, all the S_n equal to $\text{Spec}(A \times K) = \text{Spec}(A) \amalg \text{Spec}(K)$, where K is a field, A any K -algebra, and all the morphisms $u_{n,n+1}$ equal to the same morphism corresponding to the homomorphism $(x, y) \mapsto (j(y), y)$ of $A \times K$ into itself, where $j : K \rightarrow A$ is the canonical homomorphism. One easily verifies that the inductive limit of this system of rings is the canonical homomorphism corresponding to the second projection $A \times K \rightarrow K$. One thus sees that $S = \text{Spec}(K)$ is irreducible even though none of the S_n is connected.

8.5. Modules of finite presentation on a projective limit of preschemes

(8.5.1). We always keep the notations of (8.2.2); we will further restrict ourselves to the case where S_0 is one of the S_λ , to which one can always reduce.

Whenever, in this section, we consider a family $(\mathcal{F}_\lambda)$, where, for every $\lambda \in L$, $\mathcal{F}_\lambda$ is an $\mathcal{O}_{S_\lambda}$ -Module, it will be understood that this family satisfies the condition

$$(8.5.1.1) \quad \mathcal{F}_\mu = u_{\lambda\mu}^*(\mathcal{F}_\lambda) \quad \text{for } \lambda \leq \mu.$$

We will then set

$$(8.5.1.2) \quad \mathcal{F} = u_\lambda^*(\mathcal{F}_\lambda)$$

which is an $\mathcal{O}_S$ -Module not depending on the index $\lambda \in L$, by virtue of the hypothesis (8.5.1.1).

Let $(\mathcal{F}_\lambda), (\mathcal{G}_\lambda)$ be two such families of $\mathcal{O}_{S_\lambda}$ -Modules. It is clear that the maps $f_\lambda \mapsto u_{\lambda\mu}^*(f_\lambda)$ from $\text{Hom}_{\mathcal{O}_{S_\lambda}}(\mathcal{F}_\lambda, \mathcal{G}_\lambda)$ to $\text{Hom}_{\mathcal{O}_{S_\mu}}(\mathcal{F}_\mu, \mathcal{G}_\mu)$ define an inductive system of abelian groups $(\text{Hom}_{\mathcal{O}_{S_\lambda}}(\mathcal{F}_\lambda, \mathcal{G}_\lambda), u_{\lambda\mu}^*)$, and that the maps $f_\lambda \mapsto u_\lambda^*(f_\lambda)$ form an inductive system of homomorphisms of abelian groups

$$u_\lambda^* : \text{Hom}_{\mathcal{O}_{S_\lambda}}(\mathcal{F}_\lambda, \mathcal{G}_\lambda) \longrightarrow \text{Hom}_S(\mathcal{F}, \mathcal{G});$$

whence, by passing to the inductive limit, a canonical homomorphism of abelian groups

$$(8.5.1.3) \quad u_{\mathcal{F}, \mathcal{G}}^* : \varinjlim \operatorname{Hom}_{\mathcal{O}_{S_\lambda}}(\mathcal{F}_\lambda, \mathcal{G}_\lambda) \longrightarrow \operatorname{Hom}_S(\mathcal{F}, \mathcal{G}).$$

Note that when $\mathcal{F}_\lambda = \mathcal{O}_{S_\lambda}$, the condition (8.5.1.1) is satisfied, and one has $\mathcal{F} = \mathcal{O}_S$; the homomorphism (8.5.1.3) thus gives (0, 5.1.1) a canonical homomorphism of abelian groups

$$(8.5.1.4) \quad u_{\mathcal{G}} : \varinjlim \Gamma(S_\lambda, \mathcal{G}_\lambda) \longrightarrow \Gamma(S, \mathcal{G}).$$

Theorem (8.5.2). — (i) Suppose S_0 quasi-compact (resp. quasi-compact and quasi-separated) and that, for some $\lambda \in L$, $\mathcal{F}_\lambda$ is quasi-coherent and of finite type (resp. of finite presentation) and $\mathcal{G}_\lambda$ quasi-coherent. Then the homomorphism $u_{\mathcal{F}, \mathcal{G}}^*$ is injective (resp. bijective).
(ii) Suppose S_0 quasi-compact and quasi-separated. For every quasi-coherent $\mathcal{O}_S$ -Module $\mathcal{F}$ of finite presentation, there exist $\lambda \in L$ and an $\mathcal{O}_{S_\lambda}$ -Module quasi-coherent of finite presentation $\mathcal{F}_\lambda$ such that $\mathcal{F}$ is isomorphic to $u_\lambda^*(\mathcal{F}_\lambda)$.

Proof. (i) We can evidently restrict to the case where $S_0 = S_\lambda$ since the morphisms $u_{0\lambda} : S_\lambda \rightarrow S_0$ are affine, hence quasi-compact and separated. Consider first the case where $S_0 = \operatorname{Spec}(A_0)$ is affine. Then assertion (i) is equivalent to the

Lemma (8.5.2.1). — Let A_0 be a ring, $(A_\lambda)_{\lambda \in L}$ an inductive system of A_0 -algebras, $A = \varinjlim A_\lambda$; let M_0, N_0 be two A_0 -modules, and set $M_\lambda = M_0 \otimes_{A_0} A_\lambda$, $N_\lambda = N_0 \otimes_{A_0} A_\lambda$, $M = M_0 \otimes_{A_0} A = \varinjlim M_\lambda$, $N = N_0 \otimes_{A_0} A = \varinjlim N_\lambda$. If M_0 is of finite type (resp. of finite presentation), the canonical homomorphism

$$(8.5.2.2) \quad \varinjlim \operatorname{Hom}_{A_\lambda}(M_\lambda, N_\lambda) \longrightarrow \operatorname{Hom}_A(M, N)$$

is injective (resp. bijective).

We know indeed (Bourbaki, *Alg.*, chap. II, 3^e ed., §5, n° 1) that one has canonical functorial isomorphisms

$$\operatorname{Hom}_{A_\lambda}(M_\lambda, N_\lambda) \xrightarrow{\sim} \operatorname{Hom}_{A_0}(M_0, N_\lambda), \quad \operatorname{Hom}_A(M, N) \xrightarrow{\sim} \operatorname{Hom}_{A_0}(M_0, N)$$

so that the homomorphism (8.5.2.2) is none other, up to canonical isomorphisms, than the canonical homomorphism

$$(8.5.2.3) \quad \varinjlim \operatorname{Hom}_{A_0}(M_0, N_\lambda) \longrightarrow \operatorname{Hom}_{A_0}(M_0, \varinjlim N_\lambda),$$

which, to every inductive system of homomorphisms of A_0 -modules $\theta_\lambda : M_0 \rightarrow N_\lambda$, associates its inductive limit. IV-3 | 21

Now, if M_0 is of finite type (resp. of finite presentation), one has an exact sequence $A_0^m \rightarrow M_0 \rightarrow 0$ (resp. $A_0^n \rightarrow A_0^m \rightarrow M_0 \rightarrow 0$); since it is clear that (8.5.2.3) is bijective when M_0 is of the form A_0^j , it suffices to use the left-exactness of the functor $M_0 \mapsto \operatorname{Hom}_{A_0}(M_0, P)$ and the exactness of the functor $\varinjlim$ (in the category of abelian groups) to conclude.

Let us pass to the case where S_0 is quasi-compact, and let (U_i) be a finite open affine covering of S_0 ; for every λ , the $U_{i\lambda} = u_{0\lambda}^{-1}(U_i)$ form a covering of the open affine subsets of S_λ , and the $V_i = u_0^{-1}(U_i)$ a covering of the open affine subsets of S . To show that $u_{\mathcal{F}, \mathcal{G}}^*$ is injective, it suffices to show that if $f_\lambda : \mathcal{F}_\lambda \rightarrow \mathcal{G}_\lambda$ is such that $f = u_\lambda^*(f_\lambda) = 0$, then there exists $\mu \geq \lambda$ such that $u_{\lambda\mu}^*(f_\lambda) = 0$. By virtue of the lemma (8.5.2.1), for each i there exists $\mu_i \geq \lambda$ such that $f_\mu|_{U_{i\mu_i}} = 0$. It suffices therefore to take μ greater than all the μ_i .

Suppose further that S_0 is quasi-separated and $\mathcal{F}_0$ is a quasi-coherent $\mathcal{O}_{S_0}$ -Module of finite presentation. Note now that S_λ is quasi-separated and $\mathcal{F}_\lambda$ is a quasi-coherent $\mathcal{O}_{S_\lambda}$ -Module of finite presentation (0, 5.2.5); since, for every pair of indices i, j , $U_{ij\lambda} = U_{i\lambda} \cap U_{j\lambda}$ is quasi-compact and quasi-separated (1.2.7), it follows from (8.5.2.5) that for every pair of indices i, j , $u_\lambda^*(f_\lambda^{(i)}) = u_\lambda^*(f_\lambda^{(j)})$ on $U_{ij\lambda}$ by definition, it follows from what we have already seen above that there exists λ_{ij} such that $u_{\lambda\mu}^*(f_\lambda^{(i)})|_{U_{ij\lambda}} = u_{\lambda\mu}^*(f_\lambda^{(j)})|_{U_{ij\lambda}}$ for $\mu \geq \lambda_{ij}$; taking μ greater than all the λ_{ij} , one sees that $u_{\lambda\mu}^*(f_\lambda^{(i)})$ and $u_{\lambda\mu}^*(f_\lambda^{(j)})$ coincide in $U_{i\mu} \cap U_{j\mu}$ for every couple (i, j) , and therefore define an homomorphism $f_\mu : \mathcal{F}_\mu \rightarrow \mathcal{G}_\mu$ such that $f = u_\mu^*(f_\mu)$.

(ii) Before passing to the proof of (ii), let us note the following corollaries of (i): □

Corollary (8.5.2.4). — Suppose S_0 quasi-compact, $\mathcal{F}_\lambda$ quasi-coherent of finite type, $\mathcal{G}_\lambda$ quasi-coherent. Let $f_\lambda : \mathcal{F}_\lambda \rightarrow \mathcal{G}_\lambda$ be a homomorphism. For $f = u_\lambda^*(f_\lambda) : \mathcal{F} \rightarrow \mathcal{G}$ to be an isomorphism, it is necessary and sufficient that there exist $\mu \geq \lambda$ such that $f_\mu = u_{\lambda\mu}^*(f_\lambda) : \mathcal{F}_\mu \rightarrow \mathcal{G}_\mu$ be so.

Proof. We can always suppose $S_\lambda = S_0$, and we can further reduce to the case where S_0 is affine, hence also quasi-separated. Then the assertion is equivalent to the lemma (8.5.2.1), for there is by hypothesis an $\mathcal{O}_S$ -homomorphism $g : \mathcal{G} \rightarrow \mathcal{F}$ such that $g \circ f = 1_{\mathcal{F}}$ and $f \circ g = 1_{\mathcal{G}}$. Since $\mathcal{G}$ is of finite presentation, there exist by virtue of (8.5.2), (i), an index $\nu \geq \lambda$ and a homomorphism $g_\nu : \mathcal{G}_\nu \rightarrow \mathcal{F}_\nu$ such that $g = u_\nu^*(g_\nu)$ by virtue of (8.5.2), (i); one has therefore $u_\nu^*(g_\nu \circ f_\nu) = 1_{\mathcal{F}}$ and $u_\nu^*(f_\nu \circ g_\nu) = 1_{\mathcal{G}}$, whence the corollary. □

Corollary (8.5.2.5). — Suppose S_0 quasi-compact and quasi-separated. Suppose that $\mathcal{F}_\lambda, \mathcal{G}_\lambda$ are $\mathcal{O}_{S_\lambda}$ -Modules quasi-coherent of finite presentation. For $\mathcal{F}$ and $\mathcal{G}$ to be isomorphic, it is necessary and sufficient that there exist $\mu \geq \lambda$ such that $\mathcal{F}_\mu$ and $\mathcal{G}_\mu$ are isomorphic. Moreover, for every isomorphism $f : \mathcal{F} \xrightarrow{\sim} \mathcal{G}$, there exist $\nu \geq \mu$ and an isomorphism $f_\nu : \mathcal{F}_\nu \xrightarrow{\sim} \mathcal{G}_\nu$ such that $f = u_\nu^*(f_\nu)$.

Proof. This follows from (8.5.2.4) and from (8.5.2), (i) since every homomorphism $f : \mathcal{F} \rightarrow \mathcal{G}$ is of the form $u_\mu^*(f_\mu)$ for some $\mu \geq \lambda$ and some homomorphism $f_\mu : \mathcal{F}_\mu \rightarrow \mathcal{G}_\mu$. □

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Proof of (8.5.2), (ii). Consider again first the case where $S_0 = \text{Spec}(A_0)$ is affine. Then the assertion is equivalent to the lemma (8.5.7.1).

In the general case, we start from a finite open affine covering (U_i) of S_0 , and for every λ , the $U_{i\lambda} = u_{0\lambda}^{-1}(U_i)$ form an open affine covering of S_λ , and the $V_i = u_0^{-1}(U_i)$ an open affine covering of S . By the affine case, for every i , there exist an index $\lambda(i)$ and an $\mathcal{O}_{U_{i,\lambda(i)}}$ -Module quasi-coherent of finite presentation $\mathcal{F}_i$ such that $u_{\lambda(i)}^*(\mathcal{F}_i) = \mathcal{F}|_{V_i}$ (with the notations of (i)). Furthermore, since L is filtered, we can suppose that all the λ_i are equal to a single λ . Note now that S_λ is quasi-separated and $\mathcal{F}_\lambda$ is a quasi-coherent $\mathcal{O}_{S_\lambda}$ -Module of finite presentation (0, 5.2.5); since, for every pair of indices i, j , $U_{ij\lambda} = U_{i\lambda} \cap U_{j\lambda}$ is quasi-compact and quasi-separated (1.2.7), it follows from (8.5.2.5) that for every pair (i, j) , there exist an index $\lambda_{ij} \geq \lambda$ and an isomorphism $\theta_{ij} : \mathcal{F}_i|_{U_{ij\lambda}} \xrightarrow{\sim} u_{\lambda,\lambda_{ij}}^*(\mathcal{F}_j)|_{U_{ij\lambda}}$ such that $u_{\lambda_{ij}}^*(\theta_{ij})$ is the identity automorphism of $\mathcal{F}|_{(V_i \cap V_j)}$; one can further suppose all the λ_{ij} equal to λ . Changing the notations, one can therefore suppose that it holds for every pair (i, j) an isomorphism $\theta_{ij} : \mathcal{F}_i|_{U_{ij\lambda}} \xrightarrow{\sim} \mathcal{F}_j|_{U_{ij\lambda}}$ such that $u_\lambda^*(\theta_{ij})$ is the identity automorphism of $\mathcal{F}|_{(V_i \cap V_j)}$. Finally, for any three indices i, j, k , if one sets $U_{ijk,\lambda} = U_{i\lambda} \cap U_{j\lambda} \cap U_{k\lambda}$, $U_{ijk,\lambda}$ is quasi-compact, and if $\theta'_{ij}, \theta'_{jk}$, and θ'_{ik} denote the restrictions of θ_{ij}, θ_{jk} , and θ_{ik} to $U_{ijk,\lambda}$, one has $u_\lambda^*(\theta'_{ij} \circ \theta'_{jk}) = u_\lambda^*(\theta'_{ik})$. There is, hence, by virtue of (i), an index $\mu \geq \lambda$ such that $u_{\lambda\mu}^*(\theta'_{ik}) = u_{\lambda\mu}^*(\theta'_{ij} \circ \theta'_{jk})$, hence the isomorphisms $u_{\lambda\mu}^*(\theta'_{ij}) : u_{\lambda\mu}^*(\mathcal{F}_i)|_{U_{ij\mu}} \xrightarrow{\sim} u_{\lambda\mu}^*(\mathcal{F}_j)|_{U_{ij\mu}}$ satisfy the gluing condition, and define on S_μ a quasi-coherent $\mathcal{O}_{S_\mu}$ -Module $\mathcal{F}_\mu$ of finite presentation (0, 3.3.1) such that $\mathcal{F}$ is isomorphic to $u_\mu^*(\mathcal{F}_\mu)$. □

(8.5.3). The result of (8.5.2) can be expressed again by saying that if S_0 is quasi-compact and quasi-separated, the category of $\mathcal{O}_S$ -Modules quasi-coherent of finite presentation is determined up to equivalence near the datum of the categories of $\mathcal{O}_{S_\lambda}$ -Modules quasi-coherent of finite presentation, of the functors $u_{\lambda\mu}^*$ between these categories, and of the transition isomorphisms $u_{\mu\nu}^* \circ u_{\lambda\mu}^* \xrightarrow{\sim} u_{\lambda\nu}^*$. In a figurative way, one can say that the datum of a quasi-coherent $\mathcal{O}_S$ -Module of finite presentation $\mathcal{F}$ is “functorially” equivalent to the datum of a quasi-coherent $\mathcal{O}_{S_\lambda}$ -Module of finite presentation $\mathcal{F}'_\lambda$ for some sufficiently large λ and that, if a quasi-coherent $\mathcal{O}_{S_\mu}$ -Module of finite presentation $\mathcal{F}''_\mu$ also has $\mathcal{F}$ as inverse image, then $\mathcal{F}'_\lambda$ and $\mathcal{F}''_\mu$ have the same inverse image in some suitable S_ν ($\nu \geq \lambda, \nu \geq \mu$).

We will now interpret from this point of view various notions related to quasi-coherent $\mathcal{O}_S$ -Modules.

Corollary (8.5.4). — Suppose S_0 quasi-compact and quasi-separated; then, for every quasi-coherent $\mathcal{O}_{S_\lambda}$ -Module $\mathcal{G}_\lambda$, the canonical homomorphism (8.5.1.4) is bijective.

Proof. Indeed, it suffices to apply (8.5.2), (i) to the case where $\mathcal{F}_\lambda = \mathcal{O}_{S_\lambda}$, which is of finite presentation. $\square$

Proposition (8.5.5). — Suppose S_0 quasi-compact, and suppose that $\mathcal{F}_\lambda$ is an $\mathcal{O}_{S_\lambda}$ -Module quasi-coherent of finite presentation. For $\mathcal{F}$ to be locally free (resp. locally free of rank n), it is necessary and sufficient that there exist $\mu \geq \lambda$ such that $\mathcal{F}_\mu$ be so.

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Proof. The condition being trivially sufficient, let us prove that it is necessary. If $\mathcal{F}$ is locally free (resp. locally free of rank n), there exists a finite open affine covering (V_i) of S such that $\mathcal{F}|_{V_i}$ is isomorphic to $\mathcal{O}_S^{n_i}|_{V_i}$ (resp. $\mathcal{O}_S^n|_{V_i}$) for every i . By virtue of (8.2.11), there exist $\nu \geq \lambda$ and for each i a quasi-compact open subset U_{iv} of S_ν such that $V_i = u_\nu^{-1}(U_{iv})$. Since S_ν is quasi-compact, each U_{iv} is a finite union of open affines; we are thus reduced to the case where S_0 is affine and $\mathcal{F} = \mathcal{O}_S^n$; we then know that there exist $\mu \geq \lambda$ such that $\mathcal{F}_\mu$ is isomorphic to $\mathcal{O}_{S_\mu}^n$ (8.5.2.5). $\square$

Proposition (8.5.6). — Suppose S_0 quasi-compact, and consider a sequence

$$\mathcal{F}_\lambda \longrightarrow \mathcal{G}_\lambda \longrightarrow \mathcal{H}_\lambda \longrightarrow 0$$

of homomorphisms of $\mathcal{O}_{S_\lambda}$ -Modules quasi-coherent, where $\mathcal{F}_\lambda$ and $\mathcal{G}_\lambda$ are of finite type and $\mathcal{H}_\lambda$ of finite presentation. For the corresponding sequence $\mathcal{F} \rightarrow \mathcal{G} \rightarrow \mathcal{H} \rightarrow 0$ to be exact, it is necessary and sufficient that there exist $\mu \geq \lambda$ such that the sequence $\mathcal{F}_\mu \rightarrow \mathcal{G}_\mu \rightarrow \mathcal{H}_\mu \rightarrow 0$ be so (in which case it is the same for $\nu \geq \mu$).

Proof. The fact that the condition is sufficient and the last assertion result from the fact that the functor u_λ^* (resp. $u_{\mu\nu}^*$) is right-exact. To prove that the condition is necessary, note that it follows from the hypothesis and from (8.5.2), (i) that there exists $\nu \geq \lambda$ such that the composite $\mathcal{F}_\nu \rightarrow \mathcal{G}_\nu \rightarrow \mathcal{H}_\nu$ is zero. If one sets $\mathcal{H}'_\nu = \text{Coker}(\mathcal{F}_\nu \rightarrow \mathcal{G}_\nu)$, one has therefore a homomorphism $f_\nu : \mathcal{H}'_\nu \rightarrow \mathcal{H}_\nu$; by hypothesis, $u_\nu^*(f_\nu)$ is an isomorphism, and it follows from (8.5.2.4) that there exists $\mu \geq \nu$ such that $u_{\nu\mu}^*(f_\nu)$ is an isomorphism, which completes the proof. $\square$

Corollary (8.5.7). — Suppose S_0 quasi-compact, $\mathcal{F}_\lambda$ quasi-coherent of finite type, and let $f_\lambda : \mathcal{F}_\lambda \rightarrow \mathcal{G}_\lambda$ be a homomorphism. For $f = u_\lambda^*(f_\lambda) : \mathcal{F} \rightarrow \mathcal{G}$ to be surjective, it is necessary and sufficient that there exist $\mu \geq \lambda$ such that $f_\mu = u_{\lambda\mu}^*(f_\lambda) : \mathcal{F}_\mu \rightarrow \mathcal{G}_\mu$ be so.

Proof. This is the particular case of (8.5.6) applied to the sequence $0 \rightarrow \mathcal{H}_\lambda \rightarrow 0 \rightarrow 0$, where $\mathcal{H}_\lambda = \text{Coker}(f_\lambda)$, which is quasi-coherent and of finite type (taking into account the fact that one has $\mathcal{H} = \text{Coker}(f)$ and $\mathcal{H}_\mu = \text{Coker}(f_\mu)$, by virtue of the right-exactness of the functors u_λ^* and $u_{\lambda\mu}^*$). $\square$

Corollary (8.5.8). — Suppose S_0 quasi-compact and the morphisms $u_{\lambda\mu} : S_\mu \rightarrow S_\lambda$ flat. Then:

- (i) Let $\mathcal{F}_\lambda \xrightarrow{f_\lambda} \mathcal{G}_\lambda \xrightarrow{g_\lambda} \mathcal{H}_\lambda$ be a sequence of homomorphisms of $\mathcal{O}_{S_\lambda}$ -Modules quasi-coherent, such that $\text{Im } f_\lambda$ and $\text{Ker } g_\lambda$ are of finite type. For the corresponding sequence $\mathcal{F} \xrightarrow{f} \mathcal{G} \xrightarrow{g} \mathcal{H}$ to be exact, it is necessary and sufficient that there exist $\mu \geq \lambda$ such that the sequence $\mathcal{F}_\mu \xrightarrow{f_\mu} \mathcal{G}_\mu \xrightarrow{g_\mu} \mathcal{H}_\mu$ be so (in which case the same holds for $\nu \geq \mu$).
- (ii) Let $f_\lambda : \mathcal{F}_\lambda \rightarrow \mathcal{G}_\lambda$ be an $\mathcal{O}_{S_\lambda}$ -Module homomorphism quasi-coherent, and suppose that $\text{Im } f_\lambda$ is of finite type. For $f = u_\lambda^*(f_\lambda) : \mathcal{F} \rightarrow \mathcal{G}$ to be injective, it is necessary and sufficient that there exist $\mu \geq \lambda$ such that $f_\mu = u_{\lambda\mu}^*(f_\lambda) : \mathcal{F}_\mu \rightarrow \mathcal{G}_\mu$ be so.

8.6. Sub-preschemes of finite presentation of a projective limit of preschemes

(8.6.1). Given a prescheme Y , we denote in this section by $\mathfrak{Spr}(Y)$ the ordered set (I, 4.1.10) of sub-preschemes of Y that are of finite presentation (I, 1.6.1), by $\mathfrak{Spr}_o(Y)$ (resp. $\mathfrak{Spr}_f(Y)$) the subset of $\mathfrak{Spr}(Y)$ formed by the sub-preschemes induced on opens (resp. on closed sub-preschemes) of Y , of finite presentation over Y ; it amounts to the same to say that a sub-prescheme Z of Y belongs to $\mathfrak{Spr}_o(Y)$ (resp. $\mathfrak{Spr}_f(Y)$) or that it is induced on an open and that the underlying space is *retrocompact* in Y (resp. that it is closed and that the ideal $\mathcal{J}$ of $\mathcal{O}_Y$ that defines it is of finite type, which also means that $j_*(\mathcal{O}_Z) \in \Omega(\mathcal{O}_Y)$ where $j : Z \rightarrow Y$ is the canonical injection) (I, 1.6.1) and (I, 1.4.5).

(8.6.2). We always preserve the notations of (8.2.2) and suppose that S_0 is one of the S_λ . Let Y_λ be a sub-prescheme of S_λ ; then $Y_\mu = u_{\mu\lambda}^{-1}(Y_\lambda)$ (resp. $Y = u_\lambda^{-1}(Y_\lambda)$) is a sub-prescheme of S_μ for $\mu \geq \lambda$ (resp. of S); it is induced on an open (resp. it is closed) and is of finite presentation on S_μ (resp. S) if Y_λ is of finite presentation on S_λ (resp. S) (I, 4.3.2). It follows that $(\mathfrak{Spr}(S_\lambda), u_{\lambda\mu}^{-1})$ (resp. $(\mathfrak{Spr}_o(S_\lambda), u_{\lambda\mu}^{-1})$, $(\mathfrak{Spr}_f(S_\lambda), u_{\lambda\mu}^{-1})$) is an *inductive system* of sets and the maps $u_\lambda^{-1} : \mathfrak{Spr}(S_\lambda) \rightarrow \mathfrak{Spr}(S)$ (resp. $\mathfrak{Spr}_o(S_\lambda) \rightarrow \mathfrak{Spr}_o(S)$, $\mathfrak{Spr}_f(S_\lambda) \rightarrow \mathfrak{Spr}_f(S)$) form an inductive system of maps; whence, by passing to the inductive limit, canonical maps

$$(8.6.2.1) \quad \varinjlim \mathfrak{Spr}(S_\lambda) \longrightarrow \mathfrak{Spr}(S)$$

$$(8.6.2.2) \quad \varinjlim \mathfrak{Spr}_o(S_\lambda) \longrightarrow \mathfrak{Spr}_o(S)$$

$$(8.6.2.3) \quad \varinjlim \mathfrak{Spr}_f(S_\lambda) \longrightarrow \mathfrak{Spr}_f(S).$$

Recall (I, 4.1.10) that $\mathfrak{Spr}(X)$, for every prescheme X , is an ordered set by the relation “ Z is a sub-prescheme of the sub-prescheme Y ” which we write $Z \leq Y$. The maps $u_{\lambda\mu}^{-1}$ and u_λ^{-1} are *increasing* for the corresponding order relations in $\mathfrak{Spr}(S_\lambda)$, $\mathfrak{Spr}(S_\mu)$, $\mathfrak{Spr}(S)$. Furthermore, by writing that $\eta \leq \eta'$ for two elements of this set whenever there exist λ and two elements Y_λ, Y'_λ of $\mathfrak{Spr}(S_\lambda)$, of which η and η' are the canonical images, and which are such that $Y_\lambda \leq Y'_\lambda$; one easily verifies that this does not depend on the choice of representatives, and that one thus has indeed a *relation of order*. The fact that the u_λ^{-1} are increasing immediately implies that the canonical map (8.6.2.1) is *increasing*; the same is evidently true of (8.6.2.2) and (8.6.2.3).

Proposition (8.6.3). — Suppose S_0 quasi-compact (resp. quasi-compact and quasi-separated). Then the maps (8.6.2.1), (8.6.2.2), (8.6.2.3) are injective (resp. bijective).

Proof. Taking into account the remarks of (8.6.1), the assertions relative to (8.6.2.3) follow from (8.5.11) applied to $\mathcal{F}_\lambda = \mathcal{O}_{S_\lambda}$; similarly, the assertions relative to (8.6.2.2) are special cases of (8.3.5) and (8.3.11), taking into account that the S_λ and S are quasi-compact. It remains to consider the map (8.6.2.1). Let us first prove that it is surjective when S_0 is quasi-compact and quasi-separated. Let Z be a sub-prescheme of S , of finite presentation over S ; as S is of the same nature as Z , there exists an open quasi-compact U of S such that Z is a sub-prescheme of U , closed in U , of finite presentation over U . There then exist an index λ and an open quasi-compact U_λ of S_λ such that $U = u_\lambda^{-1}(U_\lambda)$ (8.2.11); as S_λ is quasi-separated, it is of the same nature as U_λ (I, 1.2.7), and consequently one can restrict to the case where $U = S$; but in this case, one is reduced to the fact that (8.6.2.3) is surjective.

Finally, to see that (8.6.2.1) is injective when S_0 is quasi-compact, it will suffice to prove the following more precise result: $\square$

Corollary (8.6.3.1). — Suppose S_0 quasi-compact and let Z'_λ, Z''_λ be two sub-preschemes of S_λ , of finite presentation. For $Z' = u_\lambda^{-1}(Z'_\lambda)$ to be majorised by $Z'' = u_\lambda^{-1}(Z''_\lambda)$ (I, 4.1.10), it is necessary and sufficient that there exist $\mu \geq \lambda$ such that $Z'_\mu = u_{\mu\lambda}^{-1}(Z'_\lambda)$ be majorised by $Z''_\mu = u_{\mu\lambda}^{-1}(Z''_\lambda)$.

Proof. It is trivial that the condition is sufficient. To see that it is necessary, note first that the underlying sets Z'_λ and Z''_λ are locally constructible in S_λ by hypothesis (I, 1.8.4), hence the hypothesis implies the existence of a $\nu \geq \lambda$ such that $Z'_\lambda \subset Z''_\lambda$ (8.3.5); replacing λ by ν , we can therefore already suppose that $Z'_\lambda \subset Z''_\lambda$. Furthermore, by hypothesis (I, 1.6.1), the sub-spaces Z'_λ and Z''_λ of S_λ are quasi-compact; for every point $x \in Z'_\lambda$, there is an open quasi-compact neighbourhood $V(x)$ in S_λ such that $V(x) \cap Z'_\lambda$ and $V(x) \cap Z''_\lambda$ are closed in $V(x)$. By covering S_λ by a finite number of such

neighbourhoods, we see that there is an open quasi-compact U_λ of S_λ containing Z'_λ and such that Z'_λ and $Z''_\lambda \cap U_\lambda$ are closed in U_λ . If we denote by Y_λ the sub-prescheme of S_λ induced by Z'_λ on $U_\lambda \cap Z''_\lambda$, it is clear that with the usual notations, Y_μ (resp. Y) is majorised by Z''_μ (resp. Z'') for $\mu \geq \lambda$ (I, 4.4.1), and as it suffices to prove that Z'_μ is majorised by Y_μ for μ large enough, we see finally that one is reduced (replacing S_λ by U_λ) to the case where Z'_λ and Z''_λ are closed sub-preschemes of S_λ . But then, this has already been proved since (8.6.2.3) is increasing and injective. $\square$

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Corollary (8.6.4). — Suppose S_0 quasi-compact, and let Z_λ be a sub-prescheme of S_λ , of finite presentation over S_λ . For $Z = u_\lambda^{-1}(Z_\lambda)$ to be the subprescheme induced on an open subset of S (resp. a closed subprescheme of S), it is necessary and sufficient that there exist $\mu \geq \lambda$ such that $Z_\mu = u_{\mu\lambda}^{-1}(Z_\lambda)$ is the subprescheme induced on an open subset of S_μ (resp. a closed subprescheme of S_μ).

Proof. Let $(U_\lambda^{(i)})$ be a finite affine open covering of S_λ , and set $U_\mu^{(i)} = u_{\mu\lambda}^{-1}(U_\lambda^{(i)})$ for $\mu \geq \lambda$ and $U^{(i)} = u_\lambda^{-1}(U_\lambda^{(i)})$. If Z is open (resp. closed) in S , $Z \cap U^{(i)}$ is open (resp. closed) in $U^{(i)}$, and conversely if each of the $Z_\mu \cap U_\mu^{(i)}$ is open (resp. closed) in $U_\mu^{(i)}$, Z_μ is open (resp. closed) in S_μ . Since L is filtered, it suffices to prove the corollary when S_λ is affine, hence quasi-separated, and the result follows from the fact that the maps (8.6.2.1), (8.6.2.2), and (8.6.2.3) are bijective. $\square$

8.7. Criteria for a projective limit of preschemes to be a reduced (resp. integral) prescheme We preserve the hypotheses and notations of (8.2.2) and suppose always that S_0 is one of the S_λ .

Proposition (8.7.1). — Suppose that S is not reduced. Then there exists λ_0 such that for $\lambda \geq \lambda_0$, S_λ is not reduced.

Proof. The question being local on S_0 , we can suppose $S_0 = \text{Spec}(A_0)$ affine, whence $S = \text{Spec}(A)$, where $A = \varinjlim A_\lambda$ is the inductive limit of an inductive system of A_0 -algebras (A_λ) . We know then (5.13.2) that the nilradical of A is the inductive limit of those of the A_λ ; if it is not zero, one of the A_λ contains therefore a nilpotent element $a_\lambda \neq 0$, and the image of a_λ in the A_μ for $\mu \geq \lambda$ is therefore a nilpotent element $\neq 0$, and the image of a_λ in A is not zero for $\mu \geq \lambda$ since the image in A is not zero. $\square$

Proposition (8.7.2). — Suppose one of the following hypotheses satisfied:

- a) S_0 is quasi-compact, the nilradical $\mathcal{N}_0$ of $\mathcal{O}_{S_0}$ is an ideal of finite type (which is satisfied for example when S_0 is Noetherian), and the morphisms $u_{\lambda\mu} : S_\mu \rightarrow S_\lambda$ are open immersions.
- b) The morphisms $u_{\lambda\mu} : S_\mu \rightarrow S_\lambda$ are faithfully flat.

Under these conditions, for S to be reduced, it is necessary and sufficient that there exist λ_0 such that S_λ be reduced for $\lambda \geq \lambda_0$.

Proof. Furthermore, in case b), if S is reduced, all the S_λ are reduced.

The last assertion follows from the fact that the morphism $S \rightarrow S_\lambda$ is then faithfully flat for every λ (8.3.8) and from (2.1.13). On the other hand, (8.7.1) proves that the condition of the statement is sufficient (without hypothesis on S_0 or on the $u_{\lambda\mu}$). It therefore remains to see that the condition is necessary in hypothesis a); then (8.2.13), the underlying space of S is identified with S (the S_λ being identified with sub-preschemes induced on opens of S_0), and the structural sheaf $\mathcal{O}_S$ is identified with the sheaf induced on S by all the $\mathcal{O}_{S_\lambda}$; in particular for every $s \in S$, the local ring $\mathcal{O}_s$ is the same for S and for all the S_λ . If $\mathcal{N}_\lambda$ is the nilradical of $\mathcal{O}_{S_\lambda}$, the nilradical $\mathcal{N}$ of $\mathcal{O}_S$ has at each point of S the same fibre $\mathcal{N}_s$ (induced on S by $\mathcal{N}_\lambda$). The hypothesis that S is reduced implies therefore $u_\lambda^*(\mathcal{N}_\lambda) = 0$; as $\mathcal{N}_0$ is supposed of finite type, the same is true of the $\mathcal{N}_\lambda$, and the conclusion follows then from (8.5.7). $\square$

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Corollary (8.7.3). — Suppose one of the following hypotheses satisfied:

- a) S_0 is a Noetherian prescheme and the morphisms $u_{\lambda\mu} : S_\mu \rightarrow S_\lambda$ are open immersions.
- b) The morphisms $u_{\lambda\mu} : S_\mu \rightarrow S_\lambda$ are faithfully flat.

Then, for S to be integral, it is necessary and sufficient that there exist λ_0 such that S_λ be integral for $\lambda \geq \lambda_0$.

Proof. To say that a prescheme is integral means that it is both reduced and irreducible; the corollary thus follows from (8.7.2) and from (8.4.3). $\square$

Remark (8.7.4). — If one makes no hypothesis on the $u_{\lambda\mu}$, it can happen that S is integral even though all the S_λ are neither reduced nor connected, as the example (8.4.4) shows, where one takes the ring A non reduced.

8.8. Preschemes of finite presentation over a projective limit of preschemes

(8.8.1). Always preserving the notations and hypotheses of (8.2.2), we shall suppose given in this section two S_α -preschemes X_α, Y_α , which defines (8.2.5) two projective systems of preschemes $(X_\lambda, v_{\lambda\mu})$ and $(Y_\lambda, w_{\lambda\mu})$ by setting $X_\lambda = X_\alpha \times_{S_\alpha} S_\lambda, v_{\lambda\mu} = 1_{X_\alpha} \times u_{\lambda\mu}, w_{\lambda\mu} = 1_{Y_\alpha} \times u_{\lambda\mu}$ (for $\alpha \leq \lambda \leq \mu$), whose projective limits are respectively $X = X_\alpha \times_{S_\alpha} S, Y = Y_\alpha \times_{S_\alpha} S$, with the canonical morphisms $v_\lambda : X \rightarrow X_\lambda$ and $w_\lambda : Y \rightarrow Y_\lambda$. For $\alpha \leq \lambda \leq \mu$, there is a canonical map $\varepsilon_{\mu\lambda} : \text{Hom}_{S_\lambda}(X_\lambda, Y_\lambda) \rightarrow \text{Hom}_{S_\mu}(X_\mu, Y_\mu)$, which to every S_λ -morphism $f_\lambda : X_\lambda \rightarrow Y_\lambda$ associates $f_\mu = f_\lambda \times 1_{S_\mu} : X_\lambda \times_{S_\lambda} S_\mu \rightarrow Y_\lambda \times_{S_\lambda} S_\mu$, and it is clear that $(\text{Hom}_{S_\lambda}(X_\lambda, Y_\lambda), \varepsilon_{\mu\lambda})$ is an inductive system of sets. Similarly, there is a canonical map $\varepsilon_\lambda : \text{Hom}_{S_\lambda}(X_\lambda, Y_\lambda) \rightarrow \text{Hom}_S(X, Y)$ which to f_λ associates $f = f_\lambda \times 1_S : X_\lambda \times_{S_\lambda} S \rightarrow Y_\lambda \times_{S_\lambda} S$ and (ε_λ) is an inductive system of maps; whence, by passing to the inductive limit, a canonical map, functorial in S_α, X_α , and Y_α :

$$(8.8.1.1) \quad \varepsilon : \varinjlim \text{Hom}_{S_\lambda}(X_\lambda, Y_\lambda) \longrightarrow \text{Hom}_S(X, Y).$$

Theorem (8.8.2). — (i) Suppose X_α quasi-compact (resp. quasi-compact and quasi-separated), and Y_α locally of finite type (resp. locally of finite presentation) over S_α . Then the map (8.8.1.1) is injective (resp. bijective).
(ii) Suppose S_0 quasi-compact and quasi-separated. For every prescheme X of finite presentation over S , there exist a $\lambda \in L$, a prescheme X_λ of finite presentation over S_λ , and an S -isomorphism $X \xrightarrow{\sim} X_\lambda \times_{S_\lambda} S$.

Proof. (i) Consider first the case where $S_0 = \text{Spec}(A_0)$, $X_\alpha = \text{Spec}(B_\alpha)$, and $Y_\alpha = \text{Spec}(C_\alpha)$ are affine; then the $S_\lambda = \text{Spec}(A_\lambda)$ and $S = \text{Spec}(A)$ are also affine, with $A = \varinjlim A_\lambda$, and the assertions of (i) are equivalent to the

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Lemma (8.8.2.1). — Let A_0 be a ring, $(A_\lambda)_{\lambda \in L}$ an inductive system of A_0 -algebras, $A = \varinjlim A_\lambda$; let B_α be an A_α -algebra, C_α an A_α -algebra of finite type (resp. of finite presentation). Then the canonical homomorphism

$$(8.8.2.2) \quad \varinjlim_{\lambda} \text{Hom}_{A_\lambda\text{-alg.}}(C_\alpha \otimes_{A_\alpha} A_\lambda, B_\alpha \otimes_{A_\alpha} A_\lambda) \longrightarrow \text{Hom}_{A\text{-alg.}}(C_\alpha \otimes_{A_\alpha} A, B_\alpha \otimes_{A_\alpha} A)$$

is injective (resp. bijective).

Proof. We know that there are functorial canonical isomorphisms

$$\text{Hom}_{A_\lambda\text{-alg.}}(C_\alpha \otimes_{A_\alpha} A_\lambda, B_\alpha \otimes_{A_\alpha} A_\lambda) \xrightarrow{\sim} \text{Hom}_{A_\alpha\text{-alg.}}(C_\alpha, B_\alpha \otimes_{A_\alpha} A_\lambda)$$

$$\text{Hom}_{A\text{-alg.}}(C_\alpha \otimes_{A_\alpha} A, B_\alpha \otimes_{A_\alpha} A) \xrightarrow{\sim} \text{Hom}_{A_\alpha\text{-alg.}}(C_\alpha, B_\alpha \otimes_{A_\alpha} A)$$

by virtue of the universal property of the tensor product of two algebras. It therefore suffices to prove the □

Lemma (8.8.2.3). — Let E be a ring, G an E -algebra of finite type (resp. of finite presentation), (F_λ) an inductive system of E -algebras. Then the canonical homomorphism

$$\varinjlim \text{Hom}_{E\text{-alg.}}(G, F_\lambda) \longrightarrow \text{Hom}_{E\text{-alg.}}(G, \varinjlim F_\lambda)$$

which, to every inductive system of homomorphisms $\theta_\lambda : G \rightarrow F_\lambda$ of E -algebras, associates its inductive limit, is injective (resp. bijective).

Proof. Suppose first the E -algebra G of finite type, and let $(t_i)_{1 \leq i \leq n}$ be a system of generators of this E -algebra; let us show that if $(\theta'_\lambda), (\theta''_\lambda)$ are two inductive systems of homomorphisms $G \rightarrow F_\lambda$, with $\varinjlim \theta'_\lambda = \varinjlim \theta''_\lambda$, there exists $\mu \geq \lambda$ such that $\theta'_\mu = \theta''_\mu$. Indeed, if $\varphi_{\mu\lambda} : F_\lambda \rightarrow F_\mu$ and $\varphi_\lambda : F_\lambda \rightarrow F = \varinjlim F_\lambda$ are the homomorphisms of the inductive system (F_λ) , by hypothesis, for each index i , there exists an index λ_i such that $\varphi_{\lambda_i}(\theta'_{\lambda_i}(t_i)) = \varphi_{\lambda_i}(\theta''_{\lambda_i}(t_i))$, and one can suppose all the λ_i equal to a single λ ; it follows in the same way the existence of $\mu \geq \lambda$ such that $\varphi_{\mu\lambda}(\theta'_\lambda(t_i)) = \varphi_{\mu\lambda}(\theta''_\lambda(t_i))$ for $1 \leq i \leq n$, that is to say $\theta'_\mu(t_i) = \theta''_\mu(t_i)$ for $1 \leq i \leq n$, whence $\theta'_\mu = \theta''_\mu$.

Suppose in the second place G of finite presentation, so that one has

$$G = E[T_1, \dots, T_n] / \mathfrak{J},$$

where $\mathfrak{J}$ is an ideal of finite type, t_i being the class of T_i modulo $\mathfrak{J}$. Let $(P_j)_{1 \leq j \leq m}$ be a system of generators of $\mathfrak{J}$. Suppose given a homomorphism of E -algebras $\theta : G \rightarrow F$; set $b_i = \theta(t_i)$; by definition, one has therefore $P_j(b_1, b_2, \dots, b_n) = \theta(P_j(t_1, \dots, t_n)) = 0$ for $1 \leq j \leq m$. Now, there exist λ and elements $x_1, \dots, x_n$ in F_λ such that $b_i = \varphi_\lambda(x_i)$ for $1 \leq i \leq n$; one has therefore $\varphi_\lambda(P_j(x_1, \dots, x_n)) = P_j(b_1, \dots, b_n) = 0$, and consequently there exists $\mu \geq \lambda$ such that $\varphi_{\mu\lambda}(P_j(x_1, \dots, x_n)) = P_j(\varphi_{\mu\lambda}(x_1), \dots, \varphi_{\mu\lambda}(x_n)) = 0$ for $1 \leq j \leq m$; one therefore deduces for every $\nu \geq \mu$ a homomorphism of E -algebras $\theta_\nu = \varphi_{\nu\mu} \circ \theta_\mu$ of G into F_ν , and it is clear that θ is the inductive limit of this inductive system of homomorphisms, which completes the proof of the lemma. $\square$

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Let us now pass to the case where X_α is quasi-compact and Y_α locally of finite type over S_α . Set $Z_\alpha = X_\alpha \times_{S_\alpha} Y_\alpha$ and introduce the projective system corresponding to $Z_\lambda = X_\lambda \times_{S_\lambda} Y_\lambda$ and its limit $Z = X \times_S Y$; the canonical bijections (I, 3.3.14) give commutative diagrams

$$\begin{array}{ccccc} \mathrm{Hom}_{S_\lambda}(X_\lambda, Y_\lambda) & \xrightarrow{\varepsilon_{\mu\lambda}} & \mathrm{Hom}_{S_\mu}(X_\mu, Y_\mu) & \xrightarrow{\varepsilon_\mu} & \mathrm{Hom}_S(X, Y) \\ \downarrow \wr & & \downarrow \wr & & \downarrow \wr \\ \mathrm{Hom}_{X_\lambda}(X_\lambda, Z_\lambda) & \xrightarrow{\varepsilon_{\mu\lambda}} & \mathrm{Hom}_{X_\mu}(X_\mu, Z_\mu) & \xrightarrow{\varepsilon_\mu} & \mathrm{Hom}_X(X, Z) \end{array}$$

and consequently we are reduced to proving that (8.8.1.1) is injective in the particular case where $S_\alpha = X_\alpha$ (taking into account (I, 1.3.4)). Moreover, as X_α is quasi-compact, hence a finite union of open affines, and as X is a union of the opens $f^{-1}(V_i)$, where the V_i are open affines of Y_α , every point of X has an open affine neighbourhood $W(x)$ in S_α such that f'_α and f''_α both map $W(x)$ into a single open affine $V_{i\alpha}$. As X_α is quasi-compact, it is covered by a finite number of open affines $W(x_k)$; this implies that for each $x \in X_\alpha$, there is an open affine neighbourhood of x such that the restrictions of f'_α and f''_α to $W(x)$ map it into the same open affine $V_{i\alpha}$. As X_α is quasi-compact, it is covered by a finite number of such open affines $W(x_k)$; by virtue of the lemma (8.8.2.1) and of the fact that L is filtered, there exists $\lambda \geq \alpha$ such that the restrictions of f'_λ and f''_λ to each of the opens $v_\alpha^{-1}(W(x_k))$ are equal, whence $f'_\lambda = f''_\lambda$.

Suppose now X_α quasi-compact and quasi-separated and Y_α locally of finite presentation over S_α , and let us prove that (8.8.1.1) is surjective. Suppose therefore given an X -morphism $f : X \rightarrow Y$. As X is quasi-compact, it is of the same nature as Y , which contains $f(X)$; there exists therefore $\lambda \geq \alpha$ and an open quasi-compact Y'_λ in Y_λ such that $Y' = w_\lambda^{-1}(Y'_\lambda)$ (8.2.11). Replacing as needed α by λ and Y_λ by Y'_λ , we can therefore restrict to the case where Y_α is quasi-compact, hence also Y and the Y_λ . As Y is quasi-compact, it is a finite union of open affines V_i , and X is a union of the opens $f^{-1}(V_i)$. As X is quasi-compact, we can cover X by a finite number of such open affines. And since X is quasi-compact, for each i , the open quasi-compact sets in X_α may, by replacing α by an index $\lambda \geq \alpha$, be supposed such that $U_i = v_\alpha^{-1}(U_{i\alpha})$ where the $U_{i\alpha}$ are open quasi-compacts in X_α ; moreover, using (8.3.4), one can moreover suppose the $(U_{i\alpha})$ form a covering of X_α .

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It will then suffice to show that there exist $\lambda \geq \alpha$ and for each i a morphism $f_{i\lambda} : U_{i\lambda} \rightarrow V_{i\lambda}$ (where $U_{i\lambda} = v_{\alpha\lambda}^{-1}(U_{i\alpha})$ and $V_{i\lambda} = w_{\alpha\lambda}^{-1}(V_{i\alpha})$) such that the morphism $f_i = \varepsilon_\lambda(f_{i\lambda}) : U_i \rightarrow V_i$ be equal to the restriction of f to U_i . Indeed, if this is so, as X_α is quasi-separated (I, 1.2.3), the $U_{i\lambda} \cap U_{j\lambda}$ are quasi-compact and the uniqueness result proved above (which applies since Y_λ is locally of finite type over S_λ) proves that there exists $\mu \geq \lambda$ such that $f_{i\mu}$ and $f_{j\mu}$ coincide on $U_{i\mu} \cap U_{j\mu}$ for every couple (i, j) , and consequently define a S_μ -morphism $f_\mu : X_\mu \rightarrow Y_\mu$ such that $\varepsilon_\mu(f_\mu) = f$.

We are thus reduced to the case where Y_α is affine, and as in addition we can also suppose that S_α is affine, we can also suppose $S_\alpha = \mathrm{Spec}(A_\alpha)$, $Y_\alpha = \mathrm{Spec}(C_\alpha)$, C_α being an A_α -algebra of finite presentation, $S = \mathrm{Spec}(A)$, $Y = \mathrm{Spec}(C)$, with $A = \varinjlim (C_\alpha \otimes_{A_\alpha} A_\lambda) = C_\alpha \otimes_{A_\alpha} A$. One has then

$$\mathrm{Hom}_S(X, Y) = \mathrm{Hom}_{A\text{-alg.}}(C, \Gamma(X, \mathcal{O}_X)) = \mathrm{Hom}_{A_\alpha\text{-alg.}}(C_\alpha, \Gamma(X, \mathcal{O}_X))$$

(I, 2.2.4) and similarly

$$\mathrm{Hom}_{S_\lambda}(X_\lambda, Y_\lambda) = \mathrm{Hom}_{A_\lambda\text{-alg.}}(C_\alpha \otimes_{A_\alpha} A_\lambda, \Gamma(X_\lambda, \mathcal{O}_{X_\lambda})) = \mathrm{Hom}_{A_\alpha\text{-alg.}}(C_\alpha, \Gamma(X_\lambda, \mathcal{O}_{X_\lambda})).$$

But as X_α is quasi-compact and quasi-separated, one knows (8.5.4) that $\varinjlim \Gamma(X_\lambda, \mathcal{O}_{X_\lambda}) = \Gamma(X, \mathcal{O}_X)$; since C_α is an A_α -algebra of finite presentation, the fact that (8.8.1.1) is bijective then follows from (8.8.2.3). $\square$

Before passing to the proof of (ii), let us note the following corollaries of (i): $\square$

Corollary (8.8.4). — Suppose S_0 quasi-compact, X_α of finite presentation over S_α , Y_α of finite type over S_α and quasi-separated over S_α (which is the case for example if Y_α is also of finite presentation over S_α). Let $f_\alpha : X_\alpha \rightarrow Y_\alpha$ be an S_α -morphism. For $f : X \rightarrow Y$ to be an isomorphism, it is necessary and sufficient that there exist $\lambda \geq \alpha$ such that $f_\lambda : X_\lambda \rightarrow Y_\lambda$ be an isomorphism. IV-3 | 32

Proof. The condition is evidently sufficient; conversely, note that since the question is local on S_0 (since L is filtered), one can always suppose S_0 affine, hence quasi-separated. There is by hypothesis an S -morphism $g : Y \rightarrow X$ such that $g \circ f = 1_X$ and $f \circ g = 1_Y$. On the other hand, it follows also from (8.8.2), (i) and from the relations $g \circ f = 1_X$ and $f \circ g = 1_Y$ that there exists $\nu \geq \mu$ such that $g_\nu \circ f_\nu = 1_{X_\nu}$ and $f_\nu \circ g_\nu = 1_{Y_\nu}$, since X_α and Y_α are of finite type over S_α and quasi-compact. This means that $f_\nu : X_\nu \rightarrow Y_\nu$ is an isomorphism, whence the corollary. $\square$

Corollary (8.8.5). — Suppose S_0 quasi-compact and quasi-separated, X_α and Y_α of finite presentation over S_α . For every S -isomorphism $f : X \xrightarrow{\sim} Y$, it is necessary and sufficient that there exist $\lambda \geq \alpha$ such that X_λ and Y_λ be S_λ -isomorphic, and an S_μ -isomorphism $f_\mu : X_\mu \rightarrow Y_\mu$ such that $f = f_\mu \times 1_S$.

Proof. The condition is evidently sufficient; conversely, note that since the question is local on S_0 (since L is filtered), one can always suppose S_0 affine, hence quasi-separated. There is by hypothesis an S -morphism $g : Y \rightarrow X$ such that $g \circ f = 1_X$ and $f \circ g = 1_Y$. As X_α is of finite presentation over S_α , and Y_α quasi-compact and quasi-separated, it results also from (8.8.2), (i) that f is of the form $f_\mu \times 1_S$ for some $\mu \geq \alpha$ and an S_μ -morphism $f_\mu : X_\mu \rightarrow Y_\mu$. Similarly for g . Applying (8.8.2.4) one obtains that there exists $\nu \geq \mu$ such that $g_\nu \circ f_\nu = 1_{X_\nu}$ and $f_\nu \circ g_\nu = 1_{Y_\nu}$, and the conclusion follows. $\square$

Proof of (8.8.2), (ii). Consider first the case where $S_0 = \text{Spec}(A_0)$ and $X = \text{Spec}(B)$ are affine; then the $S_\lambda = \text{Spec}(A_\lambda)$ and $S = \text{Spec}(A)$ are also affine, with $A = \varinjlim A_\lambda$. As X is of finite presentation over S , B is an A -algebra of finite presentation, hence by (8.8.2.3) applied to $G = B$ and $F_\lambda = A_\lambda$, there exist λ and an A_λ -algebra B_λ of finite presentation such that $B \simeq B_\lambda \otimes_{A_\lambda} A$. Setting $X_\lambda = \text{Spec}(B_\lambda)$, we obtain an S_λ -prescheme of finite presentation such that $X \simeq X_\lambda \times_{S_\lambda} S$.

To prove (ii) in the general case, note that S is quasi-compact and quasi-separated, and X is of finite presentation over S . As X is quasi-compact, it is a finite union of open affines, and by the affine case there exist λ and affine S_λ -preschemes of finite presentation that descend each piece. By (8.8.2), (i) and the fact that L is filtered, we can glue these pieces for a large enough λ to obtain X_λ of finite presentation over S_λ with $X \simeq X_\lambda \times_{S_\lambda} S$. $\square$

(8.8.3). The essential content of (8.8.2) is expressed again by saying that if S_0 is quasi-compact and quasi-separated, the category of S -preschemes of finite presentation is determined up to equivalence by the data of the categories of S_λ -preschemes of finite presentation, the functors $\rho_{\mu\lambda} : X_\lambda \mapsto X_\lambda \times_{S_\lambda} S_\mu$ (for $\lambda \leq \mu$) between these categories and the transitivity isomorphisms $\rho_{\mu\lambda} \circ \rho_{\nu\mu} \xrightarrow{\sim} \rho_{\nu\lambda}$. In the form thus stated, one can say that the data of an S_λ -prescheme of finite presentation X is equivalent “functorially” to the data of an S_μ -prescheme of finite presentation X'_μ ; if an S_μ -prescheme X''_μ is such that X is S -isomorphic to $X''_\mu \times_{S_\mu} S$, there exists ν such that $\lambda \leq \nu$ and $X'_\lambda \times_{S_\lambda} S_\nu$ and $X''_\mu \times_{S_\mu} S_\nu$ are S_ν -isomorphic. The fact that L is filtered implies in addition that if one gives oneself a finite family $(X^{(i)})_{i \in I}$ of S -preschemes of finite presentation, and a finite family $(f^{(ij)})_{(i,j) \in J}$ of S -morphisms between these preschemes $(f^{(ij)})$ being therefore of the form $X^{(\sigma(j))} \rightarrow X^{(\tau(j))}$ where σ and τ are two maps from J to I , then there is an index $\lambda \in L$, a family $(X^{(i)}_\lambda)_{i \in I}$ of S_λ -preschemes of finite presentation and a family $(f^{(ij)}_\lambda)_{(i,j) \in J}$ of S_λ -morphisms such that $X^{(i)}$ is identified with $X^{(i)}_\lambda \times_{S_\lambda} S$ and $f^{(ij)}$ with $f^{(ij)}_\lambda \times 1_S$; furthermore, if one has a relation $f^{(ij)} = f^{(ik)}$, one can suppose λ chosen such that $f^{(ij)}_\lambda = f^{(ik)}_\lambda$.

Consider in particular three S -preschemes of finite presentation X, Y, Z and two S -morphisms $f : X \rightarrow Y, g : Y \rightarrow Z$, so that the product $X \times_Z Y$ relative to these morphisms is also an S -prescheme

of finite presentation (I, 1.6.2). It then results from the preceding and from (I, 3.3.11) that one can write $X \times_Z Y = (X_\lambda \times_{Z_\lambda} Y_\lambda) \times_{S_\lambda} S$ for a convenient λ , $X_\lambda, Y_\lambda, Z_\lambda$ being S_λ -preschemes of finite presentation; one can therefore say that the determination of S -preschemes of finite presentation by the data of S_λ -preschemes of finite presentation is “compatible with fibre products”. We have seen on the other hand (8.6.3) that if g is an immersion, we can suppose the same for g_λ ; identifying Y (resp. Y_λ) with a sub-prescheme of Z (resp. Z_λ), we see that one can write, for a convenient λ , $f^{-1}(Y) = f_\lambda^{-1}(Y_\lambda) \times_{S_\lambda} S$ (I, 4.4.1); there is therefore also “compatibility with the formation of inverse images of sub-preschemes”. More particularly, if f_1, f_2 are two S -morphisms from X to Y , one calls the kernel of this pair of morphisms the inverse image N of the diagonal sub-prescheme of $Y \times_S Y$ by the S -morphism $(f_1, f_2)_S$; one deduces from what precedes that one will have for a convenient λ , $N = N_\lambda \times_{S_\lambda} S$, where N_λ is the kernel of the couple of S_λ -morphisms $(f_{1\lambda}, f_{2\lambda})$ of X_λ into Y_λ . These remarks extend to finite products and to “kernels” of arbitrary finite systems of S -morphisms from an S -prescheme of finite presentation into another; one concludes that in a general fashion the formation of S -preschemes of finite presentation by the data of S_λ -preschemes of finite presentation is “compatible with finite projective limits”, such a limit being by definition the kernel of a finite system of morphisms from one S -prescheme into a product of S -preschemes (T, 1.8).

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We shall allow ourselves in what follows to freely make the translations implied by the preceding properties (as well as by (8.3.11), (8.5.2), and (8.6.3)) without always striving to justify them as above. For example, the data of a “group prescheme” G of finite presentation over S is equivalent to the data of a group prescheme on S_λ for a large enough λ ; to write indeed the associativity condition for the S -morphism “composition law” $G \times_S G \rightarrow G$ of the group prescheme amounts to writing that the kernel of the two S -morphisms from $G \times_S G \times_S G \rightarrow G$ is equal to $G \times_S G \times_S G$ (II, 8.3.9), and the other conditions intervening in the definition of a group prescheme are interpreted similarly.

8.9. First applications to the elimination of Noetherian hypotheses

Proposition (8.9.1). — *Let A be a ring, X an A -prescheme.*

- (i) *The following conditions are equivalent:*
 - a) *X is of finite presentation over A .*
 - b) *There exist a Noetherian ring A_0 , a prescheme X_0 of finite type over A_0 , a ring homomorphism $A_0 \rightarrow A$, and an A -isomorphism $X_0 \otimes_{A_0} A \xrightarrow{\sim} X$.*
 - c) *There exist a subring A_0 of A , which is a $\mathbf{Z}$ -algebra of finite type, a prescheme X_0 of finite type over A_0 , and an A -isomorphism $X_0 \otimes_{A_0} A \xrightarrow{\sim} X$.*
- (ii) *If $\mathcal{F}$ is a quasi-coherent $\mathcal{O}_X$ -Module of finite presentation, one can suppose the subring A_0 such that there exists a coherent $\mathcal{O}_{X_0}$ -Module $\mathcal{F}_0$ such that $\mathcal{F}$ is isomorphic to $\mathcal{F}_0 \otimes_{A_0} A$; $\text{Supp}(\mathcal{F})$ is constructible and closed in X , and there is a closed sub-prescheme Z of X , having $\text{Supp}(\mathcal{F})$ as underlying space, such that the canonical immersion $Z \rightarrow X$ be of finite presentation.*
- (iii) *If Y is a second A -prescheme of finite presentation, and $f : X \rightarrow Y$ an A -morphism, one can suppose the subring A_0 of A such that there exists a prescheme Y_0 of finite type over A_0 , an A -isomorphism $Y_0 \otimes_{A_0} A \xrightarrow{\sim} Y$, and an A_0 -morphism $f_0 : X_0 \rightarrow Y_0$ (necessarily of finite type) such that f is identified with $f_0 \otimes 1_A$.*

Proof. (i) As A is the inductive limit of its subrings which are of finite type over $\mathbf{Z}$, a) implies c) by virtue of (8.8.2), (ii); c) implies b) since a $\mathbf{Z}$ -algebra of finite type is a Noetherian ring; finally, if A_0 is Noetherian, an A_0 -prescheme of finite type is of finite presentation (I, 1.6.1), hence b) implies a) by virtue of (I, 1.6.2), (iii).

The assertion (ii) follows immediately from (8.5.2), (ii), (8.3.11), and (I, 1.8.2), and the assertion (iii) from (8.8.2), (i). $\square$

(8.9.2). The proposition (8.9.1) and the results of (8.5.2) and (8.8.2) allow one to extend in numerous cases to morphisms of finite presentation $X \rightarrow Y$ the results proved by Noetherian techniques by supposing Y locally Noetherian. We shall see numerous examples in what follows and we shall limit ourselves here to giving a few results of this type.

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Proposition (8.9.3). — *Let A be a ring, M an A -module of finite presentation. Then every surjective A -endomorphism of M is bijective.*

Proof. Indeed, consider A as an inductive limit of its sub- $\mathbf{Z}$ -algebras of finite type. It results from (8.5.2.6) that there exists one of these sub-algebras A_0 and an A_0 -module M_0 of finite presentation such that M is A -isomorphic to $M_0 \otimes_{A_0} A$; furthermore, if $f : M \rightarrow M$ is a surjective A -endomorphism, one can suppose (8.5.2), (i) that there exists an A_0 -endomorphism $f_0 : M_0 \rightarrow M_0$ such that $f = f_0 \otimes 1_A$; finally (8.5.7) one can suppose that f_0 is surjective. But as A_0 is Noetherian and M_0 an A_0 -module of finite type, M_0 is a Noetherian module, hence (Bourbaki, *Alg.*, chap. VIII, §2, n° 2, lemma 3) f_0 is bijective, and it is the same for f by base change. $\square$

Proposition (8.9.4). — (“generic flatness theorem”). — *Let Y be an integral prescheme, $u : X \rightarrow Y$ a morphism of finite type and locally of finite presentation, $\mathcal{F}$ an $\mathcal{O}_X$ -Module quasi-coherent of finite presentation. Then there exists an open non-empty U of Y such that $\mathcal{F}|_{u^{-1}(U)}$ be flat over U .*

Proof. The reasoning from the beginning of (6.9.1) leads to proving the

Lemma (8.9.4.1). — *Let A be an integral ring, B an A -algebra of finite presentation, M a B -module of finite presentation. Then there exists $f \neq 0$ in A such that M_f is a free A_f -module.*

Proof. Indeed, by (8.9.1) there is a finitely generated $\mathbf{Z}$ -subalgebra A_0 of A , a finite-type A_0 -algebra B_0 , and a finite B_0 -module M_0 such that B is A -isomorphic to $B_0 \otimes_{A_0} A$ and M is B -isomorphic to $M_0 \otimes_{A_0} A$; by (6.9.2), there exists $f_0 \neq 0$ in A_0 such that $(M_0)_{f_0}$ is a free $(A_0)_{f_0}$ -module. As $M_{f_0} = (M_0)_{f_0} \otimes_{A_0} A$ and $A_{f_0} = (A_0)_{f_0} \otimes_{A_0} A$, M_{f_0} is a free A_{f_0} -module. $\square$

Corollary (8.9.5). — *Let S be a quasi-compact prescheme and quasi-separated, $u : X \rightarrow S$ a morphism of finite presentation, $\mathcal{F}$ an $\mathcal{O}_X$ -Module quasi-coherent of finite presentation. Then there exists a partition $(S_i)_{1 \leq i \leq n}$ of S into a finite number of locally closed sets in S such that, for $1 \leq i \leq n$, there exists a sub-prescheme S'_i of S , having S_i as underlying space, of finite presentation over S_i , such that if one sets $X_i = X \times_S S'_i$, the $\mathcal{O}_{X_i}$ -Module $\mathcal{F}_i = \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'_i}$ be flat over S'_i .*

Proof. Consider a finite covering $(U_j)_{1 \leq j \leq m}$ of S by open affines and define by induction $T_1 = U_1$, $T_j = U_j - \bigcup_{h < j} (U_j \cap U_h)$ for $j \geq 2$; each T_j is closed in the open affine U_j , and the T_j are pairwise disjoint; furthermore the T_j are also quasi-compact since S is quasi-separated, and consequently (since S is also quasi-compact), they are constructible in S (I, 1.8.1), and hence also each of the T_j . It will therefore evidently suffice to prove the corollary for a sub-prescheme T'_j of S having T_j as underlying space and for the morphism and the Module deduced respectively from u and $\mathcal{F}$ by the base change $T'_j \rightarrow S$. Note for this that if on U_j one takes the prescheme structure induced by that of S , the open immersion $U_j \rightarrow S$ is quasi-separated since S is quasi-separated (I, 1.2.7), hence it is of finite presentation. Then (8.9.1), with the same notations, there exist λ , a morphism of finite type $u_\lambda : X_\lambda \rightarrow S_\lambda$ and an $\mathcal{O}_{X_\lambda}$ -Module coherent $\mathcal{F}_\lambda$ such that X is isomorphic to $X_\lambda \times_{S_\lambda} S$, $\mathcal{F}$ to $\mathcal{F}_\lambda \otimes_{\mathcal{O}_{S_\lambda}} \mathcal{O}_X$. One can then apply to S_λ , X_λ , and $\mathcal{F}_\lambda$ the result of (6.9.3) and it follows that there is a partition of S_λ into sub-preschemes $S'_{\lambda i}$ of S_λ , whose underlying sets form a partition of S_λ , and which are such that $\mathcal{F}_M = \mathcal{F}_\lambda \otimes_{\mathcal{O}_{S_\lambda}} \mathcal{O}_{S'_{\lambda M}}$ be flat over $S'_{\lambda M}$. The $S'_i = S'_{\lambda i} \times_{S_\lambda} S$ are then sub-preschemes of S satisfying the condition by virtue of (2.1.4). $\square$

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8.10. Properties of permanence of morphisms by passage to the projective limit In this section, we preserve the general hypotheses and the notations of (8.8.1).

Proposition (8.10.1). — *If there exists λ such that, for $\mu \geq \lambda$, f_μ is an open morphism (resp. universally open), then f is an open morphism (resp. universally open).*

Proof. As $X = X_\lambda \times_{Y_\lambda} Y$, the assertion relative to universally open morphisms is a particular case of (2.4.3), (iii). Let us therefore suppose f_μ open for $\mu \geq \lambda$; it suffices to see that for every open quasi-compact U of X , $f(U)$ is open in Y . Now, there exists $\mu \geq \lambda$ such that $U = v_\mu^{-1}(U_\mu)$, where U_μ is an open quasi-compact in X_μ (8.2.11); one has then $f(v_\mu^{-1}(U_\mu)) = w_\mu^{-1}(f_\mu(U_\mu))$ (I, 3.4.8), hence $f(U)$ is open in Y . $\square$

Corollary (8.10.2). — *Let $f : X \rightarrow Y$ be a morphism. For f to be universally open, it suffices that, for every integer $n > 0$, if one sets $Y_n = Y \otimes_{\mathbf{Z}} \mathbf{Z}[T_1, \dots, T_n] (= \mathbf{V}_Y^n)$, and $X_n = X \times_Y Y_n$, $f_n = f \times 1_{Y_n} : X_n \rightarrow Y_n$, the projection f_n be an open morphism.*

Proof. To prove that f is universally open, it suffices to prove that it remains so for every affine U of Y ; by hypothesis, if $U_n = U \otimes_{\mathbf{Z}} \mathbf{Z}[T_1, \dots, T_n]$ is the inverse image of U in Y_n , the morphism $f_n^{-1}(U_n) \rightarrow U_n$, restriction of f_n , is open, and one sees that one can restrict to the case where $Y = \text{Spec}(A)$ is affine. Furthermore, it suffices evidently to show that for every morphism $Y' \rightarrow Y$, where $Y' = \text{Spec}(A')$ is also affine, $f' = f|_{Y'}$ is open. Let us suppose first that A' is an A -algebra of finite type, hence a quotient of a polynomial algebra $A[T_1, \dots, T_n]$; then Y' is a closed sub-prescheme of Y_n and f' the restriction of f_n to $f_n^{-1}(Y')$; but for every open V of X_n , one has $f_n(V \cap f_n^{-1}(Y')) = f_n(V) \cap Y'$, and as by hypothesis $f_n(V)$ is open in Y_n , this shows that f' is also an open morphism. When A' is arbitrary, it can be considered as an inductive limit of its sub- A -algebras of finite type A'_λ , and the fact that f' is an open morphism follows from the preceding and from (8.10.1). $\square$

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Proposition (8.10.3). — Suppose that there exists α such that: 1° S_α is quasi-compact; 2° the morphisms $X_\alpha \rightarrow S_\alpha$, $Y_\alpha \rightarrow S_\alpha$ are quasi-compact and the morphism $Y_\alpha \rightarrow S_\alpha$ quasi-separated; 3° for $\alpha \leq \lambda \leq \mu$, the morphisms $u_{\lambda\mu} : S_\mu \rightarrow S_\lambda$ are flat; 4° $f_\alpha(X_\alpha)$ is constructible in Y_α . Then, for f to be dominant, it is necessary and sufficient that there exist $\lambda \geq \alpha$ such that f_λ be dominant.

Proof. The hypotheses imply that Y_α is quasi-compact and that the morphism f_α is quasi-compact (I, 1.2.4); hence $f_\alpha(X_\alpha) = Z_\alpha$ is pro-constructible (I, 1.9.5), (vii) in Y_α . If one sets $Z_\lambda = w_{\alpha\lambda}^{-1}(Z_\alpha)$ for $\lambda \geq \alpha$ and $Z = w_\alpha^{-1}(Z_\alpha)$, we have $Z_\lambda = f_\lambda(X_\lambda)$ and $Z = f(X)$ (I, 3.4.8). As Z_λ is pro-constructible in Y_λ (8.3.11), the canonical map $\varinjlim \mathfrak{C}(S_\lambda) \rightarrow \mathfrak{C}(S)$ is injective, which proves the theorem in the case (vi). It suffices then to apply (8.3.12) by replacing S_λ , Z'_λ , and Z''_λ by Y_λ , Y_λ , and Z_λ respectively. $\square$

Proposition (8.10.4). — Suppose that there exists α such that Y_α is quasi-compact and f_α is of finite type and quasi-separated. For the morphism f to be separated, it is necessary and sufficient that there exist $\lambda \geq \alpha$ such that f_λ be separated.

Proof. The question being local on Y_α (since Y_α is quasi-compact and L filtered), one can restrict to the case where Y_α is affine, hence quasi-separated, and the hypothesis implies that X_α (hence the X_λ and X) are quasi-compact and quasi-separated. Set $X'_\lambda = X_\lambda \times_{Y_\lambda} X_\lambda$ for $\lambda \geq \alpha$ and $X' = X \times_Y X$; the first projection $X'_\alpha \rightarrow X_\alpha$ is quasi-compact and quasi-separated by hypothesis (I, 1.2.3), (iii), hence X'_λ is quasi-compact and quasi-separated. Note now that if we denote by Δ_λ (resp. Δ) the diagonal of $X_\lambda \times_{Y_\lambda} X_\lambda$ (resp. $X \times_Y X$), it results from (I, 5.3.4) and (3.4.8) that Δ_λ is the inverse image of Δ_α by the map $v'_{\lambda\alpha} : X'_\lambda \rightarrow X'_\alpha$ (resp. $v'_\lambda : X' \rightarrow X'_\alpha$). On the other hand, Δ_α is constructible in X'_α : indeed, since f_α is quasi-separated, the diagonal immersion $X_\alpha \rightarrow X'_\alpha$ is quasi-compact, and locally of finite presentation since f_α is of finite type (I, 1.4.3) and (I, 1.5.4), (iii); it results then from (I, 1.8.4.1) that Δ_λ is locally constructible, hence constructible since X'_λ is quasi-compact. One can now apply (8.3.12) by replacing S_λ and Z_λ by X'_λ and Δ_λ respectively. $\square$

Theorem (8.10.5). — Suppose S_0 quasi-compact, X_α and Y_α of finite presentation over S_α , and let $f_\alpha : X_\alpha \rightarrow Y_\alpha$ be an S_α -morphism. Consider, for a morphism, the property of being:

- (i) an isomorphism;
- (i bis) a monomorphism;
- (ii) an immersion;
- (iii) an open immersion;
- (iv) a closed immersion;
- (v) separated;
- (vi) surjective;
- (vii) radicial;
- (viii) affine;
- (ix) quasi-affine;
- (x) finite;
- (xi) quasi-finite;
- (xii) proper.

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Then, if $\mathbf{P}$ denotes one of the preceding properties, for f to possess the property $\mathbf{P}$, it is necessary and sufficient that there exist $\lambda \geq \alpha$ such that f_λ possess the property $\mathbf{P}$ (in which case f_μ also possesses it for $\mu \geq \lambda$).

If S_0 is furthermore supposed quasi-separated, the same conclusion holds when $\mathbf{P}$ is the property of being:

- (xiii) *projective*;
- (xiv) *quasi-projective*.

Proof. The case where **P** is one of the properties (i) or (v) was not inserted in the statement for the record; in these cases, the theorem follows from what has been proved respectively in (8.8.4) and (8.10.4). Furthermore, taking into account (I, 5.4.1) and (5.3.4), (v) also results from (iv). The case (i bis) follows immediately from (i), by using (I, 5.3.8) and by noting (as we have already used in (8.10.4)) that the diagonal Δ_f is deduced from Δ_{f_λ} by the base change $S \rightarrow S_\lambda$.

We will note on the other hand that (vi), (vii), and (xi) are in fact conditions on the fibres $f^{-1}(y)$ of the morphisms considered, taking into account the transitivity of the fibres by base change (I, 3.6.4): condition (vi) means indeed that all the fibres must be non-empty, condition (vii) that they must be radicial (I, 3.5.8), and condition (xi) that they must be finite (II, 6.2.2) and (6.2.3) and (I, 6.4.4), taking into account that f and the f_λ are morphisms of finite type by (I, 1.5.4), (v). The theorem in these three cases will be yet another consequence of a general result on this type of properties, which will be established in (9.3.3); we therefore postpone until then the proof of the theorem in the case (xi) (of course, the reader will be able to verify that, except for nos 8.11 and 8.12, we will make no use of the theorem in these cases until (9.3.3) and that (8.11) and (8.12) will not be used before (9.3.3)).

For the cases that remain to be proved, one can restrict to proving that the condition of the statement is necessary, all the properties **P** considered being invariant by base change (see chap. I^{er} and II for the references concerning each of these properties). One can furthermore suppose that $S_\alpha = S_0$ and that $Y_\alpha = S_\alpha$, hence $Y_\lambda = S_\lambda$ for every $\lambda \geq \alpha$. Finally, the properties (i) through (xii) are local on S_0 , hence as S_0 is a finite union of open affines and L filtered, one can restrict to proving them for $S_0 = \text{Spec}(A_0)$ affine (hence quasi-separated). We shall denote by A_λ (resp. A) the ring of S_λ (resp. S).

Cases (ii), (iii), (iv): Suppose that f is an immersion (resp. an open immersion, a closed immersion), and let X' be the sub-prescheme (resp. induced on an open, resp. closed) of S associated to f , which is therefore an S -prescheme of finite presentation. By virtue of (8.6.3), there then exists $\lambda \geq \alpha$ and a sub-prescheme X'_λ of S_λ , of finite presentation over S_λ , such that X' is isomorphic to $X'_\lambda \times_{S_\lambda} S$. For every $\mu \geq \lambda$, $X'_\mu = X'_\lambda \times_{S_\lambda} S_\mu$ is therefore a sub-prescheme (resp. induced on an open, resp. closed) of S_μ , and it results therefore from (8.8.2.4) and (8.8.5) that there exists $\mu \geq \lambda$ and an isomorphism $g_\mu : X_\mu \rightarrow X'_\mu$ such that $g = g_\mu \times 1_S$ is the isomorphism $X \rightarrow X'$ associated to f ; whence the conclusion.

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Cases (vi) and (vii): We know (I, 1.8.1) that $Z_\alpha = f_\alpha(X_\alpha)$ is constructible in S_α ; if we set $Z_\lambda = u_{\alpha\lambda}^{-1}(Z_\alpha)$ for $\lambda \geq \alpha$ and $Z = u_\alpha^{-1}(Z_\alpha)$, we have $Z_\lambda = f_\lambda(X_\lambda)$ and $Z = f(X)$ (I, 3.4.8). As Z_α is pro-constructible in S_α , and by virtue of (8.3.11) the canonical map $\varinjlim \mathcal{C}(S_\lambda) \rightarrow \mathcal{C}(S)$ is injective, the relation $f(X) = S$ implies the existence of $\lambda \geq \alpha$ such that $f_\lambda(X_\lambda) = S_\lambda$, which proves the theorem in the case (vi).

Cases (viii) and (ix): As $S = \text{Spec}(A)$ is affine, to say that $f : X \rightarrow S$ is affine (resp. quasi-affine) means that there exists an integer r and a closed immersion (resp. an immersion) $j : X \rightarrow \mathbf{V}_S^r = \text{Spec}(A[T_1, \dots, T_r])$ of S -preschemes, since f is of finite type and S quasi-compact (II, 5.1.9). As $\mathbf{V}_S^r = \mathbf{V}_{S_\lambda}^r \times_{S_\lambda} S$, and $\mathbf{V}_{S_\lambda}^r$ is an S_0 -prescheme of finite presentation, it results from (8.8.2), (i) applied to the S -morphism j , that there exists λ and an S_λ -morphism $j_\lambda : X_\lambda \rightarrow \mathbf{V}_{S_\lambda}^r$ such that $j = j_\lambda \times 1_S$; applying (iv) (resp. (ii)) to j , one deduces that there exists $\mu \geq \lambda$ such that j_μ is a closed immersion (resp. an immersion); hence f_μ is affine (resp. quasi-affine).

Case (x): By hypothesis, $X = \text{Spec}(B)$, where B is an A -algebra which is an A -module of finite presentation (I, 1.4.7); it results therefore from (8.5.2), (ii) that there is λ and an A_λ -module of finite presentation B_λ such that the A -module B is isomorphic to $B_\lambda \otimes_{A_\lambda} A$. The structure of A -algebra of B is defined by an A -homomorphism $m : B \otimes_A B \rightarrow B$; as $B \otimes_A B = (B_\lambda \otimes_{A_\lambda} B_\lambda) \otimes_A A$, there exists by (8.5.2), (i) $\mu \geq \lambda$ and an A_μ -homomorphism $m_\mu : B_\mu \otimes_{A_\mu} B_\mu \rightarrow B_\mu$ such that $m = m_\mu \otimes 1$. Considering the usual diagrams expressing the associativity and the commutativity of m , one sees by applying again (8.5.2), (i) that one can suppose μ taken large enough so that m_μ defines on B_μ an associative and commutative multiplication. In the same way one can suppose μ large enough so that B_μ thus defined admits a unit element. If $X'_\mu = \text{Spec}(B_\mu)$, it is clear then that X is S -isomorphic

to $X'_\nu \times_{S_\nu} S$, hence, by virtue of (i), there exists $\rho \geq \nu$ such that X_ρ and X'_ρ are S_ρ -isomorphic, which completes the proof in this case.

To prove the theorem in the case (xii), we shall first prove the following proposition: $\square$

Proposition (8.10.5.1). — (Chow's lemma for morphisms of finite presentation). — Let A be a ring, X, Y two A -preschemes of finite presentation, $f : X \rightarrow Y$

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8.11. Application to quasi-finite morphisms We propose in this section to prove the two following theorems:

Theorem (8.11.1). — Let $f : X \rightarrow Y$ be a proper morphism, locally of finite presentation, and quasi-finite. Then the morphism f is finite.

Theorem (8.11.2). — Let $f : X \rightarrow Y$ be a morphism locally of finite presentation, quasi-finite and separated. Then the morphism f is quasi-affine, and a fortiori quasi-projective.

Remark (8.11.3). — We will see below that one can reduce, for the proof of (8.11.1) and (8.11.2), to the case where Y is locally Noetherian; one will note that in this case one will thus obtain another proof of the theorem of Chevalley (III, 4.4.2).

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(8.11.4). The hypotheses and the conclusions of (8.11.1) and (8.11.2) are all local on Y . Indeed, this follows from (I, 1.6.1), (I, 1.2.6), (II, 5.1.1), (II, 5.4.1), and (II, 6.2.2). Therefore one can suppose $Y = \text{Spec}(A)$ affine. One knows that there then exists a subring A_0 of A , which is a $\mathbf{Z}$ -algebra of finite type, and a morphism of finite type $f_0 : X_0 \rightarrow \text{Spec}(A_0)$ such that X identifies with $X_0 \otimes_{A_0} A$ and f with $f_0 \times 1$ (8.9.1). Furthermore, A can be considered as inductive limit of its subrings containing A_0 and which are $\mathbf{Z}$ -algebras of finite type; using the method of (8.1.2), c)) as well as (8.10.5), (v), (xi) and (xii)), one sees that it suffices to prove the theorems (8.11.1) and (8.11.2) for f_0 . Suppose henceforth Y Noetherian; using now the method of (8.1.2), a)) as well as (8.10.5), (v), (ix), (x), (xi) and (xii)) one can replace Y by $\text{Spec}(\mathcal{O}_y)$, where y is an arbitrary point of Y , so one can finally suppose that $Y = \text{Spec}(A)$, where A is a local Noetherian ring. Let $\mathfrak{m}$ be the maximal ideal of A , and let $\hat{A}$ be the completion of A for the $\mathfrak{m}$ -adic topology. The ring $\hat{A}$ is Noetherian local and $\text{Spec}(\hat{A}) \rightarrow \text{Spec}(A)$ is faithfully flat and quasi-compact (0, 7.3.5); applying (2.7.1), (i), (vii), (xiv), (xv) and (xvi), we may therefore reduce to the case where A is complete. It then follows from (II, 6.2.6) that $X = X' \sqcup X''$, where X' is a Y -scheme quasi-finite and X'' a Y -scheme quasi-finite such that $X'' \cap f^{-1}(y) = \emptyset$.

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Let us first place ourselves in the hypotheses of (8.11.1); as f is proper, X' , which is closed in X , is proper over Y (II, 5.4.10)), so $f(X'')$ is closed in Y ; but y is not contained in $f(X'')$, and moreover $f(X'')$ is adherent to every point of Y , so $f(X'') = \emptyset$, and consequently X'' is empty and f is finite.

Let us now place ourselves in the hypotheses of (8.11.2) and, restricting ourselves further (as one can do from what precedes) to the case where $Y = \text{Spec}(A)$ is affine and Noetherian of finite dimension, let us moreover reason by induction on the dimension of Y . Reducing as above to the case where A is furthermore local and complete, one has $\dim(\mathcal{O}_y) = \dim(A) = \dim(Y)$ and for all $z \neq y$, $\dim(\mathcal{O}_z) < \dim(\mathcal{O}_y)$, so $\dim(Y - \{y\}) < \dim(Y)$. Now, by hypothesis one has $f(X'') \subset Y - \{y\}$ and the restriction of f to X'' is evidently a quasi-finite and separated morphism; applying the hypothesis of induction to $Y - \{y\}$ and to X'' , one sees that X'' is quasi-affine over $Y - \{y\}$; but the open subset $Y - \{y\}$ being quasi-affine over Y since Y is Noetherian, X'' is also quasi-affine over Y (II, 5.1.10), (ii)); as moreover X' is finite (and a fortiori affine) over Y , X is quasi-affine over Y (II, 4.6.17) and (II, 5.1.2), c')).

Proposition (8.11.5). — Let $f : X \rightarrow Y$ be a morphism of finite presentation. The following properties are equivalent:

- a) f is a closed immersion.
- b) f is a proper monomorphism.
- c) f is proper and for all $y \in Y$, $f^{-1}(y)$ is radicial and geometrically reduced over $k(y)$ (that is to say empty or $k(y)$ -isomorphic to $\text{Spec}(k(y))$).

It is clear that a) implies b). To see that b) implies c), let us note (I, 3.3.12) that for all $y \in Y$, the morphism $f^{-1}(y) \rightarrow \text{Spec}(k(y))$ deduced from f by extension of the base is a monomorphism,

hence injective, and consequently $f^{-1}(y)$ is empty or reduced to a point, and in every case affine. Furthermore, if A is the ring of $f^{-1}(y)$, the canonical homomorphism $A \otimes_k A \rightarrow A$ is bijective (I, 5.3.8). This evidently implies that $A = k$, for otherwise there would exist an element $a \in A$ not in k and the images of $a \otimes 1$ and of $1 \otimes a$ in A would both be equal to a , so $a \otimes 1 = 1 \otimes a$ since 1 and a form a free system over k .

It remains to prove that $c)$ implies $a)$. It follows first from (8.11.1) that f is a finite morphism, so $X = \text{Spec}(\mathcal{A})$, where $\mathcal{A}$ is a finite $\mathcal{O}_Y$ -Algebra. It suffices therefore to prove that the canonical homomorphism $\mathcal{O}_Y \rightarrow \mathcal{A}$ is surjective (II, 1.4.10), or equivalently that for all $y \in Y$, the homomorphism $\mathcal{O}_y \rightarrow \mathcal{A}_y$ is surjective. But by hypothesis $f^{-1}(y) = \text{Spec}(\mathcal{A}_y/\mathfrak{m}_y\mathcal{A}_y)$ (II, 1.5.5) is such that the corresponding homomorphism $k(y) = \mathcal{O}_y/\mathfrak{m}_y \rightarrow \mathcal{A}_y/\mathfrak{m}_y\mathcal{A}_y$ is bijective; as $\mathcal{A}_y$ is an $\mathcal{O}_y$ -module of finite type, the lemma of Nakayama shows that $\mathcal{O}_y \rightarrow \mathcal{A}_y$ is surjective, which completes the proof.

Remark (8.11.5.1). — One will note that the preceding reasoning proves that if f is a monomorphism, then, for all $y \in Y$, $f^{-1}(y)$ is empty or $k(y)$ -isomorphic to $\text{Spec}(k(y))$.

Proposition (8.11.6). — If a morphism $f : X \rightarrow Y$ of finite presentation is a universal homeomorphism, it is finite, surjective and radicial (the converse being true by (2.4.5), (i)).

Proof. Indeed, f being of finite type, universally closed, and separated by virtue of (2.4.4), is proper by definition (II, 5.4.1)). As it is evidently quasi-finite (II, 6.2.3)), it is finite by (8.11.1). One knows moreover that it is radicial (2.4.4), and evidently surjective. $\square$

8.12. New proof and generalisation of the “Main Theorem” of Zariski

Lemma (8.12.1). — Let $f : X \rightarrow Y$ be a quasi-compact and quasi-separated morphism, $\mathcal{C}$ an $\mathcal{O}_Y$ -Algebra quasi-coherent, $Z = \text{Spec}(\mathcal{C})$, which is an affine Y -prescheme over Y . Let $g : X \rightarrow Z$ be a Y -morphism, $\varphi = \mathcal{A}(g) : \mathcal{C} \rightarrow f_*(\mathcal{O}_X)$ the corresponding $\mathcal{O}_Y$ -homomorphism of $\mathcal{O}_Y$ -Algebras (II, 1.2.7). Suppose that g is an immersion. Then, for the closed image of X by g (I, 9.5.3)) to equal Z , it is necessary and sufficient that φ be injective; g is then an open immersion.

Proof. The hypothesis implies that $f_*(\mathcal{O}_X)$ is an $\mathcal{O}_Y$ -Algebra quasi-coherent (1.7.5)); furthermore, as the canonical morphism $h : Z \rightarrow Y$ is affine, hence quasi-compact and quasi-separated (1.2.2) and (1.1.2)), $g_*(\mathcal{O}_X)$ is an $\mathcal{O}_Z$ -Algebra quasi-coherent (1.7.5)). This being so, saying that the closed image of X by g equals Z signifies (I, 9.5.2)) that the canonical homomorphism $\theta : \mathcal{O}_Z \rightarrow g_*(\mathcal{O}_X)$ is injective. But one has $h_*(\mathcal{O}_Z) = \mathcal{C}$ by definition of Z (II, 1.3.1)), and $h_*(g_*(\mathcal{O}_X)) = f_*(\mathcal{O}_X)$. As Z is affine over Y , it amounts to the same to say that the homomorphism $\theta : \mathcal{O}_Z \rightarrow g_*(\mathcal{O}_X)$ is injective or that the corresponding homomorphism $\varphi = h_*(\theta) : \mathcal{C} \rightarrow f_*(\mathcal{O}_X)$ is injective (I, 1.3.9)). The fact that g is then an open immersion follows from (I, 9.5.10)) and the hypothesis. $\square$

Lemma (8.12.2). — Let Y be a prescheme quasi-compact and quasi-separated, $f : X \rightarrow Y$ a quasi-separated morphism of finite type, $\mathcal{C}$ an $\mathcal{O}_Y$ -Algebra quasi-coherent, $Z = \text{Spec}(\mathcal{C})$. Let $g : X \rightarrow Z$ be a Y -morphism, $\varphi : \mathcal{C} \rightarrow f_*(\mathcal{O}_X)$ the corresponding $\mathcal{O}_Y$ -homomorphism of $\mathcal{O}_Y$ -Algebras. Let $(\mathcal{C}_\lambda)$ be the filtered increasing family of sub- $\mathcal{O}_Y$ -Algebras quasi-coherent of finite type of $\mathcal{C}$ (where $\mathcal{C}$ is the union (I, 9.6.6) and (1.7.9))); set $Z_\lambda = \text{Spec}(\mathcal{C}_\lambda)$ and let g_λ be the morphism composed $X \xrightarrow{g} Z \rightarrow Z_\lambda$. Then the following conditions are equivalent:

- a) g is an immersion.
- b) There exists λ such that g_λ is an immersion.

Furthermore, when g_λ is an immersion, the same is true of g_μ for $\mu \geq \lambda$.

It suffices to apply (II, 3.8.4)) by replacing $\mathcal{L}$ by $\mathcal{O}_X$ and $\mathcal{S}$ by $\mathcal{C}[\mathbf{T}]$, and taking into account (II, 3.1.7)).

Proposition (8.12.3). — Let Y be a prescheme quasi-compact and quasi-separated, $f : X \rightarrow Y$ a separated morphism of finite type. Let $\mathcal{B} = f_*(\mathcal{O}_X)$, which is an $\mathcal{O}_Y$ -Algebra quasi-coherent (I, 9.2.2)); let $\mathcal{C}$ be the integral closure of $\mathcal{O}_Y$ in $\mathcal{B}$, which is an $\mathcal{O}_Y$ -Algebra quasi-coherent (II, 6.3.4)); set $Z = \text{Spec}(\mathcal{C})$, and let $g : X \rightarrow Z$ be the Y -morphism corresponding to the injection $\mathcal{C} \rightarrow \mathcal{B} = f_*(\mathcal{O}_X)$, and for all λ , let $g_\lambda : X \rightarrow Z_\lambda = \text{Spec}(\mathcal{C}_\lambda)$ be the Y -morphism corresponding to the injection $\mathcal{C}_\lambda \rightarrow \mathcal{B} = f_*(\mathcal{O}_X)$. Then the following conditions are equivalent:

- a) There exists a factorisation of f as

$$X \xrightarrow{f'} Y' \xrightarrow{u} Y$$

- a') There exists a factorisation $X \xrightarrow{f'} Y' \xrightarrow{u} Y$ of f , where f' is an open immersion and u a finite morphism.
- b) The morphism $g : X \rightarrow Z$ is an immersion.
- c) There exists λ such that $g_\lambda : X \rightarrow Z_\lambda$ is an immersion.

Furthermore, when g_λ is an immersion, g is an open immersion, $g(X)$ is dense in Z , and there exists λ such that for $\mu \geq \lambda$, g_μ is an open immersion.

where f' is an open immersion and u a finite morphism.

As the homomorphism $\varphi : \mathcal{C} \rightarrow f_*(\mathcal{O}_X)$ is injective, it follows from (8.12.1) that if g is an immersion, it is an open immersion and $g(X)$ is dense in Z , and the same is true for g_λ . The fact that a) implies $a')$ follows also from (8.12.1), applied by replacing Z by Y' and g by f' (since Y' is finite over Y , hence affine over Y): indeed if Y'' is the closed image of X by f' , Y'' is finite over Y and f' factorises as $X \xrightarrow{f''} Y'' \xrightarrow{j} Y'$, where j is the canonical injection, and f'' is an immersion (I, 4.1.10)); it follows therefore from (8.12.1) that f'' is an open immersion.

The equivalence of b) and c) follows from (8.12.2) as well as the fact that g_λ is then an immersion for λ large enough. It is clear that c) implies a), since Z_λ is finite over Y . Let us finally show that a) implies c). One has seen above that Y' is the closed image of X by f' , and it then follows from (8.12.1) that if one sets $\mathcal{B}' = u_*(\mathcal{O}_{Y'})$, the homomorphism $\varphi' : \mathcal{B}' \rightarrow \mathcal{B} = f_*(\mathcal{O}_X)$ is injective. But as by hypothesis $\mathcal{B}'$ is a finite $\mathcal{O}_Y$ -algebra, it identifies by definition of $\mathcal{B}$ with one of the $\mathcal{O}_Y$ -subalgebras $\mathcal{C}_\lambda$, which proves c).

We will say that a morphism $f : X \rightarrow Y$, where Y is quasi-compact and quasi-separated, is *pseudo-finite*, if it is of finite type and satisfies condition a) of (8.12.3) (in which case it is necessarily separated).

Corollary (8.12.4). — *Let Y be a prescheme quasi-compact and quasi-separated, $f : X \rightarrow Y$ a morphism.*

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- (i) Suppose f pseudo-finite. Then, for all morphisms $Y' \rightarrow Y$, where Y' is quasi-compact and quasi-separated, $f_{(Y')} : X' = X_{(Y')} \rightarrow Y'$ is pseudo-finite.
- (ii) Let (U_λ) be a covering of Y by quasi-compact open subsets. For f to be pseudo-finite, it is necessary and sufficient that for all λ , the restriction $f_\lambda : f^{-1}(U_\lambda) \rightarrow U_\lambda$ of f be a pseudo-finite morphism.
- (iii) Suppose furthermore Y Noetherian, and f of finite type. Then, for f to be pseudo-finite, it is necessary and sufficient that, for all $y \in Y$, the morphism $f_y = f \times 1 : X_y = X \times_Y \text{Spec}(\mathcal{O}_y) \rightarrow \text{Spec}(\mathcal{O}_y)$ be pseudo-finite.

Proof. (i) It is clear that $f_{(Y')}$ is of finite type (1.5.4). Moreover, a factorisation $X \xrightarrow{g} Z \xrightarrow{u} Y$, with g an immersion and u finite, induces a factorisation $X' \xrightarrow{g'} Z' \xrightarrow{u'} Y'$ of $f_{(Y')}$, where $Z' = Z_{(Y')}$, $g' = g_{(Y')}$ and $u' = u_{(Y')}$. The morphism g' is an immersion (I, 4.3.2) and u' is finite (II, 6.1.5); hence $f_{(Y')}$ is pseudo-finite.

(ii) The condition is necessary by virtue of (i), the U_λ being quasi-separated since Y is. To see that it is sufficient, let us observe (with the notations of (8.12.3)) that if one sets $X_\lambda = f^{-1}(U_\lambda)$, we have $\mathcal{B}|_{U_\lambda} = (f_\lambda)_*(\mathcal{O}_{X_\lambda})$, $\mathcal{C}|_{U_\lambda}$ is the integral closure of $\mathcal{O}_{U_\lambda}$ in $\mathcal{B}|_{U_\lambda}$, and consequently, if $h : Z \rightarrow Y$ is the canonical morphism, $Z_\lambda = \text{Spec}(\mathcal{C}_\lambda)$ identifies with $h^{-1}(U_\lambda) = \text{Spec}(\mathcal{C}|_{U_\lambda})$, the Y -morphism $g_\lambda : X_\lambda \rightarrow Z_\lambda = h^{-1}(U_\lambda)$ being the U_λ -morphism corresponding to f_λ . Applying the hypothesis of induction, one sees that X'' is quasi-affine over $Y - \{y\}$; but the open subset $Y - \{y\}$ being quasi-affine over Y since Y is Noetherian, X'' is also quasi-affine over Y (II, 5.1.10, (ii)); since on the other hand X' is finite (and *a fortiori* affine) over Y , X is quasi-affine over Y .

(iii) It suffices, in virtue of (ii), to prove that for all $y \in Y$ there exists a neighbourhood U of y such that the restriction $f^{-1}(U) \rightarrow U$ of f be pseudo-finite. Designate by (U_λ) the projective filtering system of affine open neighbourhoods of y , and apply the method of (8.1.2, a)). Since Y is Noetherian, the restrictions $f_\lambda : f^{-1}(U_\lambda) \rightarrow U_\lambda$ of f are of finite presentation and of the same nature as f_y . By hypothesis f_y factorises as $X_y \xrightarrow{g_y} Z_y \xrightarrow{u_y} \text{Spec}(\mathcal{O}_y)$, where u_y is finite and g_y is an open immersion. As $\mathcal{O}_y$ is Noetherian, it is of the same nature as Z_y , and as Z_y is of finite presentation over $\text{Spec}(\mathcal{O}_y)$, there exists λ and a morphism of finite presentation $u_\lambda : Z_\lambda \rightarrow U_\lambda$ such that Z_y identifies with $Z_\lambda \times_{U_\lambda} \text{Spec}(\mathcal{O}_y)$; furthermore, one can suppose λ taken such that g_λ is an open immersion and u_λ a finite morphism (8.10.5), (ii) and (x)), which proves that f_λ is pseudo-finite. $\square$

(8.12.5). We can now give the “Main theorem” of Zariski (III, 4.4.3)), a proof not using the “global” cohomological results of chap. III, but rather appealing to the finer properties of local Noetherian rings; we will furthermore generalise the statement of the theorem by removing the Noetherian hypotheses:

Theorem (8.12.6). — (“Main theorem” of Zariski). — *Let Y be a prescheme quasi-compact and quasi-separated. If a morphism $f : X \rightarrow Y$ is quasi-finite, separated and of finite presentation, there exists a factorisation of f*

$$(8.12.6.1) \quad X \xrightarrow{f'} Y' \xrightarrow{u} Y$$

where f' is an open immersion and u a finite morphism.

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Proof. By virtue of (8.12.4), (ii) and the local character (on Y) of the notions of quasi-finite morphism, separated and of finite presentation, one can restrict to the case where $Y = \text{Spec}(A)$ is affine. Applying (8.9.1), one can suppose that there is a subring A_0 of A , which is a $\mathbf{Z}$ -algebra of finite type, and an A -isomorphism $X_0 \otimes_{A_0} A \xrightarrow{\sim} X$, f identifying by this isomorphism with $f_0 \times 1$, where $f_0 : X_0 \rightarrow \text{Spec}(A_0)$ is a morphism of finite type; furthermore (8.10.5), (v) and (xi) one can suppose that f_0 is separated and quasi-finite; if one proves that f_0 is pseudo-finite, the same will be true of f by (8.12.4), (i). As A_0 is then Noetherian and the notions of morphism of finite type, separated and quasi-finite are preserved by base change, it follows from (8.12.4), (iii) that one can even suppose that A is a local ring, essentially of finite type over $\mathbf{Z}$ (1.3.8)). Set $n = \dim(A)$, and let us proceed by induction on n ; for $n = 0$, the theorem is evident, A being a field and the morphism f being already finite (II, 6.2.2)). Set $B = \Gamma(X, \mathcal{O}_X)$; denote by C the integral closure of A in B , set $Z = \text{Spec}(C)$ and let $g : X \rightarrow Z$ be the Y -morphism corresponding to the canonical A -homomorphism injective $C \rightarrow B$; by virtue of (8.12.3)), it suffices to show that g is an open immersion. Let a be the closed point of Y , and $U = Y - \{a\}$; U is Noetherian and all its local rings are essentially of finite type over $\mathbf{Z}$ and of dimension $< n$; taking into account the hypothesis of induction, one sees that the restriction $f^{-1}(U) \rightarrow U$ of f is a pseudo-finite morphism. One concludes from (8.12.3)) that, if $h : Z \rightarrow Y$ is the structural morphism, the restriction $f^{-1}(U) \rightarrow h^{-1}(U)$ of g is an open immersion. Set $A' = \hat{A}$, $Y' = \text{Spec}(A')$, $X' = X_{(Y')}$, $f' = f_{(Y')} : X' \rightarrow Y'$. As the canonical morphism $u : Y' \rightarrow Y$ is flat, it follows from (2.3.1)) that $B' = \Gamma(X', \mathcal{O}_{X'})$ identifies with the A' -algebra $B \otimes_A A'$. On the other hand, as A is a local ring excellent (7.8.3)), the morphism $u : Y' \rightarrow Y$ is regular, and consequently (6.14.4)) the integral closure of A' in B' is equal to $C' = C \otimes_A A'$. One sees therefore that $Z' = \text{Spec}(C')$ is equal to $Z_{(Y')}$ and the morphism $g' : X' \rightarrow Z'$ coming from the canonical injection $C' \rightarrow B'$ is equal to $g_{(Y')}$. As $u : Y' \rightarrow Y$ is faithfully flat and quasi-compact, to prove that g is an open immersion, it suffices to prove that g' is an open immersion (2.7.1), (x)). If $h' : Z' \rightarrow Y'$ is the canonical morphism, the fact that the restriction $f'^{-1}(U') \rightarrow h'^{-1}(U')$ of g is an open immersion implies that the same is true of the restriction $f'^{-1}(U') \rightarrow h'^{-1}(U')$ of g' . Now let us note that $u^{-1}(a)$ is reduced to the closed point a' of Y' and consequently $U' = Y' - \{a'\} = u^{-1}(U)$. If $h' : Z' \rightarrow Y'$ is the canonical morphism, it follows from (II, 6.2.4)); as A' is complete, one deduces from (II, 6.2.6)) that X' is Y' -isomorphic to a sum $X'_1 \sqcup X'_2$, where the restriction $f'|_{X'_1} : X'_1 \rightarrow Y'$ is a morphism finite, and $X'_2 \subset f'^{-1}(U')$. It follows from this that B' is the direct product of the two A' -algebras $\Gamma(X'_1, \mathcal{O}_{X'_1}) = B'_1$ and $\Gamma(X'_2, \mathcal{O}_{X'_2}) = B'_2$; one concludes immediately that the integral closure C' of A' in B' is the direct product of the integral closures C'_1, C'_2 of A' in B'_1, B'_2 respectively, whence $Z' = Z'_1 \sqcup Z'_2$, where $Z'_i = \text{Spec}(C'_i)$ ($i = 1, 2$); and the canonical morphism $g' : X' \rightarrow Z'$ is such that $g'|_{X'_i}$ is the canonical morphism $g'_i : X'_i \rightarrow Z'_i$ ($i = 1, 2$). But as B'_1 is already a finite A' -algebra, one has $C'_1 = B'_1$, and g'_1 is therefore an isomorphism. On the other hand, as $X'_2 \subset f'^{-1}(U')$ and is open in $f'^{-1}(U')$, one already knows that g'_2 is an open immersion. One concludes that g' is indeed an open immersion. $\square$

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Remark (8.12.7). — When, in (8.12.6)), one supposes that Y is an affine scheme, the proof by reduction to the Noetherian case shows that, in the factorisation (8.12.6.1)), the morphisms f' and u are also morphisms of finite presentation (1.6.2)).

Corollary (8.12.8). — *Let Y be a quasi-compact scheme such that there exists an $\mathcal{O}_Y$ -Module ample (II, 4.5.3)), $f : X \rightarrow Y$ a quasi-finite and quasi-projective morphism. Then there exists a factorisation of f as*

$$X \xrightarrow{f'} Y' \xrightarrow{u} Y$$

where f' is an open immersion and u a finite morphism.

Proof. The hypothesis implies that X identifies with a sub- Y -scheme quasi-compact of a Y -scheme of the form $Z = \mathbf{P}_Y^r$ (II, 5.3.3)). There is therefore a quasi-compact open neighbourhood U of X in Z such that X is closed in U ; as Z is a scheme, the canonical injection $U \rightarrow Z$ is a morphism of finite presentation ((1.2.7) and (1.6.2)), hence the composite morphism $g : U \rightarrow Z \rightarrow Y$ is also a morphism of finite presentation (the fact that $\mathbf{P}_Y^r$ is of finite presentation over Y following immediately from the definition (II, 4.1.1)). Let $\mathcal{J}$ be the quasi-coherent Ideal of $\mathcal{O}_U$ defining the closed sub-prescheme X ; as U is a quasi-compact scheme, $\mathcal{J}$ is the filtered inductive limit of its sub-Ideals quasi-coherent of finite type $\mathcal{J}_\lambda$ (I, 9.4.9)). If X_λ is the closed sub-prescheme of U defined by $\mathcal{J}_\lambda$, one therefore has $X = \bigcap X_\lambda$. For all $y \in Y$, one therefore has $f^{-1}(y) = \bigcap (X_\lambda \cap g^{-1}(y))$, and as the sets $X_\lambda \cap g^{-1}(y)$ are closed in the Noetherian space $g^{-1}(y)$, there exists for all y an index $\lambda(y)$ such that $f^{-1}(y) = X_{\lambda(y)} \cap g^{-1}(y)$. Denote by E_λ the set of $y \in Y$ such that the fibre $X_\lambda \cap g^{-1}(y)$ of the restriction of g to X_λ is a $k(y)$ -prescheme finite. The hypothesis that f is quasi-finite implies, by virtue of what precedes, that $Y = \bigcup_\lambda E_\lambda$. Now, each of the X_λ is, by definition, of finite presentation over Y ; it follows therefore from (9.2.3) and (9.2.6))⁽¹⁾ that the E_λ are constructible sets in the scheme Y ; as they form a filtered increasing family, there exists an index λ such that $E_\lambda = Y$ (1.9.9)), and for this λ , the morphism $f_\lambda : X_\lambda \rightarrow Y$, restriction of g to X_λ , is of finite presentation, finite and separated, hence one can apply (8.12.6)), and f_λ factorises as

$$X_\lambda \xrightarrow{j_\lambda} Y'_\lambda \xrightarrow{u_\lambda} Y$$

where j_λ is an open immersion and u_λ a finite morphism. As X is a closed sub-prescheme of X_λ , one has thus proved that f possesses the property (8.12.3, a)), whence the corollary by virtue of the equivalence of (8.12.3, a)) and (8.12.3, a')). $\square$

Corollary (8.12.9). — *Let $f : X \rightarrow Y$ be a locally quasi-finite morphism (Err_{III}, 20). For all $x \in X$ there exist an open neighbourhood U of x in X , an open neighbourhood V of $y = f(x)$ in Y , such that $f(U) \subset V$ and a factorisation*

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$$U \xrightarrow{f'} V' \xrightarrow{u} V$$

of the restriction of f to U , where f' is an open immersion and u a finite morphism.

Proof. It evidently suffices to take for V an affine neighbourhood of y in Y , for U an affine neighbourhood of x in X contained in $f^{-1}(V)$ and such that $f|U$ is quasi-finite. The morphism $U \rightarrow V$ restriction of f being then affine (hence quasi-projective), one can apply (8.12.8)). $\square$

Corollary (8.12.10). — *Let Y be an integral normal prescheme, X an integral prescheme, and $f : X \rightarrow Y$ a birational, locally quasi-finite morphism (Err_{III}, 20). Then f is an open immersion ouverte and furthermore it is necessary and sufficient that f be separated.*

Proof. The second assertion follows immediately from the first and from (I, 8.2.8)). To prove the first assertion, one may suppose X and Y affine and f quasi-finite; consider the factorisation $f = u \circ f'$ of (8.12.8), which identifies X , via f' , with the subprescheme induced on an open subset of Y' . As X is integral, one can, by virtue of (I, 5.2.3)), replace Y' by the reduced sub-prescheme of Y' having underlying space $\overline{X}$, hence one can also suppose that Y' is integral. Furthermore, since f is birational, u is also. The conclusion then follows from the following lemma: $\square$

Lemma (8.12.10.1). — *Let Y' be an integral prescheme, Y an integral prescheme and normal; then a morphism $u : Y' \rightarrow Y$ finite and birational is an isomorphism.*

¹¹(1) The reader will verify that the corollaries (8.12.8) to (8.12.11) are not used in § 9.

Proof. Set $\mathcal{A} = u_*(\mathcal{O}_{Y'})$; then $\mathcal{A}$ is a finite $\mathcal{O}_Y$ -algebra and Y' identifies with $\text{Spec}(\mathcal{A})$ (II, 1.3.6). If $R(Y)$ is the field of rational functions of Y , one therefore has, for all $y \in Y$, $\mathcal{O}_{Y,y} \subset \mathcal{A}_y \subset R(Y)$; but as the ring $\mathcal{O}_{Y,y}$ is by hypothesis integrally closed and $R(Y)$ a field of fractions, one necessarily has $\mathcal{A}_y = \mathcal{O}_{Y,y}$, whence the lemma. $\square$

Corollary (8.12.11). — *Let Y and X be integral normal preschemes, and let $f : X \rightarrow Y$ be a dominant, locally quasi-finite morphism. Let K and L (extension of K) be the fields of rational functions of Y and X respectively; and let Y' be the integral closure of Y relative to L (II, 6.3.4)); then f factorises in a unique way as $f : X \xrightarrow{f'} Y' \xrightarrow{u} Y$, where f' is birational, and correspond to the identity automorphism of L ; f' is then a local isomorphism, and for f' to be an open immersion, it is necessary and sufficient that f be separated.*

Proof. The existence and uniqueness of the factorisation of f follow from (II, 6.3.9)). It follows further from (II, 6.2.4), (v)), reducing to the affine case, that f' is locally quasi-finite; furthermore, it follows from (I, 5.5.1)) that, for f' to be separated, it is necessary and sufficient that f be, since u is affine, hence separated; the last two assertions are therefore consequences of (8.12.10)) applied to f' . $\square$

8.13. Translation in terms of pro-objects The following proposition is essentially equivalent to (8.8.2), (i): IV-3 | 49

Proposition (8.13.1). — *Let S be a prescheme, $(X_\lambda, v_{\lambda\mu})$ a projective filtering system of S -preschemes; suppose that there exists α such that $v_{\alpha\lambda}$ is an affine morphism for all $\lambda \geq \alpha$ (which implies (II, 1.6.2)) that $v_{\lambda\mu}$ is affine for $\alpha \leq \lambda \leq \mu$, so that the projective limit $X = \varprojlim X_\lambda$ exists in the category of S -preschemes (8.2.3)). Let Y be an S -prescheme, and, for all $\lambda \geq \alpha$, let $e_\lambda : \text{Hom}_S(X_\lambda, Y) \rightarrow \text{Hom}_S(X, Y)$ be the map which, to every S -morphism $f_\lambda : X_\lambda \rightarrow Y$, makes correspond $f = f_\lambda \circ v_\lambda$, where $v_\lambda : X \rightarrow X_\lambda$ is the canonical morphism. The family (e_λ) is an inductive system of maps, which therefore defines a canonical map*

$$(8.13.1.1) \quad \varinjlim \text{Hom}_S(X_\lambda, Y) \longrightarrow \text{Hom}_S(X, Y).$$

Suppose X_α quasi-compact (resp. quasi-compact and quasi-separated), and the structural morphism $Y \rightarrow S$ locally of finite type (resp. locally of finite presentation). Then the map (8.13.1.1) is injective (resp. bijective).

Proof. Setting, for $\lambda \geq \alpha$, $Z_\lambda = Y \times_S X_\lambda$, so that one has $Z_\lambda = Z_\alpha \times_{X_\alpha} X_\lambda$. Setting similarly $Z = Y \times_S X = Z_\alpha \times_{X_\alpha} X$; one knows then (8.2.5)) that, if one also sets $w_{\lambda\mu} = 1 \times v_{\lambda\mu} : Z_\mu \rightarrow Z_\lambda$ for $\alpha \leq \lambda \leq \mu$ and $w_\lambda = 1 \times v_\lambda : Z \rightarrow Z_\lambda$ for $\alpha \leq \lambda$, Z is the projective limit of the system $(Z_\lambda, w_{\lambda\mu})$ and the w_λ are the canonical morphisms. Let us note furthermore that the morphism $Z_\alpha \rightarrow X_\alpha$ is locally of finite type (resp. locally of finite presentation) (1.3.4) and (1.4.3)). Finally, one knows that one has (I, 3.3.14))

$$\text{Hom}_S(X_\lambda, Y) = \text{Hom}_{X_\lambda}(X_\lambda, Z_\lambda) \quad \text{and} \quad \text{Hom}_S(X, Y) = \text{Hom}_X(X, Z)$$

It suffices now to apply (8.8.2), (i) by setting $X_\lambda = S_\lambda$ and replacing Y_λ by Z_λ . $\square$

Corollary (8.13.2). — *With the notations of (8.13.1)), suppose X_α quasi-compact and quasi-separated, and the $v_{\alpha\lambda}$ affine for $\alpha \leq \lambda$; suppose furthermore that $Y = \varprojlim Y_\rho$, where $(Y_\rho, t_{\rho\sigma})$ is a projective filtering system of S -preschemes such that, for each ρ , the structural morphism $Y_\rho \rightarrow S$ is locally of finite presentation. Then one has a canonical bijection*

$$(8.13.2.1) \quad \text{Hom}_S(X, Y) \xleftarrow{\sim} \varprojlim_{\rho} \left(\varinjlim_{\lambda} \text{Hom}_S(X_\lambda, Y_\rho) \right).$$

Proof. Indeed, the fact that Y is the projective limit of the Y_ρ implies in particular that the canonical map $\text{Hom}_S(X, Y) \rightarrow \varprojlim \text{Hom}_S(X, Y_\rho)$ is bijective; and on the other hand, the hypotheses imply, for each ρ , the existence of a canonical bijection $\text{Hom}_S(X, Y_\rho) \xleftarrow{\sim} \varinjlim_{\lambda} \text{Hom}_S(X_\lambda, Y_\rho)$ by virtue of (8.13.1)); whence the conclusion. $\square$

(8.13.3). The preceding results allow one to interpret in the theory of preschemes the notions of “pro-variety” or of “pro-scheme” which arise in certain applications (for example in the theory of the class field body following the ideas of Serre [39] or in the theory of Néron on the reduction of abelian varieties [32]). Let us recall here rapidly the notion of *pro-object* of a category, referring to chap. V for more ample developments (we will moreover not use before chap. V the interpretation that follows, and the reader can therefore omit it until the end of this number). Given a category

$\mathcal{C}$, the category $\mathbf{Pro}(\mathcal{C})$ of *pro-objects* of $\mathcal{C}$ has for objects the *projective systems* (in the universe where one places oneself) $\mathbf{X} = (X_\mu)_{\mu \in M}$ of objects of $\mathcal{C}$ whose index sets (depending on the projective system considered) are supposed filtered preordered. Given two such pro-objects $\mathbf{X} = (X_\mu)_{\mu \in M}$, $\mathbf{X}' = (X'_{\mu'})_{\mu' \in M'}$, the *morphisms* from $\mathbf{X}$ to $\mathbf{X}'$ are by definition the elements of the set $\varprojlim_{\mu'} (\varinjlim_{\mu} \text{Hom}(X_\mu, X'_{\mu'}))$; the verification that one can take these sets as sets of morphisms is immediate, the composition of systems of morphisms $u''_\mu : X_\mu \rightarrow X'_{\mu'}$, $u''_{\mu'} : X'_{\mu'} \rightarrow X''_{\mu''}$ which are inductive and projective with respect to the lower index and upper index respectively, being carried out “argument by argument”, in other words considering the system of $u''_{\mu''} \circ u''_{\mu'}$.

(8.13.4). Consider then a prescheme quasi-compact and quasi-separated S , and denote by $\mathcal{C}$ the full subcategory of the category $(\mathbf{Sch})_S$ of S -preschemes, formed of the S -preschemes X having the following property: the structural morphism $X \rightarrow S$ factorises as $X \xrightarrow{g} X_0 \xrightarrow{f} S$, where $g : X \rightarrow X_0$ is affine and $f : X_0 \rightarrow S$ is of finite presentation; we will say for short that the preschemes of $\mathcal{C}$ are *essentially affine over S* . Consider on the other hand the full subcategory $\mathcal{C}'_0$ of $(\mathbf{Sch})_S$ formed of the S -preschemes of finite presentation, and the category $\mathbf{Pro}(\mathcal{C}'_0)$ of pro-objects of $\mathcal{C}'_0$. We will say that an object $\mathbf{X} = (X_\mu)_{\mu \in M}$ of $\mathbf{Pro}(\mathcal{C}'_0)$ is *essentially affine* if there exists a $\gamma \in M$ such that for $\mu \geq \gamma$, the transition morphism $v_{\gamma\mu} : X_\mu \rightarrow X_\gamma$ is affine (which implies that for $\gamma \leq \nu \leq \mu$, $v_{\nu\mu} : X_\mu \rightarrow X_\nu$ is affine). One will note that an object of $\mathbf{Pro}(\mathcal{C}'_0)$ isomorphic to an essentially affine object is *not* necessarily itself essentially affine. We will denote by $\mathcal{C}'$ the full subcategory of $\mathbf{Pro}(\mathcal{C}'_0)$ formed of pro-objects essentially affine.

This being so, it follows from (8.2.2) and (8.2.3) that for all objects $\mathbf{X} = (X_\mu)_{\mu \in M}$ of $\mathcal{C}'$, the S -prescheme $X = \varprojlim_{\mu} X_\mu$ exists; furthermore, as, for μ large enough, the morphism $X \rightarrow X_\mu$ is affine (8.2.2), X is *essentially affine over S* by definition. Set $X = L(\mathbf{X})$; let us show that one has thus defined a canonical functor

$$(8.13.4.1) \quad L : \mathcal{C}' \longrightarrow \mathcal{C}$$

One has indeed, for two objects $\mathbf{X} = (X_\mu)$, $\mathbf{X}' = (X'_{\mu'})$ of $\mathcal{C}'$, a canonical map

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$$\varprojlim_{\mu'} \varinjlim_{\mu} \text{Hom}_S(X_\mu, X'_{\mu'}) \longrightarrow \text{Hom}_S(\varprojlim_{\mu} X_\mu, \varprojlim_{\mu'} X'_{\mu'})$$

defined in (8.13.1.1), and on the other hand, by definition of the projective limit, a canonical bijection

$$\varprojlim_{\mu'} \text{Hom}_S(\varinjlim_{\mu} X_\mu, X'_{\mu'}) \xrightarrow{\sim} \text{Hom}_S(\varprojlim_{\mu} X_\mu, \varprojlim_{\mu'} X'_{\mu'})$$

whence a canonical map

$$(8.13.4.2) \quad \varprojlim_{\mu'} \varinjlim_{\mu} \text{Hom}_S(X_\mu, X'_{\mu'}) \longrightarrow \text{Hom}_S(\varprojlim_{\mu} X_\mu, \varprojlim_{\mu'} X'_{\mu'})$$

evidently functorial in $\mathbf{X}$ and $\mathbf{X}'$, and which completes the definition of the functor L .

Proposition (8.13.5). — *The hypotheses and notations being those of (8.13.4), the functor L is plainly faithful. If furthermore S is a Noetherian prescheme (which already implies that S is quasi-compact and quasi-separated (1.2.8)), L is an equivalence of categories.*

Proof. To say that L is plainly faithful signifies that the map (8.13.4.2) is bijective, which is a particular case of (8.13.2). Indeed, the structural morphisms $X_\mu \rightarrow S$ being of finite presentation, are in particular quasi-compact and quasi-separated, hence the X_μ are quasi-compact and quasi-separated.

To show that when S is Noetherian L is an equivalence of categories, it suffices, since one already knows that L is plainly faithful, to prove that every S -prescheme essentially affine X is *S -isomorphic* to an object of the form $L(\mathbf{X})$ where $\mathbf{X} \in \mathcal{C}'$ (0_{III}, 8.1.5). Now, by hypothesis there is a factorisation $X \xrightarrow{g} X_0 \xrightarrow{f} S$ of the structural morphism, f being of finite presentation and g affine. One can therefore write $X = \text{Spec}(\mathcal{A})$, where $\mathcal{A}$ is an $\mathcal{O}_{X_0}$ -Algebra quasi-coherent (II, 1.3.1). Now, as X_0 is Noetherian (since it is of finite type over S and S is Noetherian), $\mathcal{A}$ is the filtered inductive limit of the family $(\mathcal{A}_\lambda)$ of its sub- $\mathcal{O}_{X_0}$ -Algebras quasi-coherent of finite type (I, 9.6.6). Set $X_\lambda = \text{Spec}(\mathcal{A}_\lambda)$;

the morphisms $X_\lambda \rightarrow X_0$ are of finite type, hence of finite presentation since X_0 is Noetherian, and consequently the same is true of the composite morphisms $X_\lambda \rightarrow X_0 \rightarrow S$ (1.6.2); in other words, the X_λ belong to $\mathcal{C}'_0$ and as the morphisms $X_\lambda \rightarrow X_0$ are affine, $\mathbf{X} = (X_\lambda)$ is an object of $\mathcal{C}'$ whose projective limit exists and is S -isomorphic to X by virtue of (8.2.2). This completes the proof. $\square$

Remark (8.13.6). — It follows from (1.6.2) and (II, 1.6.2) that if X and Y are essentially affine over S , the same is true of $X \times_S Y$. One concludes for example (0_{III}, 8.2.5) that a $\mathcal{C}$ -group is nothing other than a $(\mathbf{Sch})/S$ -group which is a prescheme essentially affine over S . On the other hand, the finite products exist in the category $\mathcal{C}'$: indeed, if $\mathbf{X} = (X_\mu)_{\mu \in M}$, $\mathbf{Y} = (Y_\rho)_{\rho \in R}$ are two objects of $\mathcal{C}'$, the products $X_\mu \times_S Y_\rho$ are S -preschemes of finite presentation, and taking for transition morphisms $X_\nu \times_S Y_\sigma \rightarrow X_\mu \times_S Y_\rho$ the products of the transition morphisms $X_\nu \rightarrow X_\mu$ and $Y_\sigma \rightarrow Y_\rho$, one sees immediately that $(X_\mu \times_S Y_\rho)$ is a product of $\mathbf{X}$ and $\mathbf{Y}$ in $\mathbf{Pro}(\mathcal{C}'_0)$; furthermore (II, 1.6.2) the transition morphisms thus defined are affine for μ and ρ large enough, hence the product $\mathbf{X} \times \mathbf{Y}$ also belongs to $\mathcal{C}'$. One concludes then as above that a $\mathcal{C}'$ -group is a $\mathbf{Pro}(\mathcal{C}'_0)$ -group which is essentially affine. One deduces from (8.13.5) that the categories of the $\mathcal{C}'$ -groups and the $\mathcal{C}$ -groups are *equivalent* when S is Noetherian. It seems plausible that when S is the spectrum of a field k , the category of $\mathcal{C}$ -groups is *equivalent* to that of the K -groups, where K is the category of S -preschemes *quasi-compact*; in other words, every prescheme in groups quasi-compact over k would be essentially affine. On the other hand, if one denotes by $\mathcal{C}'_0\text{-gr}$ the category of the $\mathcal{C}'_0$ -groups, it is very likely that the category of the $\mathcal{C}'$ -groups is equivalent to the full subcategory of $\mathbf{Pro}(\mathcal{C}'_0\text{-gr})$ formed of the “proalgebraic groups essentially affine”, that is to say the projective systems $\mathbf{G} = (G_\mu)_{\mu \in M}$, where the G_μ are algebraic groups over k and the transition morphisms $G_\nu \rightarrow G_\mu$ are affine for μ large enough (which is what one can also express by saying that $\mathbf{G}$ is an extension of an algebraic group by a pro-algebraic group affine). The conjunction of these two conjectures is moreover equivalent to the following: every prescheme in groups quasi-compact over k (i.e. a prescheme in groups of finite type over k) is an extension by a prescheme in groups *affine* over k . The only pro-algebraic groups encountered in practice until now being in fact essentially affine, it would therefore be advantageous to substitute for the study of general pro-algebraic groups (introduced and studied by Serre [40]) that of schemes in groups quasi-compact over k , whose definition is conceptually much simpler.

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8.14. Characterisation of a prescheme locally of finite presentation over another, in terms of the functor it represents

(8.14.1). Given a prescheme S , we will say still, as in (8.13.4), that a projective filtering system $(X_\lambda, v_{\lambda\mu})$ of S -preschemes is *essentially affine* if there exists α such that $v_{\alpha\lambda}$ is an affine morphism for $\lambda \geq \alpha$.

The following statement, which will be especially useful in chap. V, makes precise (8.8.2), (i) by providing it with a converse:

Proposition (8.14.2). — *Let S be a prescheme, $f : X \rightarrow S$ a morphism. For every S -prescheme T , set*

$$h_X(T) = \text{Hom}_S(T, X)$$

so that h_X is a contravariant functor from the category $(\mathbf{Sch})/S$ of S -preschemes to the category $\mathbf{Ens}$ of sets (0_{III}, 8.1.1), and X an object representing this functor (0_{III}, 8.1.8). The following conditions are equivalent:

- a) *f is locally of finite presentation.*
- b) *For every projective filtering system (Z_λ) of S -preschemes, essentially affine (8.13.4) and formed of preschemes quasi-compact and quasi-separated, the canonical map (8.13.1.1)*

$$(8.14.2.1) \quad \varinjlim h_X(Z_\lambda) \longrightarrow h_X(\varprojlim Z_\lambda)$$

is bijective.

- c) *For every projective filtering system (Z_λ) of S -preschemes, such that the Z_λ are affine schemes, the map (8.14.2.1) is bijective.*
- c') *For every affine open subset U of S and every projective filtering system (Z_λ) of U -preschemes, such that the Z_λ are affine schemes, the map (8.14.2.1) is bijective.*

Proof. The fact that a) implies b) is nothing other than (8.13.1); it is trivial that b) implies c) and that c) implies c'). It remains to see that c') implies a), and as the property a) is local on S , one can restrict to the case where S is affine.

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Suppose first that X is also affine; the assertion to prove is then equivalent to the following: $\square$

Corollary (8.14.2.2). — *Let A be a ring, B an A -algebra. In order that, for every inductive filtering system (C_λ) of A -algebras, the canonical map*

$$(8.14.2.3) \quad \varinjlim \operatorname{Hom}_{A\text{-alg.}}(B, C_\lambda) \longrightarrow \operatorname{Hom}_{A\text{-alg.}}(B, \varinjlim C_\lambda)$$

be bijective, it is necessary and sufficient that B be an A -algebra of finite presentation.

Proof. It remains only to show that the condition is necessary. First, for (C_λ) take the inductive filtering system of sub- A -algebras of finite type of B , so that $\varinjlim C_\lambda = B$. The fact that (8.14.2.3) is bijective implies in particular that the identity map $1_B : B \rightarrow B$ factorises as $B \rightarrow C_\lambda \rightarrow B$ for a suitable λ , which implies that $C_\lambda = B$, hence B is an A -algebra of finite type. Write then $B = C/\mathfrak{J}$, where $C = A[T_1, \dots, T_n]$ and $\mathfrak{J}$ is an ideal of C ; setting $C_\lambda = C/\mathfrak{J}_\lambda$, where $\mathfrak{J}_\lambda$ ranges over the ideals of finite type $\mathfrak{J}_\lambda \subset \mathfrak{J}$ of C ; then $\mathfrak{J}$ is the filtered inductive limit of the ideals of finite type of $\mathfrak{J}$, and using the exactness of the inductive limit functor, one sees that B is still isomorphic to the inductive limit of the inductive system (C_λ) . There exists therefore a λ and an A -homomorphism $u : B \rightarrow C_\lambda$ such that the composite $B \xrightarrow{u} C_\lambda \xrightarrow{p_\lambda} B$ (where p_λ is the canonical homomorphism) is the identity. Let $q_\lambda : C \rightarrow C_\lambda$ be the canonical homomorphism, and set $t_i = p_\lambda(q_\lambda(T_i))$, in other words $u(t_i) = q_\lambda(T_i)$. There exists therefore a $\mu \geq \lambda$ such that the n elements $u(t_i) - q_\lambda(T_i)$ belong to $\mathfrak{J}_\mu/\mathfrak{J}_\lambda$ ($1 \leq i \leq n$); if $p_{\mu\lambda} : C_\lambda \rightarrow C_\mu$ is the canonical homomorphism, replacing λ by μ and u by $p_{\mu\lambda} \circ u$, one can therefore suppose that $u(t_i) = q_\lambda(T_i)$ for all i , and if $r = p_\lambda \circ q_\lambda$ is the canonical homomorphism $C \rightarrow C/\mathfrak{J} = B$, one can therefore write $u(r(T_i)) = q_\lambda(T_i)$ for all i , whence $q_\lambda = u \circ r$. But this implies necessarily that $\mathfrak{J} = \mathfrak{J}_\lambda$, for if $z \in \mathfrak{J}$, one has $r(z) = 0$; hence we have $B = C_\lambda$.

Let us now pass to the case where S is affine and X arbitrary; what amounts to proving is that an affine open subset V of X is of finite presentation over S , and by virtue of what has just been shown, it suffices to prove that for every projective filtering system (Z_λ) of affine S -preschemes, the map

$$(8.14.2.4) \quad \varinjlim \operatorname{Hom}_S(Z_\lambda, V) \longrightarrow \operatorname{Hom}_S(\varinjlim Z_\lambda, V)$$

is bijective. It is immediate that this map is injective, for if $(v_\lambda), (v'_\lambda)$ are two inductive systems of S -homomorphisms $v_\lambda : Z_\lambda \rightarrow V, v'_\lambda : Z_\lambda \rightarrow V$ such that the corresponding morphisms

$$Z \xrightarrow{v_\lambda} Z_\lambda \xrightarrow{v_\lambda} V, \quad Z \xrightarrow{u_\lambda} Z_\lambda \xrightarrow{v'_\lambda} V$$

are equal (u_λ being the canonical morphism), then the morphisms

$$Z \xrightarrow{u_\lambda} Z_\lambda \xrightarrow{v_\lambda} V \xrightarrow{j} X, \quad Z \xrightarrow{u_\lambda} Z_\lambda \xrightarrow{v'_\lambda} V \xrightarrow{j} X$$

(where j is the canonical injection) are equal, which implies $j \circ v_\lambda = j \circ v'_\lambda$ by hypothesis for a suitable λ , hence $v_\lambda = v'_\lambda$. $\square$

§9. CONSTRUCTIBLE PROPERTIES

9.1. The finite extension principle Let S be a prescheme, $f : X \rightarrow S$ a morphism of finite presentation (1.6.1), $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module of finite presentation. We propose in this paragraph to give criteria ensuring, for example, that the set of $s \in S$ such that the $\mathbf{k}(s)$ -prescheme $X_s = f^{-1}(s) = X \times_S \operatorname{Spec}(\mathbf{k}(s))$ has a certain property, or such that the $\mathcal{O}_{X_s}$ -Module $\mathcal{F}_s = \mathcal{F} \otimes_{\mathcal{O}_X} \mathbf{k}(s)$ has a certain property, is *locally constructible*, or at least *ind-constructible* (1.9.4); we will see that this is the case for the majority of properties that arise in algebraic Geometry. Assuming only f locally of finite presentation, we will also give (9.9) criteria for the set of points $x \in X$ where the fibre $X_{f(x)}$ (or the $\mathcal{O}_{X_{f(x)}}$ -Module $\mathcal{F}_{f(x)}$) has a certain property, to be *locally constructible*. We will see in §12 that these results, together with the additional hypothesis that f is *flat* (resp. *proper and flat*), allow us to prove that the sets considered in X (resp. in S) are even *open* in many cases.

Proposition (9.1.1). — (Finite extension principle). *Let k be a field, $\mathcal{C}$ a set of extensions of k . Suppose that the following conditions are satisfied:*

- (i) *If $K \in \mathcal{C}$ and if there exists a k -homomorphism $K \rightarrow K'$ (where K' belongs to the universe where we have placed ourselves), then $K' \in \mathcal{C}$.*

- (ii) If $K \in \mathcal{C}$, there exists a sub-extension K' of K , of finite type over k , such that $K' \in \mathcal{C}$.
- (iii) If $K \in \mathcal{C}$ is the field of fractions of an integral k -algebra of finite type A , there exists $f \in A - \{0\}$ such that for every maximal ideal $\mathfrak{m}$ of A we have $A_f/\mathfrak{m} \in \mathcal{C}$.

Let then Ω be an algebraically closed extension of k (in the universe considered). The following conditions are equivalent:

- a) $\mathcal{C}$ is non-empty.
- b) There exists $K' \in \mathcal{C}$ which is a finite extension of k .
- c) We have $\Omega \in \mathcal{C}$.

The condition (i) obviously implies that b) entails c), and c) trivially entails a); let us prove that a) entails b). By virtue of (ii) and (iii) there exists an extension $K' \in \mathcal{C}$ which is of the form $A/\mathfrak{m}$, where A is a k -algebra of finite type over k and $\mathfrak{m}$ is a maximal ideal of A . We know, by Hilbert's Nullstellensatz (Bourbaki, *Alg. comm.*, chap. V, §3, n° 1, cor. 2 of th. 1), that K' is a finite extension of k .

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Corollary (9.1.2). — Under the hypotheses (i), (ii), (iii) of (9.1.1), if k is algebraically closed and if $\mathcal{C}$ is non-empty, $\mathcal{C}$ contains all the extensions of k in the universe considered.

Indeed, as a) entails c), and the conclusion follows from the hypothesis (i).

Remark (9.1.3). — In practice, one will verify the condition (ii) of (9.1.1) by noting that K is the inductive limit of its sub-extensions of finite type K_α , and by applying the results of §8, taking into account eventually that for $K_\alpha \subset K_\beta$, K_β is faithfully flat over K_α . Frequently the set $\mathcal{C}$ is formed of fields belonging to a set $\mathcal{C}'$ of k -algebras satisfying the following condition:

- (i bis) If $A \in \mathcal{C}'$ and if there exists a homomorphism of k -algebras $A \rightarrow A'$ (where A' belongs to the universe where we have placed ourselves), then $A' \in \mathcal{C}'$.

Since this is so, the condition (i) is trivially satisfied, and one will verify generally the condition (iii) of (9.1.1) by noting that the field of fractions K of A is the inductive limit of the algebras A_f (for the relation $D(f) \supset D(g)$), and by applying the results of §8, taking into account eventually that the morphisms $D(g) \rightarrow D(f)$ are open immersions.

On the other hand, when (i bis) is not satisfied, the proof of (iii) is often more delicate, and is linked to constructibility criteria that will be developed further on.

Here are some typical examples of application of the finite extension principle:

Proposition (9.1.4). — Let k be a field, Ω an algebraically closed extension of k , X and Y two preschemes of finite type over k . The following conditions are equivalent:

- a) There exists an Ω -morphism $X_{(\Omega)} \rightarrow Y_{(\Omega)}$ (resp. an Ω -morphism possessing one of the determined properties (i) to (xiv) of (8.10.5)).
- b) There exists a finite extension k' of k and a k' -morphism $X_{(k')} \rightarrow Y_{(k')}$ (resp. a k' -morphism possessing the property considered).
- c) There exists an extension K of k and a K -morphism $X_{(K)} \rightarrow Y_{(K)}$ (resp. a K -morphism possessing the property considered).

We apply the remark (9.1.3) taking for $\mathcal{C}'$ the set of all k -algebras A (of the universe where we have placed ourselves) such that there exists an A -morphism $X \otimes_k A \rightarrow Y \otimes_k A$ (resp. a morphism having the properties of (8.10.5) that one considers). The condition (i bis) of (9.1.3) is then satisfied thanks to the fact that the properties considered in (8.10.5) are all stable under base change. The condition (iii) of (9.1.1) is thus satisfied by virtue of (8.8.2), (i) (resp. (8.10.5)), since $\text{Spec}(k)$ is quasi-compact and quasi-separated. It remains to verify the condition (ii) of (9.1.1) which also follows from (8.8.2), (i) and from (8.10.5), considering K as an inductive limit of its sub-extensions of finite type. We conclude therefore by (9.1.1).

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In particular, if there exists an extension K of k and a K -isomorphism $X_{(K)} \xrightarrow{\sim} Y_{(K)}$, we say that X and Y are geometrically isomorphic.

The following corollary generalises (II, 6.6.5):

Corollary (9.1.5). — Let k be a field, X a k -prescheme. If there exists an extension K of k such that $X_{(K)}$ is projective (resp. quasi-projective) over K , then X is projective (resp. quasi-projective) over k .

The morphism $\text{Spec}(K) \rightarrow \text{Spec}(k)$ being faithfully flat and quasi-compact, it follows already from (2.7.1), (v)) that X is of finite type over k . The hypothesis means that there exists a closed immersion (resp. an immersion) $X_{(K)} \rightarrow \mathbf{P}_K^n = \mathbf{P}_k^n \otimes_k K$ (II, 5.5.4, (ii) and 5.5.3); applying (9.1.4) for property (iv) (resp. (ii)) of (8.10.5), we deduce that for some finite extension k' of k there is a closed immersion (resp. an immersion) $X_{(k')} \rightarrow \mathbf{P}_{k'}^n$; in other words $X_{(k')}$ is projective (resp. quasi-projective) over k' . We conclude then by (II, 6.6.5).

Proposition (9.1.6). — *Let k be a field, Ω an algebraically closed extension of k , X a prescheme of finite type over k , $\mathcal{F}, \mathcal{G}$ two coherent $\mathcal{O}_X$ -Modules. Suppose that there exists an isomorphism $\mathcal{F} \otimes_k \Omega \xrightarrow{\sim} \mathcal{G} \otimes_k \Omega$. Then there exists a finite extension k' of k and an isomorphism $\mathcal{F} \otimes_k k' \xrightarrow{\sim} \mathcal{G} \otimes_k k'$.*

The reasoning is the same as in (9.1.4), applying (8.5.2), (i)) (using the property (iii) of (9.1.1), the fact that the morphisms $D(g) \rightarrow D(f)$ (with the notations of (9.1.3)) are open immersions, and *a fortiori* the morphisms are flat).

9.2. Constructible and ind-constructible properties

Definition (9.2.1). — Let $\mathbf{P}(X, \mathcal{F}, k)$ be a relation. We say that $\mathbf{P}$ is a *constructible* (resp. *ind-constructible*) property of algebraic preschemes if the two following conditions are satisfied:

- (i) If k is a field, X an algebraic prescheme over k , $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module, k' an extension of k , then for $\mathbf{P}(X, \mathcal{F}, k)$ to be true, it is necessary and sufficient that $\mathbf{P}(X_{(k')}, \mathcal{F} \otimes_k k', k')$ be true.
- (ii) Let S be an integral Noetherian prescheme, of generic point η , $u : X \rightarrow S$ a morphism of finite type, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module. For every $s \in S$, set $X_s = u^{-1}(s) = X \times_S \text{Spec}(\mathbf{k}(s))$, $\mathcal{F}_s = \mathcal{F} \otimes_{\mathcal{O}_S} \mathbf{k}(s)$. Let E be the set of $s \in S$ such that $\mathbf{P}(X_s, \mathcal{F}_s, \mathbf{k}(s))$ is true. Then one of the two sets $E, S - E$ (resp. the set E) contains a non-empty open set (and consequently is a neighbourhood of η) (resp. contains a non-empty open set if it contains η).

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Remarks (9.2.2). — (i) This is of course a convention of meta-mathematical language and not a mathematical definition properly speaking. We have analogous “definitions” for relations between k , one or more algebraic k -preschemes, k -morphisms between these preschemes, coherent Modules on these preschemes or homomorphisms between these Modules.

- (ii) We will also need to consider relations involving *constructible parts* of preschemes. For example, let $\mathbf{P}(X, Z, k)$ be a relation; we will say (by abuse of language) that $\mathbf{P}$ is a *constructible* (resp. *ind-constructible*) property of the constructible part Z of X if the following two conditions are satisfied:

- 1° If k is a field, X an algebraic prescheme over k , Z a *constructible part* of X , k' an extension of k , then for $\mathbf{P}(X, Z, k)$ to be true, it is necessary and sufficient that $\mathbf{P}(X_{(k')}, p^{-1}(Z), k')$ be true ($p : X_{(k')} \rightarrow X$ being the canonical projection).
- 2° Let S be an integral Noetherian prescheme, of generic point η , $u : X \rightarrow S$ a morphism of finite type, Z a *constructible part* of X . For every $s \in S$, set $X_s = u^{-1}(s)$, $Z_s = Z \cap X_s$. Let E be the set of $s \in S$ such that $\mathbf{P}(X_s, Z_s, \mathbf{k}(s))$ is true. Then one of the two sets $E, S - E$ (resp. the set E) contains a non-empty open set (and consequently is a neighbourhood of η) (resp. contains a non-empty open set if it contains η).

We will note that in condition 2° it suffices to suppose that Z is a constructible part of X for every s ; the first of these two properties entails the second (1.8.2), but not conversely.

- (iii) If $\mathbf{P}$ is a constructible property, it is clear that the same is true of “not $\mathbf{P}$ ”. If $\mathbf{P}, \mathbf{Q}$ are two constructible properties (resp. ind-constructible), the same is true of the properties “ $\mathbf{P}$ or $\mathbf{Q}$ ” and “ $\mathbf{P}$ and $\mathbf{Q}$ ”; indeed, under the hypotheses of (9.2.1), (ii)), E, E' are two parts of S and if each of E contains a non-empty open set, the same is true of $E \cup E'$, and if $S - E$ and $S - E'$ each contain a non-empty open set, the same is true of $S - (E \cup E') = (S - E) \cap (S - E')$.
- (iv) Let $\mathbf{P}(X, \mathcal{F}, k)$ be a relation satisfying the condition (9.2.1), (i)); let S be a prescheme, $u : X \rightarrow S$ a morphism of finite type; with the notations of (9.2.1), (ii)), let E be the set of $s \in S$ such that $\mathbf{P}(X_s, \mathcal{F}_s, \mathbf{k}(s))$ is true. Let on the other hand $g : S' \rightarrow S$ be any morphism, and set $X' = X_{(S')}$, $\mathcal{F}' = \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$; then it follows from the transitivity of fibres (I, 3.6.4) and

from the condition (9.2.1), (i)) that the set of $s' \in S'$ such that $\mathbf{P}(X'_{s'}, \mathcal{F}'_{s'}, \mathbf{k}(s'))$ is true is equal to $g^{-1}(E)$. This extends immediately to the case where there appear in $\mathbf{P}$ several preschemes, Modules, morphisms of preschemes or homomorphisms of Modules, as well as to properties of the type considered in (ii).

- (v) As we will see in the remainder of this paragraph and in the rest of chap. IV, the majority of properties $\mathbf{P}$ satisfying the condition (9.2.1), (i)) also satisfy the condition (9.2.1), (ii)). As possible exceptions, let us note the property of being *projective*, or *quasi-projective*, or *affine*, or *quasi-affine* over the base field (for an algebraic prescheme); we will see (9.6.2) that these properties are *ind-constructible*, but we will give later an example where S is a non-empty open part of $\text{Spec}(\mathbf{Z})$ (or a non-empty open part of a smooth curve over a finite field) and where all fibres X_s

except the generic fibre X_η are projective over $\mathbf{k}(s)$ (all preschemes X_s being of dimension 2).

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Proposition (9.2.3). — *Let $\mathbf{P}$ be a constructible (resp. ind-constructible) property of algebraic preschemes, S a prescheme, X a prescheme of finite presentation over S , $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module of finite presentation. Then the set E of $s \in S$ such that $\mathbf{P}(X_s, \mathcal{F}_s, \mathbf{k}(s))$ is true is locally constructible (resp. ind-constructible). Furthermore, if S is irreducible of generic point η , one of the two sets E , $S - E$ is a neighbourhood of η (resp. E is a neighbourhood of η if it contains this point).*

To prove these assertions, one can restrict to the case where $S = \text{Spec}(A)$ is affine. We know then that there exists a subring A_0 of A which is a $\mathbf{Z}$ -algebra of finite type, an A_0 -prescheme of finite type X_0 and a coherent $\mathcal{O}_{X_0}$ -Module $\mathcal{F}_0$ such that X is isomorphic to $X_0 \otimes_{A_0} A$ (8.9.1). Let $p : S \rightarrow S_0 = \text{Spec}(A_0)$ be the morphism corresponding to the injection $A_0 \rightarrow A$, and let E_0 be the set of $s_0 \in S_0$ such that

$$\mathbf{P}((X_0)_{s_0}, (\mathcal{F}_0)_{s_0}, \mathbf{k}(s_0))$$

is true; then, by virtue of (9.2.2), (iv)), we have $E = p^{-1}(E_0)$; on the other hand, one can restrict further (1.8.2) to a $\mathbf{Z}$ -algebra of finite type, hence a Noetherian scheme. Using the criterion of ind-constructibility (0_{III}, 9.2.3) (resp. the criterion of constructibility (0_{III}, 9.2.3)) (9.2.2), (iv)) and replacing S by a closed integral sub-prescheme of S , in the case where S is Noetherian and integral, our assertion then follows from (9.2.3) applied to the ind-constructible part E of S .

We will note that when S is the spectrum of a finite ring of integers of finite type, hence a Noetherian scheme, and it is clear on the other hand that the statement of (9.2.3) also applies when there appear in $\mathbf{P}$ several preschemes, Modules, morphisms of preschemes or homomorphisms of Modules. It applies also when there is a finite number of parts (in finite number) of the preschemes considered, provided one imposes on these parts the condition of being *locally constructible*. Indeed, the restriction to the case where S is affine shows that one can restrict to the case where these parts are *constructible*: applying (8.3.11) then (with the previous notations) one obtains a constructible part of X_0 for a suitable choice of A_0 .

Corollary (9.2.4). — *Let $\mathbf{P}$ be a constructible (resp. ind-constructible) property of algebraic preschemes, X, Y two S -preschemes of finite presentation, $f : X \rightarrow Y$ an S -morphism. For every $s \in S$, set $X_s = X \times_S \text{Spec}(\mathbf{k}(s))$, $Y_s = Y \times_S \text{Spec}(\mathbf{k}(s))$, $f_s = f \times 1 : X_s \rightarrow Y_s$. Then the set E of $s \in S$ such that for every $y \in Y_s$, the property $\mathbf{P}(f_s^{-1}(y), \mathbf{k}(y))$ is true, is locally constructible (resp. ind-constructible).*

Indeed, let Z be the set of $y \in Y$ such that $\mathbf{P}(f^{-1}(y), \mathbf{k}(y))$ is true. Since the fibres $f^{-1}(y)$ and $f_s^{-1}(y)$ are isomorphic, we see that E is the set of $s \in S$ such that $Y_s \subset Z$; if $g : Y \rightarrow S$ is the structural morphism, we have therefore $E = S - g(Y - Z)$.

Now, f is of finite presentation (1.6.2), (v)), hence it follows from (9.2.3) that Z is locally constructible (resp. ind-constructible) in Y , hence $Y - Z$ is locally constructible (resp. pro-constructible) in Y . As g is of finite presentation, $g(Y - Z)$ is locally constructible (resp. pro-constructible) in S , by virtue of Chevalley's theorem (1.8.4) (resp. of (1.9.5), (vii))); hence E is locally constructible (resp. ind-constructible) in S .

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Remark (9.2.5). — We will note that if $\mathbf{P}$ is a property of algebraic preschemes for which prop. (9.2.3) is true, then $\mathbf{P}$ also satisfies the condition (9.2.1), (ii)): this results indeed from the fact that in

an irreducible Noetherian space, a constructible set is either rare or contains a non-empty open set (**0**_{III}, 9.2.3).

Proposition (9.2.6). — *Let $\mathbf{P}$ denote one of the following properties of an algebraic k -prescheme X :*

- (i) X is empty.
- (ii) X is finite over k .
- (iii) X is radicial over k .
- (iv) $\dim(X)$ belongs to a given part Φ of the set $\bar{\mathbf{Z}} = \mathbf{Z} \cup \{-\infty\}$.

Then $\mathbf{P}$ is constructible.

It is clear that (i) and (ii) are particular cases of (iv), taking for Φ respectively the set $\{-\infty\}$ and the set $\{-\infty, 0\}$. We therefore need only prove (iii) and (iv). In each of these two cases the condition (i) of (9.2.1) is satisfied by virtue of (2.7.1), (xv)) and (4.1.4). On the other hand, in the case (iii), the property $\mathbf{P}$ satisfies the conclusion of (9.2.3) by virtue of (1.8.7); it therefore remains to check that the same holds in the case (iv). This will follow from the more precise proposition below:

Proposition (9.2.6.1). — *If $f : X \rightarrow S$ is a morphism of finite presentation, the function $s \mapsto \dim(f^{-1}(s))$ is locally constructible.*

The question is local on S , so one can suppose that $S = \text{Spec}(A)$ is affine and prove that for every n , the set E of $s \in S$ such that $\dim(X_s) = n$ is constructible. The same reasoning as in (9.2.3) reduces to the case where A is Noetherian and integral, and it suffices then to prove the following

Corollary (9.2.6.2). — *Let S be an integral Noetherian prescheme of generic point η , $f : X \rightarrow S$ a morphism of finite type. Then there exists a neighbourhood of η in S such that the function $s \mapsto \dim(X_s)$ is constant in this neighbourhood.*

The images by f of the irreducible components (in finite number) of X that do not meet X_η are contained in closed parts of S not containing η (since S is integral (**0**, 2.1.5)), hence (replacing S by a neighbourhood of η) one can restrict to the case where all irreducible components X_i of X meet X_η . Or, it suffices that there exists a finite covering (U_j) of X by

dense affine open sets, and the numbers $\dim((U_j)_\eta)$ are all equal to $n = \dim(X_\eta)$ (4.1.1.3); one can therefore restrict to the case where X is affine, hence also X_η . There exists then, by virtue of (4.1.2), a non-empty open set W of X such that $W \supset X_\eta$, and a $\mathbf{k}(\eta)$ -morphism of finite type surjective $h : W_\eta \rightarrow \mathbf{V}_{\mathbf{k}(\eta)}^n (= \text{Spec}(\mathbf{k}(\eta)[T_1, \dots, T_n]))$; applying (8.8.2), (i) and the method of (8.1.2), a), replacing S by a neighbourhood of η if necessary, that $h = g_\eta$, where $g : W \rightarrow \mathbf{V}_A^n (= \text{Spec}(A[T_1, \dots, T_n]))$ is an S -morphism, and one can suppose this morphism of finite type and surjective by virtue of (8.10.5), (vi) and (x)). We therefore conclude that for every $s \in S$, the morphism $g_s : W_s \rightarrow \mathbf{V}_{\mathbf{k}(s)}^n$ is of finite type and surjective, hence $\dim(W_s) = n$ (4.1.2).

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9.3. Constructible properties of morphisms of algebraic preschemes

Proposition (9.3.1). — *Let $\mathbf{P}(X, k)$ be a constructible (resp. ind-constructible) property of algebraic preschemes. Let $\mathbf{P}'(f, X, Y, k)$ denote the following relation: $f : X \rightarrow Y$ is a k -morphism of algebraic preschemes such that for every $y \in Y$, the property $\mathbf{P}(f^{-1}(y), \mathbf{k}(y))$ is true. Then $\mathbf{P}'$ is a constructible (resp. ind-constructible) property.*

Indeed, as $\mathbf{P}$ satisfies the condition (9.2.1), (i)), the same is true of $\mathbf{P}'$ by virtue of the transitivity of fibres (**I**, 3.6.4); furthermore the fact that $\mathbf{P}'$ satisfies the condition (9.2.1), (ii)) follows from (9.2.4), by reason of the remark (9.2.5).

Proposition (9.3.2). — *Let $\mathbf{P}$ denote one of the following properties of a k -morphism $f : X \rightarrow Y$ of algebraic preschemes:*

- (i) f is surjective.
- (ii) f is quasi-finite.
- (iii) f is radicial.
- (iv) For every $y \in Y$, $\dim(f^{-1}(y))$ belongs to Φ (notation of (9.2.6)).

Then $\mathbf{P}$ is constructible.

This follows immediately from (9.3.1) and (9.2.6) if one takes into account that f is of finite type (I, 3.5.8), and the characterisation of radicial morphisms (II, 6.2.2).

Proposition (9.3.3). — Suppose that the hypotheses of (8.8.1) are satisfied, and let us keep the same notations; suppose furthermore that S_α is quasi-compact, X_α and Y_α of finite presentation over S_α , and let $f_\alpha : X_\alpha \rightarrow Y_\alpha$ be an S_α -morphism. Let $\mathbf{P}$ be an ind-constructible property of morphisms of algebraic preschemes. For every $s \in S$ (resp. $s_\lambda \in S_\lambda$) set $X_s = X \times_S \text{Spec}(\mathbf{k}(s))$, $Y_s = Y \times_S \text{Spec}(\mathbf{k}(s))$, $f_s = f \times 1 : X_s \rightarrow Y_s$ (resp. $X_{\lambda, s_\lambda} = X_\lambda \times_{S_\lambda} \text{Spec}(\mathbf{k}(s_\lambda))$, $Y_{\lambda, s_\lambda} = Y_\lambda \times_{S_\lambda} \text{Spec}(\mathbf{k}(s_\lambda))$, $f_{\lambda, s_\lambda} = f_\lambda \times 1 : X_{\lambda, s_\lambda} \rightarrow Y_{\lambda, s_\lambda}$). Then, in order that for every $s \in S$ the property $\mathbf{P}(f_s)$ be true, it is necessary and sufficient that there exist $\lambda \geq \alpha$ such that for every $s_\lambda \in S_\lambda$ we have $\mathbf{P}(f_{\lambda, s_\lambda})$.

Indeed, let E (resp. E_λ) be the set of $s \in S$ (resp. $s_\lambda \in S_\lambda$) such that the property $\mathbf{P}(f_s)$ (resp. $\mathbf{P}(f_{\lambda, s_\lambda})$) is true; it follows from (9.2.2), (iv)) that $E_\mu = u_{\mu\lambda}^{-1}(E_\lambda)$ for $\lambda \leq \mu$, and $E = u_\mu^{-1}(E_\mu)$; furthermore, by virtue of (9.2.3), E (resp. E_λ) is ind-constructible in S (resp. S_λ); the proposition then follows from (8.3.4) applied to the ind-constructible part E of S .

Remark (9.3.4). — The conjunction of (9.3.3) and of (9.3.2), (ii) proves the assertion (8.10.5), (xi)).

Proposition (9.3.5). — Let $\mathbf{P}(X, Y, k)$ be the property: “ X and Y are two preschemes of finite type over the field k , and there exists an extension k' of k and a k' -morphism $g : X_{(k')} \rightarrow Y_{(k')}$ satisfying $\mathbf{Q}(g)$ ”, or $\mathbf{Q}$ is one of the properties (i) to (xiv) of (8.10.5). Then $\mathbf{P}$ is an ind-constructible property.

The definition of $\mathbf{P}$ shows indeed that the condition (9.2.1), (i) is satisfied, taking into account that the property $\mathbf{Q}(g)$ is stable under base change, and that two extensions of k can always be considered as sub-extensions of a third extension of k . To verify (9.2.1), (ii)), one can restrict to the case where $S = \text{Spec}(A)$ is affine; if $K = \mathbf{k}(\eta)$, the field of fractions of A , there exists by hypothesis and by virtue of (9.1.4) a finite extension K' of K and a K' -morphism $g : (X_\eta)_{(K')} \rightarrow (Y_\eta)_{(K')}$, satisfying $\mathbf{Q}(g')$, and K' is evidently the field of fractions of an integral A -algebra of finite type A' . If we set $S' = \text{Spec}(A')$, it follows then from (8.10.5) that there is a neighbourhood U' of the generic point η' of S' such that, if we set $X' = X \otimes_A A'$, $Y' = Y \otimes_A A'$, $Y' = Y \otimes_A A'$, there exists for every $s' \in U'$, a morphism $X'_{s'} \rightarrow Y'_{s'}$ having property $\mathbf{Q}$. But the morphism $h : S' \rightarrow S$ is of finite type, hence closed, and $h^{-1}(\eta) = \{\eta'\}$, $h(U')$ contains a neighbourhood of η in S ; for every $s \in U$, there exists therefore $s' \in U'$ such that $h(s') = s$, and as $X'_{s'} = X_s \otimes_{\mathbf{k}(s)} \mathbf{k}(s')$, $Y'_{s'} = Y_s \otimes_{\mathbf{k}(s)} \mathbf{k}(s')$, the property $\mathbf{P}(X_s, Y_s, \mathbf{k}(s))$ is true for every $s \in U$.

(9.3.6). Example. Take for example for $\mathbf{Q}$ the property of being an isomorphism. Then, combining (9.3.5) and (9.3.3), and the hypotheses being those of (8.8.1), S_α being quasi-compact, X_α and Y_α of finite presentation over S_α , we obtain the following property: using the notations of (9.3.3), in order that, for every $s \in S$, X_s and Y_s be geometrically isomorphic (9.1.4), it is necessary and sufficient that there exist $\lambda \geq \alpha$ such that, for every $s_\lambda \in S_\lambda$, X_{λ, s_λ} and Y_{λ, s_λ} be geometrically isomorphic.

We have an analogous result when the preschemes considered are endowed with “composition laws” of a certain kind (0_{III}, 8.2.1), for example “preschemes in groups”, “preschemes in rings”, etc. (0_{III}, 8.2.3). Then the previous statement is still valid when one replaces “isomorphism” by “homomorphisms” for the composition laws considered (0_{III}, 8.2.2); it suffices here to use not only (8.10.5) but also (8.8.2), (i) while noting that the notion of homomorphism for a composition law is expressed by saying that diagrams of morphisms of preschemes are commutative (it must be understood that the transition morphisms $X_\mu \rightarrow X_\lambda$ and $Y_\mu \rightarrow Y_\lambda$ for $\lambda \leq \mu$ are homomorphisms for the composition laws considered).

One can also, instead of considering morphisms of preschemes as in (9.3.5), consider homomorphisms of Modules, using (9.1.6) instead of (9.1.5).

9.4. Constructibility of certain properties of modules

(9.4.1). Notations (9.4.1). In this n° and the following ones until the end of §9, we will systematically use the following notations: given a morphism $f : X \rightarrow S$, we will set, for every $s \in S$, $X_s = f^{-1}(s) = X \times_S \text{Spec}(\mathbf{k}(s))$; for every $\mathcal{O}_X$ -Module quasi-coherent $\mathcal{F}$, $\mathcal{F}_s$ will denote the $\mathcal{O}_{X_s}$ -Module quasi-coherent $\mathcal{F} \otimes_{\mathcal{O}_S} \mathbf{k}(s)$, and for every homomorphism $u : \mathcal{F} \rightarrow \mathcal{G}$ of $\mathcal{F}$ into a quasi-coherent $\mathcal{O}_X$ -Module $\mathcal{G}$, $u_s : \mathcal{F}_s \rightarrow \mathcal{G}_s$ will be the morphism $p^*(u)$, designating by p the canonical projection $X_s \rightarrow X$. For every section g of $\mathcal{F}$ over X , we will denote by g_s the image of g by the canonical homomorphism $\Gamma(X, \mathcal{F}) \rightarrow \Gamma(X_s, \mathcal{F}_s)$. For every part Z of X , we will denote Z_s the inverse image $p^{-1}(Z) = Z \cap X_s$ (I, 3.6.1). Finally, if Y is a second S -prescheme and $h : X \rightarrow Y$ an S -morphism, we will denote h_s the morphism $h \times 1 : X_s \rightarrow Y_s$.

Proposition (9.4.2). — *Let S be an integral Noetherian prescheme of generic point η , $f : X \rightarrow S$ a morphism of finite presentation, $\mathcal{F}, \mathcal{G}, \mathcal{H}$ three quasi-coherent $\mathcal{O}_X$ -Modules of finite presentation. Let $u : \mathcal{F} \rightarrow \mathcal{G}$, $v : \mathcal{G} \rightarrow \mathcal{H}$ be two homomorphisms of $\mathcal{O}_X$ -Modules, and suppose that the sequence $\mathcal{F}_\eta \xrightarrow{u_\eta} \mathcal{G}_\eta \xrightarrow{v_\eta} \mathcal{H}_\eta$ is exact. Then there exists a neighbourhood U of η in S such that, for every $s \in U$, the sequence $\mathcal{F}_s \xrightarrow{u_s} \mathcal{G}_s \xrightarrow{v_s} \mathcal{H}_s$ is exact.*

With the general notations of (9.2.1), it is a question here of the relation $\mathbf{P}(X, \mathcal{F}, \mathcal{G}, \mathcal{H}, u, v, k)$: “ X is an algebraic prescheme over the field k , $\mathcal{F} \xrightarrow{u} \mathcal{G} \xrightarrow{v} \mathcal{H}$ is an exact sequence of quasi-coherent $\mathcal{O}_X$ -Modules”. Since, for every extension k' of k , the canonical projection $X_{(k')} \rightarrow X$ is a faithfully flat morphism (2.2.13), the condition (9.2.1), (i) is satisfied (2.2.7). By virtue of (9.2.3), one can restrict to the case where S is integral and Noetherian, in which case X is Noetherian, and $\mathcal{F}, \mathcal{G}, \mathcal{H}$ are coherent $\mathcal{O}_X$ -Modules. The hypothesis implies that there exists a neighbourhood U of η in S such that $\mathcal{F}|_{f^{-1}(U)} \rightarrow \mathcal{G}|_{f^{-1}(U)} \rightarrow \mathcal{H}|_{f^{-1}(U)}$ is exact, by virtue of (8.5.8), (i) applied following the general method of (8.1.2), a), and one can therefore already suppose that $\mathcal{F} \xrightarrow{u} \mathcal{G} \xrightarrow{v} \mathcal{H}$ is exact; it suffices evidently to prove that $\text{Ker}(v_s) = (\text{Ker } v)_s$ and $\text{Im}(u_s) = (\text{Im } u)_s$ for s near η in S . Using (8.9.1) and (8.5.2), (i), one sees that one can reduce to the case where S (hence also X) is Noetherian, and by consequence (since S is integral) and taking into account that $\mathcal{O}_\eta = \mathbf{k}(\eta)$ (so that $\text{Ker}(v_s) = (\text{Ker } v)_s$ and $\text{Im}(u_s) = (\text{Im } u)_s$), the sequence

$$0 \longrightarrow (\text{Im } u)_\eta \longrightarrow (\text{Ker } v)_\eta \longrightarrow (\text{Ker } v / \text{Im } u)_\eta \longrightarrow 0$$

is exact.

One can evidently restrict to a complex of three terms of degrees $-1, 0, +1$: $\mathcal{M} \xrightarrow{u} \mathcal{N} \xrightarrow{v} \mathcal{P}$ and to $i = 0$; the canonical homomorphism to consider is then $(\text{Ker } v / \text{Im } u)_s \rightarrow \text{Ker}(v_s) / \text{Im}(u_s)$ for every $s \in S$. As $\mathcal{F}, \mathcal{G}, \mathcal{H}$ are coherent (0_I, 5.3.4), $\text{Im}(u)$ and $\text{Ker}(v)$ are also coherent (0_I, 5.3.4) and furthermore $\text{Im}(u_s) = (\text{Im}(u))_s$ and $\text{Ker}(v_s) = (\text{Ker}(v))_s$; and from the exact sequence $0 \rightarrow \text{Im}(u) \rightarrow \text{Ker}(v) \rightarrow \text{Ker}(v) / \text{Im}(u) \rightarrow 0$, the conclusion follows from (9.4.2) applied to this exact sequence (since S is integral), which completes the proof.

Corollary (9.4.3). — *Let S be an integral Noetherian prescheme, of generic point η , $f : X \rightarrow S$ a morphism of finite presentation. Let $\mathcal{L}^\bullet = (\mathcal{L}^i)_{i \in \mathbf{Z}}$ be a complex of quasi-coherent $\mathcal{O}_X$ -Modules of finite presentation. For every i , there exist canonical homomorphisms*

$$(9.4.3.1) \quad (\mathcal{H}^i(\mathcal{L}^\bullet))_s \longrightarrow \mathcal{H}^i(\mathcal{L}_s^\bullet)$$

which are bijective for every $s \in U$.

Proposition (9.4.4). — *Let $f : X \rightarrow S$ be a morphism of finite presentation, $\mathcal{F}, \mathcal{G}, \mathcal{H}$ three quasi-coherent $\mathcal{O}_X$ -Modules of finite presentation. Let $u : \mathcal{F} \rightarrow \mathcal{G}$, $v : \mathcal{G} \rightarrow \mathcal{H}$ be two homomorphisms of $\mathcal{O}_X$ -Modules. Then the set E of $s \in S$ such that the sequence $\mathcal{F}_s \xrightarrow{u_s} \mathcal{G}_s \xrightarrow{v_s} \mathcal{H}_s$ is exact is locally constructible.*

Taking into account (9.2.3), it suffices to establish that the property $\mathbf{P}$ considered in (9.4.2) is constructible. We have already remarked in the proof of (9.4.2) that the condition (9.2.1), (i) is satisfied for this property, and it remains to verify the condition (9.2.1), (ii). Let us therefore suppose S integral Noetherian, of generic point η , and let us prove that E or $S - E$ is a neighbourhood of η . If $\eta \in E$, our assertion follows from (9.4.2), and one can therefore restrict to the case where $\eta \notin E$, in other words, the sequence $\mathcal{F}_\eta \xrightarrow{u_\eta} \mathcal{G}_\eta \xrightarrow{v_\eta} \mathcal{H}_\eta$ is not exact. Let us then distinguish two cases:

1° Set $w = v \circ u$, and suppose first that $w_\eta = v_\eta \circ u_\eta \neq 0$. As $\mathcal{F}, \mathcal{G}, \mathcal{H}$ are coherent, the same is true of $\mathcal{N} = \text{Ker}(w)$ (0_I, 5.3.4); it then follows from (9.4.2) applied to the suite exact $0 \rightarrow \mathcal{N} \rightarrow \mathcal{F} \xrightarrow{w} \mathcal{H}$ that there exists a neighbourhood U of η in S such that $\text{Ker}(w_s) = \mathcal{N}_s$; restricting S , one can therefore suppose this relation satisfied for every $s \in S$. Let j be the canonical injection $\mathcal{N} \rightarrow \mathcal{F}$, and set $\mathcal{M} = \text{Coker}(j)$; the exactness of the right of the functor $\mathcal{F} \rightarrow \mathcal{F}_s$ entails that $M_s = \text{Coker}(j_s)$ for every $s \in S$. The hypothesis $w_\eta \neq 0$ means that $M_\eta \neq 0$; as M is coherent (0_I, 5.3.4), it follows from (1.8.6) that there exists a neighbourhood U of η in S such that $M_s \neq 0$ for every $s \in U$, hence $w_s \neq 0$ for every $s \in U$, and *a fortiori* $S - E$ is a neighbourhood of η .

2° Suppose that $w_\eta = 0$; by virtue of (8.5.2), (i)), applied following the general method of (8.1.2), a)), there exists a neighbourhood U of η such that $w|_{f^{-1}(U)} = 0$; replacing S by U , one can already suppose $w = 0$ in X . Then $\mathcal{F} \xrightarrow{u} \mathcal{G} \xrightarrow{v} \mathcal{H}$ is a complex of three terms, to which one can apply (9.4.3); by hypothesis we have $(\mathcal{H}^0(\mathcal{L}^\bullet))_\eta = \mathcal{H}^0(\mathcal{L}^\bullet_\eta) \neq 0$, and $\mathcal{H}^0(\mathcal{L}^\bullet)$ is coherent (0_I, 5.3.4 and 5.3.3), hence it follows from (1.8.6) that there exists a neighbourhood U of η such that $(\mathcal{H}^0(\mathcal{L}^\bullet))_s \neq 0$ for every $s \in U$; but as one can suppose that $\mathcal{H}^0(\mathcal{L}^\bullet_s) = (\mathcal{H}^0(\mathcal{L}^\bullet))_s$ for every $s \in U$ by (9.4.3), we see again that $S - E$ is a neighbourhood of η in S .

Corollary (9.4.5). — *Let $f : X \rightarrow S$ be a morphism of finite presentation, $\mathcal{F}, \mathcal{G}$ two quasi-coherent $\mathcal{O}_X$ -Modules of finite presentation, $u : \mathcal{F} \rightarrow \mathcal{G}$ a homomorphism of $\mathcal{O}_X$ -Modules. Then the set of $s \in S$ such that u_s is injective (resp. surjective, bijective, zero) is locally constructible.*

It suffices to apply (9.4.4) to the sequences $0 \rightarrow \mathcal{F} \xrightarrow{u} \mathcal{G}, \mathcal{F} \xrightarrow{u} \mathcal{G} \rightarrow 0, 0 \rightarrow \mathcal{F} \xrightarrow{u} \mathcal{G} \rightarrow 0, \mathcal{F} \xrightarrow{u} \mathcal{G} \xrightarrow{1_{\mathcal{G}}} \mathcal{G}$.

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Corollary (9.4.6). — *Let $f : X \rightarrow S$ be a morphism of finite presentation, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module of finite presentation. Let h be a section of $\mathcal{F}$ over X ; for every $s \in S$, let h_s be the corresponding section of $\mathcal{F}_s$ over X_s (for the projection morphism $X_s \rightarrow X$). Then the set of $s \in S$ such that $h_s = 0$ is locally constructible.*

It suffices to note that h corresponds to a homomorphism $u : \mathcal{O}_X \rightarrow \mathcal{F}$ and h_s to the homomorphism u_s .

Proposition (9.4.7). — *Let $f : X \rightarrow S$ be a morphism of finite presentation, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module of finite presentation. The set E (resp. E') of $s \in S$ such that $\mathcal{F}_s$ is a locally free $\mathcal{O}_{X_s}$ -module (resp. locally free of rank n) is locally constructible.*

If X is an algebraic prescheme over a field k , $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module, k' an extension of k , then, for $\mathcal{F}$ to be locally free (resp. locally free of rank n), it is necessary and sufficient that it be so of $\mathcal{F} \otimes_k k'$, since the projection $X_{(k')} \rightarrow X$ is a faithfully flat morphism (2.2.7). In other words, the condition (9.2.1), (i) is satisfied for the properties we wish to prove constructible, and it remains to verify (9.2.1), (ii); one can therefore still suppose that S is affine, Noetherian and integral. There are then four cases to consider:

1° $\eta \in E$. By virtue of (8.5.5), applied following the general method of (8.1.2), a)), there exists a neighbourhood U of η in S such that $\mathcal{F}|_{f^{-1}(U)}$ is locally free (resp. locally free of rank n); *a fortiori* $\mathcal{F}_s$ is locally free (resp. locally free of rank n) for every $s \in U$.

2° $\eta \in E'$. Same reasoning as in 1°.

3° $\eta \in S - E$. As $\mathcal{F}_\eta$ is a coherent $\mathcal{O}_{X_\eta}$ -module, to say that it is not locally free means that it is not flat over $\mathcal{O}_{X_\eta}$ (Bourbaki, *Alg. comm.*, chap. II, §5, n° 2, cor. 2 of th. 1). The fact that $S - E$ is a neighbourhood of η will then follow from the more general lemma below:

Lemma (9.4.7.1). — *Let S be an integral Noetherian prescheme, of generic point η , X, Y two S -preschemes of finite type over S , $g : X \rightarrow Y$ an S -morphism, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module. If $\mathcal{F}_\eta$ is not g_{η} -flat, then there exists a neighbourhood U of η in S such that for every $s \in U$, $\mathcal{F}_s$ is not g_s -flat.*

Taking into account (2.1.2) and Bourbaki, *Alg. comm.*, chap. I^{er}, §2, n° 3, Remarque 1, the hypothesis means that there exists a non-empty open V of Y_η , and an injective homomorphism $v : \mathcal{M} \rightarrow \mathcal{N}$ of $\mathcal{O}_V$ -Modules coherent, such that the homomorphism $1 \otimes v : \mathcal{F}_\eta \otimes_{\mathcal{O}_V} \mathcal{M} \rightarrow \mathcal{F}_\eta \otimes_{\mathcal{O}_V} \mathcal{N}$ is not injective. We have $V = Y_\eta \cap W$, where W is open in Y (I, 3.6.1), and it follows from (8.5.2), (i) and (ii)), applied following the method of (8.1.2), a)), that there exists a neighbourhood U_0 of η

in S , two $\mathcal{O}_Z$ -Modules coherent $\mathcal{G}, \mathcal{H}$ (where $Z = W \cap h^{-1}(U_0)$, $h : Y \rightarrow S$ being the structural morphism) and an $\mathcal{O}_Z$ -homomorphism $u : \mathcal{G} \rightarrow \mathcal{H}$ such that $u_s : \mathcal{G}_s \rightarrow \mathcal{H}_s$ is injective (9.4.5). But for every $s \in U_0$, the homomorphism $1 \otimes u_s : \mathcal{F}_s \otimes_{\mathcal{O}_{Z_s}} \mathcal{G}_s \rightarrow \mathcal{F}_s \otimes_{\mathcal{O}_{Z_s}} \mathcal{H}_s$ is nothing but $(1 \otimes u)_s$; the hypothesis that $(1 \otimes u)_\eta$ is not injective therefore entails (9.4.5) the existence of a non-empty open $U \subset U_0$ such that for every $s \in U$, $(1 \otimes u)_s$ is not injective, and hence $\mathcal{F}_s$ is not $\mathcal{G}_s$ -flat for every $s \in U$.

4° $\eta \in S - E'$. It is clear that $S - E \subset S - E'$, and if $\eta \in S - E$, $S - E'$ is *a fortiori* a neighbourhood of η by 3°. Let us therefore suppose $\eta \in E$, hence $\mathcal{F}_\eta$ is locally free; these hypotheses entail that X_η is non-connected, and that the ranks of the $\mathcal{O}_{X_\eta}$ -Module locally free $\mathcal{F}_\eta$ are not the same on the various connected components of X_η . Or, it follows from (8.1.2), applied following the method of (8.1.2), a)), that one can suppose (replacing S by a neighbourhood of η) that X and X_η have the same connected components, the connected components of X_η being the intersections of X_η and the connected components of X . The conclusion then follows from the reasoning given in 2°, applied to each of the connected components of X (which are finite in number).

Remark (9.4.7.2). — The lemma (9.4.7.1) will be later generalised and freed of Noetherian hypotheses (11.2.8).

Proposition (9.4.8). — *Let S be a locally Noetherian prescheme, $f : X \rightarrow S$ a morphism of finite type, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module. Suppose that for every $s \in S$, X_s is a locally integral prescheme. Then the set E (resp. E') of $s \in S$ such that $\mathcal{F}_s$ is a torsion $\mathcal{O}_{X_s}$ -module (resp. a torsion-free $\mathcal{O}_{X_s}$ -module) is locally constructible.*

One can evidently suppose $S = \text{Spec}(A)$ affine and Noetherian and prove that E (resp. E') is constructible by using the criterion (0_{III}, 9.2.3); replacing S by the closed reduced sub-prescheme of S having as underlying space a closed irreducible part Y of S , and noting (I, 3.6.4) that for $s \in Y$, the fibre $(X_{(Y)})_s$ is canonically identified with X_s and the sheaf $(\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_Y)_s$ with $\mathcal{F}_s$, we see that one is reduced to the case where S is integral of generic point η , and to proving that E or $S - E$ (resp. E' or $S - E'$) is a neighbourhood of η in S . Furthermore, X is a finite union of affine open subsets U_i of finite type over S , and each $(U_i)_s$ is an open subprescheme of X_s , hence locally integral. If $(U_i)_\eta$ is empty (resp. nonempty), then $(U_i)_s$ is also empty (resp. nonempty) in a neighbourhood of η (9.2.6). We may therefore suppose that all $(U_i)_\eta$ are nonempty and integral; then $\mathcal{F}_s$ is torsion (resp. torsion-free) if and only if each restriction $\mathcal{F}_s|_{(U_i)_s}$ is torsion (resp. torsion-free).

We are thus reduced to the case where $X = \text{Spec}(B)$ is affine and $\mathcal{F} = \tilde{M}$, with M a finite B -module. Put $B_s = B \otimes_A \mathbf{k}(s)$ and $M_s = M \otimes_A \mathbf{k}(s)$, and suppose B_η integral. There are four cases.

1° Suppose $\eta \in E$. Then M_η is a finite torsion B_η -module, so there is an $h \neq 0$ in B_η such that $hM_\eta = 0$. By (8.5.2), (i), applied as in (8.1.2), a), after shrinking S we may write $h = g_\eta$ for some $g \in \Gamma(X, \mathcal{O}_X)$. Let $u : \mathcal{F} \rightarrow \mathcal{F}$ be multiplication by g . Since $u_\eta = 0$, (9.4.5) shows that $u_s = 0$ near η . On the other hand, multiplication by $g_\eta = h$ is injective on $\mathcal{O}_{X_\eta}$ because B_η is integral and $h \neq 0$; again by (9.4.5), multiplication by g_s on $\mathcal{O}_{X_s}$ is injective near η . Thus g_s is $\mathcal{O}_{X_s}$ -regular and annihilates $\mathcal{F}_s$, so $\mathcal{F}_s$ is torsion near η .

2° Suppose $\eta \in S - E$. If T is the torsion submodule of M_η , then $M_\eta/T \neq 0$ and, being a finite torsion-free B_η -module, it embeds into B_η^n for some n . Hence there is a nonzero homomorphism $w : M_\eta \rightarrow B_\eta^n$. By (8.5.2), (i), after shrinking S it extends to a homomorphism $v : \mathcal{F} \rightarrow \mathcal{O}_X^n$. Since $v_\eta \neq 0$, (9.4.5) gives $v_s \neq 0$ near η ; as X_s is locally integral, $\mathcal{F}_s$ is not torsion for these s .

3° Suppose $\eta \in E'$. Since M_η is a finite torsion-free B_η -module, there is an injective homomorphism $w : M_\eta \rightarrow B_\eta^n$. Using (8.5.2), (i), and (9.4.5) as in 2°, after shrinking S we obtain a homomorphism $v : \mathcal{F} \rightarrow \mathcal{O}_X^n$ such that $v_\eta = w$ and v_s is injective near η . Thus $\mathcal{F}_s$ is torsion-free near η .

4° Suppose $\eta \in S - E'$. Let T be the torsion submodule of M_η . Then $T \neq 0$ and is finite. By (8.5.2), (i) and (ii), after shrinking S there are a coherent $\mathcal{O}_X$ -module $\mathcal{G}$ and an injective homomorphism $u : \mathcal{G} \rightarrow \mathcal{F}$ such that $\mathcal{G}_\eta = \tilde{T}$ and u_η is the canonical inclusion. By 1° and (I, 8.6), near η the module $\mathcal{G}_s$ is a nonzero torsion $\mathcal{O}_{X_s}$ -module; by (9.4.5), u_s is injective near η . Therefore the torsion submodule of $\mathcal{F}_s$ is nonzero near η .

9.5. Constructibility of topological properties

Proposition (9.5.1). — *Let $f : X \rightarrow S$ be a morphism of finite presentation, Z a locally constructible part of X . Then the set E of $s \in S$ such that $Z_s \neq \emptyset$ is locally constructible.*

Proof. Indeed, we have $E = f(Z)$, and it suffices to apply the theorem of Chevalley (I, 8.4). $\square$

Corollary (9.5.2). — *If Z', Z'' are two locally constructible parts of X , the set of $s \in S$ such that $Z'_s \subset Z''_s$ (resp. $Z'_s = Z''_s$) is locally constructible.*

Proof. Indeed, the relation $Z'_s \subset Z''_s$ is equivalent to $(Z' \cap (\mathbb{C}Z''))_s = \emptyset$, and $Z' \cap \mathbb{C}Z''$ is locally constructible. $\square$

Proposition (9.5.3). — *Let $f : X \rightarrow S$ be a morphism of finite presentation, Z, Z' two locally constructible parts of X such that $Z \subset Z'$. Then the set E of $s \in S$ such that Z'_s is dense in Z_s is locally constructible in S .*

Proof. We must verify the two conditions of (9.2.2), (ii). As far as the first is concerned, consider a prescheme X algebraic over a field k , two locally closed constructible parts Z, Z' of X such that $Z \subset Z'$, and an extension k' of k . Then the projection $p : X_{(k')} \rightarrow X$ is a faithfully flat and quasi-compact morphism, and we have $p^{-1}(\overline{Z}) = \overline{p^{-1}(Z)}$ and $p^{-1}(Z') = \overline{p^{-1}(Z')}$; by virtue of (2.3.10), as p is surjective, the relation $\overline{Z} = \overline{Z'}$ is equivalent to $p^{-1}(\overline{Z}) = p^{-1}(\overline{Z'})$.

Let us verify the second condition, and suppose therefore S affine, Noetherian, integral, of generic point η . We distinguish two cases:

- 1° $\eta \in S - E$, in other words, Z_η is not dense in Z'_η ; then there exists in X an open V such that $V \cap Z_\eta = \emptyset$ and $V \cap Z'_\eta \neq \emptyset$. As X is Noetherian, V is locally constructible, and by virtue of (9.5.1), there is a neighbourhood U of η in S such that for every $s \in U$, $(V \cap Z)_s = \emptyset$ and $(V \cap Z')_s \neq \emptyset$; this implies that Z_s is not dense in Z'_s for $s \in U$, in other words $U \subset S - E$.
- 2° $\eta \in E$, so Z_η is dense in Z'_η . Let us first show that we can suppose Z' closed. Indeed, Z_η is dense in $\overline{Z'_\eta}$ (closure taken in X_η); setting $V_\eta = X_\eta - \overline{Z'_\eta}$, which is open in X_η and does not meet Z'_η , we can suppose V_η of the form $\overline{V} \cap X_\eta$, where V is open (and hence constructible) in X , and the hypothesis $V_\eta \cap Z'_\eta = \emptyset$ implies then $V_s \cap Z'_s = \emptyset$ for every s in a neighbourhood of η in S by virtue of (9.5.1). Replacing S by an open neighbourhood of η , we can therefore suppose $V \cap Z' = \emptyset$, whence $V \cap \overline{Z'} = \emptyset$ (closure taken in X), and consequently $(Z')_\eta = \overline{Z'_\eta}$, whence our assertion. The decomposition of Z' into its irreducible components being a finite union, and restricting further S to a neighbourhood of η , we can suppose that all the irreducible components Z'_i of Z' meet X_η , whence it follows that X_η contains the generic points of the Z'_i (0, 2.1.8). To say that Z_s is dense in Z'_s is equivalent to saying that each of the $(Z \cap Z'_i)_s$ is dense in $(Z'_i)_s$, and one is thus reduced to the case where Z' is irreducible. Replacing then X by the sub-prescheme having the reduced underlying space of Z' , we see that we can suppose $Z' = X$ and that X is integral and dominates S . Finally, covering X by a finite number of affine opens W_j and replacing Z by $Z \cap W_j$, we can suppose that $X = \text{Spec}(A)$, where A is a Noetherian integral ring. As X_η is Noetherian integral and Z_η is constructible and dense in X_η , Z_η contains a non-empty open of X_η of the form $(D(t))_\eta$, where t is a nonzero element of A . Replacing S by a neighbourhood of η in S if necessary, by virtue of the relation $(D(t))_\eta \subset Z_\eta$, we can suppose in addition that $D(t) \subset Z$ (9.5.2). Finally, since the homothety of ratio t_s in $\mathcal{O}_{X_s}$ is injective, it follows from (9.4.5) that for s close to η , t_s is $\mathcal{O}_{X_s}$ -regular, so $(X_s)_{t_s}$ is dense in X_s , and *a fortiori* it is the same for Z_s which contains $(X_s)_{t_s}$.

$\square$

Corollary (9.5.4). — *Let $f : X \rightarrow S$ be a morphism of finite presentation, Z a locally constructible part of X . The set E of $s \in S$ such that Z_s is closed (resp. open, resp. locally closed, resp. locally constructible) in X_s is locally constructible in S .*

Proof. To say that Z_s is open in X_s means that $(X - Z)_s = X_s - Z_s$ is closed in X_s , and as $X - Z$ is locally constructible, we can limit ourselves to considering the set of $s \in S$ such that Z_s is closed and the set of $s \in S$ such that Z_s is locally closed.

Let us verify in each case the two conditions of (9.2.2), (ii). The first results from the fact that $p : X_{(k')} \rightarrow X$ is faithfully flat and quasi-compact, and from (2.3.12) and (2.3.14). Let us therefore verify the second condition, S being supposed affine, Noetherian, integral, of generic point η . Set $Z' = \bar{Z}$; the same reasoning as in (9.5.3) shows that Z'_η is equal to the closure of Z_η in X_η ; by virtue of (9.5.3), there is thus a neighbourhood U of η such that for $s \in U$, Z_s is dense in Z'_s , the latter being closed. To say that Z_s is closed in X_s means that $Z''_s = \emptyset$, where $Z'' = Z' - Z$; it follows from (9.5.1) that the set E of $s \in S$ where $Z''_s = \emptyset$ is such that E or $S - E$ contains a neighbourhood of η . To say that Z_s is locally closed in X_s means that Z''_s is closed in X_s ; it suffices therefore to apply the preceding result replacing Z by Z'' , which is locally constructible in X . $\square$

Proposition (9.5.5). — *Let $f : X \rightarrow S$ be a morphism of finite presentation, Z a locally constructible part of X such that for every $s \in S$, Z_s is locally closed in X_s . For every $s \in S$, let $D(s) \subset \mathbf{Z} \cup \{-\infty\}$ be the set of dimensions of the irreducible components of Z_s . Then the function $s \mapsto D(s)$ is locally constructible in S .*

Proof. Let Φ be a finite part of $\mathbf{Z} \cup \{-\infty\}$; it is a question of showing that the set of $s \in S$ such that $D(s) = \Phi$ is locally constructible; taking into account (9.2.3), we have once more to verify the two conditions of (9.2.2), (ii).

As far as the first is concerned, it is a question of seeing that if X is a prescheme algebraic over a field k , Z a locally closed part of X , k' an extension of k , $p : X_{(k')} \rightarrow X$ the canonical projection, then the set of dimensions of the irreducible components of Z is the same as that of the dimensions of the irreducible components of $p^{-1}(Z)$; taking into account the existence of a sub-prescheme of X having Z as underlying space (4.2.8), this follows from (I, 5.2.1).

For the second condition of (9.2.2), (ii), we are in the case where S is Noetherian, integral, of generic point η , and $f : X \rightarrow S$ is a morphism of finite type, so that X is Noetherian. The sub-space Z_η of the Noetherian space X_η is by hypothesis locally closed, therefore has a finite number of irreducible components $Z_{i\eta}$, which are locally closed in X . It follows for each index i that there exists a locally closed part Z_i of X such that $(Z_i)_\eta = Z_{i\eta}$, so if $Z' = \bigcup_i Z_i$, we have $Z'_\eta = Z_\eta$. But as Z and Z' are locally constructible, we can, replacing S by a neighbourhood of η , suppose that $Z' = Z$ (9.5.2). Furthermore, for $i \neq j$, $Z_{i\eta} \cap Z_{j\eta}$ is rare and closed in $Z_{i\eta}$; therefore (9.5.3) and (9.5.4), we can also suppose, restricting S , that for $j \neq i$, $(Z_i)_s \cap (Z_j)_s$ is rare and closed in $(Z_i)_s$. As Z_i is locally closed in X , there is a subprescheme of X having Z_i as underlying space (and still denoted Z_i), which is of finite type over S (I, 6.3.5). Set $U_i = Z_i - \bigcup_{j \neq i} (Z_i \cap Z_j)$, which is open in Z_i and such that $(U_i)_s$ is open and everywhere dense in $(Z_i)_s$; furthermore, by construction, the U_i have no two common points. As $(Z_i)_s$ is a $k(s)$ -prescheme algebraic, the set of dimensions of the irreducible components of $Z_s = \bigcup_i (Z_i)_s$ is the same as the set of dimensions of the irreducible components of the reunion of the $(U_i)_s$ (4.1.1.3), each of these components being already an irreducible component of one of the $(U_i)_s$. We can therefore limit ourselves to the case where $Z = U_i$; furthermore, as Z_η is then irreducible, there is only a single irreducible component of Z meeting X_η , and one can evidently, restricting Z , suppose Z irreducible. We are thus finally reduced to proving the following particular case of (9.5.5): $\square$

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Corollary (9.5.6). — *Let S be a Noetherian prescheme, integral, of generic point η , X an irreducible prescheme, $f : X \rightarrow S$ a dominant morphism of finite type. Then there exists a neighbourhood U of η in S such that, for every $s \in U$, all the irreducible components of X_s are of dimension $n = \dim(X_\eta)$.*

Proof. We can evidently reduce to the case where $S = \text{Spec}(A)$ is affine, A being therefore Noetherian; replacing f by f_{red} , which is of finite type (I, 5.4), (vi), we can suppose A integral and X reduced, therefore integral. We know (4.1.2) that there exists an open W dense in X_η and a $k(\eta)$ -morphism finite and surjective $h : W \rightarrow \mathbf{V}((k(\eta))^n) = \text{Spec}(B')$, where $B' = k(\eta)[T_1, \dots, T_n]$ (T_i indeterminates). If V is an open of X such that $V \cap X_\eta = W$, we know (9.5.3) that for s close to η in S , V_s is an open dense in X_s , and one can therefore (4.1.3) reduce to the case where $V = X$, $W = X_\eta$. Set $B = A[T_1, \dots, T_n]$ and $Y = \text{Spec}(B) = \mathbf{V}_A^n$, so that $\text{Spec}(B') = Y_\eta$; it follows from (8.8.2), (i) and (8.10.5), (vi) and (x), applied following the method of (8.1.2), a), that replacing S by a neighbourhood of η , we can suppose that $h = g_\eta$, where $g : X \rightarrow Y$ is a finite surjective morphism; in other words, we have $X = \text{Spec}(C)$, where C is a finite B -algebra and the homomorphism $B \rightarrow C$ corresponding

to g is *injective*; as C is an integral ring, C is therefore a B -module of finite type *without torsion*, and $C_\eta = C \otimes_A k(\eta)$ is a module of finite type without torsion over $B_\eta = B' \otimes_A k(\eta) = k(\eta)[T_1, \dots, T_n]$ (being a ring of fractions whose denominators lie in $A - \{0\} \subset B - \{0\}$). It follows from (9.4.8) that there is a neighbourhood U of η in S such that for $s \in S$, $C_s = C \otimes_A k(s)$ is a module of finite type *without torsion* over $B_s = B \otimes_A k(s) = k(s)[T_1, \dots, T_n]$, and in particular the homomorphism $g_s : B_s \rightarrow C_s$ is injective. As no element of B_s is a divisor of zero in C_s , for every prime ideal $\mathfrak{p}_i$ of C_s (the elements being divisors of zero of C_s), we necessarily have $\mathfrak{p}_i \cap B_s = 0$, whence the canonical homomorphism $B_s \rightarrow C_s/\mathfrak{p}_i$ is injective. We deduce that for each irreducible component $Z_i = \text{Spec}(C_s/\mathfrak{p}_i)$ of X_s , the restriction to Z_i of g is a finite and dominant morphism $Z_i \rightarrow Y_s$, therefore *surjective* (II, 6.1.10). We conclude then from (4.1.2) that $\dim Z_i = n$, which completes the proof. $\square$

Remark (9.5.7). — One should note that under the hypotheses of (9.5.6), it may happen that X_s is irreducible for *no* $s \neq \eta$ in a neighbourhood of η , in other words, the property “ X is an irreducible k -prescheme algebraic” is *not constructible*. Take for example $S = \text{Spec}(k[T])$, where k is an algebraically closed field, T an indeterminate; we then have $k(\eta) = K = k(T)$. Let L be a finite separable extension of K of degree > 1 , and let X be the *integral closure* of S in L (II, 6.3.4); we then have $X = \text{Spec}(B)$, where B is the integral closure of $k[T]$ in L . We know that B is a Dedekind ring, and that all but finitely many maximal ideals of $k[T]$ are unramified in B . Moreover, the residue field of every maximal ideal of B is necessarily k , since it is a finite extension of the algebraically closed field k . Thus, for almost all maximal ideals $\mathfrak{j}_s$ of $k[T]$, the ring $B_s = B/\mathfrak{j}_s B$ is a direct product of $[L : K]$ copies of k ; in particular X_s is not irreducible, although $X_\eta = \text{Spec}(L)$ is. The same example shows that the property “ X is an integral k -prescheme algebraic” is not constructible. Finally, the property “ X is a reduced k -prescheme algebraic” is also not constructible. It suffices for this to take again $S = \text{Spec}(k[T])$, where this time k is an algebraically closed field of characteristic $p > 0$, and for X the integral closure of S in $L = K^{1/p}$ (where $K = k(T)$), so that $X = \text{Spec}(B)$ with $B = k[T^{1/p}]$ (k being perfect); every maximal ideal of $k[T]$ is of the form $(T - \alpha)$ with $\alpha \in k$; the unique ideal of B over the ideal $(T - \alpha)$ is the principal ideal $(T^{1/p} - \alpha^{1/p})$ and it is immediate that the quotient ring $B/(T - \alpha)B$ has nilpotent elements; in other words, X_s is not reduced for any $s \neq \eta$, whilst $X_\eta = \text{Spec}(L)$ is integral.

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We shall see a little further on (9.7) that one obtains on the other hand constructible properties when one considers the “geometric” notions corresponding to the notions of irreducible, reduced, or integral prescheme (4.5) and (4.6).

9.6. Constructibility of certain properties of morphisms

Proposition (9.6.1). — *Let X, Y be two S -preschemes of finite presentation, $f : X \rightarrow Y$ an S -morphism. Let E be the set of $s \in S$ for which f_s has one of the following properties: being*

- (i) *surjective;*
- (ii) *dominant;*
- (iii) *separated;*
- (iv) *proper;*
- (v) *radicial;*
- (vi) *finite;*
- (vii) *quasi-finite;*
- (viii) *an immersion;*
- (ix) *a closed immersion;*
- (x) *an open immersion;*
- (xi) *an isomorphism;*
- (xii) *a monomorphism.*

Then E is locally constructible in S .

Proof. The assertions concerning (i), (v), and (vii) are included only for reference, having already been established in (9.3.2).

(ii): Let us note first that f is of finite presentation (I, 6.2), (v), therefore, by virtue of the theorem of Chevalley (I, 8.4), $f(X)$ is locally constructible in Y . Furthermore, we have $f_s(X_s) = (f(X))_s$

(I, 3.6.1); the set of s such that f_s is dominant is the set of s such that $(f(X))_s$ is dense in Y_s ; the conclusion follows therefore from (9.5.3).

(iii): As $f : X \rightarrow Y$ is of finite presentation, the diagonal immersion $\Delta_f : X \rightarrow X \times_Y X$ is of finite presentation (I, 6.2), (iv) and (v); it follows from (1.8.4.1) that $\Delta_f(X) = \Delta_X$ is a locally constructible part of $X \times_Y X$. To say that f_s is separated means (taking into account (I, 5.3.4)) that $(\Delta_X)_s$ is closed in $X_s \times_{Y_s} X_s = (X \times_Y X)_s$; the conclusion follows from (9.5.4).

(iv): Let us show that the property “ X and Y are preschemes algebraic over a field k , and $f : X \rightarrow Y$ a proper k -morphism” is constructible. The condition (9.2.1), (i) is verified by virtue of (2.7.1), (vii). We can therefore reduce to the case where S is affine, Noetherian, integral, of generic point η , and one must prove that E or $S - E$ is a neighbourhood of η . Suppose first that $\eta \in E$; it follows from (8.10.5), (xii), applied following the method of (8.1.2), *a*), that replacing S by a neighbourhood of η , we can suppose that f is itself a proper morphism; one then knows that it is the same for f_s for every $s \in S$ (II, 5.4.2), (iii). Suppose on the contrary that $\eta \in S - E$, and we distinguish two cases:

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- 1° Suppose that f_η is not separated; then it follows from (iii) that f_s is not separated (and *a fortiori* not proper) in a neighbourhood of η .
- 2° Suppose f_η separated; Y is a finite union of affine opens V_i , and for f_s to be proper, it suffices that each of its restrictions $f_s^{-1}((V_i)_s) \rightarrow (V_i)_s$ be so (II, 5.4.1); we can therefore reduce to the case where Y is *affine*, hence a scheme. To say that f_η is not proper means (II, 5.6.3) that there exists a morphism of finite type $h : Z \rightarrow Y_\eta$ such that the morphism $(f_\eta)_{(Z)} : X_\eta \times_{Y_\eta} Z \rightarrow Z$ is not closed. As Y is a scheme, we deduce from (8.8.2), (i) and (ii) (restricting to S if needed a neighbourhood of η) that there exists a morphism of finite type $g : Y' \rightarrow Y$ such that Z is isomorphic to Y'_η and that $g_\eta = h$; we therefore have a closed part M' of X'_η such that $f'_\eta(M')$ is not closed in Y'_η . Now M' is the trace on X'_η of a closed part N' of X' ; as X' is Noetherian and f' of finite type, $f'(N')$ is constructible in Y' (I, 8.4) and by hypothesis $(f'(N'))_\eta = f'_\eta(N'_\eta) = f'_\eta(M')$ is not closed in Y'_η . We conclude from (9.5.4) that there exists a neighbourhood U of η in S such that for $s \in U$, $(f'(N'))_s = f'_s(N'_s)$ is not closed in Y'_s ; in other words, the morphism f'_s is not closed, and *a fortiori* the morphism f_s is not proper.

(vi): The property is a conjunction of properties (iv) and (vii) (8.11.1), therefore the proposition follows from what has already been proved.

(ix): The verification of condition (9.2.1), (i) follows from (2.7.1), (xi). We can therefore reduce to the case where S is affine, Noetherian, integral, of generic point η , and it suffices to prove that E or $S - E$ is a neighbourhood of η . If $\eta \in E$, we can (replacing S by a neighbourhood of η) suppose that f is a closed immersion by virtue of (8.10.5), (iv), and then it is clear that f_s is a closed immersion for every $s \in S$ (I, 4.3.2). Suppose therefore that $\eta \in S - E$, and we distinguish two cases:

- 1° f_η is not a finite morphism. Then it follows from (vi) that in a neighbourhood of η , f_s is not finite, nor *a fortiori* a closed immersion.
- 2° f_η is finite; then, by virtue of (8.10.5), (x), we can suppose (restricting S if needed) that f is itself a *finite* morphism. In this case $\mathcal{A} = \mathcal{A}(X) = f_*(\mathcal{O}_X)$ is an $\mathcal{O}_Y$ -Module coherent (II, 6.1.3) and $f = \mathcal{A}(u)$, where $u : \mathcal{O}_Y \rightarrow \mathcal{A}$ is a homomorphism of $\mathcal{O}_Y$ -Algebras (II, 1.1.2); as f_η is not a closed immersion by hypothesis, u_η is not surjective (II, 1.4.10), therefore (9.4.5) there exists a neighbourhood of η in which u_s is not surjective, and consequently (I, 4.2.3) f_s is not a closed immersion.

(viii): The verification of condition (9.2.1), (i) is done as in (ix), using this time (2.7.1), (x) and the fact that every immersion of one Noetherian prescheme into another is quasi-compact. We are therefore reduced to the case where S is affine, Noetherian, integral, of generic point η , and it suffices to prove that E or $S - E$ is a neighbourhood of η . If $\eta \in E$, we conclude as in (ix) with the help of (8.10.5), (ii). If $\eta \in S - E$, we distinguish again two cases:

- 1° $f_\eta(X_\eta)$ is not a locally closed part of Y_η . As $f(X)$ is constructible in Y (I, 8.4) and $f_s(X_s) = (f(X))_s$, we deduce from (9.5.4) that for s close to η , $f_s(X_s)$ is not locally closed in Y_s , and *a fortiori* f_s is not an immersion.
- 2° $f_\eta(X_\eta)$ is locally closed in Y_η . As $f(X)$ is constructible in Y (I, 8.4) and since the same holds for $\overline{f(X)}$ since Y is Noetherian, it follows from (8.3.11) that restricting to S if necessary, we can suppose that $f(X)$ is locally closed in Y . There is then an open V of Y containing $f(X)$

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and in which $f(X)$ is closed. Since Y is Noetherian, V is of finite type over S ; replacing Y by V , we may therefore reduce to the case where $f(X)$ is closed in Y . But then $f_s(X_s) = (f(X))_s$ is closed in Y_s for every $s \in S$, and saying that f_s is an immersion is equivalent to saying that it is a closed immersion. We are therefore reduced to case (ix).

(x): Using this time (2.7.1), (ix) and (8.10.5), (iii), we are reduced to the case where S is affine, Noetherian, integral, of generic point η , and where $\eta \in S - E$. We distinguish three cases:

- 1° $f_\eta(X_\eta)$ is not open in Y_η . As $f(X)$ is constructible in Y (I, 8.4), we deduce from (9.5.4) that $f_s(X_s)$ is not open in Y_s for s close to η , and *a fortiori* f_s is not an open immersion.
- 2° $f_\eta(X_\eta)$ is open in Y_η but f_η is not an immersion. It follows then from (viii) that for s close to η , f_s is not an immersion, nor *a fortiori* an open immersion.
- 3° $f_\eta(X_\eta)$ is open in Y_η and f_η is an immersion. As $f(X)$ is constructible in Y (I, 8.4) and since the same holds for $\overline{f(X)}$ since Y is Noetherian, it follows from (8.3.11) that restricting to S if necessary, we can already suppose that $f(X)$ is open in Y . But then $f_s(X_s) = (f(X))_s$ is open in Y_s for every $s \in S$, so we can reduce to the case where f is *surjective* and f is a closed immersion, in other words, that X is a closed sub-prescheme of Y defined by a coherent ideal $\mathcal{J}$ of $\mathcal{O}_Y$. By hypothesis f_η is not an isomorphism, hence $\mathcal{J}_\eta \neq 0$; we conclude (9.4.5) that in a neighbourhood of η , $\mathcal{J}_s \neq 0$, and consequently f_s is not an open immersion.

(xi): The property is a conjunction of properties (i) and (x) and therefore follows from what has been proved.

(xii): By virtue of (I, 5.3.8), to say that f_s is a monomorphism means that $\Delta_{(f)_s} = (\Delta_f)_s : X_s \rightarrow X_s \times_{Y_s} X_s = (X \times_Y X)_s$ is an isomorphism, and as $X \times_Y X$ is an S -prescheme of finite presentation (I, 6.2), (iv), the conclusion follows from (xi). $\square$

Proposition (9.6.2). — *Let X, Y be two S -preschemes of finite presentation, $f : X \rightarrow Y$ an S -morphism.*

I) *Let E be the set of $s \in S$ for which f_s has one of the following properties: being*

- (i) *affine;*
- (ii) *quasi-affine;*
- (iii) *projective;*
- (iv) *quasi-projective.*

Then E is ind-constructible in S .

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II) *Let $\mathcal{L}$ be an $\mathcal{O}_X$ -Module invertible. Then the set E' of $s \in S$ such that $\mathcal{L}_s$ is an $\mathcal{O}_{X_s}$ -Module ample (resp. very ample) relatively to f_s , is ind-constructible in S .*

Proof. I) Let us verify conditions (i) and (ii) of (9.2.1). As far as the condition (9.2.1), (i) is concerned, it follows, for properties (i) and (ii), from (2.7.1), (xiii) and (xiv); for properties (iii) and (iv), it follows from (9.1.5). Let us then verify condition (9.2.1), (ii), supposing therefore S Noetherian, integral, of generic point η , and that $f_\eta : X_\eta \rightarrow Y_\eta$ has one of the properties (i) to (iv) of the statement. Applying (8.10.5), (viii), (ix), (xiii), and (xiv) following the method of (8.1.2), a), we see first that there exists a neighbourhood open U of η such that, if V and W are the respective inverse images of U in X and Y , the restriction $V \rightarrow W$ of f to the one of the properties (i) to (iv) that one considers. The conclusion follows then from the stability of the properties considered under base change (II, 4.6.13) and (4.4.10).

II) We proceed in the same way. For condition (9.2.1), (i) it follows this time from (2.7.2). For condition (9.2.1), (ii), with the same notations as in I), it follows from (8.10.5.2) that for a neighbourhood U of η in S , the restriction $\mathcal{L}|_V$ is ample (resp. very ample) relatively to f . The conclusion follows again from the stability of the properties considered under base change (II, 4.6.13). $\square$

We can improve (9.6.2), II) under certain conditions:

Proposition (9.6.3). — *Let X, Y be two S -preschemes of finite presentation, $f : X \rightarrow Y$ a proper S -morphism, $\mathcal{L}$ an $\mathcal{O}_X$ -Module invertible. Then the set E (resp. E') of $s \in S$ such that $\mathcal{L}_s$ is an $\mathcal{O}_{X_s}$ -Module ample (resp. very ample) relatively to $f_s : X_s \rightarrow Y_s$ is locally constructible in S .*

Proof. Let k be a field, Z, T two preschemes algebraic over k , $\mathcal{M}$ an $\mathcal{O}_Z$ -Module invertible, $g : Z \rightarrow T$ a k -morphism of finite type; then, if $\mathbf{P}(Z, T, \mathcal{M}, g, k)$ denotes the relation “ $\mathcal{M}$ is ample (resp. very

ample) relatively to g'' , we have already noted in (9.6.2) that $\mathbf{P}(Z, T, \mathcal{M}, g, k)$ satisfies condition (9.2.1), (i) by virtue of (2.7.2). We know on the other hand that E and E' are ind-constructible. It remains therefore to see that if S is Noetherian, integral, of generic point η and if $\eta \in S - E$ (resp. $\eta \in S - E'$), then $S - E$ (resp. $S - E'$) contains a neighbourhood of η . We shall separately consider the cases E and E' .

Case of E' . — This amounts to saying that: *either* the canonical homomorphism $(f_s)^*((f_s)_*(\mathcal{L}_s)) \rightarrow \mathcal{L}_s$ is not surjective; *or* the preceding homomorphism is surjective and the canonical morphism $r : X_s \rightarrow \mathbf{P}((f_s)_*(\mathcal{L}_s))$ is not an immersion. Taking into account the isomorphism (9.6.3.1), these conditions are written respectively in the form: 1° the canonical homomorphism $(f^*(f_*(\mathcal{L})))_s \rightarrow \mathcal{L}_s$ is not surjective; 2° the preceding homomorphism is surjective and the canonical morphism $X_s \rightarrow \mathbf{P}((f_*(\mathcal{L}))_s)$ is not an immersion.

Suppose first that the canonical homomorphism $(f^*(f_*(\mathcal{L})))_\eta \rightarrow \mathcal{L}_\eta$ is not surjective. As $f_*(\mathcal{L})$ is coherent, so is $f^*(f_*(\mathcal{L}))$, and then it follows from (9.4.5) that for every s in a neighbourhood of η , the homomorphism $(f^*(f_*(\mathcal{L})))_s \rightarrow \mathcal{L}_s$ is not surjective, which proves in this case that $S - E'$ is a neighbourhood of η .

Suppose next that the canonical homomorphism $(f^*(f_*(\mathcal{L})))_\eta \rightarrow \mathcal{L}_\eta$ is surjective but that the morphism $X \rightarrow \mathbf{P}((f_*(\mathcal{L}))_\eta)$ is not an immersion. Then the same reasoning as above shows that for s sufficiently close to η , the homomorphism $(f^*(f_*(\mathcal{L})))_s \rightarrow \mathcal{L}_s$ is surjective; on the other hand, by virtue of (9.6.1), (viii), the morphism $X_s \rightarrow \mathbf{P}((f_*(\mathcal{L}))_s)$ is not an immersion for s in a neighbourhood of η , which completes the proof in the case of E' .

Case of E . — Consider first the particular case where $Y = S$:

Corollary (9.6.4). — *Let $f : X \rightarrow S$ be a proper morphism of finite presentation, $\mathcal{L}$ an $\mathcal{O}_X$ -Module invertible. Then the set E of $s \in S$ such that $\mathcal{L}_s$ is ample (relatively to f_s) is open in S , and $\mathcal{L}|_{f^{-1}(E)}$ is ample relatively to the restriction $f^{-1}(E) \rightarrow E$ of f .*

Proof. As condition (9.2.1), (i) is verified by the property $\mathbf{P}$ defined above, the result of (9.2.2), (iv) and the reasoning of (9.2.3) show that it suffices to reduce to the case where S is Noetherian; but then the result follows from (III, 4.7.1) and the stability of amplitude under base change (II, 4.6.13). $\square$

Corollary (9.6.5). — *Under the hypotheses of (9.6.4), for $\mathcal{L}$ to be ample relatively to f , it is necessary and sufficient that, for every $s \in S$, $\mathcal{L}_s$ be ample relatively to f_s .*

(9.6.6) End of the proof of (9.6.3). — Let us return to the general case, S being Noetherian, integral, of generic point $\eta \in S - E$. As f is proper, it follows from (9.6.4) that the set V of $y \in Y$ such that $\mathcal{L}_y$ is an $\mathcal{O}_{X_y}$ -Module ample, relatively to the morphism $f_y : X_y \rightarrow \text{Spec}(k(y))$, is open, and therefore $F = Y - V$ is closed in Y . As f_s is proper and, for every $y \in Y$ above s , $\mathcal{L}_y = (\mathcal{L}_s)_y$, this being the case since f_s is proper, we have $F_s \neq \emptyset$. It suffices to show that $F_s \neq \emptyset$ for every s in a neighbourhood of η in S , and as F is closed in Y and $F_\eta \neq \emptyset$, it follows from (9.5.1) that the set of $s \in S$ such that $F_s \neq \emptyset$ is a neighbourhood of η in S . $\square$

Remarks (9.6.7). — (i) For each of the properties $\mathbf{P}$ considered in (9.6.1), proposition (9.3.3) is applicable, and these properties (for the morphisms f_s) are therefore “stable” under passage to a projective limit of a projective system of preschemes X_λ essentially affine (8.13.4) one over another.

(ii) Let Z be a locally constructible part of X such that, for every $s \in S$, $Z \cap X_s$ is open in X_s , and denote by Z_s the sub-prescheme of X_s induced on the open $Z \cap X_s$. Then, in propositions (9.6.1) and (9.6.2), (I), we can everywhere replace f_s by its restriction $f_s|_{Z_s} : Z_s \rightarrow Y_s$ without changing the conclusions. Indeed, the verification of (9.2.1), (i) proceeds as in (9.6.1) and (9.6.2). On the other hand, in the reduction to the Noetherian case, made in (9.2.3), if $Z = q^{-1}(Z_0)$, where Z_0 is the canonical projection and Z_0 a constructible part of X_0 (8.3.11), the fact that $(Z_0)_{s_0}$ is open in $(X_0)_{s_0}$, for $s_0 = p(s)$, follows from (2.4.10) and from the fact that the projection $X_s \rightarrow (X_0)_{s_0}$ is surjective. Now, as Z_η is open in X_η , there is an open $Z' \subset X$ such that $Z_\eta = Z'_\eta \cap X_\eta$; as X is Noetherian, Z' is constructible, and since it has the same properties as Z , we conclude from (9.5.2) and from (0_{III}, 9.2.2) that there is a neighbourhood U of η in S such that $Z_s = Z'_s$ for $s \in U$. Replacing S by U , we can therefore reduce to the case where Z is open in X , and then the result has been proved in (9.6.1) and (9.6.2).

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9.7. Constructibility of properties of geometric separability, geometric irreducibility, and geometric connectedness

Lemma (9.7.1). — *Let S be an irreducible prescheme of generic point η , $f : X \rightarrow S$ a morphism of finite presentation. If X_η has n irreducible components (resp. n' connected components), there exists a neighbourhood open U of η in S such that for every $s \in U$, X_s has at least n irreducible components (resp. n' connected components).*

Proof. We can reduce to the case where S is affine, so X is quasi-compact and quasi-separated. Let V_i ($1 \leq i \leq n$) be the interiors of the irreducible components of X_η , which are pairwise disjoint and quasi-compact, and whose reunion is dense in X_η ; by virtue of (8.2.11), applied following the method of (8.1.2), a), there exists for each i a quasi-compact open W_i in X such that $W_i \cap X_\eta = V_i$; as X is quasi-separated, the intersections $W_i \cap W_j$ are quasi-compact (I, 2.7), and therefore, replacing S by a neighbourhood of η , we can suppose that $W_i \cap W_j = \emptyset$ for $i \neq j$ by virtue of (8.3.3). Furthermore, as the W_i are constructible, there is a neighbourhood U of η in S such that for every $s \in U$ the $(W_i)_s$ are non-empty (9.5.1) and (9.2.3) and the reunion of the $(W_i)_s$ is dense in X_s (9.5.3) and (9.2.3). This being the case, as the irreducible components of each $(W_i)_s$ (being finite in number) are also irreducible components of the reunion of the $(W_i)_s$ (0, 2.1.7), the closures in X_s of these components are the irreducible components of X_s (0, 2.1.6) and their number is evidently $\geq n$.

Let now C_j ($1 \leq j \leq n'$) be the connected components of X_η , which are quasi-compact opens of X_η , pairwise disjoint. If we replace V_i by C_j in the preceding reasoning, we see (using twice (8.3.3)) that we can suppose X a reunion of n' quasi-compact opens M_j ($1 \leq j \leq n'$), pairwise disjoint, and such that, in a neighbourhood of η , the $(M_j)_s$ are non-empty. As the reunion of the $(M_j)_s$ is X_s , the connected components of the $(M_j)_s$ are the connected components of X_s , and therefore their number is $\geq n'$. $\square$

Lemma (9.7.2). — *Let S be an irreducible prescheme of generic point η , $f : X \rightarrow S$ a morphism of finite presentation. If X_η is not reduced, there exists a neighbourhood open U of η in S such that, for every $s \in U$, X_s is not reduced.*

Proof. Indeed, let $\mathcal{N}$ be the Nilradical of $\mathcal{O}_X$; it follows from (8.2.13), applied following the method of (8.1.2), a), that $\mathcal{N}_\eta$ is the nilradical of $\mathcal{O}_{X_\eta}$, and the hypothesis is that $\mathcal{N}_\eta \neq 0$. We conclude from (9.4.5) and (9.2.3) that there is a neighbourhood U of η such that, for every $s \in U$, $\mathcal{N}_s$ identifies with an Ideal of $\mathcal{O}_{X_s}$ and $\mathcal{N}_s \neq 0$; as $\mathcal{N}$ is evidently contained in the Nilradical of $\mathcal{O}_{X_s}$, we see that X_s is not reduced for $s \in U$. $\square$

(9.7.3) Given a polynomial $F \in A[T_1, \dots, T_n]$, where A is a ring and the T_i are indeterminates, for a ring homomorphism $\rho : A \rightarrow B$, we denote by $F_{(\rho)}$ or $F_{(B)}$ the polynomial in $B[T_1, \dots, T_n]$ obtained by replacing each coefficient of F by its image under ρ . If k is a field, $F \in k[T_1, \dots, T_n]$ a non-constant polynomial, and $X = \text{Spec}(k[T_1, \dots, T_n]/(F))$, to say that X is integral (or that the ideal (F) is prime) means that F is *irreducible* (i.e. that in every factorisation $F = F_1 F_2$ in polynomials of $k[T_1, \dots, T_n]$, F_1 or F_2 is of degree 0); this follows from the fact that the polynomial ring $k[T_1, \dots, T_n]$ is factorial. We deduce immediately from (4.6.2) and (4.5.2):

Lemma (9.7.4). — *Let k be a field, Ω an algebraically closed extension of k , F a non-constant polynomial of $k[T_1, \dots, T_n]$. The following conditions are equivalent:*

- a) $X = \text{Spec}(k[T_1, \dots, T_n]/(F))$ is geometrically integral.
- b) $F_{(K)}$ is irreducible for every extension K of k .
- c) $F_{(\Omega)}$ is irreducible.

We shall say in this case that F is geometrically irreducible.

Lemma (9.7.5). — *Let A be an integral ring, K the field of fractions of A , F a non-constant polynomial of $A[T_1, \dots, T_n]$ of degree d such that $F_{(K)}$ is geometrically irreducible. Then there exists $f \neq 0$ in A such that for every $x \in D(f)$, $F_{(k(x))}$ is geometrically irreducible.*

Proof. Set $F = \sum_{\alpha} c_{\alpha} T^{\alpha}$ as usual, with $c_{\alpha} = c_{\alpha_1 \alpha_2 \dots \alpha_n}$, $T^{\alpha} = T_1^{\alpha_1} T_2^{\alpha_2} \dots T_n^{\alpha_n}$, $|\alpha| = \alpha_1 + \alpha_2 + \dots + \alpha_n \leq d$ (at most the c_{α} such that $|\alpha| = d$ being not zero). As the nonzero c_{α} are invertible in K , we can suppose, replacing A by A_g ($g \neq 0$ in A), that the c_{α} are nonzero and invertible in A . It then follows that for every $x \in \text{Spec}(A)$, $F_{(k(x))}$ is of degree d . $\square$

Proof (continued). We shall prove the lemma by means of a preliminary lemma.

Lemma (9.7.5.1). — *Let A be an integral ring, $S = \operatorname{Spec}(A)$, η the generic point of S , B the polynomial ring $A[T_1, \dots, T_m]$, $(P_i)_{i \in I}$ a finite family of elements of B , $\mathfrak{a}$ the ideal of B generated by the P_i . For every $s \in S$, let $\mathfrak{a}_s$ be the ideal of $k(s)[T_1, \dots, T_m]$ generated by the polynomials $(P_i)_{(k(s))}$ ($i \in I$). Then, if $V(\mathfrak{a}_\eta) = \emptyset$, there exists a neighbourhood U of η in S such that $V(\mathfrak{a}_s) = \emptyset$ for every $s \in U$.*

Proof. Indeed, let $Y = \operatorname{Spec}(B)$, and let Z be the closed part $V(\mathfrak{a})$ of Y ; as $\mathfrak{a}$ is an ideal of finite type in B , Z is constructible in Y (0, 9.1.5) and, with the notations introduced in (9.4.1), we have $V(\mathfrak{a}_s) = Z_s$ for every $s \in S$; as the structural morphism $Y \rightarrow S$ is of finite presentation, the conclusion of the lemma follows from (9.5.1) and (9.2.3). $\square$

Let (p, q) be a pair of integers > 0 such that $p + q = d$; we introduce indeterminates T'_β, T''_γ for all systems of integers β, γ such that $|\beta| \leq p$ and $|\gamma| \leq q$; for every system of integers α such that $|\alpha| \leq d$, consider the polynomial of $B = A[T'_\beta, T''_\gamma \mid |\beta| \leq p, |\gamma| \leq q] : P_\alpha(T'_\beta, T''_\gamma) = \sum_{\delta+\gamma=\alpha} T'_\delta T''_\gamma - c_\alpha$. Let Ω be an algebraic closure of K ; to say that there exist two polynomials F_1, F_2 of $\Omega[T_1, \dots, T_n]$, of respective degrees p and q , and such that $F_1 F_2 = F_{(\Omega)}$ means that the system of equations $P_\alpha(t'_\beta, t''_\gamma) = 0$ ($|\alpha| \leq d$) admits a solution (t'_β, t''_γ) formed of elements of Ω . Let $\mathfrak{a}$ be the ideal of B generated by the P_α ; the preceding interpretation and the theorem of the zeros of Hilbert show that the hypothesis on $F_{(K)}$ implies that $V(\mathfrak{a}_\eta) = \emptyset$, designating by η the generic point of $\operatorname{Spec}(A)$; lemma (9.7.5.1) then proves that in a neighbourhood of η , we have $V(\mathfrak{a}_s) = \emptyset$, and consequently for these values of x , $F_{(k(x))}$ does not admit any factorisation $F_{(k(x))} = G_1 G_2$, where G_1, G_2 are polynomials of respective degrees p and q , whose coefficients lie in an algebraic closure of $k(x)$. It suffices to apply this result for all pairs of integers (p, q) such that $p > 0, q > 0$ and $p + q = d$, to obtain the conclusion of lemma (9.7.5). $\square$

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Proposition (9.7.6). — *Let S be a Noetherian prescheme, integral, of generic point η , $f : X \rightarrow S$ a morphism of finite type, $\mathcal{F}$ an $\mathcal{O}_X$ -Module coherent. If $\mathcal{F}_\eta$ is without associated prime cycle, there exists a neighbourhood U of η in S such that, for every $s \in U$, $\mathcal{F}_s$ is without associated prime cycle.*

Proof. As $\mathcal{F}_\eta$ is Noetherian, there exists an injective homomorphism $v : \mathcal{F}_\eta \rightarrow \bigoplus_{i=1}^m \mathcal{G}_i$, where each $\mathcal{G}_i$ is irredundant and $\operatorname{Ass}(\mathcal{F}_\eta)$ is the set of the x_i , with $\{x_i\} = \operatorname{Ass}(\mathcal{G}_i)$ (3.2.6); if Z_i is the closure of $\{x_i\}$ in X_η , then $Z_i = \operatorname{Supp}(\mathcal{G}_i)$ (3.1.4), and by hypothesis the Z_i are the irreducible components of $Z = \operatorname{Supp}(\mathcal{F}_\eta)$. It follows from (8.5.2), (i) and (ii) (restricting to S if necessary a neighbourhood of η) and (8.5.8) that there exist $\mathcal{O}_X$ -Modules coherent $\mathcal{F}_i$ and an injective homomorphism $u : \mathcal{F} \rightarrow \bigoplus_{i=1}^m \mathcal{F}_i$ such that $(\mathcal{F}_i)_\eta = \mathcal{G}_i$ for every i and $v = u_\eta$. If $Y_i = \operatorname{Supp}(\mathcal{F}_i)$, we have $(Y_i)_s = \operatorname{Supp}((\mathcal{F}_i)_s)$ for every $s \in S$ (I, 9.1.13) and in particular $(Y_i)_\eta = Z_i$ for every i . The hypothesis implies that for $i \neq j$, $Z_i - (Z_i \cap Z_j)$ is open and dense in Z_i ; we deduce therefore from (9.5.3) and (9.5.4) that in a neighbourhood of η , $(Y_i)_s - ((Y_i)_s \cap (Y_j)_s)$ is open and dense in $(Y_i)_s$. Suppose that one has proved that each of the $(\mathcal{F}_i)_s$ is without associated prime cycle for $s \in U$. Then the elements of $\operatorname{Ass}((\mathcal{F}_i)_s)$ are the maximal points of $(Y_i)_s$; no two of them can therefore belong to $(Y_j)_s$ for $i \neq j$, and the decomposition is irredundant. We may moreover suppose that $\mathcal{F}_\eta$ is irredundant; in addition, we may reduce to the case where $S = \operatorname{Spec}(A)$ is affine and $f : X \rightarrow S$ is dominant. Then A is a Noetherian integral ring, B is a finite-type A -algebra containing A , and $\mathcal{F} = \tilde{M}$ with M a finite B -module. By hypothesis, if K is the field of fractions of A , the $B_{(K)}$ -module $M_{(K)}$ is irredundant. Let $\mathfrak{q}$ be the unique element of $\operatorname{Ass}(M_{(K)})$, and let $\mathfrak{p}$ be the prime ideal of B which is the inverse image of $\mathfrak{q}$ by the canonical map $B \rightarrow B_{(K)}$. We know (5.11.1.1) that there exists a finite filtration $(N_h)_{0 \leq h \leq m}$ of $M_{(K)}$ such that $N_m = 0$, and N_h / N_{h+1} is isomorphic to a nonzero $B_{(K)}$ -sub-module $\neq 0$ of $B_{(K)} / \mathfrak{q}$. Let M_h be the inverse image of N_h by the canonical map $M \rightarrow M_{(K)}$. It follows from (9.4.4) that for s sufficiently close to η , $(M_h)_{(k(s))}$ identifies with a sub- $B_{(k(s))}$ -module of $M_{(k(s))}$, and the quotient $(M_h)_{(k(s))} / (M_{h+1})_{(k(s))}$ to a nonzero $B_{(k(s))}$ -sub-module $\neq 0$ of $B_{(k(s))} / \mathfrak{p}_{(k(s))}$. Taking into account (3.1.7), with the same notations, it remains to prove, that if B is integral, then $B_{(k(s))}$ has no associated prime cycle for s close to η . Now, replacing A by A_g (where g is a nonzero element $\neq 0$ of A), we can suppose that B contains a polynomial sub-algebra $C = A[T_1, \dots, T_n]$ such that B is a C -module of finite type (Bourbaki, *Alg. comm.*, chap. V, § 3, n° 1, cor. 1 of th. 1). As $B_{(K)}$ is a torsion-free $C_{(K)}$ -module, we may apply the preceding argument once more, replacing B, M , and

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$\mathfrak{q}$ by C , B , and (0) respectively; it suffices to show that for s close to η , $C_{(k(s))}$ has no embedded associated prime. But this is obvious since $C_{(k(s))} = k(s)[T_1, \dots, T_n]$ is an integral ring. $\square$

Theorem (9.7.7). — *Let $f : X \rightarrow S$ be a morphism of finite presentation, and let E be the set of $s \in S$ for which X_s has one of the following properties: being*

- (i) *geometrically irreducible;*
- (ii) *geometrically connected;*
- (iii) *geometrically reduced;*
- (iv) *geometrically integral.*

Then E is locally constructible in S .

Proof. To see that the properties considered are constructible, let us note first that they trivially satisfy condition (9.2.1), (i), by virtue of (4.5.6), (i) and (4.6.5), (i). It remains therefore to verify (9.2.1), (ii), so we can suppose $S = \text{Spec}(A)$ affine, Noetherian, integral, of generic point η . By virtue of (4.6.8), there exists a finite extension K' of $K = k(\eta)$ such that $(X_\eta)_{(K')}$ has its irreducible components (resp. connected components) geometrically irreducible (resp. geometrically connected) and that $((X_\eta)_{(K')})_{\text{red}}$ is geometrically reduced. Furthermore, there exists a base of K' over K formed of elements integral over A , the integral ring A' generated by these elements is finite over A and has K' as field of fractions. Set $S' = \text{Spec}(A')$; the morphism $g : S' \rightarrow S$ is finite and dominant, therefore surjective (II, 6.1.10), so that $X' = X_{(S')}$; let η' be the generic point of S' , $X'_{\eta'} = (X_\eta)_{(K')}$; the set E' of $s' \in S'$ such that $X'_{s'}$ has one of the properties (i), (ii), (iii), or (iv) (E corresponding well to the same property) (9.2.2), (iv); and $S - E = g(S' - E')$ and $g^{-1}(\eta) = \{\eta'\}$ since A' is integral and finite over A , therefore the image by g of every neighbourhood of η' is a neighbourhood of η . The theorem will therefore be proved if one proves that E' or $S' - E'$ is a neighbourhood of η' . In other words, we can henceforth suppose that the irreducible components (resp. connected components) of X_η are geometrically irreducible (resp. geometrically connected) and that $(X_\eta)_{\text{red}}$ is geometrically reduced.

Suppose first that $\eta \in S - E$ for one of the properties (i) to (iv). If X_η is not geometrically irreducible (resp. geometrically connected), it is not irreducible (resp. connected) by virtue of the preceding hypothesis, and therefore it is the same for X_s for s in a neighbourhood of η (9.7.1), and *a fortiori* in this neighbourhood X_s is not geometrically irreducible (resp. geometrically connected). On the other hand, if X_η is not geometrically reduced, it is not reduced (since otherwise it would be equal to $(X_\eta)_{\text{red}}$ which is geometrically reduced by hypothesis); therefore, in a neighbourhood of η , X_s is not reduced (9.7.2) and hence not geometrically reduced. Finally, if X_η is not geometrically integral, either it is not reduced (nor *a fortiori* integral) in a neighbourhood of η ; or X_η is reduced (therefore geometrically reduced by hypothesis), and then it is not geometrically irreducible, and we have just seen above that it is then the same for X_s for s close to η ; *a fortiori* X_s is not geometrically integral for these values of s .

We shall now suppose that $\eta \in E$ and examine separately each of the properties considered.

1° Suppose that X_η is geometrically integral. Let L be the field of rational functions on X_η ; the hypothesis on X_η implies that L is a separable extension of K (4.6.3), therefore a finite separable extension of a pure extension $K(T_1, \dots, T_n)$ (T_i indeterminates); there is then an element $x \in L$, integral over the ring $K[T_1, \dots, T_n]$, and such that $L = K(T_1, \dots, T_n)(x)$; let $G \in K[T_1, \dots, T_n, T_{n+1}]$ be its minimal polynomial. There exists $g \neq 0$ of A such that all the coefficients $\neq 0$ of G (which all belong to K) belong to A_g ; replacing A by A_g (which amounts to replacing S by a neighbourhood of η), we can therefore suppose that G has its coefficients in A ; designating by F the polynomial G considered as element of $A[T_1, \dots, T_n, T_{n+1}]$, we then have $G = F_{(k(\eta))}$. Set $Y = \text{Spec}(A[T_1, \dots, T_{n+1}]/(F))$, so that $Y_\eta = \text{Spec}(K[T_1, \dots, T_{n+1}]/(G))$; Y_η is an integral scheme having L as field of rational functions. As X_η and Y_η are Noetherian, there exists a non-empty open $V \subset Y_\eta$ and an immersion (necessarily dominant) $v : V \rightarrow X_\eta$ (I, 6.5.1), (ii) and (6.5.4), (ii). Let W be an open of Y such that $W \cap Y_\eta = V$; applying (8.8.2), (i) and (8.10.5), (iii) following the method of (8.1.2), a), replacing S by a neighbourhood of η , we can suppose that $v = w_\eta$ where $w : W \rightarrow X$ is an open immersion.

Since we have seen (4.6.3) that the criterion for a prescheme algebraic integral to be geometrically integral depends only on its field of rational functions; as X_η is geometrically integral by hypothesis,

it is the same for Y_η , and the definition of G implies therefore that this polynomial is geometrically irreducible (9.7.4). Applying (9.7.5), we see that there is a neighbourhood U of η in S such that $F_{(k(s))}$ is geometrically irreducible for every $s \in U$, therefore Y_s is geometrically integral for $s \in U$ (9.7.4); furthermore we can also suppose that for $s \in U$, W_s is not empty (9.5.1), and therefore is geometrically integral (4.6.3); finally, we can also suppose that for $s \in U$, $w_s : W_s \rightarrow X_s$ is an open immersion (9.6.1), (x), and that its image in X_s is everywhere dense (9.5.3). In other words, for $s \in U$, there is in X_s an open everywhere dense W'_s which is geometrically integral; the criterion (4.5.9), c) therefore shows that X_s is already geometrically irreducible for $s \in U$. Finally, as X_η is reduced and without associated prime cycle (3.2.1), we can also suppose that for $s \in U$, X_s has no associated prime cycle (9.7.6); this being the case (for $s \in U$ fixed), let Ω be an algebraically closed extension of $k(s)$, and let $p : (X_s)_{(\Omega)} \rightarrow X_s$ be the canonical projection; $p^{-1}(W'_s)$ is an open dense in $(X_s)_{(\Omega)}$ (2.3.10) and is integral by hypothesis; furthermore $(X_s)_{(\Omega)}$ has no associated prime cycle (4.2.7), therefore we conclude from (3.2.1) that $(X_s)_{(\Omega)}$ is reduced; we conclude therefore from (3.2.1) and (4.6.1) that X_s is geometrically integral (4.6.1).

2° Suppose that X_η is geometrically irreducible; as X_{red} is also of finite type over S (I, 5.1.8), taking into account (I, 5.4), (vi), we can, by replacing X by X_{red} , then X_η is also integral, and as by hypothesis X_η is geometrically irreducible, it is geometrically integral, and we are brought back to the conditions of 1° and we conclude (returning to the initial hypotheses) that X_s is geometrically irreducible for s close to η .

3° Suppose X_η geometrically connected, and let Z_i ($1 \leq i \leq n$) be the irreducible components of X_η ; there exists (by virtue of (5.10.8.1) applied to $\mathfrak{T} = \{\emptyset\}$) a surjective map $j \mapsto v(j)$ of an interval $[1, m]$ of $\mathbf{N}$ onto $[1, n]$ such that $Z_{v(j)} \cap Z_{v(j+1)} \neq \emptyset$ for $1 \leq j \leq m$. For every i , let X_i be the closure of Z_i in X , and let Y be the reunion of the X_i ; as $Y_\eta = X_\eta$ by definition, we can, by virtue of (9.5.1), suppose that $Y_s = X_s$ for every $s \in S$. Now, considering the reduced sub-preschemes of X having the X_i as underlying spaces, we see by 2° that there exists a neighbourhood U of η in S such that for $s \in U$ the $(X_i)_s$ are geometrically irreducible (since the $(X_i)_\eta$ can be supposed to be geometrically irreducible, as we saw at the beginning). Furthermore, we can also suppose that for $s \in U$, $(X_{v(j)})_s \cap (X_{v(j+1)})_s \neq \emptyset$ (9.5.1) for $1 \leq j \leq m$; we conclude immediately that X_s is connected, therefore (4.5.13.1) geometrically connected for $s \in U$.

4° Suppose X_η geometrically reduced; let Z_i, W_i be the irreducible components of X_η , where W_i is the interior of Z_i in X_η ; there is for each i an open V_i of X such that $W_i = V_i \cap X_\eta$ for every i ; as the W_i are opens, pairwise disjoint, and their reunion is dense in X_η , (9.5.1), (9.5.3), and (9.5.4) allow us to suppose that for s close to η , the $(V_i)_s$ are opens, pairwise disjoint, in X_s and their reunion is dense in X_s . Furthermore, as the W_i are geometrically reduced and have been supposed at the beginning geometrically irreducible, it follows from 1° that for s close to η , the $(V_i)_s$ are geometrically integral, and *a fortiori* reduced. On the other hand, we deduce from (9.7.6) that X_s has no associated prime cycle, since it is already reduced (3.2.1); we conclude therefore from (3.2.1) and (4.6.1) that X_s is reduced, and from (4.6.1) that X_s is geometrically reduced. $\square$

9.8. Primary decomposition in the neighbourhood of a generic fibre

Proposition (9.8.1). — *Let $f : X \rightarrow S$ be a morphism of finite presentation, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module of finite presentation. Then the set E of $s \in S$ such that $\mathcal{F}_s$ is an $\mathcal{O}_{X_s}$ -Module without associated immersed prime cycle is locally constructible.*

Proof. We know that for quasi-coherent Modules on algebraic preschemes, the property of being without associated immersed prime cycle is invariant by change of base field (4.2.7); we can thus restrict to the case where S is affine, Noetherian and integral, and to proving that in this case E or $S - E$ is a neighbourhood of η . We saw in (9.7.6) that if $\eta \in E$, E is a neighbourhood of η ; it remains thus to consider the case where $\mathcal{F}_\eta$ has associated immersed prime cycles. We can restrict to the case where X is affine, and where there is a sub- $\mathcal{O}_{X_\eta}$ -Module coherent $\mathcal{H}$ of $\mathcal{F}_\eta$ whose support Z' is non-empty and rare with respect to the support Z of $\mathcal{F}$ (3.1.3); then, by virtue of (8.5.2), (i) and (ii) and (8.5.8), applied following the method of (8.1.2), a), there exists a sub- $\mathcal{O}_X$ -Module coherent $\mathcal{G}$ of $\mathcal{F}$ such that $\mathcal{G}_\eta = \mathcal{H}$; if $Y = \text{Supp}(\mathcal{F})$, $Y' = \text{Supp}(\mathcal{G})$, we have by extension $Y_s = \text{Supp}(\mathcal{F}_s)$, $Y'_s = \text{Supp}(\mathcal{G}_s)$ for all $s \in S$ (I, 9.1.13) and in particular $Z = Y_\eta$, $Z' = Y'_\eta$; as $Z - Z'$ is dense in Z and $Z' + \emptyset$, there is a neighbourhood U of η such that for $s \in U$, $Y_s - Y'_s$ is dense in Y_s , and $Y'_s \neq \emptyset$ (9.5.1)

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and (9.5.3); by considering a generic point of an irreducible component of Y'_s and a neighbourhood sufficiently small of this point in X_s , we deduce immediately from (3.1.3) that $\mathcal{F}_s$ admits associated immersed prime cycles.

(9.8) Let S be a Noetherian prescheme, *integral*, of generic point η , $f : X \rightarrow S$ a morphism of finite type, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module. Consider an irredundant reduced decomposition $(\mathcal{G}_i)_{i \in I}$ of $\mathcal{F}_\eta$ (3.2.6); the $\mathcal{G}_i$ are thus quotients of $\mathcal{F}_\eta$ and there is an injective homomorphism $h : \mathcal{F}_\eta \rightarrow \bigoplus \mathcal{G}_i$; furthermore $\text{Ass}(\mathcal{G}_i)$ is reduced to a point x_i . Let Z_i be the closure of $\{x_i\}$ in X , so that $(Z_i)_\eta = \text{Supp}(\mathcal{G}_i)$. Denote by $\mathcal{J}_i$ the coherent Ideal of $\mathcal{O}_X$ defining the closed sub-prescheme of X of underlying space Z_i (sub-prescheme still denoted Z_i). By virtue of (8.5.2) and (8.5.8), applied following the method of (8.1.2, a)), we can (by restricting S to a neighbourhood of η if needed) suppose that there exist quotients $\mathcal{F}_i$ of $\mathcal{F}$ ($i \in I$) such that $\mathcal{G}_i = (\mathcal{F}_i)_\eta$ for all i , and a homomorphism $g : \mathcal{F} \rightarrow \bigoplus \mathcal{F}_i$ such that $h = g_\eta$. Furthermore (I, 9.3.5), there exists an integer m such that $(\mathcal{J}_i^m \mathcal{F}_i)_\eta = (\mathcal{J}_i)_\eta^m \mathcal{G}_i = 0$ and by restricting S again, we can suppose that $\mathcal{J}_i^m \mathcal{F}_i = 0$ (8.5.2.5), so that the support of $\mathcal{F}_i$ is contained in Z_i ; but as it is closed and contains x_i , it is necessarily equal to Z_i . $\square$

Proposition (9.8.3). — *Under the conditions of (9.8), for all $s \in S$, and all $i \in I$, denote by $x_{i\alpha}$ ($\alpha \in J_{s,i}$) the maximal points of $(Z_i)_s$. There exists a neighbourhood U of η in S such that, for all $s \in U$, the $x_{i\alpha}$ (for $i \in I$ and, for each i , $\alpha \in J_{s,i}$) are pairwise distinct and that $\text{Ass}(\mathcal{F}_s)$ is the set of $x_{i\alpha}$ (in other words, the prime cycles associated to $\mathcal{F}_s$ are the irreducible components of $(Z_i)_s$). Furthermore, one can take U such that, for the closure of $\{x_{i\alpha}\}$ in X_s to be a maximal associated prime cycle of $\mathcal{F}_s$, it is necessary and sufficient that $(Z_i)_\eta$ (closure of $\{x_i\}$ in X_η) be a maximal associated prime cycle of $\mathcal{F}_\eta$.*

Proof. It follows from (3.1.3, c')) that for each i , there exists an open W_i in X_η such that $W_i \cap (Z_i)_\eta$ is non-empty, and a coherent $\mathcal{O}_{X_\eta}$ -Module $\mathcal{H}_i$, of support $W_i \cap (Z_i)_\eta$, such that there is an injective homomorphism $v_i : \mathcal{H}_i \rightarrow \mathcal{F}_\eta|_{W_i}$. Let V_i be an open of X such that $V_i \cap X_\eta = W_i$; applying, as in (9.8.2), the results of (8.5.2) and (8.5.8), one can (by restricting S) suppose that there exists a coherent Module $\mathcal{F}'_i$, of support $V_i \cap Z_i$ and a homomorphism $u_i : \mathcal{F}'_i \rightarrow \mathcal{F}|_{V_i}$ such that $\mathcal{H}_i = (\mathcal{F}'_i)_\eta$ and $v_i = (u_i)_\eta$. We will show that there is a neighbourhood U of η such that for $s \in U$, the following properties are satisfied:

- (i) The $(\mathcal{F}_i)_s$ are without associated immersed prime cycle.
- (ii) The homomorphism $g_s : \mathcal{F}_s \rightarrow \bigoplus (\mathcal{F}_i)_s$ is injective.
- (iii) $(Z_i)_s \cap (V_i)_s$ is dense in $(Z_i)_s$, $(\mathcal{F}'_i)_s$ has support $(Z_i)_s \cap (V_i)_s$, and $(u_i)_s : (\mathcal{F}'_i)_s \rightarrow \mathcal{F}_s|_{(V_i)_s}$ is injective.

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- (iv) For $i \neq j$, every irreducible component of $(Z_i)_s$ is distinct from every irreducible component of $(Z_j)_s$.

Now, (i) has already been seen (9.7.6); (ii) is a particular case of (9.4.5); (iii) follows from (9.5.3) and (9.4.5). Finally, if $i \neq j$, $(Z_i)_\eta \cap (Z_j)_\eta = (Z_i \cap Z_j)_\eta$ is rare in $(Z_i)_\eta$ or in $(Z_j)_\eta$; suppose for example that $(Z_i \cap Z_j)_\eta$ is rare in $(Z_j)_\eta$; then it follows from (9.5.3) and (9.5.4) that for s in a neighbourhood of η , $(Z_i \cap Z_j)_s = (Z_i)_s \cap (Z_j)_s$ is rare in $(Z_j)_s$, which shows that no irreducible component of $(Z_j)_s$ can be contained in an irreducible component of $(Z_i)_s$, nor *a fortiori* be equal to it.

This being so, it follows from (ii) and (3.1.7) that for $s \in U$, we have

$$\text{Ass}(\mathcal{F}_s) \subset \bigcup \text{Ass}((\mathcal{F}_i)_s),$$

from (i) that $\text{Ass}((\mathcal{F}_i)_s)$ is the set of $x_{i\alpha}$ ($\alpha \in J_{s,i}$). On the other hand, by virtue of (iii) and the criterion (3.1.3, c')), each of the $x_{i\alpha}$ ($\alpha \in J_{s,i}$, $i \in I$) belongs to $\text{Ass}(\mathcal{F}_s)$. Finally (iv) signifies that the $x_{i\alpha}$ are pairwise distinct.

It remains to prove the last assertion of the statement. Now, it follows from (i) that for s and for given i , none of the sets $\{x_{i\alpha}\}$ can be contained in another for $\alpha \in J_{s,i}$. On the other hand, if $(Z_j)_\eta \subset (Z_i)_\eta$, we can suppose U taken such that $(Z_j)_s \subset (Z_i)_s$ for $s \in U$ (9.5.1), so each $x_{j\alpha}$ belongs to the closure of a $x_{i\alpha}$; on the contrary, if $(Z_j)_\eta \cap (Z_i)_\eta$ is rare in $(Z_j)_\eta$, we have seen by proving (iv) that $(Z_j)_s \cap (Z_i)_s$ is rare in $(Z_j)_s$, so none of the $x_{j\alpha}$ is adherent to a $x_{i\alpha}$. In particular, if $(Z_j)_\eta$ is maximal, this is equivalent to saying that $(Z_j)_\eta$ is rare in $(Z_i)_\eta$ for all $i \neq j$, and we conclude that $(Z_j)_s \cap (Z_i)_s$ is rare in $(Z_j)_s$ for all $i \neq j$, so that every $x_{j\alpha}$ ($\alpha \in J_{s,j}$) is *maximal*. $\square$

Corollary (9.8.4). — *The notations and hypotheses being those of (9.8), there exists a neighbourhood U of η in S such that, for all $s \in U$, each of the $(\mathcal{F}_i)_s$ is without associated immersed prime cycle; furthermore, if $(\mathcal{F}_{i\alpha})_{\alpha \in J_{s,i}}$ is the irredundant reduced decomposition of $(\mathcal{F}_i)_s$, then, for all $s \in U$, the family $(\mathcal{F}_{i\alpha})_{i \in I, \alpha \in J_{s,i}}$ is an irredundant reduced decomposition of $\mathcal{F}_s$.*

Proof. The first assertion follows from (9.8.1) and the definition of $\mathcal{F}_i$. On the other hand, we have seen in (9.8.3) that the homomorphism $\mathcal{F}_s \rightarrow \bigoplus (\mathcal{F}_i)_s$ is injective and by definition it is the same for each of the homomorphisms $(\mathcal{F}_i)_s \rightarrow \bigoplus \mathcal{F}_{i\alpha}$, so the homomorphism $\mathcal{F}_s \rightarrow \bigoplus_{i,\alpha} \mathcal{F}_{i\alpha}$ is injective. As we can suppose (9.4.5) that for $s \in U$ each of the $(\mathcal{F}_i)_s$ is a quotient of $\mathcal{F}_s$, the $\mathcal{F}_{i\alpha}$ are quotients of $\mathcal{F}_s$, and it remains to verify (3.2.5) that the $x_{i\alpha}$ are pairwise distinct and belong to $\text{Ass}(\mathcal{F}_s)$, which has been proved in (9.8.3). $\square$

Proposition (9.8.5). — *Let $f : X \rightarrow S$ be a morphism of finite presentation, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module of finite presentation. For all $s \in S$, let $E(s)$ (resp. $E'(s)$) be the set (finite subset of $\mathbf{Z} \cup \{-\infty\}$) of dimensions of the maximal prime cycles associated to $\mathcal{F}_s$ (resp. of the maximal prime cycles, i.e., the irreducible components of $\text{Supp}(\mathcal{F}_s)$). Then the functions $s \mapsto E(s)$ and $s \mapsto E'(s)$ are locally constructible in S .*

Proof. It follows from (4.2.7) and (4.2.8) that the condition (9.2.1), (i) is satisfied for the properties whose constructibility we wish to show. We are thus reduced to the case where S is affine, Noetherian and integral, and to proving that E and E' are constant in a neighbourhood of the generic point η of S ; but this follows from (9.8.3) and (9.8.6). $\square$

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Proposition (9.8.6). — *With the hypotheses and notations of (9.8), let $i \in I$ be such that $(Z_i)_\eta$ is maximal (in other words, an irreducible component of $\text{Supp}(\mathcal{F}_\eta)$). Then there exists a neighbourhood U of η in S such that for all $s \in U$ and all $\alpha \in J_{s,i}$, the geometric length of $\mathcal{F}_s$ at $x_{i\alpha}$ (relative to $\mathbf{k}(s)$) (4.7.5) is equal to the geometric length of $\mathcal{F}_\eta$ at x_i (relative to $\mathbf{k}(\eta)$).*

Proof. We can evidently restrict to the case where $S = \text{Spec}(A)$ is affine; let us first show that the sub-prescheme $(Z_i)_\eta$ of X_η , which is reduced, is geometrically integral. Indeed there is a finite extension K' of $K = \mathbf{k}(\eta)$ such that $((Z_i)_\eta)_{(K')}$ is geometrically reduced and that the irreducible components of $((Z_i)_\eta)_{(K')}$ are geometrically irreducible (4.6.8). We proceed as in the proof of (9.7.7) by considering a sub- A -algebra A' of K' that is a field of fractions and finite over A . We set $S' = \text{Spec}(A')$ and consider the finite surjective morphism $g : S' \rightarrow S$; let $X' = X_{(S')}$ and $\mathcal{F}' = \mathcal{F} \otimes_g \mathcal{O}_{S'}$ and let η' be the generic point of S' . For all $s' \in S'$, i.e., $s = g(s')$; if T is an irreducible component of $\text{Supp}(\mathcal{F}_s)$, the irreducible components T'_j of $T_{(\mathbf{k}(s'))}$ are irreducible components of $\text{Supp}(\mathcal{F}'_{s'})$ and dominate T (4.2.7) and the radicial multiplicities of T with respect to $\mathcal{F}_s$ and of each T'_j with respect to $\mathcal{F}'_{s'}$ are the same (4.7.9). The reasoning from the first part of (9.7.7) then shows that we can restrict to proving the proposition for X' and $\mathcal{F}'$; and by virtue of the choice of K' , the reduced sub-preschemes having for underlying spaces the irreducible components of $((Z_i)_\eta)_{(K')}$ are geometrically integral (4.6.1).

Suppose now that $(Z_i)_\eta$ is geometrically integral; then (9.7.7), it is the same for (Z_i) , for s in a neighbourhood of η ; the definition (4.7.5) shows that it will therefore suffice to prove that the length of the $\mathcal{O}_{X_\eta, x_i}$ -module $(\mathcal{F}_\eta)_{x_i}$ is equal to that of the $\mathcal{O}_{X_s, x_{i\alpha}}$ -module $(\mathcal{F}_s)_{x_{i\alpha}}$ (where we have dropped the index α , now unnecessary by hypothesis). The question being evidently local on X , we can suppose (by restricting to a neighbourhood of x_i) that $\mathcal{F} = \mathcal{F}_i$, so that $\mathcal{F}_\eta$ is irredundant on $X = \text{Spec}(B)$ affine, and one writes Z instead of Z_i , and x the generic point of Z (and of Z_η). The Noetherian ring B contains A as sub-ring, and $\mathcal{F} = \tilde{M}$, where M is a B -module of finite type; furthermore, if K is the field of fractions of A , the $B_{(K)}$ -module $M_{(K)}$ is $\neq 0$ and irredundant. Let q be the unique element of $\text{Ass}(M_{(K)})$ and let $\mathfrak{p}$ be the prime ideal of B , inverse image of q . Using (5.11.1.1) as in the proof of (9.7.6), we reduce to the case where B is integral and M a sub-module $\neq 0$ of B ; then $\mathcal{F}_\eta$ is a sub- $\mathcal{O}_{X_\eta}$ -Module non-zero of $\mathcal{O}_{Z_\eta}$, and by virtue of (9.4.5), for s in a neighbourhood of η , $\mathcal{F}_s$ is isomorphic to a sub- $\mathcal{O}_{X_s}$ -Module non-zero of $\mathcal{O}_{Z_s}$; as Z_s is geometrically integral, the lengths of $(\mathcal{F}_\eta)_x$ and $(\mathcal{F}_s)_{x_s}$ are both equal to 1, which finishes the proof. $\square$

Corollary (9.8.7). — *Let $f : X \rightarrow S$ be a morphism of finite presentation, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module of finite presentation. For all $s \in S$, let $M(s)$ be the set of pairs (d, m) such that there exists an irreducible component of $\text{Supp}(\mathcal{F}_s)$ of dimension d and of radicial multiplicity m for $\mathcal{F}_s$ (4.7.8). Then the function $s \mapsto M(s)$ is locally constructible in S .*

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Proof. It follows from (4.2.7) and (4.7.9) that the condition (9.2.1), (i) is satisfied for the property whose constructibility we wish to show. We are thus reduced to the case where S is affine, Noetherian and integral, and to proving that M is constant in a neighbourhood of the generic point of S ; but this follows from the proposition, namely (9.8.3) and (9.8.6). $\square$

Proposition (9.8.8). — *Let $f : X \rightarrow S$ be a morphism of finite presentation, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module of finite presentation. For all $s \in S$, let $l(s)$ be the sum of the total multiplicities for $\mathcal{F}_s$ (relative to $\mathbf{k}(s)$) of the generic points of the irreducible components of $\text{Supp}(\mathcal{F}_s)$ (4.7.12). Then the function $s \mapsto l(s)$ is locally constructible in S .*

Proof. Taking into account (4.7.12), the condition (9.2.1), (i) is satisfied for the property whose constructibility we wish to show, and we are thus reduced to the case where S is affine, Noetherian and integral, and to showing that l is constant in a neighbourhood of the generic point η , and constant. Using the notations of (9.8.2), this follows from the definition (4.7.12), from the fact that the geometric number of irreducible components of each $(Z_i)_s$ is constant in a neighbourhood of η (9.7.8), that the geometric length of $\mathcal{F}_s$ at $x_{i\alpha}$ is also equal to that of $\mathcal{F}_\eta$ at x_i for each i such that $(Z_i)_\eta$ is maximal (9.8.6) and finally from the fact that the closure of $\{x_{i\alpha}\}$ is a maximal associated prime cycle of $\mathcal{F}_s$ if and only if $(Z_i)_\eta$ is a maximal associated prime cycle of $\mathcal{F}_\eta$ (9.8.3). $\square$

Remark (9.8.9). — One can make precise in various ways the preceding propositions; let us limit ourselves to a statement given as an example. Let us say that a finite subset P of a k -prescheme algebraic X is *saturated* if, for every pair of points x, y of P , the generic points of the irreducible components of $\{x\} \cap \overline{\{y\}}$ also belong to P ; for every finite subset Q of X , there exists a smallest finite subset P of X containing Q and saturated; we say that P is the *saturate* of Q . For every coherent $\mathcal{O}_X$ -Module $\mathcal{F}$, we will call *primary skeleton* of $\mathcal{F}$ the system (P, Q, ω, d, m) where $Q = \text{Ass}(\mathcal{F})$, P is the saturate of Q , ω the ordering relation $\overline{\{x\}} \subset \overline{\{y\}}$ on P , d the function $x \mapsto \dim \overline{\{x\}}$ on P , m the function $x \mapsto \text{length}_\omega \mathcal{F}_x$ defined on the set of maximal elements of Q for the relation ω . We call *virtual skeleton* any system (P, Q, ω, d, m) where P is a set, Q a subset of P , ω a relation of order on P , d a map from P to $\mathbf{N}$, m a map from $\mathbf{N}$ to the set of maximal elements of Q ; one defines in an obvious way the notion of *isomorphism* of two virtual skeletons. Finally, with the preceding notations, we call *primary type* of $\mathcal{F}$ the class (for the relation of isomorphism of virtual skeletons) of the primary skeleton of $\mathcal{F}$. It follows from (4.2.6), (4.2.7), (4.2.8), (4.5.1) and (4.7.9) that if, for an algebraically closed extension k' of k , one sets $X' = X \otimes_k k'$ and $\mathcal{F}' = \mathcal{F} \otimes_k k'$, the *primary type* of $\mathcal{F}'$ is independent of the algebraically closed extension k' of k considered; we say that this is the *geometric primary type* of $\mathcal{F}$. Given the definitions, the statement we have in view is the following:

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(9.8) Let $f : X \rightarrow S$ be a morphism of finite presentation, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module of finite presentation. For all $s \in S$, let $T(s)$ be the geometric primary type of $\mathcal{F}_s$. Then the function $s \mapsto T(s)$ is locally constructible in S .

Taking into account the remarks that precede, we are reduced as usual to proving that if S is affine, Noetherian and integral, and of generic point η , $T(s)$ is constant in a neighbourhood of η . Reasoning as at the beginning of (9.7.7), we can suppose that all the irreducible components of X_η that intervene are geometrically irreducible, and then the proposition follows from (9.5.1), (9.5.5), (9.8.3) and (9.8.6); we leave the details to the reader. One could generalise by considering several coherent Modules and defining their “simultaneous primary skeleton”, etc. The general conclusion of this paragraph is that for all the properties of the type considered (and for an S -irreducible prescheme) *the properties valid on the “generic fibre” are also valid on all neighbouring fibres*.

9.9. Constructibility of local properties of fibres

Proposition (9.9.1). — *Let $f : X \rightarrow S$ be a morphism locally of finite presentation, Z a locally constructible subset of X such that for all $s \in S$, Z_s is closed in X_s , Φ a finite subset of $\mathbf{Z} \cup \{\pm\infty\}$. Then the following subsets of X are locally constructible:*

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- (i) *The set of $x \in X$ such that $\dim_x(Z_{f(x)})$ belongs to Φ .*
- (ii) *The set of $x \in X$ such that $\operatorname{codim}_x(Z_{f(x)}, X_{f(x)})$ belongs to Φ .*
- (iii) *The set of $x \in X$ such that the local ring $\mathcal{O}_{X_{f(x)}, x}$ is equidimensional.*

We note that properties (i) and (ii) can also be expressed by saying that the functions $x \mapsto \dim_x(Z_{f(x)})$ and $x \mapsto \operatorname{codim}_x(Z_{f(x)}, X_{f(x)})$ are locally constructible in X (0_{III}, (9.3.1)).

Proof. The questions being local on X , we can restrict to the case where $S = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affine and where f is a morphism of finite presentation; there exists then a sub-ring A_0 of A that is a $\mathbf{Z}$ -algebra of finite type, an A_0 -prescheme of finite type X_0 and a constructible subset Z_0 of X_0 such that $X = X_0 \otimes_{A_0} A$ and $Z = h^{-1}(Z_0)$, where $h : X \rightarrow X_0$ is the canonical projection ((8.9.1) and (8.3.11)). Furthermore, for all $s \in S$, if s_0 is the projection of s in $S_0 = \operatorname{Spec}(A_0)$, and h_s the projection $X_s \rightarrow (X_0)_{s_0}$, we have $Z_s = h_s^{-1}((Z_0)_{s_0})$. As the morphism h_s is faithfully flat and quasi-compact, the hypothesis that Z_s is closed in X_s implies that $(Z_0)_{s_0}$ is closed in $(X_0)_{s_0}$ (2.3.12).

This being so, the transitivity of fibres (I, 3.6.4) and the prop. (4.2.7) imply that the set of dimensions of the irreducible components of Z_s containing x is the same as the set of dimensions of the irreducible components of $(Z_0)_{s_0}$ containing $x_0 = h_s(x)$. In particular, we have $\dim_x(Z_s) = \dim_{x_0}((Z_0)_{s_0})$. On the other hand, if $Z_s^{(\beta)}$ are the irreducible components of Z_s containing x and $X_s^{(\alpha)}$ the irreducible components of X_s containing x , we have $\operatorname{codim}_x(Z_s, X_s) = \inf_{\beta, \alpha} (\sup(\operatorname{codim}(Z_s^{(\beta)}, X_s^{(\alpha)})))$, (α, β) varying over the set of pairs such that $x \in Z_s^{(\beta)} \in X_s^{(\alpha)}$ (0, (14.2.6)). As the algebraic preschemes are biequidimensional (5.2.1), we can write, by virtue of (0, (14.3.3.1))

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$$(9.9.1.1) \quad \operatorname{codim}_x(Z_s, X_s) = \inf_{\beta, \alpha} (\sup(\dim(X_s^{(\alpha)}) - \dim(Z_s^{(\beta)})))$$

with the same choice of pairs (α, β) . As h_s is faithfully flat and quasi-compact, for every pair formed of an irreducible component $(X_0)_{s_0}^{(\alpha)}$ of $(X_0)_{s_0}$ containing x_0 and of an irreducible component $(Z_0)_{s_0}^{(\beta)}$ of $(Z_0)_{s_0}$ containing x_0 and contained in $(X_0)_{s_0}^{(\alpha)}$, there exists a pair $(X_s^{(\alpha)}, Z_s^{(\beta)})$ of the type described above and such that $X_s^{(\alpha)}$ dominates $(X_0)_{s_0}^{(\alpha)}$ (2.3.5). The formula (9.9.1.1) (and the analogous formula applied to $(X_0)_{s_0}$) shows then, by virtue of (4.2.7), that we have

$$\operatorname{codim}_x(Z_s, X_s) = \operatorname{codim}_{x_0}((Z_0)_{s_0}, (X_0)_{s_0}).$$

We thus see that if E (resp. E_0) is the set of $x \in X$ (resp. $x_0 \in X_0$) satisfying one of the conditions (i), (ii), (iii) of the statement (resp. the same condition), we have $E = h^{-1}(E_0)$, and by virtue of (1.8.2), we see that one can restrict to considering the case where A is Noetherian, as well as that of (9.9.1.1), and we have (0_{III}, (9.2.3)), and also of (9.9.1), (iii).

By hypothesis Z_η is closed in X_η ; as Z is constructible, it follows from (8.3.11), applied following the method of (8.1.2, a)), that we can, by replacing S by an open neighbourhood of η , suppose that Z is closed in X . Suppose then that the irreducible components of X (resp. Z) are X_i (resp. Z_j) the irreducible components of X_η (resp. Z_η); by virtue of (0_I, (2.1.8)), the $X_i \cap X_\eta$ (resp. $Z_j \cap X_\eta$) are the irreducible components of X_η (resp. Z_η) containing x ; by virtue of (9.5.1), we can suppose further, by restricting S to a neighbourhood, that for all $s \in S$, the X_i (resp. Z_j) are exactly the irreducible components of X (resp. Z) meeting $\overline{\{x\}}$ and that the $(X_i)_s \cap \overline{\{x\}}$ and $(Z_j)_s \cap \overline{\{x\}}$ are non-empty for all $s \in S$. This being so, it follows still from (9.7.1) that we can suppose, by restricting S , that the couples (i, j) such that $(Z_j)_s \subset (X_i)_s$ are the same for all $s \in S$. The conclusion follows then from (9.5.6): for all s sufficiently close to η , all the irreducible components of $(X_i)_s$ (resp. $(Z_j)_s$) have the same dimension equal to that of $(X_i)_\eta$ (resp. $(Z_j)_\eta$). Furthermore, if (i, j) is a pair such that $Z_j \neq X_i$, $(X_i)_\eta$ does not contain the generic point of $(Z_j)_\eta$, hence $\dim((X_i \cap Z_j)_\eta) < \dim((Z_j)_\eta)$; consequently, the common dimension of the irreducible components of $(X_i)_s \cap (Z_j)_s$ is, for all $s \in S$, strictly less than the common dimension of the irreducible components of $(Z_j)_s$, which proves that

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none of the irreducible components of $(Z_i)_s$ is contained in an irreducible component of $(X_i)_s$ for $s \in S$. $\square$

Proposition (9.9.2). — *Let $f : X \rightarrow S$ be a morphism locally of finite presentation, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module of finite presentation, Φ a finite subset of $\mathbf{Z} \cup \{\pm\infty\}$. The following subsets of X are locally constructible:*

- (i) *The set of $x \in X$ such that $\mathcal{F}_{f(x)}$ is equidimensional at the point x (5.1.12).*
- (ii) *The set of $x \in X$ such that the geometric lengths of $\mathcal{F}_{f(x)}$ relative to $\mathbf{k}(f(x))$, at the generic points of the irreducible components of $\text{Supp}(\mathcal{F}_{f(x)})$ containing x (4.7.5), belong to Φ .*
- (iii) *The set of $x \in X$ such that the dimensions of the prime cycles associated to $\mathcal{F}_{f(x)}$ and containing x belong to Φ .*
- (iv) *The set of $x \in X$ such that $\mathcal{F}_{f(x)}$ is geometrically reduced at the point x (4.6.17).*
- (v) *The set of $x \in X$ such that $\mathcal{F}_{f(x)}$ is geometrically integral at the point x (4.6.22).*
- (vi) *The set of $x \in X$ such that $\dim \cdot \text{proj}((\mathcal{F}_{f(x)})_x) \in \Phi$.*
- (vii) *The set of $x \in X$ such that $\text{coprof}((\mathcal{F}_{f(x)})_x) \in \Phi$.*
- (viii) *The set of $x \in X$ such that $\mathcal{F}_{f(x)}$ possesses property (S_n) at the point x (5.7.2).*
- (ix) *The set of $x \in X$ such that $\mathcal{F}_{f(x)}$ is a Cohen-Macaulay $\mathcal{O}_{X_{f(x)}}$ -Module at the point x (5.7.1).*

Proof. (i) The support Z of $\mathcal{F}$ is locally constructible and closed in X (8.9.1), and by considering a sub-prescheme of X having Z for underlying space and which is of finite presentation over S (8.9.1), we see that the assertion (i) is a particular case of (9.9.1), (iii)).

(ii) All the properties considered are local on X , and we will therefore restrict to the case where $X = \text{Spec}(B)$ and $S = \text{Spec}(A)$ are affine and f a morphism of finite presentation. We keep the notations from the beginning of the proof of (9.9.1), and we suppose furthermore A_0 chosen so that there exists a coherent $\mathcal{O}_{X_0}$ -Module $\mathcal{F}_0$ such that $\mathcal{F}$ is isomorphic to $\mathcal{F}_0 \otimes_{A_0} A$. Then ((4.2.7) and (4.7.9)) the set of geometric lengths of $(\mathcal{F}_0)_{f(x_0)}$ at the generic points of the irreducible components of $\text{Supp}((\mathcal{F}_0)_{f(x_0)})$ which contain x_0 is the same as the analogous set for $\mathcal{F}_{f(x)}$; in other words, if E (resp. E_0) is the set of $x \in X$ (resp. $x_0 \in X_0$) satisfying the condition (ii) of the statement, we have $E = h^{-1}(E_0)$, and by virtue of (1.8.2), we see that one can restrict to considering the case where A is Noetherian; in other words, say that S is integral and $\eta = f(x)$ its generic point. Let S' be the reduced closed subscheme of S whose underlying space is $\overline{\{f(x)\}}$, and put $X' = f^{-1}(S')$. The fibres of X and X' at the points of S' are the same, so we may replace S by S' and X by X' ; thus we may suppose that S is integral and that $\eta = f(x)$ is its generic point.

Now, if one sets $Z = \text{Supp}(\mathcal{F})$, the same reasoning as in the proof of (9.9.1) shows that, if Z_i are the irreducible components of Z containing x , we can suppose that they are exactly the irreducible components of Z meeting $\{x\}$ and that $(Z_i)_s \cap \{x\} \neq \emptyset$ for all $s \in S$. The conclusion follows from (9.8.3) and (9.8.6).

(iii) We reason as in (iii), using this time (4.7.11). With the same notations as in (iii), the prime cycles associated to $\mathcal{F}_s$ that are irreducible components of $(Z'_i)_s$ are immersed if and only if $(Z'_i)_\eta$ is an associated immersed prime cycle of $\mathcal{F}_\eta$. We conclude already that if x belongs to a (resp. does not belong to any) associated immersed prime cycle of $\mathcal{F}_\eta$, the set of $x' \in \overline{\{x\}}$ such that x' belongs to a (resp. does not belong to any) associated immersed prime cycle of $\mathcal{F}_{f(x')}$ is a neighbourhood of x in $\overline{\{x\}}$. The conclusion follows from this remark, from the characterisation of the points where a Module is geometrically reduced (4.7.10), and of (ii).

(iv) Taking into account (4.7.11), we reduce still again to the case where S is Noetherian and integral and where $\eta = f(x)$ is its generic point. Let Y be a closed sub-prescheme of X having $\text{Supp}(\mathcal{F})$ for underlying space. Reasoning as at the beginning of the proof of (9.7.7), we see that there exists a finite integral A -algebra A' (such that if $S' = \text{Spec}(A')$, the morphism $S' \rightarrow S$ is finite and surjective) such that if $Y' = Y_{(S')}$ and η' is the generic point of S' , the irreducible components of $Y_{\eta'}$ are geometrically irreducible. On the other hand, if $X' = X_{(S')}$, the morphism projection $h : X' \rightarrow X$ is finite and surjective; consequently, if x' is a point of X' above x , we have $h(\{x'\}) = \{x\}$ and the image by h of a neighbourhood of x' in $\{x'\}$ is a neighbourhood of x in $\{x\}$. Taking into account (4.7.11) and setting $\mathcal{F}' = \mathcal{F}_{(S')}$, we are thus reduced to proving that if $\mathcal{F}_{\eta'}$ is (resp. is not) geometrically integral at the point x' , the set of $y' \in \overline{\{x'\}}$ such that $\mathcal{F}'_{f'(y')}$ (where

$f' = f_{(S')} : X' \rightarrow S'$ is (resp. is not) geometrically integral at the point y' is a neighbourhood of x' in $\overline{\{x'\}}$. We can thus restrict to the case where $X' = X$ and where the irreducible components of Y_η are *geometrically irreducible*; if one denotes by Z_k the closed subsets of X such that $Z_k \cap X_\eta$ are the irreducible components of Y_η , it follows from (9.7.7), (9.7.8) and (9.5.3) that for all s sufficiently close to η , the $(Z_k)_s$ are the irreducible components of Y_s and that they are *geometrically irreducible*. To say that at a point $y \in X_s$, $\mathcal{F}_s$ is geometrically integral means that $\mathcal{F}_s$ is geometrically reduced at this point and furthermore that y belongs to a *single* of the $(Z_i)_s$ (4.6.22). The conclusion follows thus from part (iv) and of (9.5.1) applied to the intersection of $\overline{\{x\}}$ and of each Z_i .

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(v) Taking into account (4.7.11), we reduce once more to the case where S is Noetherian and integral and where $\eta = f(x)$ is its generic point. Let Y be a closed sub-prescheme of X having $\text{Supp}(\mathcal{F})$ for underlying space. Reasoning as at the beginning of the proof of (9.7.7), we see that there exists a finite integral A -algebra A' (such that if $S' = \text{Spec}(A')$, the morphism $S' \rightarrow S$ is finite and surjective) such that if $Y' = Y_{(S')}$ and if η' is the generic point of S' , the irreducible components of $Y_{\eta'}$ are *geometrically irreducible*. On the other hand, if $X' = X_{(S')}$, the morphism projection $h : X' \rightarrow X$ is finite and surjective; consequently, if x' is a point of X' above x , we have $h(\overline{\{x'\}}) = \overline{\{x\}}$ and the image by h of a neighbourhood of x' in $\overline{\{x'\}}$ is a neighbourhood of x in $\overline{\{x\}}$. Taking into account (4.7.11) and setting $\mathcal{F}' = \mathcal{F}_{(S')}$, we are thus reduced to proving that if $\mathcal{F}_{\eta'}$ is (resp. is not) geometrically integral at the point x' , the set of $y' \in \overline{\{x'\}}$ such that $\mathcal{F}'_{f'(y')}$ (where $f' = f_{(S')} : X' \rightarrow S'$) is (resp. is not) geometrically integral at the point y' is a neighbourhood of x' in $\overline{\{x'\}}$. We can thus restrict to the case where $X' = X$ and where the irreducible components of Y_η are *geometrically irreducible*; if one denotes by Z_k the closed subsets of X such that $Z_k \cap X_\eta$ are the irreducible components of Y_η , it follows from (9.7.7), (9.7.8) and (9.5.3) that for all s sufficiently close to η , the $(Z_k)_s$ are the irreducible components of Y_s and that they are *geometrically irreducible*. To say that at a point $y \in X_s$, $\mathcal{F}_s$ is geometrically integral means that $\mathcal{F}_s$ is geometrically reduced at this point and furthermore that y belongs to a *single* of the $(Z_i)_s$ (4.6.22). The conclusion follows thus from part (iv) and of (9.5.1) applied to the intersection of $\overline{\{x\}}$ and of each Z_i .

(vi) Keeping the notations of (ii), (6.2.1) gives

$$\dim \cdot \text{proj}((\mathcal{F}_s)_x) = \dim \cdot \text{proj}(((\mathcal{F}_0)_{s_0})_{x_0}),$$

so we may again restrict to the case where A is Noetherian. It remains to show that for every $x \in X$ there is a neighbourhood V of x in $\overline{\{x\}}$ such that

$$\dim \cdot \text{proj}((\mathcal{F}_{f(x')})_{x'}) = \dim \cdot \text{proj}((\mathcal{F}_{f(x)})_x)$$

for every $x' \in V$. As above, we may suppose that S is integral and that $\eta = f(x)$ is its generic point, so $(\mathcal{F}_\eta)_x = \mathcal{F}_x$. By generic flatness (6.9.1), after replacing S by an open neighbourhood of η , we may suppose that f and $\mathcal{F}$ are flat over S . Then (6.2.3) gives

$$\dim \cdot \text{proj}((\mathcal{F}_{f(x')})_{x'}) = \dim \cdot \text{proj}(\mathcal{F}_{x'})$$

for every $x' \in X$. By (6.11.1), after replacing X by an open neighbourhood of x , we may suppose that

$$\dim \cdot \text{proj}(\mathcal{F}_{x'}) \leq \dim \cdot \text{proj}(\mathcal{F}_x)$$

for every $x' \in X$.

If $\dim \cdot \text{proj}(\mathcal{F}_x) = n$, there is a finite $\mathcal{O}_x$ -module M such that $\text{Ext}_{\mathcal{O}_x}^n(\mathcal{F}_x, M) \neq 0$ (0, 17.2.4). After replacing X by an open neighbourhood of x , there exists a coherent $\mathcal{O}_X$ -module $\mathcal{G}$ such that $M = \mathcal{G}_x$ (0, 5.3.8); by virtue of (T, (4.2.2)), we have $(\mathcal{E}x_{\mathcal{O}_X}^n(\mathcal{F}, \mathcal{G}))_x \neq 0$. But $\mathcal{E}x_{\mathcal{O}_X}^n(\mathcal{F}, \mathcal{G})$ is a coherent $\mathcal{O}_X$ -Module (0_{III}, (12.3.3)), hence its support Y is closed (0_I, (5.2.2)); as it contains x , it contains also $\overline{\{x\}}$, whence we conclude (by applying again (T, (4.2.2))) that $\dim \cdot \text{proj}(\mathcal{F}_{x'}) \geq n$ for all $x' \in \overline{\{x\}}$, which achieves the proof of the assertion in case (vi).

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(vii) As B is an A -algebra of finite type, X is S -isomorphic to a *closed sub-scheme* of an S -scheme of the form $Y = \text{Spec}(A[T_1, \dots, T_r])$; if $j : X \rightarrow Y$ is the canonical injection, and $\mathcal{G}_s = (j_s)_*(\mathcal{F}_s)$ for all $s \in S$, and, by virtue of (0, (16.4.11)), we can restrict to proving the assertion for Y and $\mathcal{G}$. In other words, we can suppose that $B = A[T_1, \dots, T_r]$, so that the schemes $X_s = \text{Spec}(\mathbf{k}(s)[T_1, \dots, T_r])$ are

regular (0, (17.3.7)). Let $W = \text{Supp}(\mathcal{F})$ and $W_s = \text{Supp}(\mathcal{F}_s)$ (I, 9.1.13); we have, by (6.11.2.1)

$$(9.9.2.1) \quad \text{coprof}((\mathcal{F}_{f(x)})_x) = \dim \cdot \text{proj}((\mathcal{F}_{f(x)})_x) - \text{codim}_x(W_{f(x)}, X_{f(x)}).$$

But as W is constructible (8.9.1) and each W_s closed, it follows from (9.9.1), (ii) that the two functions from the right-hand side of (9.9.2.1) are constructible, which finishes the proof of the proposition in the case (vii).

(viii) Let U_n be the set of $x \in X$ such that $\text{coprof}((\mathcal{F}_{f(x)})_x) \leq n$, and set $Z_n = X - U_n$; it follows from (vii) that the Z_n are constructible; furthermore, as the function $x \mapsto \dim_x(W_{f(x)})$ takes only a finite number of values, the numbers $\dim(W_{f(x)})$ have a finite upper bound m when x ranges over X ; as $\text{coprof}((\mathcal{F}_{f(x)})_x) \leq \dim((\mathcal{F}_{f(x)})_x) \leq \dim(W_{f(x)})$, we have $Z_n = \emptyset$ for $n \geq m$. By (5.7.4), the set of $x \in X$ where $(\mathcal{F}_{f(x)})_x$ possesses property (S_k) is the set of $x \in X$ satisfying all the relations

$$(9.9.2.2) \quad \text{codim}_x((Z_n)_{f(x)}, W_{f(x)}) > n + k$$

for all $n \geq 0$; as this relation is automatically satisfied for $n \geq m$, we need only consider the relations (9.9.2.2) for $0 \leq n < m$. But by virtue of (9.9.1), (ii), the set $V_{n,k}$ of x satisfying (9.9.2.2) is constructible, and so is the intersection of these sets for $0 \leq n < m$.

(ix) The assertion follows immediately from (vii), the set of $x \in X$ where $(\mathcal{F}_{f(x)})_x$ is a Cohen-Macaulay module being defined by the relation $\text{coprof}((\mathcal{F}_{f(x)})_x) = 0$. $\square$

Corollary (9.9.3). — Let $f : X \rightarrow S$ be a morphism of finite presentation, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module of finite presentation, **P** one of the properties (i) to (ix) of (9.9.2). Then the set of $s \in S$ such that the property **P** is true at all the points $x \in X_s$, is locally constructible in S .

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Proof. Indeed, its complement is the image by f of the complement of the set E of points where **P** is true. As E is locally constructible in X , so is $X - E$, and hence $f(X - E)$ is locally constructible in S by virtue of the theorem of Chevalley (1.8.4). $\square$

Proposition (9.9.4). — Let $f : X \rightarrow S$ be a morphism locally of finite presentation. The set of $x \in X$ such that $X_{f(x)}$ has at the point x one of the (determined) following properties:

- (i) being geometrically regular;
- (ii) possessing property (R_k) geometric;
- (iii) being geometrically normal;
- (iv) being geometrically reduced (i.e. separable);
- (v) being geometrically pointwise integral;

is a locally constructible subset of X .

Proof. For properties (iv) and (v), this follows from (9.9.2), (iv) and (v)) applied to $\mathcal{F} = \mathcal{O}_X$. For the other properties, taking into account (6.7.8), we reduce, as at the beginning of (9.9.2), (ii) to the case where S is Noetherian and integral and of generic point $\eta = f(x)$. Furthermore, by virtue of the generic flatness theorem (6.9.1), we can, by replacing S by a neighbourhood open of η , suppose that f is a flat morphism. To say that $X_{f(y)}$ is geometrically regular at the point y means that f is regular at the point y (6.8.1), and we know that the set L of these points is open in X (6.8.7), which proves the proposition in case (i).

To prove case (ii), set $F = X - L$, which is closed in X . To say that X_s satisfies the geometric property (R_k) at the point $y \in f^{-1}(s)$ means either that $y \in L$, or that the generic points z_i of the irreducible components of the closed set F_s which contain y satisfy the relation $\dim(\mathcal{O}_{X_{z_i}, s_i}) \geq k + 1$ (taking into account (4.2.7) and (5.2.3)); in other words, the points $y \in F_s$ where X_s satisfies property (R_k) geometric are those such that $\text{codim}_y(F_s, X_s) \geq k + 1$ (5.1.2). The conclusion follows thus from (i) and (9.9.1), (ii).

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Finally, in case (iii), the conclusion follows from (ii), from (9.9.2), (viii), from the fact that property (S_k) is stable by extension of the base field (6.7.1) and finally from the criterion of Serre (5.8.6). $\square$

Corollary (9.9.5). — Let $f : X \rightarrow S$ be a morphism of finite presentation, and denote by **P** one of the properties (i) to (v) of (9.9.4). Then the set of $s \in S$ such that the property **P** is true at all the points $x \in X_s$, is locally constructible in S .

Proof. The proof from (9.9.4) is the same as that of (9.9.3) from (9.9.2). $\square$

Proposition (9.9.6). — *Let $f : X \rightarrow S$ be a morphism locally of finite presentation, $\mathcal{L}_\bullet$ a complex formed of $\mathcal{O}_X$ -Modules quasi-coherent of finite presentation; for all $s \in S$, let $(\mathcal{L}_\bullet)_s$ be the complex $\mathcal{L}_\bullet \otimes_{\mathcal{O}_S} \mathbf{k}(s)$ of $\mathcal{O}_{X_s}$ -Modules coherent of finite type. Then, for a given integer n , the set of $x \in X$ such that $(\mathcal{H}_n((\mathcal{L}_\bullet)_{f(x)}))_x = 0$ is locally constructible in X .*

Proof. We can restrict to the case where $\mathcal{L}_i = 0$ except for $i = 0, 1$ or 2 , and where $n = 1$. Furthermore, the question being local on X , we can restrict to the case where $S = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affine, B being a finitely presented A -algebra. There exists then a Noetherian sub-ring A_0 of A , a finite type A_0 -prescheme X_0 and a complex $\mathcal{L}_\bullet^{(0)}$ of $\mathcal{O}_{X_0}$ -Modules coherent, zero except in dimensions $0, 1$ and 2 and such that $X = X_0 \otimes_{A_0} A$ and $\mathcal{L}_\bullet = \mathcal{L}_\bullet^{(0)} \otimes_{A_0} A$. For all $s \in S$, if s_0 is the projection of s in $S_0 = \operatorname{Spec}(A_0)$, and h_s the projection $X_s \rightarrow (X_0)_{s_0}$, we have $(\mathcal{L}_\bullet)_s = (\mathcal{L}_\bullet^{(0)})_{s_0} \otimes_{\mathbf{k}(s_0)} \mathbf{k}(s)$, and consequently, by virtue of (1.8.2), if $x \in X$ is such that $(\mathcal{H}_n((\mathcal{L}_\bullet)_{f(x)}))_x = 0$ (resp. $\neq 0$), there exists a neighbourhood open V of x in $\overline{\{x\}}$ such that, for all $x' \in V$, we have $(\mathcal{H}_n((\mathcal{L}_\bullet)_{f(x')}))_{x'} = 0$ (resp. $\neq 0$). Replacing S by the reduced closed subprescheme having $\overline{\{f(x)\}}$ as its underlying space, we may suppose that S is integral and that $f(x) = \eta$ is its generic point. Then, by restricting S to a neighbourhood open of η , we can suppose that for all $s \in S$, on a $(\mathcal{H}_n(\mathcal{L}_\bullet))_s = \mathcal{H}_n((\mathcal{L}_\bullet)_s)$ (9.4.3), and by extension, if Z is the support of $\mathcal{H}_n(\mathcal{L}_\bullet)$, the support of $\mathcal{H}_n((\mathcal{L}_\bullet)_s)$ is $Z_s = Z \cap X_s$ (I, 9.1.13.1). The hypothesis is that $Z_\eta \cap \overline{\{x\}} = \emptyset$ (resp. $\neq \emptyset$). As $Z \cap \overline{\{x\}}$ is closed in the Noetherian space X , we conclude from (9.5.1) that there is a neighbourhood U of η in S such that, for all $s \in U$, we have $Z_s \cap \overline{\{x\}} = \emptyset$ (resp. $\neq \emptyset$). The set $V = f^{-1}(U) \cap \overline{\{x\}}$ thus answers the question. $\square$

§10. JACOBSON PRESCHMES

10.1. Very dense subsets of a topological space

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(10.1.1). We say that a subset of a topological space X is *quasi-constructible* if it is a finite union of locally closed subsets of X . We say that a subset T of X is *locally quasi-constructible* if for every $x \in X$, there exists an open neighbourhood V of x such that $T \cap V$ is quasi-constructible in V . It is clear that every quasi-constructible subset of X is locally quasi-constructible; the converse is true if X is quasi-compact. The argument of (III, 9.1.3), where one removes the word “retrocompact”, shows that the set of quasi-constructible (resp. locally quasi-constructible) subsets of X , which we will denote by $\Omega c(X)$ (resp. $\mathfrak{L}qc(X)$), is stable under finite union, finite intersection, and passage to complements. If $f : X \rightarrow Y$ is a continuous map, it follows immediately from these definitions that, for every quasi-constructible (resp. locally quasi-constructible) subset Z of Y , $f^{-1}(Z)$ is quasi-constructible (resp. locally quasi-constructible) in X .

The constructible (resp. locally constructible) subsets of X are evidently quasi-constructible (resp. locally quasi-constructible); the converse is true when X is Noetherian (resp. locally Noetherian).

In what follows, we will denote respectively by $\mathfrak{O}(X)$, $\mathfrak{F}(X)$, $\mathfrak{L}(X)$, $\mathfrak{C}(X)$, $\mathfrak{Lc}(X)$ the set of subsets of X that are respectively open, closed, locally closed, constructible, locally constructible.

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Proposition (10.1.2). — *Let X be a topological space, X_0 a subset of X . The following conditions are equivalent:*

- a) *For every locally closed subset $Z \neq \emptyset$ of X , we have $Z \cap X_0 \neq \emptyset$.*
- a') *For every closed subset Z of X , we have $Z = \overline{Z \cap X_0}$.*
- b) *For every locally quasi-constructible subset $Z \neq \emptyset$ of X , we have $Z \cap X_0 \neq \emptyset$.*
- b') *For every locally quasi-constructible subset Z of X , we have $Z \subset \overline{Z \cap X_0}$ (in other words, $Z \cap X_0$ is dense in Z).*
- c) *The map $U \mapsto U \cap X_0$ of $\mathfrak{O}(X)$ into $\mathfrak{O}(X_0)$ is injective (hence bijective).*
- c') *The map $F \mapsto F \cap X_0$ of $\mathfrak{F}(X)$ into $\mathfrak{F}(X_0)$ is injective (hence bijective).*
- c'') *The map $Z \mapsto Z \cap X_0$ of $\Omega c(X)$ into $\Omega c(X_0)$ is injective (hence bijective).*
- c''') *The map $Z \mapsto Z \cap X_0$ of $\mathfrak{L}qc(X)$ into $\mathfrak{L}qc(X_0)$ is injective.*

Furthermore, when these conditions are satisfied, the map $Z \mapsto Z \cap X_0$ of $\mathfrak{L}qc(X)$ into $\mathfrak{L}qc(X_0)$ is bijective.

Proof. Let us note that the assertions of surjectivity of c) and c') are trivial; they mean that every locally closed subset of X_0 is the trace on X_0 of a locally closed subset of X , and that the map $\Omega c(X) \rightarrow \Omega c(X_0)$ defined in c'') is also surjective.

We will prove the implications

$$c''') \Rightarrow c'' \Rightarrow c' \Rightarrow c \Rightarrow b \Rightarrow b' \Rightarrow a' \Rightarrow a \Rightarrow c''').$$

The first three are trivial. To see that c) implies b), let us first note that c) means that X_0 is *dense* in X . Replacing X and X_0 by U and $U \cap X_0$ respectively, one can therefore restrict to the case where Z is locally closed in X , hence $Z = V \cap \mathbb{C}W$, where V and W are open in X ; the hypothesis $Z \neq \emptyset$ means that $V \not\subset W$, or equivalently $V \cup W \neq W$. By virtue of c), we have $(V \cup W) \cap X_0 \neq W \cap X_0$, hence $V \cap X_0 \not\subset W$, and consequently $(V \cap \mathbb{C}W) \cap X_0 \neq \emptyset$.

To see that b) implies b'), it suffices to apply b) to $Z \cap U$, where U is an open neighbourhood of an arbitrary point of Z . As $Z \cap X_0 \subset \bar{Z}$, it is trivial that b') implies a'). To show that a') implies a), let us note that if Z is locally closed in X , we can write $Z = F - F'$ where F and F' are closed in X and $F' \subset F$; whence $Z \cap X_0 = (F \cap X_0) - (F' \cap X_0)$. If $Z \cap X_0 = \emptyset$, then, by virtue of a'), we would have $F \cap X_0 = F' \cap X_0$, hence $F = F'$, i.e. $Z = \emptyset$.

To see that a) implies c''), it suffices to show that if $Z \cap X_0 \neq \emptyset$, then $Z' \cap X_0 = Z'' \cap X_0$ is equivalent to $((Z' \cup Z'') - (Z' \cap Z'')) \cap X_0 = \emptyset$. In other words, it suffices to prove that a) implies b); indeed, replacing X and X_0 by U and $U \cap X_0$ respectively, one is reduced to the case where Z is locally closed in X , whence the conclusion.

It remains to show that the map $\Omega qc(X) \rightarrow \Omega qc(X_0)$ is surjective. So let Z_0 be a locally quasi-constructible subset of X_0 : there is a covering (V_α) of X_0 by open subsets of X_0 such that $Z_0 \cap V_\alpha$ is quasi-constructible in V_α (and hence also in X_0). By virtue of c), there is for each α a unique quasi-constructible subset Z_α in X such that $Z_0 \cap V_\alpha = X_0 \cap Z_\alpha$. If α and β are any two indices, we have $Z_\beta \cap U_\alpha \cap X_0 = U_\alpha \cap X_0 = V_\alpha \cap Z_0 \cap V_\beta = Z_\alpha \cap X_0 \cap V_\beta \subset Z_0 \cap X_0$; as $Z_\beta \cap U_\alpha$ and Z_α are quasi-constructible in X , it follows from c'') that $Z_\beta \cap U_\alpha \subset Z_\alpha$. If we set $Z = \bigcup_\alpha Z_\alpha$, then $Z \cap U_\alpha = Z_\alpha$ for every α ; moreover, since the V_α cover X_0 , it follows from c) that the U_α cover X , and we see therefore that Z is locally quasi-constructible in X and $Z_0 = Z \cap X_0$. $\square$

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Definition (10.1.3). — When a subset X_0 of a topological space satisfies the equivalent conditions of (10.1.2), we say that X_0 is *very dense* in X .

We have already seen in the course of the proof of (10.1.2) that X_0 is then dense in X .

Corollary (10.1.4). — If X_0 is very dense in X and U an open subset of X , $U \cap X_0$ is very dense in U . Conversely, if (U_α) is an open covering of X such that $U_\alpha \cap X_0$ is very dense in U_α for every α , then X_0 is very dense in X .

As every locally closed subset of U is locally closed in X , the first assertion follows from criterion a) of (10.1.2); and the same is true of the second, since if $Z \neq \emptyset$ is locally closed in X , $Z \cap U_\alpha$ is locally closed in U_α for every α , and $Z \cap U_\alpha \neq \emptyset$ for at least one α .

10.2. Quasi-homeomorphisms

Proposition (10.2.1). — Let X_0, X be two topological spaces, $f : X_0 \rightarrow X$ a continuous map. The following conditions are equivalent:

- a) The map $U \mapsto f^{-1}(U)$ of $\mathfrak{D}(X)$ into $\mathfrak{D}(X_0)$ is bijective.
- a') The map $F \mapsto f^{-1}(F)$ of $\mathfrak{F}(X)$ into $\mathfrak{F}(X_0)$ is bijective.
- b) The topology of X_0 is the inverse image by f of that of X , and $f(X_0)$ is very dense (10.1.3) in X .
- c) The functor $\mathcal{F} \mapsto f^*(\mathcal{F})$ from the category of sheaves of sets (resp. sheaves of abelian groups) on X to the category of sheaves of sets (resp. sheaves of abelian groups) on X_0 is an equivalence of categories.

Proof. It is clear that a) and a') are equivalent, and a) implies that the topology of X_0 is the inverse image by f of that of X . On the other hand, if $f(X_0)$ is not very dense in X , there are two distinct open subsets U_1, U_2 of X such that $U_1 \cap f(X_0) = U_2 \cap f(X_0)$, and hence $f^{-1}(U_1) = f^{-1}(U_2)$, which completes showing that a) implies b). Conversely, the condition b) implies that the maps $U \mapsto U \cap f(X_0)$ of $\mathfrak{D}(X)$ into $\mathfrak{D}(f(X_0))$ and $V \mapsto f^{-1}(V)$ of $\mathfrak{D}(f(X_0))$ into $\mathfrak{D}(X_0)$ are bijective, hence so is the composite $U \mapsto f^{-1}(U)$.

To see that $a)$ implies $c)$, it suffices to apply the definition of $f^*(\mathcal{F})$ (0, 3.7.1) and the axioms for sheaves: $a)$ means that for every open U of X , the canonical map $\Gamma(U, \mathcal{F}) \rightarrow \Gamma(f^{-1}(U), f^*(\mathcal{F}))$ is a functorial bijection in $\mathcal{F}$, whence $c)$. It remains to show that $c)$ implies $b)$.

Suppose first that $f(X_0)$ is not very dense in X ; there then exist two distinct closed subsets $Y \supset Y'$ of X such that $Y \cap f(X_0) = Y' \cap f(X_0)$. Let $\mathcal{F}$ (resp. $\mathcal{F}'$) be the sheaf of abelian groups on X that is the direct image by the canonical injection $Y \rightarrow X$ (resp. $Y' \rightarrow X$) of the simple sheaf associated to the constant presheaf $\mathbf{Z}$ on Y (resp. Y'); the definition of f^* (0, 3.7.1) shows that $f^*(\mathcal{F})$ is isomorphic to $f^*(\mathcal{F}')$, but $\mathcal{F}$ is not isomorphic to $\mathcal{F}'$, hence condition $c)$ is not satisfied. (One notes that the functor f^* is not then even *faithful*, for if u and v are the identity automorphism and the zero endomorphism of $\mathcal{F}/\mathcal{F}'$, $f^*(u)$ and $f^*(v)$ are equal.)

Let us now show that $c)$ implies that the topology of X_0 is the inverse image of that of X by f . Let us note that if condition $c)$ is satisfied for the category of sheaves of sets, it is also satisfied for the category of sheaves of abelian groups, for it follows immediately from the definitions (III, 8.2.5) that the category of *objects in abelian groups* in the category of sheaves of sets is none other than the category of sheaves of abelian groups. It will therefore suffice to prove our assertion supposing $c)$ satisfied for the categories of sheaves of abelian groups on X and X_0 . Now, if $\mathbf{Z}_X$ denotes the simple sheaf on X associated to the constant presheaf $\mathbf{Z}$, there is an isomorphism canonical up to sign $\Gamma(X, \mathcal{F}) \xrightarrow{\sim} \text{Hom}(\mathbf{Z}_X, \mathcal{F})$ functorial in $\mathcal{F}$ (by the same argument as in (0, 5.1.1)); as it is clear by definition (0, 3.7.1) that $f^*(\mathbf{Z}_X) = \mathbf{Z}_{X_0}$, the hypothesis that f^* is an equivalence means that the canonical homomorphism $\Gamma(X, \mathcal{F}) \rightarrow \Gamma(X_0, f^*(\mathcal{F}))$ is *bijective* for every sheaf of abelian groups $\mathcal{F}$ on X . Now, let Y_0 be a closed subset of X_0 ; let $\mathcal{F}_0$ be the sheaf on X that is the direct image by the canonical injection $Y_0 \rightarrow X_0$ of the simple sheaf associated to the constant presheaf $\mathbf{Z}$ on Y_0 . As f^* is an equivalence, $\mathcal{F}_0$ is isomorphic to a sheaf of the form $f^*(\mathcal{F})$, where $\mathcal{F}$ is a sheaf of abelian groups on X . Consider then a section s_0 of $\mathcal{F}_0$ over X_0 such that $(s_0)_{x_0} = 1_{x_0}$ if $x_0 \in Y_0$, $(s_0)_{x_0} = 0_{x_0}$ if $x_0 \notin Y_0$. The preceding remarks show that there exists a section s of $\mathcal{F}$ over X such that $(s_0)_{x_0} = s_{f(x_0)}$ for every $x_0 \in X_0$ (the fibres $(\mathcal{F}_0)_{x_0}$ and $\mathcal{F}_{f(x_0)}$ being canonically identified (0, 3.7.1)); this means that Y_0 is the inverse image by f of the set Y of $x \in X$ such that $s_x \neq 0_x$, and Y is closed in X . $\square$

Definition (10.2.2). — When a continuous map $f : X_0 \rightarrow X$ satisfies the equivalent conditions of (10.2.1), we say that f is a *quasi-homeomorphism* of X_0 into X .

By virtue of (10.2.1), $b)$, to say that a subset X_0 of a topological space X is very dense in X means that the canonical injection $X_0 \rightarrow X$ is a quasi-homeomorphism.

Corollary (10.2.3). — *The composite of two quasi-homeomorphisms is a quasi-homeomorphism.*

This follows immediately from (10.2.1), $a)$.

Corollary (10.2.4). — *Let $f : X \rightarrow Y$ be a quasi-homeomorphism, Y' a locally quasi-constructible subset of Y , $X' = f^{-1}(Y')$; then the restriction $f' = f|_{X'} : X' \rightarrow Y'$ is a quasi-homeomorphism.*

Proof. It is clear by virtue of (10.2.1), $b)$ that the topology induced on X' by that of X is the inverse image by f' of the topology induced on Y' by that of Y . On the other hand, let $Z \neq \emptyset$ be a closed subset in Y' , and $z \in Z$; there is an open neighbourhood U of z in Y such that $U \cap Y'$ is the union of a finite number of closed subsets Y'_j of U ; if j is an index such that $z \in Y'_j$, $Z \cap Y'_j$ is therefore closed in U . Since $f(X) \cap U$ is very dense in U (10.1.4), $Z \cap Y'_j \cap f(X)$ is not empty (10.1.2), $a)$; but as $Z \subset Y'$, we have $Z \cap f(X) = Z \cap f'(X')$; the criterion (10.1.2), $a)$ shows that $f'(X')$ is very dense in Y' , and we conclude by (10.2.1), $b)$. $\square$

Corollary (10.2.5). — *Let $f : X \rightarrow Y$ be a continuous map, (V_α) an open covering of Y . If, for every α , the restriction $f^{-1}(V_\alpha) \rightarrow V_\alpha$ of f is a quasi-homeomorphism, then f is a quasi-homeomorphism.*

This follows immediately from the criterion (10.2.1), $b)$ and from (10.1.4).

Corollary (10.2.6). — *Let $f : X \rightarrow Y$ be a quasi-homeomorphism, Y' a locally quasi-constructible subset of Y , $X' = f^{-1}(Y')$. For Y' to be quasi-compact (resp. Noetherian, resp. retrocompact in Y), it is necessary and sufficient that X' be quasi-compact (resp. Noetherian, resp. retrocompact in X).*

Proof. Let us prove the first two assertions; by virtue of (10.2.4), one can restrict to the case where $Y' = Y$. To say that X is quasi-compact (resp. Noetherian) means that for every increasing filtered family (U_α) in $\mathfrak{O}(X)$ (resp. for every increasing filtered family (U_α) in $\mathfrak{O}(X)$) having X as greatest element, there exists γ such that $U_\alpha = U_\gamma$ for $\alpha \geq \gamma$. As $U \mapsto f^{-1}(U)$ is a bijection of $\mathfrak{O}(Y)$ onto $\mathfrak{O}(X)$, our assertion follows immediately from the preceding remark.

The quasi-compact open subsets of X are therefore the sets of the form $f^{-1}(U)$ where U is an open quasi-compact subset of Y , by virtue of (10.2.1), *a*) and of what precedes. For X' to be retrocompact in X , it is necessary and sufficient that for every quasi-compact open U in Y , $f^{-1}(U) \cap X' = f^{-1}(U \cap Y')$ be quasi-compact (III, 9.1.1); the first part of the proof shows that this amounts to saying that $U \cap Y'$ is quasi-compact for every open quasi-compact U , i.e. that Y' is retrocompact in Y . $\square$

Proposition (10.2.7). — *Let $f : X \rightarrow Y$ be a quasi-homeomorphism. Then the map $Z \mapsto f^{-1}(Z)$ of $\mathfrak{P}(Y)$ into $\mathfrak{P}(X)$ defines by restriction the bijections (cf. (10.1.1) for the notation):*

$$\begin{aligned} \mathfrak{O}(Y) &\xrightarrow{\sim} \mathfrak{O}(X) \\ \mathfrak{F}(Y) &\xrightarrow{\sim} \mathfrak{F}(X) \\ \mathfrak{L}\mathfrak{f}(Y) &\xrightarrow{\sim} \mathfrak{L}\mathfrak{f}(X) \\ \mathfrak{Q}\mathfrak{c}(Y) &\xrightarrow{\sim} \mathfrak{Q}\mathfrak{c}(X) \\ \mathfrak{L}\mathfrak{q}\mathfrak{c}(Y) &\xrightarrow{\sim} \mathfrak{L}\mathfrak{q}\mathfrak{c}(X) \\ \mathfrak{C}(Y) &\xrightarrow{\sim} \mathfrak{C}(X) \\ \mathfrak{L}\mathfrak{c}(Y) &\xrightarrow{\sim} \mathfrak{L}\mathfrak{c}(X). \end{aligned}$$

Proof. For the first two, this is none other than the definition (10.2.2); as the topology of X is the inverse image by f of that of $f(X)$, the five remaining maps, where one replaces Y by $f(X)$, are bijective. It therefore suffices (by (10.2.1), *b*)) to restrict to the case where X is a very dense subspace of Y , and that the maps $\mathfrak{L}\mathfrak{f}(Y) \rightarrow \mathfrak{L}\mathfrak{f}(X)$, $\mathfrak{Q}\mathfrak{c}(Y) \rightarrow \mathfrak{Q}\mathfrak{c}(X)$, $\mathfrak{L}\mathfrak{q}\mathfrak{c}(Y) \rightarrow \mathfrak{L}\mathfrak{q}\mathfrak{c}(X)$ are bijective has already been proved (10.1.2). Every locally constructible subset being locally quasi-constructible, the maps $\mathfrak{C}(Y) \rightarrow \mathfrak{C}(X)$ and $\mathfrak{L}\mathfrak{c}(Y) \rightarrow \mathfrak{L}\mathfrak{c}(X)$ are injective; furthermore, for every open $U \subset Y$, the restriction $f^{-1}(U) \rightarrow U$ of f is a quasi-homeomorphism (10.2.4), hence if we show that $\mathfrak{C}(Y) \rightarrow \mathfrak{C}(X)$ is surjective, the same will be true of $\mathfrak{L}\mathfrak{c}(Y) \rightarrow \mathfrak{L}\mathfrak{c}(X)$. But by virtue of (10.2.6), every retrocompact open subset Z in X is of the form $f^{-1}(T)$, where T is retrocompact open in Y ; this evidently proves the surjectivity of $\mathfrak{C}(Y) \rightarrow \mathfrak{C}(X)$. $\square$

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Remarks (10.2.8). — (i) We saw in the proof of (10.2.1) that if f is a quasi-homeomorphism, the canonical map

$$(10.2.8.1) \quad \Gamma(Y, \mathcal{F}) \longrightarrow \Gamma(X, f^*(\mathcal{F}))$$

is a functorial isomorphism in $\mathcal{F}$ in the category of sheaves of abelian groups on Y . As f^* is exact in this category (0, 3.7.2), this implies that the canonical homomorphisms (T, 3.2.2)

$$H^i(Y, \mathcal{F}) \xrightarrow{\sim} H^i(X, f^*(\mathcal{F}))$$

are bijective for every i .

(ii) If $f : X \rightarrow Y$ is a continuous map and $\mathcal{F}, \mathcal{G}$ two sheaves of abelian groups on Y , we have $f^*(\mathcal{F} \otimes_{\mathbf{Z}_Y} \mathcal{G}) = f^*(\mathcal{F}) \otimes_{\mathbf{Z}_X} f^*(\mathcal{G})$, an isomorphism canonical up to sign (0, 4.3.3). We conclude that if f is a quasi-homeomorphism, f^* is also an *equivalence* of the category of sheaves of rings on Y , and of the category of sheaves of rings on X ; the data of a structure of ringed space on Y is therefore equivalent to that of a structure of ringed space on X .

Given two ringed spaces $(X, \mathcal{O}_X)$, $(Y, \mathcal{O}_Y)$, we will say that a morphism of ringed spaces $f = (\psi, \theta) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ is a *quasi-isomorphism* if ψ is a quasi-homeomorphism of X into Y and $\theta^\sharp : \psi^*(\mathcal{O}_Y) \rightarrow \mathcal{O}_X$ an isomorphism of sheaves of rings; when this is so, the ringed space $(X, \mathcal{O}_X)$ is entirely determined, up to isomorphism, by $(Y, \mathcal{O}_Y)$, the space X , and the quasi-homeomorphism ψ (which can be chosen arbitrarily). When f is a quasi-isomorphism of ringed spaces, the functor

$$\mathcal{F} \longmapsto f^*(\mathcal{F})$$

is an *equivalence* of the category of $\mathcal{O}_Y$ -Modules and of that of $\mathcal{O}_X$ -Modules, since $f^*(\mathcal{F})$ is canonically identified here with $\psi^*(\mathcal{F})$. We conclude for example isomorphisms of bi- ∂ -functors

$$\mathrm{Ext}_{\mathcal{O}_Y}^i(\mathcal{F}, \mathcal{G}) \xrightarrow{\sim} \mathrm{Ext}_{\mathcal{O}_X}^i(f^*(\mathcal{F}), f^*(\mathcal{G})).$$

In general, one can say that the usual constructions of the theory of sheaves and of homological algebra, carried out on the ringed space Y or the ringed space X , are equivalent.

10.3. Jacobson spaces

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Definition (10.3.1). — We say that a topological space X is a *Jacobson space* if the set X_0 of closed points of X is very dense in X (in other words, if the canonical injection $X_0 \rightarrow X$ is a quasi-homeomorphism).

This means therefore (10.1.2) that every locally closed (or only locally quasi-constructible) subset $Z \neq \emptyset$ of X contains a closed point of X , or equivalently that *every closed subset of X is the closure of the set of its closed points*.

Proposition (10.3.2). — *Let X be a Jacobson space, Z a quasi-constructible subset of X ; then the subspace Z is a Jacobson space, and for a point of Z to be closed in Z , it is necessary and sufficient that it be closed in X .*

Proof. If X_0 is the set of closed points of X , $Z \cap X_0$ is very dense in Z by virtue of (10.2.4) applied to the injection $i : X_0 \rightarrow X$; as the set Z_0 of closed points of Z evidently contains $Z \cap X_0$, it is *a fortiori* very dense in Z , hence Z is a Jacobson space. Let us now show that we have $Z_0 = Z \cap X_0$; let $x \in Z$ be a closed point in Z ; let $\overline{\{x\}}$ be its closure in X and $\overline{\{x\}} \cap Z = \{x\}$, which is therefore a locally quasi-constructible subset of X , and as its intersection with X_0 is not empty (10.1.2), we have $x \in X_0$. $\square$

Proposition (10.3.3). — *Let X be a topological space, (U_α) an open covering of X . For X to be a Jacobson space, it is necessary and sufficient that each of the subspaces U_α be a Jacobson space.*

Proof. The condition is necessary by virtue of (10.3.2). Conversely, let us show first that the hypothesis that the U_α are Jacobson spaces implies that for a point $x \in U_\alpha$ to be closed in X , it suffices that it be closed in U_α . It suffices indeed to see that this condition implies that x is also closed in each of the U_β that contain it; but $U_\alpha \cap U_\beta$ is open in U_α , hence x is closed in $U_\alpha \cap U_\beta$, and by virtue of (10.3.2), x is also closed in U_β , which completes the proof. $\square$

10.4. Jacobson preschemes and Jacobson rings

Definition (10.4.1). — We say that a prescheme X is a *Jacobson prescheme* if the underlying topological space of X is a Jacobson space. We say that a ring A is a *Jacobson ring* if $\mathrm{Spec}(A)$ is a Jacobson prescheme.

Every closed subset of $X = \mathrm{Spec}(A)$ is of the form $Z = V(\mathfrak{a})$, where $\mathfrak{a}$ is an ideal equal to its radical, and the set X_0 of closed points of X is the set of maximal ideals of A ; to say that $Z \cap X_0$ is dense in Z means therefore that $\mathfrak{a}$ is an *intersection of maximal ideals* (I, 1.1.4); as $\mathfrak{a}$ is an intersection of prime ideals, this amounts to saying that every prime ideal of A is an intersection of maximal ideals; by virtue of (10.3.1) and (10.1.2), the usual definition of Jacobson rings (Bourbaki, *Alg. comm.*, chap. V, §3, n° 4, def. 1) thus coincides with the definition (10.4.1).

Proposition (10.4.2). — *Let X be a prescheme, (U_α) a covering of X formed by affine open subsets. For X to be a Jacobson prescheme, it is necessary and sufficient that the rings $\Gamma(U_\alpha, \mathcal{O}_X)$ of the U_α be Jacobson rings.*

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This follows from (10.3.3) and the definition (10.4.1).

(10.4.3). A discrete prescheme is a Jacobson prescheme; an Artinian ring is therefore a Jacobson ring. Every principal ideal ring having an infinity of maximal ideals (for example $\mathbf{Z}$) is a Jacobson ring; a Noetherian local ring A is a Jacobson ring if and only if its maximal ideal is its only prime ideal, i.e. if A is Artinian. Every sub-prescheme of a Jacobson prescheme is a Jacobson prescheme by virtue of (10.3.2).

Proposition (10.4.4). — *Let B be an integral ring. The following conditions are equivalent:*

- a) *There exists $f \neq 0$ in B such that B_f is a field.*

- b) The field of fractions of B is a B -algebra of finite type.
- c) There exists a field K containing B and which is a B -algebra of finite type.
- d) The generic point of $\text{Spec}(B)$ is isolated.

Proof. It is clear that d) is equivalent to a), since d) means that there exists $f \neq 0$ in B such that $D(f)$ is reduced to the generic point of $\text{Spec}(B)$. It is trivial that a) implies b) and that b) implies c). Finally, c) implies a), by virtue of Bourbaki, *Alg. comm.*, chap. V, § 3, n° 1, cor. 2 of th. 1. $\square$

Proposition (10.4.5). — *Let A be a ring. The following conditions are equivalent:*

- a) A is a Jacobson ring.
- b) For every non-maximal prime ideal $\mathfrak{p}$ of A and every $f \neq 0$ in $B = A/\mathfrak{p}$, B_f is not a field.
- b') Every A -algebra of finite type K which is a field is a finite A -algebra.

Proof. We know that a) implies b') (Bourbaki, *Alg. comm.*, chap. V, § 3, n° 4, cor. 3 of th. 3). Furthermore, the kernel of the homomorphism $A \rightarrow K$ is then a maximal ideal $\mathfrak{m}$ of A , and K is a finite extension of $A/\mathfrak{m}$ (*loc. cit.*). It is trivial that b') implies b), since $A/\mathfrak{p} = B$ is not a field and B_f is a B -algebra of finite type. It remains to show that b) implies a). We will use the following lemma: $\square$

Lemma (10.4.5.1). — *Let X be a topological space having the following property: for every locally closed subset $Z \neq \emptyset$ of X , there exists a subset Z' of Z , locally closed in X (or in Z , which amounts to the same), and a point $x \in Z'$, closed in Z' . Then, for X to be a Jacobson space, it is necessary and sufficient that no non-closed point x of X be isolated in $\overline{\{x\}}$.*

Proof. If X is a Jacobson space, a non-closed point x of X cannot be isolated in $\overline{\{x\}}$, for this would mean that there exists an open U of X such that $U \cap \overline{\{x\}} = \{x\}$; but $U \cap \overline{\{x\}}$ is locally closed in X and would contain no closed point in X , which contradicts the hypothesis that X is a Jacobson space. Conversely, suppose the conditions of the statement are satisfied; then the set X_0 of closed points of X is identical to the set of $x \in X$ that are isolated in $\overline{\{x\}}$; but it follows from (5.1.10.1) that this set is very dense in X , hence X is a Jacobson space by definition. $\square$

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Let us recall (5.1.10) that the hypothesis made on X in (10.4.5.1) is always satisfied when the underlying space of X is a prescheme.

Returning to the proof of (10.4.5), condition b) means that for every non-closed point x of $\text{Spec}(A)$, the generic point x of $\text{Spec}(A/\mathfrak{j}_x) = \overline{\{x\}}$ is not isolated in $\overline{\{x\}}$, hence b) implies a) by the lemma (10.4.5.1) and (10.4.4).

Corollary (10.4.6). — *Every algebra of finite type B over a Jacobson ring A is a Jacobson ring, and the inverse image in A of every maximal ideal of B is a maximal ideal of A . In particular, every algebra of finite type over a field or over $\mathbf{Z}$ is a Jacobson ring.*

Proof. A B -algebra of finite type K which is a field is also an A -algebra of finite type, hence an A -module of finite type and *a fortiori* a B -module of finite type, whence the first assertion; the second was proved in the course of the proof of (10.4.5), applied to $K = B/\mathfrak{m}$. $\square$

Corollary (10.4.7). — *If X is a Jacobson prescheme, $f : Y \rightarrow X$ a morphism locally of finite type, then Y is a Jacobson prescheme and the image by f of every closed point in Y is a closed point in X .*

Proof. The question being local on X and on Y , one reduces to the case where X and Y are affine, and the corollary then follows from (10.4.6). $\square$

Corollary (10.4.8). — *If k is an algebraically closed field, X a k -prescheme locally of finite type, the set of rational points of X over k is very dense in X .*

Proof. Indeed, X is a Jacobson prescheme (10.4.7) and the closed points are exactly the rational points over k (I, 6.4.2). $\square$

(10.4.9). The fact that preschemes locally of finite type over a field or over $\mathbf{Z}$ are Jacobson preschemes is particularly important, because of the possibility of reducing to this case in many questions of algebraic geometry (8.1.2), c). We will give two examples:

(10.4.10). Applications (10.4.10) : I : Proof of (6.15.9). Let k be a separably closed field, X a k -prescheme locally of finite type over k and *unibranch*. We know that the integral closure of a k -algebra of finite type A in an extension of finite type of its field of fractions is a finite A -algebra (Bourbaki, *Alg. comm.*, chap. V, §3, n° 2, th. 2), hence every k -algebra of finite type is a universally Japanese ring; we conclude that the set of points $x \in X$ where X is geometrically unibranch is locally constructible (9.7.10). But the hypothesis and the lemma (6.15.8) imply that this set contains all the *closed* points of X . The conclusion therefore follows from (10.4.6), (10.3.1) and the bijectivity of the canonical map $\mathcal{L}c(X) \rightarrow \mathcal{L}c(X_0)$ (where X_0 is the set of closed points of X) (10.2.7).

(10.4.11). Applications : II : Proposition (10.4.11). Let S be a prescheme, X an S -prescheme of finite type. Every S -endomorphism of X which is radicial is surjective (hence bijective). IV-3 | 104

Proof. Let $f : X \rightarrow S$ be the structural morphism, $g : X \rightarrow X$ the S -endomorphism considered, and, for every $s \in S$, let g_s be the morphism deduced from g by the base change $\text{Spec}(k(s)) \rightarrow S$, which is a $k(s)$ -endomorphism of the fibre $f^{-1}(s) = X \times_S \text{Spec}(k(s))$.

To prove that g is surjective, it suffices to prove that g_s is surjective for every $s \in S$, hence one can (by virtue of (I, 3.5.7)) reduce to the case where $S = \text{Spec}(k)$ is the spectrum of a field k , in which case f is a morphism of finite presentation, since S is Noetherian. Applying (8.9.1) and (8.10.5), (vii), one reduces to the case where $S = \text{Spec}(A)$, where A is a sub- $\mathbf{Z}$ -algebra of finite type of k . But then X is a Jacobson prescheme (10.4.7), and $g(X)$ is constructible in X (I, 1.8.5), hence, to prove that $g(X) = X$, it suffices to show that $g(X)$ contains all the *closed* points of X (10.3.1).

Lemma (10.4.11.1). — *Let Y be a $\mathbf{Z}$ -prescheme of finite type.*

- (i) *For a closed point $y \in Y$, it is necessary and sufficient that $k(y)$ be a finite field.*
- (ii) *For every prime number p and every integer $d \geq 1$, the set of points $y \in Y$ such that $k(y)$ is an extension of $\mathbf{F}_p$ of degree dividing d is finite.*

The assertion (i) follows from the fact that the image of a closed point $y \in Y$ in $\text{Spec}(\mathbf{Z})$ is a closed point (10.4.7), in other words a prime number, and from (I, 6.4.2). On the other hand, as Y is a finite union of affine open subsets of finite type over $\mathbf{Z}$, one can restrict to proving (ii) in the case where $Y = \text{Spec}(C)$, where C is a $\mathbf{Z}$ -algebra of finite type. Now, the points $y \in Y$ such that the degree of $k(y)$ over $\mathbf{F}_p$ divides d correspond bijectively to the homomorphisms $C \rightarrow \mathbf{F}_{p^d}$; but if (t_i) is a system of generators of the $\mathbf{Z}$ -algebra C , every homomorphism of C is determined by its values on the elements t_i , and there are therefore only a finite number of homomorphisms of C into a finite field.

This being so, since X is a $\mathbf{Z}$ -prescheme of finite type, let $T_{p,d}$ be the set of closed points $z \in X$ such that $k(z)$ is an extension of $\mathbf{F}_p$ of degree dividing d ; it follows from (10.4.11.1) that the set $T_{p,d}$ is finite and that the set of closed points of X is the union of the $T_{p,d}$. Furthermore, if $z \in T_{p,d}$ and if h is any endomorphism of X , $k(h(z))$ is isomorphic to a subfield of $k(z)$, hence $h(z) \in T_{p,d}$; in other words $T_{p,d}$ is *stable* under every endomorphism of X . As g is by hypothesis injective, its restriction to $T_{p,d}$ is *bijective* since $T_{p,d}$ is finite, which completes proving the proposition.

We will see further on (17.9.7) that when one further supposes that X is an S -prescheme of finite presentation, and on the other hand that g is a monomorphism, then one can assert that g is an *automorphism* of X . □

10.5. Noetherian Jacobson preschemes

Proposition (10.5.1). — *Let B be a Noetherian integral ring. The conditions equivalent to a) through d) of (10.4.4) are then also equivalent to the following:*

- e) $\text{Spec}(B)$ is finite.
- f) B is a semi-local ring of dimension ≤ 1 .

Proof. It follows indeed from the Artin–Tate theorem (0, 16.3.3) that conditions e) and f) are equivalent. Condition f) implies that $\text{Spec}(B)$ is the union of the set of its closed points and of its generic point, hence f) implies e) without assuming A to be Noetherian. Finally, e) implies d) without assuming A to be Noetherian, for the generic point x of X is the complement of the union of the closures $\overline{\{y\}}$, where y ranges over the set of points $y \neq x$, and as these points are finite in number, $\{x\}$ is the complement of a closed set in X . □

Corollary (10.5.2). — *For a Noetherian ring A to be a Jacobson ring, it is necessary and sufficient that there exist no prime ideal $\mathfrak{p}$ of A such that $A/\mathfrak{p}$ is a semi-local ring of dimension 1.*

Proof. This follows immediately from (10.5.1) and from condition b) of (10.4.5), the prime ideals $\mathfrak{p}$ of A such that $A/\mathfrak{p}$ is semi-local of dimension 0 being the maximal ideals of A . $\square$

Corollary (10.5.3). — *Let X be an irreducible Noetherian prescheme. The following conditions are equivalent:*

- a) *The generic point of X is isolated.*
- b) *X is finite.*
- c) *X is of dimension ≤ 1 and the set of its closed points is finite.*

Proof. There exists by hypothesis a finite number of irreducible affine open subsets U_i ($1 \leq i \leq n$) covering X , and each of which contains the generic point of X ; it suffices to prove the equivalence of a), b), and c) for each of the $(U_i)_{\text{red}}$ (taking into account (0, 14.1.7)). But this equivalence follows from (10.5.1). $\square$

Remark (10.5.4). — A Noetherian prescheme X satisfying the equivalent conditions of (10.5.3) is not necessarily an affine scheme; in fact it can even be *non-separated*, as one sees by replacing, in the example (I, 5.5.11) of the “affine line with doubled point”, X_1 and X_2 by the spectrum of the ring of the discrete valuation $(K[s])_{(s)}$, U_{12} and U_{21} by the open subset reduced to the generic point in X_1 and X_2 respectively; the non-separated prescheme X that one obtains has exactly 3 points.

Proposition (10.5.5). — *Let X be a Noetherian prescheme.*

- (i) *The set X_0 of points $x \in X$ such that $\overline{\{x\}}$ is finite is very dense in X .*
- (ii) *For X to be a Jacobson prescheme, it is necessary and sufficient that there exist no sub-prescheme of X isomorphic to the spectrum of a semi-local integral ring of dimension 1.*

Proof. The condition that $\overline{\{x\}}$ is finite is equivalent, here (10.5.3), to the fact that x is isolated in $\overline{\{x\}}$, it being the underlying space of a sub-prescheme (Noetherian) of X ; the assertion (i) therefore follows from (10.4.5.1). Similarly, taking into account (10.4.5.1), to show the assertion (ii), let us note that for a non-closed point x of X to belong to X_0 , it is necessary and sufficient that the closed sub-prescheme Y of X having $\overline{\{x\}}$ as underlying space be of dimension 1 and finite, hence a finite union of sub-preschemes which are open (in Y) and affine U_i which are spectra of semi-local integral rings of dimension 1. Conversely, if there is a sub-prescheme Z of X which is the spectrum of a semi-local integral ring of dimension 1, Z is not a Jacobson prescheme (10.5.2), hence neither is X (10.3.2). $\square$

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Remark (10.5.6). — The assertion (ii) of (10.5.5) is still valid when X is *locally Noetherian*: indeed, if (V_α) is a covering of X by affine open subsets (Noetherian), every sub-prescheme of one of the V_α is a sub-prescheme of X ; conversely, if a sub-prescheme Z of X is isomorphic to the spectrum of a semi-local integral ring of dimension 1, there is an α such that $V_\alpha \cap Z$ contains an affine open U of Z which is non-reduced to the generic point of Z , and which is therefore also the spectrum of a semi-local integral ring of dimension 1. We conclude by (10.3.3).

Proposition (10.5.7). — *Let X be a locally Noetherian prescheme, Y a closed subset of X such that every non-empty closed subset of X meets Y . Then the subprescheme induced on the open subset $X - Y$ is a Jacobson prescheme.*

Proof. Applying the criterion (10.5.5), (ii), and supposing that there were in $X - Y$ a sub-prescheme Z which is the spectrum of a semi-local integral ring of dimension 1, the generic point z of Z being isolated in Z (or in the closure $\overline{Z}$ of Z in X), and Z being distinct from $\{z\}$. Let $y \neq z$ be a point of Z ; as it does not belong to Y , it is not a closed point in X , and its closure $\overline{\{y\}}$ in X meets Y at a point $x \neq y$, which is therefore a specialisation of y . The existence of the chain $\{x\} \subset \overline{\{y\}} \subset \overline{\{z\}}$ then shows that the dimension of $\overline{\{z\}}$ would be ≥ 2 , and the same would be true of the dimension of $U \cap \overline{\{z\}}$, where U is an affine (hence Noetherian) open neighbourhood of x in X . But this contradicts the fact that the generic point z of $U \cap \overline{\{z\}}$ is isolated in $U \cap \overline{\{z\}}$ (10.5.3). $\square$

Corollary (10.5.8). — *Let A be a Noetherian ring; for every element f of the radical $\mathfrak{R}$ of A , the ring A_f is a Jacobson ring, and the open subset $\text{Spec}(A) - V(\mathfrak{R})$ is a Jacobson scheme.*

Proof. If $X = \text{Spec}(A)$, $Y = V(\mathfrak{J})$, where $\mathfrak{J}$ is an ideal of A , to say that every non-empty closed subset of X meets Y is equivalent (0, 2.1.3) to saying that Y contains all the closed points of X , or equivalently that $\mathfrak{J}$ is contained in the radical $\mathfrak{R}$ of A . If $f \in \mathfrak{R}$, the open $D(f)$ does not meet $V(\mathfrak{R})$, and is a Jacobson space by virtue of (10.5.7). $\square$

Corollary (10.5.9). — *Let A be a Noetherian local ring, $\mathfrak{m}$ its maximal ideal, $X = \text{Spec}(A)$, $Y = X - \{\mathfrak{m}\}$; then Y is a Jacobson scheme, whose closed points are the prime ideals $\mathfrak{p} \in \text{Spec}(A)$ such that $\dim(A/\mathfrak{p}) = 1$.*

Proof. The first assertion is a particular case of (10.5.8); on the other hand the closed points of Y are the prime ideals $\mathfrak{p}$ of A which are maximal elements in the set of prime ideals $\neq \mathfrak{m}$, which, by definition of the dimension, means that $\dim(A/\mathfrak{p}) = 1$. $\square$

Proposition (10.5.10). — *Let A be a Noetherian local ring, reduced, complete, and not a field. Then the finite intersections of the kernels of the local homomorphisms of A into discrete valuation rings V , which are A -algebras of finite type, form a basis of filter tending to 0 for the adic topology of A .*

Proof. It suffices (Bourbaki, *Alg. comm.*, chap. III, §2, n° 7, prop. 8) to prove that the intersection of the kernels considered in the statement is reduced to 0. Suppose first that A is integral and of dimension 1; by virtue of the Nagata theorem (0, 23.1.5) and (23.1.6), the integral closure A' of A is a complete local integral ring and integrally closed and of dimension 1; it is therefore a discrete valuation ring, and the proposition follows immediately in this case.

Let us pass to the general case; let x be the closed point of $X = \text{Spec}(A)$, and set $Y = X - \{x\}$; we know (10.5.9) that Y is a Jacobson prescheme. Let us show that the intersection of the prime ideals $\mathfrak{p}$ of A such that $\dim(A/\mathfrak{p}) = 1$ is reduced to 0. But to say that the intersection of these prime ideals is reduced to 0 means that the set of these ideals is dense in X , or equivalently in Y (since Y is dense in X), and this follows immediately from (10.5.9). $\square$

10.6. Dimension in Jacobson preschemes The results of this section give precision in certain cases and generalise results from §5.

Proposition (10.6.1). — *Let S be a locally Noetherian prescheme, satisfying in addition the conditions: 1° S is a Jacobson prescheme; 2° for every $s \in S$, $\mathcal{O}_{S,s}$ is universally catenary (5.6.2); 3° every irreducible component S' of S is equidimensional (in other words, for every closed point s of S' and every sub-prescheme S' having S' as underlying space, we have $\dim(S') = \dim(\mathcal{O}_{S',s})$). We then have the following properties:*

- (i) *For every morphism locally of finite type $g : X \rightarrow S$, X satisfies conditions 1°, 2°, and 3° above. In particular, if X is equidimensional (for example if X is irreducible), X is biequidimensional (in other words X is catenary and for every closed point x of X , we have $\dim(\mathcal{O}_{X,x}) = \dim(X)$ (0, 14.3.3)).*
- (ii) *Let X, Y be two S -preschemes locally of finite type over S , $f : X \rightarrow Y$ an S -morphism; suppose X irreducible and f dominant. If ξ (resp. η) is the generic point of X (resp. Y) and $e = \dim(f^{-1}(\eta)) = \deg. \text{tr}_{k(\eta)} k(\xi)$, we have*

$$(10.6.1.1) \quad \dim(X) = \dim(Y) + e.$$

- (iii) *Let X, Y be two S -preschemes locally of finite type over S , $f : X \rightarrow Y$ an S -morphism, n a non-negative integer such that $\dim(f^{-1}(y)) \geq n$ (resp. $\dim(f^{-1}(y)) \leq n$) for every $y \in Y$. Then we have*

$$(10.6.1.2) \quad \dim(X) \geq \dim(Y) + n$$

(resp.

$$(10.6.1.3) \quad \dim(X) \leq \dim(Y) + n).$$

Proof. (i) The property 1° for X follows from (10.4.7). For every $x \in X$, $\mathcal{O}_{X,x}$ is the local ring of an $\mathcal{O}_{S,g(x)}$ -algebra of finite type at a prime ideal, and the homomorphism $\mathcal{O}_{S,g(x)} \rightarrow \mathcal{O}_{X,x}$ is local; hence (5.6.3), (iv) $\mathcal{O}_{X,x}$ is universally catenary. To show that X satisfies condition 3°, consider several cases:

a) X is a closed irreducible sub-prescheme of S ; let S' be an irreducible component of S containing X , ξ the generic point of X , x a closed point of X ; for every $s \in S'$, $\mathcal{O}_{S',s}$, a quotient of $\mathcal{O}_{S,s}$, is catenary

(5.6.1), hence conditions 2° and 3° imply that S' is biequidimensional (0, 14.3.3). By virtue of (5.1.2) and of (0, 14.3.3.2), we therefore have

$$\dim(\mathcal{O}_{X,x}) = \dim(\mathcal{O}_{S',x}) - \dim(\mathcal{O}_{S',\xi}) = \dim(S') - \dim(\mathcal{O}_{S',\xi})$$

which proves that $\dim(\mathcal{O}_{X,x})$ does not depend on the closed point x considered, whence the assertion in this case (5.1.4).

b) X is irreducible and g dominant. Then, for every closed point x of X , $g(x)$ is closed in S by (10.4.7); as $\mathcal{O}_{S,g(x)}$ is universally catenary, it follows from (5.6.5.3) that we have

$$\dim(\mathcal{O}_{X,x}) = \dim(\mathcal{O}_{S,g(x)}) + e$$

where $e = \dim(g^{-1}(\zeta))$, ζ being the generic point of S . As $\dim(S) = \dim(\mathcal{O}_{S,g(x)})$ by virtue of condition 3° for S , we have $\dim(\mathcal{O}_{X,x}) = \dim(S) + e$ for every closed point $x \in X$; this proves condition 3° for X (5.1.4), and at the same time the formula

$$(10.6.1.4) \quad \dim(X) = \dim(S) + e.$$

c) *General case.* Considering a reduced sub-prescheme X' of X having for underlying space an irreducible component of X , one reduces to the case where X is integral; using (I, 5.2.2) and the case a) proved above, one can then replace S by the reduced sub-prescheme having $g(X)$ as underlying space; one is then brought to case b), and this completes proving (i).

(ii) The morphism f being locally of finite type (I, 1.3.4), one can apply the results of (i) replacing S by Y ; furthermore, as X is irreducible and f dominant, one can also replace S by Y in (10.6.1.4), which gives (10.6.1.1).

(iii) The relative assertion where $\dim(f^{-1}(y)) \leq n$ for every $y \in Y$ has already been proved under more general hypotheses in (5.6.7). Suppose that $\dim(f^{-1}(y)) \geq n$ for every $y \in Y$, and consider a generic point η of an irreducible component Y' of Y ; there exists at least one irreducible component Z of $f^{-1}(\eta)$ of dimension $\geq n$; if ξ is the generic point of Z , ξ is also a generic point of an irreducible component X' of X such that $f(X')$ is dense in Y' (0, 2.1.8). Consider the reduced sub-preschemes of X, Y having respectively for underlying spaces X', Y' , and the restriction $X' \rightarrow Y'$ of f (I, 5.2.2); it then follows from (ii) that $\dim(X') \geq \dim(Y') + n$, and *a fortiori* $\dim(X) \geq \dim(Y) + n$; this being true for every irreducible component Y' of Y , we conclude the inequality (10.6.1.2). $\square$

Corollary (10.6.2). — Suppose that X satisfies conditions 1°, 2°, and 3° of (10.6.1). Then, for every dense open U in X , we have $\dim(U) = \dim(X)$.

Proof. We know indeed that $\dim(U) \leq \dim(X)$ (0, 14.1.4); furthermore, as X is a Jacobson prescheme, U contains a closed point of every irreducible component of X , hence $\dim(U) = \dim(X)$ by virtue of (10.6.1) and (0, 14.1.2.1). $\square$

Proposition (10.6.3). — Suppose that X satisfies conditions 1°, 2°, and 3° of (10.6.1), and let Y be a closed subset of X . For every $x \in Y$, and every open neighbourhood U of x in X not meeting the irreducible components of Y that do not contain x , we have

$$(10.6.3.1) \quad \dim_x(Y) = \dim(U \cap Y) = \dim \overline{\{x\}} + \operatorname{codim}(\overline{\{x\}}, Y) = \sup_i \dim(Y_i)$$

where Y_i ($1 \leq i \leq m$) are the irreducible components of Y containing x .

Proof. Considering a closed sub-prescheme of X having Y as underlying space, one can restrict, by virtue of (10.6.1), (i), to the case where $Y = X$. By the choice of U , U is the union of the $U \cap X_i$, hence $\dim(U) = \sup_i \dim(U \cap X_i)$; but by (10.6.2) applied to a closed sub-prescheme of X having X_i as underlying space, we have (taking into account (10.6.1), (i)), $\dim(U \cap X_i) = \dim(X_i)$; this proves that the second and the fourth terms of (10.6.3.1) are equal. As $U \cap X_i$ is biequidimensional (10.6.1), (i) (since the immersion $U \cap X_i \rightarrow X_i$ is of finite type (I, 6.3.5)), we have

$$(10.6.3.2) \quad \dim(X_i) = \dim(U \cap X_i) = \dim(U \cap \overline{\{x\}}) + \operatorname{codim}(U \cap \overline{\{x\}}, U \cap X_i)$$

(0, 14.3.5). By (10.6.2) applied to a closed sub-prescheme of X having $\overline{\{x\}}$ as underlying space, $\dim(U \cap \overline{\{x\}}) = \dim(\overline{\{x\}})$; as we also have, for the same reasons,

$$(10.6.3.3) \quad \dim(X_i) = \dim(\overline{\{x\}}) + \operatorname{codim}(\overline{\{x\}}, X_i)$$

we obtain

$$\dim(U) = \dim(\overline{\{x\}}) + \sup_i \operatorname{codim}(\overline{\{x\}}, X_i) = \dim(\overline{\{x\}}) + \operatorname{codim}(\overline{\{x\}}, X)$$

by definition of the codimension (0, 14.2.1). This shows that $\dim(U)$ is *independent* of the open neighbourhood U of x satisfying the conditions of the statement, hence is equal to $\dim_x(X)$, by (0, 14.1.4.1). $\square$

Corollary (10.6.4). — *Under the hypotheses of (10.6.3), let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -Module, Y its support. For every $x \in Y$, we have*

$$(10.6.4.1) \quad \dim(\overline{\{x\}}) + \dim(\mathcal{F}_x) = \dim_x Y.$$

Proof. Indeed, this follows from (10.6.3.1) and from the formula $\dim(\mathcal{F}_x) = \operatorname{codim}(\overline{\{x\}}, Y)$ (5.1.12.2). $\square$

10.7. Examples and counter-examples

(10.7.1). Let S be a locally Noetherian prescheme of dimension ≤ 1 and suppose that S is a *Jacobson prescheme*; when S is Noetherian, this amounts to saying that the irreducible components of S of dimension 1 are infinite, for every $x \in S$ which is not a closed point is the generic point of such a component (10.4.5) and (10.5.4). Then S also satisfies conditions 2° and 3° of (10.6.1): indeed, every local ring $\mathcal{O}_{S,s}$ is of dimension 0 or 1, and hence universally catenary (5.6.3), (7.2.9); on the other hand, an irreducible component S' of S is, either reduced to a closed point s of S' , or biequidimensional.

From these remarks and from (10.6.1) it follows that every prescheme locally of finite type over S also satisfies the properties 1° , 2° , and 3° ; it is in particular the case of preschemes *locally of finite type over a field or over $\mathbf{Z}$* . IV-3 | 110

(10.7.2). Let A be a Noetherian local ring *universally catenary* and let S be the complementary of the closed point a in $X = \operatorname{Spec}(A)$. Then S satisfies conditions 1° , 2° , and 3° of (10.6.1): indeed, one has already seen that S is a Jacobson prescheme (10.5.9); as A is universally catenary, so are all the local rings $A_{\mathfrak{p}}$ at the prime ideals of A (5.6.3). On the other hand, an irreducible component S' of S is the complement of a in an irreducible component X' of X . For every closed point x of S , the closure of x in X is $\{x, a\}$, so $\dim(\overline{\{x\}}) = 1$. Since X' is biequidimensional (0, 14.3.3), it follows that $\dim(\mathcal{O}_{X',x}) = \dim(X') - 1$ (0, 14.3.2), proving that S' is equicodimensional.

(10.7.3). Let A be a discrete valuation ring; let us show that in $B = A[T_1, \dots, T_n]$, with n indeterminates, there exist two maximal ideals $\mathfrak{m}$, $\mathfrak{n}$ of *respective heights* n and $n + 1$. We have seen (5.2.5), (i) for $n = 1$; let us prove it by induction on n . As $A[T_1, \dots, T_{n+1}]$ is a free, hence faithfully flat, $A[T_1, \dots, T_n]$ -module, there are maximal ideals $\mathfrak{m}'$ and $\mathfrak{n}'$ of $A[T_1, \dots, T_{n+1}]$ lying respectively over $\mathfrak{m}$ and $\mathfrak{n}$ (0, 6.5.1); by (5.5.3), these ideals have respective heights $n + 1$ and $n + 2$, proving the assertion. Suppose $n \geq 2$ in what follows. Let $\mathfrak{J}$ be the ideal $\mathfrak{m} \cap \mathfrak{n} = \mathfrak{m}\mathfrak{n}$, which is contained in the radical of B ; if one sets $B' = R^{-1}B$, the ideal $\mathfrak{J}' = R^{-1}\mathfrak{J}$ is contained in the radical of B' . As a topological space, $X' = \operatorname{Spec}(B')$ is identified with a subspace of $X = \operatorname{Spec}(B)$, and, for the points x of X' , the local rings $\mathcal{O}_{X',x}$ and $\mathcal{O}_{X,x}$ are the same (10.5.7); one knows (10.5.7) that S is a Jacobson prescheme, evidently irreducible and Noetherian; furthermore all the local rings $\mathcal{O}_{S,s}$ are universally catenary for every $s \in S$ by virtue of (5.6.3), since A is universally catenary. However, there exist two closed points a, b of S such that $\mathcal{O}_{S,a}$ and $\mathcal{O}_{S,b}$ do not have the same dimension, otherwise S would not be biequidimensional (5.6.4). To see this, consider the two maximal ideals $\mathfrak{m}' = R^{-1}\mathfrak{m}$, $\mathfrak{n}' = R^{-1}\mathfrak{n}$ of B' , which are of heights n and $n + 1$ respectively; we have $\mathfrak{J}' = \mathfrak{m}' \cap \mathfrak{n}'$, and $\mathfrak{J}'$ is contained in no prime ideal of B' distinct from $\mathfrak{m}'$ and $\mathfrak{n}'$, which are therefore the only maximal ideals of B' . Let a', b' be the two closed points of X' corresponding to $\mathfrak{m}'$ and $\mathfrak{n}'$; the closures of a' in X' are $\{a, a'\}$; as $B'_{\mathfrak{m}'}$ is biequidimensional, $\mathfrak{p}'$ is then an ideal of B' of height n such that its closure in X' is $\{a, a'\}$; one constructs in the same way an ideal $\mathfrak{q}'$ of B' of height n such that the closure of b in X' is $\{b, b'\}$. This being so, a and b are closed points in S , hence the question is answered.

10.8. Rectified depth

Definition (10.8.1). — Let X be a locally Noetherian prescheme, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module. For every $x \in X$, we call *rectified depth* of $\mathcal{F}$ at the point x and denote $\text{prof}_x^*(\mathcal{F})$ the number (integer ≥ 0 or $+\infty$) equal to

$$(10.8.1.1) \quad \text{prof}_x^*(\mathcal{F}) = \text{prof}(\mathcal{F}_x) + \dim(\overline{\{x\}})$$

where $\overline{\{x\}}$ is the closure of the point x in X . For every closed subset Z of X , we call *rectified depth of $\mathcal{F}$ along Z* and denote $\text{prof}_Z^*(\mathcal{F})$ the number

$$(10.8.1.2) \quad \text{prof}_Z^*(\mathcal{F}) = \inf_{x \in Z} \text{prof}_x^*(\mathcal{F}).$$

In other terms, for every integer n , the relation $\text{prof}_Z^*(\mathcal{F}) \geq n$ is equivalent to $\text{prof}_x^*(\mathcal{F}) \geq n$ for every $x \in Z$. If $Z = X$, one writes $\text{prof}^*(\mathcal{F})$ instead of $\text{prof}_X^*(\mathcal{F})$.

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Remarks (10.8.2). — (i) At every closed point $x \in X$, the rectified depth equals the depth.

(ii) Let Y be a closed sub-prescheme of X , $j : Y \rightarrow X$ the canonical injection. We know (5.7.3), (vi) that $\text{prof}(\mathcal{G}_x) = \text{prof}(j_*(\mathcal{G})_x)$ for every $x \in Y$; we therefore deduce that $\text{prof}_x^*(\mathcal{G}) = \text{prof}_x^*(j_*(\mathcal{G}))$ for every $x \in Y$.

(iii) The notion of rectified depth is of interest only when it is of a *local* character, i.e. when it does not change when one replaces X by a neighbourhood open U of x . This obviously requires that x not be isolated in $\overline{\{x\}}$ when x is not a closed point of X , i.e. most often, that X be a *Jacobson prescheme* (10.4.5.1); furthermore, it will also be necessary to know that $\dim(U) = \dim(X)$ for every dense open U in X , and one must therefore also suppose that X satisfies conditions 2° and 3° of (10.6.1).

Lemma (10.8.3). — Let X be a regular and biequidimensional prescheme, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module. Then, for every $x \in X$, we have

$$(10.8.3.1) \quad \text{prof}_x^*(\mathcal{F}) = \dim(X) - \dim \cdot \text{proj}(\mathcal{F}_x).$$

Proof. Indeed, as X is biequidimensional, we have (0, 14.3.5.1)

$$\dim(\overline{\{x\}}) = \dim(X) - \text{codim}(\overline{\{x\}}, X) = \dim(X) - \dim(\mathcal{O}_{X,x})$$

by virtue of (5.1.2). On the other hand, as X is regular, we have by (0, 17.3.4)

$$\text{prof}(\mathcal{F}_x) = \dim(\mathcal{O}_{X,x}) - \dim \cdot \text{proj}(\mathcal{F}_x)$$

whence the lemma. $\square$

Corollary (10.8.4). — Under the hypotheses of (10.8.3), the function $x \mapsto \text{prof}_x^*(\mathcal{F})$ is lower semi-continuous.

Proof. This follows from (10.8.3.1), since $x \mapsto \dim \cdot \text{proj}(\mathcal{F}_x)$ is upper semi-continuous (6.11.1). $\square$

Proposition (10.8.5). — Let S be a prescheme locally Noetherian, X a prescheme locally of finite type over S , $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module. Suppose that S satisfies the conditions: 1° S is a Jacobson prescheme; 2° S is regular; 3° the irreducible components of S are equicodimensional. Then the function $x \mapsto \text{prof}_x^*(\mathcal{F})$ is semi-continuous inferiorly; in other terms, for every integer n , the set U_n of $x \in X$ such that $\text{prof}_x^*(\mathcal{F}) \geq n$ is open.

Proof. As the local rings $\mathcal{O}_{S,s}$ of S are regular, they are universally catenary (5.6.4); in other words, S satisfies conditions 1°, 2°, and 3° of (10.6.1), hence the same is true of X (10.6.1), (i). The notion of rectified depth being then of local character (10.8.2), (iii), one can restrict to the case where $S = \text{Spec}(A)$ and $X = \text{Spec}(B)$ are affine, A being a regular ring and B a quotient of A -algebra of finite type, hence a quotient of a polynomial ring $C = A[T_1, \dots, T_n]$, and this last ring is regular (0, 17.3.7). One can therefore suppose that X is a closed sub-prescheme of a regular prescheme Y also satisfying conditions 1°, 2°, and 3° of (10.6.1); taking into account the remark (10.8.2), (ii), one is thus brought to the case where X is in addition regular and Noetherian. But as the local rings of X are then integral, the irreducible components of X are open (I, 6.1.10), and one can therefore also suppose X irreducible. Then, as the local rings of X are catenary (0, 16.5.12), the hypothesis 3°

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of (10.6.1) implies that X is biequidimensional ((5.1.5) and (0, 14.3.3)); it therefore suffices to apply (10.8.4). $\square$

One notes that if S is the spectrum of a field or of $\mathbf{Z}$, it satisfies the conditions of (10.8.5).

Corollary (10.8.6). — *The hypotheses being those of (10.8.5), for every $x \in X$, the number $\text{prof}_x^*(\mathcal{F})$ is the unique integer n having the following property: there exists an open neighbourhood U of x such that for every point $x' \in U \cap \overline{\{x\}}$, closed in U , we have $\text{prof}(\mathcal{F}_{x'}) = n$. In particular, for every x such that $\text{prof}_x^*(\mathcal{F}) \geq m$ (resp. $\text{prof}_x^*(\mathcal{F}) \leq m$), it is necessary and sufficient that there exist an open V of X such that, for every point $x' \in V \cap \overline{\{x\}}$, closed in V , we have $\text{prof}(\mathcal{F}_{x'}) \geq m$ (resp. $\text{prof}(\mathcal{F}_{x'}) \leq m$).*

Proof. Indeed, one can restrict to the case where x is not a closed point; if $\text{prof}_x^*(\mathcal{F}) = n$, the set of $y \in X$ such that $\text{prof}_y^*(\mathcal{F}) \leq n$ is closed by virtue of (10.8.5), hence contains $\overline{\{x\}}$, and by virtue of the lower semi-continuity of $\text{prof}_y^*(\mathcal{F})$, there exists an open neighbourhood U of x such that $\text{prof}_y^*(\mathcal{F}) = n$ for every $y \in U \cap \overline{\{x\}}$, hence $\text{prof}(\mathcal{F}_y) = n$ if $x' \in U \cap \overline{\{x\}}$ is closed (since the notion of rectified depth is local). $\square$

For preschemes satisfying the hypotheses of (10.8.5), the notion of rectified depth can therefore be defined by means of the values of the depth at the closed points of X (the latter forming a set which is very dense in every closed subset of X).

Proposition (10.8.7). — *Let S be a prescheme locally Noetherian satisfying conditions 1° , 2° , and 3° of (10.6.1). Let X be a prescheme locally of finite type over S , $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module, $Y = \text{Supp}(\mathcal{F})$. Then, for every $x \in Y$, we have*

$$(10.8.7.1) \quad \text{prof}_x^*(\mathcal{F}) = \dim_x(Y) - \text{coprof}(\mathcal{F}_x).$$

Proof. Indeed, by definition, $\text{coprof}(\mathcal{F}_x) = \dim(\mathcal{F}_x) - \text{prof}(\mathcal{F}_x)$, and it follows from (10.6.4) that $\dim_x(Y) = \dim(\overline{\{x\}}) + \dim(\mathcal{F}_x)$; whence (10.8.7.1) by definition of $\text{prof}_x^*(\mathcal{F})$. $\square$

Corollary (10.8.8). — *Under the hypotheses on f and $\mathcal{F}$ being those of (9.9.1), the function $x \mapsto \text{prof}_x^*(\mathcal{F}_{f(x)})$ is constructible.*

Proof. For every $s \in S$, the fibre X_s , being locally of finite type over $k(s)$, is a Jacobson prescheme (10.4.7). Since $\text{Spec}(k(s))$ satisfies the conditions of (10.6.1), one has

$$\text{prof}_x^*(\mathcal{F}_{f(x)}) = \dim_x(\text{Supp}(\mathcal{F}_{f(x)})) - \text{coprof}((\mathcal{F}_{f(x)})_x)$$

by (10.8.7). Now, if $Z = \text{Supp}(\mathcal{F})$, we have $Z_{f(x)} = \text{Supp}(\mathcal{F}_{f(x)})$ (I, 9.1.13) and Z is locally constructible (8.9.1); therefore the functions $x \mapsto \dim_x(\text{Supp}(\mathcal{F}_{f(x)}))$ and $x \mapsto \text{coprof}((\mathcal{F}_{f(x)})_x)$ are locally constructible ((9.9.1) and (9.9.3)), which proves the proposition. $\square$

10.9. Maximal spectra and ultra-preschemes The results of this section will not be used in what follows.

(10.9.1). Let X be a Jacobson prescheme, and let $S(X)$ be the *ringed space* whose underlying space is the subspace of closed points of X , and the sheaf of rings the sheaf induced on this subspace by $\mathcal{O}_X$, in other words the sheaf $\psi^*(\mathcal{O}_X)$, where $\psi : S(X) \rightarrow X$ denotes the canonical injection. As ψ is a quasi-homeomorphism, we have seen (10.2.8), (ii) that if $\theta : \mathcal{O}_X \rightarrow \psi_*(\mathcal{O}_{S(X)})$ is the homomorphism of sheaves of rings such that $\theta^\# : \psi^*(\mathcal{O}_X) \rightarrow \mathcal{O}_{S(X)}$ is the identity, then $j_x = (\psi, \theta)$ is a *quasi-isomorphism* of ringed spaces, and $\mathcal{F} \mapsto j_X^*(\mathcal{F})$ is an equivalence of the category of $\mathcal{O}_X$ -Modules and that of $\mathcal{O}_{S(X)}$ -Modules. It is clear that under this equivalence, locally free (resp. coherent) $\mathcal{O}_X$ -modules correspond to locally free (resp. coherent) $\mathcal{O}_{S(X)}$ -modules. Furthermore, if $\mathcal{L}$ is a locally free $\mathcal{O}_X$ -module and U is an open subset of X such that $\mathcal{L}|_U \simeq (\mathcal{O}_X|_U)^n$, then

$$j_X^*(\mathcal{L})|_{U \cap S(X)} \simeq (\mathcal{O}_{S(X)}|_{U \cap S(X)})^n.$$

(10.9.2). Let X, Y be two Jacobson preschemes and $f = (p, \lambda) : X \rightarrow Y$ a morphism *locally of finite type*. One has seen (10.4.7) that $p(S(X)) \subset S(Y)$ and by restriction of p to $S(X)$, one defines therefore

a continuous map $S(p) : S(X) \rightarrow S(Y)$. For every open V of Y , one defines by composition a homomorphism of rings

$$\Gamma(V \cap S(Y), \mathcal{O}_{S(Y)}) \longrightarrow \Gamma(V, \mathcal{O}_Y) \xrightarrow{\lambda_V} \Gamma(f^{-1}(V), \mathcal{O}_X) \longrightarrow \Gamma(f^{-1}(V) \cap S(X), \mathcal{O}_{S(X)})$$

where the two extreme isomorphisms have been defined in (10.9.1); it is clear that this defines a homomorphism of sheaves of rings $S(\lambda) : S(\mathcal{O}_{S(Y)}) \rightarrow \rho_*(\mathcal{O}_{S(X)})$ (recalling that the open subsets of $S(X)$ (resp. $S(Y)$) correspond bijectively to those of X (resp. Y) (10.2.1)); one obtains thus a morphism of ringed spaces $S(f) = (S(p), S(\lambda)) : (S(X), \mathcal{O}_{S(X)}) \rightarrow (S(Y), \mathcal{O}_{S(Y)})$ such that the diagram

$$\begin{array}{ccc} S(X) & \xrightarrow{S(f)} & S(Y) \\ j_X \downarrow & & \downarrow j_Y \\ X & \xrightarrow{f} & Y \end{array}$$

is commutative; furthermore, if Z is a third Jacobson prescheme and $g : Y \rightarrow Z$ a morphism locally of finite type, $S(g \circ f) = S(g) \circ S(f)$. One has thus defined a *covariant functor* $S : \mathbf{C} \rightarrow \mathbf{C}'$, where $\mathbf{C}$ is the category of *ringed spaces in local rings*, and $\mathbf{C}'$ the category whose objects are the *Jacobson preschemes* and the morphisms are the morphisms locally of finite type between Jacobson preschemes.

(10.9.3). Let us now determine the subcategory $\mathbf{C}''$ of $\mathbf{C}'$ formed by the ringed spaces isomorphic to $S(X)$. Suppose first that $X = \text{Spec}(A)$, where A is a Jacobson ring; then $S(X)$ is the set of *maximal* ideals of A , equipped: 1° with the topology induced by that of X , so that a base for this topology is formed by the $D^m(h) = D(h) \cap S(X)$, i.e. the set of maximal ideals $\mathfrak{m}$ of A such that $h \notin \mathfrak{m}$; 2° with the sheaf of rings $\mathcal{O}_{S(X)}$ such that $\Gamma(D^m(h), \mathcal{O}_{S(X)}) = A_h$. We will say that this ringed space is the *maximal spectrum* of the Jacobson ring A and denote it $\text{Spm}(A)$.

One notes that $j : D(h) \rightarrow X$ is the canonical injection, the ringed space induced on $D^m(h)$ by $S(X)$ and $S(D(h))$ are canonically isomorphic, and that the canonical injection $D^m(h) \rightarrow S(X)$ of ringed spaces is equal to $S(j)$.

Let B be a second Jacobson ring, $Y = \text{Spec}(B)$, $\varphi : B \rightarrow A$ a ring homomorphism making A a B -algebra of *finite type*, $f = ({}^a\varphi, \tilde{\varphi}) : X \rightarrow Y$ the corresponding morphism of preschemes, and

$$S(f) : \text{Spm}(A) \longrightarrow \text{Spm}(B)$$

the morphism of ringed spaces corresponding to f . It is clear that $S(f) = (\psi, \theta)$ is a morphism of ringed spaces in local rings, i.e. (Err₁₁, 1.8.2) for every $x \in \text{Spm}(A)$, $\theta_x^\sharp$ is a *local* homomorphism. Conversely:

Proposition (10.9.4). — *Let A, B be two Jacobson rings. If $u = (\psi, \theta) : \text{Spm}(A) \rightarrow \text{Spm}(B)$ is a morphism of ringed spaces in local rings such that $\Gamma(\theta) : B \rightarrow A$ makes A a B -algebra of finite type, there exists a morphism of preschemes $f : \text{Spec}(A) \rightarrow \text{Spec}(B)$ and a unique one such that $u = S(f)$.*

Proof. The uniqueness of f is evident, since if $f = ({}^a\varphi, \tilde{\varphi})$, one must have $\varphi = \Gamma(\theta)$; it remains to show that $S(f)$ is defined and that $u = S(f)$. Now, the first assertion follows from the fact that φ is supposed to make A a B -algebra of finite type, the fact that $\theta_x^\sharp$ is a local homomorphism for every $x \in \text{Spm}(A)$ allows one to repeat the argument of (I, 1.7.3), restricting to points x of $\text{Spm}(A)$, to show thus successively that $\psi(x) = {}^a\varphi(x)$ for every $x \in \text{Spm}(A)$, then that $\theta_x^\sharp = \varphi_x : B_{(x)} \rightarrow A_x$, which completes proving that $u = S(f)$. $\square$

(10.9.5). Consider now a ringed space $(X, \mathcal{O}_X)$; we will say that an open subset U of X is *ultra-affine* if the ringed space induced $(U, \mathcal{O}_X|_U)$ is isomorphic to a maximal spectrum $\text{Spm}(A)$, where A is a Jacobson ring. We will say that X is an *ultra-prescheme* if every point of X admits an ultra-affine open neighbourhood. One shows, as in (I, 2.1.3) and (2.1.4), that the ultra-affine open subsets form a *base* for the topology of X and that X is a Kolmogoroff space. If Y is a second ultra-prescheme, we will say that a morphism of ringed spaces $f : X \rightarrow Y$ is a *morphism of ultra-preschemes* if it satisfies the following conditions: 1° f is a morphism of ringed spaces in local rings; 2° for every $x \in X$, there is an ultra-affine open neighbourhood V of $f(x)$ in Y and an ultra-affine open U of x in X such

that $f(U) \subset V$ and the homomorphism $\Gamma(V, \mathcal{O}_Y) \rightarrow \Gamma(U, \mathcal{O}_X)$ corresponding to f makes $\Gamma(U, \mathcal{O}_X)$ a $\Gamma(V, \mathcal{O}_Y)$ -algebra of finite type.

It is immediate that one defines well the composite of two such morphisms, being a third such by the final remark of (10.9.3). It is clear that the category $\mathbf{C}_0''$ thus defined is a subcategory of $\mathbf{C}'$ that contains $\mathbf{C}''$; we now propose to show that $\mathbf{C}'' = \mathbf{C}_0''$, in other words:

Proposition (10.9.6). — *The functor $X \mapsto S(X)$ of $\mathbf{C}$ into $\mathbf{C}_0''$ is an equivalence of categories.*

Proof. 1° Let us first show that the functor $X \mapsto S(X)$ is *fully faithful*, in other words that for X, Y in $\mathbf{C}$, the canonical map

$$\mathrm{Hom}_{\mathbf{C}}(X, Y) \longrightarrow \mathrm{Hom}_{\mathbf{C}_0''}(S(X), S(Y))$$

is *bijective*. In the first place it is *injective*: let f, g be two morphisms locally of finite type of X into Y , and suppose that $S(f) = S(g)$; we have $f^{-1}(V) \cap S(X) = g^{-1}(V) \cap S(X)$ for every open V of Y , hence (10.3.1) and (10.2.7) $f^{-1}(V) = g^{-1}(V)$; it suffices therefore to prove that for every affine open V of Y , f and g coincide in $f^{-1}(V) = g^{-1}(V)$; in other words, one reduces to the case where $Y = \mathrm{Spec}(B)$ is the spectrum of a Jacobson ring B . It suffices (I, 2.2.4) to show that the homomorphisms of rings $B \rightarrow \Gamma(X, \mathcal{O}_X)$ corresponding to f and g are the same. Now, for every $s \in B$, the images of s under these homomorphisms are two sections of $\mathcal{O}_X$ over X which, by hypothesis, induce the *same* section over $S(X)$; we know (10.2.8) that these sections are identical, whence our assertion. In the second place let us prove that every morphism $h : S(X) \rightarrow S(Y)$ (for the category $\mathbf{C}_0''$) is of the form $S(f)$, where $f : X \rightarrow Y$ is a morphism locally of finite type. By hypothesis, there exists an ultra-affine open covering (U') (resp. (V'_μ)) of $S(X)$ (resp. $S(Y)$) such that for every λ , $h(U'_\lambda)$ is contained in one of the V'_μ and that it corresponds to the morphism $h_\lambda : U'_\lambda \rightarrow V'_\mu$ making the second ring an algebra of finite type over the first. For every couple of indices α, β , consider the open subset $U_{\alpha\beta}$ of U_α whose trace on $S(X)$ is $U'_\alpha \cap U'_\beta$; by virtue of 1°, the identical self-morphism of $U'_\alpha \cap U'_\beta$ is of the form $S(\theta_{\alpha\beta})$, where $\theta_{\alpha\beta} : U_{\alpha\beta} \rightarrow U_{\beta\alpha}$ is an isomorphism of preschemes. We see immediately (by virtue of 1°) that the family $(\theta_{\alpha\beta})$ satisfies the glueing condition of recollement (0, 4.1.7), and that this family defines a prescheme X' such that we have $X' = S(X)$; it is clear then that the U_α are identified with affine open subsets of X , such that the U'_α are the restrictions of $S(X)$ to the U_α .

2° It remains to prove that every ultra-prescheme X' is of the form $S(X)$ for a Jacobson prescheme X (which will necessarily be unique to isomorphism near, by virtue of 1°). There are ultra-affine open subsets (U'_α) of X' , each of which is of the form $S(U_\alpha)$, U_α being the spectrum of a Jacobson ring. For every couple of indices α, β , consider the open $U_{\alpha\beta}$ of U_α whose trace U'_α is $U'_\alpha \cap U'_\beta$; by virtue of 1°, the identical self-morphism of $U'_\alpha \cap U'_\beta$ is of the form $S(\theta_{\alpha\beta})$, where $\theta_{\alpha\beta} : U_{\alpha\beta} \rightarrow U_{\beta\alpha}$ is an isomorphism of preschemes. One easily verifies (by virtue of 1°) that the family $(\theta_{\alpha\beta})$ satisfies the glueing condition (0, 4.1.7), and that this family defines a prescheme X ; it is clear then that we have $X' = S(X)$, which completes the proof. $\square$

10.10. Algebraic spaces of Serre

(10.10.1). The language introduced by Serre in (FAC) is sometimes convenient, notably in questions where the main interest is attached to the rational points over the base field k (algebraically closed by hypothesis) of the “algebraic varieties” over k that one considers. We will sketch here this language by linking it to the considerations that precede, to allow the reader to translate the statements of Serre in the language of schemes. It is moreover possible to develop the language of Serre also for preschemes over a non-algebraically closed field (and even over an Artinian ring); but this introduces considerable technical complications, and moreover, on a field of arbitrary base, the advantages (above all psychological ones) of the point of view of Serre disappear; therefore we will confine ourselves to the framework fixed by Serre. The present section, just like the preceding one, will not be used in the sequel of this Treatise, and we will therefore content ourselves with brief indications.

(10.10.2). Given an ultra-prescheme R , one can naturally (as in the whole category) define the notion of *R -ultra-prescheme*. Consider in particular an algebraically closed field k ; $\mathrm{Spm}(k)$ is then identical to

$\text{Spec}(k)$; we will say that a $\text{Spm}(k)$ -ultra-prescheme X' is a *prealgebraic space over k* : it is a ringed space in local rings whose every point has an open neighbourhood isomorphic to the maximal spectrum of a k -algebra of finite type; it follows that it is the same as saying (by (10.9.6)) that $X' = S(X)$, where X is a k -prescheme locally of finite type. If X, Y, Z are three k -preschemes locally of finite type, it is the same as to say of $X \times_k Y$ (I, 1.3.4), hence the same is true of $X' \times_k Y'$, in the category of prealgebraic spaces over k (and also of the “fibre product” $X' \times_{k'} Y'$, where X', Y', Z' are three prealgebraic spaces over k). One can therefore define the diagonal morphism $\Delta_{X'} : X' \rightarrow X' \times_k X'$ (which is none other than $S(\Delta_X)$); one says that X' is an *algebraic space over k* if X is a scheme, and it amounts to the same as saying that the image of $\Delta_{X'}$ is a *closed* subset of $X' \times_k X'$.

(10.10.3). The simplifications that come from the hypothesis that k is algebraically closed amount first of all to this: for every k -prescheme X locally of finite type, there is a bijective correspondence between *closed points* of X_k (i.e. the *points of X with values in k* (I, 3.4.4)) and *rational points of X over k* (I, 3.4.5), by virtue of (I, 6.4.2). This shows in particular (I, 3.4.3.1) that for two k -prealgebraic spaces X', Y' , the underlying set product $X' \times Y'$ is identical to the underlying set of the product of the underlying topological spaces of X' and Y' (but it is well understood that the topology of the latter is not the *product* topology of the topologies of the underlying spaces, in general it is strictly finer than the latter).

On the other hand, the local rings $\mathcal{O}_x$ at the points of a prealgebraic space X' over k are k -algebras whose residue field is *isomorphic to k* ; if A and B are two such k -algebras, a k -homomorphism $\varphi : A \rightarrow B$ is necessarily *local*: indeed, if an element x of the maximal ideal A were such that $\varphi(x)$ is invertible, there would exist $\lambda \in k$ nonzero such that $\varphi(x - \lambda.1)$ belongs to the maximal ideal of B , which is absurd since $x - \lambda.1$ is invertible. We conclude immediately from this that if $X' = S(X)$, $Y' = S(Y)$ are two prealgebraic spaces over k , every morphism $X' \rightarrow Y'$ of k -ringed spaces is also a morphism of ringed spaces *in local rings*, hence every morphism of k -preschemes $X' \rightarrow Y'$ of k -ringed spaces is also, in the notations above, of the form $S(f)$, where $f : X \rightarrow Y$ is a morphism of k -preschemes.

Finally, for every open U of X' , and every $x \in U$, $s(x)$ (0, 5.5.1) is identified with an element of U in k , in other words a map $\bar{s} : x \rightarrow s(x)$ of U in k ; as the map $h_U : s \rightarrow \bar{s}$ is evidently a homomorphism of rings $\Gamma(U, \mathcal{O}_{X'}) \rightarrow \Gamma(U, \mathcal{A}(X'))$ (the *germs of maps of X' in k*) and commutes with restrictions to a neighbourhood $V \subset U$, the h_U define a homomorphism of *sheaves* of rings $h : \mathcal{O}_{X'} \rightarrow \mathcal{A}(X')$. If one takes U to be ultra-affine, $\text{Spm}(A)$, where A is a Jacobson ring, to say that $\bar{s} = 0$ means that for every maximal ideal $\mathfrak{m}$ of A , s belongs to $\mathfrak{m}$, or equivalently that s is in the *radical* of A ; but as A is a Jacobson ring, its radical is its nilradical, hence $\bar{s} = 0$ if and only if s is nilpotent.

(10.10.4). One says that the k -prealgebraic space $X' = S(X)$ is *reduced* if it is reduced at each of its points, i.e. the set of points $x \in X$ where X is reduced is open (0, 5.2.2), its complement contains at least one closed point x' of X ; as each of the local rings $\mathcal{O}_{x'}$ is reduced, one sees in (10.10.3) that for the homomorphism $h : \mathcal{O}_{X'} \rightarrow \mathcal{A}(X')$ to be *injective*, it is necessary and sufficient that X' be *reduced*. In (FAC), Serre restricts himself in fact to prealgebraic *reduced* spaces, which allows him to define $\mathcal{O}_{X'}$ as a *sub-sheaf* of $\mathcal{A}(X')$. One notes that if X' and Y' are two k -spaces prealgebraic reduced, it is the same of $X' \times_k Y'$: indeed, this amounts to saying that if A and B are two reduced k -algebras of finite type, $A \otimes_k B$ is also reduced; but we have just seen that the radicals of A and B are 0, and as k is algebraically closed, A and B are “separable” algebras over k in the sense of Bourbaki (Alg., chap. VIII, §7, n° 5, prop. 5); hence $A \otimes_k B$ is without radical (*loc. cit.*, n° 6, cor. 3 of th. 3), and since it is a Jacobson ring, it is reduced. However, if Z' is a third prealgebraic space reduced over k , the “fibre product” $X' \times_{Z'} Y'$ of two prealgebraic reduced spaces over Z' is not in general reduced, which implies that the category of these spaces is insufficient in many questions (notably in the theory of algebraic groups). But as we have seen above, one can keep the language of Serre without restricting oneself, as this latter (who in addition considers only quasi-compact *prealgebraic* spaces), to the case of reduced prealgebraic spaces.